

SECTION **7.0**

GENERAL SURGERY

CHAPTER 7.1

Esophageal Surgery

SURGEONS

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ESOPHAGOSTOMY

SURGICAL CONSIDERATIONS

Description

Cervical esophagostomy is performed to divert oral secretions away from the esophagus to a stoma in certain types of more distal esophageal perforation. To perform an esophagostomy, the esophagus is approached through a left cervical incision. The sternocleidomastoid muscle and carotid sheath are retracted laterally and the thyroid medially, exposing the cervical esophagus ([Fig. 7.1-1](#)). The esophagus is mobilized with care being taken not to injure the left recurrent laryngeal nerve, which typically lies in the tracheoesophageal groove. The esophagus is brought to the skin surface as a loop or end stoma and sutured to the skin with absorbable sutures.

Variant procedure or approaches

The procedure is usually performed via a left cervical approach; however, the right side is an alternative.

Usual preop diagnosis

Distal esophageal perforation; distal esophageal obstruction

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

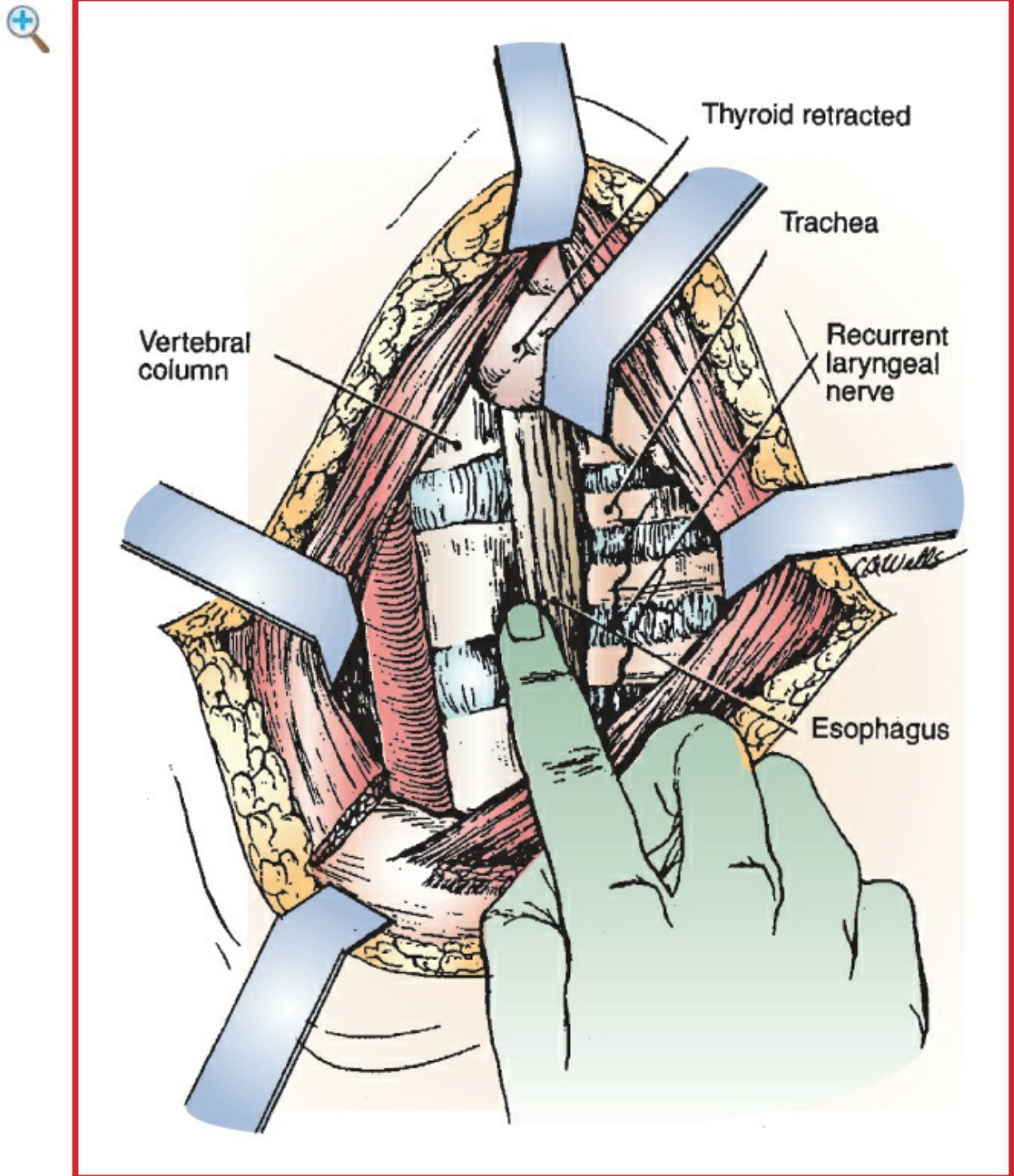


Figure 7.1-1 Surgical anatomy for cervical esophagostomy. (Reprinted from Nora PF, ed: Operative Surgery Principles and Techniques, 3rd edition. WB Saunders, Philadelphia: 1990. Copyright © 1990 Elsevier. With permission.)

■ SUMMARY OF PROCEDURE

Position	Supine, with head rotated to right
Incision	Cervical
Antibiotics	Cefazolin 2 g iv preop
Surgical time	45 min
EBL	25–50 mL
Postop care	Stoma pouch to collect saliva PACU → ward
Mortality	<0.1%
Morbidity	Skin irritation: 15–20% Saliva leakage: 5–10% Wound infection: <5% Recurrent laryngeal nerve injury: 1–2%
Pain score	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–60 yr
Male:Female	1:1
Incidence	Not uncommon
Etiology	Surgically created
Associated conditions	Esophageal perforation; pharyngeal cancer

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Esophageal Surgery following Esophagectomy.

Suggested Readings

1. Carrott PW Jr, Low DE: Advances in the management of esophageal perforation Thorac Surg Clin 2011; 21(4):541–55.
2. Sdralis EIK, Petousis S, Rashid F, et al: Epidemiology, diagnosis, and management of esophageal perforations: systematic review. Dis Esophagus 2017; 30(8):1–6.

ESOPHAGEAL DIVERTICULECTOMY

SURGICAL CONSIDERATIONS

Description

Esophageal diverticula are divided into three anatomic types: pharyngoesophageal (Zenker's), midesophageal, and epiphrenic. Structurally, they are either “true” diverticula—meaning they consist of all three layers of the esophageal wall (mucosa, submucosa, and muscularis)—or “false” diverticula consisting of only mucosa (or mucosa and submucosa). Pharyngoesophageal diverticula are the most common, accounting for 60–65% of all cases. These are false diverticula that originate in Killian triangle, a weak point in the posterior wall of the esophagus, just proximal to the transverse fibers of the cricopharyngeal muscle ([Fig. 7.1-2A](#)). They are associated with incomplete, or discoordinate, upper esophageal sphincter relaxation, and the resultant increased hypopharyngeal pressure produces a narrow-mouthed posterior diverticulum. These diverticula frequently present in the seventh decade of life and are 2–3 times more common in men. Symptoms depend on the stage of the disease. Early on, patients may complain of vague pharyngeal sensations, dysphagia, cough, and excessive salivation. Later, more severe symptoms—such as severe (or frequent) dysphagia, regurgitation of food, halitosis, voice changes, aspiration, and odynophagia (painful swallowing)—may occur.

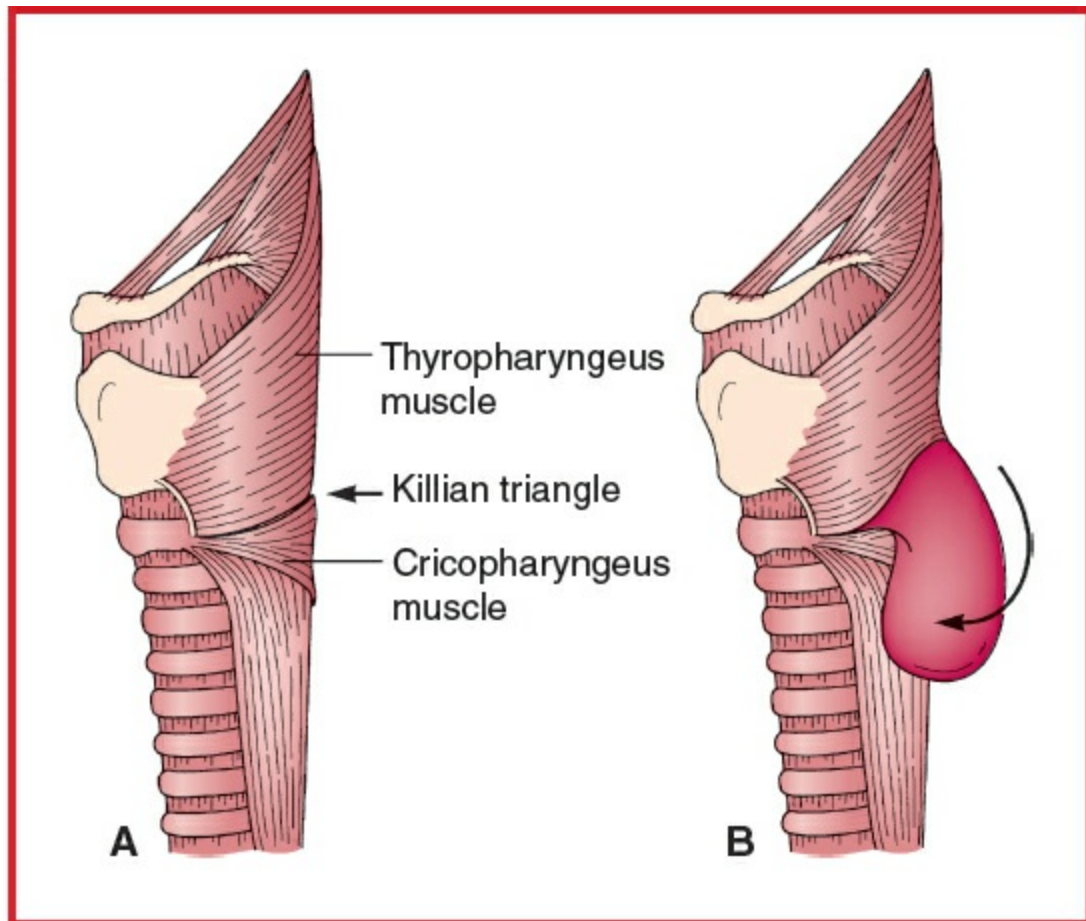


Figure 7.1-2 Formation of Zenker diverticulum. **A:** Herniation of the pharyngeal mucosa and submucosa occurs at the point of potential weakness (Killian triangle [arrow]) between the oblique fibers of the thyropharyngeus muscle and the more horizontal fibers of the cricopharyngeus muscle. **B:** As the diverticulum enlarges, it drapes over the cricopharyngeus sphincter and descends into the superior mediastinum in the prevertebral space. (Reprinted from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice* 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001. After Orringer MB: Diverticula and miscellaneous conditions of the esophagus. In: Sabiston DC Jr, ed. *Textbook of Surgery*, 13th edition. WB Saunders, Philadelphia: 1986. Copyright © 1986 Elsevier. With permission.)

Surgery is the only effective therapy for Zenker diverticulum. Respiratory complications (aspiration) or nutritional deficiencies (weight loss) may be directly attributable to the diverticulum and should not be contraindications to surgery. Multiple different operative approaches are advocated: diverticulectomy alone, cricopharyngeal myotomy, diverticulectomy with myotomy, and myotomy with suspension of the diverticulum. Myotomy alone, which corrects the underlying physiologic abnormality, is up to 78% effective and may be considered for patients with small (<2 cm) diverticula. Diverticulectomy or suspension should be added if the diverticulum itself is large or dependent. Both procedures are performed via a left cervical incision (Fig. 7.1-3 inset) and are associated with a low rate of recurrence and complications. The upper esophagus is exposed by retracting the sternocleidomastoid muscle and carotid sheath

laterally and the thyroid gland medially. The diverticulum is located in the prevertebral space. Care is taken not to injure the recurrent laryngeal nerve. Following excision of the diverticulum, a cricopharyngeal myotomy may be performed, starting on the upper esophagus and extending across the cricopharyngeal muscle near the neck of the diverticulum and on to the inferior pharyngeal constrictor muscle.

Recent emphasis has been placed on endoscopic treatment of Zenker diverticulum (Dohlman procedure). In this procedure, a modified laryngoscope, tissue coagulator, and endoscopic stapler are used to divide the common wall between diverticulum and true esophageal lumen. This is advantageous because it does not require an incision and it does not injure the recurrent laryngeal nerve. It has similar results to open procedures.

Midesophageal diverticula, by definition, occur in the middle 3rd of the esophagus. These “true” diverticula typically arise in the setting of mediastinal granulomatous disease whereby a fibrotic reaction around inflamed mediastinal lymph nodes results in traction on the muscular wall of the esophagus. Diverticula usually arise within 4–5 cm of the carina and comprise an estimated 10–17% of all esophageal diverticula. Most midesophageal diverticula are asymptomatic and do not require surgical intervention. In cases that require intervention because of either regurgitation or development of an esophagobronchial fistula, the approach is through a right thoracotomy with excision of the inflammatory mass. Primary closure of the fistula and the interposition of viable tissue, such as muscle, should be performed.

Epiphrenic diverticula arise in the distal 10 cm of the esophagus and may be related to an underlying esophageal motility disorder. These false diverticula are most commonly present in the 6th decade. The clinical presentation is variable; most patients have symptoms related to their underlying dysmotility syndrome: dysphagia, chest pain, or regurgitation. Most patients with epiphrenic diverticula are asymptomatic, and there appears to be no relation between size of the diverticulum and symptoms. Surgery for epiphrenic diverticula typically consists of diverticulectomy with myotomy either through a left thoracotomy (Fig. 7.1-4) or via laparoscopy. With the transthoracic approach, a low, left thoracotomy is used, the esophagus is mobilized and encircled, and the diverticulum is mobilized and excised. A myotomy should be performed opposite the diverticulectomy and should extend proximally above the diverticulum and distally onto the stomach. Because there is, by definition, an underlying motility disorder, the myotomy should be carried onto the stomach, and a nonobstructing fundoplication may be added to prevent significant postoperative reflux.

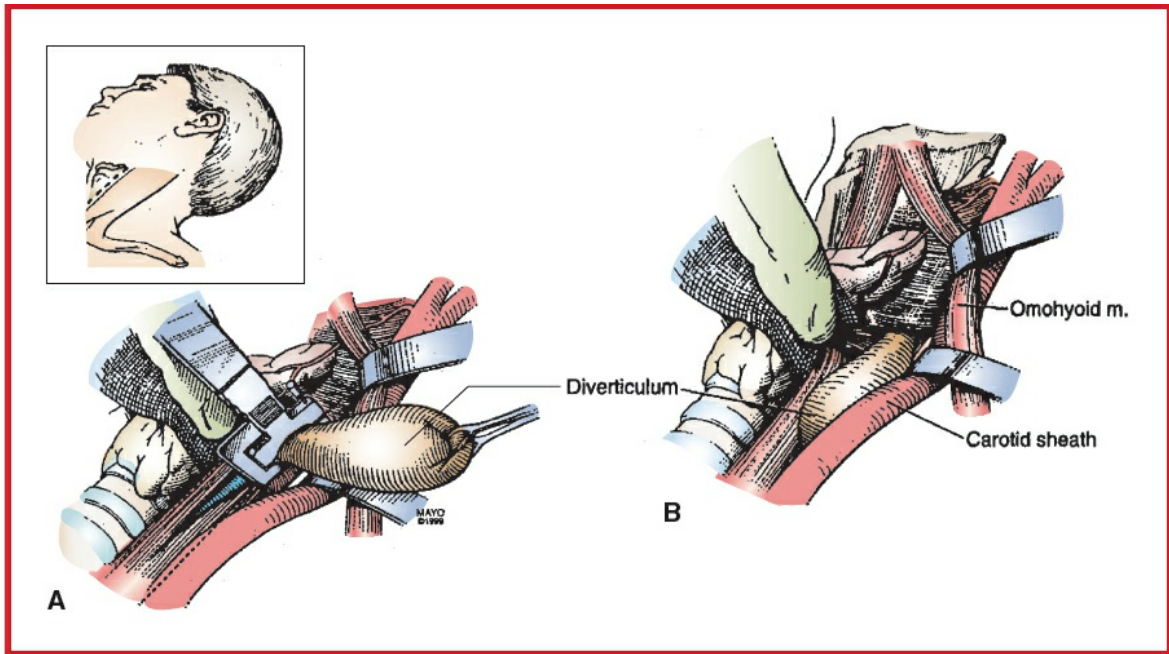


Figure 7.1-3 Zenker diverticulum approached through a left cervical incision (**inset**) . **A:** The diverticulum is grasped, and a cricopharyngeal myotomy is extended onto the upper esophagus. **B:** The base of the diverticulum is stapled, and the diverticulum is resected. (Reproduced with permission from Shields TW, LoCicero J III, Ponn RB General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

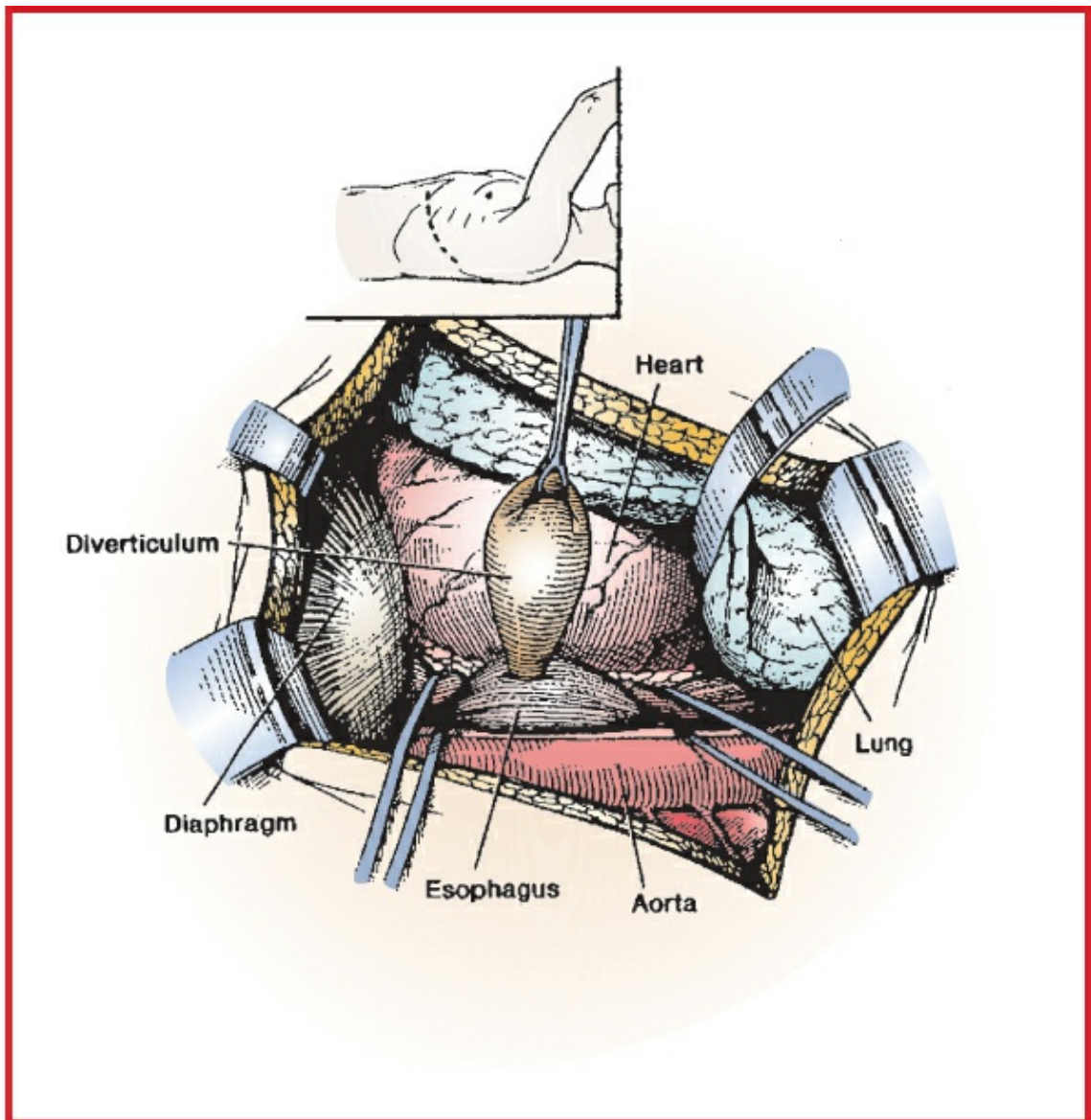


Figure 7.1-4 Epiphrenic diverticulum. Through a left thoracotomy, the diverticulum is mobilized and resected. A contralateral myotomy is then created and extended distally to the stomach to eliminate any functional obstruction secondary to the preexistent dysmotility. (Reproduced with permission from Shields TW, LoCicero J III, Ponn RB General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

Variant procedure or approaches

Laparoscopic diverticulectomy and myotomy has gained increasing acceptance, and reported outcomes are similar to those obtained with the open procedure. The surgical approach is similar to that used during laparoscopic fundoplication. Dissection of the diverticula may be facilitated by the passage of a bougie or video endoscope. After the diverticulum is amputated using an endoscopic stapler, a myotomy is performed opposite the diverticula, and a partial fundoplication is fashioned. Diverticulectomy can also be performed using a robotic approach through the abdomen or through the chest.

Usual preop diagnosis

Esophageal diverticulum

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Hypopharyngeal	Epiphrenic
Position	Supine	Right lateral decubitus
Incision	Left, right cervical, endoscopic	Left thoracotomy
Special instrumentation	None	Chest retractor
Unique considerations	Care not to injure recurrent laryngeal nerve	10 cm myotomy
Antibiotics	Cefazolin 2 g iv preop	⇐
Surgical time	1–2 h	⇐
Closing considerations	Inspect for perforation	⇐
EBL	50–100 mL	100–200 mL
Postop care	PACU → ward	ICU × 1–2 d
Mortality	<1%	⇐
Morbidity	Recurrent nerve paralysis: <5% Temporary phonetic problems: <5% Esophageal stricture: <3% Esophageal fistula: <2%	Atelectasis: 5–10% Esophageal perforation: <2% <2%

Pain score 6–7 (<2 with endoscopic) 7–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	35–90 yr (50% of patients >70 yr)
Male:Female	2:1
Incidence	Uncommon
Etiology	Uncoordinated cricopharyngeal muscle and lower esophageal sphincter; weakness of esophageal wall
Associated conditions	Cachexia (25–30%); hiatus hernia with or without reflux (25%); chronic pulmonary infection (15–20%); aspiration (30–40%)

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Esophageal Surgery following Esophagectomy.

Suggested Readings

1. Dzeletovic I, Ekborn DC, Baron TH: Flexible endoscopic and surgical management of Zenker's diverticulum. *Expert Rev Gastroenterol Hepatol* 2012; 6(4):449–65.
2. Ferreira LE, Simmons DT, Bain TH: Zenker's diverticula: pathophysiology, clinical presentation, and flexible endoscopic management. *Dis Esophagus* 2008; 21:1–8.
3. Ishaq S, Sultan H, Siau K, et al: New and emerging techniques for endoscopic treatment of Zenker's diverticulum: state-of-the-art review. *Dig Endosc* 2018; 30(4):449–60.
4. Johnson CM, Postma GN: Zenker diverticulum—which surgical approach is superior? *JAMA Otolaryngol Head Neck Surg* 2016; 142(4):401–3.

MANAGEMENT OF ESOPHAGEAL PERFORATION

■ SURGICAL CONSIDERATIONS

Description

Esophageal perforation may be spontaneous, instrumental (iatrogenic), traumatic, or 2° intrinsic esophageal disease. Spontaneous (or emetogenic) perforation most commonly occurs in the lower 3rd of the esophagus. Instrumental perforations may occur at any level, but are most common just above the cardia and in the cervical esophagus. The

level of traumatic perforation depends on the location of the penetrating wound. Symptoms of esophageal perforation at the cricopharyngeal sphincter include neck pain, fever, and crepitations in the substernal and neck areas. Perforation in the mediastinum may result in hydropneumothorax, mediastinitis, fever, and substernal pain. Cervical perforations are managed with antibiotics and drainage in the cervical area. Therapy for intrathoracic perforation generally requires emergent operation. Surgical options include primary repair, drainage and diversion, placement of temporary covered stents, and esophageal resection. The optimal choice depends on the nature and duration of the perforation as well as the clinical condition of the patient. Spontaneous perforations are often amenable to primary repair—either through the abdomen or the left chest. Patients suffering from iatrogenic perforation incurred during dilation of a malignant or nondilatable stricture may require urgent esophagectomy. Patients with delayed recognition of a perforation may be hemodynamically unstable and may only tolerate drainage and diversion (generally through a cervical esophagostomy).

Variant procedure or approaches

Cervical or right thoracic drainage is indicated when the perforation occurs in the neck or high in the mediastinum.

Usual preop diagnosis

Esophageal perforation

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Left Thoracotomy	Cervical or Thoracic Drainage
Position	Right lateral decubitus	Supine or right lateral decubitus
Incision	Left thoracotomy	Cervical or right chest
Antibiotics	Zosyn (piperacillin and tazobactam) 3.375 g iv q6h	⇐

VTE prophylaxis	Heparin 5000 units sq	⇐
Surgical time	2–4 h	1 h
Closing considerations	Chest drain	Cervical or thoracic drain
EBL	100–200 mL	50–100 mL
Postop care	Chest tube to suction; PACU → ward	⇐
Mortality	5–10%	2–5%
Morbidity	Pneumonia: 5–10% Esophageal leak: 2–5% Pericarditis: 1–3%	— 10% —
Pain score	8–10	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	Variable: 20–80 yr
Male:Female	1:1
Incidence	1 in 8000 admissions
Etiology	Instrumental (endoscopy, dilatation, intubation); traumatic (penetrating, foreign body, caustic agents); intrinsic disease (carcinoma, peptic ulceration); spontaneous
Associated conditions	Esophageal stricture (75%); cancer (25%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Esophageal Surgery following Esophagectomy.

Suggested Readings

1. Carrott PW Jr, Low DE: Advances in the management of esophageal perforation Thorac Surg Clin 2011; 21(4):541–55.
2. Schembre DB: Recent advantages in the use of stents for esophageal disease Gastrointestinal Endosc Clin N Am 2010; 20:103.

ESOPHAGOMYOTOMY

SURGICAL CONSIDERATIONS

Description

Esophagomyotomy is performed for achalasia and other motility disorders to facilitate esophageal emptying into the stomach. Most authors recommend incising the muscular layer of the distal esophagus and continuing down across the gastroesophageal junction for at least 1 cm (Heller myotomy). The muscle is dissected back from the mucosa so that roughly 180° is exposed (Fig. 7.1-5). The distal esophagus is mobilized either from below the diaphragm or via a left thoracic approach. Care is taken not to injure the vagus nerves.

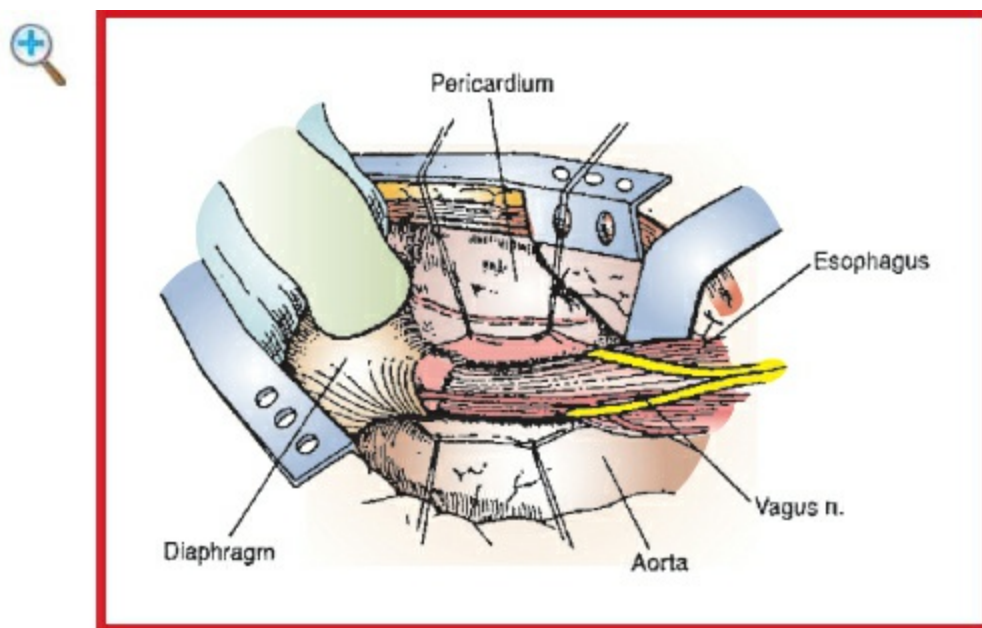


Figure 7.1-5 Surgical anatomy for esophagomyotomy. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

When approached from the abdomen, the esophagus is exposed by incising the gastroesophageal ligament. The distal esophagus is mobilized and pulled downward to perform the myotomy. The procedure is usually done laparoscopically.

Variant procedure or approaches

The procedure usually is performed through a left thoracotomy, but some surgeons prefer a transabdominal approach. Laparoscopic or thoracoscopic approaches can be used to perform esophagomyotomy.

Peroral endoscopic myotomy (POEM) for achalasia has been developed somewhat

recently as an endoscopic alternative to surgical myotomy. This approach involves no external surgical incisions. The procedure is done completely endoscopically, and involves incising the proximal esophageal mucosa, creating a submucosal tunnel that is carried distally and past the gastroesophageal junction, and then performing a myotomy with division of the circular fibers of the muscularis propria. The intentionally created proximal mucosal defect is then closed with sutures or clips. This approach has been shown to be done with very low morbidity even early in a clinician's learning curve, with significant improvement in dysphagia or regurgitation symptoms that are at least similar to surgical myotomies. The main downside of the procedure is a high incidence of postoperative reflux.

Usual preop diagnosis

Achalasia; diffuse esophageal spasm; nutcracker esophagus

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES				
	Thoracic Approach	Abdominal Approach	Laparoscopic Approaches	Endoscopic Approaches
Position	Right lateral decubitus	Supine	Supine	Supine
Incision	Left thoracotomy in 6th interspace	Midline upper abdomen	Multiple small abdominal incisions	None
Special instrumentation	Chest retractor	Denier retractor	Laparoscopic instrumentation	Endoscopic instrumentation
Unique considerations	Care should be taken to avoid extending gastric myotomy too far to prevent esophageal reflux.	←	Legs spread apart (laparoscopic approach)	None
Antibiotics	Cefazolin 2 g iv preop	←	←	←
VTE prophylaxis	Heparin 5000 units sq	←	←	←

Surgical time	1–2 h	⇐	⇐	90 min for experienced surgeons; 2–4 h early in a clinician's experience
Closing considerations	✓ for perforation. Consider fundoplication to prevent esophageal reflux.	⇐	⇐	⇐
EBL	150–200 mL	100–150 mL	50 mL	Minimal
Postop care	ICU × 1–2 d	PACU → room	⇐	⇐
Mortality	<0.5%	⇐	⇐	
Morbidity	Esophagitis: ≤20% Transient dysphagia/ reflux: 5% Esophageal leak: 1–2%	⇐ ⇐ ⇐	⇐ ⇐ ⇐	Perforation 2% Reflux 20%
Pain score	7–9	6–8	3–4	Minimal

■ PATIENT POPULATION CHARACTERISTICS

Age range	30–50 yr
Male:Female	1:1
Incidence	0.6/100,000
Etiology	Neuromuscular disorder of unknown etiology often characterized by absence of ganglion cells of Auerbach plexus
Associated conditions	Predisposition to development of carcinoma (1–20%); pulmonary complications 2° aspiration (5–10%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Esophageal Surgery following Esophagectomy.

Suggested Readings

- Schlottmann F, Luckett DJ, Fine J, et al: Laparoscopic Heller myotomy versus peroral endoscopic myotomy (POEM) for achalasia: a systematic review and meta-analysis. *Ann Surg* 2018; 267(3):451–60.
- Zhu ZJ, Chen LQ, Duranceau A: Long-term results of total vs. partial fundoplication after esophagomyotomy for primary esophageal motor disorders. *World J Surg* 2008; 32:401–7.

ESOPHAGOGASTRIC FUNDOPLASTY

SURGICAL CONSIDERATIONS

Description

Esophagogastric fundoplasty represents a variety of operations designed to prevent esophageal reflux by wrapping the fundus of the stomach around a 3- to 4-cm segment of the lower esophagus. This fundal wrapping acts to reinforce the lower esophageal sphincter. Surgery may be performed transabdominally, transthoracically, or laparoscopically, depending on surgeon's preference. The most common approach is the open or laparoscopic Nissen fundoplication, wherein the anterior and posterior walls of the stomach are sutured together around the lower esophagus with nonabsorbable sutures (Fig. 7.1-6A). This is accomplished by incising the gastrosplenic ligament and ligating three or four short gastric vessels. Care must be taken not to injure the spleen or vagus nerves during the repair.

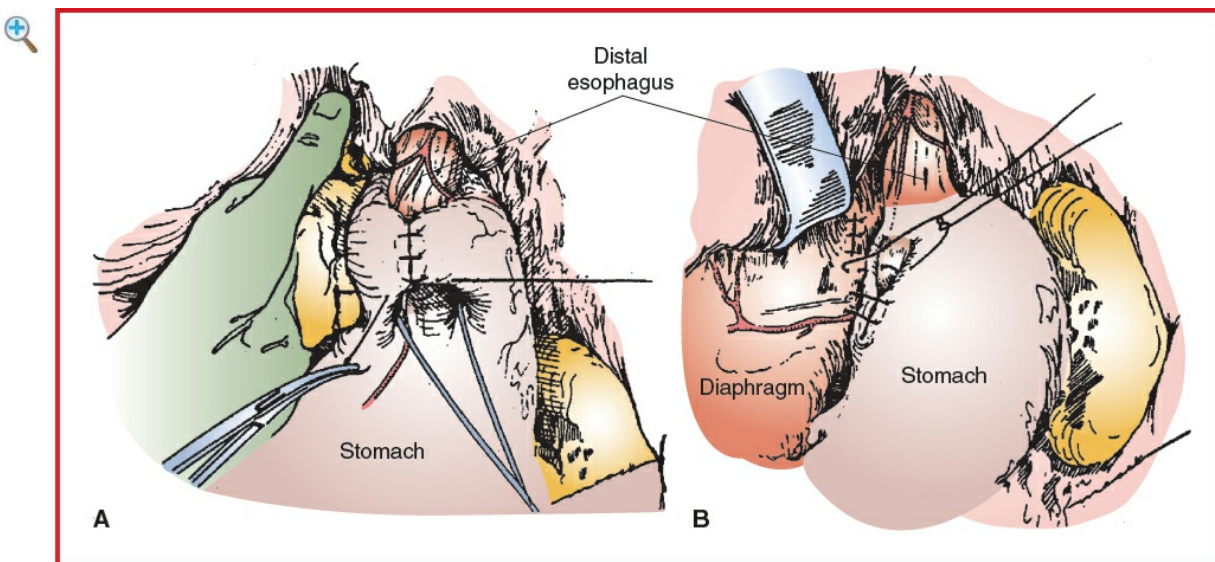


Figure 7.1-6 **A:** Nissen fundoplication may be performed via either the transabdominal or transthoracic approach. **B:** Hill repair is performed through the abdomen. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Variant procedure or approaches

Modifications of the **Nissen** fundoplication include the **Toupet** procedure, a posterior partial fundoplication, and the **Hill** procedure, in which the gastroesophageal junction is sutured to the median arcuate ligament of the diaphragm or to the preaortic fascia (Fig. 7.1-6B). Compared to the Nissen fundoplication, which involves a complete 360° wrap, these alternative techniques involve a partial fundoplication of 270° or less. Partial

wraps can be more appropriate than a complete wrap in patients with esophageal dysmotility, where a complete wrap will prevent reflux but also may be associated with postprocedure dysphagia. Another modification is the **Belsey Mark IV** repair done through the left chest, in which there is a 240° semifundoplication between the stomach and esophagus, making it easier for the patient to overcome the resistance of the wrap. There are proponents of each repair, although the Nissen fundoplication remains the standard to which others are compared. The laparoscopic approach is widely used, although the left transthoracic approach provides excellent exposure for either Nissen or Belsey repairs. The Belsey repair is associated with excellent outcomes but requires significantly more recovery than a laparoscopic Nissen and is now generally reserved for patients who have recurrent reflux after a previous repair or repairs. In some patients who are felt to have a shortened esophagus associated with their reflux, an esophageal lengthening procedure with gastroplasty will often be performed with fundoplasty to reduce the chance of postsurgical reflux recurrence.

Usual preop diagnosis

Sliding hiatus (hiatal) hernia; paraesophageal hiatus hernia; or free reflux

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

■ SUMMARY OF PROCEDURES

	Nissen (Toupet) Fundoplication	Laparoscopic Nissen	Hill Procedure	Belsey Mark IV
Position	Supine	Supine, legs apart	⇐	Right lateral decubitus
Incision	Midline abdominal or laparoscopic ports	Laparoscopic ports	Midline abdominal	Left posterolateral thoracotomy
Special instrumentation	No. 40–50 Hurst dilators; NG tube	Laparoscopic instrumentation	NG tube	Chest retractor; NG tube
Unique considerations	Fundoplication should be loose; parietal cell vagotomy performed if peptic ulcer disease present. Fundoplication may be limited to 180–280° posterior wrap (Toupet).	CO ₂ insufflation		⇐ 240° semifundoplication
Antibiotics	Cefazolin 2 g iv	⇐	⇐	⇐
VTE prophylaxis	Heparin 5000 units sq	⇐	⇐	⇐
Surgical time	1–2 h	⇐	⇐	⇐
Closing considerations	Inspect spleen for bleeding	⇐	⇐	⇐
EBL	100–150 mL	50 mL	100–150 mL	100–200 mL
Postop care	PACU → ward	⇐	⇐	ICU × 1–2 d
Mortality	<0.5%	⇐	⇐	⇐
Morbidity	Recurrent hernia: 20% Gas-bloat syndrome: 10–20% Temporary dysphagia: 5–10% Gastric fistula: <2%	⇐ ⇐ ⇐ ⇐	⇐ <5% 5% ⇐	⇐ ⇐ 2% ⇐
Pain score	6–8	2–4	7–8	7–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	46–60 yr
Male:Female	1:2
Incidence	Not uncommon
Etiology	Esophagogastric reflux (100%); esophageal hiatus hernia (80–90%)
Associated	Diverticulosis of colon (30–35%); cholelithiasis (25–

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Esophageal Surgery following Esophagectomy.

Suggested Readings

1. Joubert KD, Betzold RD, Steliga MA: Successful treatment of esophageal necrosis secondary to acute type B aortic dissection. *Ann Thorac Surg* 2016; 102(6):e547–9.
2. Patel A, Young LB, Rundback JH: Percutaneous esophagogastrostomy creation for gastric bypass reversal. *J Vasc Interv Radiol* 2016; 27(10):1552–3.
3. Weber C, Davis CS, Shankaran V, et al: Hiatal hernias: a review of the pathophysiologic theories and implication for research. *Surg Endosc* 2011; 25(10):3149–53.

ESOPHAGECTOMY

SURGICAL CONSIDERATIONS

Description

Esophagectomy is most commonly performed for malignant disease of the middle and lower 3rd of the esophagus and gastric cardia. This procedure also may be indicated for intractable benign stricture, Barrett esophagus with in situ multifocal adenocarcinoma, and end-stage achalasia. There are several surgical options for esophageal resection, including the Ivor Lewis approach, which involves a laparotomy and right thoracotomy; the transhiatal approach, whereby the esophagus is mobilized through abdominal and neck incisions; the left thoracoabdominal approach; and the McKeown or three-field approach that involves esophageal mobilization in the right chest along with abdominal and neck incisions ([Fig. 7.1-7](#)). Although there are advantages and disadvantages to each, the final result is to use a portion of the stomach to replace the esophagus. In all approaches, the stomach is mobilized while preserving its blood supply from the right gastroepiploic and gastric arteries. The stomach is then transposed into the chest, and a gastroesophageal anastomosis is fashioned either in the chest (Ivor Lewis and left chest approaches) or in the neck (transhiatal approach and three-field approach). To avoid delayed gastric emptying, a pyloroplasty/pyloromyotomy or pyloric Botox® injection is often added, as is placement of a temporary jejunal feeding tube. Regardless of the surgical technique, patients with larger or locally advanced tumors may have received preoperative chemotherapy or radiation. In these patients, the combination of less

distinct tissue planes and radiation-induced inflammation tends to lead to increased bleeding and/or increased insensible fluid losses.

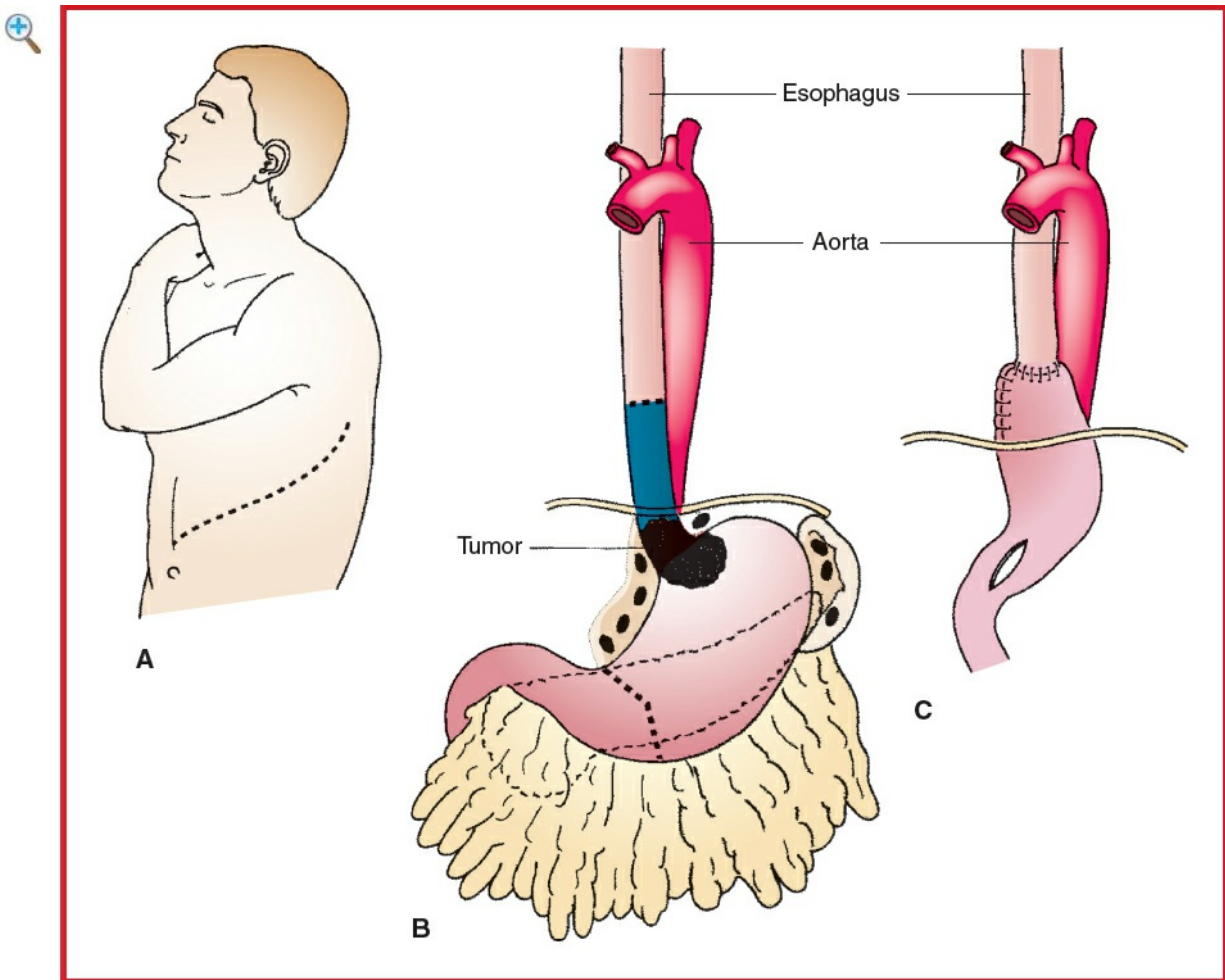


Figure 7.1-7 Standard thoracoabdominal esophagogastrectomy for carcinomas of the distal esophagus and cardia. **A:** Thoracoabdominal incision. **B:** Tissue to be resected (darker area). **C:** Completed reconstruction after intrathoracic esophagogastric anastomosis and either pyloromyotomy or pyloroplasty to prevent postvagotomy pylorospasm. (Reproduced with permission from Greenfield LJ, Mulholland, MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

The surgical approach for esophageal cancer depends on tumor location, patient comorbidities, and surgeon experience. These approaches up until recently have been performed through open incisions, with postoperative major morbidity rates, particularly pulmonary morbidity, ranging from 27% to 57% and mortality rates ranging from 2% to 14%, depending on hospital volume. Recent advances in laparoscopic and thoroscopic technologies have given rise to the minimally invasive esophagectomy (MIE), which has the potential benefit of reducing perioperative morbidity and increasing overall utilization of surgical resection of esophageal cancer. However, the benefits of MIE have not been very well characterized. Most single-center or

collaborative multicenter studies that have compared morbidity, mortality, and oncologic efficacy of MIE relative to open esophagectomies have shown that MIE slightly decreases intensive care and total hospital lengths of stay, reduces rates of major morbidity, and produces equivalent or reduced mortality relative to open approaches, though some studies find that MIE patients are more likely to require surgical reintervention. Even with improvements over traditional open techniques, minimally invasive approaches still have mortality rates as high as 6% and morbidity rates as high as 46%.

The previously mentioned variants of esophagectomy all involve the use of the stomach as an esophageal replacement. When the stomach is not available (as with prior resection or caustic injury), the colon can be used as an esophageal substitute. Both the left and right colon can be used, with the vascular supply to the grafts based on either the ascending branch of the left colic artery or the right colic artery. Colon interpositions typically have higher complication rates than esophagectomies using gastric conduits. Supercharged jejunum has also been used with excellent success. Supercharged means an arterial and venous anastomosis is also made in the neck.

Usual preop diagnosis

Carcinoma of esophagus or gastroesophageal junction; Barrett esophagus; benign strictures

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

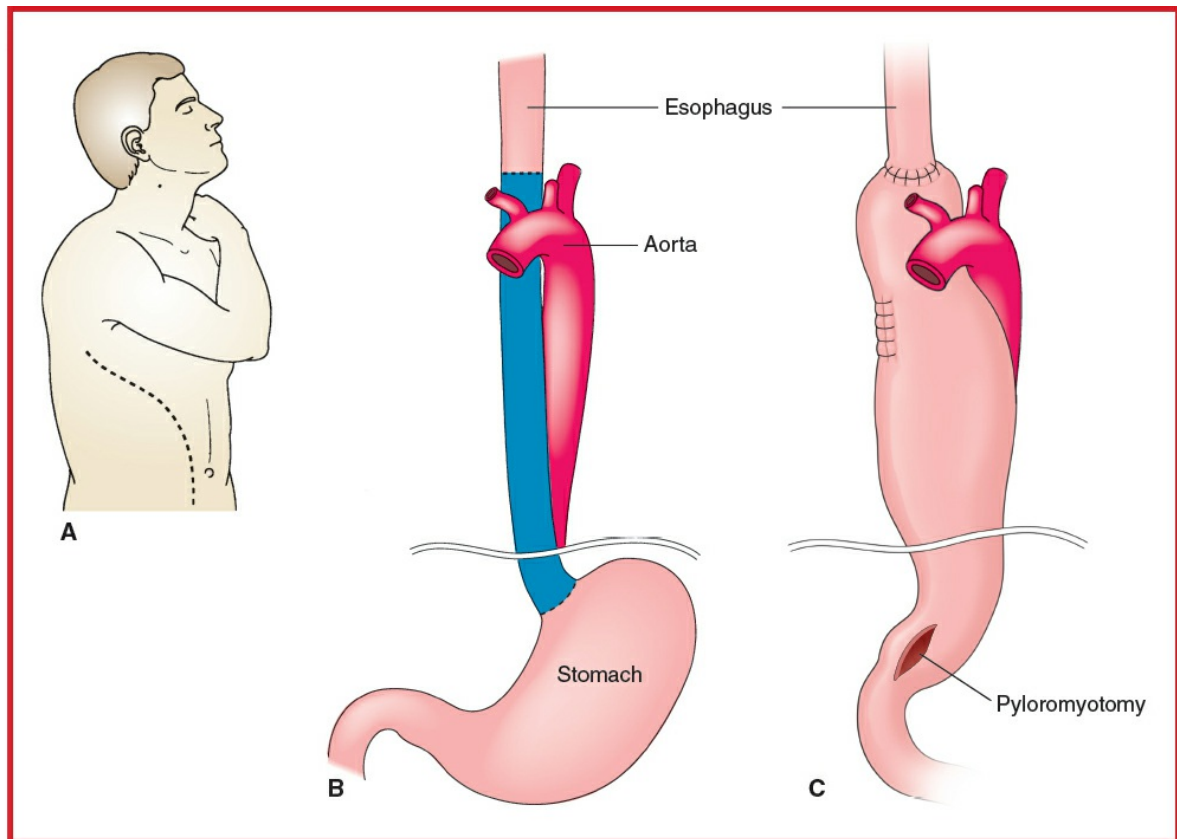


Figure 7.1-8 Standard thoracoabdominal esophagogastrectomy for tumors of the upper and middle 3rd of the thoracic esophagus. **A:** Either a continuous thoracoabdominal incision or separate thoracic and abdominal incisions are used. **B:** Portion of esophagus to be resected (blue) . **C:** Completed reconstruction with high intrathoracic esophagogastric anastomosis and gastric drainage procedure. (Reproduced with permission from Greenfield LJ, Mulholland, MW, Oldham KT, et al., eds Surgery: Scientific Principles and Practice, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ SUMMARY OF PROCEDURES

SUMMARY OF PROCEDURES

	Ivor Lewis	Transhiatal Esophagectomy	Three-Field Approach	Thoracoabdominal	Total Esophagectomy + Colonic Interposition or Supercharged Jejunum
Position	Supine and left lateral decubitus	Supine	Left lateral decubitus and then supine	Right lateral decubitus	Supine and right lateral decubitus
Incision	Midline abdominal + right chest and/or cervical incisions	Cervical and midline abdominal	Right chest and then midline abdominal and left neck	Thoracoabdominal across costal margin (see Fig. 7.1-8)	Midline + right thoracic and cervical
Unique considerations	DLT	None	DLT for the chest portion	DLT	⇐
Antibiotics	Cefoxitin 2 g iv preop	⇐	⇐	⇐	⇐
VTE prophylaxis	Heparin 5000 units sq	⇐	⇐	⇐	⇐
Surgical time	3–4 h	4–5 h	5–6 h	3–4 h	5–6 h
Closing considerations	Lung reexpansion	Pneumothorax	Lung reexpansion	Lung reexpansion	Vascular integrity of colonic interposition
EBL	300–800 mL	⇐	⇐	⇐	⇐
Postop care	Step-down unit vs. ICU × 1 d	⇐	⇐	⇐	ICU × 2–3 d
Mortality	4–5%	⇐	⇐	⇐	5–8%
Morbidity	Respiratory complications: 15–20%	⇐	⇐	⇐	⇐
	Anastomotic leakage: <5%	5–10%	⇐	<5%	10%
	Anastomotic stricture: <5%	⇐	⇐	⇐	10%
	Wound infection: <5%	⇐	⇐	⇐	⇐
Pain score	7–9	6–8	7–9	7–9	7–9

PATIENT POPULATION CHARACTERISTICS

Age range	40–80 yr
Male:Female	2:1 for carcinoma

Incidence	1–2% of malignant disease
Etiology	Alcohol and tobacco; dietary factors—hot spicy foods; lye burns
Associated conditions	Barrett esophagus; hiatus hernia; reflux esophagitis; radiation esophagitis; caustic burns

Suggested Readings

1. Grimmer PP, Goense L, Gockel I, et al: Diagnosis, assessment, and management of surgical complications following esophagectomy. *Ann N Y Acad Sci* 2018; 1434(1):254–73.
2. Seto Y, Mori K, Aikou S: Robotic surgery for esophageal cancer: merits and demerits. *Ann Gastroenterol Surg* 2017; 1(3):193–8.
3. Taurchini M, Cuttitta A: Minimally invasive and robotic esophagectomy: state of the art. *J Vis Surg* 2017; 3:125.

ANESTHETIC CONSIDERATIONS FOR ESOPHAGEAL SURGERY

(Procedures covered: esophagostomy; esophageal diverticulectomy; closure of esophageal perforation; esophagomyotomy; esophagogastric fundoplasty; esophagectomy; colonic interposition.)

Patients presenting for esophageal surgery typically are those with carcinoma, motility disorders, strictures, hiatal hernia, reflux esophagitis, diverticula, and perforation. Patients tend to be elderly, with a history of tobacco abuse and alcohol consumption, and may have associated cardiopulmonary disease. Obesity and ischemic heart disease are increasingly frequent comorbid conditions. Esophageal disorders such as hiatal hernia, Barrett esophagus, and carcinoma with incomplete obstruction predispose the patient to recurrent aspiration pneumonitis from intermittent reflux of gastric contents. Patients with esophageal diverticula, stricture, achalasia, and carcinoma with obstruction may present with retained food and oral secretions. Therefore, patients may need to be considered nonfasted. Minimally invasive techniques for esophagectomy (with laparoscopy and thoracoscopy) have potential benefits of decreased tissue injury, and hence pain, with earlier recovery and reduced hospital stay. These advantages are not yet clearly established. The anesthetic considerations are broadly similar with the additional issues that accompany laparoscopic and thoracoscopic (VATS) procedures.

PREOPERATIVE

Respiratory

Many patients with esophageal cancer have a long history of smoking resulting in COPD. Mediastinal lymphadenopathy and tumors of the upper esophagus may lead to tracheal/bronchial compression or erosion that may be revealed by CT and bronchoscopy. A history of gastric reflux and recurrent aspiration pneumonia → ↓ pulmonary reserve and ↑ risk of regurgitation/aspiration during anesthetic induction. (See Premedication, below.) If thoracotomy is a part of the procedure, the patient should be evaluated to ensure that OLV will be tolerated (see below). Determine if patient has been exposed to radiation and chemotherapeutic agents that may cause pulmonary fibrosis. Some agents, such as Bleomycin, may cause pulmonary toxicity (above a total dose of 300 U) that is worsened by high concentrations of O₂, which should be avoided during the perioperative period.

Tests: Consider PFTs (FEV1, FVC, DLCO), ABG, and flow-volume loops, if indicated by H&P. They can be helpful in predicting the likelihood of perioperative pulmonary complications and the possible need for postoperative ventilatory support. Patients with baseline hypoxemia/hypercarbia on room air have a greater likelihood of postop complications and ventilatory insufficiency. Severe restrictive or obstructive lung disease also will increase the chance of pulmonary morbidity in the periop period. A-P, lateral CXR (if suggestive of tracheal compression obtain MRI/CT).

Cardiovascular

Elderly patients may have coexisting CAD. In the setting of a recent angioplasty/stent, continued management with antiplatelet agents (clopidogrel) may preclude the use of epidural techniques. Patients may be hypovolemic and malnourished from dysphagia or anorexia. Preop chemotherapy treatment with agents such as doxorubicin may cause acute dysrhythmias and chronic

	cardiomyopathy (seen increasingly in patients if total dose is $>550 \text{ mg/m}^2$). Some esophageal tumors are related to alcohol consumption, and chronic abuse may produce a toxic cardiomyopathy. Preop optimization of cardiac status is important. Tests: ECG to r/o dysrhythmias, myocardial ischemia, or prior MI. ECHO or dobutamine stress ECHO will provide important functional information about the heart at rest or under stress, respectively. Reversible ischemia and significant cardiomyopathy should prompt further cardiac consultation.
Hematologic	Tests: CBC with differential
Laboratory	PT & PTT, basic metabolic profile and tests as indicated from H&P
Premedication	Consider midazolam 1–2 mg iv. Consider H_2 antagonists (e.g., ranitidine 50 mg iv), metoclopramide (10 mg iv 1 h preop), and Na citrate (30 mL po 10 min preop) in patients with reflux and partial obstruction. Patients with esophageal motility disorders and strictures/tumors with complete obstruction may not tolerate oral antacids.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po and gabapentin 600 mg with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA (with or without epidural for intraop/postop analgesia). If the patient is clinically hypovolemic, restore intravascular volume before epidural placement and induction of GA. Surgeries involving only a cervical or endoscopic approach typically

do not require epidural analgesia. If epidural analgesia is planned, placement and testing of the catheter before anesthetic induction is recommended. This is accomplished by injecting 5–7 mL of 1–2% lidocaine with 1:200,000 epinephrine via the epidural catheter and eliciting a segmental block. If a thoracic or thoracoabdominal approach is performed, placement of a DLT is indicated to provide OLV for optimal surgical conditions. (For additional discussion of DLT placement and OLV, see Anesthetic Considerations in Lobectomy, Pneumonectomy in Thoracic Surgery.)

Induction	Patients with esophageal disease are often at risk for pulmonary aspiration; therefore, the trachea should be intubated after rapid-sequence induction with cricoid pressure. In cases with a difficult airway needing a DLT, placement of a single lumen tube should be accomplished (a) with the aid of a bougie, (b) via an LMA, or (c) with FOB. The single lumen tube is then changed to a DLT using a tube exchange catheter.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Depending on the procedure, consider regional nerve blocks and ketamine or lidocaine infusions as appropriate for multimodal analgesia.
Maintenance	Standard maintenance, with or without N ₂ O (for OLV). Alternatively, a combined technique may be used. A local anesthetic (e.g., 2% lidocaine with 1:200,000 epinephrine or 0.25% bupivacaine) can be infused or injected into a thoracic (3–5 mL) or lumbar (5–10 mL) epidural catheter to provide both anesthesia and optimal surgical conditions (contracted bowel and profound muscle relaxation). Continuous infusion of local anesthetic generally provides better hemodynamic stability than hourly bolus injections. To enhance the effect of epidural analgesia, a loading dose of opiates (e.g., hydromorphone 0.4–1 mg [thoracic] or 1–1.5 mg [lumbar]) can be administered early during the surgery and at least 1 h before conclusion of the case. Esophagectomy procedures can cause significant third-space losses, thus, the maintenance of euolemia with close monitoring of BP and UO. Esophageal procedures

involving transient compression of the myocardium (i.e., thoracic approach for esophagomyotomy or esophagectomy) can induce dysrhythmias and/or ↓ BP. Monitoring of BP with an arterial catheter is recommended. When OLV is necessary there can be ventilation-induced lung injury, oxygen toxicity, atelectasis, and ischemia-reperfusion injury. The use of PEEP, lower tidal volumes, and low ventilatory pressures may be protective ventilator strategies. The goals of fluid management are as follows: (a) optimize the tissue blood flow and oxygen delivery and (b) avoid excessive fluid administration, which could lead to anastomotic edema with impaired wound healing and pulmonary edema with increased cardiopulmonary morbidity. A balance of colloid + crystalloid and blood products when necessary helps achieve this goal-directed fluid therapy.

Emergence

Generally, tracheal extubation should be anticipated at the end of the case. The decision to keep the patient intubated postop depends on cardiopulmonary status and the site and extent of the surgical procedure. Patients with significant intraop fluid shift may develop airway edema that can → airway obstruction if extubated prematurely. This may also occur in operations involving high anastomoses in the neck. For those who may require prolonged postop ventilation, the DLT should be exchanged to a single lumen ETT before transport to ICU. The use of an airway exchange catheter is safe, and direct laryngoscopy during the exchange prevents inadvertent seepage of blood or silent aspiration. Weaning from mechanical ventilation should begin when the patient is awake and cooperative, able to protect the airway, hemodynamically stable, and has adequate return of pulmonary function (as measured by VC ≥ 10 mL/kg; MIF of >30 cm H₂O; rapid, shallow breathing index of ≤100 [RR ÷ TV(L)]; respiratory rate <25; and ABG that demonstrates adequate gas exchange). Discuss with surgeon local anesthetic wound infiltration. Consider iv acetaminophen if po dose is given

	>4–6 hr ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14–16 ga × I–2 BSS @ up to 10 mL/kg/h Warm fluids Consider T&C for 2–4 U PRBC	Goal-Directed Fluid Management Plt, FFP, and cryoprecipitate (if required) should be administered according to lab tests (Plt count, PT, PTT, DIC screen, thromboelastography [TEG]). Maintain euvolemia based on estimates of blood loss, fluid shifts, fluids administered, and HR, BP, UO, base deficit, and invasive monitoring when employed. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors Urinary catheter ± Arterial line ± CVP	CVP cannulation site determined by surgical approach. Attempt to prevent hypothermia during long operations. Consider forced-air warmer, heated humidifier, warming blanket, warming the OR, keeping patient covered until ready for prep, etc.
Positioning	If lateral decubitus position, use axillary roll, airplane arm holder. ✓ pressure points, including ears, eyes, and genitals ✓ radial pulses to ensure correct placement of axillary roll (if misplaced, will compromise distal	Problems that can arise include brachial plexus injuries and damage to soft tissues, ears, eyes, and genitals from malpositioning. Check down eye at frequent intervals. Placing the oximeter probe on the down arm may assist in monitoring adequacy of

Complications	pulses) Hypoxemia	perfusion. Hypoxemia during OLV most commonly results from luminal obstruction (by blood or pulmonary secretions) of the DLT, ↑ shunting, or malposition of DLT, which can move during surgical manipulation. Rx: suction DLT and ✓ position; PEEP to ventilated lung (but may ↑ shunting); CPAP to nonventilated lung; return to double-lung ventilation. Temporary clamping of the PA (or inflate the balloon of the PA catheter, if available) may be necessary to improve shunting and oxygenation.
	Hypercarbia Dysrhythmia	Ensure adequate TV and RR. ✓ for mechanical compression of heart or great vessels. Check K+.
Complications (cont'd)	Hypotension	✓ volume status and cardiac function. Consider phenylephrine for BP support if ↓ BP 2° epidural anesthetic. Avoid epidural bolus dosing (↓ BP). Maintaining an adequate BP is important for adequate perfusion and hence integrity of a newly formed anastomosis.
	VTE prophylaxis	See Appendix B: Standard Adult Anesthetic Protocols .

Complications	Aspiration Atelectasis	For patients at risk for atelectasis or aspiration, recover in the 45° head up or Fowler position (semisitting)
	Hemorrhage	For hemorrhage, ✓ coags; replace factors as necessary.
	Pneumothorax Hemothorax	Dx for pneumothorax and hemothorax: wheezing, coughing, dyspnea, ↓ PO ₂ , ↓ PCO ₂ . Confirm by CXR. Rx: chest tube drainage as necessary. In emergency (e.g., tension pneumothorax), use needle aspiration. Supportive Rx: O ₂ , vasopressors, volume, ± ETT, and IPPV.
	Hypoxemia Hypoventilation	For hypoxemia and hypoventilation; Rx: adequate analgesia, minimize sedation, supplemental O ₂ may require IPPV.
	Recurrent laryngeal nerve injury	For laryngeal nerve injury, Dx: indirect visualization of vocal cords; patient usually will be hoarse.
	Esophageal anastomotic leak	Surgical repair Rx for esophageal anastomotic leak.
	VTE prophylaxis	Esophageal cancer, like other cancers, predisposes patients to VTE with ↑ risk following the commonly used chemotherapeutic agents cisplatin and 5-FU.
	SVT/AFib	Treat underlying cause, and correct electrolyte

		abnormalities. Adenosine (6 mg iv, push and repeat to 12 mg iv) may be used for SVT. Most postop SVTs are 2° catecholamine surge. AFib may resolve spontaneously. Hemodynamically unstable patients will require cardioversion. β-Blockers, amiodarone, Ca++ channel blockers, and overdrive cardiac pacing are effective in patients with stable AFib.
Multimodal pain management	Lumbar-thoracic epidural analgesia: hydromorphone (0.5–1.5 mg load, 0.1–0.3 mg/h infusion) + local anesthetic PCEA PCA NSAID	Patient should recover in ICU or hospital ward that is accustomed to treating side effects of epidural opiates (e.g., respiratory depression, hypotension, breakthrough pain, nausea, pruritus). Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia (see Appendix K, p. K-6). When an epidural cannot be used, a PCA with an opiate such as fentanyl or morphine/hydromorphone may be used for pain control. Acetaminophen and ketorolac are helpful as adjuvant therapy for postop pain management.
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Consider DVT prophylaxis.
Chewing gum may promote
return of bowel function.

Tests

CBC, basic metabolic
profile
ABG
CXR (r/o pneumothorax,
atelectasis)

Suggested Readings

1. Findlay JM, Gillies RS, Millo J, et al: Enhanced recovery for esophagectomy: systematic review and evidence-based guidelines. *Ann Surg* 2014; 259(3):413–31.
2. Hashimi S, Smith M: Medical evaluation of patients preparing for an esophagectomy *Surg Clin North Am* 2012; 92(5):1127–33.
3. Kingma BF, Steenhagen E, Ruurda JP, et al: Nutritional aspects of enhanced recovery after esophagectomy with gastric conduit reconstruction. *J Surg Oncol* 2017; 116(5):623–9.
4. Pisarska M, Małczak P, Major P, et al: Enhanced recovery after surgery protocol in oesophageal cancer surgery: systematic review and meta-analysis. *PLoS One* 2017; 12(3):e0174382.
5. Raymond D: Complications of esophagectomy. *Surg Clin North Am* 2012; 92(5):1299–313.
6. Veelo DP, Geerts BF: Anaesthesia during oesophagectomy. *J Thorac Dis* 2017; 9(Suppl 8):S705–12.
7. Visser E, Marsman M, van Rossum PSN, et al: Postoperative pain management after esophagectomy: a systematic review and meta-analysis. *Dis Esophagus* 2017; 30(10):1–11.

CHAPTER 7.2

Stomach Surgery

SURGEONS

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GASTRIC RESECTIONS

SURGICAL CONSIDERATION

Description

A **total or partial gastrectomy** is performed most commonly for gastric cancers (adenocarcinomas or gastrointestinal stromal tumors [GIST]) and includes **omentectomy, lymph node dissection**, and occasionally resection of adjacent organs such as the **spleen** or **colon**, depending on the pathology, location, and extent of the tumor. Historically, there have been other indications for gastrectomy such as ulcer disease, Zollinger-Ellison syndrome, and uncontrollable hemorrhagic gastritis, but these operations are rare if ever performed for those indications in current surgical practice. Rarely, patients who suffer from severe and uncontrollable gastroparesis postgastric surgery may require total gastrectomy.

In an open gastric resection, the abdomen is entered through an upper midline incision, and the lateral segment of the left lobe of the liver is retracted anteriorly and to the patients right, exposing the esophagogastric junction. The omentum is taken off of the

colon and left attached to the greater curvature of the stomach. The spleen may be removed if involved by tumor or if an unplanned splenic injury occurs. The vessels to the stomach are individually ligated and divided. The short gastric vessels high on the greater curvature are challenging to reach and are a source of potential blood loss. This is also the most likely time that a splenic injury may occur by traction or tearing of the capsule while exposing the short gastric arteries. Currently, most surgeons are using various devices to ligate the arteries, and these occasionally have a technical failure, which can result in blood loss. In the lesser sac, the left gastric artery as it branches from the celiac axis and vein needs to be divided and can be another point of potential unexpected blood loss. A total gastrectomy is performed for more proximal cancers and a partial resection for distal cancers. In all cases of gastric cancer, the antrum and pylorus are resected, leaving a duodenal stump and requiring **gastroenterostomy** to restore intestinal continuity. One area of potential complication when the stomach is resected is the accidental stapling of the gastric tube, which remains undetected until too late. To prevent this, the tube should be pulled back well into the esophagus, preferably with manual confirmation by the surgeon that the tube is no longer present in the stomach.

After completion of the gastric resection, reestablishment of intestinal continuity is performed. In the case of a total gastrectomy, a Roux limb of jejunum is brought up to the distal esophagus; in partial gastrectomies, a Roux limb or loop of jejunum (**Billroth II**) is connected to the remnant stomach. With a Roux limb, the jejunum is divided just beyond the ligament of Treitz, and the distal end is brought up through a hole in the mesentery of the colon and anastomosed to the esophagus or stomach. Intestinal continuity is established by anastomosing the biliary pancreatic limb of the proximal jejunum to the Roux limb of jejunum, ~60 cm distal to the anastomosis with the esophagus. A drain is then placed near the closed end of the duodenum. The anastomosis can either be stapled or hand sewn depending on the preference of the surgeon. At completion of the anastomosis, an NG tube can be advanced across the proximal anastomosis, and the abdomen is irrigated. A number of surgeons then place a **feeding jejunostomy** tube into the jejunum, which adds a few minutes to the procedure prior to fascial closure. Total gastrectomy traditionally has been associated with a morbidity and mortality out of proportion to the operation's apparent magnitude. This is most likely a consequence of the patient's underlying condition, which often includes advanced malignancy and, almost invariably, some degree of malnutrition. Venous thromboembolism (VTE) is a significant concern in these patients because of their increased hypercoagulable state from the cancer and an operation of >1 h in duration. Patients undergoing gastric resection should receive sequential compression devices on the lower extremities and subcutaneous heparin thromboprophylaxis.

Variant procedure or approaches

Occasionally, gastric cancer can have extensive local involvement of adjacent organs requiring an **en bloc resection of the stomach** in addition to the **colon, spleen, or pancreas**. For some distal gastric cancers, a combined gastric resection with a **Whipple procedure (pancreaticoduodenectomy)** may be necessary. The need for en bloc resection of the stomach in combination with other organs increases the complexity of the surgery, risk of blood loss, and postoperative morbidity and mortality. In general, exposure for a **partial gastrectomy** is similar to, but less extensive than, that required for a total gastrectomy. The same cancer principles are followed for either a partial or total resection, including >5 cm proximal margin, **lymphadenectomy**, and **omentectomy**. A partial gastrectomy is a simpler resection. The blood supply to the distal stomach is divided, and the duodenum is divided just beyond the pylorus. The body of the stomach is divided with a stapler (care should be taken not to staple the gastric tube) at a level appropriate for the pathology. Reconstruction after resection may be either to the duodenum (**Billroth I**), loop of jejunum (**Billroth II**; [Fig. 7.2-1](#)), or a **Roux-en-Y limb of jejunum**. The anastomoses may be stapled or sewn; then the abdomen is closed. Like many operations, gastric resections are increasingly performed using minimally invasive techniques. The **laparoscopic approach** has the advantages of diminished postoperative pain and quicker recovery, but longer operative time requires a pneumoperitoneum and reverse Trendelenburg positioning.

Tumors (usually adenocarcinoma) of the gastroesophageal (GE) junction are increasing in frequency and may be of either gastric or esophageal origin. If the tumor is associated with Barrett esophagus (intestinal metaplasia in the esophagus, seen on endoscopy), surgery consists of either an **Ivor Lewis** with esophagogastric anastomosis in the mediastinum (combined abdominal and transthoracic approach) or **transhiatal esophagectomy** with GE anastomosis in the neck. Bulky GE junction tumors that encompass the upper stomach may limit the extent of esophageal resection if stomach pull-up is used. Postop pain can be severe, and most patients benefit from continuous epidural analgesia.

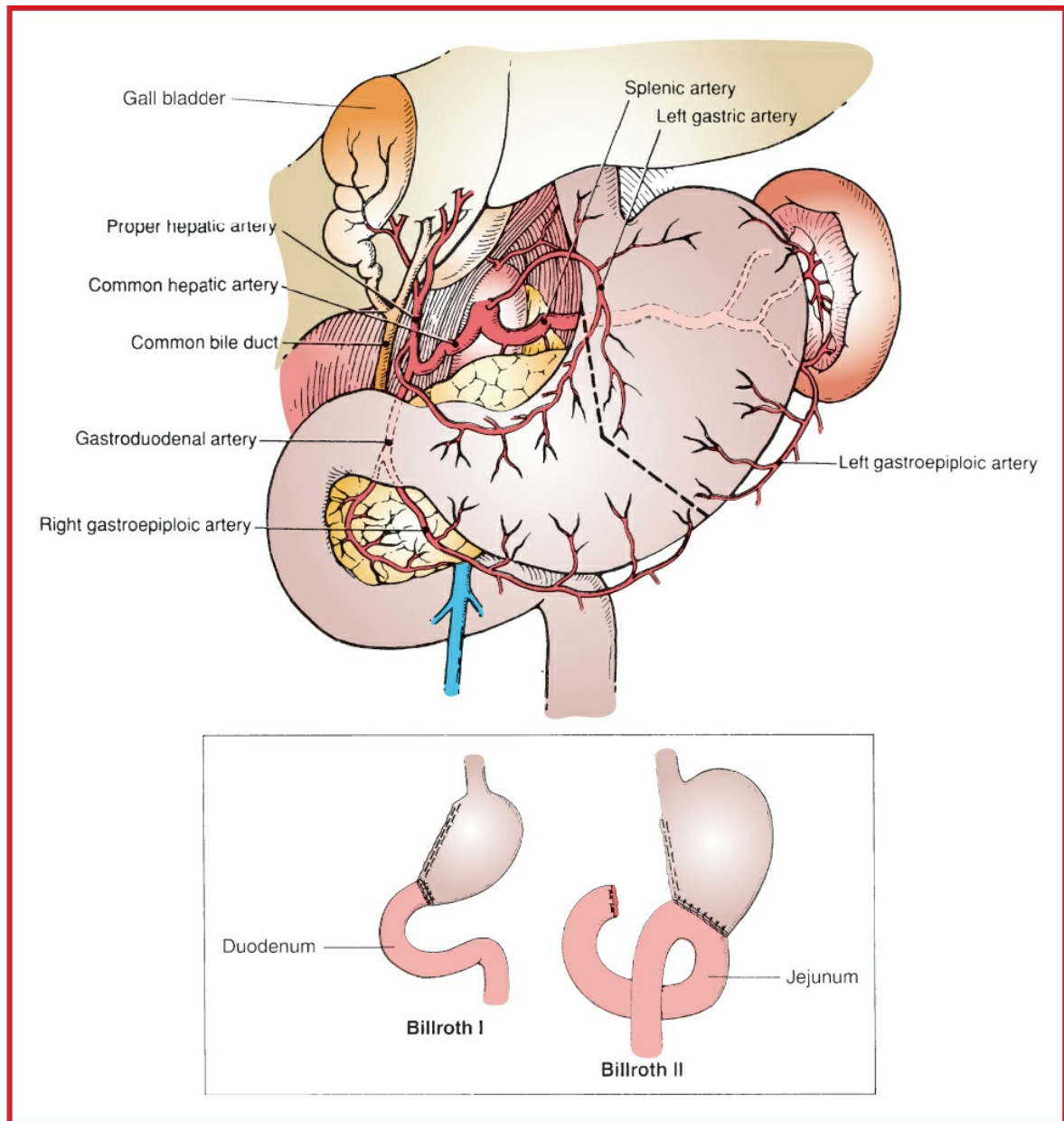


Figure 7.2-1 Anatomy of duodenostomy (Billroth I) and gastrojejunostomy (Billroth II). (Reproduced with permission from Scott-Conner CEH, Dawson DLOperative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

Usual preop diagnosis

Total gastrectomy; gastric cancers; GIST; gastroparesis; partial gastrectomy; gastric cancer; GIST; gastric ulcers

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Total Gastrectomy (Open vs. Laparoscopic)	Partial Gastrectomy
Position	Supine	⇐
Incision	Open: Upper midline or bilateral subcostal; Lap: Multiport incisions	Upper midline or right subcostal
Special instrumentation	Open: Upper hand or other self-retaining costal retractor; Lap: Endoscopic instruments	Upper hand or other costal retractor
Antibiotics	1st (e.g., cefazolin) or 2nd (e.g., cefotetan) generation cephalosporin 2 g iv	⇐
Surgical time	2–4 h	1.5–2 h
Closing considerations	Muscle relaxation required; ±NG suction	⇐
EBL	Open: 500+ mL, with potential for significantly more; Lap: 100–500 mL	100–500 mL
Postop care	PACU → ward or ICU	⇐
Mortality	0–22%	0–1.8% (may be >10% if emergent)
Morbidity	Pulmonary complications: — 15% ⇐ VTE — Reoperation: 0–5% ⇐ Anastomotic leak — Sepsis ⇐ Wound infection — Late anastomotic stricture	

Pain score

7–8

7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Mostly >50 yr
Male:Female	Male predominance
Incidence	In the United States, people of Korean, Japanese, Latino, and African American race have a higher incidence than Caucasians.
Etiology	Gastric adenocarcinoma, gastrointestinal stromal tumors associated with race, age, alcohol and tobacco use, geographic location, and gastric ulcers
Associated conditions	Weight loss (common); anemia (common); malnutrition (common); VTE

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Operations for Peptic Ulcer Disease.

Suggested Readings

Lee J-H, Lee CM, Son S-Y, et al: Laparoscopic versus open gastrectomy for gastric cancer: long-term oncologic results. *Surgery* 2014; 155(1):154–64.

Minter RM, Nathin H: Gastric neoplasms. In: Greenfield LJ, et al., eds *Surgery: Scientific Principles and Practice*, 6th edition. Lippincott Williams & Wilkins, Philadelphia: 2017, 754–67.

Selby LV, DeMatteo RP, Tholey RM, et al: Evolving application of minimally invasive cancer operations at a tertiary cancer center. *J Surg Oncol* 2017; 115(4):365–70.

GASTRIC OR DUODENAL PERFORATION

SURGICAL CONSIDERATIONS

Description

Patients who present with gastric or duodenal perforations require emergency surgery because they usually have peritonitis at presentation. The primary reason for perforations that present to the emergency room is ulcer disease most likely from *Helicobacter pylori* infection or from NSAID use. Another cause of gastric perforation

is trauma, but that is not in the scope of this section. Simple closure with an omental patch, also called a **Graham patch**, is the most commonly performed operation to treat a duodenal perforation. It is unusual to perform a definitive ulcer operation unless the patient has failed medical treatment regimens aimed at eradication of *H. pylori* infection along with proton pump inhibitors to suppress acid production.

Duodenal ulcers are most often caused by a benign process, whereas gastric ulcer perforation without a history of NSAID use is suspicious of a malignant process. For this reason, surgeons may elect to treat a perforated gastric ulcer by resection. Occasionally in patients who have failed maximal medical management of their peptic ulcer disease and who are not systemically ill at the time of operation, some surgeons may perform **vagotomy and pyloroplasty** or **highly selective vagotomy** at the time of closure of the perforation ([Fig. 7.2-2](#)).

Many surgeons routinely perform simple closure with **Graham patch** via a **laparoscopic** approach. This requires a few trocars and a pneumoperitoneum and can be combined with abdominal washout and irrigation. If an open approach is used, an upper midline incision is used. The liver is retracted superiorly and the area of perforation identified. An NGT to suction minimizes ongoing leakage from the perforation and decompresses the stomach. Perforation of the stomach may be handled either by resection (see Gastric Resections) or by biopsy and simple suture closure. Perforation of the duodenum is usually repaired by simple suturing of the site. Omentum often is used to buttress (Graham patch) the area of closure of the stomach or duodenum. The abdomen is irrigated and closed.

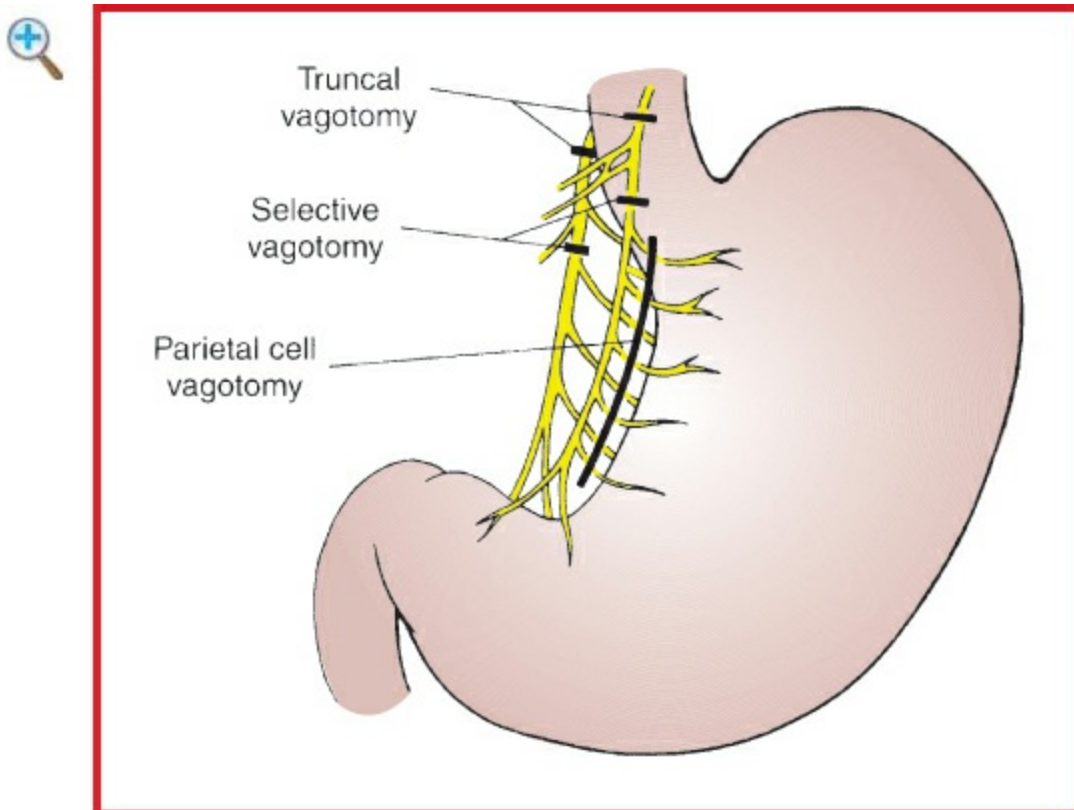


Figure 7.2-2 Types of vagotomy. Heavy lines indicate where vagal trunks are cut.

Variant procedure or approaches

In some patients, **nonoperative management** of perforated ulcer may be appropriate if they have had an upper GI study that shows an ulcer cavity and no extravasation. These patients are treated with NG decompression, antibiotics, and proton pump inhibitors. In general, this has a relatively high likelihood of success if the candidates are chosen well.

Usual preop diagnosis

Free air under diaphragm and peritonitis, perforated peptic ulcer

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position

Supine

Incision	Midline or 3–4 trocars for laparoscopic approach
Special instrumentation	Costal retractor or laparoscopic equipment
Unique considerations	Patients usually have peritonitis.
Antibiotics	Cefotetan 2 g iv or extended spectrum PCN
Surgical time	1 h
Closing considerations	Muscle relaxation required; NG suction
EBL	Minimal
Postop care	PACU → ward
Mortality	5–15%, mostly dependent on patient population
Morbidity	Pneumonia Intra-abdominal abscess Sepsis Wound infection Reperforation
Pain score	7 for open procedure, 3 for laparoscopic

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adult, more common in socially disadvantaged populations
Male:Female	Previous heavy male predominance still exists for duodenal ulcer, but there has been a significant increase in incidence in gastric perforation in women >65.
Incidence	Fairly common. Stable incidence, but with change in the distribution especially more elderly women.
Etiology	Peptic ulcer disease (PUD); nonsteroidal medications; malignancy (if gastric)
Associated conditions	Malignancy (if perforation is gastric); nonsteroidal medications; steroid use, especially during pulse

therapy; other risk factors for PUD (e.g., alcoholism, smoking)

Suggested Reading

1. Mulholland MW: Gastroduodenal ulceration. In: Greenfield LJ, et al., eds. *Surgery: Scientific Principles and Practice*, 6th edition. Lippincott Williams & Wilkins, Philadelphia: 2017, 725–32.

ANESTHETIC CONSIDERATIONS

(Procedures covered: gastric resections; oversew gastric/duodenal perforation operations for peptic ulcer disease; duodenotomy.)

PREOPERATIVE

Patients presenting for gastric surgery generally comprise two groups: (a) those presenting for emergency surgery following GI bleeding or perforation and (b) those presenting with gastric carcinoma. Patients in the first group are often hemodynamically unstable and require rapid preop assessment and appropriate fluid resuscitation.

Respiratory	Patients with GI bleeding are at ↑ risk for aspiration of blood and gastric contents. Tests: Consider CXR and ABG if aspiration occurred.
Cardiovascular	Consider hypovolemia if active hemorrhage or peritonitis present. Tests: Orthostatic vital signs; ECG, if indicated from H&P
Renal	GI fluid loss can lead to renal and electrolyte abnormalities. Tests: Consider electrolytes; BUN; Cr.
Hematologic	Consider GI blood loss occurring acutely or chronically. Tests: CBC with Plts, PT/PTT if coagulopathy suspected
Laboratory	Other tests as indicated from H&P
Premedication	Consider midazolam 1–2 mg. Consider H ₂ -antagonist (ranitidine 50 mg iv slowly), metoclopramide (10 mg

	iv 1 h preop), and Na citrate (30 mL po 10 min preop).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural for postop analgesia (if hemodynamically stable and no coagulopathy). If postop epidural analgesia is planned, insertion of catheter prior to anesthetic induction is helpful to establish correct placement in the epidural space (accomplished by injecting 5–7 mL of 1% lidocaine via the epidural catheter, eliciting a segmental block).

Induction	The patient with gastric disease or upper GI bleeding is often at risk for pulmonary aspiration, and the trachea should be intubated with a rapid-sequence induction with cricoid pressure. If the patient is clinically hypovolemic, restore intravascular volume (colloid, crystalloid, or blood products) before induction, and titrate induction dose of sedative/hypnotic agents.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. Balanced anesthesia with inhalational agents and/or propofol infusion and multimodal analgesia (see Appendix B, p. B-3). Maintain muscle relaxation based on nerve stimulator response. Discuss with surgeon the need for postop NG tube. If not needed, place OG tube to evacuate stomach contents intraop. Combined epidural/GA: A local anesthetic (2%

lidocaine with 1:200,000 epinephrine) can be injected into a thoracic (2–4 mL) or lumbar (3–6 mL q60–90min) epidural catheter to provide both anesthesia and optimal surgical exposure (contracted bowel and profound muscle relaxation). A continuous infusion of local anesthetic (e.g., 2% lidocaine or 0.25% bupivacaine at 2–5 mL/h [lumbar]; 2–4 mL/h [thoracic]) may enhance hemodynamic stability, compared to an intermittent bolus. Be prepared to treat ↓ BP with fluid and vasopressors. GA is administered to supplement regional anesthesia and for amnesia. Systemic sedatives should be minimized during epidural opiate administration as they increase the likelihood of postop respiratory depression. Treat ↓ BP with fluid and vasopressors. If epidural opiates are used for postop analgesia, a loading dose (e.g., hydromorphone 0.6–1.0 mg [lumbar]; 0.4–0.6 mg [thoracic]) may be administered at least 1 h before conclusion of surgery.

Emergence

The decision to extubate at the end of surgery depends on the patient's underlying cardiopulmonary status and extent of the surgical procedure. Patient should be hemodynamically stable, normothermic, cooperative, and fully reversed from any muscle relaxants and without pulmonary compromise before extubation. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements

Anticipate large third-space losses
IV: 14–16 ga × 1–2
BSS @ 8–12 mL/kg/h
Fluid warmer

T&S, consider T&C for 2–4 U PRBC. Plts, FFP, and cryoprecipitate should be administered according to lab tests (Plt count, PT, PTT, DIC screen, thromboelastography). Expect higher fluid

		requirements if epidural used (2° sympathectomy → vasodilation). Maintaining euvolemia is important goal. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors UO ± Arterial line ± CVP	Consider others as indicated by patient's status. Prevent hypothermia: Use forced air warmer, fluid warmer, and warming blanket, warm room temperature, keeping patient covered until ready for prep, etc.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Acute hemorrhage Hypoxemia	

POSTOPERATIVE

Complications	PONV Hemorrhage VTE Ileus Hypothermia	
Multimodal pain management	Epidural analgesics PCA	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia and/or TAP blocks (see Appendix C, p. C-4).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-	Encourage early enteral nutrition, Foley catheter

	6)	removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	CXR	If CVL placed periop

Suggested Readings

1. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
2. Kostakis ID, Alexandrou A, Armeni E, et al.: Comparison between minimally invasive and open gastrectomy for gastric cancer in Europe: a systematic review and meta-analysis. *Scand J Surg* 2017; 106(1):3–20.
3. Søreide K, Thorsen K, Søreide JA: Strategies to improve the outcome of emergency surgery for perforated peptic ulcer. *Br J Surg* 2014; 101(1):e51–64.

OPEN OPERATIONS FOR MORBID OBESITY

SURGICAL CONSIDERATIONS

Description

Several studies have demonstrated the superiority of laparoscopic operations for weight loss over open operations. In selected patients (e.g., those with previous upper abdominal surgery), however, open procedures may be appropriate. Open techniques devised to promote weight loss are of two fundamental types: (a) **restrictive procedures**, which work by decreasing the size of the gastric pouch, thereby limiting the amount of food that can be consumed at one time; and (b) **malabsorptive procedures**, which work by bypassing most of the small bowel and creating a state of chronic malabsorption. Of these two general types of procedures, the partitioning procedures may be less effective at promoting weight loss than the malabsorptive procedures but are much more popular because they are associated with far fewer serious side effects.

The **Roux-en-Y gastric bypass** (Fig. 7.2-3) is the most common weight loss operation in the United States. The abdomen is entered through an upper midline incision, and the esophagogastric junction is exposed either by retracting the liver superiorly or by taking down the ligamentous attachments of the lateral segment of the left lobe of the liver and retracting this down and to the patient's right. The upper stomach is mobilized, and two rows of staples are used to partition the stomach into a small proximal and large distal

pouch. A Roux segment of jejunum is then anastomosed to the small proximal pouch to provide drainage. An NG tube is placed, and the abdomen is closed with drains.

Vertical banded gastroplasty (Fig. 7.2-4) has been replaced by the adjustable gastric band. The abdomen is entered similarly to the gastric bypass operation. The pars flaccida is opened by dividing the gastrohepatic ligament. The base of the right crus is identified, and a tunnel behind the stomach is created. The band is then placed and secured. Silastic tubing from the band is then attached to a port placed in the subcutaneous tissue.

An alternative open procedure is the biliopancreatic diversion with duodenal switch, which involves a longitudinal gastric resection with a proximal small bowel bypass. Another alternative open procedure is a **sleeve gastrectomy**, where a gastric tube is created from the GE junction to the pylorus over a 36°F dilator. The remainder of the stomach, based on the greater curvature, is resected. Both of these operations are gaining popularity in the bariatric surgery community.

Variant procedure or approaches

The **jejunoileal (JI) bypass** is of historical interest (Fig. 7.2-5). One of the early bariatric operations in the 1950s, the JI bypass was constructed by connecting the proximal jejunum to the terminal ileum. This procedure has been abandoned mainly because of the many associated complications, including cirrhosis, osteoporosis, kidney stones, and intractable diarrhea.

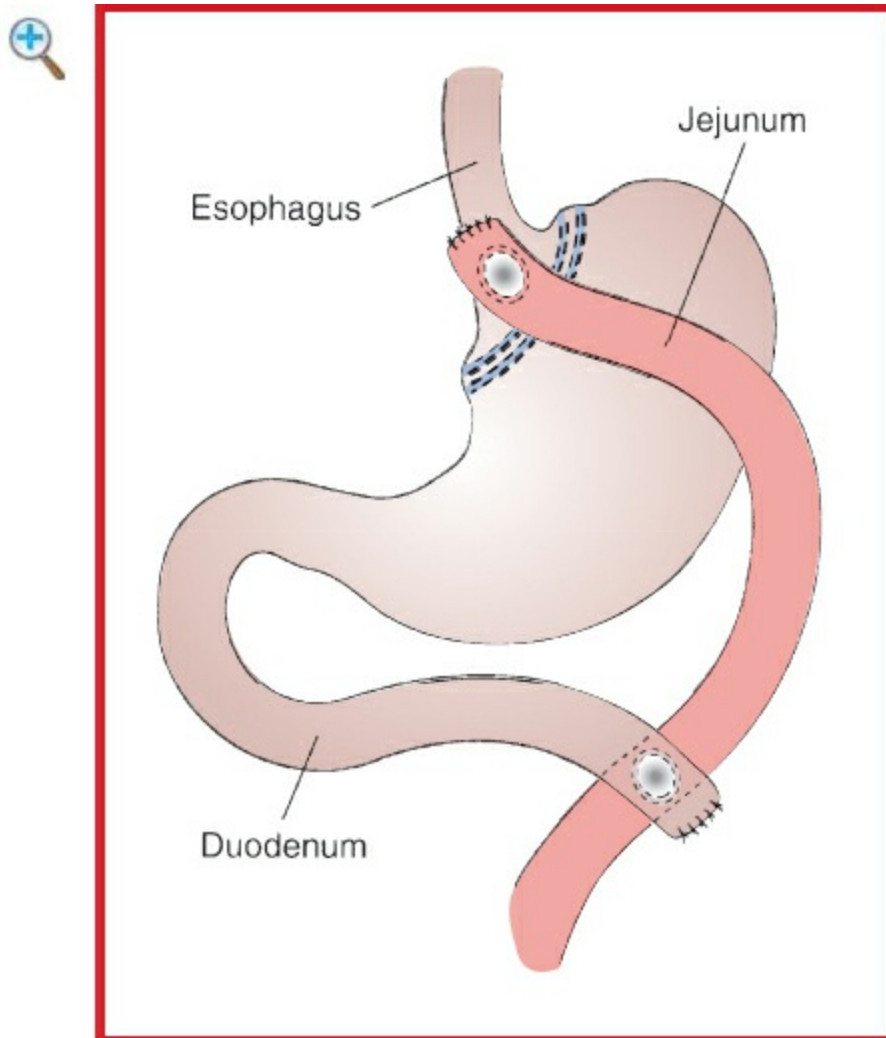


Figure 7.2-3 Proximal Roux-en-Y gastric bypass. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice* 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Morbid obesity (BMI > 40 kg/m² or >35 kg/m² with associated comorbidities). Common obesity-related comorbidities include obstructive sleep apnea, hyperlipidemia, osteoarthritis, diabetes mellitus, GERD, liver disease, respiratory insufficiency, CHF, hypertension, and CAD

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

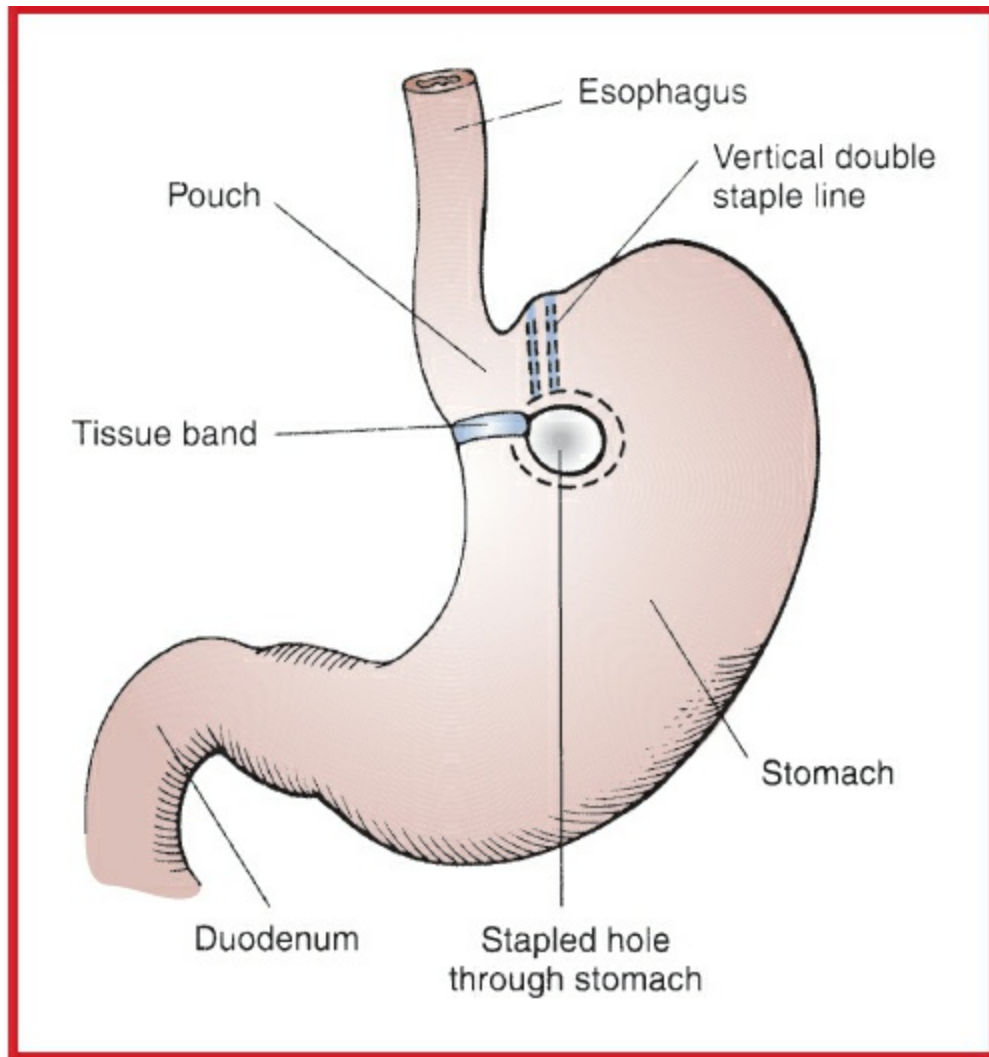


Figure 7.2-4 Vertical banded gastroplasty.

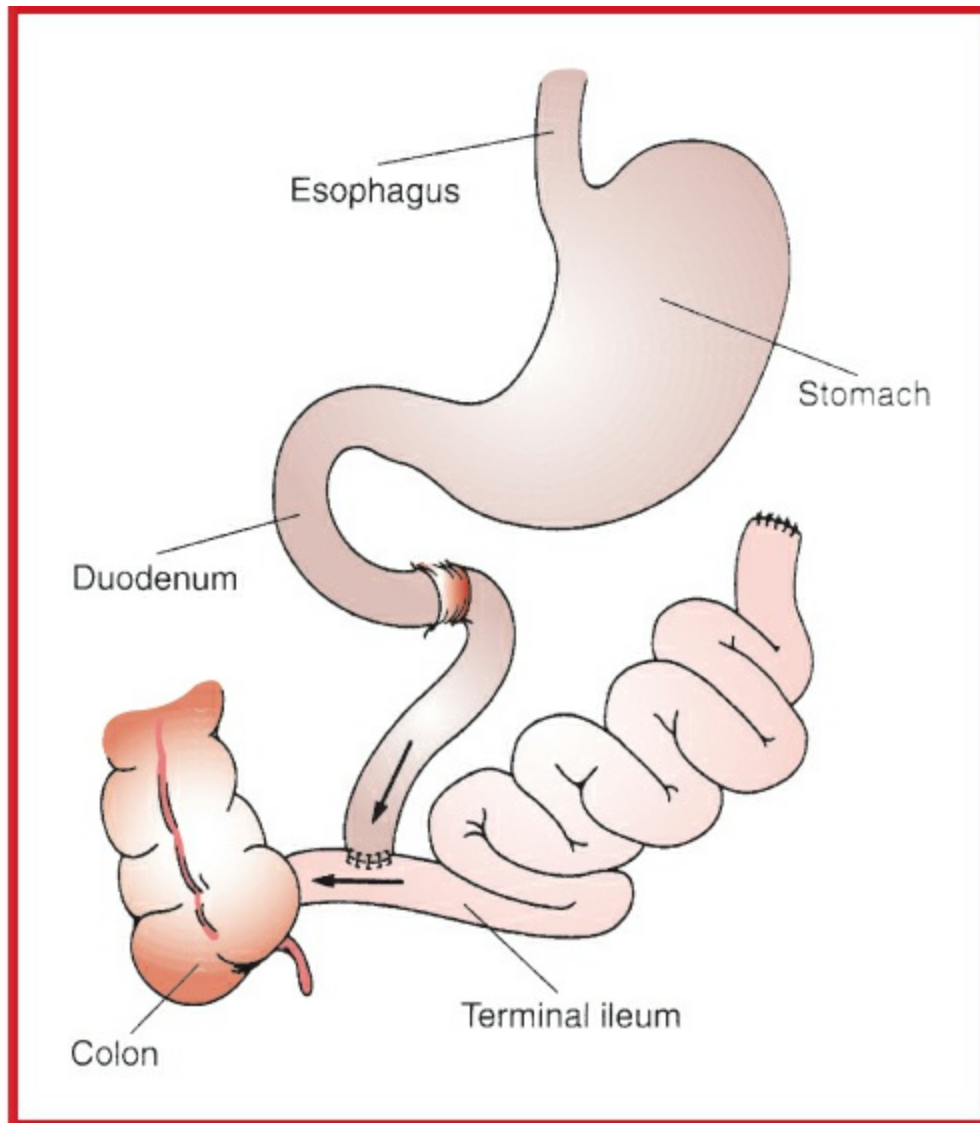


Figure 7.2-5 Schematic representation of jejunioileal bypass. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ SUMMARY OF PROCEDURE (SLEEVE GASTRECTOMY OR ROUX-Y)

Position	Supine (Fig. 7.2-6) with reverse Trendelenburg
Incision	Midline
Special instrumentation	Large OR tables; heavy-duty retractors; special stapling devices
Unique considerations	Prophylactic cholecystectomy often advocated in symptomatic patients. Pneumatic compression boots may not be large enough; heparin (5000 U sc 2 h

	preop, then q12h) used commonly. ↑↑ aspiration risk + potentially difficult airway
Antibiotics	Cefazolin 3 g iv (or 2 g iv if total body weight <120 kg)
Surgical time	2–3 h
Closing considerations	Anticipate 20 min closure time
EBL	<500 mL
Postop care	Postop ventilation may be necessary; VTE precautions
Mortality	0.5–1.6%
Morbidity	Wound infection: 4–8% Anastomotic leak: 3% Dehiscence: 1.6% PE: 1–1.6%
Pain score	7

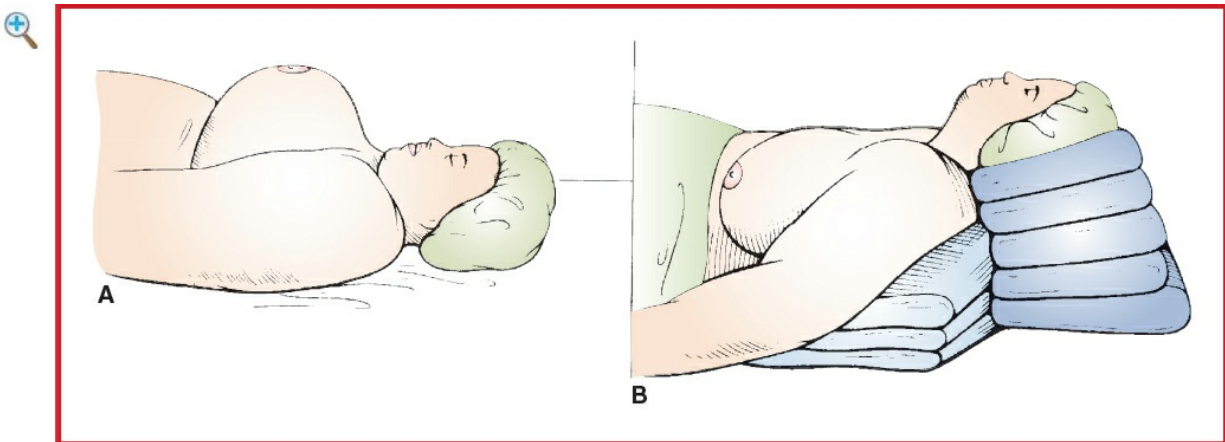


Figure 7.2-6 **A:** In standard supine position, the atlanto-occipital gap of morbidly obese patient is obliterated by fat, and access for laryngoscope is hindered by large breasts. **B:** By elevating the shoulders and occiput so that head is in the “sniffing” position, airway access is greatly facilitated.

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adult
Male:Female	<1:1 for disease but 80% of patients who undergo surgery are female

Incidence	20% US population considered morbidly obese
Etiology	Multifactorial
Associated conditions	Sleep apnea ± CO ₂ retention; CAD/CHF/cardiomyopathy; pulmonary HTN/systemic HTN; diabetes; unusually high risk of DVT and PE; GERD

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Obesity is associated with many chronic medical problems. Morbidity and mortality rise sharply with increasing age and BMI. Obesity, and length of exposure to obesity, increase the risk of hospital admission and lengthen hospital stay. Medical comorbidities must be optimized before elective bariatric surgery. Evaluate any patient who has had previous bariatric surgery for metabolic changes that can include protein, vitamin, iron, and calcium deficiencies.

Review a list of all current medications the patient is taking, including nonprescription appetite suppressors and diet drugs. Many of these drugs can have important side effects. For example, the medication Contrave® is a combination of naltrexone, which is a potent, long-acting opiate antagonist, and the antidepressant bupropion and should be held several days prior to surgery. Another weight loss medication, sibutramine, works in the brain by inhibiting the reuptake of norepinephrine, serotonin, and dopamine, producing a feeling of “anorexia,” which limits food intake. Sibutramine has been implicated as a cause of dysrhythmias and hypertension. Orlistat® blocks digestion and absorption of dietary fat by binding lipases in the gastrointestinal tract and can cause deficiencies in fat-soluble vitamins (A, D, E, K). A reduction in vitamin K levels can increase the anticoagulation effects of coumadin (also see Anesthetic Considerations for Laparoscopic Bariatric Surgery).

Respiratory	Increased O ₂ consumption and CO ₂ production (e.g., ↑ basal metabolic rate), ↓ chest-wall compliance (↓ 20–60%) with normal lung compliance. ↓ ERV and FRC; so tidal breathing may fall within the range of closing capacity → V/Q abnormalities. Supine position further ↓ FRC → worsening hypoxemia. MV is required to remain normocarbic. There is a normal
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	response to CO₂ unless patient develops the obesity hypoventilation (Pickwickian) syndrome (↑ PaCO₂, ↓ PaO₂, loss of hypercarbic drive, sleep apnea, hypersomnolence, polycythemia, pulmonary HTN, CHF). Tests: CXR; PFTs (FVC, FEV₁, MMEF₂₅₋₇₅ ± bronchodilators; room-air ABG)
Cardiovascular	Blood volume and CO ↑ with increasing weight. HTN is very common (use correct size BP cuff). LV dysfunction may be present; patient unable to increase CO or tolerate ↑ blood volume. Pulmonary HTN may be present in OSA. Obesity is a risk factor for CAD and sudden death. Anticipate problems with vascular access. Tests: ECG; others as indicated from H&P. (Patient with SOB may require ECHO for LV function, as SOB can have a cardiac or pulmonary etiology.)
Endocrine	Glucose intolerance and diabetes mellitus (DM) common. Tests: Fasting glucose
Hepatic	Liver function is often abnormal. Drug metabolism can be significantly affected. Combined with altered pharmacokinetics, many drugs (e.g., midazolam and vecuronium) may have unpredictably prolonged action.
Gastrointestinal	↑ intra-abdominal pressure, gastric volume, and acidity, with ↑ incidence of hiatal hernia, make this patient population at risk for pulmonary aspiration of gastric contents. There is particular risk in patients with previous restrictive bariatric surgery.
Airway	Obese patients can have airway problems including frequent difficult mask ventilation. Careful airway examination (e.g., mouth opening, Mallampati classification, thyromental distance, neck ROM) is paramount. Awake fiber-optic intubation should be considered if difficult airway access is anticipated.

	Establish availability of OR table large enough to accommodate morbidly obese patient. Special tables are now available.
Hematologic	Polycythemia may occur 2° chronic hypoxemia. Tests: CBC
Laboratory	Other tests as indicated from H&P
Premedication	Sedatives are best avoided for patients with OSA. A small dose of iv midazolam (0.5–1.0 mg) may be appropriate for the especially anxious patient. In the bariatric patient, intramuscular medications can be erroneously injected into adipose tissues. Consider anticholinergics if performing awake fiber-optic intubation (glycopyrrolate 0.2 mg iv 30 min preop). Consider full-stomach precautions, including a nonparticulate antacid (3 M Na citrate) 30 mL, 10 min prior to induction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural for postop analgesia in open surgeries. If using a combined anesthetic approach, placement of an epidural catheter should be accomplished before induction, with the patient in the sitting position. A bilateral sensory block (using 5–7 mL 2% lidocaine) helps confirm correct placement of the catheter within the epidural space. Verification of placement is particularly important in this population because regional anesthesia (e.g., transversus abdominis plane block) in the obese patient may be technically more difficult.

Induction	A rapid-sequence induction with propofol (2–2.5 mg/kg LBW) and succinylcholine (1 mg/kg TBW) combined with
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cricoid pressure is optimal. Mask ventilation and ET intubation may be difficult (2° to increased oxygen consumption and rapid desaturation, redundant tissue, cervical immobility). Following successful intubation and induction of anesthesia, an OG/NG tube should be placed and the stomach contents suctioned. Lipophilic drugs have a greater apparent volume of distribution; however, since cardiac output governs drug distribution and is related to lean body weight (LBW), induction doses of hypnotics and opioids should be given based upon LBW (e.g., male LBM = $1.1[\text{weight}] - 128[\text{weight/height}]^2$).

ERAS

Verify preincision antibiotics. Assess volume status (see [Appendix K](#)). Maintain normovolemia and normothermia.

Maintenance

Patients should be ventilated with an FiO_2 of 0.5–1.0 and tidal volumes of 10–12 mL/kg IBW (ideal body weight [kg] = height [cm] – 100). Inhalational agents (desflurane, sevoflurane, isoflurane) may be given. There are only modest differences in inhalational agent pharmacokinetics between obese and nonobese patients. Open Procedures—**Combined epidural/GA:** A local anesthetic (2% lidocaine with 1:200,000 epinephrine) can be injected into a thoracic (3–5 mL) or lumbar (5–10 mL q60min) epidural catheter to provide both anesthesia and improved surgical exposure (contracted bowel and profound muscle relaxation). A continuous infusion of local anesthetic (e.g., 2% lidocaine or 0.25% bupivacaine at 5–10 mL/h [lumbar], 3–5 mL/h [thoracic]) may enhance hemodynamic stability. The dose of local anesthetic given via epidural catheter should be decreased to 75% of normal dose. Be prepared to treat ↓ BP with fluid and vasopressors (ephedrine 5–10 mg iv, phenylephrine 50–100 mcg iv). GA is administered to supplement regional anesthesia and for amnesia. Sedative drugs (opiates, benzodiazepines, etc.) should be minimized in the presence of epidural opiates, as they increase the likelihood of postop respiratory depression. If epidural

	opiates are used for postop analgesia, a loading dose (e.g., hydromorphone 0.6–1.0 mg with lumbar catheter and 0.4–0.6 mg for thoracic) should be administered 1–2 h before end of surgery.	
Emergence	The decision to extubate at the end of open surgery depends on patient's underlying cardiopulmonary status and the extent of the surgical procedure. Patients should be hemodynamically stable, normothermic, alert, cooperative, and fully reversed from any muscle relaxants before extubation. Elective ICU admission for postop care may be appropriate. Following laparoscopic surgery, in contrast, patients typically are extubated and admitted to PACU. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Anticipate large fluid loss. IV: 14–16 ga × 1–2 BSS @ 10–15 mL/kg/h T&S for 2 U PRBC. Warm fluids.	Third-space losses greatly exceed blood loss. Guide fluid management by UO, filling pressure. Euvolemia is an important goal. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors UO ± Arterial line	Invasive monitoring as clinically indicated. Arterial line if noninvasive BP unreliable 2° extremity size
Positioning	Supine position = ↓ FRC ✓ and pad pressure points ✓ eyes NB: Avoid Trendelenburg	Supine positioning → ↓ lung volumes, which may ↑ V/Q mismatch → hypoxemia. This is exacerbated by use of the Trendelenburg position, which usually is not well tolerated by morbidly obese

Complications	Hypoxemia 2° ↓ FRC	patients. 100% O ₂ → absorption atelectasis
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▼ POSTOPERATIVE

Complications	Hypoxemia Hypercarbia VTE Atelectasis	Recover patient in sitting position to improve ventilatory mechanics. Give supplemental O ₂ . Verify VTE prophylaxis.
Multimodal pain management	Epidural analgesia: Hydromorphone (0.6–1.0 mg with lumbar cath and 0.4–0.6 mg for thoracic cath load then 0.1–0.3 mg/h infusion) PCA	If epidural is used for postop analgesia, pain service management recommended. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia and/or TAP blocks (see Appendix C, p. C-4).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	ABG CXR	Others as clinically indicated. CXR for central line placement

Suggested Readings

1. Alvarez A, Goudra BG, Singh PM: Enhanced recovery after bariatric surgery. *Curr Opin Anaesthesiol* 2017; 30(1):133–9.
2. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
3. Brodsky JB, Lemmens HJM: Regional anesthesia and obesity. *Obes Surg* 2007;

17(9):1146–9.

4. Brolin RE: Bariatric surgery and long-term control of morbid obesity. JAMA 2002; 288(22):2793–6.
5. Choban PS, Jackson B, Poplawski S, et al: Bariatric surgery for morbid obesity why, who, when, how, where, and then what? Cleve Clin J Med 2002; 69(11):897–903.
5. Ingrande J, Lemmens HJM: Dose adjustments of anaesthetics in the morbidly obese Br J Anaesth 2012; 105 (Suppl 1):16–23.
7. le Roux CW, Heneghan HM: Bariatric surgery for obesity. Med Clin North Am 2018; 102(1):165–82.
3. Livingston EH: Obesity and its surgical management. Am J Surg 2002; 184(2):103–13.
9. Lukosiute A, Karmali A, Cousins JM: Anaesthetic preparation of obese patients current status on optimal work-up. Curr Obes Rep 2017; 6(3):229–37.
9. Ogunnaike BO, Jones SB, Jones DB, et al: Anesthetic considerations for bariatric surgery. Anesth Analg 2002; 95(6):1793–805.
1. Perilla V, Sollazzi L, Bozza P, et al: The effects of the reverse Trendelenburg position on respiratory mechanics and blood gases in morbidly obese patients during bariatric surgery. Anesth Analg 2000; 91:1520–5.

GASTROSTOMY PLACEMENT

SURGICAL CONSIDERATIONS

Description

A **gastrostomy** is a tube placed through the abdominal wall directly into the stomach. Such tubes can be used for gastric decompression or for feeding, and they may be permanent or temporary. Patients undergoing gastrostomy placement often have neurologic impairment that compromises their ability to handle oral secretions and increases their risk of aspiration. **Percutaneous endoscopic gastrostomy (PEG)**, in contrast to the other techniques, most commonly is performed using iv sedation and local anesthesia.

Variant procedure or approaches

The traditional **Stamm gastrostomy** usually is placed at the time of a **laparotomy** performed for another purpose; or it may be performed through a separate, small laparotomy incision in patients in whom endoscopic placement is not possible for technical reasons. The incision may be upper midline or transverse directly over the

stomach. The anterior wall of the stomach is identified, and two purse-string sutures are placed in the stomach around the site where the tube will enter. The gastrostomy tube is introduced through the abdominal wall directly over the intended site of entry into the stomach. A small hole is made in the stomach in the center of the purse-string sutures, the tube is introduced into the stomach, and the purse-strings are tied securely around the tube. The wound is then closed. GA usually is preferred, but the operation may be performed under local anesthesia in thin patients.

The **Janeway gastrostomy**, a technical modification, also requires a laparotomy. The greater curvature of the stomach is identified and a stapler placed across a portion of this, creating a tube that arises from the main body of the stomach. The staple line may be oversewn, and then the end of the tube is brought through the abdominal wall and matured to the skin as a small stoma. This allows for permanent access to the stomach with removal of the tube between feedings and is useful in patients with long-term dependence on gastrostomy access. The Janeway gastrostomy is rarely used, although young patients with neurologic impairment who are expected to need lifetime gastrostomy feeding are good candidates.

For a **PEG**, the stomach is intubated endoscopically and the gastric and abdominal walls punctured under endoscopic guidance. The gastrostomy tube is introduced through the mouth and passed through the stomach and abdominal wall from inside out. In most centers, this has become the most common technique of gastrostomy placement due to its simplicity, and, in many patients, it can be performed under local anesthesia with MAC. Previous gastric operations may make endoscopic placement difficult or dangerous, as may some obstructing lesions of the esophagus or pharynx. To avoid damage to the back wall of the stomach the anesthesiologist may be asked to inject air forcefully into the stomach.

Usual preop diagnosis

Temporary gastrostomies often are used after major abdominal surgery as an alternative to NG suction. Percutaneous gastrostomies often are placed in patients with advanced malignancy and intestinal obstruction or inadequate oral intake and in patients with neurologic impairment and difficulty eating

ERAS: Preop counseling by the surgeon/anesthesiologist should include if appropriate limiting NPO duration, alcohol and, smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Stamm	Janeway	PEG
Position	Supine	⇐	⇐
Incision	Midline or transverse	⇐	Puncture
Special instrumentation	None	⇐	Endoscope, percutaneous gastrostomy kit
Antibiotics	± Cefazolin 1–2 g iv	⇐	⇐
Surgical time	45 min	1 h	0.5–1 h
Closing considerations	Muscle relaxation for closure	⇐	None
EBL	Minimal	⇐	⇐
Mortality	Minimal	⇐	⇐
Morbidity	Wound infection: 2.1–9%	⇐	—
	Hemorrhage: 0.9–1.1%	⇐	⇐
	Aspiration pneumonia: 2.2%	⇐	1.6%
	Failure to function: 2.2%	⇐	—
Pain score	4–5	5	1–2

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages, though with peaks in infancy and the elderly
Male:Female	~1:1
Incidence	Common
Etiology	See Preop Diagnosis, above.
Associated conditions	Gastrostomy placed at time of laparotomy, when NG drainage is anticipated for prolonged period. For feeding in the neurologically impaired or in those with complex upper digestive difficulties. Advanced malignancy (for either feeding or palliative decompression)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Ostomy Procedures in Intestinal Surgery.

Suggested Readings

1. Pennington C: To PEG or not to PEG. Clin Med 2002; 2(3):250–5.
2. Schrag SP, Sharma R, Jaik NP, et al: Complications related to percutaneous endoscopic gastrostomy (PEG) tubes. A comprehensive clinical review. J Gastrointest Liver Dis 2007; 16(4):407–18.

We gratefully acknowledge the past contributions of surgeon, Myriam J. Curet, MD, FACS.

CHAPTER 7.3

Intestinal Surgery

SURGEON

Jeffrey A. Norton, MD

ANESTHESIOLOGIST

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DUODENOTOMY

SURGICAL CONSIDERATIONS

Description

A **duodenotomy** is performed to ligate a bleeding vessel at the base of a duodenal ulcer or to perform some procedure on the ampulla of Vater or the duct of Santorini or to remove a low-grade duodenal tumor. It is important, therefore, to be familiar with the anatomy of the proximal duodenum in relation to the major and minor pancreatic duct orifices (Fig. 7.3-1). The duodenotomy may be made longitudinally or transversely, depending on the surgeon's preference. A transverse opening allows one to close the duodenotomy without tension; however, it must be made carefully to ensure adequate exposure. Therefore, a longitudinal duodenotomy is preferred. With mobilization, it can be closed transversely. Bleeding vessels at the base of an ulcer must be secured with suture ligatures. Care must be taken to avoid perforating the duodenum when performing a sphincterotomy.

Usual preop diagnosis

Duodenal ulcer; impacted common duct stone; chronic pancreatitis 2° alcoholism, gallstones, pancreatic divisum, or other obstructions of the main pancreatic duct; benign ampullary tumor; duodenal neuroendocrine tumor

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting nothing by mouth (NPO) duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, deep venous thrombosis (DVT) prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery](#), p. K-1.

■ SUMMARY OF PROCEDURES

Position	Supine
Incision	Midline abdominal or subcostal
Unique considerations	Magnifying glasses, if operation involves lesser pancreatic sphincter
Antibiotics	Cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	1–2 h
Closing considerations	Secure closure of the duodenum without tension usually two layers
EBL	Minimal
Postop care	±NG decompression
Mortality	<0.5%
Morbidity	Duodenal leak: <2%; postop pancreatitis: <3%
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1
Incidence	Not uncommon
Etiology	Duodenal ulcer; impacted common duct stone; villous tumors of ampulla; chronic pancreatitis; pancreatic divisum
Associated	Bleeding duodenal ulcer (50–60%); chronic

conditions

pancreatitis (20–25%); impacted common duct stones (10–15%); tumor (2%)

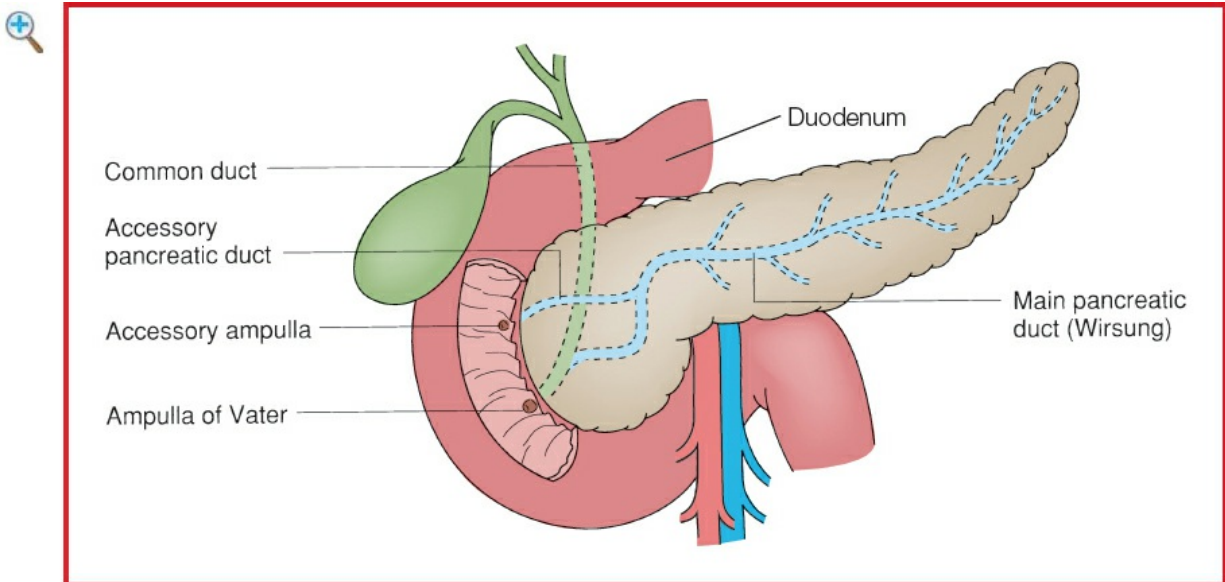


Figure 7.3-1 Anatomy of pancreatic ductal system. In 30% of patients, the accessory duct ends blindly. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Operations for Peptic Ulcer Disease, Stomach Surgery.

Suggested Readings

1. Cisco RM, Norton JA: Surgery for gastrinoma. *Adv Surg* 2007; 41:165–76.
2. Norton JA, Foster DS, Ito T, et al: Gastrinomas: medical or surgical treatment. *Endocrinol Metab Clin North Am* 2018; 47(3):577–601.

OPEN APPENDECTOMY

SURGICAL CONSIDERATIONS

Description

Open appendectomy is performed for appendicitis or suspected appendicitis; however, it has been largely replaced by the **laparoscopic** approach (see p. 667). The negative laparotomy rate has been reduced by the judicious use of preop computed tomography (CT) examination. Through a right lower quadrant (RLQ) (McBurney's) o

right paramedian incision, the cecum is exposed and pulled into the wound (Fig. 7.3-2). The appendix is then delivered through the wound; and the mesoappendix is clamped, cut, and ligated. The appendix is removed by crushing, ligating, and then transecting the base. The appendiceal stump may be invaginated into the wall of the cecum or left alone. In some instances, it may be easier to divide the base of the appendix before delivering the appendix into the wound. The wound should be left open, and soft drains are used in cases of a perforated appendix. In children, the appendix may be inverted and allowed to slough off internally. In laparoscopic procedures, the appendix is usually transected with an endoscopic stapling device. This method is also being used in open procedures.

Variant procedure or approach

Laparoscopic appendectomy

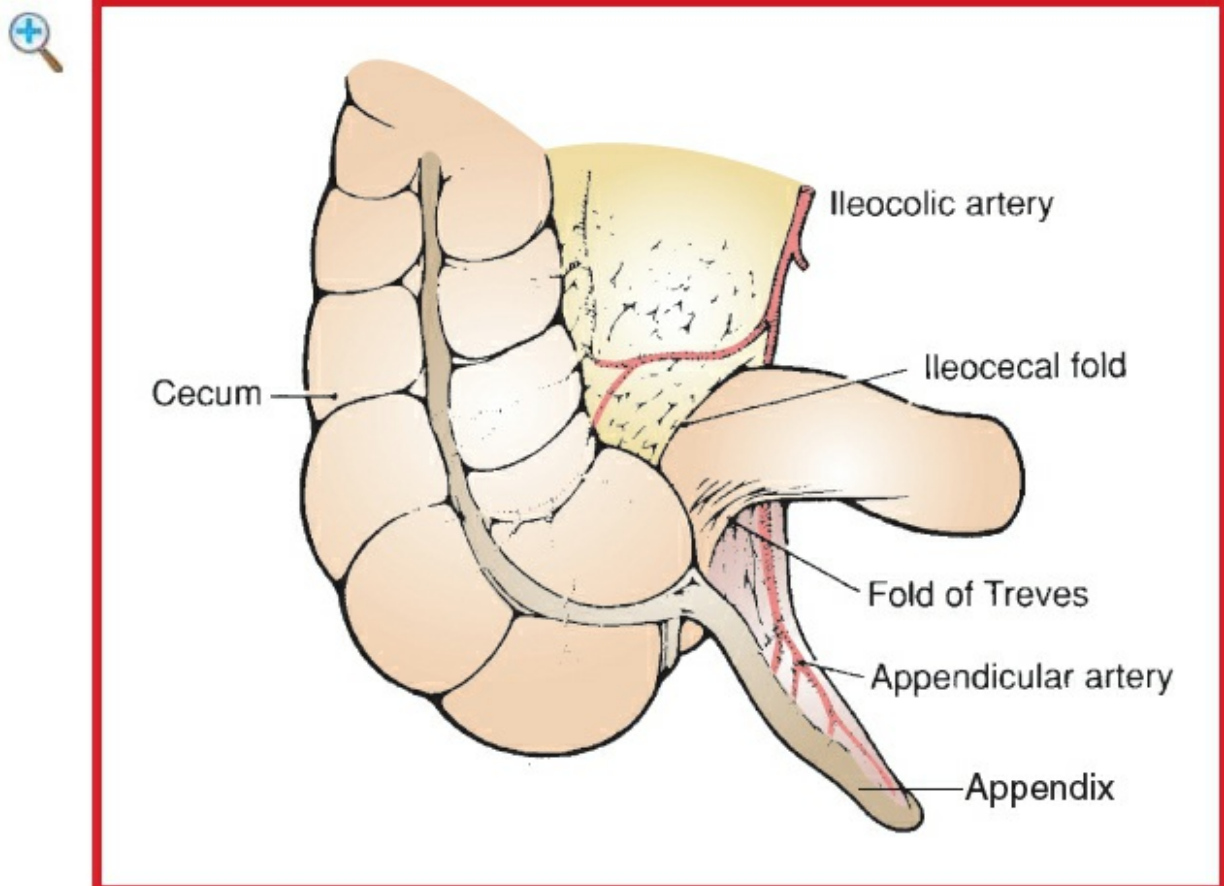


Figure 7.3-2 Relevant anatomy for appendectomy. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

Usual preop diagnosis

Appendicitis

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine
Incision	RLQ (McBurney's) or right paramedian
Unique considerations	Variation in stump closure; NG tube if prolonged ileus expected
Antibiotics	Cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	1 h
Closing considerations	Skin wound should not be closed when the appendix is perforated. Drain in the presence of well-defined abscess cavity.
EBL	<75 mL
Postop care	Wound care when left open
Mortality	Perforated: 2% Nonperforated: <0.1%
Morbidity	Pelvic, subphrenic, or intra-abdominal abscess (perforation): 20% Wound abscess: <5% Fecal fistula: <1% Wound hematoma: <0.5% Ileus: variable
Pain score	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1

Incidence	1/15 persons
Etiology	Obstruction (80–90%); fecaliths (75%); carcinoid tumors (<5%)
Associated conditions	None

ANESTHETIC CONSIDERATIONS

See Anesthetic Consideration following Excision of Meckel Diverticulum.

Suggested Readings

1. Ikard RW, Federspiel CF: Laparoscopic versus open appendectomy. *N Engl J Med* 1995; 333(13):881–2.
2. See Suggested Readings following Excision of Meckel diverticulum, [pp. 579](#).

EXCISION OF MECKEL DIVERTICULUM

SURGICAL CONSIDERATIONS

Description

Meckel diverticulum is a true congenital diverticulum, usually arising within 100 cm of the ileocecal valve. It was first described by Meckel in 1809. Excision of Meckel diverticulum is indicated for bleeding, obstruction, perforation, inflammation, intussusception, and when there is a palpable mass near the base of the diverticulum. Ectopic mucosa is present in roughly 50% of symptomatic patients, with gastric mucosa the most frequent. After entering the peritoneal cavity, the distal ileum, along with the diverticulum, is delivered into the wound. The diverticulum is excised, and the wound is closed in two layers. Following excision of the diverticulum, care must be taken not to narrow the bowel lumen during closure. If a diagnosis can be made preop, a **laparoscopic** approach may be used (see Laparoscopic Bowel Resection).

Usual preop diagnosis

Meckel diverticulum

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options,

physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ PATIENT POPULATION CHARACTERISTICS

Position	Supine
Incision	Midline abdominal or RLQ (McBurney's)
Antibiotics	Cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	1–1.5 h
EBL	<100 mL
Mortality	<0.5%
Morbidity	Wound infection: 5% Pulmonary complication: <5% Anastomotic leak: <1%
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	<40 yr
Male:Female	3:1
Incidence	0.3–2.5%
Etiology	Congenital
Associated conditions	Exomphalos; esophageal atresia; anorectal atresia; gross malformations of the CNS or CV system

ANESTHETIC CONSIDERATIONS

(Procedures covered: open appendectomy; excision of Meckel diverticulum)

PREOPERATIVE

This patient population is generally fit and healthy, apart from their acutely presenting illness. These patients are at risk for aspiration pneumonitis due to delayed gastric emptying, ileus, or frank bowel obstruction → full-stomach precautions are appropriate.

Surgery for appendicitis is one of the most common nonobstetric procedures performed on the pregnant patient (~1/1500 pregnancies).

Respiratory	Respiratory impairment may occur 2° the acute abdominal pain and splinting. Tachypnea and hyperpnea can be heralding Sx of appendiceal perforation and sepsis. Patients with an acute abdomen should be treated as if they have full stomachs. Tests: As indicated from H&P
Cardiovascular	May be dehydrated from fever, emesis, and ↓ oral intake → ↑ HR + ↑ BP (2° pain) or ↓ BP (sepsis, hypovolemia). Assess volume status appropriately, and hydrate adequately before proceeding with anesthetic induction. Tests: ECG, if indicated from H&P
Gastrointestinal	Patient typically has abdominal pain with N/V. Muscular resistance to palpation of the abdominal wall frequently parallels the severity of the inflammatory process. With spreading peritoneal irritation (as with perforation), the patient will develop abdominal distension and paralytic ileus. Consider electrolyte abnormalities 2° N/V. Tests: Electrolytes
Hematologic	Leukocytosis (10,000–18,000) with moderate left shift. Look for hemoconcentration which may signify dehydration. Tests: CBC
Laboratory	Other tests as indicated from H&P
Premedication	Full-stomach precautions. Consider midazolam 1–2 mg iv. Opiate medications often delayed or minimized until diagnosis made. Opiate analgesics may be used during the evaluation of an acute abdomen including appendicitis.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with the surgeon the possibility of a

transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with a sip of water. Consider aprepitant 40 mg po with a sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA, with rapid-sequence iv induction, followed by ET intubation (see full-stomach precautions).

Induction	Rapid-sequence induction of anesthesia. Restore intravascular volume before anesthetic induction if the patient is hypovolemic.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Standard maintenance. Evacuate stomach with OG or NG tube. Maintain muscle relaxation based on nerve stimulator response.	
Emergence	Patient should be extubated awake after return of airway reflexes. Discuss with the surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 5–8 mL/kg/h	See Goal-Directed Fluid Management, Appendix K , p. K-4
Monitoring	Standard monitors	Others, as indicated by the patient's status
Positioning	✓ pad pressure points	

Complications	✓ eyes Bleeding Open technique may follow unsuccessful laparoscopic approach.
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▼ POSTOPERATIVE

Complications	Sepsis (possible with appendiceal rupture) Paralytic ileus Atelectasis PONV	Adequate antibiotic coverage Adequate pain control, incentive spirometry, early ambulation
Multimodal pain management	Multimodal approach	Consider combining PCA or fixed dose po/iv opiates with acetaminophen, po or iv NSAIDs, and gabapentin. Emphasize a multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia and/or TAP blocks (see Appendix C, p. C-4).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	As indicated clinically	

Suggested Readings

1. Liu Q, Jiang H, Xu D, et al: Effect of gum chewing on ameliorating ileus following

colorectal surgery: a meta-analysis of 18 randomized controlled trials. *Int J Surg* 2017; 47:107–15.

2. Reitman E, Flood P: Anaesthetic considerations for non-obstetric surgery during pregnancy. *Br J Anaesth* 2011; 107:i72–8.

ENTEROSTOMY

SURGICAL CONSIDERATIONS

Description

Enterostomy is performed for stenting the small intestine with a long tube for feeding purposes, for bypassing small or large bowel obstructions, and as a permanent or temporary procedure following **proctocolectomy**. An intestinal tube is either purse-stringed into the small bowel and brought through the abdominal wall, or the intestine itself is brought to the exterior and fashioned into a stoma. Different tubes are used for feeding, according to surgeon's preference. After purse-stringing the tube in the bowel, the seromuscular layer of the jejunum is sutured over the tube for a distance of 3–4 cm before exiting through the abdominal wall. This is called a **Witzel feeding jejunostomy**. The **Brooke ileostomy** is created by bringing a 2-inch segment of the ileum through an abdominal wall stab wound. The ileum is folded back on itself and sutured to the skin edge or dermis ([Fig. 7.3-3](#)). Some surgeons secure the ileum to the underlying peritoneum and fascia, but this is not necessary.

Variant procedure or approaches

There are various intestinal or drainage tubes that may be inserted into the bowel, depending on the function required. For example, certain tubes are used for feeding, whereas others may be used for drainage or decompression.

Usual preop diagnosis

Intestinal obstruction due to extensive adhesions; following removal of the large intestine (including the rectum); for enteral feedings

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

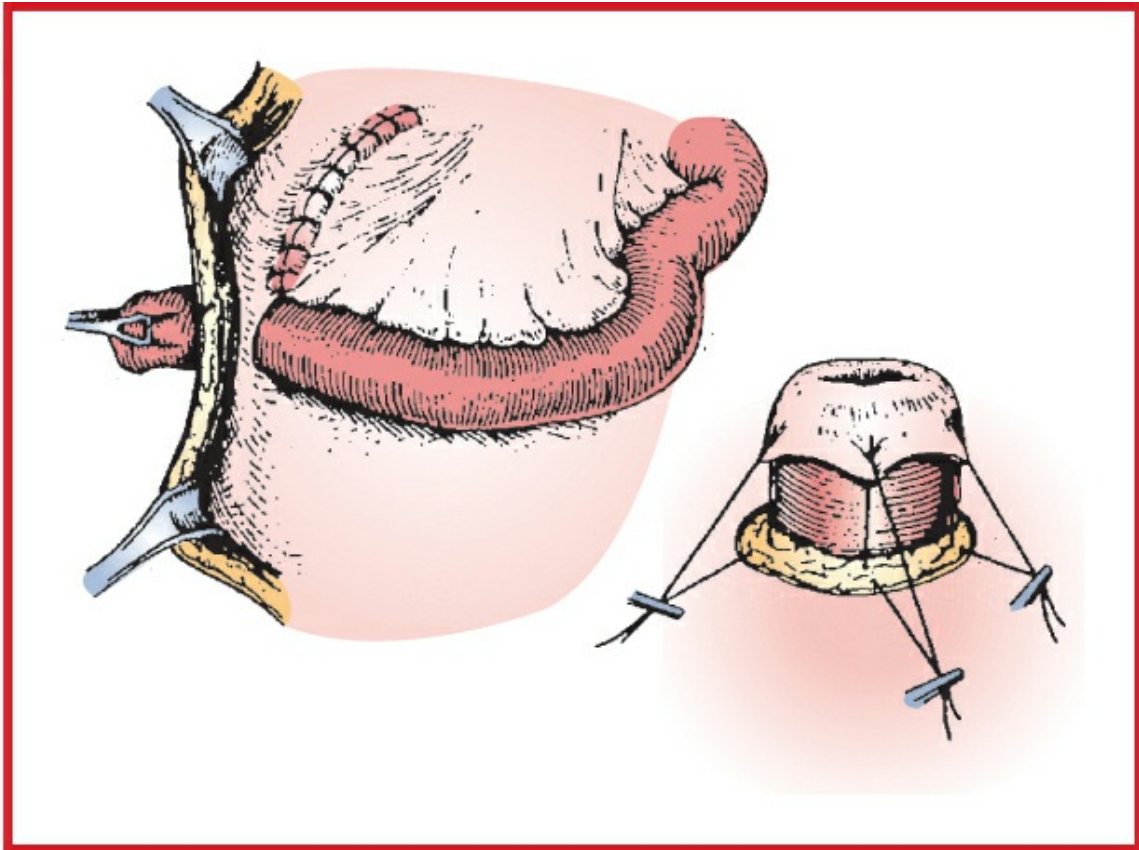


Figure 7.3-3 Brooke ileostomy. (Reproduced with permission from Hardy JD:Rhoad's Textbook of Surgery, 5th edition. JB Lippincott, Philadelphia: 1977.)

■ SUMMARY OF PROCEDURES

	Enterostomy	Ileostomy
Position	Supine	⇐
Incision	Midline abdominal	⇐
Antibiotics	Cefoxitin 2 g iv preop	⇐
VTE prophylaxis	Heparin 5000 units sq	⇐
Surgical time	1–1.5 h	⇐
Closing considerations	Securing tube to abdominal wall	Viable stoma
EBL	<100 mL	⇐
Postop care	Tube irrigation	Stoma care
Mortality	<0.5%	⇐

Morbidity	Ileus: 60–70%	⇐
	Wound infection: <5%	⇐
Pain score	5–6	Stoma necrosis: <2% 5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–65 yr
Male:Female	1:1
Incidence	Common
Etiology	Intestinal obstruction (60–70%); diseases resulting in total proctocolectomy (10–15%); inability to eat (5–10%)
Associated conditions	Inflammatory bowel disease (IBD); intestinal adhesions; inability to eat

■ ANESTHETIC CONSIDERATIONS FOR OSTOMY PROCEDURES

(Procedures covered: enterostomy; continent ileostomy; gastrostomy; gastrojejunostomy)

▲ PREOPERATIVE

This patient population is very diverse and includes those with IBD and cancer and those presenting post-CVA and trauma. Thus, the population ranges from the otherwise healthy to the chronically and critically ill. Many of these patients have abnormal protective airway reflexes and are at risk of aspiration of gastric contents.

Respiratory	Patients post-CVA or head trauma may have abnormal laryngeal reflexes and difficulty swallowing, making them prone to aspiration of gastric contents and associated pneumonitis (evaluate gag reflex). Tests: Consider CXR to r/o pneumonia. Consider ABG.
Cardiovascular	Patients may be hypovolemic 2° chronically poor po intake.

	Tests: ECG; orthostatic vital signs
Musculoskeletal	Patients often frail and debilitated (e.g., post-CVA)
Gastrointestinal	Patients often malnourished and prone to electrolyte abnormalities 2° poor po intake Tests: Electrolytes; BUN; Cr
Laboratory	CBC with differential; others as indicated from H&P
Premedication	Depends on patient status. Titrate small doses of benzodiazepines (midazolam 0.25–0.5 mg iv) or opiate (fentanyl 25–50 mcg iv).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with the surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with a sip of water. Consider aprepitant 40 mg po with a sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA is appropriate for most ostomy procedures; MAC with local anesthesia may be appropriate in selected patients.

Induction	Patient may be at risk for pulmonary aspiration. If GA is planned, consider rapid-sequence induction with cricoid pressure. If patient is hypovolemic, restore intravascular volume (colloid, crystalloid, or blood products) before induction and titrate induction doses of sedative/hypnotic agents.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	MAC: Titration of sedatives (e.g., propofol 25–150 mcg/kg/min) and analgesics (fentanyl 25–50 mcg

	iv); GA : standard maintenance	
Emergence	The trachea should be extubated after the return of protective laryngeal reflexes, if the patient is at risk for aspiration of gastric contents. Discuss with the surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18–20 ga × 1 BSS @ 5–8 mL/kg/h	See Goal-Directed Fluid Management, Appendix K, p. K-4
Monitoring	Standard monitors	Others as clinically indicated
Positioning	✓ and pad pressure points ✓ eyes	

▼ POSTOPERATIVE

Complications	PONV Aspiration VTE	
Multimodal pain management	Consider combining PCA or fixed dose po/iv opiates with acetaminophen, po or iv NSAIDs, and gabapentin.	Emphasize a multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia and/or TAP blocks (see Appendix C, p. C-4).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6).	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Hendren S, Hammond K, Glasgow SC, et al: Clinical practice guidelines for ostomy surgery. *Dis Colon Rectum* 2015; 58(4):375–87.
2. Liu Q, Jiang H, Xu D, et al: Effect of gum chewing on ameliorating ileus following colorectal surgery: a meta-analysis of 18 randomized controlled trials. *Int J Surg* 2017; 47:107–15.

CONTINENT ILEOSTOMY POUCH (KOCK)

SURGICAL CONSIDERATIONS

Description

A **Kock pouch** consists of an internal reservoir fashioned from the distal ileum and an intussuscepted nipple valve used to provide continence. Approximately 45 cm of the small bowel are required for construction of the pouch and valve. After suturing two limbs of the ileum together over a distance of 15 cm, the distal segment is intussuscepted over itself to form the nipple valve. The pouch is then sutured closed and mounted beneath the abdominal wall stoma site ([Fig. 7.3-4A](#)). The stoma is made flush with the skin for cosmetic reasons and left intubated for 1 mo with a special plastic catheter. The pouch remains decompressed for 1 mo before intermittent catheterization is initiated. The continent ileostomy reservoir has been modified by **Barnett** to include the construction of an isoperistaltic valve with an intestinal collar around its base to prevent intussusception and valve prolapse ([Fig. 7.3-4B](#)). These procedures are typically performed following a total proctocolectomy or to replace conventional ileostomies.

Usual preop diagnosis

Inflammatory bowel disease; familial polyposis or malfunctioning ileostomies

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE (KOCK OR BARNETT POUCH)

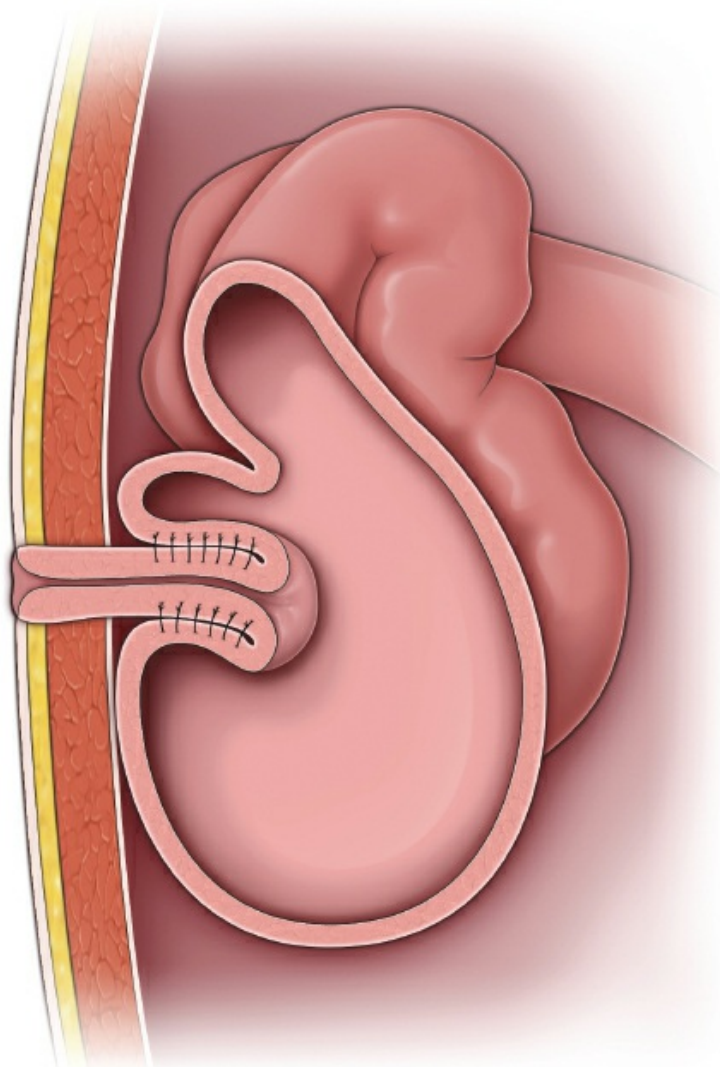
Position

Supine

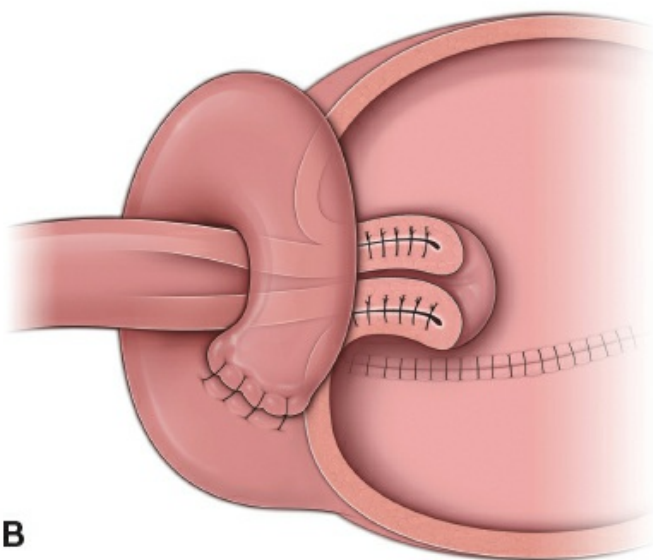
Incision	Midline abdominal
Special instrumentation	GIA or TA staplers
Antibiotics	Usual bowel prep with antibiotics; cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	3–4 h
Closing considerations	Valve vascularity
EBL	200–300 mL
Postop care	Maintain pouch decompression
Mortality	<1%
Morbidity	Intestinal ileus: 5% Wound infection: <5% Intestinal obstruction: 2–3% Pouch fistula: 1–3% Valve necrosis: <0.5%
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–80 yr
Male:Female	1:1
Incidence	Common
Etiology	Ileostomy (50%); proctocolectomy (5%)
Associated conditions	Extracolonic inflammatory bowel manifestations (10%)



A



B

Figure 7.3-4 A: Continent ileostomy or Kock pouch. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)**B:** Barnett modification. The constricting action of the intestinal collar increases with increasing pressure in the pouch.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Ostomy Procedures.

Suggested Reading

1. Malik T, Lee MJ, Harikrishnan AB: The incidence of stoma related morbidity—a systematic review of randomised controlled trials. *Ann R Coll Surg Engl* 2018; 100(7):501–8.

SMALL-BOWEL RESECTION WITH ANASTOMOSIS

SURGICAL CONSIDERATIONS

Description

Small bowel resection is performed for several diseases (listed below). This is being done by laparoscopy with increasing frequency. After entering the peritoneal cavity, the involved small bowel is delivered into the wound and the lesion resected between bowel clamps ([Fig. 7.3-5](#)). Varying amounts of mesentery are included, depending on the diagnosis. More extensive resections are indicated for malignant disease, including regional lymph nodes. Reanastomosis may be accomplished by various suturing techniques or stapling. The peritoneal cavity may be accessed through vertical or transverse incisions or laparoscopy. Operative techniques include **open end-to-end; closed end-to-end; side-to-side; or stapled, functional end-to-end anastomoses**.

Variant procedure or approaches

Laparoscopic small bowel resections are being performed more frequently.

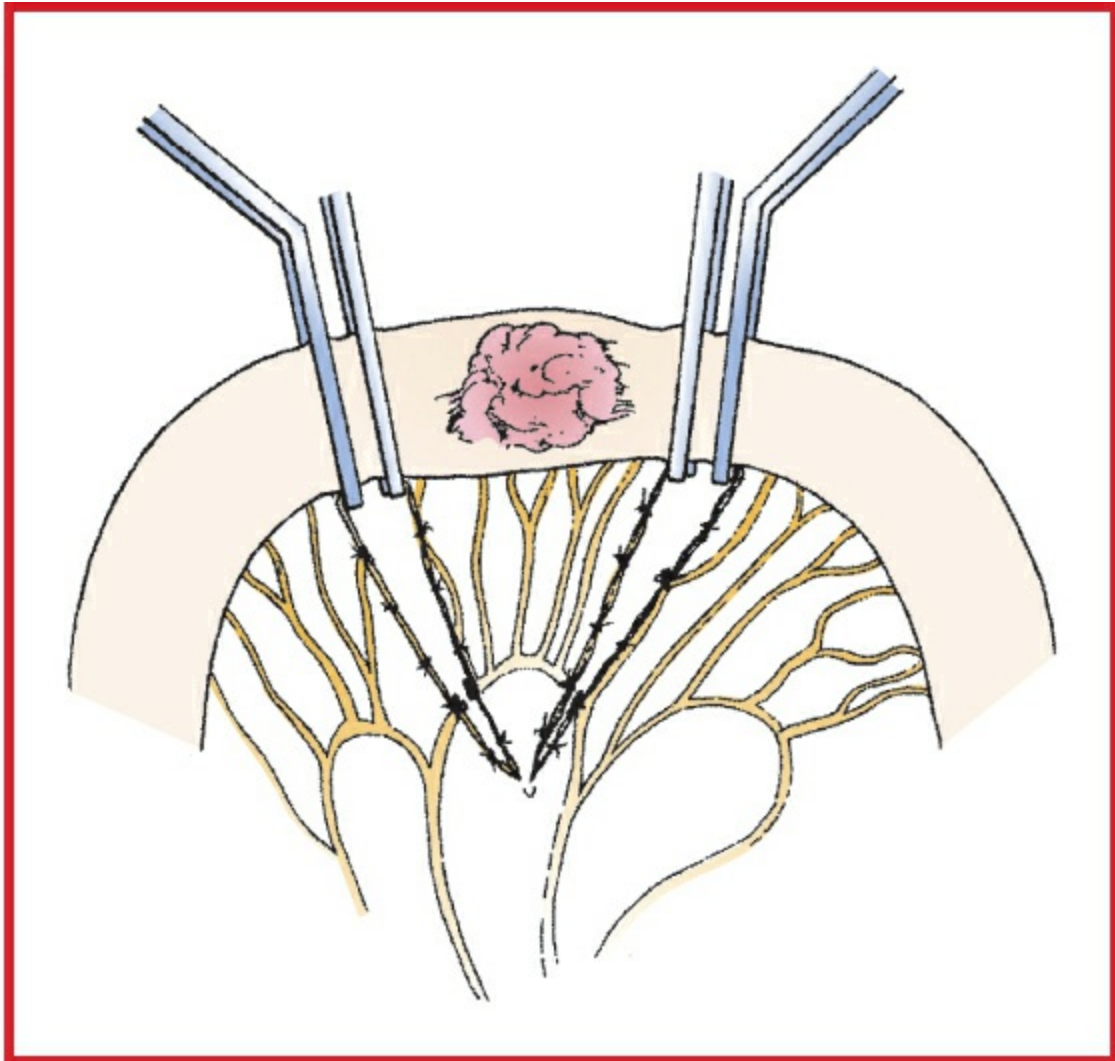


Figure 7.3-5 Block-Potts bowel clamps are applied from the antimesenteric to mesenteric border to avoid twisting. A Kocher clamp is applied on the specimen side, and the bowel is transected with a scalpel. (Reproduced with permission from Baker RJ, Fischer JE, eds *Mastery of Surgery*, Vol. II, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Intestinal obstruction; complicated by intestinal gangrene due to adhesions; internal hernia; volvulus; intussusception; mesenteric vascular occlusion; Crohn disease; radiation enteritis; intestinal fistulae; small bowel tumors; trauma; and carcinoid tumors

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Position	Supine
Incision	Vertical or transverse
Unique considerations	Adequate fluid resuscitation; NG tube
Antibiotics	Cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	1–3 h
EBL	50–100 mL
Postop care	NG or long intestinal tube decompression
Mortality	Varies according to etiology: 1–5%
Morbidity	Atelectasis: <10% Intestinal ileus: <10% Wound infection: <5% Intestinal leak, fistula: <3%
Pain score	7–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–90 yr
Male:Female	1:1
Incidence	Common
Etiology	Interference with blood supply (obstruction, strangulated hernia, volvulus, mesenteric thrombosis); trauma; tumors; Crohn disease
Associated conditions	Multiple, depending on etiology (see Preop diagnosis)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Intestinal and Peritoneal Procedures.

Suggested Readings

1. Aparicio T, Zaanan A, Mary F, et al: Small bowel adenocarcinoma. *Gastroenterol Clin North Am* 2016; 45(3):447–57.
2. Bower KL, Lollar DI, Williams SL, et al: Small bowel obstruction *Surg Clin North Am* 2018; 98(5):945–71.

ENTEROLYSIS

SURGICAL CONSIDERATIONS

Description

Enterolysis consists of separating loops of bowel adhered to other loops or the abdominal wall by sharp dissection and by excising adhesive bands. Care must be taken to avoid producing enterotomies. Covering potential adhesion sites with a hyaluronic carboxymethylcellulose membrane may lessen the formation of intraperitoneal adhesions. This operation can also be done laparoscopically with fewer recurrent adhesions.

Usual preop diagnosis

Intra-abdominal adhesions; intestinal obstruction

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES

Position	Supine
Incision	Midline abdominal
Special instrumentation	A long intestinal tube may be necessary for decompression and fixation of bowel loop.
Unique considerations	Bowel decompression
Antibiotics	Cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq

Surgical time	1–4 h
Closing considerations	Adequate decompression to permit wound closure
EBL	150–500 mL
Postop care	PACU; continued intestinal decompression 2–5 d
Mortality	1–3%
Morbidity	Wound abscess: 15–20% Prolonged ileus: 10–20% Fistula formation: <10% Pulmonary complications: 5–10% Recurrent intestinal obstruction: 5–8%
Pain score	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1
Incidence	Common
Etiology	Previous intra-abdominal operative procedure (>90%); malignant tumors (15–20%); hernias (10–15%); volvulus (5–10%); inflammatory bowel disease (5%); gallstone ileus (<5%); intussusception (<5%); radiation

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Intestinal and Peritoneal Procedures.

CLOSURE OF ENTERIC FISTULAE

SURGICAL CONSIDERATIONS

Description

Enteric fistulae may occur between the bowel and abdominal wall (enterocutaneous), between loops of the intestine (enteroenteric or enterocolic), or between the bowel and

bladder or vagina (enterovesical or enterovaginal). Surgical repair is usually reserved for fistulae to the abdominal wall, bladder, and vagina and consists of excising the fistula and repairing the bowel and the other organ separately. Most fistulae are characterized by the adherence of the two visceral organs with a communication between their lumens.

The organs involved are separated by blunt-sharp dissection and repaired locally after excision of the indurated margins of the defect. In the case of both the small and large intestines, it may be necessary to resect a segment of the bowel with the defect and to perform an end-to-end anastomosis. If the repair sites involved lie close together, it is important to interpose tissue, such as the omentum, between the viscera to minimize chance of recurrence. Occasionally, a fistula may be bypassed rather than surgically resected.

Usual preop diagnosis

Enteric fistula

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, the possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES	
Position	Supine
Incision	Midline abdominal
Unique considerations	Preop nutritional support and fistula wound care
Antibiotics	Cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	2–4 h
Closing considerations	Separation of repairs by interposition of the omentum and other tissues
EBL	50–300 mL
Postop care	NG decompression until bowel function returns; TPN

	support.
Mortality	0–5%
Morbidity	Ileus: 60–70% Pulmonary complications: 10% Recurrent fistula: 5–10% Wound infection: 5–10%
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1
Incidence	Common
Etiology	Anastomotic leaks (60–70%); carcinoma (10–15%); Crohn disease (5–10%); iatrogenic bowel injury (5–10%); perforative diverticulitis (5–10%); radiation enteritis (5%); foreign body perforation (<5%)
Associated conditions	Malnutrition (30%); inflammatory bowel disease (25%); cancer (15%)

Suggested Reading

1. Schechter WP, Hirshberg A, Chang DS, et al: Enteric fistulas: principles of management. J Am Coll Surg 2009; 209(4):484–91.

ANESTHETIC CONSIDERATIONS FOR INTESTINAL AND PERITONEAL PROCEDURES

(Procedures covered: small bowel resection; enterolysis; closure of enteric fistulae; excision of intra-abdominal and retroperitoneal tumor; drainage of subphrenic abscess)

PREOPERATIVE

Patients requiring exploratory laparotomy present both electively and emergently for a very wide range of conditions. As a result of their abdominal pathology, these patients are often at high risk for the pulmonary aspiration of gastric contents. Precautions to prevent this are necessary to help ensure safe patient outcome.

Respiratory	Respiratory insufficiency can be present due to intra-abdominal pathology (e.g., ascites, large tumor, free blood, bowel distension, pain); ↓ FRC → ↑ A-a gradient and arterial hypoxemia; diaphragmatic impairment and splinting → ↑ respiratory insufficiency. Tests: Consider CXR; ABG.
Cardiovascular	Patients for emergency surgery are often critically ill and unstable and should be evaluated for the presence of hypovolemia (hypotension, tachycardia) and should receive adequate volume replacement before anesthetic induction. Elective patients may be hypovolemic 2° bowel prep. Tests: ECG; consider orthostatic vital signs.
Musculoskeletal	Abdominal rigidity may be present; abdominal pain is common.
Gastrointestinal	Diarrhea, vomiting, and prolonged NPO status can lead to electrolyte abnormalities. Malnutrition may be present. Tests: Electrolytes, consider PT/PTT.
Renal	Renal insufficiency/failure may be present, especially in elderly and chronically ill patients and in those who are hypovolemic. Tests: Consider BUN; Cr; electrolytes.
Laboratory	CBC with differential; Plt count
Premedication	Consider midazolam 1–2 mg iv and Na citrate (30 mL po 10 min preop).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with the surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with a sip of water. Consider aprepitant 40–80 mg po with a sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

GETA± **epidural for postop analgesia.** If postop epidural analgesia is planned, placement of catheter before anesthetic induction is helpful to establish correct placement in the epidural space (accomplished by injecting 5–7 mL of 2% lidocaine via the epidural catheter and confirming segmental block).

Induction	The patient with abdominal pathology is often at risk for pulmonary aspiration. Consider rapid-sequence iv induction with cricoid pressure. (See Rapid-Sequence Induction, Appendix B, p. B-5 .) If patient is clinically hypovolemic, restore intravascular volume before induction and titrate induction dose of sedative/hypnotic agents.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Balanced anesthesia. Maintain neuromuscular blockade based on nerve stimulator response. Place gastric tube to evacuate stomach contents. Combined epidural and GA: Local anesthetic (2% lidocaine with 1:200,000 epinephrine 5–15 mL initial dose and then 3–5 mL q60min) can be injected into the epidural catheter to provide both anesthesia and optimal surgical exposure (contracted bowel and muscle relaxation). A continuous infusion of local anesthetic (e.g., 2% lidocaine or 0.25% bupivacaine) at 3–5 mL/h may be used in place of bolus redosing. Be prepared to treat hypotension with fluid and vasopressors. GA is administered to supplement regional anesthesia and for amnesia. If epidural opiates are used for postop analgesia, a loading dose (e.g., hydromorphone 0.5–1.0 mg) should be administered at least 1 h before the conclusion of surgery. The use of systemic analgesics may be minimized during this type of anesthetic with the

	benefit of decreasing the likelihood of postop respiratory depression.	
Emergence	The decision to extubate at the end of surgery depends on the patient's underlying cardiopulmonary status and the extent of the surgical procedure. Patients should be hemodynamically stable, warm, cooperative, and fully reversed from any muscle relaxants before extubation. If the above criteria are not met, consider postop ventilation in the ICU setting. Discuss with the surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Anticipate large fluid shift. IV: 14–16 ga × 1–2 T&S, consider T&C BSS @ 10–15 mL/kg/h Fluid warmer	Plts, FFP, and cryoprecipitate should be administered according to lab tests (Plt count, PT, PTT, DIC screen, thromboelastography [TEG]). Strive to maintain euvolemia based on estimated blood loss and fluid shifts, HR, BP, UO, ABG, and invasive monitors when used. Consider Goal-Directed Fluid Management Appendix K, p. K-4.
Monitoring	Standard monitors UO ± Arterial line ± CVP/PA catheter ± TEE	Invasive monitors, as indicated by the patient's status. Prevent hypothermia: use forced-air warmer; consider warming blanket, warming iv fluids, warm room temperature, keeping the patient covered until ready for prep.

Positioning	✓ pad pressure points	
Complications	✓ eyes Hemorrhage Sepsis Atelectasis Hypoxemia VTE Hemorrhage Ileus PONV	Acute septic shock may require aggressive hemodynamic support. A PA catheter, or TEE may help guide management in the unstable patient. Pulmonary function abnormalities may persist for 1 wk postop (↓ vital capacity and ↓ FRC).
Multimodal pain management	Epidural analgesia PCA	Emphasize a multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia and TAP blocks (see Appendix C, p. C-4). Consider combining PCA or fixed dose po/iv opiates with acetaminophen, po or iv NSAIDs, and gabapentin.
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	CBC; CXR (if central line placed); electrolytes; glucose	Others as directed by intraop course.

POSTOPERATIVE

Suggested Readings

1. Becker HP, Willms A, Schwab R: Small bowel fistulas and the open abdomen. Scand

J Surg 2007; 96:263–71.

2. Liu Q, Jiang H, Xu D, et al: Effect of gum chewing on ameliorating ileus following colorectal surgery: a meta-analysis of 18 randomized controlled trials. Int J Surg 2017; 47:107–15.

CHAPTER 7.4

Colorectal Surgery

SURGEON

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ANESTHESIOLOGIST

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ENHANCED RECOVERY AFTER SURGERY (ERAS)

Fast-track or enhanced recovery programs after elective surgery incorporate a variety of evidence-based interventions in an attempt to maintain normal physiologic function, enhance recovery, and prevent or minimize postoperative complications. They provide the patients with optimal means to counteract or minimize the deleterious effects of surgery. These interventions encompass all phases of a patient’s periop care from the preop phase, through the intraop phase, and to the end of the postop care. These programs are increasingly used in the perioperative management of patients with colorectal conditions. The anesthesiologist is responsible for several key elements that can affect outcomes after surgery: stress reactions to the surgery, fluid therapy, analgesia, and avoidance of PONV. Typical components of ERAS programs applicable to colorectal surgery are noted below. Additional information can be found in [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

Preop	Intraop	Postop
Avoid preop fasting; allow clear liquids until 2 h before induction of anesthesia.	Standard anesthetic protocol allowing rapid awakening	Minimize excessive postop fluids.
Preop carbohydrate loading	Use of midthoracic epidural for open	Early resumption of enteral feeding

surgery		
Avoid long-acting sedative medications before surgery.	Prevention of PONV	Prevention of postop ileus
DVT prophylaxis with compression stockings and LMWH or unfractionated heparin	Maintenance of normothermia	Postop glucose control
Prophylactic antibiotics given 30–60 min before incision	Avoid excessive intraop fluids.	Early mobilization
Preop optimization (smoking cessation, avoidance of alcohol, preop nutritional assessment; correction of preop anemia, prehab)	Use of laparoscopic surgical techniques	Avoid NG tubes. Multimodality postop analgesia minimizing narcotics
	Opioid-sparing analgesia	Opioid-sparing analgesia

Several studies have demonstrated that the ERAS programs compared with traditional postop care are associated with earlier recovery and discharge after colorectal surgery with decreased morbidity.

Suggested Readings

1. Grant MC, Pio Roda CM, Canner JK, et al: The impact of anesthesia-influence process measure compliance on length of stay: results from an enhanced recovery after surgery for colorectal surgery cohort. *Anesth Analg* 2019; 128(1):68–74.
2. Gustafsson UO, et al: Guidelines for perioperative care in elective colonic surgery: Enhanced Recovery After Surgery (ERAS®) Society recommendations: 2018. *World J Surg* 2019; 43(3):659–95.
3. Tiernan JP, et al: Enhanced recovery after surgery. Recent developments in colorectal surgery. *Surg Clin North Am* 2018; 98(6):1241–9.

LAPAROSCOPIC COLORECTAL SURGERY

Laparoscopic surgery has changed the face of general surgery with the widespread use of laparoscopic cholecystectomy, appendectomy, and other surgical procedures. Although across the country, many colorectal procedures continue to be done in the standard open fashion, laparoscopic techniques are being used more and more for

procedures on the colon and rectum. From 2010 to 2014, the number of minimally invasive surgery (MIS) segmental colectomies increased by 157% at academic hospitals, 151% at comprehensive hospitals, and 153% at community hospitals. All of the following procedures can be done, and have been done, laparoscopically. Advantages to the patient include smaller incisions, less postop discomfort, a decreased in-hospital stay, and decreased wound-related complications with early return to work and regular activity. Steep positional changes are often used to facilitate retraction of the small bowel out of the operative field. The patient is often placed on a beanbag to prevent movement. The term **laparoscopic-assisted** is more appropriate for colorectal procedures because the colon often is mobilized laparoscopically. A small incision is then made, through which the bowel is exteriorized, the mesentery is divided, and an anastomosis is created. **Hand-assisted laparoscopy** involves placement of a hand port through a 5- to 10-cm incision. The abdominal cavity is then insufflated as in standard laparoscopy, but the surgeon's hand is used alongside the other laparoscopic instruments. Laparoscopic techniques continue to advance with the development of single-incision laparoscopic surgery (SILS) for resection of the colon and rectum **natural orifice transluminal endoscopic surgery (NOTES)** techniques in which the resected specimen is removed from a natural orifice such as the vagina, and the use of the da Vinci robot for the laparoscopic treatment of rectal cancer as well as for the treatment of benign and malignant conditions of the colon.

Laparoscopic resections of the colon or rectum are done using a pneumoperitoneum with its associated physiologic changes. Laparoscopic surgery involves the insufflation of CO₂ into the peritoneal cavity at a rate of 4–6 L/min to a pressure of 15–20 mm Hg. The raised intra-abdominal pressure (IAP) and the effects of CO₂ can cause changes in the physiology of the respiratory and cardiovascular systems. Increased IAP affects venous return, systemic vascular resistance, and myocardial function. Pneumoperitoneum also results in cephalad shift of the diaphragm, decreasing FRC possibly to values less than the closing volume. This can result in airway collapse, atelectasis, V/Q mismatch, potential hypoxemia, and hypercarbia.

Because of the inability to use retractors as in open surgery, retraction of the bowel out of the operating field is usually accomplished through changes in the patient's position on the operating table, frequently requiring steep degrees of Trendelenburg position. Because of this, the patient must be secured to the operating table. These position changes can further affect the cardiovascular and respiratory systems in addition to those caused by the pneumoperitoneum itself.

An important consideration for any laparoscopic procedure is that the surgeon might

need to convert to an **open laparotomy**. This may occur secondarily for failure to progress or in an emergently for technical difficulties.

Suggested Readings

1. Andolfi C, Umanskiy K: Appraisal and current considerations of robotics in color and rectal surgery. *J Laparoendosc Adv Surg Tech A* 2019; 29(2):152–8.
2. Chang J, Rattner DW: History of minimally invasive surgical oncology. *Surg Oncol Clin N Am* 2019; 28(1):1–9.
3. Cirotchi R, Fearnhead N, Vettoretto N, et al: The role of emergency laparoscopic colectomy for complicated sigmoid diverticulitis: a systematic review and meta-analysis. *Surgeon* 2018.
4. Fransgaard T, Pinar I, Thygesen LC, et al: Association between robot-assisted surgery and resection quality in patients with colorectal cancer. *Surg Oncol* 2018; 27(2):177–84.
5. Mege D, Garrett K, Milsom J, et al: Changing trends in surgery for abdominal Crohn's disease. *Colorectal Dis* 2019; 21(2):200–7.
5. Ng JL, Lai JH, Li HH, et al: Totally-laparoscopic versus laparoscopic-assisted low anterior resection for rectal cancer: are outcomes different? *ANZ J Surg* 2018; 88(12):E818–23.
7. Stewart CL, Ituarte PHG, Melstrom KA, et al: Robotic surgery trends in general surgical oncology from the National Inpatient Sample. *Surg Endosc* 2018.

TOTAL PROCTOCOLECTOMY

SURGICAL CONSIDERATIONS

Description

A **total proctocolectomy** involves the removal of the entire colon, rectum, and anus (Fig. 7.4-1). Indications for this operation include ulcerative colitis (UC), Crohn disease (CD), and familial adenomatous polyposis (FAP). Inflammatory bowel disease (IBD) can be diagnosed at any age, but there are peaks in diagnosis in the teens and the 20s and the 60s and 70s. The most common indication for total proctocolectomy in the setting of UC or CD is intractable symptoms despite maximal medical therapy. Patients who come to surgery are often chronically or acutely ill and may be malnourished or anemic. They are frequently on immunosuppressive drugs such as corticosteroids, 6-mercaptopurine, or azathioprine or biologic inhibitors of tumor necrosis factor alpha (infliximab, adalimumab, golimumab, and certolizumab pegol) or anti-integrin molecules (natalizumab and vedolizumab). These medications all predispose these

patients to an increased risk of postop infections and complications due to poor wound healing. Another important indication for proctocolectomy in patients with UC is the presence of dysplasia or cancer.

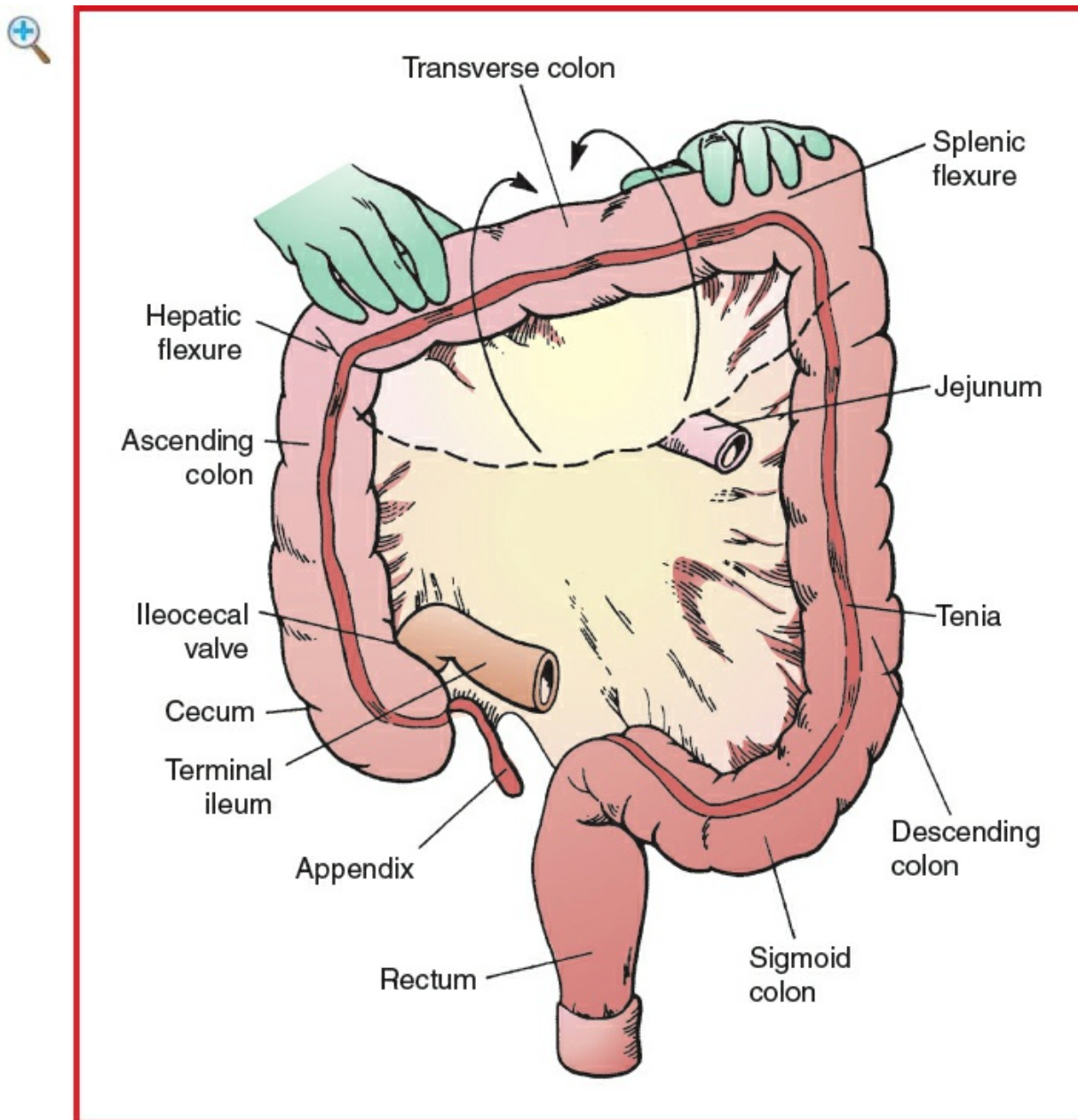


Figure 7.4-1 Anatomy of the colon. (Reproduced with permission from Hardy JD:Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Baltimore: 1988.)

FAP is an autosomal-dominant disease resulting in colon cancer. Patients develop hundreds to thousands of adenomatous polyps throughout their colon and rectum, as well as elsewhere in the GI tract. Colorectal cancer is inevitable unless proctocolectomy is performed. This is typically done in the late teens or 20s. In contrast to patients with CD or UC, patients with FAP are usually healthy without other medical comorbidities.

Sequential compression stockings are used for thromboprophylaxis. Patients with CD and UC are at ↑ risk for the development of DVT and should be given subcutaneous unfractionated heparin or low-molecular-weight heparin. Patients taking chronic corticosteroids are given stress dose steroids before the procedure. Broad-spectrum antibiotics covering gram-negative rods and anaerobes are given before the incision.

Total proctocolectomy with **end ileostomy**, total proctocolectomy with **continent ileostomy (Koch pouch)**, and restorative **proctocolectomy with ileal pouch anal anastomosis (IPAA)** all involve complete removal of the colon and rectum, down to the level of the pelvic floor or levator ani muscles. They differ in the fate of the anal canal, creation of a stoma, or construction of an anastomosis. The patient is placed in a lithotomy position in padded Allen stirrups. A Foley catheter is placed. If the procedure is done using laparoscopic techniques, a small incision is made in the periumbilical region and the suprapubic region or as a Pfannenstiel incision to extract the specimen. If done as an open procedure, it is commonly performed through a midline incision. The abdomen is explored for evidence of unexpected malignancy or, in the case of FAP, for desmoid tumors. The right colon is mobilized first, and then the small bowel mesentery is mobilized to allow for creation of an ileostomy. The transverse colon may be mobilized by separating it from the greater omentum, or the greater omentum may be resected along with the specimen. The sigmoid and descending colon are mobilized, and the splenic flexure is taken down. The ileum is then divided flush with the cecum, and the vessels in the colon mesentery are ligated. At this point, the entire abdominal colon has been resected. An avascular fascial envelope surrounds the rectum and its mesentery, the mesorectum. It is possible to circumferentially dissect the rectum down to the level of the pelvic floor without ligating any vessels. There may be significant blood loss if an inadvertent injury to the spleen occurs during mobilization of the splenic flexure. Massive blood loss may occur if the presacral venous plexus is entered during posterior rectal mobilization.

Total proctocolectomy with ileostomy

For patients with CD, elderly patients with UC, or FAP patients with low rectal cancer complete removal of the colon, rectum, and anus is the procedure of choice. After completing the abdominal mobilization of the colon and rectum, the perineal phase of the operation begins. Ideally, two teams of surgeons participate in the operation simultaneously. The abdominal surgeon can create the ileostomy and close the abdomen, while the perineal surgeon finishes removal of the rectum and anus. A circumferential incision is made at the anal verge, and the intersphincteric plane is identified. The dissection proceeds cephalad until the abdominal dissection is encountered, and the specimen is removed. The levator ani muscle, external anal sphincter, and skin are

closed. While this is being done, the abdominal surgeon makes a circular incision over the previously marked ileostomy site. A muscle-splitting incision is carried through the rectus fascia. The terminal ileum is then brought through this site. After the fascia and skin are closed, the ileostomy is matured. Some surgeons prefer to do the perineal phase of a proctocolectomy in the prone jackknife position. In this case, after completing the abdominal phase of the operation, the abdomen is closed, the stoma is matured, and the patient is flipped prone to finish the procedure.

Total proctocolectomy with continent ileostomy (Koch pouch)

Because of frequent complications and the development of alternative procedures (see below), this procedure has been largely abandoned.

Restorative proctocolectomy with ileal J-pouch anal anastomosis

IPAA is the procedure most performed for patients with FAP and UC. In this operation the colon and rectum are removed down to the level of the pelvic floor; however, the anal canal and anal sphincter complex are preserved. The rectum is stapled and divided at the level of the surgical anal canal, ~1–1.5 cm above the dentate line. An ileal reservoir is constructed by anastomosing the distal 30 cm of the ileum in a side-to-side fashion, creating a J-pouch. The apex of the pouch is then anastomosed to the anal canal using a circular stapling device (see Fig. 7.4-2). Rarely, a hand-sewn anastomosis is created. A temporary diverting loop ileostomy may or may not be created, depending on the clinical situation.

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Total Proctocolectomy	IPAA	Laparoscopic
Position	Modified lithotomy	⇐	⇐
Incision	Midline	Midline	Suprapubic or Pfannenstiel
Special instruments	Two-table setup (separate	Two-table setup; anal retractors	⇐

	abdominal and perineal instruments); deep pelvic instruments		
Unique considerations	Patients frequently on chronic corticosteroids	⇐; epinephrine solutions injected under the rectal mucosa to facilitate dissection and reduce bleeding	⇐
Antibiotics	Cefazolin 2 g iv and metronidazole 500 mg iv or second-generation cephalosporin (e.g., cefotetan 2 g iv)	⇐	⇐
Surgical time	3–4 h	3–4 h	4–5 h
Closing considerations	Ileostomy completed after skin is closed (15–30 min).	Temporary ileostomy often used, completed after skin closure	⇐
EBL	100–1000 mL (most blood loss during pelvic and perineal dissection)	⇐	⇐
Postop care	Transient inability to void common	Small bowel mesentery lengthened by dissection around duodenum. May lead to PONV; transient inability to void	⇐
Morbidity	Dyspareunia: 30% Stoma	5–10% NA	⇐

	complications:	⇐	
	20%	⇐	
	SBO: 10–15%	Nocturnal incontinence	
	Erectile	Poor function: 5%	
	dysfunction: 2–4%	Pelvic sepsis: 0–4%	
Mortality	2–5% (older patients and those with underlying chronic medical problems)	<1%	⇐
Pain score	8	8	6

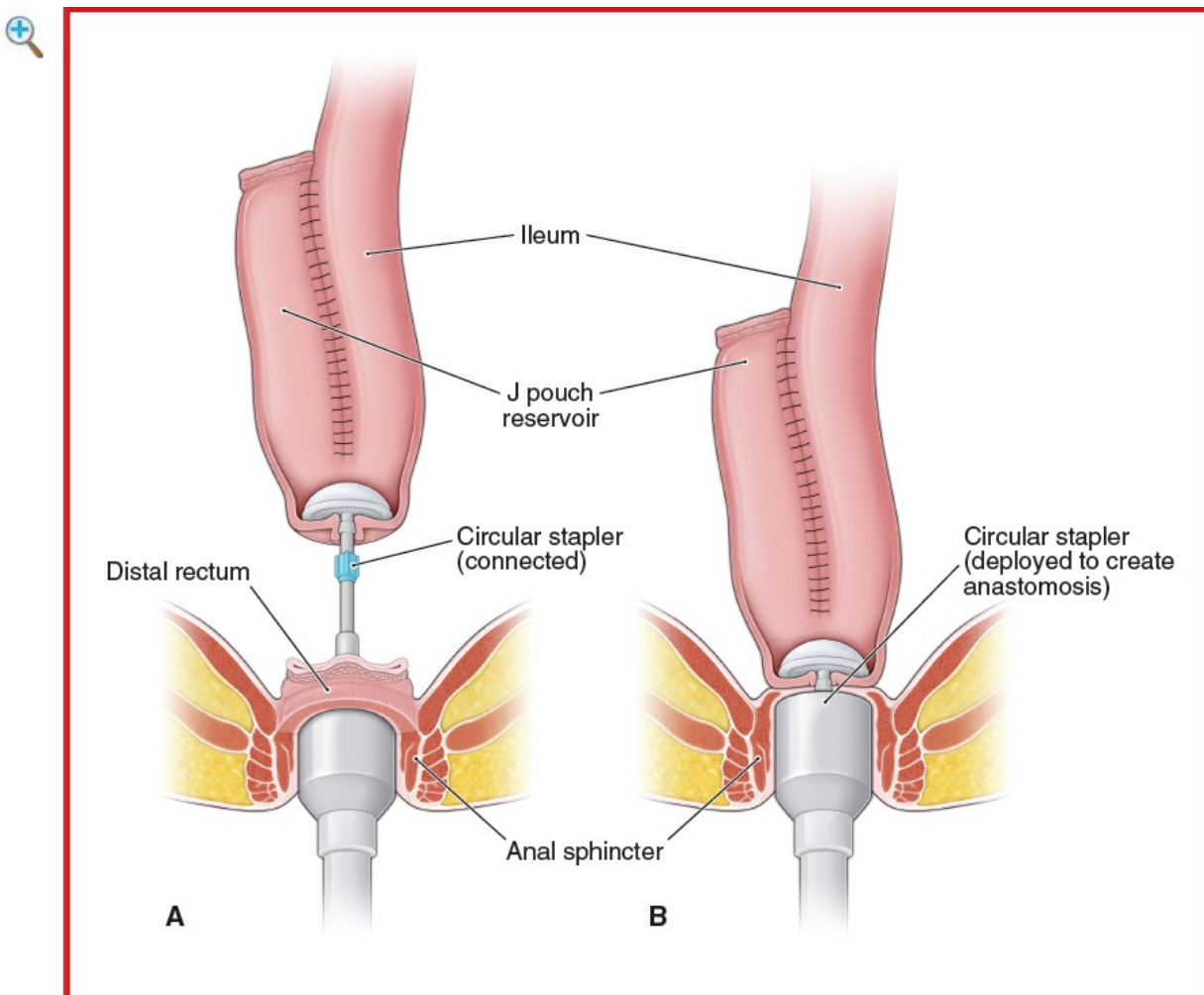


Figure 7.4-2 Creation of a J-pouch using a circular stapler.

■ PATIENT POPULATION CHARACTERISTICS

	Ulcerative Colitis	Familial Adenomatous Polyposis
Age range	3rd–5th decade	2nd–4th decade
Male:Female	1:1	⇐
Incidence	6–10/100,000	100–150 cases/yr

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Large Bowel Surgery.

Suggested Readings

1. Cohen BL, Sachar DB: Update on anti-tumor necrosis factor agents and other new drugs for inflammatory bowel disease. *BMJ* 2017; 357:j2505.
2. Ferrari L, Krane MK, Fichera A: Inflammatory bowel disease surgery in the biological era. *World J Gastrointest Surg* 2016; 8(5):363–70.
3. Gajendran M, Loganathan P, Catinella AP, et al: A comprehensive review and update on Crohn's disease. *Dis Mon* 2018; 64(2):20–57.
4. Holder-Murray J, Marsicovetere P, Holubar SD: Minimally invasive surgery for inflammatory bowel disease. *Inflamm Bowel Dis* 2015; 21(6):1443–58.
5. Masaki T, Kishiki T, Kojima K, et al: Recent trends (2016-2017) in the treatment of inflammatory bowel disease. *Ann Gastroenterol Surg* 2018; 2(4):282–8.
5. Neumann PA, Rijcken E: Minimally invasive surgery for inflammatory bowel disease: review of current developments and future perspectives. *World J Gastrointest Pharmacol Ther* 2016; 7(2):217–26.
7. Nickerson TP, Merchea A: Perioperative considerations in Crohn disease and ulcerative colitis. *Clin Colon Rectal Surg* 2016; 29(2):80–4.
3. Øresland T, Faerden AE: Surgery in the age of biological treatment. *Scand J Gastroenterol* 2015; 50(1):121–7.
9. Shung DL, Abraham B, Sellin J, et al: Medical and surgical complications of inflammatory bowel disease in the elderly: a systematic review. *Dig Dis Sci* 2015; 60(5):1132–40.

SEGMENTAL (PARTIAL) COLECTOMY

SURGICAL CONSIDERATIONS

Description

Segmental colectomy involves removal of a portion of the colon (Fig. 7.4-1) with the creation of an **anastomosis** or a **stoma**. The most common indications for the operation in the Western world are colon cancer and diverticulitis. Less common indications include traumatic perforation, gastrointestinal hemorrhage, ischemic colitis, volvulus, and IBD. Both colon cancer and diverticular disease occur most often in patients >50 yr. Patients may have any of the comorbid medical conditions associated with aging, as well as complications related to the disease requiring colon resection. Free perforation of the colon can occur from a variety of conditions, including diverticulitis, cancer, and ischemia. Patients may be hypovolemic and have systemic sepsis. **Emergent laparotomy** should follow a period of resuscitation and administration of antibiotics. The involved segment of bowel is resected, the abdomen is irrigated, and a stoma is created.

Colon cancer is the second most common cancer in the United States with 153,000 new cases diagnosed annually. Patients may be entirely asymptomatic, with the diagnosis being made only as the result of a screening exam. Because of the large caliber of the colon and the liquid nature of stool, patients with cancers of the right colon are more likely to present with large cancers and anemia. The caliber of the left colon is smaller and the stool more solid. When symptomatic, symptoms of obstruction and change in bowel habits predominate for left-sided lesions.

Colonic diverticula occur in up to 60–70% of people over 50 yr of age in the United States. A colonic diverticulum is a herniation of the mucosa and submucosa through the relative weakening that occurs in the muscular wall of the bowel at the site of penetrating blood vessels. This occurs predominantly in the sigmoid colon. Most people with colonic diverticula are completely asymptomatic and will never experience any complications related to diverticulosis. Diverticulitis occurs when a microscopic or macroscopic perforation of a colonic diverticulum occurs, resulting in a pericolic inflammatory and infectious process. The severity of the attack depends on the degree of perforation and how well the body is able to wall it off. This ranges from minor inflammation around the sigmoid colon that can be managed with antibiotics to an intra-abdominal or pelvic abscess requiring **percutaneous drainage** to free perforation with purulent or feculent peritonitis requiring emergency surgery. Repeated bouts of diverticulitis eventually can result in fibrosis of the colon, stricture formation, and obstruction.

Ideally, surgery on the colon is performed in an elective setting; however, perforation with peritonitis or complete obstruction of the colon may require emergency surgery. Most patients presenting for elective colon resection undergo preop bowel preparation

that consists of a combination of mechanical cleaning of the colon as well as oral antibiotics to decrease the bacterial load in the colon. As a result, they are frequently hypovolemic and hypokalemic when they come to the OR.

The patient is positioned either supine or in the modified lithotomy position, depending on the segment of the colon to be removed. Sequential compression stockings are used for thromboprophylaxis, and chemoprophylaxis for DVT with either unfractionated heparin or low-molecular-weight heparin is given. Intravenous antibiotics covering gram-negative rods and anaerobes should be given before the incision with redosing as appropriate for the antibiotic used.

Segmental colectomy

Segmental resection of the colon may be performed laparoscopically or via midline or transverse abdominal incisions, depending on the underlying disease, portion of the colon to be resected, and the surgeon's preference. For open surgery, midline incisions are preferred when a high-lying splenic flexure must be mobilized, IBD is present, or the extent of the colon resection is not known preoperatively. Transverse incisions are most commonly reserved for resections of the right colon. The most commonly performed partial colon resections are **right hemicolectomy**, **sigmoid colectomy**, **left hemicolectomy**, and **abdominal colectomy** with an **ileorectal anastomosis**. The sequence of steps in a **partial colectomy** is the same for all parts of the colon. The right colon and left colon are retroperitoneal structures, whereas the transverse colon and sigmoid colon are primarily intraperitoneal. The first step is mobilization of the colon and its mesentery. Care must be taken not to injure the left ureter during mobilization of the sigmoid colon or the duodenum during mobilization of the right colon. Proximal and distal sites for resection are selected, and the intervening mesentery is divided. The **anastomosis** may be hand sewn or stapled, which is a decision based primarily on the surgeon's preference.

There is no clear advantage to either anastomotic technique. Creation of a **diverting stoma** rather than an anastomosis may be necessary in patients who are hemodynamically unstable or when intra-abdominal conditions, such as inflammation, make an anastomosis unsafe. There may be significant blood loss if an inadvertent injury to the spleen occurs during mobilization of the splenic flexure. Excessive traction of the hepatic flexure can result in difficult-to-control venous bleeding.

Obstruction of the colon most commonly occurs as a result of cancer of the sigmoid colon or repeated bouts of diverticulitis. Patients present with abdominal distension, obstipation, and N/V. Patients are treated with NG tube decompression and correction of hypovolemia. An attempt may be made to stent the obstructing lesion endoscopically

preop to allow decompression and preparation of the colon. If this is not possible, surgical options include segmental resection with a **colostomy**, segmental resection with primary anastomosis, and an on-table colonic lavage or **subtotal colectomy** with an ileorectal anastomosis.

Usual preop diagnosis

Colon cancer; diverticular disease; Crohn disease; ulcerative colitis; trauma; ischemic colitis; lower GI hemorrhage; intractable constipation; colon volvulus

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of NG tube and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine or modified lithotomy
Incision	Transverse, vertical midline, laparoscopic
Unique considerations	Bowel prep, or underlying disease may → dehydration, electrolyte abnormalities, or anemia.
Antibiotics	Cefazolin 2 g iv and metronidazole 500 mg iv or second-generation cephalosporin (e.g., cefotetan 2 g iv)
Surgical time	1–3 h
Closing considerations	Colostomy or ileostomy matured after wound is closed (requires 10–20 min).
EBL	100–300 mL (500–2000 mL if splenic injury, loss of major vascular pedicle, or repeat operation for cancer, Crohn disease)
Postop care	ICU for underlying disease; NG may be used in the setting of obstruction. Avoid use of long-lasting anticholinergic effects.
Mortality	0.5–2% (mostly related to underlying disease)
Morbidity	SBO: 5–10% Wound infection: 4–10% Anastomotic leak: 2–4%

Wound dehiscence: 1–2%
Bleeding: 1%
Splenic injury: 1%

Pain score 8

PATIENT POPULATION CHARACTERISTICS

	Crohn Disease	Colon Cancer	Trauma	Diverticulitis
Age range	2nd–4th decade	5th–7th decade	2nd–4th decade	>40 yr
Male:Female	1:1	1.3:1	3:1	1:1
Incidence	1–6/100,000	30/100,000	1–2/100,000	10/100,000
Etiology	Unknown	Genetic: 5%	Trauma	Western countries: low-fiber diet
Associated conditions	Malnutrition; anemia; intra-abdominal sepsis; intestinal fistulae; peri-anal disease; nephrolithiasis; sclerosing and ankylosing spondylitis	Iron deficiency; colonic obstruction; colonic perforation	Liver fracture; spleen fracture; rib fracture; closed-head injury; penetration of viscera adjacent to colon injury	Hemorrhoids; chronic constipation

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Large Bowel Surgery.

Suggested Readings

- Atallah C, Efron JE: Laparoscopy for rectal cancer. *Clin Colon Rectal Surg* 2017; 30(2):104–11.
- Baer C, Menon R, Bastawrous S, et al: Emergency presentations of colorectal cancer. *Surg Clin North Am* 2017; 97(3):529–45.
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- Liu X, Li JB, Shi G, et al: Systematic review of single-incision versus conventional multiport laparoscopic surgery for sigmoid colon and rectal cancer. *World J Surg Oncol* 2018; 16(1):220.
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STOMA CLOSURE OR PERISTOMAL HERNIA REPAIR

SURGICAL CONSIDERATIONS

Description

A **temporary colostomy** or **ileostomy** can be made for a variety of staged procedures. An **end colostomy** is often created after resection of obstructing or perforated lesions of the left colon. A proximal **loop ileostomy or colostomy** is often created to protect a “high-risk” anastomosis, such as a low pelvic colorectal or ileoanal anastomosis. A patient with a permanent stoma may develop a hernia at the stoma site. This may result in complications, such as obstruction or strangulation of the bowel, or problems with proper fitting of the stoma appliance. The extent of procedure depends on the type of stoma created.

Closure of loop stoma

Closure of a loop stoma is performed through a circular incision, placed just outside the mucocutaneous junction of the stoma and the skin. The proximal and distal ends of the bowel are separated from the subcutaneous tissue and anterior fascia, and then the posterior fascia. The bowel is cleaned of adherent skin, and the previously opened antimesenteric border of the bowel is simply closed with sutures. Alternatively, the previously exteriorized portion of bowel is resected, and the two ends are anastomosed with sutures or staples. On rare occasions, it is necessary to extend the incision transversely through the abdominal wall to safely perform an anastomosis. The fascia is closed in the standard fashion. Loop ileostomies are most commonly performed for patients who have had surgery for rectal cancer and who have a low colorectal anastomosis. These patients have frequently received chemotherapy and radiation therapy. A loop ileostomy is also created as part of a **restorative proctocolectomy**

with an **ileal J-pouch anal anastomosis**. These patients tend to be significantly younger.

Closure of end stoma

Closure of an end stoma can be attempted as a laparoscopic procedure. Dense adhesions or difficulty defining the intra-abdominal anatomy may necessitate conversion to an open operation. After entering the peritoneal cavity, the stoma is freed from the abdominal wall. The most common indication for closure of an end colostomy is restoration of intestinal continuity after emergent surgery for perforated diverticulitis or an obstructing left-sided cancer. The severity and indication for the original procedure can have a significant impact on the difficulty of the colostomy takedown. Not infrequently, this procedure begins with **cystoscopy** and **placement of bilateral ureteral stents** by an urologist given the risk of ureteral injury in **reoperative pelvic surgery**. Adhesions are lysed, and the dysfunctioned distal segment of the bowel is identified. It may be necessary to mobilize the proximal bowel to provide a tension-free anastomosis. An anastomosis can then be created using either hand-sewn or stapled techniques.

Parastomal hernia repair

Repair of a stoma hernia may be performed as an open operation, as a laparoscopic operation, or as a robotic procedure. After freeing up and dissecting out any intra-abdominal contents from the hernia, the hernia defect is typically closed using prosthetic or biologic mesh. When done as an open procedure, the abdomen may be entered via a midline or a peristomal incision. In some cases, repair of a stoma hernia requires relocation of the stoma to an alternate site on the abdomen.

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Closure of Loop Stoma	Closure of End Stoma	Repair of Parastomal Hernia
Position	Supine	Modified lithotomy	Supine
Incision	Circumstomal	Midline or laparoscopic	Midline circumstomal or laparoscopic

Special instruments	Anastomotic staplers	Anastomotic staplers; may undergo cystoscopy and stents before procedure	Prosthetic or biologic mesh
Antibiotics	Cefazolin 2 g iv and metronidazole 500 mg iv or second-generation cephalosporin (e.g., cefotetan 2 g iv)	⇐	⇐
Surgical time	45 min–1.5 h	1.5–3 h	⇐
Closing considerations	Requires only a few fascial sutures; the skin may be left open.	Paralysis for closure of midline fascial incision and stoma site	Stoma matured after fascial and skin closure if it is relocated to a new site.
EBL	25 cc	100–500 cc	<100 cc
Mortality	<1%	1–3%	1%
Morbidity	Anastomotic leak SBO Wound infection	⇐ — —	Recurrent hernia formation — —
Pain score	5	8	6

■ PATIENT CHARACTERISTICS

	Closure of Loop Stoma	Closure of End Stoma	Repair of Parastomal Hernia
Age range	Variable; older for patient with cancer, younger	Variable; tend to be older	⇐

	for patients with IBD		
Male:Female	1:1	⇐	⇐
Etiology	Protection of anastomosis after surgery for rectal cancer; IBD; or as treatment of an anastomotic leak	Prior emergent surgery for colon perforation with peritonitis or obstruction	Prior surgery for rectal cancer or inflammatory bowel disease
Associated conditions	Colorectal cancer, ulcerative colitis, diverticulitis	Colorectal cancer, diverticulitis, trauma	Colorectal cancer, Crohn disease, ulcerative colitis

Suggested Readings

1. Colvin J, Rosenblatt S: Surgical management of parastomal hernias. *Surg Clin North Am* 2018; 98(3):577–92.
2. DeAsis FJ, Lapin B, Gitelis ME, et al: Current state of laparoscopic parastomal hernia repair: a meta-analysis. *World J Gastroenterol* 2015; 21(28):8670–7.
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ANESTHETIC CONSIDERATIONS FOR LARGE BOWEL SURGERY

(Procedures covered: total proctocolectomy; partial colectomy with anastomosis; colostomy; stoma closure; and peristomal hernia repair.)

PREOPERATIVE

Patients undergoing colorectal procedures present a heterogeneous group regarding their age, nature of the disease, and implications of the surgery. Indications for large bowel

surgery include cancer and IBD, either of which can be elective or emergent.

Respiratory	In addition to the common pulmonary diseases such as COPD, asthma, chronic bronchitis, restrictive lung disease, etc., patients with IBD may have inflammatory lung involvement with frequently negligible clinical consequences. In emergent cases due to obstruction, perforation, or peritonitis (metabolic acidosis, electrolytes abnormalities), pain, cramping, and distension are the predominate features. Limited diaphragmatic excursion ($\downarrow$ TV, $\downarrow$ FRC) leads to rapid desaturation after induction. In cases of large bowel obstruction, sufficient preoxygenation and RSI are advisable. In addition, patients with IBD may have arthritis and reduced neck ROM. Tests: Electrolytes, ABG in septic cases
Cardiovascular	Even in elective cases, hypovolemia 2° poor PO intake, vomiting, and diarrhea may be encountered. Antihypertensive drugs can exacerbate the problem (ACE inhibitors). Hemodynamic instability is common in cases of sepsis and peritonitis ($\uparrow$ HR, $\downarrow$ BP). Vasopressor agents may be indicated. Tests: ABG, ECG
Renal	Electrolyte abnormalities 2° vomiting or gastric tube suctioning (metabolic alkalosis) are common. Diarrhea and mechanical bowel preparation can cause metabolic acidosis. Tests: Electrolytes
Gastrointestinal	Most patients do not have any acute GI symptoms; some may have nausea or abdominal pain without obvious implications for anesthesia. Patients with obstruction either have already an indwelling NG tube or may benefit from decompression of the stomach. Risk of aspiration is prevalent requiring RSI. Pulling the NG tube before the induction of anesthesia is controversial. Patients with IBD may have hepatic function impairments and altered drug

	metabolism.
Hematologic	Chronic bowel inflammation and cancer often associated with chronic and at times acute GI blood loss. Look for anemia. Consider dehydration as well causing hemoconcentration that may mask anemia. Low albumin levels influence the amount of free circulating drugs in the blood. Consider coagulopathy 2° malabsorption. Tests: CBC, PLTs, INR
Laboratory	CBC, BMP, T&S vs. T&C. Other tests as indicated by H&P
Premedication	Standard premedication is usually appropriate. Patients with IBD on chronic steroid therapy should receive their usual daily dose of steroids throughout the periop period and may require full stress-dose steroids.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water, possibly in combination with gabapentin 300 mg and/or celecoxib 200 mg. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural for intra- and postop analgesia for open procedures provided there are no contraindications. The majority of colorectal procedures are performed laparoscopically, with a relatively small incision for the hand port. The postop pain after laparoscopic cases can be treated with a regimen of iv and po analgesics in majority of patients. For open procedures, thoracic epidural is associated with improved postop pain control, earlier return of bowel function, early intake of food, and out-of-bed mobilization. Intraop use of low-concentration bupivacaine or ropivacaine blunts the surgical stress and reduces the intraop opioid requirements.

Induction	RSI with cricoid pressure should be performed for induction in patients with an acute abdomen to ↓ risk of pulmonary aspiration. An NG tube may not interfere with cricoid pressure and may be left in place or removed immediately before induction after suctioning as it may obscure the airway anatomy during laryngoscopy. If patient is clinically hypovolemic, restoring intravascular volume (colloid, crystalloid) before induction may minimize hypotension. Even with titration of induction agents in elderly patients or those with acute abdomen/sepsis, profound hypotension may occur despite fluid resuscitation, requiring the use of vasopressors to blunt the effect of induction agents. RSI is not indicated for elective, fasted patients, without other indications. Preop hypovolemia is less likely with liberalized NPO guidelines commonly part of ERAS protocols.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance without N ₂ O. Combined epidural/GA and local anesthetic (e.g., 0.125% bupivacaine 3–5 mL q45–60min or 1.5–2% lidocaine with 1:200,000 epinephrine 5–10 mL q60–90min) can be injected into the epidural catheter to provide both anesthesia and optimal surgical exposure (contracted bowel and profound muscle relaxation). A continuous infusion of local anesthetic (e.g., 0.125% bupivacaine at 4–8 mL/h) may enhance hemodynamic stability compared to bolus technique. A loading dose of epidural opiate (e.g., hydromorphone 0.4–0.8 mg) should be administered 1–2 h before conclusion of surgery. Patients receiving epidural opiates should be monitored for development of delayed postop respiratory depression, and systemic sedatives (e.g., opiates, benzodiazepines) should be minimized. If placed intraop, gastric tubes are increasingly removed in the OR at the conclusion of surgery.

Emergence	The decision to extubate at the end of surgery depends on the patient's volume status, underlying cardiopulmonary status, and extent of surgical procedure. Patients should be hemodynamically stable, warm, alert, and cooperative and fully reversed from any muscle relaxants before extubation and have adequate return of pulmonary function. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis usually includes dexamethasone and ondansetron.	
Blood and fluid requirements	18 ga iv × 1 for elective cases: Consider additional and larger iv for emergency cases or hemodynamically unstable patients. BSS at 5–10 mL/kg/h Warm fluids Consider T&S or T&C.	Maintain euvolemia based on EBL and estimates of fluid shifts, UO, HR, BP, base deficit, lab studies, and invasive monitoring when used. Third-space losses may be significant in the setting of acute abdomen and systemically ill patients. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors UO ± Arterial line ± CVP Temperature	Arterial line and CVP, as indicated by patient's status. Hypothermia may delay healing and predispose patients to wound infections. Avoid hypothermia with forced-air warming blanket, warming room temperature, and keeping patient covered until ready for prep.
Positioning	Trendelenburg/reverse Trendelenburg ± lithotomy and pad	Both Trendelenburg and reverse Trendelenburg may be used during laparoscopic

	pressure points ✓ eyes	bowel surgery. Position, pad, and secure patient accordingly. Arms usually tucked for laparoscopic cases. ✓ IV access before draping.
Complications	Septic shock Aspiration of gastric contents Hemorrhage Injuries to the spleen, bowel, and ureters Unresectability of tumor	Hemodynamic instability 2° sepsis, hemorrhage, especially during manipulation of necrotic bowel

POSTOPERATIVE

Complications	Atelectasis Hemodynamic instability Sepsis VTE (see Appendix B, p. B-7) Injuries to bowel and ureters PONV (see Appendix B, p. B-6)	Patients may have considerable third-space losses that require invasive monitoring, ICU admission, and vasopressors. If placed intraop, gastric tubes are often removed in the OR at the conclusion of surgery.
Multimodal pain management	Emphasize multimodal approach combining nonopioid ± bolus/PCA opioid analgesics or epidural analgesia and/or nerve blocks (see Appendix C, p. C-4).	Consider use of non-opiate adjuncts: NSAIDs, ketamine, gabapentin, acetaminophen. PCA
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6) DVT prophylaxis	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Chewing gum may promote

return of bowel function.

Tests

CBC; CXR (if central line placed), electrolytes, glucose

Suggested Readings

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2. Grant MC, Pio Roda CM, Canner JK, et al: The impact of anesthesia-influence process measure compliance on length of stay: results from an enhanced recovery after surgery for colorectal surgery cohort. *Anesth Analg* 2019; 128(1):68–74.
3. Gustafsson UO, Scott MJ, Hubner M, et al: Guidelines for perioperative care in elective colorectal surgery: Enhanced Recovery After Surgery (ERAS®) Society Recommendations: 2018. *World J Surg* 2019; 43(3):659–95. doi: 10.1007/s00268-018-4844-y
4. Helander EM, Webb MP, Bias M, et al: Use of regional anesthesia techniques analysis of institutional enhanced recovery after surgery protocols for colorectal surgery. *J Laparoendosc Adv Surg Tech A* 2017; 27(9):898–902.
5. Liu Q, Jiang H, Xu D, et al: Effect of gum chewing on ameliorating ileus following colorectal surgery: a meta-analysis of 18 randomized controlled trials. *Int J Surg* 2017; 47:107–15.

OPERATIONS FOR RECTAL PROLAPSE

SURGICAL CONSIDERATIONS

Description

Rectal prolapse (procidentia) is intussusception of the full thickness of the rectal wall beyond the anal canal. It must be distinguished from rectal mucosal prolapse, caused by elongation of the mucosal attachments to the underlying sphincter muscle, and internal intussusception, where the upper rectum folds into the lower rectum but does not descend through the sphincter mechanism. Rectal mucosal prolapse is treated as part of the spectrum of hemorrhoidal disease, and mild-to-moderate intussusception does not benefit from surgery. Procidentia is associated frequently with anal incontinence. The surgical approaches to procidentia are determined by patient age, concurrent medical disease, sphincter function, and prior operative history, including previous abdominal surgery and prior attempts at surgical repair. Surgical treatment of procidentia may be

undertaken through an abdominal or a perineal approach. Abdominal repair of rectal prolapse can be performed in either an open, laparoscopic fashion or robotic fashion. The abdominal approaches have a lower recurrence rate and because they do not diminish the capacity of the rectal reservoir and are generally preferable for maintaining fecal continence. **Rectopexy** is an abdominal approach in which the rectum is mobilized in the posterior plane from the sacral promontory to the levator muscles. The rectum is then pulled cephalad and sutured to the presacral fascia with multiple nonabsorbable sutures or tacked to the presacral fascia using surgical tacks. Mesh may or may not be used to fix the rectum to the sacrum depending on the operative preference of the surgeon. Many surgeons routinely perform **sigmoid resection** along with rectopexy. They believe that removal of the redundant sigmoid further diminishes the chance of late recurrence and may alleviate constipation. A number of approaches have been described, the most popular being the **Ripstein procedure**. **Sling procedures** have equivalent recurrence rates but higher complication rates. As in rectopexy, the rectum is mobilized along the presacral plane down to the level of the levators. A band of Marlex mesh is sewn to the presacral fascia at the sacral promontory, upward traction is placed on the rectum, and the mesh is sutured to the rectum. **Transabdominal repair** of rectal prolapse is the favored approach for younger patients with rectal prolapse or older patients with few medical comorbidities due to the significantly lower rate of recurrence of the prolapse.

Not uncommonly, however, patients with rectal prolapse are of advanced in age or have significant medical comorbidities. In these higher-risk patients, prolapse may be repaired via a perineal approach. The most common of these is **perineal rectosigmoidectomy (or Altemeier procedure)**. The prolapsed rectum is withdrawn through the anal canal to its full extent, and a circumferential incision is made in the outer tube of the prolapsed bowel just proximal to the dentate line. This exposes the inner tube of prolapsed bowel and mesentery. Redundant bowel is mobilized from the distal end, up to the point that no additional bowel can be delivered into the operative field. The redundant bowel is transected, and a primary anastomosis is fashioned between the cut ends of the inner and outer bowel. Prior to anastomosis, the levator muscles are often approximated in the midline in an effort to aid continence. When the volume of prolapsed tissue is small, or a previous abdominal approach makes blood supply to the rectum questionable, the **Delorme procedure** often is performed. During this procedure, the mucosa is stripped off the prolapsed rectum, and the prolapsed rectal muscle is foreshortened by plication until it resides above the sphincters. **Perineal repair** of rectal prolapse is most commonly done under a general anesthetic; as the patient is typically positioned prone, the procedure can take 60–90 min; and in the case of a **perineal rectosigmoidectomy**, the peritoneal cavity is entered. However, in very

high-risk patients for whom intubation or general anesthesia is a concern, the procedure can be done under an epidural or local anesthesia with sedation.

Usual preop diagnosis

Full-thickness rectal prolapse

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES				
	Rectopexy	Rectopexy with Sigmoid Resection	Perineal Rectosigmoidectomy	Delorme Procedure
Position	Lithotomy	⇐	Prone jackknife; lithotomy	⇐
Incision	Low transverse; low midline; laparoscopic or robotic	⇐	No external incision	⇐
Special instrumentation	Deep pelvic instruments; mesh if sling planned	Deep pelvic instruments	Hip roll for jackknife position; anastomosis may be created with EEA stapler.	Hip roll for jackknife position
Unique considerations	Presacral venous plexus bleeding may occur; bowel prep may cause dehydration or hypokalemia.	⇐	Epinephrine solutions may be used to diminish bleeding; bowel prep may cause dehydration or hypokalemia.	⇐
Antibiotics	Ertapenem 1 g iv or second-generation cephalosporin (e.g., cefotetan 2 g iv)	⇐	⇐	⇐
Surgical time	1–2 h	⇐	⇐	⇐
EBL	<100 mL; more if reoperation	100–300 mL; more if reoperation	100–200 mL	100 mL
Postop care	No rectal probes or medications	⇐	⇐	⇐
Mortality	0–2%	0–4%	1–4%	0–1%
Morbidity	Rectal stricture (with sling): 5–10% Recurrent prolapse: 2–8% Pelvic infection: 5%	— 2–5% — Anastomotic leak: 2–4%	— 20–40% — —	— 5–10% — —
Pain score	7	7	2	2

■ PATIENT POPULATION CHARACTERISTICS

Age range	Women: peak incidence in 6th–7th decade; men: evenly distributed through age range
Male:Female	1:4
Incidence	Unknown
Etiology	Decreased pelvic muscular support; congenital deficiency of rectal support; pudendal neuropathy; chronic constipation and straining; multiparity; myelomeningocele; spina bifida; cystic fibrosis (children); acute parasitic diarrheal illness (children)
Associated conditions	Fecal incontinence; urinary stress incontinence; rectocele; cystocele

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Operations for Fecal Incontinence.

Suggested Readings

1. Barfield LR: Perineal approaches to rectal prolapse. Clin Colon Rectal Surg 2017; 30(1):12–5.
2. Bordeianou L, Paquette I, Johnson E, et al: Clinical practice guidelines for the treatment of rectal prolapse. Dis Colon Rectum 2017; 60(11):1121–31.
3. Emile SH, Elfeki H, Shalaby M, et al: Perineal resectional procedures for the treatment of complete rectal prolapse: a systematic review of the literature. Int J Surg 2017; 46:146–54.
4. Hori T, Yasukawa D, Machimoto T, et al: Surgical options for full-thickness rectal prolapse: current status and institutional choice. Ann Gastroenterol 2018; 31(2):188–97.
5. Joubert K, Laryea JA: Abdominal approaches to rectal prolapse. Clin Colon Rectal Surg 2017; 30(1):57–62.
6. Rickert A, Kienle P: Laparoscopic surgery for rectal prolapse and pelvic floor disorders. World J Gastrointest Endosc 2015; 7(12):1045–54.
7. Vogler SA: Rectal prolapse. Dis Colon Rectum 2017; 60(11):1132–5.

RECTAL SURGERY

SURGICAL CONSIDERATIONS

Description

Many lesions within the distal two-thirds of the rectum can be excised through a **transanal approach** (Fig. 7.4-3). The most common benign tumors treated by local excision are adenomas. Lesions such as carcinoid tumor, endometrioma, and solitary rectal ulcer also may be locally excised. Transanal excision of benign lesions may be performed in the submucosal plane, whereas suspected malignancies are excised by removing the entire thickness of the rectal wall. Full antibiotic and mechanical bowel prep are given. Transanal excision usually is performed in the prone jackknife position, although the lithotomy position may be used when the lesion is located on the posterior rectal wall. A local anal block, usually 0.25% bupivacaine with 1:200,000 epinephrine, is performed to relax the sphincter mechanism and minimize sphincter injury, aid in hemostasis, and diminish postoperative pain. An **anoscope** is inserted into the anal canal. Stay sutures may be placed adjacent to the area of resection. On occasion, lesions may be prolapsed all the way through the anus and excised outside the body. Generally, the dissection starts at the distal end of the lesion and proceeds proximally. The **proctotomy** may be closed with running or interrupted sutures. These sutures are then grasped and used for further traction. When the specimen is removed, a few final sutures are needed to close the proximal-most incision. **Rigid proctoscopy** is performed to confirm preservation of an adequate lumen.

Variant procedure or approaches

TEM (transanal endoscopic microsurgery) and TAMIS (transanal minimally invasive surgery)

Another technique for the removal of larger lesions or lesions more proximally located in the rectum is **transanal endoscopic microsurgery** and transanal minimally invasive surgery. In these techniques, the patient is placed on the operating table in a position that puts the lesion to be removed posterior. Because of this, the patient may be prone, supine in lithotomy, or in the right or left lateral decubitus position. The procedure is performed through a larger operating proctoscope or transanally placed port. The colon is insufflated using carbon dioxide to provide an operating space. The carbon dioxide will be absorbed through the mucosa of the colon and be exhaled. Laparoscopic-like instruments are passed through the proctoscope, and the surgeon views the operation either through a stereo viewer attached to the proctoscope or on a video screen. The lesion in question is removed and the defect in the rectum sutured closed.

Transsacral Resection

The **transsacral (Kraske) approach** to rectal tumors offers wider exposure than the

transanal approach but is more painful and has a substantially higher likelihood of complications (wound infection, fecal fistula, incontinence). A transsacral approach may be advantageous when the lesion is located behind the rectum (retrorectal tumors) and when resection of the lower sacrum or coccyx is anticipated. Transsacral resection generally is performed in the prone jackknife position. An incision is made from the posterior commissure of the anus to the base of the sacrum. The sphincter muscles are spared, but the levator muscles are divided to expose the posterior wall of the rectum. The coccyx may be disarticulated and removed to improve exposure. It is also possible to remove the lower sacral segments through this approach, but increasing morbidity accrues as the sacral nerve roots are sacrificed. For a posterior wall lesion, the posterior wall of the rectum is opened, and the lesion, along with a full-thickness disc of rectal wall, is excised. If the lesion is on the anterior wall, two proctotomy incisions are necessary. The proctotomy incisions are closed with standard anastomotic techniques. The levator muscles are reapproximated, and the skin is closed. A drain may be placed within the retrorectal space before closing. The transsacral approach may be combined with an **abdominal approach (abdominal-transsacral resection)** in some cases of low rectal cancer.

Transsphincteric

The **transsphincteric (York-Mason)** approach to rectal lesions also gives wider exposure than does the transanal approach, but at the expense of a substantially higher risk of fecal incontinence. Transsphincteric excision is performed with the patient in the prone jackknife position. An incision is made at the posterior commissure of the anus and is extended along the lateral border of the coccyx and sacrum. The external sphincter, internal sphincter, and levator ani muscles are sequentially transected in the posterior midline. As each muscle is cut, the cut edges are tagged with sutures to facilitate accurate reapproximation. The rectal wall is incised, and the lesion is excised. The proctotomy incision is closed via standard anastomotic suturing techniques, and the individual components of the sphincter muscle are reapproximated with interrupted sutures. The overlying skin is closed in a standard fashion.

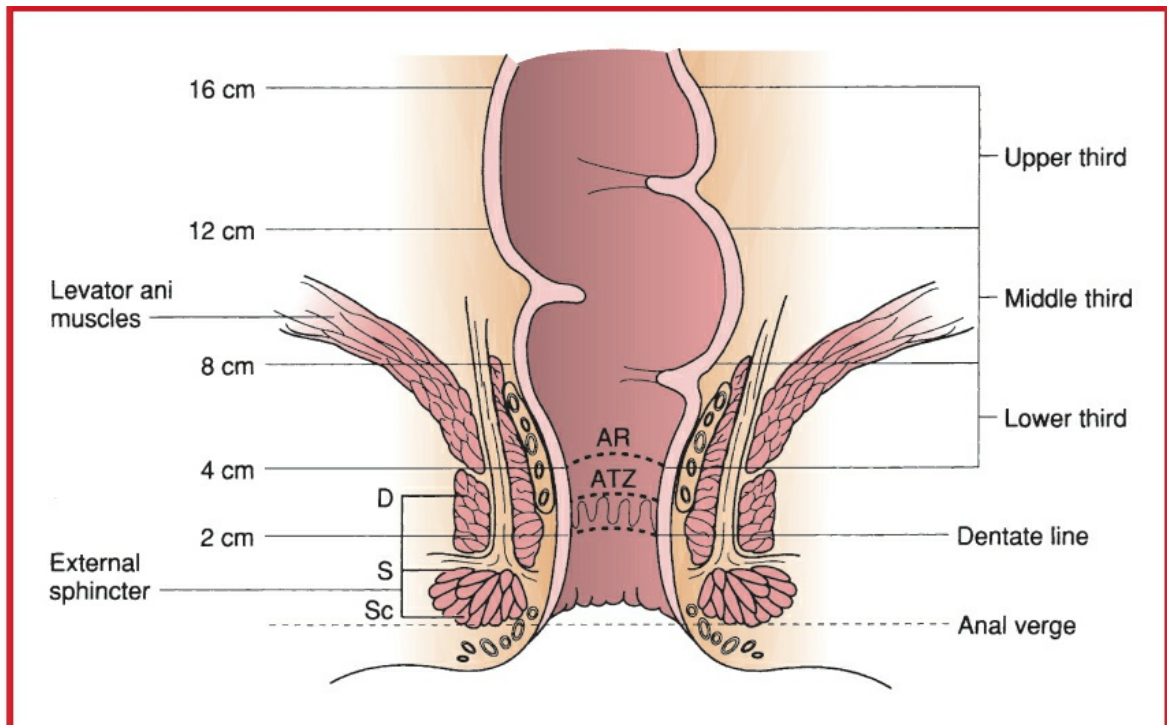


Figure 7.4-3 Anorectal anatomy with important landmarks. Approximate measurements are relative to the anal verge. D, deep; S, superficial; Sc, subcutaneous; AR, anorectal ring; ATZ, anal transition zone. (Reproduced with permission from Yahanda AM, Chang AE: Colorectal cancer. In: Greenfield LJ, Mulholland MW, Oldham KT, et al., eds. *Surgery: Scientific Principles and Practice* 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of urinary drain postop. See [Appendix K](#), p. K-1.

SUMMARY OF PROCEDURES

	Transanal Excision	Transsacral Excision	Transsphincteric Excision	TEM or TAMIS
Position	Prone jackknife or lithotomy	Prone jackknife	⇐	Prone jackknife, lithotomy, or lateral decubitus
Incision	Intrarectal	Anus to lateral sacral wall	⇐	Intrarectal
Special instrumentation	Rigid proctoscope; headlight and/or fiber-optic retractors; Foley catheter	Gigli or power saw if sacral resection contemplated; headlight and/or fiber-optic retractors; Foley catheter	Headlight and/or fiber-optic retractors; Foley catheter	Proctoscope (40-mm diameter), removable faceplate, optical stereoscope, light cord, pressure transducer, laparoscopic camera, insufflator, instruments (5-mm diameter), Martin arm to hold equipment
Unique considerations	Bowel prep may → dehydration and hypokalemia.	⇐	⇐	⇐
Antibiotics	Cefazolin 2 g iv and metronidazole 500 mg iv or cefotetan 2 g iv	⇐	⇐	⇐
Surgical time	15–120 min	1–2 h	⇐	⇐
EBL	<100 mL	<100 mL (500 mL if sacral resection)	⇐	⇐
Postop care	No rectal temperatures, suppositories, or enemas	⇐	⇐	⇐
Mortality	0–2%	⇐	⇐	0.3–2%
Morbidity	Tumor recurrence: 5–50% Urinary retention: 10–20% Bleeding: 2–5% Pelvic sepsis: 0–4% Ureteral injury: < 1% (minimized by use of Foley)	50% ⇐ ⇐ ⇐ ⇐ Fecal fistula: 10–30% Fecal incontinence: 5–10%	5–50% ⇐ ⇐ ⇐ ⇐ 10–40%	4–20% ⇐ 2% 0.8% Intraperitoneal entry: 1–4% 2%
Pain score	3	7	7	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Rectal adenomas, 5th–7th decades; rectal adenocarcinoma, 6th–9th decades; endometrioma, 2nd–4th decades; solitary rectal ulcer syndrome, 4th–8th decades; carcinoid tumors, 5th–8th decades
Male:Female	1:1
Incidence	Varies with disease; not uncommon
Etiology	Varies with disease
Associated conditions	Preexisting anorectal pathology, such as fecal incontinence, may require concurrent treatment.

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Operations for Fecal Incontinence.

Suggested Readings

1. Beets GL, Figueiredo NF, Beets-Tan RG: Management of rectal cancer without radical resection. *Annu Rev Med* 2017; 68:169–82.
2. deBeche-Adams T, Nassif G: Transanal minimally invasive surgery. *Clin Colon Rectal Surg* 2015; 28(3):176–80.
3. Kelley KA, Tsikitis VL: Endoluminal therapy in colorectal cancer. *Clin Colon Rectal Surg* 2016; 29(3):216–20.
4. Leong KJ, Evans J, Davies MM, et al: Transanal endoscopic surgery: past, present and future. *Br J Hosp Med (Lond)* 2016; 77(7):394–402.
5. Rai V, Mishra N: Transanal approach to rectal polyps and cancer. *Clin Colon Rectal Surg* 2016; 29(1):65–70.
6. Saur NM, Wexner SD: An update on minimally invasive techniques in colorectal surgery for neoplasia. *Minerva Chir* 2016; 71(6):377–81.

ANORECTAL SURGERY

■ GENERAL CONSIDERATIONS

Anorectal procedures are common general surgical operative procedures. The most common anorectal procedures are done for symptomatic hemorrhoids, perianal abscesses and fistulae, and anal fissures. Many of these procedures can be

accomplished under local anesthesia with an anal canal block. This provides anesthesia for the procedure and also provides for postoperative pain control if a long-acting local anesthetic is used. Performing the block can be painful and is usually performed in conjunction with sedation provided by the anesthesiologist. These procedures can be more extensive, and reoperative procedures may require general anesthesia. Anorectal procedures can be safely performed in the prone jackknife position as well as in the lithotomy position at the preference of the surgeon. One of the most common complications after anorectal surgery is urinary retention. This fact should be kept in mind when determining the volume of intravenous fluid to be given during the procedure.

SURGERY FOR ANAL FISTULAE

SURGICAL CONSIDERATIONS

Description

The majority of perianal fistulae are due to infection in the anal glands within the anal canal that communicate with crypts located at the dentate line (cryptoglandular fistula). Fistulae also may be the result of trauma, Crohn disease (CD), inflammatory processes within the peritoneal cavity, neoplasms, or as a consequence of radiation therapy. The ultimate treatment of fistula-in-ano is determined by the etiology and the anatomic course of the fistula. The principle behind treatment of cryptoglandular fistulae is to ablate the offending gland and lay open the tract. Fistulae that track above the majority of the sphincter mechanism must be treated by procedures that either do not cut the overlying sphincter or cut the sphincter and repair it. A fistula may be treated at the time of drainage of a perianal abscess or as a separate, elective operation. The route of a fistula tract is best determined by exploration at the time of operation. Although local anesthesia is acceptable for most fistulae, a few fistula operations require regional or general anesthesia because the ultimate route and depth of the fistula are unknown. Special consideration is given to fistulae that arise in the setting of CD. Poor wound healing and the importance of sphincter function in patients with chronic diarrhea dictate that only the most superficial fistulae are laid open. The primary goal is palliation, specifically abscess drainage and recurrence prevention. This is often accomplished by placing a silastic seton (a loose ligature around sphincter muscles) around the fistula tract and leaving it in place indefinitely. In the absence of active CD in the rectum and anus, attempts at fistula cure may be undertaken.

VARIANT PROCEDURE OR APPROACHES

Fistulotomy

Fistulotomy involves cutting all tissues superficial to a fistula so that the fistula tract is brought to the skin level. The opened, fibrotic fistula wall is often sewn to the skin edge (marsupialized). Fistulectomy involves excision of the entire fistula tract. When conventional fistulotomy would cause incontinence, a seton may be used. Other approaches that may be used to avoid fecal incontinence are complete fistulotomy with immediate reconstruction of the sphincter and excision of the internal opening by an endorectal advancement flap technique.

Fistula plug

The fistula plug involves the use of a bioabsorbable xenograft made of lyophilized porcine intestinal submucosa that is reconstituted in saline and placed in the fistula tract. This method may be preferred for fistulae that would require transection of a significant portion of the anal sphincters such that the surgeon is concerned about the impact on continence. It is used for both cryptoglandular fistulae and CD-related fistulae.

LIFT procedure

A LIFT procedure (ligation of the intersphincteric fistula tract) is a third technique used to treat fistulae that cannot be treated by fistulotomy. An incision is made in the anatomic plane between the internal and external sphincters. After the fistula tract is divided, the openings in the anal canal as well as in the internal and external anal sphincters are sutured closed.

Usual preop diagnosis

Fistula-in-ano

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES

	Fistulotomy or Fistulectomy	Fistulotomy with Seton	Endorectal Advancement Flap	Fistula Plug
Position	Prone jackknife; rarely lithotomy	⇐	Prone jackknife	⇐
Incision	Perianal	⇐	⇐	None
Antibiotics	None	None	Ertapenem 1 g iv or cefotetan 2 g iv	None
Surgical time	10–30 min	⇐	60–90 min	5–10 min
EBL	<50 mL	⇐	⇐	Negligible
Mortality	Minimal	⇐	⇐	⇐
Morbidity	Fecal incontinence: 0–30% Nonhealing or recurrent fistula: 5%	10–30% 10–20% (This procedure is used only in complex fistulae, so complication rate appears higher.)	0–10% 10–40% (This procedure is used only in complex fistulae, so complication rate appears higher.)	0% 25–50%
Pain score	6	6	6	0–1

PATIENT POPULATION CHARACTERISTICS

Age range	2nd–7th decades
Male:Female	2:1
Incidence	Common
Etiology	Infection within anal glands located at dentate line (cryptoglandular fistula); trauma; Crohn disease; inflammatory processes within the peritoneal cavity; neoplasms; consequence of radiation therapy

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Operations for Fecal Incontinence.

Suggested Readings

1. Adegbola SO, Sahnan K, Pellino G, et al: Short-term efficacy and safety of three novel sphincter-sparing techniques for anal fistulae: a systematic review. *Tech Coloproctol* 2017; 21(10):775–82.

2. Akiba RT, Rodrigues FG, da Silva G: Management of complex perineal fistula disease. *Clin Colon Rectal Surg* 2016; 29(2):92–100.
3. Jamshidi R: Anorectal complaints: hemorrhoids, fissures, abscesses, fistulae. *Clin Colon Rectal Surg* 2018; 31(2):117–20.
4. Lee MJ, Heywood N, Adegbola S, et al: Systematic review of surgical intervention for Crohn's anal fistula. *BJS Open* 2017; 1(3):55–66.
5. Rozalén V, Parés D, Sanchez E, et al: Advancement flap technique for anal fistula in patients with Crohn's disease: a systematic review of the literature. *Cir Esp* 2017; 95(10):558–65.

HEMORRHOIDECTOMY/STAPLED HEMORRHOIDOPEXY

SURGICAL CONSIDERATIONS

Description

Hemorrhoids are normally occurring vascular tissues located in discrete aggregations known as hemorrhoidal cushions within the distal rectum and anus. Thought to play a role in the fine control of enteric continence, they are only treated if they cause a symptom that persists after conservative therapy. Hemorrhoids may bleed, prolapse, and cause mucous drainage, itching, or pain (when thrombosed). The primary pathophysiologic event in the development of symptomatic hemorrhoids is thought to be mucosal prolapse 2° degeneration of the fibroelastic tissue that tethers vascular cushions and overlying mucosa to the submucosa. Many modern treatments diminish prolapse by fixing the mucosa to the submucosa with scar tissue. Hemorrhoids are classified as internal (when above the dentate line) or external (when below) (see [Fig. 7.4-3](#)). Internal hemorrhoids are further classified by symptoms: I, bleed; II, bleed prolapse, and spontaneously reduce; III, bleed, prolapse, and require manual reduction; and IV, bleed, prolapse, and cannot be reduced. The most common symptom from external hemorrhoids is severe pain caused by thrombosis. Surgical treatment involves excision of the thrombosed hemorrhoid, often under local anesthetic, in the office. Internal hemorrhoids may be treated by nonexcisional or excisional techniques. Nonexcisional techniques generally are used in the office or outpatient clinic. They do not require an anesthetic because their use is limited to the insensate tissues above the dentate line. Nonexcisional treatments include rubber-band ligation, infrared coagulation, sclerotherapy, and cryotherapy.

Hemorrhoidectomy

Surgical **hemorrhoidectomy** may be performed in the lithotomy or prone jackknife position. An **anoscope** is placed in the anal canal, and a hemorrhoid column is grasped and tented up into the lumen. A suture is placed at the internal apex of the complex. An incision is started at the external apex of the hemorrhoidal complex, and a plane is developed deep to the hemorrhoidal tissue and superficial to the sphincter muscles. When the internal sphincter is identified, the dissection proceeds in the avascular space along its luminal surface. The dissection is continued up into the rectum to the transfixing suture. Lateral incisions along the redundant tissue are completed to excise the hemorrhoid. Care is taken to leave healthy bridges of mucosa between adjacent hemorrhoidal columns. Hemostasis is obtained with cautery, and the mucosal defect may be closed with a running, absorbable suture. It also is acceptable to leave the mucosal wound open. The procedure is repeated over the other enlarged, symptomatic hemorrhoidal complexes, removing redundant tissue and leaving long, vertical scars to prevent further mucosal prolapse.

VARIANT PROCEDURE OR APPROACHES

Stapled hemorrhoidectomy

A **stapled hemorrhoidectomy** or **hemorrhoidopexy** is used for the correction of mucosal hemorrhoidal prolapse. This procedure is performed in the same position and under the same anesthesia as a conventional hemorrhoidectomy. A circular anal dilator is anchored to the skin with heavy sutures, an obturator is passed, and a purse-string is created with absorbable 2-0 suture 3–4 cm above the dentate line. The stapler head is passed above the purse-string, and the sutures are pulled while the stapler is closed. In women, the vaginal wall must be examined to ensure that it has not been incorporated. The stapler is fired, cutting the incorporated mucosal prolapse, pulling the internal and external hemorrhoids proximally. Thus, the hemorrhoids themselves are not removed. Hemostasis is obtained with sutures as needed. Cautery should be used with caution, because of the staples. This procedure is associated with a higher rate of disease recurrence and cost.

Transanal Hemorrhoidal Dearterialization

Transanal hemorrhoidal dearterialization (THD) is an effective treatment for hemorrhoidal disease. The ligation of hemorrhoidal arteries located with a Doppler probe can provide a significant reduction of the arterial overflow to the hemorrhoidal piles. Plication of the redundant rectal mucosa/submucosa (called “mucopexy”) can provide a repositioning of prolapsing tissue to the anatomical site. In long-term follow-up, there is resolution of symptoms in the vast majority of patients. The most common complication is transient tenesmus, which sometimes can result in rectal discomfort or

pain. Rectal bleeding occurs in small number of patients.

Usual preop diagnosis

Symptomatic hemorrhoids; bleeding; and/or prolapse

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Surgical Hemorrhoidectomy	Stapled Hemorrhoidopexy
Position	Prone jackknife, lithotomy, or left lateral decubitus	Prone jackknife, lithotomy
Incision	Series of vertical incisions from anal verge to top of hemorrhoid columns	None
Special instrumentation	Headlight or lighted anoscope; operating anoscope	Circular anal dilator, obturator, stapler
Antibiotics	None	None
Surgical time	30–90 min	10–20 min
EBL	<100 mL	100 mL
Postop care	Sitz baths, oral fluids, fiber supplements	⇐
Mortality	Rare	⇐
Morbidity	Urinary retention: 15–30% Incontinence: 1–6% Bleeding: 2–5% Stricture: 2–5% Infection: 1–2%	⇐
Pain score	9	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Peak prevalence 45–65 yr
Male:Female	1:1
Incidence	Prevalence 75/1000
Etiology	Low-fiber diet; genetic; pregnancy
Associated conditions	Constipation

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Operations for Fecal Incontinence.

Suggested Readings

1. Emile SH, Elfeki H, Sakr A, et al: Transanal hemorrhoidal dearterialization (THD) versus stapled hemorrhoidopexy (SH) in treatment of internal hemorrhoids: a systematic review and meta-analysis of randomized clinical trials. *Int J Colorectal Dis* 2019; 34(1):1–11.
2. Jacobs DO: Hemorrhoids: what are the options in 2018? *Curr Opin Gastroenterol* 2018; 34(1):46–9.
3. Rakinic J: Benign anorectal surgery: management. *Adv Surg* 2018; 52(1):179–204.
4. Song Y, Chen H, Yang F, et al: Transanal hemorrhoidal dearterialization versus stapled hemorrhoidectomy in the treatment of hemorrhoids: a PRISMA-compliant updated meta-analysis of randomized control trials. *Medicine (Baltimore)* 2018; 97(29):e11502.

OPERATIONS FOR FECAL INCONTINENCE

■ SURGICAL CONSIDERATIONS

Description

In the majority of patients, fecal incontinence is caused by a combination of pudendal neuropathy and atrophy of pelvic floor muscles. Only when an anatomic defect in the sphincter mechanism can be identified is surgery likely to be beneficial. **Sphincteroplasty** is performed in the prone jackknife position after a full mechanical bowel prep. An incision is made at the anal verge, centered over the area of injured sphincter, and extended sufficiently around the anus to reach the retracted, cut edges of

the sphincter. The anoderm and rectal mucosa are dissected off the internal surface of the sphincter. The external surface of the sphincter mechanism is then dissected free to the level of the pelvic diaphragm. Care must be taken not to injure the inferior hemorrhoidal nerves during dissection around the posterolateral sphincter. The fibrotic portion linking the two ends of sphincter is cut, and the ends are overlapped and secured in place with two layers of interrupted horizontal mattress sutures. In women with obstetric injuries, the transverse perineal muscles are reapproximated. The skin may be reapproximated at the anal verge and along the reconstructed perineum or left open. The remainder of the skin is closed as completely as possible.

VARIANT PROCEDURES OR APPROACHES

Parks postanal repair

The surgical options for patients without anatomic defects in their sphincters are generally unsuccessful. The **posterior anoplasty of Parks** was designed to passively enhance continence by increasing the normally occurring angle between the rectum and the anal canal and to increase the mechanical efficiency of weak sphincter muscle by shortening the fiber length. Lack of efficacy has limited its use, although some surgeons still perform it in the setting of continued incontinence after abdominal repair of rectal prolapse. The operation is performed in the prone jackknife position after standard bowel prep. A hemispherical incision is made at the level of the intersphincteric groove over the posterior half of the anus. The plane between the internal and external anal sphincters is identified and developed proximally to above the puborectalis musculus. The puborectalis fibers are “reefed,” or pulled together, as far as possible with nonabsorbable sutures. The external sphincter is plicated together in the midline with a series of nonabsorbable sutures that start at the deep external sphincter and progress to the subcutaneous sphincter. Skin is closed with absorbable sutures.

Thiersch procedure

The **Thiersch operation (pinch graft)** has a high complication rate and should be considered primarily in debilitated patients with symptomatic rectal prolapse or fecal incontinence. As initially described the anal canal was encircled with a silver wire, which served as a passive obstacle to prolapse or defecation. In more recent years, an elastic sheet of Dacron-impregnated Silastic mesh has been used. Two small incisions are made on opposite sides of the anal verge. A pathway around the anal canal is created by blunt dissection, and a 1.5-cm-wide piece of mesh is led around the anal canal. The ends of the mesh are overlapped in one of the incisions and either sutured or stapled together at an appropriate level of tension. The wounds are irrigated with antibiotic solution, and the incisions are closed.

Usual preop diagnosis

Fecal (enteric) incontinence

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, and physiotherapy plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Overlapping Sphincteroplasty	Parks Repair	Modified Thiersch Procedure
Position	Prone jackknife	⇐	Prone jackknife or lithotomy
Incision	Circumanal	⇐	Two small incisions lateral to the anus
Special instrumentation	Headlight	⇐	Headlight; Silastic mesh
Unique considerations	Urinary catheter preop; standard bowel prep	⇐	None
Antibiotics	Ertapenem 1 g iv or second-generation cephalosporin	⇐	⇐
Surgical time	1–2 h	1 h	30–45 min
EBL	<100 mL	⇐	⇐
Postop care	Early: Sitz baths Late: fiber supplement, stool softener	⇐ ⇐	⇐ ⇐
Mortality	Rare	⇐	⇐
Morbidity	Unimproved incontinence: 20% Improved but minor incontinence: 30%	60–80%	20% Erosion of prosthesis: 30–60%

	Prolonged wound healing: 20% Infection: 1–2%		Obstructed defecation: 20–40% Infection
Pain score	8	7	6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Bimodal: 3rd–5th decades for obstetric injury, fistulotomy, and perineal trauma; 6th–8th decades for pudendal neuropathy/pelvic floor atrophy
Male:Female	1:4
Incidence	Not uncommon
Etiology	Pudendal neuropathy; pelvic floor atrophy; obstetric injury; injury during anal surgery (fistulotomy, sphincterotomy, hemorrhoidectomy); perineal trauma; neurologic disease; congenital anomalies
Associated conditions	Urinary incontinence; chronic constipation; multiparity

ANESTHETIC CONSIDERATIONS

(Procedures covered: excision or repair of rectal prolapse; rectal surgery; anal fistulotomy/fistulectomy; anal sphincterotomy/sphincteroplasty; hemorrhoidectomy; operations for fecal incontinence.)

Rectal surgery is frequently performed in prone or jackknife positions. Many procedures do not require general anesthesia and can be performed under MAC. Coordination with the surgeon is recommended to determine the type and timing of anesthesia.

PREOPERATIVE

Depending on the institutional preference, rectal procedures are frequently performed in prone or jackknife position. Minor rectal surgeries do not necessarily require general endotracheal anesthesia and can be managed with sedation/deep sedation alone.

Respiratory	A careful evaluation of patient's respiratory status is important. If patient has ↓ reserve, the lithotomy position may be better tolerated than the prone or jackknife positions. Tests: As indicated by H&P
Musculoskeletal	Rectoperineal area is very sensitive; thus, it is often difficult for the patient to stay in supine or sitting position. Consider sufficient pain control preinduction and choose lateral decubitus position if spinal anesthesia is planned.
Hematologic	Acute and chronic GI blood loss is not common in this patient population. Tests: Consider CBC
Laboratory	Other tests as indicated by H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a lumbar epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Sedation, GA, spinal techniques may be used. Epidural anesthesia is less common and may spare the anoperineal region.

General anesthesia

Induction	General (LMA vs. ETT): Standard induction. Procedures planned in prone or jackknife position may require careful airway management. Consider ET intubation if airway concerns exist. Management of the prone patient is more challenging than the supine position.
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ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	Standard emergence. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Regional anesthesia

MAC	iv sedation combined with local perirectal infiltration of local anesthesia by surgeon (such as four-quadrant pudendal nerve block) is appropriate for many outpatient procedures. Perirectal injection of local anesthetic is painful; rapid-acting agents such as propofol, remifentanyl, alfentanil, and ketamine (hypersalivation may require anticholinergic agents) should be administered to minimize patient discomfort during the injection. As drug effect will vary widely among patients, careful titration of the drugs to desired level of sedation is warranted. Consider need to return patient to supine position quickly (keep bed/gurney at the ready).	
Spinal	Patient may be in either sitting, lateral decubitus, prone, or jackknife position for placement of a subarachnoid block. Doses of local anesthetics should be adequate to provide a high lumbar level (L1-L2) of sensory anesthesia (e.g., bupivacaine 8–12 mg). Patients should remain in relative head-up position with hyperbaric solutions and in head-down position with hypobaric solutions to limit cephalad spread of block.	
Blood and fluid requirements	iv: 18–20 ga × 1+ (depending on scope of rectal procedure) BSS at 3–5 mL/kg/h	Blood products rarely needed. See Goal-Directed Fluid Management, Appendix K , p. K-4.
Monitoring	Standard monitors	Others as clinically indicated

Positioning	Prone, jackknife, or lithotomy may be used. ✓ and pad pressure points ✓ eyes and arm positioning ✓ empty stoma bags before turning prone	Chest support or bolsters to optimize ventilation in the jackknife position; care in positioning the patient's extremities and genitals after turning into jackknife position. Avoid pressure on eyes and ears after turning patient.
Complications	Apnea Peripheral nerve injury	May need to convert emergently from MAC to GA. Lithotomy position can damage the peroneal nerve, resulting in foot drop. Improper arm positioning can lead to axillary plexus stretch and nerve damage.

POSTOPERATIVE

Complications	Urinary retention	Catheterize until return of urinary function especially after spinal anesthesia and may be prevented by restricting the volume of iv fluids administered.
Multimodal pain management	Emphasize multimodal approach combining nonopioid ± bolus/PCA opioid analgesics with epidural analgesia (see Appendix C, p. C-4).	Consider use of nonopioid adjuncts: NSAIDs, ketamine, gabapentin, acetaminophen
ERAS	PONV protocol (Appendix B, p. B-4 and B-6) Consider DVT prophylaxis.	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Chewing gum may promote return of bowel function.

Tests	As indicated by patient status
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Suggested Readings

1. Baldini G, Fawcett WJ: Anesthesia for colorectal surgery. *Anesthesiol Clin* 2015; 33(1):93–123.
2. Helander EM, Webb MP, Bias M, et al: Use of regional anesthesia techniques analysis of institutional enhanced recovery after surgery protocols for colorectal surgery. *J Laparoendosc Adv Surg Tech A* 2017; 27(9):898–902.
3. Kim AJ, Yong RJ, Urman RD: The role of transversus abdominis plane blocks in enhanced recovery after surgery pathways for open and laparoscopic colorectal surgery. *J Laparoendosc Adv Surg Tech A* 2017; 27(9):909–14.
4. Liu Q, Jiang H, Xu D, et al: Effect of gum chewing on ameliorating ileus following colorectal surgery: a meta-analysis of 18 randomized controlled trials. *Int J Surg* 2017; 47:107–15.

CHAPTER 7.5

Hepatic Surgery

SURGEON

Samuel K. S. So, MD, FACS

ANESTHESIOLOGIST

Hendrikus J. M. Lemmens, MD, PhD

HEPATIC RESECTION

SURGICAL CONSIDERATIONS

Description

Liver resections usually are performed to remove primary or metastatic tumors to the liver. The most common malignant primary liver tumor is hepatocellular carcinoma (HCC), usually caused by chronic hepatitis B or C, and cirrhosis due to chronic alcohol abuse. Liver resection is also performed for an enlarging hepatic adenoma, which is a benign primary tumor that is susceptible to rupture. The most common secondary tumors are metastases from colorectal cancer. In rare cases, it may be necessary to resect a devitalized area of the liver following trauma. The mortality and morbidity following liver resection depend on the extent of the surgery, experience of the surgeon, and the patient's hepatic function. In general, the risk of resection is higher in patients with primary HCC where the uninvolved part of the liver frequently is cirrhotic or diseased from chronic hepatitis B or C. Cirrhotic patients with Plt counts <80,000, portal HTN with varices, ascites, albumin <3.5 g/L, and prolonged international normalized ratio (INR) are generally unsuitable candidates for major liver resection because of the high risk of postop liver failure.

Intraop blood loss is the most important predictor of short-term survival. Bleeding is mostly from intrahepatic branches of portal and hepatic veins injured during the

dissection, potentially leading to massive blood loss within minutes. Liver resection performed by an experienced liver surgeon using modern dissection techniques often can be accomplished without the need for blood transfusions (cell salvage techniques should not be used when operating on cancer patients). The mortality rate of major liver resection should be <2–5%. In my experience, most patients do not require postop ICU care and are usually discharged within 4–5 d. Improved outcomes result from better surgical exposure and mobilization of the liver combined with the standard adoption of new dissection tools to minimize blood loss. These include new ablation devices to coagulate along the planned line of a resection combined with the use of high-pressure water jet (Hydrojet) or ultrasonic pulses (CUSA) to expose the intrahepatic vessels and bile ducts. Intraop ultrasound is important for two reasons: (a) in planning the line of resection and mapping out its relationship with the large intrahepatic portal and hepatic veins and (b) in surveying the entire liver to look for multifocal lesions.

Anatomic versus nonanatomic liver resection

Until the last decade, most liver surgeons performed **anatomic liver resections** in which the porta hepatis is dissected and the corresponding extrahepatic branches of the hepatic artery, portal vein, bile duct, and hepatic vein are mobilized and ligated before resection of the liver parenchyma (Fig. 7.5-1). In **nonanatomic liver resections**, only the tumor with a margin of 1–2 cm is removed instead of the entire anatomic lobe or segment. This approach is particularly appropriate in patients with cirrhosis or chronic hepatitis, in whom removing too much of the liver predisposes them to hepatic decompensation, and in patients with liver metastases where the risk of recurrence remains high. When nonanatomic resection is performed, dissection of the porta hepatis is unnecessary. Instead, branches of the vessels and hepatic ducts are ligated and resected as they are encountered during the resection of the liver parenchyma. All patients undergoing liver resection should also be electrically grounded with the appropriately placed pads prior to draping for possible RF ablation of lesions found not to be suitable for resection.

Laparoscopic liver resection is an emerging approach that is increasingly used for selected cases by liver surgeons experienced in laparoscopic surgery. Some have also begun exploring the use of robotic-assisted liver resection. For minor liver resection by experienced surgeons, a laparoscopic approach has been reported to be associated with shorter hospital stay, less or comparable blood loss, less postop pain, and similar outcome compared with open liver resection. The role and safety of a laparoscopic approach for major liver resection have not been determined and should not be undertaken lightly. A CO₂ pneumoperitoneum of 10–14 mm Hg and lowering the central venous pressure (CVP) to <5 mm Hg are recommended to reduce venous bleeding

associated with laparoscopic liver resection. For severe bleeding, briefly increasing the pneumoperitoneum pressure and decreasing the airway pressure with a pause in artificial ventilation are approaches that can be used to facilitate hemostasis.

Temporary occlusion of the hepatoduodenal ligament that contains the main portal vein, hepatic artery, and common bile duct (**Pringle maneuver**) can be used during resection to minimize blood loss. Most patients will tolerate this maneuver for 15–20 min. In some patients, it may be necessary to repeat the maneuver twice to complete a major resection. However, with good surgical exposure and use of newer dissection tools to minimize blood loss, the Pringle maneuver is rarely necessary.

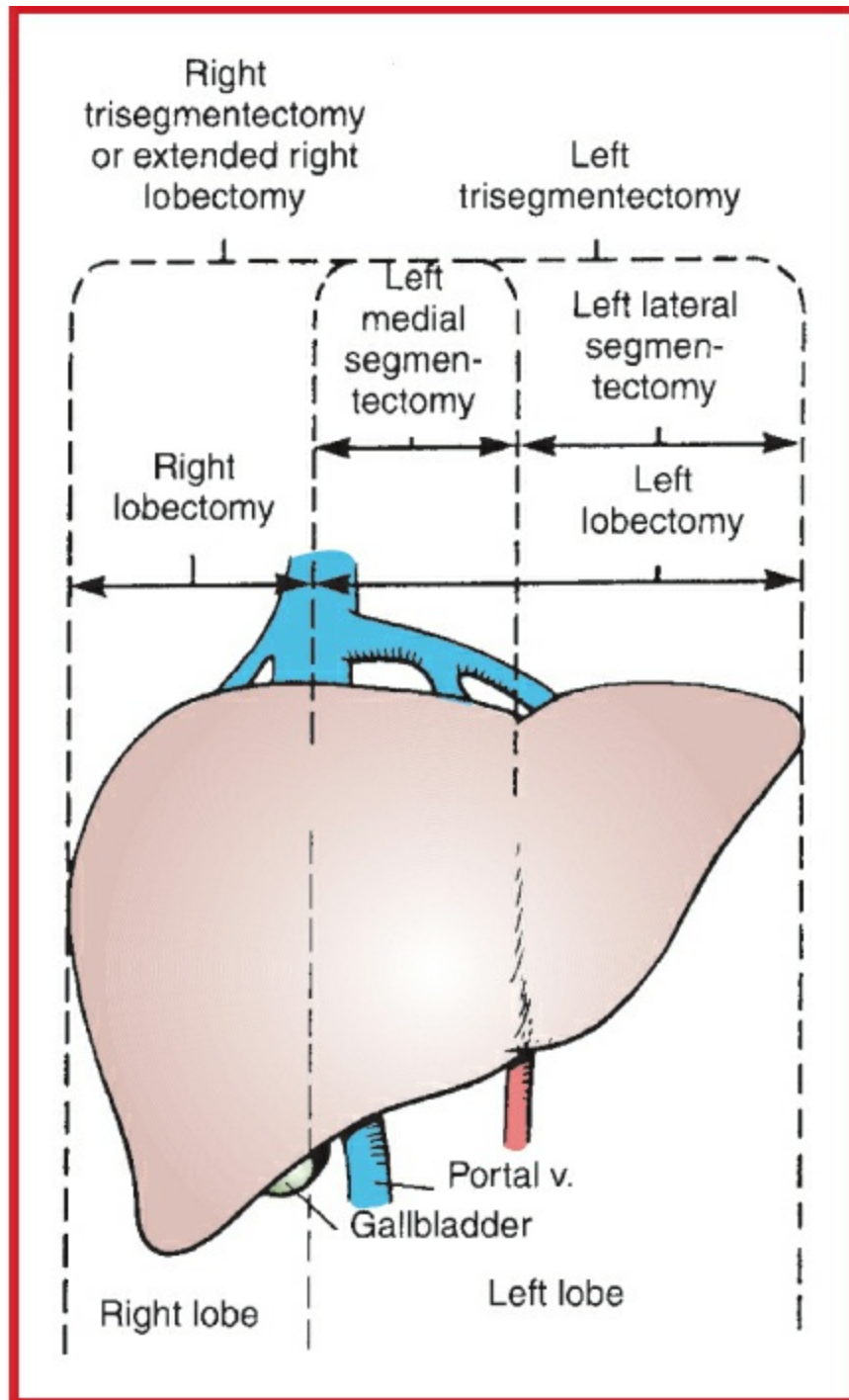


Figure 7.5-1 Types of traditional anatomic liver resection. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Usual preop diagnosis

Benign and malignant primary or metastatic tumors of the liver

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration and alcohol and smoking cessation. A review of nutritional status should include the possible benefit of oral nutritional supplements for 1–2 wk

preop in malnourished patients. Additional discussion of carb loading, deep venous thrombosis (DVT) prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. Oral bowel prep is usually avoided. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Right/Left Lobectomy/Trisegmentectomy	Partial Right Lobectomy	Left Lateral Segmentectomy
Position	Supine	⇐	⇐
Incision	Upper midline, extending to right subcostal (Lexus incision)	⇐	Upper midline
Special instrumentation	Thompson liver retractor; InLine™, Helix Hydrojet™, or CUSA™; TissueLink™ dissecting sealer; argon beam laser; tumor ablation device; intraop ultrasound; microwave or RF tumor ablation devices	⇐	⇐
Unique considerations	Maintain normal core temp. Most intrahepatic bleeding can be controlled temporarily by compression. Note the duration of Pringle maneuver, if applied. Give mannitol furosemide for hemoglobinuria associated with extensive microwave or radiofrequency ablation of the liver to reduce the risk of postoperative acute tubular necrosis.	⇐	⇐
Antibiotics	Vancomycin/cefazolin iv preop	⇐	⇐
Surgical time	3–8 h	4–6 h	<3 h
Closing considerations	Secure meticulous hemostasis; place two to three subphrenic or	⇐	⇐

	perihepatic Jackson-Pratt flat bulb suction drains.		
EBL	Variable (50–800 mL)	50 mL	50 mL
Postop care	PACU → ward	⇐	⇐
Mortality	<2–5%	⇐	<1%
Morbidity	Ascites: 20–30%	10–20%	<10%
	Wound infection: 5–10%	⇐	⇐ <2%
	Liver failure: <5%	⇐	⇐
	Bile leak: <5%	<2–5%	⇐
	Postop bleeding: <2–5%	⇐	—
Pain score	7–8	7–8	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	19–85 yr
Male:Female	4:1 in primary HCC 2:1 in liver metastases
Incidence	Metastatic colorectal cancer is the most common disease requiring liver resection in the United States. The incidence of primary HCC is rising 2° the prevalence of chronic hepatitis C; however, because of delayed Dx and associated hepatitis C cirrhosis, many are not suitable resection candidates. Primary HCC is a common cancer in Asian Americans because of the high prevalence of chronic hepatitis B. Many young Asians (30–50 yr) with HCC 2° chronic hepatitis B may not have cirrhosis.
Associated conditions	Cirrhosis, chronic hepatitis, and, rarely, hemochromatosis; Hx of colorectal surgery in patients with liver metastases

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Hepatorrhaphy.

Suggested Readings

1. Farges O, Goutte N, et al: ACHBT-French Hepatectomy Study Group: Incidence and risks of liver resection: an all-inclusive French nationwide study. *Ann Surg* 2012; 256(5):697–704; discussion 704–5.
2. Hughes MJ, McNally S, Wigmore SJ: Enhanced recovery following liver surgery: systematic review and meta-analysis. *HPB (Oxford)* 2014; 16(8):699–706.
3. Joliat G-R, Labgaa I, Hübner M, et al: Cost–benefit analysis of the implementation of an enhanced recovery program in liver surgery. *World J Surg* 2016; 40(10):2441–50.
4. Melloul E, Hübner M, Scott M, et al: Guidelines for perioperative care for liver surgery: enhanced recovery after surgery (ERAS) society recommendations. *World J Surg* 2016; 40(10):2425–40.
5. Tan MCB, Jarnagin WR: Surgical management of non-colorectal hepatic metastasis. *J Surg Oncol* 2014; 109(1):8–13.
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HEPATORRHAPHY

SURGICAL CONSIDERATIONS

Description

Although most liver lacerations have stopped bleeding by the time a surgeon sees them or after selective hepatic arterial embolization by interventional radiology, others may require suturing, packing, or **partial liver resection** to control bleeding. Various techniques are available to control hemorrhage in the OR including packing, suturing, inflow occlusion, and resection. Small lacerations that have stopped bleeding require no specific therapy other than possible drainage. Lacerations that continue to bleed usually can be sutured and drained. Extensive tears of the liver that are actively bleeding may require temporary occlusion of the porta hepatis, containing the hepatic artery, portal vein, and bile duct (**Pringle maneuver**) to excise devitalized parenchyma and control bleeding with sutures, clips, and coagulators. Occasionally, the hepatic vein draining the involved lobe will require clamping to control back-bleeding. When the bleeding cannot be controlled, it is expedient to pack the wound, drain the abdomen, and close. The pack can be removed without much risk of rebleeding within 48–72 h. Correction of hypothermia is important to prevent coagulopathy.

Usual preop diagnosis

Trauma with CT evidence of hepatic laceration

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration and alcohol and smoking cessation. A review of nutritional status should include the possible benefit of oral nutritional supplements for 1–2 wk preop in malnourished patients. Additional discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. Oral bowel prep is usually avoided. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Midline abdominal
Special instrumentation	Denier retractor
Antibiotics	Cefazolin 2 g iv
Surgical time	1–2 h
EBL	~300–2000 mL
Mortality	1–2%
Morbidity	Continued bleeding: 2–3% Biliary fistula Perihepatic abscess Intrahepatic hematoma Arterioportal fistula Hepatic and renal failure
Pain score	7–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Variable
Male:Female	1:1
Incidence	Rare
Etiology	Trauma (blunt vs. penetrating); surgical; hepatic adenomas; needle biopsies of liver

ANESTHETIC CONSIDERATIONS FOR HEPATIC PROCEDURES

(Procedures covered: hepatic resection; hepatorrhaphy.)

PREOPERATIVE

Patients presenting for hepatic surgery may have primary or metastatic tumors from GI and other sources. Liver function may be entirely normal in these patients. HCC is seen commonly in males >50 yr and is associated with chronic active hepatitis B or C and cirrhosis. The preop considerations listed below describe patients without cirrhosis. (See Preoperative Considerations for Surgery for Portal Hypertension for evaluation of patients with cirrhosis).

Respiratory	Respiratory function is typically normal; however, patients with ascites may have respiratory compromise. Tests: CXR; others as clinically indicated
Cardiovascular	Patients may be hypovolemic, and volume status should be carefully assessed before induction of anesthesia (skin turgor, UO, orthostatic BP, and HR). Tumors may surround major vascular structures and impede venous return. Consider evaluation with CT/MRI scan.
Hepatic	Liver resection can be indicated for hemangiomas, hydatid cysts, and tumors. It is important to determine the size of the tumor and involvement of vascular structures preop to be adequately prepared for major intraop blood and fluid losses. Tests: LFTs; albumin; ultrasound/CT/MRI
Hematologic	The liver produces all clotting factors except VIII, and the degree of hepatic insufficiency determines the extent of any coagulopathy. T&C 2–4 U PRBCs. Tests: CBC; PT; PTT; Plt count; others as indicated from H&P
Laboratory	Tests as indicated from H&P. NB: For elective cases, if abnormal LFTs are present on preop labs, it is

	important to perform a complete medical workup. This can include reviewing old lab data, hepatitis serology, and an abdominal ultrasound to r/o cholestatic causes of liver dysfunction. Surgery and anesthesia in the presence of acute hepatitis are associated with high mortality.
Premedication	Consider midazolam 1–2 mg iv. If PT significantly prolonged, consider 10 mg vitamin K iv/po 24 h before surgery. Consider FFP for rapid correction of PT.
ERAS	DVT prophylaxis usually includes LMWH started 2–12 h preop. Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane for postop pain management. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: For major liver resections, the surgeon may attempt to reduce blood loss by applying intermittent vascular inflow occlusion or total vascular exclusion. As the result of ischemia induced by vascular occlusion and the loss of liver mass during resection, liver function may be significantly abnormal in the postop period. Coagulation abnormalities may exist; consequently, an epidural catheter for postop pain relief may be associated with ↑ risk of hematoma formation. Hypovolemia (CVP < 5 mm Hg) prevents liver congestion and results in less bleeding but may increase the likelihood of postop renal failure. Euvolemia (CVP 7–10 mm Hg) results in more bleeding and a higher transfusion rate.

Induction	Standard induction. Restore intravascular volume before induction. Trauma patients or those with ascites require rapid-sequence induction with cricoid pressure until intubation has been confirmed. If patient is hemodynamically unstable, consider etomidate (0.2–0.4 mg/kg) or ketamine (1–3 mg/kg) in place of propofol.
ERAS	Verify preincision antibiotics. Assess volume status (see

	Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Standard maintenance; N ₂ O can be used if bowel distention will not impede surgical exposure/closure. If total vascular occlusion is used, elevate CVP to at least 12 mm Hg by rapid fluid administration before cross-clamping. Have phenylephrine and epinephrine infusions ready to treat ↓ BP. After major resections, significant hemodynamic changes occur. CO and HR increase and SVR decreases.	
Emergence	For major hepatic resections, the patient is best cared for in an ICU. After major blood loss, consider keeping the patient mechanically ventilated. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen (liver function permitting). PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Anticipate large blood loss. IV: 14–16 ga × 2 NS/BSS @ 10–20 mL/kg/h Fluid warmer Humidify gases Consider utilizing rapid-transfusion device.	Blood loss can be significant; keep at least 2 U PRBC ahead. Lobectomies often are associated with more blood loss than wedge resections. Massive transfusions (e.g., 10+ u RBC/24 h) may be required, and appropriate blood products should be available (e.g., PRBC + FFP + Plt). If procedure does not involve cancer, blood salvage devices may be used. See Goal-Directed Fluid Management, Appendix K, p. K-4 . CVP < 5 cm H ₂ O is recommended to ↓ hepatic venous congestion.

Control of blood loss	Surgical control Pringle maneuver Total vascular exclusion	Surgical occlusion of the main blood vessels entering the hilar area (Pringle maneuver) or total vascular occlusion. Total vascular exclusion is accomplished by complete occlusion of liver inflow and outflow.
Monitoring	Standard monitors UO Arterial line ± CVP ± TEE	Others as clinically indicated. If the extent of the resection is not known at the beginning of surgery, appropriate monitoring (CVP, arterial line, additional iv's) should be established before beginning resection. Forced-air warmer
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Massive hemorrhage	Ensure adequate vascular access. Consider rapid-transfusion device.

POSTOPERATIVE

Complications	✓ liver function - Hemorrhage - Electrolyte imbalance - Hypoglycemia - Hypothermia, shivering - DIC - Pulmonary insufficiency	Patients with normal liver function preop may have significant postop impairment of liver function 2° loss of liver mass or ischemic injury induced by vascular occlusion. >90% of patients will develop some form of respiratory complication (atelectasis, effusion,
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		pneumonia).
Multimodal pain management	PCA	Patient should be recovered in ICU or hospital ward that is accustomed to treating the side effects of opiates (e.g., respiratory depression, breakthrough pain, nausea, pruritus). Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with TAP blocks (see Appendix C, p. C-4).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6) Maintain normoglycemia.	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	ABG; CXR; others as clinically indicated	

Suggested Readings

1. Clavien PA, Petrowsky H, DeOliveira ML, et al: Strategies for safer liver surgery and partial liver transplantation. *N Engl J Med* 2007; 356(15):1545–59.
2. Clavien Pierre-A, Barkun J: Consensus conference on laparoscopic liver resection A jury-based evaluation. *Ann Surg* 2015; 261(4):630–1.
3. Egger ME, Gottumukkala V, Wilks JA, et al: Anesthetic and operative considerations for laparoscopic liver resection. *Surgery* 2017; 161(5):1191–202.
4. Hughes MJ, McNally S, Wigmore SJ: Enhanced recovery following liver surgery: systematic review and meta-analysis. *HPB (Oxford)* 2014; 16(8):699–706.
5. Niemann CU, Roberts JP, Ascher NL, et al: Intraoperative hemodynamics and liver function in adult-to-adult living liver donors. *Liver Transpl* 2002; 8:1126–32.
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7. Wakabayashi G, Cherqui D, Geller DA, et al: Recommendations for laparoscopic liver resection: a report from the second international consensus conference held in Morioka. *Ann Surg* 2015; 261(4):619–29.

3. Xiong J-J, Altaf K, Javed MA, et al: Meta-analysis of laparoscopic vs open live resection for hepatocellular carcinoma. *World J Gastroenterol* 2012; 18(45):6657–68.

CHAPTER 7.6

Biliary Tract Surgery

SURGEON

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OPEN CHOLECYSTECTOMY AND COMMON BILE DUCT EXPLORATION

SURGICAL CONSIDERATIONS

Description

With the advent of laparoscopic cholecystectomy, the traditional **open cholecystectomy** has become a rarity and is generally reserved for gallbladders that are expected to be difficult to remove due to inflammation, previous operations, and adhesions or because of other medical problems, such as coagulopathy or cirrhosis. In most institutions, fewer than 10% of cholecystectomies begin as open procedures, and approximately 5% of **laparoscopic cholecystectomies** are converted to open cholecystectomies during the course of the operation due to technical difficulties, complications, or unexpected findings. Because of its rarity, the open cholecystectomy may be a more challenging operation for both surgeon and anesthesiologist than it was in previous decades. A small number of open cholecystectomies are performed urgently following a complication of an attempted laparoscopic cholecystectomy and may be associated with significant instability from hemorrhage or sepsis related to an iatrogenic injury of abdominal or retroperitoneal structures.

The technical aspects of open cholecystectomy have not changed since its original

description over 100 years ago. The operation can be performed through a right subcostal (**Kocher**), paramedian, or midline incision. Upward traction is applied to the liver or gallbladder, whereas downward traction on the duodenum exposes the region of the cystic duct and artery and common duct. Adequate exposure is critical to performing a safe operation. Depending on local conditions and the surgeon's preference, the gallbladder may be removed from the top down, excising the gallbladder from the liver bed and isolating the cystic duct and artery as the final stage of the operation. The cystic duct and artery may be isolated and divided first, and the gallbladder removed retrograde from the gallbladder bed as the final step of the procedure. The anatomy of the biliary tree is quite variable, with the classic anatomy present in only 30% of patients, and few surgeons always remove the gallbladder in the same way every time. (Fig. 7.6-1 shows exposure of the gallbladder.)

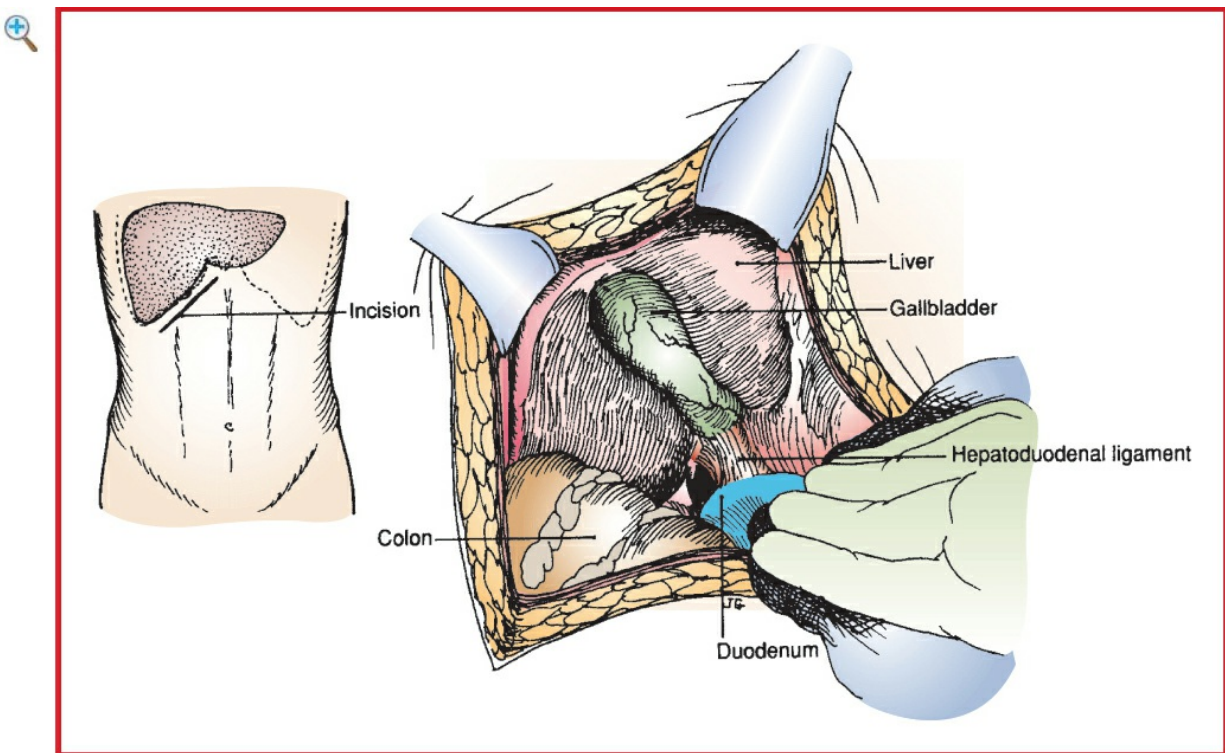


Figure 7.6-1 Incision and exposure of the gallbladder. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

Cholangiography may be performed at the discretion of the surgeon. Some surgeons perform it in all patients, and others perform it only in patients in whom there is some clinical evidence of choledocholithiasis. The cystic duct is opened, and a catheter is placed into the duct and secured with a ligature, tie, or special cholangiogram clamp. Dye is injected into the biliary tree via the catheter, and x-rays are taken. If stones are found, a **common duct exploration** may be performed. Alternatively, an **endoscopic retrograde cholangiogram (ERCP)** with stone extraction may be carried out postop.

Cholangiography usually adds 10–15 min to the procedure.

Choledochotomy, or “**common duct exploration**,” is the opening and exploration of the common duct for the purpose of extracting stones. Once commonly performed, it is a procedure reserved chiefly for patients who have failed management of common duct stones with endoscopic (ERCP) or laparoscopic techniques. Common duct stones are visualized by operative cholangiography to determine the number of stones, position, and the anatomy of the duct. Ducts smaller than 5 mm in diameter are at greater risk of injury with common duct exploration and should be managed endoscopically. An extensive **Kocher maneuver** is performed to allow exposure and palpation of the entire duct, including the intrapancreatic portion. A longitudinal incision is made in the duct, and exploration is carried out through this incision. The duct may be irrigated with NS, balloon catheters may be passed, and various instruments may be introduced to grasp, remove, or crush retained stones. The duct may be biopsied by this approach, and **choledochoscopy**—the direct visualization of the duct’s interior using a small flexible scope—can be performed. Rarely, an impacted stone may require electrohydraulic or laser lithotripsy through the choledochoscope, adding considerable time to the operation in centers equipped to perform the procedure. The choledochotomy is closed over a T-tube to allow decompression of the edematous duct, and later extraction of stones missed at the initial exploration. In the past, **transduodenal sphincteroplasty** was utilized for stones impacted near the sphincter of Oddi, but this procedure has largely been replaced by endoscopic or percutaneous techniques at specialized centers. It is reserved for highly unusual cases, such as a patient with a previous **Billroth II gastrectomy**. Depending on the complexity of the findings, a common duct exploration can be expected to add from 30 min to over 1 h to the cholecystectomy. In general, the mortality of patients undergoing common duct exploration is ~2–5 times that of a simple cholecystectomy. This difference can be explained by the fact that patients undergoing common duct exploration tend to be older and sicker or suffering from concomitant cholangitis—the opening of the duct itself is not necessarily a significant physiologic insult.

Variant procedure or approaches

Cholecystectomy remains the mainstay of treatment for symptomatic biliary stone disease. **Nonsurgical treatment of cholelithiasis**, particularly by oral dissolution and/or **lithotripsy**, has very limited usefulness and is rarely used in clinical practice. **Tube cholecystostomy** can be performed as an open procedure or percutaneously. It is generally reserved as a temporary measure in patients too ill to tolerate a more extensive procedure.

Usual preop diagnosis

Symptomatic cholelithiasis; acute cholecystitis; chronic cholecystitis; biliary dyskinesia; gallbladder polyps or carcinoma; choledocholithiasis

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting nothing by mouth (NPO) duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, deep venous thrombosis (DVT) prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery](#), p. K-1.

■ SUMMARY OF PROCEDURES		
	Cholecystectomy	Cholecystectomy/Common Duct Exploration
Position	Supine	⇐
Incision	Right subcostal or midline	Right subcostal or midline
Special instrumentation	Costal margin retractor; cholangiocatheter	Choledochotomy instruments; choledochoscope
Unique considerations	Requires intraop x-ray for cholangiogram	May include choledochoscopy
Antibiotics	Ampicillin, piperacillin, or mezlocillin (1–3 g iv); ± gentamicin; or cefotetan (2 g iv)	⇐
Surgical time	45–90 min	1–2.5 h
Closing considerations	Muscle relaxation	⇐
EBL	Minimal to 250 mL	⇐
Postop care	PACU → ward	⇐
Mortality	0.1% under 50 yr; 0.5% over 50 yr	0–1.5% under 60 yr; 5% in advanced age
Morbidity	Postop bile leak: 0–9%	⇐
	Pancreatitis: 0–4.6%	2–5%

	Bile duct injury: 0–0.25%	—
	Cardiac and respiratory complications: Rare, but the leading cause of death	—
	Hemorrhage: rare	—
Pain score	6	6–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Mostly adult; increases with age
Male:Female	1:2–3
Incidence	600,000/yr in the United States; ≥90% performed laparoscopically
Etiology	See associated conditions (below).
Associated conditions	Ileal disease; cirrhosis; hemolytic disorders; choledocholithiasis; cholangitis; or active pancreatitis

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Biliary Tract Surgery.

Suggested Readings

1. Coccolini F, Catena F, Pisano M, et al: Open versus laparoscopic cholecystectomy in acute cholecystitis. Systematic review and meta-analysis. *Int J Surg* 2015; 18:196–204.
2. Thangavelu A, Rosenbaum S, Thangavelu D: Timing of cholecystectomy in acute cholecystitis. *J Emerg Med* 2018; 54(6):892–7.

BILIARY DRAINAGE PROCEDURES

■ SURGICAL CONSIDERATIONS

Description

Biliary drainage procedures may be performed for malignant and nonmalignant

indications, and the type of drainage procedure performed depends on factors such as the nature of the biliary obstruction, the patient's overall condition and prognosis, the need for other surgical procedures, and institutional expertise. Most drainage procedures of the biliary tree are performed with **endoscopic** or **transhepatic techniques**. These techniques allow concomitant treatment of biliary stone disease, decompression of obstructive jaundice, relief of cholangitis, and delineation of the anatomy of the biliary tree. There remain a significant number of patients, however, for whom a traditional surgical procedure is the most appropriate. With the advent of laparoscopic cholecystectomy, surgical bile duct injuries have become the most common reason for surgical biliary drainage procedures. In general, the biliary drainage procedures' complexity of the different operations that may be performed and the morbidity attendant to these have more to do with the indications for operation than with the procedure that is performed.

All of these operations are performed under GA through an upper midline or right subcostal incision. Self-retaining retractors are used to retract the liver superiorly to expose the region of the porta hepatis. If the gallbladder will not be used for the bypass procedure (**cholecystojejunostomy**), then it usually is removed as the first step in the procedure (see **Open Cholecystectomy**). If the patient has had previous upper right quadrant surgery, the complexity and duration of the procedure and blood loss may increase significantly. Any associated hepatic cirrhosis may make the procedure particularly demanding. Most patients undergoing surgical biliary drainage procedures have had several nonsurgical instrumentations of the bile duct prior to presenting for surgery. Indwelling stents result in colonization of the biliary tract with any number of bacterial or fungal organisms from which the patient may have already suffered bouts of cholangitis. It is not unusual for the patient to develop bacteremia during the operation while these stents are being manipulated.

Transduodenal sphincteroplasty for benign obstruction of the ampulla of Vater or extensive choledocholithiasis has largely been abandoned in favor of the more well-tolerated and less-invasive **endoscopic sphincterotomy**. It remains the treatment of choice for rare cases of early ampullary carcinoma. Endoscopic sphincterotomy or placement of an internal stent is the most commonly performed technique for opening the ampulla and usually is performed by gastroenterologists outside the operating room (OR) with iv sedation. **Open sphincteroplasty** is usually reserved for patients in whom **endoscopic retrograde cholangiopancreatogram (ERCP)** has been unsuccessful or in whom a laparotomy is required for other reasons. For these open procedures, the second portion of duodenum is incised over the region of the ampulla, and the ampulla is cannulated. A longitudinal incision is made over the course of the ampulla, and the

mucosa of the ampulla is sutured to the mucosa of the duodenum with fine interrupted sutures with care being taken not to compromise the pancreatic duct. The duodenum is closed with suture, a small closed suction drain is placed, and the wound is closed. A postop stay of 5–7 d can be expected.

Cholecystojejunostomy usually is performed as palliation for malignant obstruction of the distal bile duct. Its advantage is that it does not require dissection of the portal triad, but the long-term results are poor as biliary obstruction typically recurs as the malignancy advances. The abdomen is opened as described above and the region of the porta hepatis examined to ensure that the cystic duct is not imminently compromised by tumor. The jejunum is then brought up to the gallbladder, usually bypassing the jejunum through the transverse mesocolon. The anastomosis may be performed either to an intact loop of jejunum (**loop cholecystojejunostomy**) or a Roux-en-Y loop of jejunum (**Roux-en-Y cholecystojejunostomy**) and is carried out with one or two layers of sutures, depending on surgeon's preference. If a Roux-en-Y is created, a second jejunojejunal anastomosis must be performed. This procedure has largely been replaced by transhepatic and endoscopic techniques.

Choledochoduodenostomy is an archaic procedure in which the bile duct is incised longitudinally and anastomosed directly to the adjacent duodenum. This was performed historically in patients with gallstones impacted at the ampulla. However, loss of the normal sphincter mechanism at the ampulla allows reflux of duodenal contents directly into the bile duct. Patients may suffer from repeated episodes of cholangitis or obstructive jaundice from debris occluding the anastomosis. Secondary biliary cirrhosis may occur, and the author has seen one case proceed to liver transplantation in which a cast of the biliary tree comprised of fibrous food material was extracted from the choledochoduodenal anastomosis. **Roux-en-Y choledochojejunostomy or hepaticojejunostomy** remains the gold standard by which all other biliary drainage procedures are measured. An anastomosis is fashioned between the common bile duct, common hepatic duct, or even the lobar or segmental bile ducts and a Roux-en-Y loop of jejunum. This is often a relatively demanding operation because it requires dissection deep in the porta hepatis to gain access to the bile duct. Furthermore, many patients have had previous surgery in the porta hepatis with extensive formation of adhesions. Long-term results are more reliable, however, and thus surgical drainage is preferred to less-invasive procedures for benign disease. Exposure of the biliary tree is the same as for the above procedures. The gallbladder is always removed (if it is still present). The bile duct is dissected free from the surrounding structures in the porta hepatis and traced proximally into the liver to healthy tissue above the level of obstruction.

Variant procedure or approaches

Endoscopic or transhepatic placement of temporary or permanent **biliary stents** is an increasingly common alternative to surgical drainage in patients with incurable pancreatic or biliary tract disease.

Usual preop diagnosis

Transduodenal sphincteroplasty: extensive choledocholithiasis, often after failed attempt at ERCP; rarely, malignant disease. Cholecystojejunostomy: malignant obstruction of the distal common bile duct, usually due to pancreatic cancer. Choledochojejunostomy or hepaticojejunostomy: benign strictures of the distal bile duct; long-standing stone disease; pancreatitis; iatrogenic injury; Oriental cholangiohepatitis; after resection of some tumors of the pancreas or bile duct

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES

	Cholecysto- jejunostomy	Choledocho- jejunostomy	Choledocho- duodenostomy	Transduodenal sphincteroplasty
Position	Supine	⇐	⇐	⇐
Incision	Midline or right subcostal	⇐	⇐	⇐
Special instrumentation	Costal retractor	⇐	⇐	⇐
Unique considerations	Consider usual prep for patients with obstructive jaundice.	⇐	⇐	⇐
Antibiotics	Ampicillin, piperacillin or mezlocillin (1–3 g iv) ± gentamicin; or cefotetan (2 g iv)	⇐	⇐	⇐
Surgical time	1 h; may be significantly longer for resection of tumor	1.5–3 h	⇐	⇐
Closing considerations	Muscle relaxation; NG suction	⇐	⇐	⇐
EBL	<250 mL (greater if tumor resection performed, or in presence of portal HTN)	250–500 mL (greater if tumor resection performed, or in presence of portal HTN)	⇐	<250 mL (transfusion rarely needed)
Postop care	PACU	⇐	⇐	⇐
Mortality	≤35% in advanced pancreatic cancer; otherwise, <10%	≤29%, with pancreatic cancer; 0.5% for benign disease	0–5.4% (usually for benign disease)	0–7%
Morbidity	Sepsis Respiratory complications Renal failure Hemorrhage Wound infection Breakdown of biliary anastomosis Thrombotic complications	⇐ ⇐ ⇐ ⇐ ⇐ ⇐ ⇐	⇐ ⇐ ⇐ ⇐ ⇐ ⇐ ⇐	Pancreatitis: rare Recurrent Cholangitis: rare
Pain score	7	7	7	6

PATIENT POPULATION CHARACTERISTICS

Age range	5th–7th decade; may be younger for repair of laparoscopic cholecystectomy injuries
Male:Female	1.1:2.1

Incidence	Procedure-related incidence not available, but declining in favor of techniques by interventional gastroenterology and radiology
Etiology	See Usual Preop Diagnosis, above.
Associated conditions	Jaundice (very common); fat-soluble vitamin deficiencies; malignancy, especially pancreatic (common); malnutrition (common)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Biliary Tract Surgery.

Suggested Readings

1. Pavlidis ET, Pavlidis TE: Pathophysiological consequences of obstructive jaundice and perioperative management. *Hepatobiliary Pancreat Dis Int* 2018; 17(1):17–21.
2. Quencer KB, Tadros AS, Marashi KB, et al: Bleeding after percutaneous transhepatic biliary drainage: incidence, causes and treatments. *J Clin Med* 2018; 7(5). pii: E94.

EXCISION OF BILE DUCT TUMOR

SURGICAL CONSIDERATIONS

Description

Primary tumors (cholangiocarcinomas) of the extrahepatic bile ducts, including the hepatic bifurcation (Klatskin tumors), are uncommon malignancies, with the only curative treatment for them being surgical excision. Such tumors are usually classified as being proximal bile duct tumors, involving the hepatic bifurcation and above; middle bile duct tumors, involving the midportion of the common hepatic and common bile duct; and distal bile duct tumors, which involve the distal bile duct, including the intrapancreatic or intraduodenal portion of the bile duct ([Fig. 7.6-1](#)). The gallbladder usually will be removed in any such operation.

Distal bile duct tumors, which carry a significantly higher cure rate than either proximal bile duct or pancreatic tumors, may be treated by **pancreaticoduodenectomy**. (See [p. 691](#) for discussion of this operation.)

Mid-bile duct tumors usually are excised by removing a generous portion of the mid

bile duct, resecting the ducts up to the hepatic bifurcation, and sometimes performing a **pancreaticoduodenectomy**. Biliary drainage usually is established by anastomosing the proximal bile duct to a Roux loop of jejunum. For proximal bile duct tumors, most of the extrahepatic bile ducts are excised, and biliary drainage is established by anastomosis of the right and left hepatic ducts or even multiple segmental ducts to a Roux loop of jejunum. These are often technically demanding operations with the potential for major blood loss. It may be necessary to perform a major hepatic resection at the same time, and the possibility of this should always be assumed when an operation of this sort is carried out. Often, a transhepatic catheter will have been placed radiographically preoperatively to provide relief of jaundice and to facilitate identification of the bile ducts. Colonization of the biliary tract with enteric bacteria or yeast is common and may result in bacteremia during surgical manipulation. Prolonged cholestasis may lead to fat-soluble vitamin deficiencies, in particular vitamin K deficiency, which may cause coagulopathy. Long-standing biliary obstruction may cause moderate atrophy and portal venous compromise.

Surgical exposure for any of these operations usually is achieved through a long midline or transverse subcostal incision with midline extension and the use of self-retaining retractors. The liver and gallbladder are retracted superiorly while downward traction is placed on the duodenum. If the gallbladder is still in place, a **cholecystectomy** is performed. For proximal bile duct tumors and mid-bile duct tumors not requiring pancreaticoduodenectomy, the bile duct is divided distally, just above the duodenum, and the pancreatic portion of the bile duct is oversewn. The bile duct is then resected proximally to the level of the bifurcation of the hepatic ducts. A Roux-en-Y loop of jejunum is anastomosed to the hepatic ducts to establish biliary drainage. Drains are placed, and most surgeons place an NG tube for such cases.

Variant procedure or approaches

Endoscopic or transhepatic stenting of areas of stricture often is used as a palliative alternative to surgical excision. These are usually performed radiographically and do not require GA. They may be used as an alternative to resection or in preparation for surgery.

Usual preop diagnosis

Cholangiocarcinoma (common); benign strictures of the bile ducts (infrequent); sclerosing cholangitis (rare)

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status,

discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Midline or subcostal
Special instrumentation	Costal retractor
Unique considerations	Many cases prove unresectable at operation. Intraoperative radiation therapy may be used.
Antibiotics	Ampicillin, piperacillin, or mezlocillin, 1–3 g iv, ± gentamicin; or cefotetan 2 g iv; fluconazole 200 mg iv; antibiotics may be adjusted based on preop bile cultures
Surgical time	3–8 h
Closing considerations	Muscle relaxation required for closure; NG suction
EBL	500–5000 mL, depending on need for liver resection and presence of portal HTN
Postop care	ICU postoperatively
Mortality	5–10%
Morbidity	Sepsis; hemorrhage; anastomotic leakage; wound infection; liver failure; VTE
Pain score	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	50–70
Male:Female	Male > female
Incidence	4500 cases of bile duct cancer/yr in the United States
Etiology	Multifactorial

Associated conditions

Ulcerative colitis; sclerosing cholangitis; typhoid carrier state; Clonorchis sinensis; choledochal cyst; Caroli disease; gallstones

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Biliary Tract Surgery.

Suggested Readings

1. Lidsky ME, Jarnagin WR: Surgical management of hilar cholangiocarcinoma at Memorial Sloan Kettering Cancer Center. Ann Gastroenterol Surg 2018; 2(4):304.
2. Madadi-Sanjani O, Petersen C, Ure B: Minimally invasive hepatobiliary surgery Clin Perinatol 2017; 44(4):805–18.
3. Najran P, Lamarca A, Mullan D, et al: Update on treatment options for advanced bile duct tumours: radioembolisation for advanced cholangiocarcinoma. Curr Oncol Rep 2017; 19(7):50.
4. Razumilava N, Gores GJ: Cholangiocarcinoma. Lancet 2014; 383(9935):2168–79.

CHOLEDOCHAL CYST EXCISION OR ANASTOMOSIS

SURGICAL CONSIDERATIONS

Description

This rare congenital anomaly includes various types of dilatation of the biliary tree, and patients may present with cystolithiasis, cholangitis, pancreatitis or, rarely, malignancy. Adults with long-standing obstructive jaundice from a choledochal cyst may present with secondary biliary cirrhosis. Although four types of cysts are commonly recognized, the vast majority consists of fusiform dilatation of much or most of the extrahepatic biliary tree. Although the traditional description of choledochal cyst is that of an infant with a palpable abdominal mass and jaundice or cholangitis, this is a relatively rare presentation today. Today, many cysts are found in adults undergoing evaluation for symptoms thought to be due to gallbladder disease. These patients may present with biliary colic, pancreatitis, or cholangitis. Recommended treatment consists of **excision of the cyst** when technically safe. **Cyst-enteric bypass**, usually to a Roux loop of jejunum, is rarely performed today because of the small but real risk of developing malignancy in these cysts. Only in an elderly patient under unusual technical

circumstances would this be appropriate. The operation is performed through a midline or right subcostal incision. The liver is retracted superiorly and the duodenum inferiorly, exposing the biliary tree. The gallbladder is excised, along with as much of the cyst as possible. Intraoperative cholangiogram demonstrates the transition from cyst to normal biliary tract. The duct is divided as distally as possible, just above the duodenum, and the cyst reflected superiorly. The entire cyst should be excised to prevent the development of malignancy in the remnant. This not infrequently requires excision to the hepatic bifurcation, and an anastomosis is performed at this level, often between the common orifice of the right and left hepatic ducts and a Roux loop of jejunum. Diffuse involvement of the intrahepatic bile ducts (Caroli disease) may require liver resection or transplantation.

Reoperative cases are increasingly common; most follow a cyst-enteric bypass. These cases may be significantly more difficult than first-time operations.

Variant procedure or approaches

There is an increasing tendency among gastroenterologists to perform **endoscopic sphincterotomy** in these patients, rather than to refer them for surgical resection, particularly in older patients. It remains to be seen if these patients will develop cancer in the retained cysts.

Usual preop diagnosis

Choledochal cyst, the most common type involving fusiform enlargement of the entire extrahepatic biliary tree

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Midline or right subcostal
Special instrumentation	Costal retractor
Antibiotics	Ampicillin, piperacillin, or mezlocillin (1–3 g iv); ±

	gentamicin; or cefotetan (2 g iv)
Surgical time	2–4 h
Closing considerations	NG suction
EBL	250 mL, with potentially greater blood loss in reoperations
Postop care	PACU → ward
Mortality	Very rare
Morbidity	Anastomotic leak Wound infection Pulmonary complications Pancreatitis
Pain score	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Classically, 60% <10 yr of age, although this may be changing; also adults of all ages
Male:Female	1:3
Incidence	Rare in the United States; more common in Japan
Etiology	Unclear
Associated conditions	Jaundice; pancreatitis; malignancy within the cyst

Suggested Readings

1. Ronnekleiv-Kelly SM, Soares KC, Ejaz A, et al: Management of choledochal cysts. *Curr Opin Gastroenterol* 2016; 32(3):225–31.
2. Soares KC, Arnaoutakis DJ, Kamel I, et al: Choledochal cysts: presentation, clinical differentiation, and management. *J Am Coll Surg* 2014; 219(6):1167.

ANESTHETIC CONSIDERATIONS FOR BILIARY TRACT SURGERY

(Procedures covered: open cholecystectomy; cholangiography; choledochotomy; biliary drainage procedures; transduodenal sphincterotomy or sphincteroplasty; cholecystojejunostomy; excision of bile duct tumor; choledochal cyst excision/anastomosis.)

PREOPERATIVE

Patients presenting for biliary tract surgery are an extremely diverse group, ranging from the otherwise healthy to the extremely ill. With the increasing popularity of laparoscopic surgery, open cholecystectomies will be performed rarely or when it is not possible to complete the laparoscopic procedure. Cirrhosis, even of a mild degree, substantially increases the risk of cholecystectomy, with hemorrhage and postoperative liver failure being the greatest danger. Patients with bile duct tumors are usually jaundiced at presentation and have undergone transhepatic or endoscopic studies for diagnostic purposes. Often an external transhepatic biliary drain or common bile duct stent placed endoscopically may be present, and jaundice may have been relieved. Rarely, a hepatic resection may be performed as part of the procedure. Prior operation or the presence of portal HTN may substantially increase the duration, complexity, and blood loss of the procedure.

Respiratory

Pain 2° an acute abdominal process may impair respiratory function ($\downarrow$ FRC, hypoventilation, atelectasis). For patients undergoing laparoscopic cholecystectomy, intra-abdominal CO₂ insufflation may $\rightarrow$ atelectasis, $\downarrow$ FRC, $\uparrow$ PIP, and $\uparrow$ PaCO₂. Tachypnea, hyperpnea, and acute respiratory alkalosis can be signs of sepsis or due solely to pain associated with inflammation of the gallbladder.
Tests: Consider CXR and others as indicated from H&P.

Cardiovascular

Patients may be dehydrated from fever, vomiting, and decreased oral intake; assess hemodynamic status by evaluating BP and HR in the supine and standing positions. Fluid resuscitate if patient shows Sx of orthostatic $\downarrow$ SBP until hemodynamic status improves. Patients undergoing laparoscopic cholecystectomy may experience hemodynamic compromise 2° positioning (reverse Trendelenburg

	→ excessive intra-abdominal pressure with subsequent impairment of venous return). Epigastric discomfort is common with biliary tract disease and can mimic symptoms of myocardial ischemia. Tests: ECG; others as indicated from H&P
Renal	In patients with obstructive jaundice, preop administration of bile salts po may prevent renal insufficiency following surgery. Tests: UA and others as indicated from H&P
Gastrointestinal	Patients with peritonitis will exhibit guarding and may develop abdominal distention and paralytic ileus. Therefore, full-stomach precautions are warranted. Laparoscopic approach is contraindicated in these patients. Ensure adequate hydration and consider administration of lactulose or bile salts. Tests: Bilirubin; AST (SGOT); ALT (SGPT); alkaline phosphatase; albumin
Hematologic	Leukocytosis is often present with a moderate left shift. ✓ coags. Administer vitamin K as needed (10 mg iv/SC). Tests: CBC, with differential and Plt. Consider T&S.
Laboratory	Other tests as indicated from H&P
Premedication	Meperidine (0.5–0.6 mg/kg iv) is thought to cause less sphincter of Oddi spasm than other opiates. Sphincter spasm can interfere with intraop cholangiograms and cause pain; reverse opiate-induced spasm with naloxone in 40 mcg increments. Atropine (0.4–0.6 mg im or iv) or glycopyrrolate (0.2–0.3 mg im or iv) may help decrease spasm of the sphincter and can be given in combination with the opiate. Parenteral vitamin K is indicated if PT is prolonged (10 mg/d po or im for 3 d). Administer H₂ antagonists (ranitidine 50 mg iv), if patient is at risk for gastric aspiration.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

Discuss with surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: In patients at risk for aspiration, ET intubation should be accomplished following a rapid-sequence iv induction.

Induction	Standard induction if no aspiration risk. In patients at risk of aspiration, a rapid-sequence induction should be performed.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Standard maintenance. Muscle relaxants facilitate surgery and are indicated.	
Emergence	If there is a risk for aspiration of gastric contents, patient should be extubated awake after return of protective airway reflexes; otherwise, no special considerations. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Minimal blood loss	Blood products usually not required. Anticipate that patient may be dehydrated and require generous iv hydration (e.g., 10–15 mL/kg) before anesthetic induction. See Goal-Directed

	Possible 3rd-space loss IV: 16–18 ga × 1–2 BSS @ 5–8 mL/kg/h	Fluid Management, Appendix K, p. K-4 . IV size and number according to risk of significant blood loss
Monitoring	Standard monitors	Others as clinically indicated
Positioning	✓ and pad pressure points ✓ eyes	A steep, reverse Trendelenburg position may be required, causing cardiorespiratory impairment: venous return → ↓ CO.
Complications	Atelectasis 2° surgical retraction ↓ BP	These complications are unique to laparoscopic procedures.

▼ POSTOPERATIVE

Complications	Ventilatory impairment Pneumothorax Atelectasis PONV VTE Subcutaneous emphysema	Monitor patients for hypoxemia in the postop period. Administer supplemental O ₂ , and consider a portable CXR to aid in the diagnosis Appendix B, p. B-4 and B-6.
Multimodal pain management	PCA Shoulder pain	Prolonged PCA meperidine is associated with ↑ normeperidine → CNS disorder, seizures. 2° subdiaphragmatic gas trapping Emphasize multimodal approach combining

		nonopioid and bolus/PCA opioid analgesics with epidural analgesia and/or TAP blocks (see Appendix C, p. C-4).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
2. Lee A, Chiu CH, Cho MW, et al: Factors associated with failure of enhanced recovery protocol in patients undergoing major hepatobiliary and pancreatic surgery: a retrospective cohort study. *BMJ Open* 2014; 4(7):e005330.
3. Mitra S, Khandelwal P, Roberts K, et al: Pain relief in laparoscopic cholecystectomy—a review of the current options. *Pain Pract* 2012; 12(6):485–96.
4. Rondelli F, Manina G, Agnelli G, et al: Venous thromboembolism after laparoscopic cholecystectomy: clinical burden and prevention. *Surg Endosc* 2013; 27(6):1860–4.
5. Snowden C, Prentis J: Anesthesia for hepatobiliary surgery. *Anesthesiol Clin* 2015; 33(1):125–41.

CHAPTER 7.7

Laparoscopic General Surgery

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LAPAROSCOPIC REPAIR OF PERFORATED PEPTIC ULCER

SURGICAL CONSIDERATIONS

Description

Ulcer perforation occurs in 5–10% of peptic ulcer patients and is responsible for more than 70% of deaths associated with peptic ulcer disease (PUD). These patients can present in shock and often are extremely volume depleted. **Open repair** is indicated if the patient has a “hostile” abdomen, if there is simultaneous bleeding and perforation, if the patient is hemodynamically unstable, if the patient has significant cardiovascular or respiratory risk factors that would make them not tolerate a pneumoperitoneum, or if the surgeon has inadequate experience with the laparoscopic approach. Also trained operating room (OR) personnel and equipment must be available. Risk factors associated with unsuccessful **laparoscopic repair** include shock on admission, delayed presentation (>24 h), underlying comorbidities, age >70 yr, and ASA III–IV. For

laparoscopic repair, the patient is often placed in a modified lithotomy position with the surgeon standing between the legs. In this case, both arms can be left out. Alternatively, the patient may be in a supine position with the surgeon standing on the patient's left side. The patient should be placed in a reverse Trendelenburg position (RTP). Access to the abdomen is obtained at the umbilicus or left upper quadrant (LUQ) Palmer point with a Veress needle and/or optical trocar entry. Alternatively, a **Hasson technique** (peritoneal entry through a ~1-cm skin incision) can be employed at the umbilicus. Two to four additional ports are placed. Generally, the perforation site can be easily identified by laparoscopy. Occasionally, the surgeon may ask the anesthesiologist to insufflate air through an orogastric or nasogastric tube to help localize the perforation site. Several methods of repair have been reported. The most common involves suture closure of the perforation with **omentopexy**. Air leak testing post leak closure can be done either via the OR or NG tube, or some surgeons will perform **endoscopy** to ensure adequate closure. The peritoneal cavity is then irrigated. The position of the operating table is frequently changed during irrigation to allow better access to the entire abdominal cavity. Adequate irrigation can take 20–30 min. Conversion rates are quite high (10–15%) with inability to localize the perforation the most common reason for conversion to open surgery.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting nothing by mouth (NPO) duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading deep venous thrombosis (DVT) prophylaxis, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine, low lithotomy
Incision	3–5 ports
Antibiotics	Extended spectrum PCN (e.g., ticarcillin, piperacillin), 2nd-generation cephalosporin (e.g., cefotetan, cefoxitin) given preincision
Surgical time	1–1.5 h
Closing considerations	Port closure only Placement of a drain
Special instrumentation	Gastroscope; oro- or nasogastric tube

EBL	<75 mL
Postop care	2–4-d hospital stay; liquids on postop day 1; consider upper GI.
Mortality	Typically <0.1% (as high as 3% in some series)
Morbidity	6–10% overall, including: Abscess (2–8%) Leak (~5%) Ileus (3–6%) Pneumonia
Pain score	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–60 yr
Male:Female	May be slightly more common in males
Incidence	~10% lifetime risk for perforation in PUD
Etiology	May be related to <i>Helicobacter pylori</i> infection; NSAID use; stress; corticosteroid use
Associated conditions	Gastrinoma; MEN type 1; COPD; CRF; up to 20% may be in shock at the time of admission.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Laparoscopic Cholecystectomy.

Suggested Readings

1. Søreide K, Thorsen K, Søreide JA: Strategies to improve the outcome of emergency surgery for perforated peptic ulcer. *Br J Surg* 2014; 101(1):e51–64.
2. Wilhelmsen M, Møller MH, Rosenstock S: Surgical complications after open and laparoscopic surgery for perforated peptic ulcer in a nationwide cohort. *Br J Surg* 2015; 102(4):382–7.

LAPAROSCOPIC ESOPHAGEAL FUNDOPLICATION

SURGICAL CONSIDERATIONS

Description

Approximately 40% of Americans suffer from heartburn, and most cases are treated medically. Indications for **esophageal fundoplication** (to ↑ lower esophageal sphincter pressure) include complications of gastroesophageal reflux disease (GERD), such as stricture, respiratory problems, esophageal ulcerations, and Barrett esophagus (a premalignant condition). Other indications include failure of medical management or unwillingness to submit to a lifetime of medication. Most patients with GERD are treated laparoscopically; those undergoing **laparoscopic fundoplication** have the benefits of a minimally invasive approach—decreased pain, earlier return of GI function, earlier ambulation, earlier discharge, quicker return to normal activities, and reduced incidence of wound infections and hernias. The most common fundoplication is the **Nissen (360°) wrap**, and its variations include the **Toupet (270° posterior) wrap** and the **Dor (anterior) wrap**. Before surgery, patients with GERD should have documented esophageal hyperacidity (by either pH probe or by esophagitis revealed on upper endoscopy) and also should have a hypotensive sphincter (demonstrated on manometry). Typically, they will have been treated with proton pump inhibitors (e.g., lansoprazole) and sometimes with prokinetic agents (e.g., metoclopramide). The patient is placed supine in a low lithotomy position, with the surgeon standing between the legs. The abdomen is entered with a Veress needle ~2 cm above the umbilicus or in the left upper quadrant (LUQ) Palmer point or with an open Hassan placement 2 cm above the umbilicus. A total of five trocars are inserted—two in the LUQ, two in the right upper quadrant (RUQ), and one at the umbilicus. The stomach should be decompressed either with an OG tube or with a gastroscope. The liver is elevated with a liver retractor, and the gastroesophageal (GE) junction and both diaphragmatic crura are dissected out. The hiatus should be closed with one or two sutures. The esophagus is encircled ([Fig. 7.7-1A](#)), and the vagal nerves are identified and preserved. The short gastrics are then taken down to decrease tension on the wrap. The fundus is brought in through the retroesophageal window ([Fig. 7.7-2](#)). An esophageal dilator (56–60 Fr) generally is placed (often by the anesthesiologist) to calibrate the wrap. Passage of the dilator is possibly the most hazardous part of the procedure because it may cause perforation of the esophagus at the GE junction. As the dilator approaches the stomach, it is important to watch the junction on the video monitors to ensure that the dilator is not being held at an angle that will risk perforation. The anesthesiologist should stop passing the dilator immediately if any resistance is felt. Communication between the surgeon and the anesthesia team is critical during the passage of the dilator.

The dilator is withdrawn. An NG tube may be placed at this time.

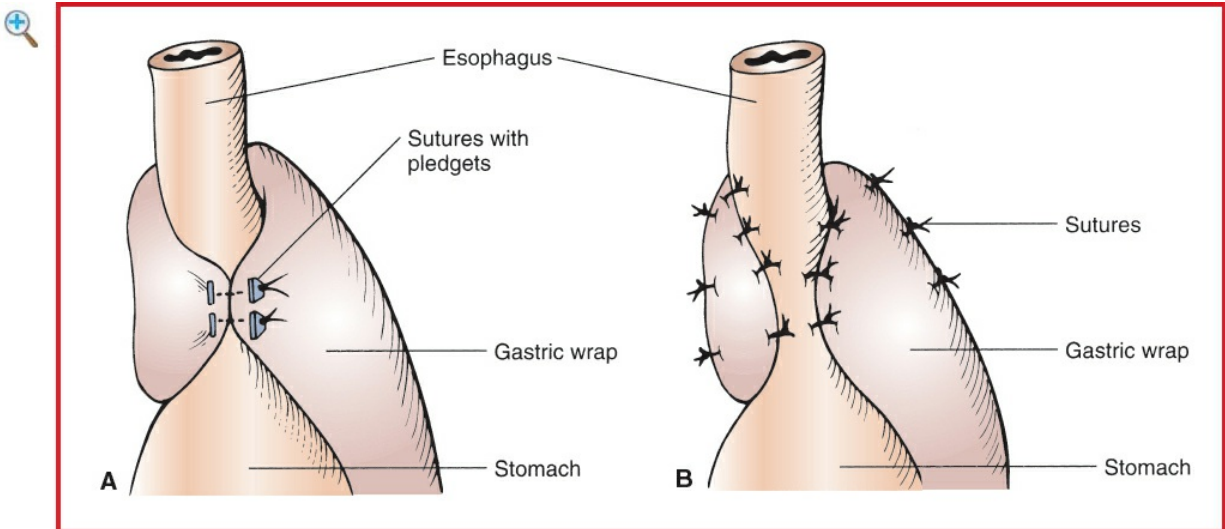


Figure 7.7-1 A: Nissen fundoplication. B: Partial fundoplication (Toupet procedure).

Variant procedure or approaches

Open surgery may be indicated if the patient has had previous gastric surgery or if there is a complication with an ongoing laparoscopic procedure. Some surgeons may elect not to divide the short gastrics, which decreases operative time. The Dor fundoplication is an anterior hemifundoplication in which the fundus is wrapped and sutured to the left and right sides of the esophagus and the left and right crura, but anteriorly. A Dor fundoplication is often used in combination with a **Heller myotomy** for treatment of achalasia. A Toupet procedure ([Fig. 7.7-1B](#)) is a posterior hemifundoplication in which the two walls of the fundus do not meet. The stomach is sewn to the left and right walls of the esophagus and then is anchored to the right and left crura. In the past, these procedures were performed commonly in patients with a high risk of postop dysphagia (e.g., those who have impaired esophageal peristalsis or a preop stricture). Now, because of their higher failure rate, they are seldom performed except in patients with severe esophageal dysmotility, such as patients with scleroderma. These procedures are identical to the Nissen fundoplication except for the wrap itself.

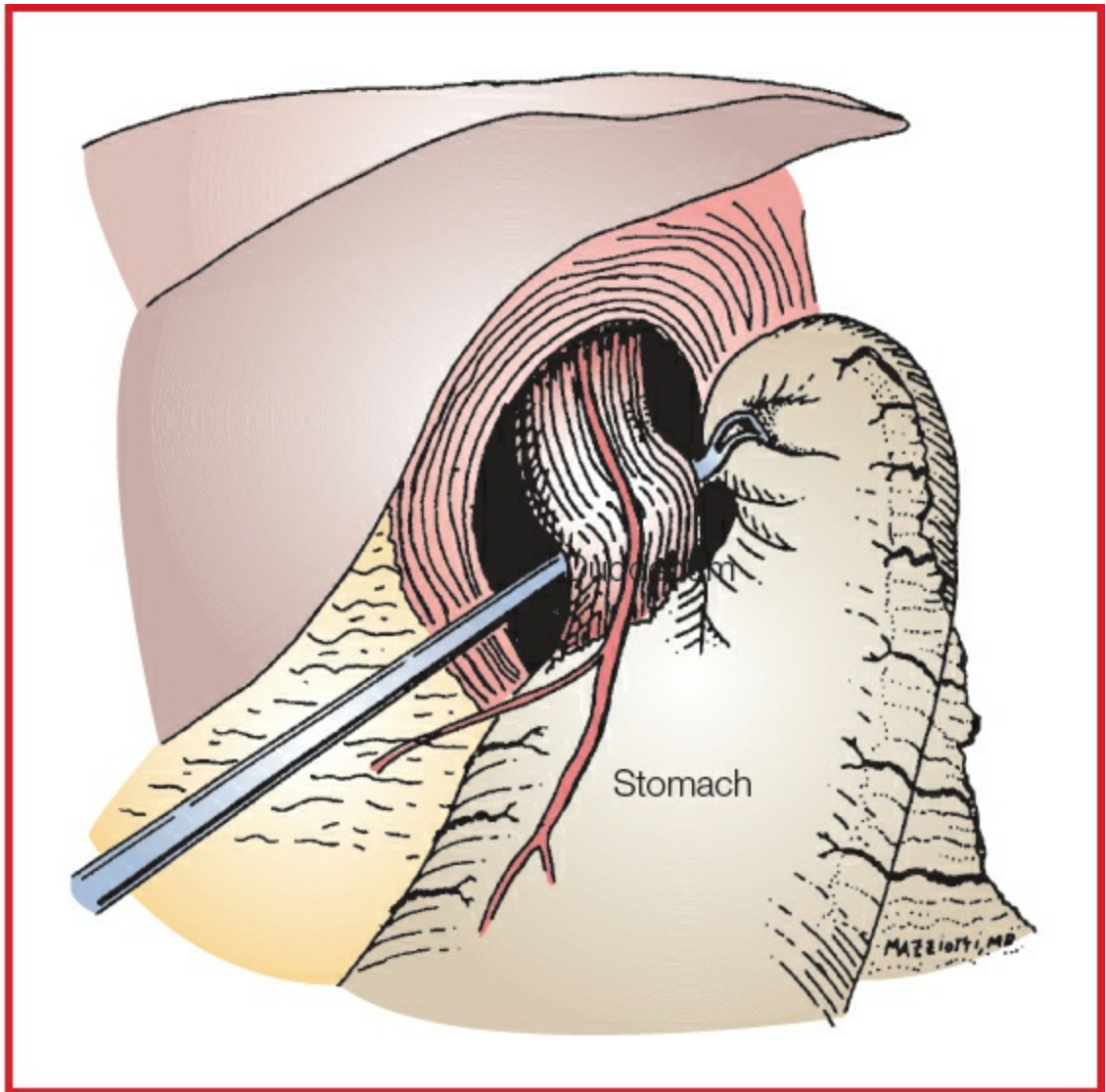


Figure 7.7-2 Gastric fundus is pulled posteriorly and to the right of the esophagus, with the fundus itself used as a retractor, reaching from right to left behind the esophagus to grasp the fundus and retract it back to the right side behind the esophagus. Placing caudad traction on the wrapped fundus, the GE junction, and distal esophagus can be brought further into the abdominal cavity. (Reproduced with permission from Baker RJ, Fischer JE, eds *Mastery of Surgery*, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

GERD; with or without esophagitis; esophageal stricture; or Barrett esophagus

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K](#), p. K-1.

■ SUMMARY OF PROCEDURE

Position	Supine low lithotomy
Incision	Five ports, upper abdomen
Special instrumentation	Harmonic scalpel; bipolar cautery; endoscope. Esophageal dilators should be available.
Antibiotics	Cefazolin 2 g iv preincision (weight-based dosing recommended)
Surgical time	1.5–3 h
Closing considerations	Port closures only; muscle relaxation may be needed for fascial closure.
EBL	<75 mL
Postop care	1–2-d hospital stay; liquids on pod 1; consider a postoperative upper GI.
Mortality	<0.1%
Morbidity	Overall: 8% Atelectasis: common Esophageal or gastric perforation: rare (most commonly associated with passage of NG tube or esophageal dilator) Hemorrhage from short gastrics or splenic tear Pneumothorax (does not usually require treatment) Postop dysphagia
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages (incidence ↑ with age). May be performed in infants; next peak occurs in adulthood; rare in children and adolescents
Male:Female	1:1
Incidence	1% (increasing significantly over the past 5 yr)
Etiology	Unknown
Associated	Aspiration risk; reactive airway disease; occasionally,

conditions

respiratory problems associated with chronic aspiration

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Laparoscopic Cholecystectomy.

Suggested Readings

1. Broeders JA, Roks DJ, Jamieson GG, et al: Five-year outcome after laparoscopic anterior partial versus Nissen fundoplication: four randomized trials. *Ann Surg* 2012; 255(4):637–42.
2. Cowgill SM, Gillman R, Kraemer E, et al: Ten year follow up after laparoscopic Nissen fundoplication for gastroesophageal reflux disease. *Am Surg* 2007; 73(8):748–53.
3. Rebecchi F, Allaix ME, Cinti L, et al: Comparison of the outcome of laparoscopic procedures for GERD. *Updates Surg* 2018; 70(3):315–21.

LAPAROSCOPIC HELLER MYOTOMY ± ANTIREFLUX PROCEDURE

SURGICAL CONSIDERATIONS

Description

Laparoscopic and thoracoscopic esophageal myotomies have become much more common. These procedures are performed for achalasia, an uncommon condition in which the lower esophageal sphincter fails to relax with swallowing and in which the body of the esophagus is aperistaltic and dilates. Patients typically complain of dysphagia with chest pain and may experience regurgitation. The etiology of this condition remains unknown.

Treatment options consist of pneumatic dilation, botulinum toxin injection of the lower esophageal sphincter, or surgical myotomy. **Pneumatic dilation** remains the most common procedure performed for achalasia and is effective in ~60% of patients. Many gastroenterologists are reluctant to perform this procedure, however, because of the risk of esophageal perforation, which is generally considered to be ~3–5%, but may be as high as 10%. **Botulinum toxin injection** is a newer technique that is effective to some degree in most patients, but the duration of its effectiveness is short. Most patients require retreatment within 1.5 yr, and the efficacy of retreatment may diminish over

time. In many centers, **surgical myotomy (Heller operation)** has become the procedure of choice for the treatment of achalasia.

The patient is positioned as for a laparoscopic antireflux procedure—supine in the lithotomy position with reverse Trendelenburg. The abdomen is usually entered with a **Veress needle** at the umbilicus or left upper quadrant Palmer point or **Hasson trocar** at the umbilicus, and five laparoscopic ports are placed across the upper abdomen—two beneath the left costal margin, two beneath the right costal margin, and one in the midline either at the umbilicus or midway between the umbilicus and the xiphoid. The liver is elevated, and the ligamentous attachments anterior to the esophagus are divided. The esophagus is not usually encircled, and, as hiatal hernias are uncommon with achalasia, crural repair is seldom necessary.

The myotomy is begun at the gastroesophageal junction using monopolar cautery, bipolar scissors, or a harmonic scalpel (ultrasonic cutting/coagulation). It is carried proximally until normal musculature is encountered. Some surgeons perform intraop **esophagoscopy** to ensure that the myotomy has been carried proximally enough and that there has not been a mucosal perforation. Generally, a myotomy of 5–8 cm is adequate, although sometimes a longer myotomy may be necessary. Some surgeons then perform a very loose, **partial fundoplication** to prevent reflux. This can be performed as an anterior or posterior fundoplication, bringing the fundus either anterior (Dor fundoplication—our preference) or posterior (Toupet fundoplication) to the esophagus. The stomach is secured to the esophageal wall and crura with sutures, and an NG tube may be passed. The ports are removed, and port closure is carried out.

Variant procedure or approaches

Although most of these procedures are being performed laparoscopically, they also can be performed by **thoracoscopy or thoracotomy**. If a thoracoscopy or thoracotomy is performed, a double lumen tube (DLT) is placed to allow collapse of the left lung.

Usual preop diagnosis

Achalasia

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine lithotomy, reverse Trendelenburg
Incision	Five ports, upper abdomen
Special instrumentation	Laparoscopic instrumentation ± gastroscope
Antibiotics	Cefazolin 2 g iv pre-incision (weight-based dosing)
Surgical time	1.5–2 h
Closing considerations	None
EBL	Minimal
Postop care	2-d hospital stay; liquids on pod 1
Mortality	Rare
Morbidity	Insignificant atelectasis: Common Esophageal or gastric perforation: Rare Pneumothorax (does not usually require treatment) Gastroesophageal reflux: >15%
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–40 yr
Male:Female	1:1
Incidence	Achalasia: 1 in 20,000 adults
Etiology	Unknown
Associated conditions	None. Chagas disease may produce identical esophageal findings.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations Following Laparoscopic Cholecystectomy.

Suggested Readings

1. Bhayani NH, Kurian AA, Dunst CM, et al: A comparative study on comprehensive objective outcomes of laparoscopic Heller myotomy with per-oral endoscopic

- myotomy (POEM) for achalasia. *Ann Surg* 2014; 259(6):1098–103.
2. Bonatti H, Hinder RA, Klocker J, et al: Long-term results of laparoscopic Heller myotomy with partial fundoplication for the treatment of achalasia. *Am J Surg* 2005; 190(6):874–8.
 3. Jeansonne LO, White BC, Pilger KE, et al: Ten-year follow-up of laparoscopic Heller myotomy for achalasia shows durability. *Surg Endosc* 2007; 21(9):1498–1502.
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LAPAROSCOPIC CHOLECYSTECTOMY, ± COMMON DUCT EXPLORATION

SURGICAL CONSIDERATIONS

Description

This operation is usually performed for symptomatic gallstones, gallstone pancreatitis, or acute cholecystitis. A laparoscopic approach is preferred over an open cholecystectomy because of its minimally invasive nature, which allows earlier recovery and return to normal activities. **Laparoscopic cholecystectomy** may be contraindicated for patients with uncorrectable coagulopathy, severe chronic obstructive pulmonary disease (COPD), or severe cardiac disease (unable to tolerate ↑ intra-abdominal pressure). In addition, patients with prior abdominal surgery or with acute cholecystitis are at a higher risk for conversion to open surgery. The operation begins with access to the abdominal cavity at the umbilicus, either with a **Veress needle** (closed technique: blind placement) or a **Hasson trocar** (open technique). The patient will need an OG tube to decompress the stomach before proceeding. CO₂ is insufflated to an intra-abdominal pressure of 15 mm Hg (or up to 18 mm Hg in super obese patients). If the patient develops ventilatory or hemodynamic problems, consider decreasing the intra-abdominal pressure to 10–12 mm Hg. A total of four trocars are used—one at the umbilicus and three in the RUQ. The patient is placed in a RTP and rotated to the left to move the stomach, duodenum, and transverse colon away from the operative field. The cystic artery and cystic duct (with hepatic duct = triangle of Calot) are clipped and cut ([Fig. 7.7-3](#)). The gallbladder is then dissected off the liver with monopolar cautery, placed in a bag, and brought out, usually through the umbilical port or subxiphoid port sites. Hemostasis is then achieved, the area is irrigated, and the 10-mm trocar sites are closed. The rate of conversion to an open operation is 2–3% for

elective gallbladder surgery and ~10% for acute cholecystitis. Should this occur, the operation is then converted to an **open cholecystectomy**.

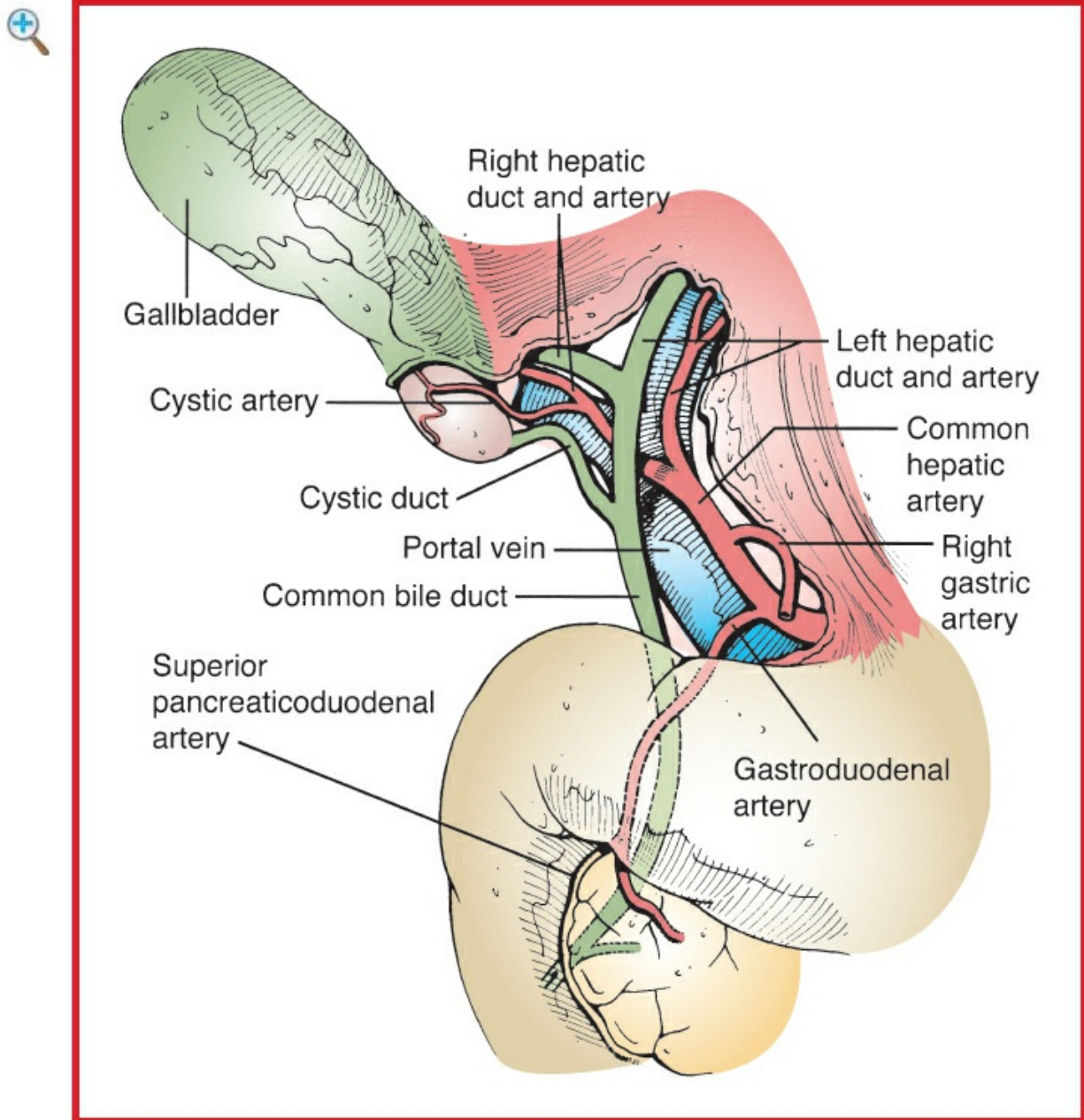


Figure 7.7-3 Surgical anatomy for laparoscopic cholecystectomy and common duct exploration.

Cholangiography can be added easily to the laparoscopic cholecystectomy. A clip is placed high on the cystic duct; then, a small incision is made in the duct just beneath the clip. A cholangiocatheter may be introduced, and dye injected into the biliary tree. X-rays—either fluoroscopy or—are used to assess the biliary anatomy and to look for stones within the ductal system. Cholangiography carries few risks, generally adds about ~15 min to the procedure, and can be used to identify the important anatomy ~85% of the time. Some surgeons perform it routinely during all cholecystectomies;

others perform it selectively and only if there is evidence that the patient has had common duct stones or if the anatomy is in question.

Newer technology allows for less invasive imaging of the biliary tree using indocyanine green dye imaging 45 min before imaging, and often before the first incision, 3 mg of ICG is administered intravenously. A specialized camera is used to visualize the biliary tree when ICG is taken up in the bile glows and gives an outline of the anatomy.

In rare circumstances, **laparoscopic common duct exploration** may be carried out to treat common duct stones. A number of techniques have been used, most employing a thin fiberoptic choledochoscope passed through the cystic duct into the common duct. The anesthesiologist may be asked to administer 1-mg glucagon to relax the sphincter of Oddi. This procedure is performed in only a relatively few centers. More commonly, **endoscopic retrograde cholangiopancreatography (ERCP)** will be used, either preop or postop, to demonstrate the presence or absence of stones and to remove any stones that are found.

Variant procedure or approaches

Open cholecystectomy remains the major alternative to laparoscopic cholecystectomy, either because the cholecystectomy is predicted to be technically difficult or because of the illness of the patient. Approximately 5% or less of laparoscopic cholecystectomies are converted to open cholecystectomy intraop because of difficulties with the procedure. **ERCP** remains the most common means of treating choledocholithiasis in the industrialized world with laparoscopic techniques used in a relatively small number of centers and by a small number of surgeons. **Open common duct exploration** is occasionally necessary for stones not retrievable by ERCP (generally <5%).

Usual preop diagnosis

Acute or chronic cholecystitis, usually with cholelithiasis, with or without choledocholithiasis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Four upper abdominal ports

Special instrumentation	Routine laparoscopic instruments. May require fluoroscopy and choledochoscope for cholangiography, common duct exploration; OG tube.
Antibiotics	Cefazolin 2 g iv preincision (weight-based dosing)
Surgical time	0.5–2 h; may be longer for common duct exploration
Closing considerations	Muscular relaxation helpful for closure of 10-mm port site
EBL	Minimal
Postop care	Usual discharge from PACU or within 24 h
Mortality	<1/1000
Morbidity	Bile leak: 1% Common duct injury: 0.1–0.5% Hemorrhage (requiring transfusion) Infection Injury to bowel Major vascular injury: Uncommon
Pain score	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Typically adult, increasing in the pediatric patient population
Male:Female	Female > male
Incidence	Common
Etiology	Stone disease. Risk factors = female gender, age, parity, obesity
Associated conditions	Hemolytic anemia for pigmented stones

ANESTHETIC CONSIDERATIONS

(Procedures covered: repair of perforated peptic ulcer; laparoscopic esophageal fundoplication; laparoscopic Heller myotomy and antireflux surgery; laparoscopic

cholecystectomy.)

 PREOPERATIVE: LAPAROSCOPIC FUNDOPLICATION/HELLER MYOTOMY

Indications for surgery include failed optimal medical management, high-volume reflux, severe esophagitis, and benign stricture. Severe reflux may lead to airway and pulmonary symptoms, as well.

Respiratory	Patients may have a history of gastroesophageal reflux resulting in aspiration, particularly when recumbent. Recurrent aspiration may result in bronchospasm, pneumonia, and chronic pneumonitis. A prolonged history of reflux may give rise to premalignant Barrett esophagus, which may result in stricture formation and weakness of the esophageal wall. Achalasia may result in severe regurgitation and aspiration. Pulmonary physiologic changes include both restrictive and obstructive lung defects and bronchoalveolar injury. Physical exam may show wheezing and rhonchi. Tests: Consider PA and lateral CXR. Consider baseline arterial blood gases and pulmonary function tests if significant compromise of pulmonary function is present.
Cardiovascular	Patients with both reflux and achalasia may present with chest pain. Typically, the pain of reflux is sharp and substernal. Chest pain in achalasia may radiate to the neck, back, and arms and may be difficult to differentiate from pain of cardiac origin. Differentiation should be made before surgery both by careful history taking and appropriate tests. Tests: Consider ECG, exercise, or chemical stress testing if source of chest pain remains unclear. Other tests as indicated from H&P.
Laboratory	Patients with severe inflammation and erosion may suffer from an iron deficiency anemia. Consider CBC.

Premedication	Otherwise as indicated from H&P. These patients may be at risk for aspiration and should be treated with full-stomach precautions.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with a sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

PREOPERATIVE: LAPAROSCOPIC CHOLECYSTECTOMY

The preop evaluation of patients undergoing open cholecystectomy is discussed under Anesthetic Considerations for Biliary Tract Surgery.

Respiratory	Patients presenting for laparoscopic cholecystectomy may present electively or acutely. Increasingly, cholecystectomy is performed in patients with acute cholecystitis. Patients presenting acutely may experience severe abdominal pain, leading to diaphragmatic splinting and basal lung atelectasis. During the procedure, intra-abdominal CO ₂ insufflation → atelectasis, ↓ FRC, ↑ PIP, ↑ PaCO ₂ , and ↓ PaO ₂ ; therefore, laparoscopic procedures may be contraindicated in patients with severe respiratory or cardiovascular disease. Postop respiratory function, however, is less impaired (e.g., ↓ FRC 30% vs. 50%) and is recovered more quickly (24 vs. 72 h) in patients undergoing laparoscopic cholecystectomy than in the open procedure.
Systemic	Patients presenting for laparoscopic cholecystectomy with acute cholecystitis may have evidence of mild to moderate sepsis. They may be febrile, tachycardic, and hypotensive and need resuscitation before surgery.
Laboratory	CBC. Consider T&S in patients with recurrent and chronic gallbladder disease. They may have liver adhesions. During the acute inflammatory stage, there may be ↑ vascularity → bleeding.

Premedication	Consider midazolam 1–2 mg iv unless the patient is unstable or septic. If urgent surgery, check for recent administration of analgesics.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Laparoscopic fundoplication and Heller myotomy: Consider full stomach precautions and rapid-sequence intubation technique. Additional iv access may be required if the arms are inaccessible. Laparoscopic cholecystectomy: Standard induction. In acutely unwell patients, despite resuscitation, patients may be more susceptible to the effects of anesthesia and positive pressure ventilation. Consider additional iv access, invasive arterial pressure monitoring, and central access in unstable patients. Etomidate may be a useful induction agent in this setting. OG/NGT before incision to ensure stomach decompressed.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Continue muscle relaxation. Intra-abdominal CO ₂ insufflation will → ↑ intra-abdominal pressure, which will predispose to passive regurgitation of gastric contents. Also, intra-abdominal pressure >15 mm Hg → ↓ venous return + ↑ SVR → ↓ CO. Controlled ventilation will minimize the possibility of hypercarbia from absorbed CO ₂ .
Emergence	Given the high incidence of PONV, prophylactic

	antiemetic Rx (e.g., ondansetron 4 mg iv) recommended and should be given 30–60 min before the end of the case. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 8–12 mL/kg/h Fluid warmer	Blood loss typically minimal with these procedures, but potential for major bleeding exists. Consider additional iv access, conversion to open procedure if risk factors for increased bleeding present. See Zero Balance Fluid Management, Appendix K , p. K-2 and K-4.
Monitoring	Standard monitors Urinary catheter OG/NG tube	Others as clinically indicated. Prevent hypothermia (forced-air warmer, warming blanket, warm OR, etc.)
Positioning	✓ and pad pressure points ✓ eyes	Initially in Trendelenburg position (↑ venous return, ↓ lung volumes, potential for mainstem intubation) for trocar placement; then reverse Trendelenburg (↓ venous return, ↑ lung volumes) during subsequent portions of the surgical procedure
Complications	Respiratory: Pneumoperitoneum Hypercarbia/hypoxemia Pneumothorax	Pneumoperitoneum with CO ₂ allows the surgeon to operate laparoscopically. This creates cephalad displacement of the diaphragm with ↓ FRC, ↓ pulmonary compliance, and atelectasis. This can manifest as ↑ PIP, ↓ PO ₂ , and ↑ PCO ₂ .

Pneumomediastinum	Ventilation should be controlled during the operation to minimize the effects of pneumoperitoneum and hypercarbia. An increase in MV is appropriate.
Endobronchial intubation	Pneumothorax can occur 2° retroperitoneal dissection of insufflated CO ₂ . Pneumothorax will manifest as ↓ PO ₂ , ↑ PIP, hemodynamic instability (↑ HR, ↓ BP), and possibly subcutaneous emphysema. The position of the ETT may change with altered patient position → endobronchial intubation.
Cardiovascular: ↓ BP	↓ BP can occur 2° patient positioning (reverse Trendelenburg) and from ↓ venous return 2°
Hemorrhage	pneumoperitoneum (↑ intra-abdominal pressure >15 mm Hg).
Dysrhythmias	Hemorrhage can result from inadvertent injury to blood vessels (during trocar placement). Vascular injection of CO ₂ (air embolism) can cause ↓ BP, dysrhythmias, and even cardiovascular collapse. If cardiopulmonary compromise occurs, the pneumoperitoneum can be released to allow for differential diagnosis and treatment.

	Visceral injury	Injury to the viscera may necessitate an open procedure or may go undiagnosed and → other postop complications, depending on the organ that is injured. Always be prepared to convert to an open procedure.
	Hypothermia	2 ^o dry gas insufflations
	Subcutaneous emphysema	Keep CO ₂ insufflation pressure <12 mm Hg.

POSTOPERATIVE

Complications	PONV (common)	
	Shoulder pain	From pneumoperitoneum; usually self-limited
	Respiratory	Typical postop complications are secondary to pneumoperitoneum and pain.
Multimodal pain management	PCA; consider multimodal analgesia.	Consider combining PCA or fixed dose po/iv opiates with acetaminophen, po or iv NSAIDs, and gabapentin. Emphasize nonopioid analgesics (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization. Consider need for DVT prophylaxis.
Tests	As indicated from H&P	

Suggested Readings

1. Abeles A, Kwasnicki RM, Darzi A: Enhanced recovery after surgery: current

- research insights and future direction. *World J Gastrointest Surg* 2017; 9(2):37–45.
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 7. Li Z, Zhao Q, Bai B, et al: Enhanced recovery after surgery programs for laparoscopic abdominal surgery: a systematic review and meta-analysis. *World J Surg* 2018; 42(11):3463–73.

LAPAROSCOPIC SPLENECTOMY

SURGICAL CONSIDERATIONS

Description

Laparoscopic-assisted splenectomy is best suited for normal or slightly enlarged spleens (e.g., idiopathic thrombocytopenic purpura [ITP]). Laparoscopic splenectomy usually is contraindicated in patients with large hilar lymph nodes or portal hypertension. Massive splenomegaly, with spleens >30 cm in the longitudinal axis, can be considered a relative contraindication with some surgeons electing to have the splenic artery embolized the day before surgery to cause shrinkage and lower the risk of hemorrhage. There is a high conversion rate (to open surgery) if the size of the spleen is between 20 and 30 cm. Sometimes, a hand-assisted approach may be helpful for spleens in this size range. Conversion rates are higher in patients with perisplenitis and morbid myeloproliferative disorders. For the procedure, patients should be placed on a beanbag in a 45° lateral decubitus position or a full lateral decubitus position. The advantage of the 45° right lateral decubitus position is that it is easy to rotate the table and place the patient in a supine position if there is an urgent need for conversion. With the 45° lateral decubitus or the full lateral decubitus position, the kidney rest should be elevated and the OR table should be flexed to increase the area between the costal

margin and the superior iliac crest. All pressure points should be padded, and the patient should be secured firmly to the table. The first trocar site is generally in the LUQ (see Fig. 7.7-5). Access can be with a closed (Veress needle or Optiview trocar) or open (Hasson trocar) approach. The initial approach may be to the hilum, short gastrics, or inferior pole. Most surgeons use the harmonic scalpel or LigaSure to take down the various ligaments and to dissect out the hilum, which generally is stapled with an endo gastrointestinal anastomosis (GIA). The short gastrics are either taken down with the harmonic scalpel or are LigaSure. The lateral and superior attachments are taken down last. The spleen is then placed in a bag. Removal of the spleen can be done in several ways. Some surgeons remove it with a morcellizer placed through one of the trocar sites. Others enlarge one trocar site slightly and remove the spleen in chunks. For very large spleens, some surgeons make a Pfannenstiel incision and extract it through the pelvis.

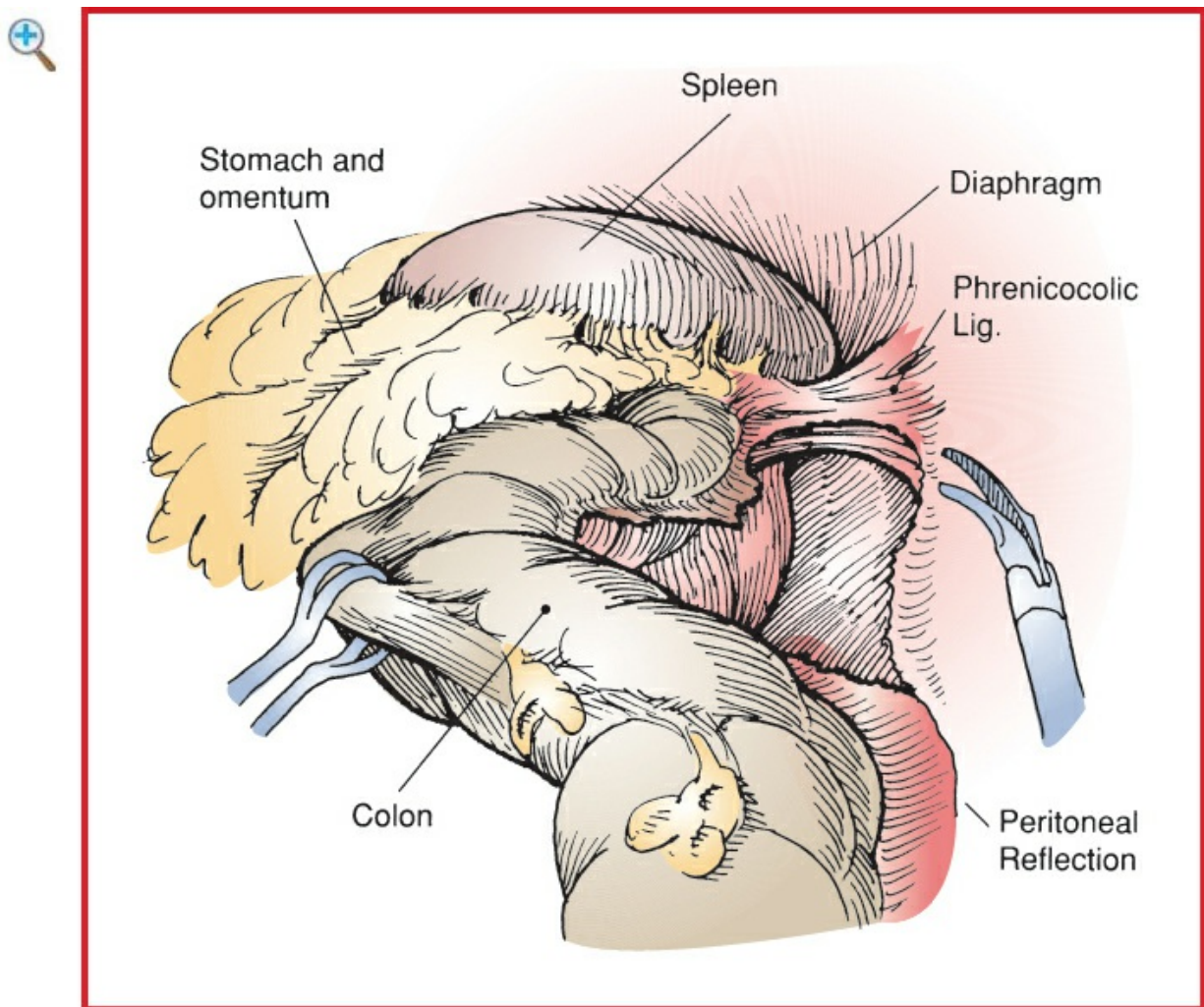


Figure 7.7-4 Anatomy of the spleen. (Reproduced with permission from Wind GG: The spleen. In: Wind GG, Dudai M, eds. Applied Laparoscopic Anatomy: Abdomen and Pelvis Williams & Wilkins, Baltimore: 1997.)

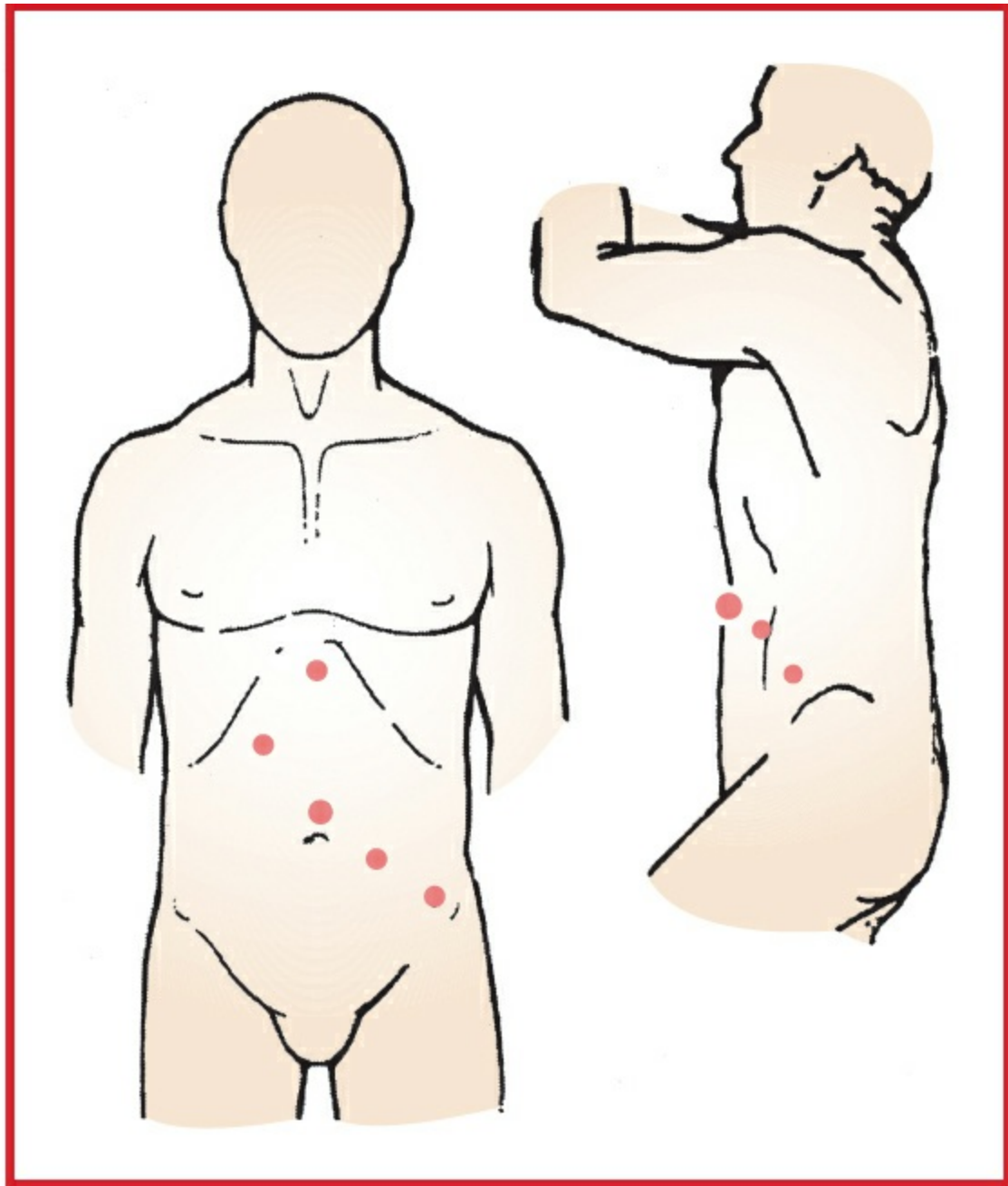


Figure 7.7-5 Trocar location for splenectomy. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

Variant procedure or approaches

Open splenectomy (described on p. 714). A hand-assisted device can be used for larger, and a larger incision (~5 cm) is made either in the midline or the pelvis. A hand is inserted to assist in retracting and dissecting the spleen while improving visualization.

Usual preop diagnosis

Idiopathic thrombocytopenic purpura (ITP) (60%); hereditary spherocytosis (10%)

hemolytic anemia (5%); thrombotic thrombocytopenic purpura (TTP) (5%); lymphoma (5%); hypersplenism (5%)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Usually 45° lateral decubitus, OR table flexed, beanbag
Incision	3–5 trocar ports; 1 enlarged to extract spleen
Special instrumentation	Vascular staplers; Harmonic scalpel; LigaSure instruments
Unique considerations	Patients should have received vaccinations (pneumococcal, meningococcal, Haemophilus influenzae) preop. DVT prophylaxis; ± stress steroids; ± Plt transfusion
Antibiotics	Cefazolin 2 g iv preop (weight-based dosing)
Surgical time	1–3 h
Closing considerations	Rapid closure not requiring muscle relaxation (following search for accessory spleen)
EBL	<100 mL; if significant enough to require transfusion, may require conversion to laparotomy
Postop care	PACU → ward. Generally, begin liquids pod 0, and discharge pod 1 or 2.
Mortality	0.1% (predominately related to underlying disease)
Morbidity	5% Hemorrhage Pulmonary complications Injury to the pancreas, stomach, or splenic flexure of the colon
Pain score	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages; more common in adults
Male:Female	1:1
Incidence	Uncommon
Etiology	Hematologic disorders (e.g., ITP, TTP); hemolytic anemia, tumor of the spleen (e.g., Hodgkin disease); hypersplenism
Associated conditions	Steroid dependence; neutropenia; thrombocytopenia; hemolytic anemias

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients present for laparoscopic splenectomy with a variety of diseases, including ITP, lymphomatous disease (Hodgkin's and non-Hodgkin's), autoimmune hemolytic anemia, TTP, hereditary spherocytosis, Evans syndrome, hairy cell leukemia, hypersplenism 2° portal HTN, sarcoidosis, polycythemia vera, and myelofibrosis. Open splenectomy is usually reserved for traumatic laceration of the spleen. Laparoscopic cases tend to take longer than open splenectomies. Patients who have been treated with chemotherapeutic drugs will require careful preop exam to evaluate for potentially toxic side effects. See Anesthetic Considerations for (open) Splenectomy. Patients should receive pneumococcal, meningococcal, and H. influenzae vaccinations 2 wk preoperatively. If they do not receive preop vaccines, these should be given 2 wk postop.

Respiratory	Splenomegaly may cause left lower-lobe atelectasis. As intra-abdominal CO ₂ insufflation → further atelectasis, ↓ FRC, ↑ PIP, ↑ PaCO ₂ , and ↓ PaO ₂ ; therefore, laparoscopic procedures may be contraindicated in patients with advanced lung disease. Tests: Consider PA and lateral CXR; ABG and PFTs, if clinically indicated.
Cardiovascular	Cardiovascular changes caused by the pneumoperitoneum include ↓ venous return → ↓ CO, and ↑ SVR.

Hematologic	Cytopenias are very common. Tests: CBC with differential and Plt count, T&S, consider T&C. Antibodies may be present → allow ↑ time for T&S/T&C. Platelet counts are typically low, but preop platelet transfusion is usually not helpful due to immediate splenic sequestration.
Hepatic	Consider the presence of coexisting hepatic dysfunction due to primary disease or therapy for it. Tests: Hepatic panel, PT/PTT
Laboratory	See above. As indicated from H&P.
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard induction. In patients with coexisting liver disease, modify medication choice and dosage accordingly. Place OG/NGT before incision to ensure the stomach is decompressed.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	No special considerations. Prophylactic antiemetics are appropriate. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 18–16 ga × 1 BSS @ 8–12 mL/kg/h	Blood loss should be 100 cc. If Plt transfusion is necessary,

	Fluid warmer	it should be given after ligation of splenic vessels (↓ sequestration). Be prepared to obtain more iv access if bleeding is excessive. Consider extra iv access if arms are inaccessible at the start of the case. See Zero Balance Fluid Management, Appendix K, p. K-2 and K-4.
Monitoring	Standard monitors Urinary catheter OG/NGT	Others as indicated by patient status. Prevent hypothermia (forced-air warmer, heated and humidified inspired gases, warming blanket, warm OR, etc.)
Positioning	✓ and pad pressure points ✓ eyes	Careful positioning and padding of patient are essential.
General considerations		If the spleen is large (>20 cm), there is a higher rate of conversion to an open procedure. In particular, intraoperative pain will be greater; hence, analgesic requirements will be higher.
Complications	Respiratory: Cardiovascular: ↓ BP Hemorrhage Visceral injury Hypothermia Subcutaneous emphysema	The complications of laparoscopy are discussed in Anesthetic Considerations for Laparoscopic Cholecystectomy (see p. 652). Conversion to open may be associated with significant hemorrhage. Plt transfusion may be necessary.

POSTOPERATIVE

Complications	Shoulder pain	2° pneumoperitoneum; usually self-limited. Ketorolac (30 mg iv) or other NSAIDs effective
	PONV	
	Atelectasis	Usually left lower lobe
Multimodal pain management	Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see Appendix K, p. K-5).	Consider combining PCA or fixed dose po/iv opiates with acetaminophen, po or iv NSAIDs, and gabapentin.
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition and mobilization. Consider the need for DVT prophylaxis.
Tests	As indicated by patient status	

Suggested Readings

1. Abeles A, Kwasnicki RM, Darzi A: Enhanced recovery after surgery: current research insights and future direction. *World J Gastrointest Surg* 2017; 9(2):37–45.
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3. Grahn SW, Alvarez J, Kirkwood K: Trends in laparoscopic splenectomy for massive splenomegaly. *Arch Surg* 2006; 141(8):755–62.
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7. Sampath S, Meneghetti AT, MacFarlane JK, et al: An 18-year review of open and laparoscopic splenectomy for idiopathic thrombocytopenic purpura. *Am J Surg* 2007;

LAPAROSCOPIC ADRENALECTOMY

SURGICAL CONSIDERATIONS

Description

Laparoscopic adrenalectomy typically is performed for a variety of adrenal problems including Conn disease, functioning adenoma, pheochromocytoma, Cushing disease, hyperplasia, virilizing or feminizing tumors, and an enlarging, nonfunctioning adenoma. Urologists generally use a retroperitoneal approach, whereas other surgeons use a transabdominal approach. The patient should be on a beanbag in the full lateral decubitus position ([Fig. 7.7-6](#)). The kidney rest should be elevated, and the table should be flexed to open up the area between the costal margin and iliac crest. On the left side, three trocars are required along the costal margin. Surgery begins by immobilizing the spleen and the tail of the pancreas medially, and the colon laterally, which allows the spleen to fall away completely from the adrenal gland. The adrenal is then seen behind the hilum of the spleen. Care should be taken to minimize manipulation of a pheochromocytoma to prevent sudden, unexpected HTN during the operation. In general, most surgeons prefer to clip and cut the adrenal vein first. After the adrenal is completely mobilized, it is removed through one of the port sites. For a right-sided approach, four trocars generally are required along the right costal margin. The fourth trocar is used to retract the liver. The first step on the right side is to mobilize the lateral attachments of the liver to expose the adrenal. An enlarged right adrenal may be difficult to mobilize enough to see the adrenal vein because the right adrenal vein empties directly into the inferior vena cava (IVC). The right adrenal vein may be transected with a stapler or divided between clips.

For a pheochromocytoma, seek preoperative consultation with an endocrinologist >4 wk in advance to manage alpha-blockade and possible beta-blockade to avoid sympathetic overstimulation during surgery (see Anesthetic Considerations for Pheochromocytoma, [p. 755](#)).

Variant procedure or approaches

Open procedure or retroperitoneal approach

Usual preop diagnosis

Indeterminate adrenal mass (nonfunctional adenoma); also can be performed for

functional adenomas, rarely for pheochromocytoma. Contraindicated for known carcinoma

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

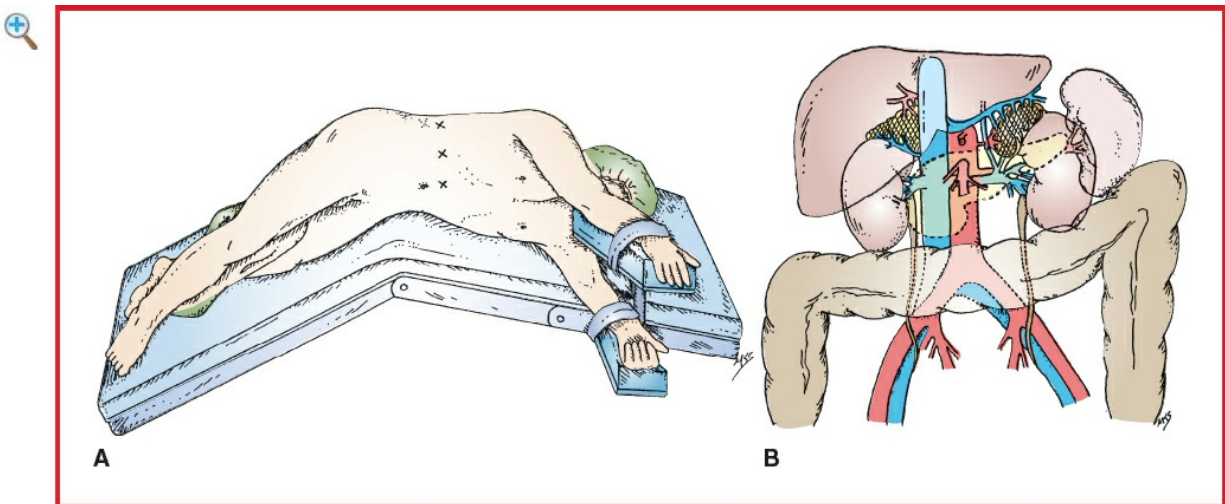


Figure 7.7-6 Laparoscopic adrenalectomy. **A:** Patient positioning and port locations. **B:** Anatomic relationships of adrenals (sutured) to adjacent and overlying structures. (Reproduced with permission from Scott-Conner CEH, Dawson DL Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

■ SUMMARY OF PROCEDURE

Position	Lateral decubitus. A patient with bilateral disease may need repositioning between sides, or surgeon may use a supine approach.
Incision	Three ports on the left side; four ports on the right side
Antibiotics	Cefazolin 2 g iv preincision
Surgical time	1–2 h
Closing considerations	Brief closure; muscle relaxation helpful for fascial repair
EBL	<75 mL, although occasionally significant blood loss may occur

Postop care	Patient is usually discharged on pod 1 or 2, depending on BP control.
Mortality	<0.1%
Morbidity	Hemorrhage/hematoma Injury to the pancreas or kidney UTI DVT
Pain score	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages
Male:Female	1:1
Incidence	Varies with underlying disorder
Etiology	Pheochromocytoma (25%); nonfunctioning adenomas, aldosteronoma (21%); cortisol-producing adenoma (15%); Cushing disease (8%)
Associated conditions	Depending on the function of tumor (e.g., HTN for aldosteronoma, Cushing disease)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The preop evaluation of patients undergoing laparoscopic adrenalectomy is discussed under Anesthetic Considerations for (open) Adrenalectomy.

Cardiovascular	Cardiovascular changes caused by the pneumoperitoneum include ↓ venous return → ↓ CO and ↑ SVR. Decreased blood flow to the splanchnic and renal circulations (→ ↓ UO) may result from high intra-abdominal pressures.
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INTRAOPERATIVE

Anesthetic technique

GETA: An epidural should not be necessary for postop analgesia if the procedure is performed laparoscopically. If the surgical team feels that there is a high likelihood of conversion to an open procedure, consider placement of an epidural catheter for postop analgesia.

Induction	See induction under (open) Adrenalectomy.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	See maintenance under (open) Adrenalectomy.	
Emergence	See emergence under (open) Adrenalectomy. The case is likely to take longer if performed laparoscopically, but usually has a less painful postop course. Prophylactic antiemetics are appropriate. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–14 ga × 1 BSS @ 3–5 mL/kg/h	Warming techniques important (forced-air warmers, fluid warmers, humidified inspired gases, etc.). See Zero Balance Fluid Management, Appendix K , p. K-2 and K-4 .
Monitoring	Standard monitors arterial line ±CVP catheter ±PA catheter UO TEE	A PA catheter may be useful in the management of patients with pheochromocytoma. TEE may be helpful to evaluate cardiac function and filling.
Positioning	✓ and pad pressure points ✓ eyes	Careful support and padding of extremities and torso are very important.
Complications:	Respiratory:	The complications of

laparoscopic	Pneumoperitoneum Hypercarbia/hypoxemia Pneumothorax Pneumomediastinum Endobronchial intubation Cardiovascular: ↓ BP Hemorrhage Dysrhythmias Visceral injury Subcutaneous emphysema Hypothermia	laparoscopy are discussed in Anesthetic Considerations for Laparoscopic Cholecystectomy (p. 652).
Complications: endocrine	Pheochromocytoma: BP lability Myocardial dysfunction	Rx: intraop HTN with SNP or phentolamine (2.5–5 mg q5min); ↑ HR with esmolol; ↓ BP with phenylephrine or dopamine. (See Anesthetic Considerations for Adrenalectomy, p. 661.)
	Conn syndrome: CHF 2° hypervolemia ↓ BP Electrolyte disturbances Hyperglycemia Cushing syndrome: ↓ BP Acute adrenal insufficiency	See Anesthetic Considerations for Adrenalectomy. Continue replacement steroids. (See Anesthetic Considerations for Adrenalectomy, p. 661.)

POSTOPERATIVE

Complications	PONV Shoulder pain	Consider ketorolac 30 mg iv.

Multimodal pain management	PCA (see Appendix C, p. C-4)	See Anesthetic Considerations for Adrenalectomy (see p. 661). Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition and mobilization. Consider the need for DVT prophylaxis.
Tests	As indicated	

Suggested Readings

1. Abeles A, Kwasnicki RM, Darzi A: Enhanced recovery after surgery: current research insights and future direction. *World J Gastrointest Surg* 2017; 9(2):37–45.
2. Ciriác J, Weizman D, Urbach DR: Laparoscopic adrenalectomy for the management of benign and malignant adrenal tumors. *Expert Rev Med Devices* 2006; 3(6):777–86.
3. Colen TY, Mihm FG, Mason TP, Roberson JB: Catecholamine-secreting paragangliomas: recent progress in diagnosis and perioperative management. *Skull Base* 2009; 19(6):377–85.
4. Kim AJ, Yong RJ, Urman RD: The role of transversus abdominis plane blocks in enhanced recovery after surgery pathways for open and laparoscopic colorectal surgery. *J Laparoendosc Adv Surg Tech A* 2017; 27(9):909–14.
5. Li Z, Zhao Q, Bai B, et al: Enhanced recovery after surgery programs for laparoscopic abdominal surgery: a systematic review and meta-analysis. *World J Surg* 2018; 42(11):3463–73.
6. Livingstone M, Duttchen K, Thompson J, et al: Hemodynamic stability during pheochromocytoma resection: lessons learned over the last two decades. *Ann Surg Oncol* 2015; 22(13):4175–80.
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9. Taskin H, Berber E: Robotic adrenalectomy. *J Surg Oncol* 2012; 106(5):622–5.

LAPAROSCOPIC AND ROBOTIC BOWEL RESECTION

SURGICAL CONSIDERATIONS

Description

Laparoscopic and robotic bowel surgeries have increased in acceptance and are now considered routine. There are clear advantages to a laparoscopic approach to bowel resection, including decreased pain, earlier return of GI function, earlier ambulation, and earlier discharge from the hospital.

For a **left-sided colon resection**, the patient is placed in a low lithotomy position, while in other bowel resections, a supine position is used. The patient's arms should be tucked to improve access to all four quadrants. Generally, four to five ports are required, one in each quadrant ([Fig. 7.7-7](#)). For a **laparoscopic-assisted approach**, one of these ports will be enlarged slightly for removal of the specimen. Very often, the operating table will need to be tilted or rotated throughout the course of the procedure to help move the small intestines away from the surgical dissection site. The procedure typically begins with localization of the pathology. This may require an intraop **sigmoidoscopy** if the lesion has not already been marked on colonoscopy or if it is not grossly apparent. The involved section of the intestine is then mobilized. Often, division of the mesentery is done intracorporeally. Occasionally, surgeons will exteriorize the bowel and do extracorporeal division of the mesentery and extracorporeal division of the bowel. For right-sided lesions, the anastomosis typically is done extracorporeally; however, for left-sided lesions, once the bowel is removed, the pneumoperitoneum will be reinsufflated, and the anastomosis will be performed intracorporeally with an end-to-end stapler placed through the anus.

Some surgeons evaluate blood flow to anastomoses, either routinely or selectively, using indocyanine green imaging. 3 mg of ICG should be administered intravenously approximately before evaluation. When an appropriate camera is utilized, blood flow to the anastomosis appears green, ensuring that it is perfused.

Unipolar and bipolar cautery, the Harmonic scalpel, LigaSure™, and many different surgical staplers may be used during these procedures. These generally require longer operative times than the corresponding open procedures, but typically, they are associated with shorter hospitalization stays and earlier return to work than traditional laparotomy.

Variant procedure or approaches

Open surgery, robotic, laparoscopic-assisted, or totally laparoscopic procedures. Some surgeons also have recommended a hand-assisted procedure where a pneumoperitoneum is still used, but the port site of extraction is enlarged at the beginning of the operation. The surgeon's hand can be placed through a special sleeve that maintains pneumoperitoneum, but also allows the surgeon to retract and dissect manually. Occasionally, the specimen is removed via the anus or through a vaginotomy.

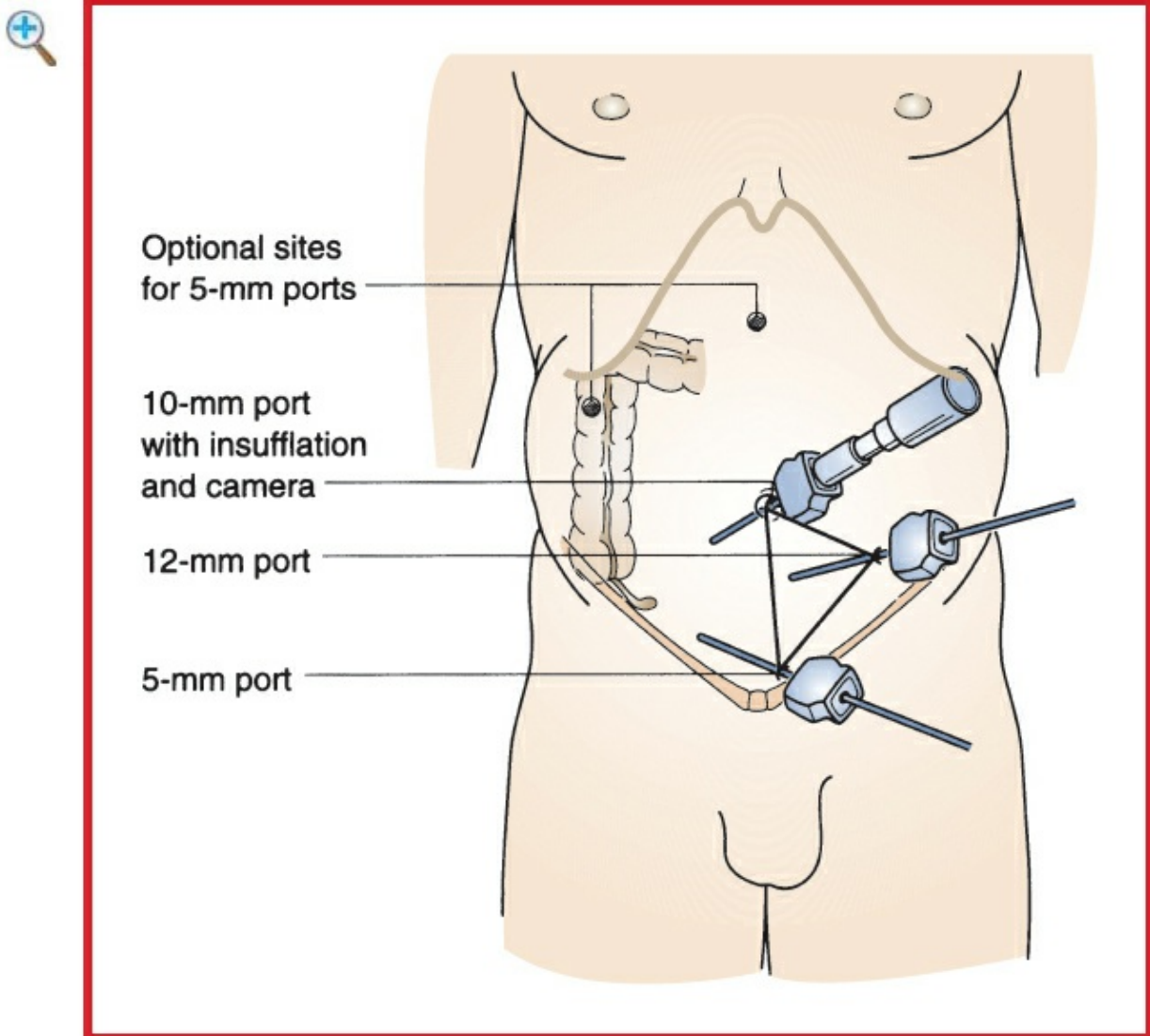


Figure 7.7-7 Trocar placement for a right colectomy. The general principle is to place the camera port so that the surgeon's visual axis is parallel to the telescope and to place the working ports so that the operative site is at the apex of an isosceles triangle. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al, eds. *Surgery: Scientific Principles and Practice*, 2nd edition. Lippincott-Raven Publishers, Philadelphia: 1997.)

Usual preop diagnosis

Intestinal obstruction; inflammatory conditions (e.g., diverticulitis, bleeding, neoplasms); cancer (controversial)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. ERAS programs to standardize perioperative care have resulted in reduced narcotic requirements, shorter length of stay, and reduced complications in colectomy patients. The main anesthesia adjustments in these programs are the use of multimodality pain control, including epidural analgesia, regional blocks, and nonnarcotic iv analgesia to supplement narcotics. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Usually supine; sometimes lithotomy for left-sided procedures
Incision	4–5 trocar sites, one in each quadrant. One site will be enlarged for specimen removal and anastomosis in laparoscopic-assisted procedures.
Special instrumentation	Harmonic scalpel, various laparoscopic stapling instruments
Unique considerations	The operating table will be moved quite frequently to help with visualization.
Antibiotics	2nd-generation cephalosporin (e.g., cefotetan, or cefoxitin) or cephalexin and Flagyl preincision
Surgical time	1.5–2 h (small bowel resection); 3–4 h (colonic resection). Surgical times will ↑ in the presence of acute inflammation and as the bowel resection becomes more extensive.
Closing considerations	Usually requires closure of several port sites and extraction site. Muscle relaxation helpful
EBL	<200 mL
Postop care	NG tube seldom used. Most patients start a full liquid diet on pod 0; discharge home generally 1–4 d
Mortality	<0.1%

Morbidity	<5% (similar in magnitude to those seen with open operations) Complications include: Hemorrhage Infection Anastomotic leak Intestinal obstruction
Pain score	6–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages (increases with age)
Male:Female	1:1
Incidence	Varies with disease process
Etiology	Diverticulitis; polyps; inflammatory bowel disease (IBD); cancer
Associated conditions	None

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

For respiratory, cardiovascular, musculoskeletal, gastrointestinal, and renal considerations, see Anesthetic Considerations for Intestinal and Peritoneal Procedures.

Additional considerations	Laparoscopic bowel resection is performed for both malignant and benign diseases. Patients presenting may have a long history of inflammatory bowel disease such as ulcerative colitis or Crohn disease. Typically, they have received medical therapy, surgical treatments, or both. Medical therapies include 5-aminosalicylic acid compounds and immunosuppressants, such as corticosteroids and methotrexate. Side effects of medical therapies include interstitial nephritis, pancreatitis, pleuropericarditis (5-aminosalicylic acids), immunosuppression, truncal obesity, diabetes, and
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	hypertension (corticosteroids). Patients who have had previous surgery or severe disease may have adhesions making the impending laparoscopic surgery more technically challenging. Inflammatory bowel disease is associated with a number of coexisting conditions including arthritis, uveitis, and biliary stasis, which may impact on anesthetic management.
Tests	CBC, basic metabolic panel, T&S. Others as per history and physical exam.
Premedication	Midazolam 1–2 mg. Consider po acetaminophen 650–1000 mg, gabapentin 300–600 mg, and celecoxib 200 mg.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard induction. Standard rapid-sequence induction if at risk for aspiration. OG/NGT placed before incision for gastric decompression.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance
Emergence	No special considerations unless the patient is at risk for aspiration; extubation should then occur after the patient is fully awake and has protective airway reflexes. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.

Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 8–12 mL/kg/h Fluid warmer	Blood loss should be <200 cc. See Zero Balance Fluid Management, Appendix K, p. K-2 .
Monitoring	Standard monitors Urinary catheter	Others as indicated by patient status. Prevent hypothermia (forced-air warmer, warming blanket, warm room, etc.)
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Respiratory: Pneumoperitoneum Hypercarbia/hypoxemia Pneumothorax Pneumomediastinum Endobronchial intubation Cardiovascular: ↓ BP Dysrhythmias Hemorrhage Visceral injury Hypothermia Subcutaneous emphysema	These complications (associated with the preperitoneal approach), as well as the general complications of laparoscopy, are discussed in Anesthetic Considerations for Laparoscopic Cholecystectomy (p. 652).

POSTOPERATIVE

Complications	PONV Shoulder pain	Ketorolac (30 mg iv) or other NSAIDs
Multimodal pain management	PCA and multimodal analgesia	Consider combining PCA or fixed dose po/iv opiates with acetaminophen, po or iv NSAIDs, and gabapentin. Consider iv ketamine or iv

		lidocaine. Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition and mobilization. Consider the need for DVT prophylaxis.
Tests	As indicated by patient status	

Suggested Readings

1. Aarts MA, Rotstein OD, Pearsall EA, et al: Postoperative ERAS interventions have the greatest impact on optimal recovery: experience with implementation of ERAS across multiple hospitals. *Ann Surg* 2018; 267(6):992–7.
2. Abeles A, Kwasnicki RM, Darzi A: Enhanced recovery after surgery: current research insights and future direction. *World J Gastrointest Surg* 2017; 9(2):37–45.
3. COST (Clinical Outcomes of Surgical Therapy Study Group): A comparison of laparoscopically assisted and open colectomy for colon cancer. *N Engl J Med* 2004; 350(20):2050–9.
4. Favuzza J, Delaney CP: Outcomes of discharge after elective laparoscopic colorectal surgery with transversus abdominis plane blocks and enhanced recovery pathway. *J Am Coll Surg* 2013; 217(3):503–6.
5. Feldheiser A, Aziz O, Baldini G, et al: Enhanced Recovery After Surgery (ERAS) for gastrointestinal surgery, part 2: consensus statement for anaesthesia practice. *Acta Anaesthesiol Scand* 2016; 60(3):289–334.
6. Keshava A, Young CJ, Richardson GL, et al: A historical comparison of single incision and conventional multiport laparoscopic right hemicolectomy. *Colorectal Dis* 2013; 15(10):e618–22.
7. Kim AJ, Yong RJ, Urman RD: The role of transversus abdominis plane blocks in enhanced recovery after surgery pathways for open and laparoscopic colorectal surgery. *J Laparoendosc Adv Surg Tech A* 2017; 27(9):909–14.
8. Li Z, Zhao Q, Bai B, et al: Enhanced recovery after surgery programs for laparoscopic abdominal surgery: a systematic review and meta-analysis. *World J Surg* 2018; 42(11):3463–73.
9. Liu V, Rosas E, Hwang J, et al: Enhanced recovery after surgery program implementation in 2 surgical populations in an integrated health care delivery system.

JAMA Surg 2017; 152(7):e171032. doi:10.1001/jamasurg.2017.1032

9. Scott MJ, Baldini G, Fearon KCH, et al: Enhanced Recovery After Surgery (ERAS) for gastrointestinal surgery, part 1: pathophysiologic considerations. *Acta Anaesthesiol Scand* 2015; 59(10):1212–31.

LAPAROSCOPIC APPENDECTOMY

SURGICAL CONSIDERATIONS

Description

Appendectomy generally is performed for suspected appendicitis. The patient is placed in the supine position with the left arm tucked. An OG/NG tube should be inserted to decompress the stomach. Access is obtained at the umbilicus, either through a closed (**Veress needle**) technique or open (**Hasson trocar**) technique. Two additional trocars are inserted—one suprapubic and the other in the left lower quadrant (LLQ). Occasionally, a fourth trocar may be placed in the RUQ to help with mobilization and retraction. The table will then be rotated to the left side, and the surgeon may ask for it to be placed in Trendelenburg or reverse Trendelenburg position, depending on the location of the cecum. The base of the cecum is identified and the appendix is mobilized ([Fig. 7.7-8B](#)). This may require dissection of the peritoneal edge along the right gutter. After the appendix is mobilized, a window is created in the mesoappendix, and the base of the appendix is stapled with an endo GIA. The mesoappendix is then stapled with a vascular-cartridge endo GIA. The appendix generally is placed in a bag before delivering it, or it may be brought directly through the 10-/12-mm trocar (at the umbilicus or other location). The fascia is closed, and the skin is closed. When unexpected pathology is identified, it can be dealt with by laparoscopy or by laparotomy, with incision placement dependent on findings.

Variant procedure or approaches

Open appendectomy via RLQ or midline incision

Usual preop diagnosis

Acute abdomen; possible acute appendicitis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

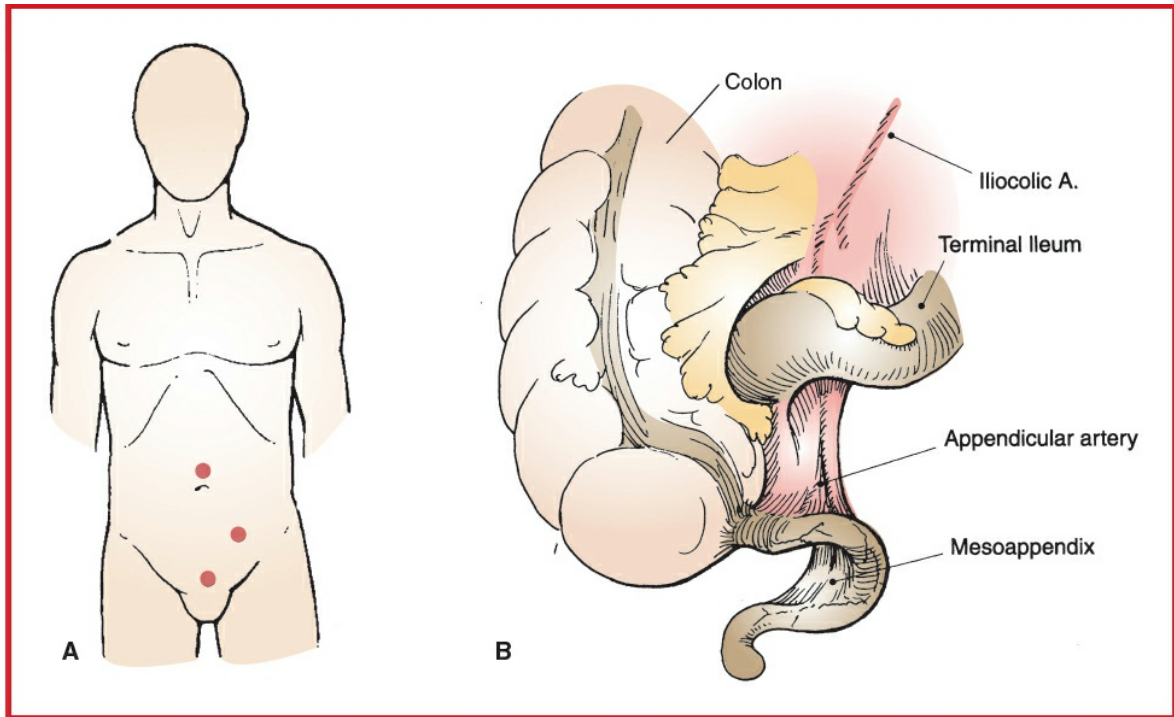


Figure 7.7-8 Setup and initial view for laparoscopic appendectomy. **A:** Reproduced with permission from Scott-Conner CEH, Dawson DL *Operative Anatomy*, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003. **B:** Reproduced with permission from Wind GG: The spleen. In: Wind GG, Dudai M, eds. *Applied Laparoscopic Anatomy: Abdomen and Pelvis*. Williams & Wilkins, Baltimore: 1997.

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	3–4 ports (see Fig. 7.7-8A)
Antibiotics	Broad-spectrum antibiotics such as cefotaxime (1-2 g iv), Zosyn (piperacillin and tazobactam)
Surgical time	30–90 min
Closing considerations	Muscular relaxation helpful for umbilical closure; NG tube if prolonged ileus anticipated
EBL	<75 mL
Postop care	Patients are hospitalized for varying lengths of time, depending on their need for antibiotics. Many with simple acute appendicitis can go home from the PACU or within 24 h.
Mortality	Perforated: 2–5% (nonperforated: <0.1%); increased in elderly patients and infants

Morbidity	Overall: 3% if nonperforated; 20–50% with perforation Intra-abdominal abscess Hematoma Ileus Risk of intra-abdominal abscess, mostly dependent on pathology (e.g., early acute inflammation vs. perforation) Hemorrhage Infection Stump leak: exceedingly rare Conversion to open procedure: dependent on pathology and experience of surgeon
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages; commonly, 19–25 yr
Male:Female	1:1
Incidence	Common (7%)
Etiology	Acute appendicitis; etiology unknown
Associated conditions	Consider other pelvic pathology in women of childbearing age.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

This patient population is generally fit and healthy, apart from their acutely presenting illness. Full-stomach precautions are appropriate in these patients.

Respiratory	Respiratory impairment can occur 2° acute abdominal pain and splinting. Patients with acute abdomen should be treated as if they have full stomachs. Tests: As indicated from H&P
Cardiovascular	Cardiovascular changes caused by the pneumoperitoneum include ↓ venous return → ↓ CO

	and ↑ SVR. Decreased blood flow to the splanchnic and renal circulations (→ ↓ UO) may result from high intra-abdominal pressures. May be dehydrated from fever, emesis, and decreased oral intake. Assess volume status with VS and hydrate adequately before anesthetic induction. Tests: As indicated from H&P
Gastrointestinal	Patient typically has abdominal pain with N/V. Muscular resistance to palpation of abdominal wall frequently parallels the severity of the inflammatory process. With spreading peritoneal irritation (as with perforation), patient will develop abdominal distension and paralytic ileus. Tests: Renal panel; consider pregnancy test.
Hematologic	Moderate leukocytosis (10,000–18,000) with left shift. Hemoconcentration likely if patient dehydrated Tests: CBC
Laboratory	Others as indicated from H&P
Premedication	Patients with acute appendicitis should be treated as if they have full stomachs. Ascertain if patient received opiates during ED eval and management. Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA, with rapid-sequence iv induction

Induction	Preoxygenate patient and have an assistant apply cricoid pressure. Etomidate 0.1–0.4 mg/kg or propofol 1.5–3 mg/kg + succinylcholine 1.5 mg/kg, for intubation.
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	OG/NGT placed before incision to decompress the stomach.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Standard maintenance, without N ₂ O	
Emergence	Extubate when the patient is awake and with active laryngeal protective reflexes. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 3–5 mL/kg/h	See Zero Balance Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors. Urinary catheter	Others as indicated by patient status. Prevent hypothermia (forced-air warmer, warming blanket, warm room, etc.)
Positioning	✓ and pad pressure points ✓ eyes Secure or tuck arms.	Trendelenburg position with elevation of the right side of the abdomen improves surgical exposure.
Complications	Respiratory: Pneumoperitoneum Hypercarbia/hypoxemia Pneumothorax Pneumomediastinum Endobronchial intubation Cardiovascular: Hypotension Dysrhythmias Visceral injury Hypothermia Subcutaneous emphysema	These complications (associated with the preperitoneal approach), as well as the general complications of laparoscopy, are discussed in Anesthetic Considerations for Lap Cholecystectomy.

POSTOPERATIVE

Complications	PONV Urinary retention	Straight catheterization of the bladder
Multimodal pain management	PCA Oral analgesics Ketorolac	See Appendix C, p. C-4 . Acetaminophen 650–1000 mg po q6h or Percocet: 1–2 tab po q6h 30 mg iv. Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition and mobilization. Consider the need for DVT prophylaxis.
Tests	As indicated by patient status	

Suggested Readings

1. Abeles A, Kwasnicki RM, Darzi A: Enhanced recovery after surgery: current research insights and future direction. *World J Gastrointest Surg* 2017; 9(2):37–45.
2. Bennett J, Boddy A, Rhode M: Choice of approach for appendectomy: a meta-analysis of open versus laparoscopic appendectomy. *Surg Laparosc Endosc Percutar Tech* 2007; 17(4):245–55.
3. Cash CL, Frazee RC, Abernathy SW, et al: A prospective treatment protocol for outpatient laparoscopic appendectomy for acute appendicitis. *J Am Coll Surg* 2012; 215(1):101–5.
4. Kim AJ, Yong RJ, Urman RD: The role of transversus abdominis plane blocks in enhanced recovery after surgery pathways for open and laparoscopic colorectal surgery. *J Laparoendosc Adv Surg Tech A* 2017; 27(9):909–14.
5. Li Z, Zhao Q, Bai B, et al: Enhanced recovery after surgery programs for laparoscopic abdominal surgery: a systematic review and meta-analysis. *World J Surg* 2018; 42(11):3463–73.

LAPAROSCOPIC INGUINAL HERNIA REPAIR

SURGICAL CONSIDERATIONS

Description

Laparoscopic inguinal hernia repair is clearly preferred in patients with recurrent or bilateral hernias (Fig. 7.7-9). In patients with first-time unilateral hernias, there are no clear advantages in terms of operative time, postop pain, or time to discharge, compared with tension-free open repair. There are three types of laparoscopic hernia repairs. The first, **ONLAY**, generally has been discarded because of high recurrence rates. The other two are the **totally extraperitoneal (TEP)** and the **transabdominal preperitoneal (TAPP)** techniques. Of these, the TEP is somewhat more difficult to learn, but is associated with lower recurrence rates and complications than is the TAPP. For both the TEP and the TAPP, the patient is placed in the supine position with both arms tucked. Some surgeons insert a Foley catheter, but many do not. A small incision is made at the umbilicus. For a **TEP**, this incision only goes to the preperitoneal space, and a balloon is then used for dissection. Two additional ports are placed in the midline—one suprapubic and one halfway between the umbilicus and the suprapubic port. Further dissection is required to identify the hernia defects, which are then reduced. The cord structures are completely freed, and any cord lipomas are dissected. Mesh is placed to cover the entire area and may be tacked to Cooper ligament. For a **TAPP** repair, the umbilical incision extends into the abdomen. Two additional ports are placed at the level of the iliac crest—one in the RLQ and one in the LLQ. A peritoneal flap over the hernia defect is created, and the preperitoneal space is entered. The rest of the dissection and the placement of the mesh are the same as with the TEP repair. At the end of the TAPP procedure, the peritoneal flap is placed over the mesh and tacked into place to prevent exposure of the bowel to the mesh. Increasingly, the robotic surgical platform is being used for the TAPP approach.

Variant procedure or approaches

Open hernia repair (see Inguinal Herniorrhaphy)

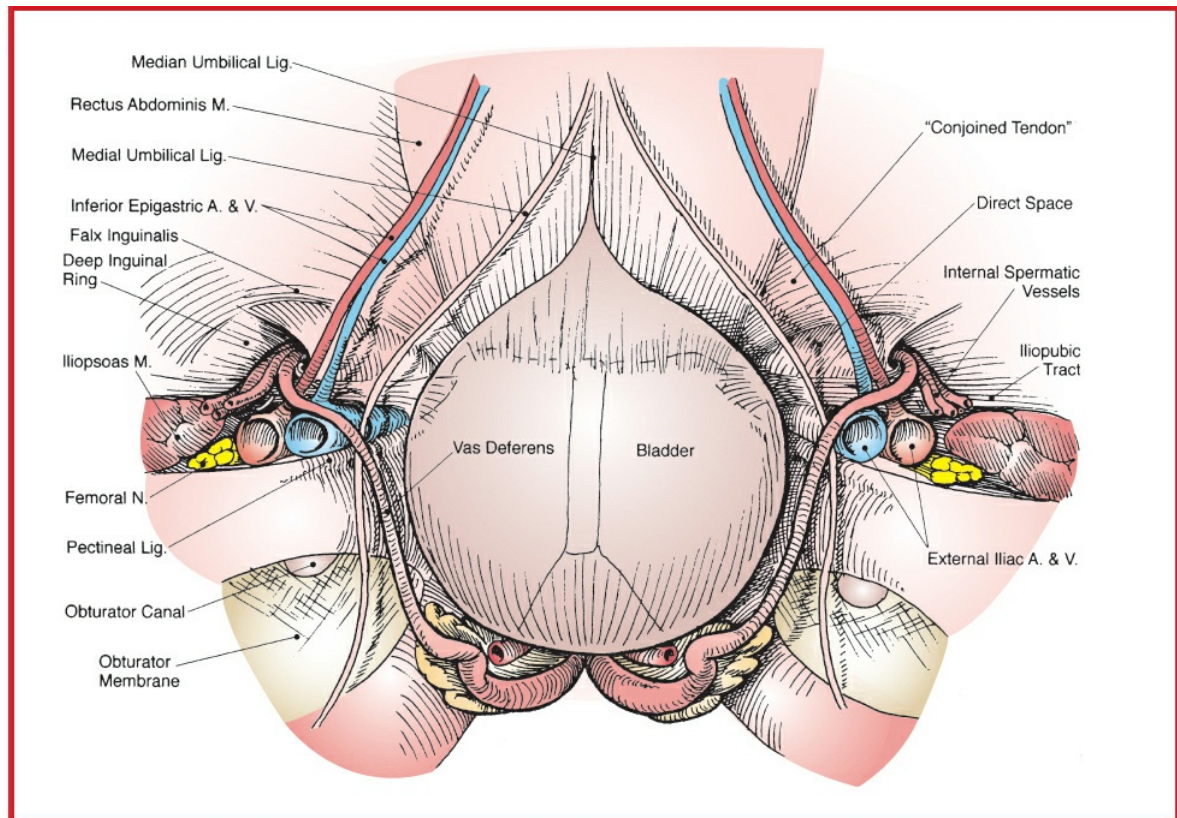


Figure 7.7-9 The inguinal region. (Reproduced with permission from Wind GG: The spleen. In: Wind GG, Dudai M, eds. Applied Laparoscopic Anatomy: Abdomen and Pelvis. Williams & Wilkins, Baltimore: 1997.)

Usual preop diagnosis

Inguinal hernia

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine, arms tucked
Incision	3 ports—1 at the umbilicus; 2 either in the midline or RLQ or LLQ, depending on the laparoscopic approach
Special instrumentation	Balloon dissector and structural balloon for TEP repair
Antibiotics	Cefazolin 2 g iv preop (weight-based dosing)

Surgical time	1–2 h
Closing considerations	Minimal time; muscle relaxation may be helpful.
EBL	<50 mL
Postop care	1–2 h in PACU/holding area → home
Mortality	<0.1%
Morbidity	2% Orchialgia, neuralgia Recurrence of hernia: significant learning curve Bowel obstruction Bladder injury: rarely reported
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Mostly adults
Male:Female	Male > female
Incidence	Common
Etiology	Most hernias congenital; chronically increased intra-abdominal pressure (e.g., chronic cough, obesity)
Associated conditions	None important

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The preop evaluation of patients undergoing laparoscopic hernia repair is discussed in Anesthetic Considerations for Inguinal Hernia. Patients presenting for this procedure are generally healthy. Laparoscopic repair of inguinal hernia is usually associated with less pain and earlier return to preop function when compared to the open procedure. Patients with strangulated or incarcerated hernias usually require emergent open procedures.

Cardiovascular	Cardiovascular changes caused by the pneumoperitoneum include ↓ venous return → ↓ CO
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	and ↑ SVR. Decreased blood flow to the splanchnic and renal circulations (→ ↓ UO) may result from high intra-abdominal pressures.
Laboratory	Hgb/Hct (healthy patients); otherwise, as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard induction. OG/NG tube before incision to decompress the stomach	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	No special considerations except to minimize coughing on emergence. Consider 1 mg/kg iv lidocaine for cough suppression. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 18 ga × 1 BSS @ 3–5 mL/kg/h	Blood loss should be minimal. See Zero Balance Fluid Management, Appendix K , p. K-2 and K-4 .
Monitoring	Standard monitors Urinary catheter	Others as indicated by patient status Prevent hypothermia (forced-air warmer, heated and humidified gases, warming

		blanket, warm OR, etc.)
Positioning	✓ and pad pressure points ✓ eyes	Careful positioning and padding of the patient are essential.
Complications	Hemorrhage from trocar insertion. Subcutaneous emphysema	These complications (associated with the preperitoneal approach), as well as the general complications of laparoscopy, are discussed in Anesthetic Considerations for Laparoscopic Cholecystectomy.

▼ POSTOPERATIVE

Complications	PONV Urinary retention	See p. B-6 . Straight catheterization of the bladder
Multimodal pain management	Consider multimodal analgesia.	Consider combining PCA or fixed dose po/iv opiates with acetaminophen, po or iv NSAIDs, and gabapentin. Consider iv ketamine or iv lidocaine. Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see p. C-2).
ERAS	PONV protocol (see p. B-1)	Encourage early enteral nutrition and mobilization. Consider need for DVT prophylaxis.
Tests	As indicated by patient status	

Suggested Readings

1. Abeles A, Kwasnicki RM, Darzi A: Enhanced recovery after surgery: current research insights and future direction. *World J Gastrointest Surg* 2017; 9(2):37–45.
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LAPAROSCOPIC BARIATRIC SURGERY

SURGICAL CONSIDERATIONS

Description

Currently, nearly two-thirds of Americans are overweight (BMI 25–30 kg/m²) or obese (BMI > 30 kg/m²). According to the United States Centers for Disease Control and Prevention, 35% of American adults are obese. Many obese patients have associated comorbidities, including HTN, diabetes, and sleep apnea, which they have been unable to correct by dieting and exercise. In 1998, the NIH recommended bariatric surgery as the primary treatment for morbid obesity (BMI > 40 kg/m² or >35 kg/m² with associated comorbidities). Surgical treatment results in weight loss of ~2/3–3/4 of excess body weight, usually with consequent correction of comorbidities. Operations for morbid obesity are classified as restrictive, such as the **adjustable gastric banding** and **vertical banded gastroplasty**; malabsorptive, such as a **jejunoileal bypass**; or a combination, such as the **Roux-en-Y gastric bypass**. The 1991 NIH consensus statement identified the Roux-en-Y bypass as the best surgical treatment for morbid obesity. In general, this operation is approached laparoscopically in most patients because of the decreased pain, earlier ambulation, earlier discharge from the hospital, quicker return to regular activity, and decreased wound complication rates, when compared with an open approach. Open approaches, though very rare, are undertaken in

patients with previous upper abdominal surgery, patients who may not tolerate an increased intra-abdominal pressure (e.g., CHF, severe CAD, severe pulmonary disease), patients undergoing revision bariatric surgery, and occasionally patients who fall into a super morbidly obese group ($\text{BMI} > 60 \text{ kg/m}^2$). Surgery is indicated in patients with a BMI of 35–40 kg/m^2 , if they have associated comorbidities, or $>40 \text{ kg/m}^2$, if they have no associated comorbidities.

For laparoscopic procedures, the patient generally is placed in reverse Trendelenburg-head-elevated laryngoscopic position (RT-HELP). Some surgeons prefer a split-leg table, with the surgeon standing between the legs. For a laparoscopic Roux-en-Y bypass, the first incision is made in the midline, approximately halfway between the umbilicus and the xiphoid; 4–5 additional ports are then placed in the LUQ and the midline (Fig. 7.7-10). Most surgeons begin by performing a jejunojejunostomy. During this time, the patient is placed in a RTP to drop the small intestines into the pelvis. The omentum is placed in the upper abdomen, and the ligament of Treitz is identified. The jejunum is divided with an endo GIA stapler ~20 cm from the ligament of Treitz. A Roux limb of 100 cm is then measured. (Occasionally, this Roux limb may be 150 cm, if the patient's BMI is $>50 \text{ kg/m}^2$.) The jejunojejunostomy is created with a stapler, and the enterotomy is closed. Some surgeons prefer a **retrocolic approach**, wherein a passage is made through the transverse mesocolon. Other surgeons prefer an **antecolic approach**, in which the omentum is divided to allow for a place where the Roux limb can pass without tension. At this point, the liver retractor is placed, and the gastric pouch is created. Before stomach stapling, everything in the stomach, OG tubes and calibrating tubes, should be removed. The gastric pouch is then stapled and cut. Often, a calibrating tube is placed after the first two staple firings to help maintain the size of the pouch and the anastomosis. Some surgeons hand-sew the gastrojejunostomy, and some staple it with a linear stapler. Some surgeons staple the anastomosis and place the anvil of the end-to-end anastomotic stapler through the mouth (rarely done). Other surgeons place the anvil through a separate gastrotomy before complete division of the pouch. At the end, the gastrojejunostomy typically is tested by inflating air into the gastric pouch, either through an OG tube, the sizing laparoscopic bariatric surgery tube, or a gastroscope. Generally, the NG or OG tube is removed at the end of the procedure. Often, at this point, surgeons will close the defect of the transverse mesocolon.

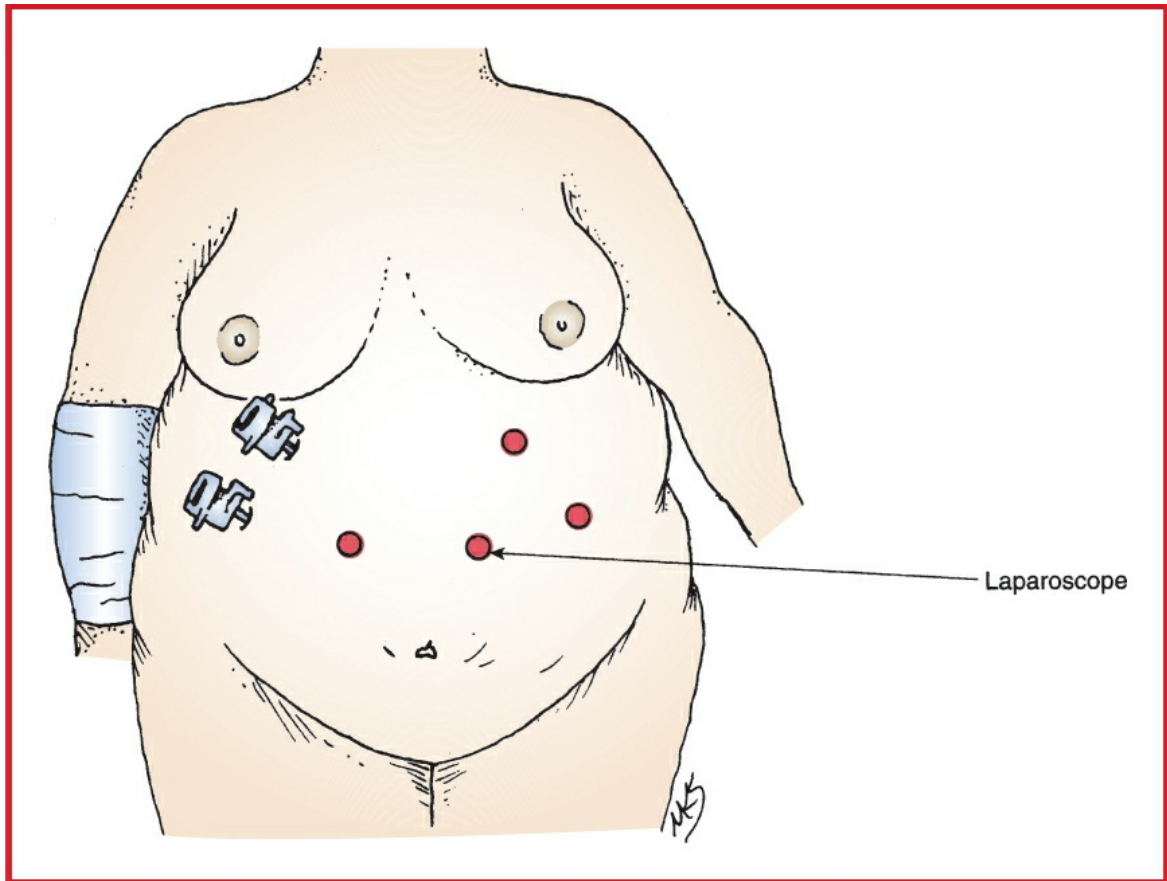


Figure 7.7-10 Patient position and trocar position for bariatric laparoscopic surgery. (Reproduced with permission from Scott-Connors CEH, Dawson DL Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

Variant procedure or approaches

Open bypass

Usual preop diagnosis

Morbid obesity generally in combination with a medical condition(s) felt to be worsened by the obesity (e.g., osteoarthritis, diabetes mellitus, respiratory insufficiency, CHF, hypertension, sleep apnea); BMI > 40 kg/m² or BMI > 35 kg/m² if associated comorbidities present

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine or split leg (see Fig. 7.2-6 in Open Operations)
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	for Morbid Obesity for suggested positioning)
Incision	Five to six port sites
Unique considerations	Patients are at high risk for aspiration, and airway management may be difficult. Patients are at high risk for thromboembolic events and VTE prophylaxis is critical. Patients are at risk for postop opioid-induced respiratory depression.
Antibiotics	Cefoxitin 3 g iv (or 2 g iv if total body weight <120 kg)
Surgical time	2–4 h
Closing considerations	Typically, fascial defects are not closed.
EBL	<50 mL
Postop care	Most patients receive an upper GI study the next day. Clear liquids are begun if the study is normal, and most patients are discharged within 48 h.
Mortality	<1%
Morbidity	Overall: 10% Stricture: 10% Leak: 1–2% VTE Pneumonia Hemorrhage Hernia Obstruction Infection
Pain score	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Generally, between 18 and 60 yr
Male:Female	Female > male
Incidence	5%

Etiology

Genetic; environmental

Associated conditions

Sleep apnea; HTN; diabetes; GERD; osteoarthritis

 ANESTHETIC CONSIDERATIONS **PREOPERATIVE**

Obesity is associated with many chronic medical problems. Morbidity and mortality rise sharply with increasing age and BMI. Obesity and length of exposure to obesity increase the risk of hospital admission and lengthen hospital stay. Medical comorbidities must be optimized before elective bariatric surgery. Evaluate any patient who has had previous bariatric surgery for metabolic changes that can include protein, vitamin, iron, and calcium deficiencies.

Review a list of all current medications the patient is taking, including nonprescription appetite suppressors and diet drugs. Many of these drugs can have important side effects. For example, the combination of phentermine and fenfluramine (“phen-fen”), which is no longer prescribed in the United States, is associated with persistent, serious, heart, and lung problems. Another weight loss medication, sibutramine, works in the brain by inhibiting the reuptake of norepinephrine, serotonin, and dopamine, producing a feeling of “anorexia,” which limits food intake. Sibutramine has been implicated as a cause of dysrhythmias and hypertension. Orlistat blocks digestion and absorption of dietary fat by binding lipases in the gastrointestinal tract and can cause deficiencies in fat-soluble vitamins (A, D, E, K). A reduction in vitamin K levels can increase the anticoagulation effects of warfarin.

Respiratory

Adipose tissue is metabolically active. Oxygen consumption and CO₂ production rise with increasing weight 2° ↑ metabolic demands. The work of breathing is increased, while respiratory muscle performance is impaired. Mass loading of the thoracic and abdominal walls causes abnormalities in both lung volume and gas exchange, especially when the patient is supine. The increased total respiratory resistance and decreased compliance associated with extreme obesity result in shallow, rapid breathing. Decreased FRC 2° ↓ ERV → ↓ TLC. Airways close during normal ventilation → ↑ pulmonary shunt →

hypoxemia. These changes increase in direct proportion with increasing BMI. General anesthesia and the supine position further reduce FRC. Preop PFTs show a restrictive breathing pattern. For symptomatic patients, an ABG can be obtained while the patient breathes room air to establish a baseline. However, this is rarely necessary. Younger obese patients have an increased ventilatory response to hypoxia and a relatively decreased response to hypercapnia. Their arterial blood sample often demonstrates alveolar hyperventilation (PaCO_2 30–35 mm Hg) and relative hypoxemia (PaO_2 70–90 mm Hg) while breathing air. With increasing age, sensitivity to CO_2 decreases, so that PaCO_2 rises and PaO_2 falls further.

Preop assessment of a patient's face, neck, and upper airway is required because mask ventilation and tracheal intubation can be a challenge in some obese patients. A large neck circumference and high Mallampati score may predict difficult intubation. A review of the patient's previous anesthetic records will reveal whether airway problems had been encountered during previous surgical procedures. A thorough preparation requires that multiple laryngoscopy blades and sizes, a gum elastic bougie, and laryngeal mask airway are readily available before laryngoscopy. Video laryngoscopes and fiberoptic intubation are increasingly used for known or anticipated problematic laryngoscopy.

Tests: As indicated from H&P

Cardiovascular

Cardiac output rises proportionally with increased weight. Lean body weight, which increases with increasing obesity, is significantly related to cardiac output. $\uparrow \text{CO}$ and normal PVR $\rightarrow \uparrow \text{BP}$. Mild-to-moderate hypertension is seen in most morbidly obese patients. The increased left ventricular wall stress caused by increased stroke volume and the resultant ventricular dilation leads to cardiac

hypertrophy. Left ventricular dysfunction is often present in young, asymptomatic patients. Even normotensive patients have increased preload and afterload, $\uparrow$ PAP (which may be exacerbated with hypercarbia and hypoxia) and elevated right and left ventricular stroke work. Because these patients are often not physically active, they may appear to be asymptomatic even in the presence of significant cardiovascular disease. Signs of pulmonary HTN (exertional dyspnea, fatigue, syncope) should be sought and TEE obtained in symptomatic patients. RV failure is common in older patients. A medical consultation with a cardiologist may be indicated before bariatric surgery. The ECG may show increased rate, changes in QRS voltage, left QRS axis shift, slowed conduction, and evidence of ischemia or previous myocardial infarction. Polycythemia suggests chronic hypoxemia.

Tests: ECG and others as indicated from H&P

Obstructive sleep apnea (OSA) syndrome

Many obese patients maintain a normal PaCO_2 during the day, but have CO_2 retention, sleep disturbances, intermittent airway obstruction with hypoxemia, pulmonary hypertension, and cardiac dysrhythmias at night. OSA syndrome is characterized by frequent episodes of apnea (>10 -sec cessation of airflow, despite continuous respiratory effort against a closed airway) and hypopnea (50% reduction in airflow or reduction associated with a decrease of $\text{SpO}_2 > 4\%$). OSA is frequently undiagnosed in patients scheduled for bariatric surgery. OSA occurs more often in patients with large necks and high Mallampati (III and IV) scores. The patient may not be aware of symptoms, so it is important to interview the patient's sleeping partner. If OSA is present, the partner will describe loud snoring followed by silence as airflow ceases with obstruction and then gasping or choking as the patient awakes and airflow restarts. The STOP-BANG questionnaire is a useful

screening tool for OSA. A recent study found that a STOP-BANG score of 5–8 identified patients with a high probability of moderate/severe OSA. However, a definitive diagnosis of OSA can only be confirmed by polysomnography. Because of fragmented sleep patterns, OSA patients may complain of daytime sleepiness and headaches. Chronic sleep apnea leads to secondary polycythemia, hypoxemia, and hypercapnia, all of which increase the risk of cardiac and cerebral vascular disease. Patients with a history of snoring or OSA are often difficult to ventilate by mask, and their tracheas may be more difficult to intubate than those of similar weight patients without OSA. OSA patients who use CPAP devices at home should be instructed to bring them to the hospital to use following surgery. A patient known or even suspected of having OSA should be continuously monitored by pulse oximetry in the postop period, even following a completely uneventful operation. Patients are at highest risk for hypoxic respiratory failure 72 h after tracheal extubation.

**Obesity
hypoventilation
syndrome**

A small number of patients have obesity hypoventilation syndrome (OHS), characterized by somnolence, cardiac enlargement, polycythemia, hypoxemia, and hypercapnia. OHS patients tend to be older, are super obese ($\text{BMI} > 50 \text{ kg/m}^2$), and have more restricted pulmonary function than other patients with OSA. Hypoventilation that is central and independent of intrinsic lung disease is probably caused by the respiratory center's progressive desensitization to hypercapnia from nocturnal sleep disturbances. In the most severe form of OHS, the "Pickwickian syndrome," hypersomnolence, hypoxia, hypercapnia, pulmonary hypertension, right ventricular enlargement, and hypervolemia occur. These patients rely on a hypoxic ventilatory drive

and may hypoventilate or even become apneic following emergence from general anesthesia after being given 100% O₂ to breathe.

Gastrointestinal

It is widely believed that morbidly obese patients are at greater risk for acid aspiration during induction of general anesthesia. Recently, this belief has been challenged. One study reported that fasting obese patients actually had a lower incidence of high-volume, low-pH gastric fluid than lean patients, whereas another found no differences in gastric volume or pH between lean and moderately obese surgical patients. Obese patients without GERD symptoms have relatively normal gastroesophageal sphincter tone. Obese patients at special risk for gastric acid aspiration may be those with diabetes and gastroparesis. Nonalcoholic steatohepatitis (NASH, "fatty hepatitis"), with or without liver dysfunction, is extremely common. Histologic abnormalities are present in the livers of as many as 90% of morbidly obese patients. Preop liver function tests should be obtained, but they often do not reflect the actual severity of liver dysfunction. Alanine aminotransferase (ALT) is the most frequently elevated liver enzyme. Hepatic metabolism of many anesthetic agents is usually not clinically altered with NASH.

Renal

There is ↑ RBF and ↑ GFR associated with obesity. Renal clearance of drugs may be greater than in the normal weight patient. The most common renal abnormality seen is proteinuria.

Tests: As indicated from H&P

Premedication

Sedative drugs should be avoided whenever possible. For the very anxious patient, small amounts of midazolam (1–2 mg, iv) can be given. If a fiberoptic airway intubation is planned, atropine or glycopyrrolate will decrease oral secretions. Most medications for chronic HTN are continued before

surgery. An exception are the angiotensin-converting enzyme (ACE) inhibitor and angiotensin II receptor blocker (ARB) classes of antihypertensives, which should be stopped preop because they can cause ↓↓ BP following induction of anesthesia. Diabetic medications (insulin, oral hypoglycemics) are usually withheld on the morning of surgery, but blood sugar levels must be closely monitored in the perioperative period. Antibiotics and heparin (VTE prophylaxis) are usually administered before surgery at the surgeon's request. For protection against acid aspiration an H₂-receptor antagonist can be given the night before and again on the morning of surgery.

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

■ INTRAOPERATIVE

Anesthetic technique

Despite conflicting evidence that morbidly obese patients are at greater risk for acid aspiration, it remains prudent to establish a secure airway as quickly and safely as possible. Patients cannot tolerate the supine position and should be preoxygenated in the reverse Trendelenburg position (RTP) until SPO₂ is 100% for several min. With apnea, expect rapid desaturation because FRC is reduced and O₂ reserves are limited. Preoxygenation in the RTP can increase O₂ reserves, but may cause pooling of blood and → ↓ BP. The increase in adipose tissue seen in obese patients increases apparent volume of distribution of lipophilic anesthetic agents. However, drug distribution is dependent on cardiac output, which is strongly related to lean body mass. Administration of intravenous induction agents and opioids should be based on lean body weight (LBW) to avoid drug overdose.

Tracheal intubation is necessary for controlled ventilation and airway protection. High Mallampati score and large neck circumference are the most reliable predictors of potential intubation difficulties. If a problem is anticipated preop, an “awake intubation” with a fiberoptic bronchoscope or video laryngoscope is recommended.

Appropriate nerve blocks and topical anesthesia to the airway are applied, and sedative drugs are kept to a minimum. It is important that the patient breathes supplemental O₂ during the intubation procedure. Successful direct laryngoscopy requires proper patient position. The patient must be placed with the head, upper body, and shoulders significantly elevated (“stacked” or “ramped”) so that the ear is level with the sternum (head-elevated laryngoscopic position, HELP; see [Fig. 7.2-6](#)). When a morbidly obese patient is in this position, the endoscopist’s view during direct laryngoscopy is significantly improved.

Induction	For most patients, a rapid-sequence induction with propofol (2–2.5 mg/kg LBW) and succinylcholine (1 mg/kg TBW), combined with cricoid pressure (~10 lb), is the best means for securing the airway. Bag and mask ventilation is often difficult 2° upper airway obstruction and ↓ pulmonary compliance. Gastric insufflation during ineffective mask ventilation will further increase the risk of regurgitation and acid aspiration. A second person experienced with airway management, preferably another anesthesiologist, must always be present to assist when difficulty is anticipated. Aids for difficult intubation, including a short laryngoscope handle, a variety of laryngoscope blades, video laryngoscopes, a gum elastic bougie, and equipment for cricothyroidotomy and transtracheal jet ventilation, should be available. An LMA can serve as a “bridge” until an endotracheal tube is placed when difficulty is encountered.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Patients should be ventilated with a FiO ₂ of 0.5–1.0 and a tidal volume 10–12 mL/kg IBW, preferably in the RTP. Larger tidal volumes will only marginally improve oxygenation while producing hypocapnia and potentially causing lung trauma. PEEP

superimposed on a large tidal volume can actually worsen hypoxemia by $\downarrow$ CO, which in turn will $\downarrow$ O₂ delivery. Placement of subdiaphragmatic packs or retractors or changing to lithotomy or Trendelenburg positions will also impair ventilation.

The patient is ventilated with an inhalational anesthetic (isoflurane, sevoflurane, or desflurane) with a FiO₂ of 50–100% O₂. All inhalational anesthetics, including isoflurane, are rapidly eliminated. With appropriate timing, there are no clinical differences in the recovery time after general anesthesia with any inhalational anesthetic agent. N₂O can be used because it does not dilate the bowel during laparoscopic bariatric surgery, but its role is limited due to the high oxygen demand of many patients. For analgesia, incremental boluses of fentanyl (50–100 mcg) or sufentanil (5–10 mcg) or a remifentanyl infusion (0.05–0.2 mcg/kg/min) are commonly used. Fentanyl and sufentanil infusions have also been used successfully in this patient population. Dexmedetomidine may be used as a supplemental infusion as it may minimize opioid requirements. Morphine and hydromorphone, which have prolonged onset and duration of action, should be avoided.

The anesthesiologist is usually responsible for proper placement of the gastric tube to decompress the stomach and to help size the gastric pouch. The anesthesiologist may be asked to help perform leak tests for anastomotic integrity, either by insufflation of the gastric tube or placement of saline or dye down the tube. It is extremely important that the gastric tube be completely withdrawn before the gastric pouch is stapled. Esophageal or nasopharyngeal temperature probes are not recommended as they may be inadvertently stapled if not completely withdrawn. Heat loss may be exaggerated by the CO₂ pneumoperitoneum and by

cold irrigating fluids. Warming blankets and other devices should be employed intraop, and warmed iv and irrigating fluids are occasionally needed if there is a significant drop in temperature.

Loss of pneumoperitoneum may indicate incomplete paralysis. Because nondepolarizing muscle relaxants are hydrophilic, there is limited distribution to adipose tissue and no clinical advantage between any of the commonly used agents. Neuromuscular recovery time is similar in obese and nonobese patients with atracurium, vecuronium, or rocuronium. Relaxants should be administered in incremental doses based on IBW, and neuromuscular blockade must be completely reversed before extubation of the trachea. This can be done with either neostigmine/glycopyrrolate or sugammadex. If sugammadex is given to a woman of childbearing potential, postoperative counseling regarding the interaction of sugammadex and oral contraceptives should be given.

Emergence

The inhalation agent is discontinued several min before surgery is completed. Once the patient is spontaneously ventilating, fentanyl may be titrated according to respiratory rate. If a remifentanyl infusion is being used, it should be turned off just before extubation to allow spontaneous ventilation. The patient should be extubated when fully awake. In most cases, a nasal trumpet may be inserted to minimize upper airway obstruction after extubation. If hemodynamically stable, the patient's airway should be extubated with the upper body elevated 30–45°. Then, the patient should be transferred from the OR in that position. Postop admission to an ICU and mechanical ventilation is rarely needed. Factors that may necessitate ventilatory support include extremes of age, super obesity, coexisting cardiac disease or pulmonary disease and CO₂ retention, fever or infection, an uncooperative or extremely anxious patient, or incomplete reversal of

	neuromuscular blockade. The need for postop admission to an ICU is relatively common after open bariatric procedures, but rare after laparoscopic surgery. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 × 1 BSS @ 8–12 mL/kg/h Fluid warmer	Intraop fluid requirements are usually greater than would be anticipated in a normal weight patient. 2–3 L of crystalloid should be given during a laparoscopic bariatric operation. It is essential that adequate amounts of iv fluid be given to reduce postop renal failure and to avoid other rare but serious complications such as rhabdomyolysis.
Monitoring	±CVP Standard monitors	Noninvasive cuff pressure may be inaccurate if the anatomy of the upper arm does not allow a proper fit. Cuff pressures can be obtained from the forearm or ankle. A central line should only be considered if reliable peripheral venous access cannot be obtained. Note that the length of a standard iv catheter placed percutaneously in the neck of a very large patient may not be long enough to reach an intrathoracic location.

Positioning	RTP ✓ and pad pressure points ✓ eyes SCDs	In the supine position, FRC is ↓↓ → V/Q mismatch + ↑ O ₂ consumption, ↑ CO, and ↑ PAP. A left lateral tilt will prevent inferior vena cava compression in the supine patient. The Trendelenburg and lithotomy positions further decrease lung volumes. If possible, the patient should always be in the reverse Trendelenburg position (RTP) during surgery because in this position, the diaphragm is “unloaded” and FRC is maximized.
CO₂ pneumoperitoneum	↑ PIP + ↓ TV ↑ ETCO ₂ + PaCO ₂ ↑ HR + ↑ MAP	The physiologic effects of the CO ₂ pneumoperitoneum are usually well tolerated by the patient and require no intervention. Maintaining the patient in the RTP minimizes the restriction of respiratory mechanics from the CO ₂ insufflation. All changes return to normal once the pneumoperitoneum is relieved. However, complications can occur.
Complications	Hypoxemia Massive gas embolism Pneumothorax Pneumomediastinum	The pneumoperitoneum can displace the diaphragm cephalad, causing the position of the

endotracheal tube to change. Occasionally, the endotracheal tube's tip can enter a bronchus, so always consider tube displacement in the differential diagnosis of hypoxemia during laparoscopy.

▼ POSTOPERATIVE

Position and oxygenation	Semirecumbent and RTP	Maximize oxygenation by allowing the diaphragm to fall and FRC to increase. Patients can become hypoxemic if supplemental O ₂ is withheld in the immediate recovery period. Restoration of normal pulmonary function after open abdominal surgery may take several days.
	CPAP/BiPAP	In theory, CPAP could distend the gastric pouch, but its use following bariatric surgery has not been associated with anastomotic leaks. Patients who use nocturnal CPAP/BiPAP should be encouraged to bring their equipment the day of surgery for use during the postoperative period.
Complications	Venous thromboembolism	VTE prophylaxis should always be considered in the postop period, even for patients with epidural

		catheters. A vena cava umbrella is occasionally placed preop in older and high-risk patients (BMI > 60 kg/m², history of DVT/hypercoagulable state). Early ambulation significantly minimizes the risk of VTE and must be encouraged.
Complications (cont'd)	PONV	Multimodal intraop prophylaxis with several antiemetic agents will reduce but not eliminate PONV. Dexamethasone (4–8 mg) should be part of the therapeutic regimen.
Multimodal pain management	PCA	PCA with opioid dose based on LBW is usually satisfactory. Avoid morphine and hydromorphone as their prolonged onset and duration of action increase the risk of postoperative opioid-induced respiratory depression. The use of nonopioid analgesic adjuncts should be instituted early. Dexmedetomidine, which has no respiratory depressant effects, is a useful alternative or supplement to opioids. IV acetaminophen can be a useful adjunct in patients with minimal liver disease. Nonsteroidal anti-inflammatory drugs are helpful initially, but should be discontinued within a day or two to avoid the potential

		complication of gastric ulceration. Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see p. C-2).
ERAS	PONV protocol (see p. B-1)	Encourage early enteral nutrition and mobilization. Consider the need for DVT prophylaxis.
Tests	As indicated by patient status	

Suggested Readings

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ANESTHESIA FOR LAPAROSCOPY IN PREGNANCY

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Approximately 0.75–2% of pregnant woman require nonobstetric surgery, accounting for more than 80,000 anesthetics for nonobstetric surgery per year, and 10–20% of those will require intra-abdominal surgery (diagnostic or therapeutic), most commonly for appendectomy, cholecystectomy, trauma, or gynecologic indications. If surgery cannot be avoided, it is best carried out in the second trimester. Warn patients about possible fetal loss (3–12% in first trimester) and premature labor (5–8% in second and 25–40% in third trimesters). Anesthesia and surgery are associated with increased spontaneous abortion, growth retardation, and perinatal mortality; however, no increase in congenital abnormalities has been found. Rates of fetal loss, premature labor, and maternal mortality are higher among sicker patients. It is unclear whether adverse outcomes after surgery relate to the disease process itself, disturbances in nutrition, the surgical procedure, exposure to radiation, or drugs. No correlation has been found between outcome and any specific anesthetic technique or agent (including N₂O).

Laparoscopy is the procedure of choice for many surgeries in nonpregnant patients. Historically, its use in pregnancy was controversial, although significant experience now supports its safety and efficacy. Advantages of laparoscopy include more rapid recovery, shorter hospital stays, less postop narcotic use (→ ↓ fetal depression), lower risk of wound infection, less postoperative ileus, and better cosmesis. No differences in fetal outcome or incidence of preterm labor have been found when compared to open procedures. However, regardless of the method of abdominal access, the laparoscopic or open appendectomy and cholecystectomy population is at high risk (37–41%) for perinatal complications. Risks of laparoscopy include difficult surgical access and potential uterine injury with Veress needles or trocars. Current recommendations suggest that either an open technique (e.g., Hasson cannula) or Veress needle/optical trocar is appropriate for obtaining laparoscope access with adjustments for fundal

height.

Note that acute appendicitis and cholecystitis often present with advanced or complicated disease because of difficulty diagnosing the “acute abdomen” and a reluctance to use radiation-based diagnostic tests in pregnancy.

Respiratory	Pregnant patients have $\uparrow$ MV with respiratory alkalosis ($\text{PaCO}_2 = 32\text{--}34$ mm Hg) and are subject to the rapid onset of hypoxemia if ventilation is compromised due to alterations in lung and metabolic functions, including a 20% $\downarrow$ in FRC and a 20% $\uparrow$ in O_2 consumption. Avoid hyperventilation ($\text{PaCO}_2 < 30$), which causes decreased uterine artery blood flow. Uptake of and sensitivity to inhalational anesthetics are enhanced ($\downarrow$ FRC, $\downarrow$ MAC, and hyperventilation). During laparoscopy, peritoneal insufflation and head-down positioning may further compromise lung function ($\downarrow$ FRC and $\downarrow$ lung compliance). Mucosal capillary engorgement in upper airways may necessitate a smaller ETT and mandates careful airway suctioning to avoid bleeding. Airway management is more difficult in this patient population. Tests: As indicated from H&P
Cardiovascular	Pregnancy $\rightarrow \uparrow$ CO, $\downarrow$ MAP, $\uparrow$ HR, $\uparrow$ blood volume, and $\downarrow$ oncotic pressure $\rightarrow \uparrow$ risk of pulmonary edema. Supine hypotensive syndrome: To minimize aortocaval compression and $\rightarrow \downarrow \downarrow$ BP (>20-wk gestation), the supine position should be modified by the use of a left lateral pelvic tilt to displace the uterus. Moderate abdominal insufflation pressures (10–15 mm Hg) should be used to minimize further caval compression and $\downarrow$ uterine blood flow. Tests: As indicated from H&P
Hematologic	WBC count is elevated during pregnancy (8000–12,000/mm) and may delay the diagnosis of concurrent infections (e.g., appendicitis). Iron deficiency anemia often is superimposed on the

	dilutional anemia of pregnancy (Hct 33%). Pregnant patients are hypercoagulable, and antiembolic compression devices should be used. Tests: Hgb/Hct. T&S if significant blood loss is anticipated. Coag studies and Plt count if + PIH
Gastrointestinal	Emergency surgery, ↑ GERD, ↑ intragastric pressure, ↓ lower esophageal sphincter tone, and gastric hyperacidity → ↑ risk of aspiration pneumonitis. All pregnant patients (>16–20 wk) should be considered to have full stomachs and should receive a nonparticulate antacid (e.g., 0.3 M Na citrate 30 mL) immediately before GA, as well as iv metoclopramide 10 mg and ranitidine 50 mg 30–60 min before surgery.
Renal	Pregnancy → ↑ renal blood flow and ↑ creatinine, clearance and ↓ serum creatinine and ↓ BUN. Dependent edema results from increased water and sodium retention and ↓ oncotic pressure. Tests: As indicated from H&P
Laboratory	Others tests as indicated from H&P
Premedication	Full-stomach precautions (see Gastrointestinal, above); midazolam (1–2 mg) given as appropriate to decrease anxiety
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

After 16-wk gestation, anticipate ↑ risks of aspiration and difficult intubation. Plan ahead for management of a potentially difficult airway. GA is the preferred technique because laparoscopy and abdominal surgery are poorly tolerated under regional anesthesia. A lead shield should be used to protect the fetus during fluoroscopy.

Fetal monitoring

The primary obstetric care provider should be notified, and a qualified person should be readily available to interpret the fetal heart rate (FHR) and perform cesarean delivery if indicated. If the fetus is considered to be viable (~24-wk gestation), FHR and contraction monitoring should be performed at a minimum before and after the procedure to confirm fetal well-being and the absence of uterine contractions. If feasible, continuous intraoperative monitoring of FHR during surgery (electronic FNF monitoring, transvaginal Doppler) is recommended. Each case requires a team approach and a management plan for evaluation and action if a nonreassuring trace develops. Surgery should only be done at an institute with adequate facilities and neonatal services.

General anesthesia

If difficult intubation is anticipated, an awake fiberoptic intubation or video laryngoscopy is recommended. In any event, emergency airway management equipment (e.g., LMA, transtracheal jet ventilator) must be immediately accessible in the OR. Communication with the surgeon and obstetrician regarding maternal and fetal condition is essential.

Induction	Left uterine displacement is best achieved by elevating the right hip with a wedge, towel, or equivalent device. Standard rapid-sequence induction with cricoid pressure as appropriate. To optimize intubation, place patient in maximal “sniff” position and consider ramping obese patients (Fig. 7.2-6). If tracheal intubation is unsuccessful, follow the difficult intubation/ventilation drill. After ETT is secured, pass OG tube and decompress the stomach to minimize injury from the Veress needle or trocar.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	0.8–1% isoflurane or 1.5–2.0% sevoflurane in air/O ₂ (50%) mixture. Avoid N ₂ O → distension of the bowel and PONV. Administer an opioid (e.g., fentanyl 50–200 mcg iv) or midazolam (1–2 mg) to ↓ volatile requirements and ↓ maternal awareness. Administer muscle relaxants

	(e.g., vecuronium 0.1 mg/kg). Control ventilation, avoiding hypocapnia ($\text{PCO}_2 < 30$ mm Hg) $\rightarrow$ $\downarrow$ uterine blood flow, and hypercapnia ($\text{PCO}_2 > 36$ mm Hg) $\rightarrow$ fetal acidosis. Maintain ETCO_2 or PaCO_2 at 32–36 mm Hg by $\uparrow$ MV. Use low insufflation cutoff pressure (8–12 mm Hg, not 15 mm Hg as for nonpregnant patient). Maintain adequate placental perfusion by keeping maternal blood pressure at preoperative baseline levels. Fluids and vasopressors (phenylephrine, ephedrine) should be utilized as indicated.	
Emergence	Reverse muscle relaxation with neostigmine 0.05 mg/kg and glycopyrrolate 0.01 mg/kg or atropine 0.02 mg/kg. Anticholinergics (glycopyrrolate/atropine) given before neostigmine may minimize possible $\uparrow$ uterine tone 2° neostigmine. Delay extubation until patient is fully awake and muscle strength has returned to normal. O_2 by mask and transport in the lateral position. Consider metoclopramide (10 mg iv) or ondansetron (4 mg iv) PONV prevention. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga $\times$ 1 BSS @ 3–5 mL/kg/h	Laparoscopy is associated with minimal blood loss; however, inadvertent vascular injury may occur. See Zero Balance Fluid Management, p. K-2.
Monitoring	Standard monitors $\pm$ Arterial line (major surgery) FHR monitor (see above) Uterine contraction monitor	Maternal and neonatal outcome good, and severe acidosis absent when maternal ventilation controlled by PaCO_2 or ETCO_2 measurement. Clinical experience indicates no long-term adverse neonatal effects.
Positioning	✓ and pad pressure	Changes in position have

	points ✓ eyes ±Left uterine displacement	marked effects on respiration (↓ compliance, ↑ pressure, hypoxemia) and hemodynamics. Head-up position + GA + peritoneal insufflation → 50% ↓ CO in the nonpregnant patient and even greater decrease in late pregnancy. Position changes should be gradual to minimize adverse effects. If >20-wk gestation, left uterine displacement to avoid aortocaval compression
CO₂ peritoneum: Maternal effects	↓ ventilation ↓ BP ↑ GI reflux (↑ abdominal pressure) CO₂ absorption	Maternal hypoxemia and respiratory acidosis ↓ venous return → ↓ BP. → ↑ MAP and ↑ SVR
CO₂ peritoneum: Fetal effects	↓ uterine blood flow ↑ preterm labor Fetal acidosis Fetal ↑ HR/↑ BP	Up to 60% (in animal studies) → fetal hypoxia. 2° ↑ abdominal pressure 2° absorbed CO₂
Complications	Uterine injury Bowel/organ injury Venous air embolism Thromboembolism/PE Pneumothorax	Dx: ↓ ETCO₂, hypoxemia, hypotension. Rx: 100% O₂, stop insufflation immediately, volume and pressors, attempt CVP air aspiration. Prevention: Pneumatic compression devices, ± thromboprophylaxis (e.g., heparin, LMWH) Rx: Supportive

Dx: ↑ airway pressure, ↓ breath sounds, and hyperresonance. Rx: 100% O₂, stop insufflation, tube thoracostomy

▼ POSTOPERATIVE

Multimodal pain management	Standard pain management	Avoid NSAIDs (e.g., ibuprofen) → pulmonary and CV fetal effects (closure of ductus venosus). Consider intravenous acetaminophen 1000 mg. Pre-emptive or rescue transversus abdominis plane blocks may be appropriate in select patients. Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see p. K-5).
ERAS	PONV protocol (see p. B-1)	Encourage early enteral nutrition and mobilization. Consider the need for DVT prophylaxis.
Tests	Hct/CBC Others as indicated by operative course	

Suggested Readings

1. Abeles A, Kwasnicki RM, Darzi A: Enhanced recovery after surgery: current research insights and future direction. *World J Gastrointest Surg* 2017; 9(2):37–45.
2. Cheek TG, Baird E: Anesthesia for nonobstetric surgery: maternal and fetal considerations. *Clin Obstet Gynecol* 2009; 52:535–45.
3. COG Committee Opinion Number 696, April 2017; Nonobstetric Surgery During Pregnancy. <https://www.acog.org/Clinical-Guidance-and-Publications/Committee-Opinions/Committee-on-Obstetric-Practice/Nonobstetric-Surgery-During-Pregnancy>

4. Cohen-Kerem R, Railton C, Oren D, et al: Pregnancy outcome following non obstetric surgical intervention. *Am J Surg* 2005; 190:467–73.
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6. Erekson E, Brousseau E, Dick-Biascoechea M, et al: Maternal postoperative complications after nonobstetric antenatal surgery. *J Matern Fetal Neonatal Med* 2012; 25(12):2639–44.
7. Kim AJ, Yong RJ, Urman RD: The role of transversus abdominis plane blocks in enhanced recovery after surgery pathways for open and laparoscopic colorectal surgery. *J Laparoendosc Adv Surg Tech A* 2017; 27(9):909–14.
8. Li Z, Zhao Q, Bai B, et al: Enhanced recovery after surgery programs for laparoscopic abdominal surgery: a systematic review and meta-analysis. *World J Surg* 2018; 42(11):3463–73.
9. Pearl J, Price R, Richardson W, Fanelli R: Guidelines for diagnosis, treatment, and use of laparoscopy for surgical problems during pregnancy. *Surg Endosc* 2011; 25:3479–92.
10. Pearl JP, Price RR, Tonkin AE, et al: Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) Guidelines for the Use of Laparoscopy during Pregnancy. May 2017. <https://www.sages.org/publications/guidelines/guidelines-for-diagnosis-treatment-and-use-of-laparoscopy-for-surgical-problems-during-pregnancy/>
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CHAPTER 7.8

Pancreatic Surgery

SURGEON

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ANESTHESIOLOGIST

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OPERATIVE DRAINAGE FOR PANCREATITIS

SURGICAL CONSIDERATIONS

Description

Surgical treatment for pancreatitis is indicated for drainage or debridement of infected necrotic peripancreatic tissue or pancreatic necrosis. Pancreatic necrosis is defined by areas of poor perfusion in the pancreas on CT angiogram. Infected pancreatic necrosis occurs with the presence of gas and/or bacterial invasion. Pancreatic abscesses usually develop in the lesser sac but may spread to the subphrenic spaces or into the pericolic gutters. Fistulization into adjacent organs, particularly the transverse colon and duodenum, may occur. Severe intra-abdominal hemorrhage from erosion into major arteries lying adjacent to the pancreas is uncommon but may occur before, during, or after operative drainage. Intraop, exploration of the peritoneal cavity is performed before opening the lesser sac. Areas lateral to the left and right sides of the colon, as well as the base of the transverse mesocolon and the subhepatic areas, should be palpated to identify fluid or abscess collections. The gastrocolic ligament is then incised to approach the pancreas through the lesser sac. There are different operative approaches, depending on the location of involved tissue and surgeon's preference. Upper midline or transverse abdominal incisions are used most often. Posterior drainage through the bed of the 12th rib, or retroperitoneal lateral approaches, may be used ([Fig. 7.8-1](#)).

VARIANT PROCEDURE OR APPROACHES Minimally invasive approaches to the distal pancreas (laparoscopic pancreatectomy) and the retroperitoneal space are being used for peripancreatic drainage and debridement. In minimally invasive retroperitoneal pancreatectomy (**MIRP**), a large catheter is placed by interventional radiology and used as an avenue to débride necrotic pancreatic tissue. Recently, a laparoscopic opening between the stomach and retroperitoneum has also been used for drainage and debridement in select cases.

Usual preop diagnosis

Severe pancreatitis 2° to gallstones; alcohol; or post ERCP

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting nothing by mouth (NPO) duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, deep venous thrombosis (DVT) prophylaxis, postop pain management options, physiotherapy plan, possibility of NG/jejunal feeding tubes, and urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

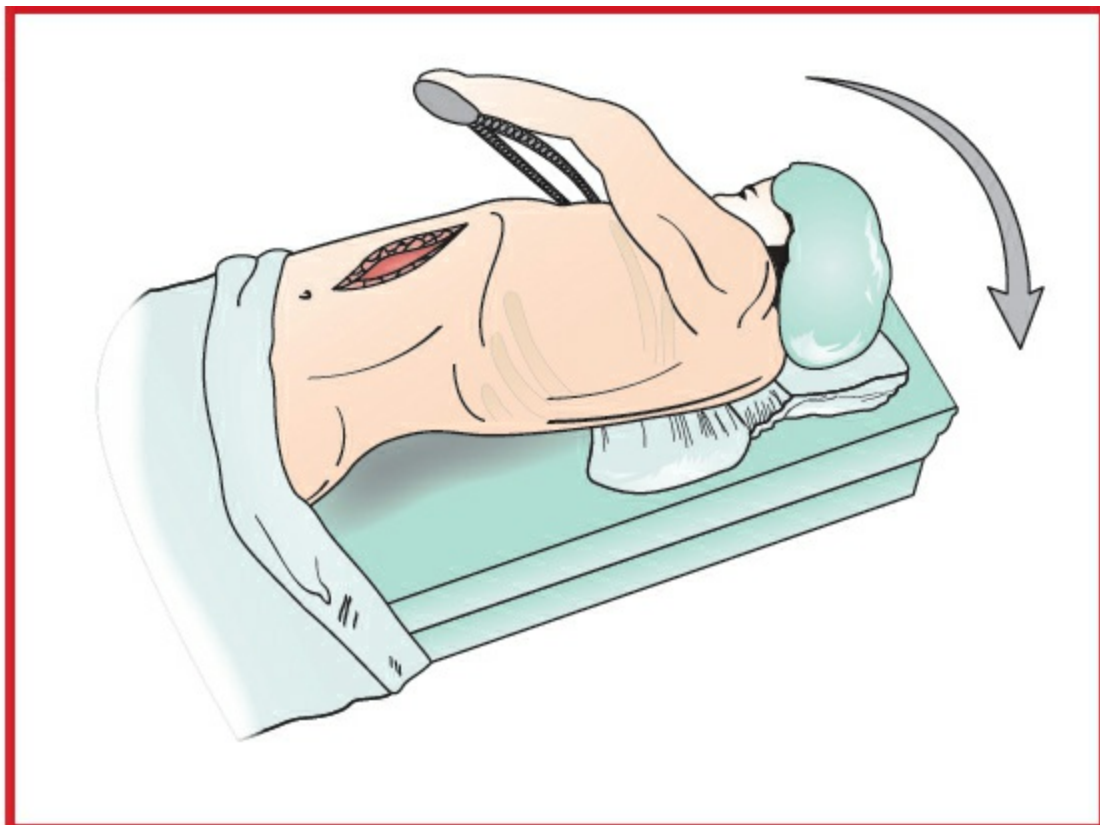


Figure 7.8-1 Incision for anterior and posterior drainage in pancreatitis. Note that bed or table is rotated until patient is almost supine. (Reproduced with permission from Berne TV, Donovan AJ: Synchronous anterior and posterior drainage of pancreatic abscess. Arch Surg 1981; 116:527–33. Copyright © 1981 American Medical Association. All rights reserved.)

■ SUMMARY OF PROCEDURE

Position	Supine or rotated slightly (Fig. 7.8-1) for posterior approach
Incision	Midline, transverse, flank, or synchronous anterior and posterior
Unique considerations	Must perform adequate debridement of necrotic tissue and provide adequate drainage of abdomen; NG tube; jejunal feeding tube
Antibiotics	Cefoxitin 2 g 30 min preop or antibiotic directed at cultured organisms
VTE prophylaxis	Heparin 5000 units sq
Surgical time	1–2 h
Closing considerations	Adequate drainage of pancreatic bed and fluid resuscitation of patient
EBL	300–750 mL
Postop care	Routine wound and drain care; usually ICU and intubated; often, subsequent operative procedures are required.
Mortality	8–30%
Morbidity	Fistulae formation: 18–55% Delayed gastric emptying: 50% Unremitting sepsis: 10–30% Atelectasis: 5–10% Respiratory deterioration: 5% Hemorrhage Bowel perforations
Pain score	7–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	30–60 yr
	1:1

Male:Female	
Incidence	10–30% of patients with pancreatitis
Etiology	Alcoholism (30–50%); postop pancreatitis (15–40%); biliary tract disease (20–30%); idiopathic pancreatitis (15–20%)
Associated conditions	Malnutrition; glucose intolerance; multiorgan dysfunction

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pancreatic Surgery.

Suggested Readings

1. da Costa DW, Boerma D, van Santvoort HC, et al: Staged multidisciplinary step-up management for necrotizing pancreatitis. *Br J Surg* 2014; 101(1):e65–79.
2. Jegatheeswaran S, Puleston JM, Duggan S, et al: The practical management of chronic pancreatitis: a multidisciplinary symposium held at the Annual Meeting of the Pancreatic Society of Great Britain and Ireland, Manchester 2016. *Dig Surg* 2019; 36(1):41–5.
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DRAINAGE OF PANCREATIC PSEUDOCYST

SURGICAL CONSIDERATIONS

Description

Internal **drainage of a pancreatic pseudocyst** may be accomplished by anastomosing the cyst to the stomach, duodenum, or other small bowel via a Roux-en-Y loop of jejunum. The procedure of choice for internal decompression depends on the location of the pseudocyst in relation to the portion of the GI tract that will provide maximal dependent drainage of the cyst. Open, laparoscopic, and endoscopic procedures have

been used with success. The three approaches are equivalent. Operation is reserved for patients with refractory symptoms. At operation, the abdomen is entered via a midline incision or laparoscopically. The pseudocyst is localized by palpation with or without intraop ultrasound. If the pseudocyst lies behind the stomach (or duodenum), it is approached anteriorly, through the posterior wall of the stomach (or duodenum). A portion of the posterior wall is incised, allowing entry into the cyst cavity, which is then drained. An anastomosis is created between the cyst and stomach (or duodenum). The anterior wall of the stomach (or duodenum) is then closed. If the cyst presents inferior to the stomach, it is anastomosed in a similar fashion to a Roux-en-Y loop of jejunum (Fig. 7.8-2). Drains are placed; external drainage is sometimes necessary, especially in the setting of infection. Spontaneous resolution of pancreatic pseudocyst may be expected in most patients with cysts <6 cm. If infection of the pseudocyst occurs with clinical signs of sepsis, percutaneous drainage under CT guidance can be performed, although subsequent operative drainage is often necessary.

Usual preop diagnosis

Pancreatic pseudocyst 2° to acute pancreatitis refractory to medical management

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Midline abdominal or transverse
Unique considerations	Location of pseudocyst in relation to GI tract; NG tube
Antibiotics	Cefoxitin 2 g iv 30 min preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	1–2 h
Closing considerations	Adequate drainage
EBL	100–300 mL

Postop care	± NG decompression
Mortality	5–10%
Morbidity	Bleeding: 5–7% Sepsis: <5% Recurrence: 2–3%
Pain score	6–8

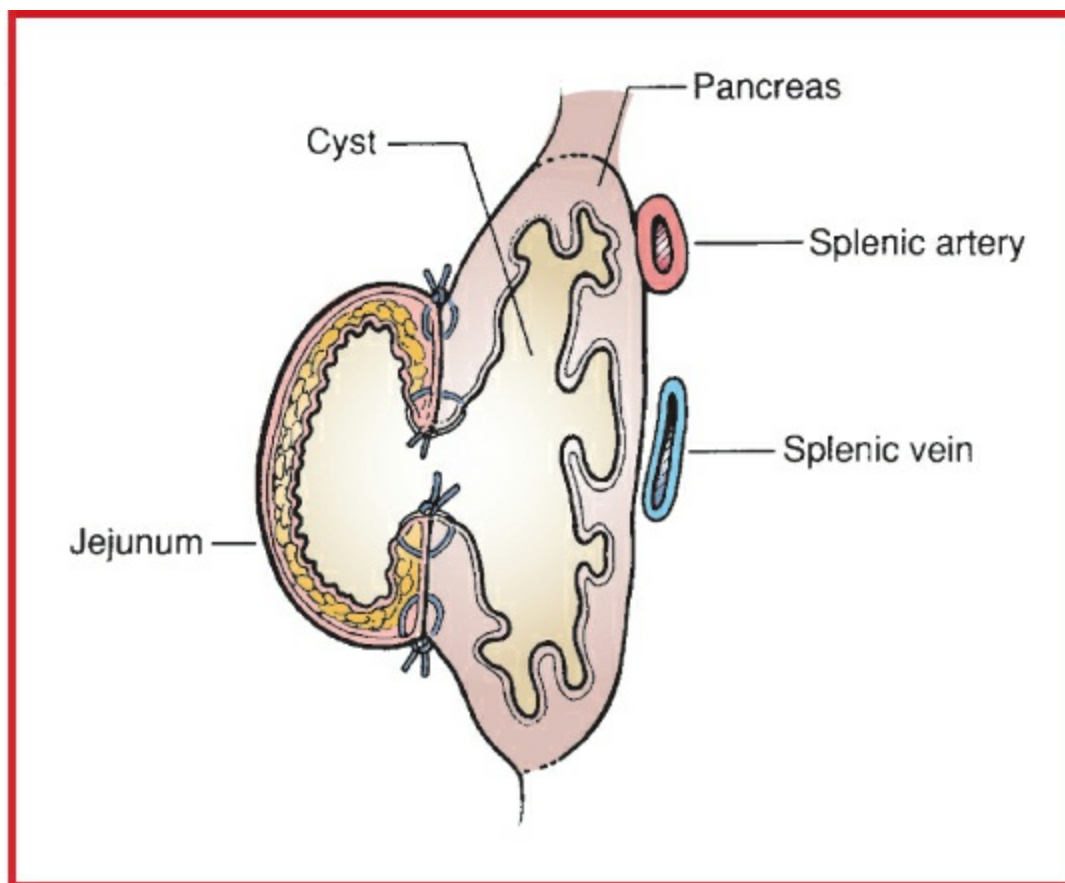


Figure 7.8-2 Roux-en-Y drainage of a pseudocyst. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–80 yr
Male:Female	1:1
Incidence	Rare
Etiology	Acute pancreatitis; trauma
Associated	Acute pancreatitis: 90%; gallstones; alcohol use;

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pancreatic Surgery.

Suggested Readings

1. Andalib I, Dawod E, Kahaleh M: Modern management of pancreatic fluid collections. *J Clin Gastroenterol* 2018; 52(2):97–104.
2. Beck WC, Bhutani MS, Raju GS, et al: Surgical management of late sequelae in survivors of an episode of acute necrotizing pancreatitis. *J Am Coll Surg* 2012; 214(4):682–90.
3. Matsuoka L, Alexopoulos SP: Surgical management of pancreatic pseudocysts. *Gastrointest Endosc Clin N Am* 2018; 28(2):131–41.

LONGITUDINAL PANCREATICOJEJUNOSTOMY

SURGICAL CONSIDERATIONS

Description

Pancreaticojejunostomy, as described by **Puestow**, is typically done for chronic pancreatitis with pancreatic ductal obstruction. It consists of a longitudinal opening of the pancreatic duct, which is then anastomosed to a Roux-en-Y loop of jejunum ([Fig. 7.8-3](#)). This approach is necessary to ensure adequate drainage of a duct with multiple strictures and dilations. Through a midline or transverse abdominal incision, the pancreas is exposed by mobilizing the duodenum (**Kocher maneuver**), exposing the head of the pancreas and opening the lesser sac to visualize the body and tail. The pancreatic duct may be aspirated to identify its location, and intraop ultrasound is commonly used, and then it is incised longitudinally. A Roux-en-Y loop of jejunum is then brought up to the pancreas and anastomosed to the opened duct. A drain is left along the anastomosis, and the wound is closed in the usual fashion.

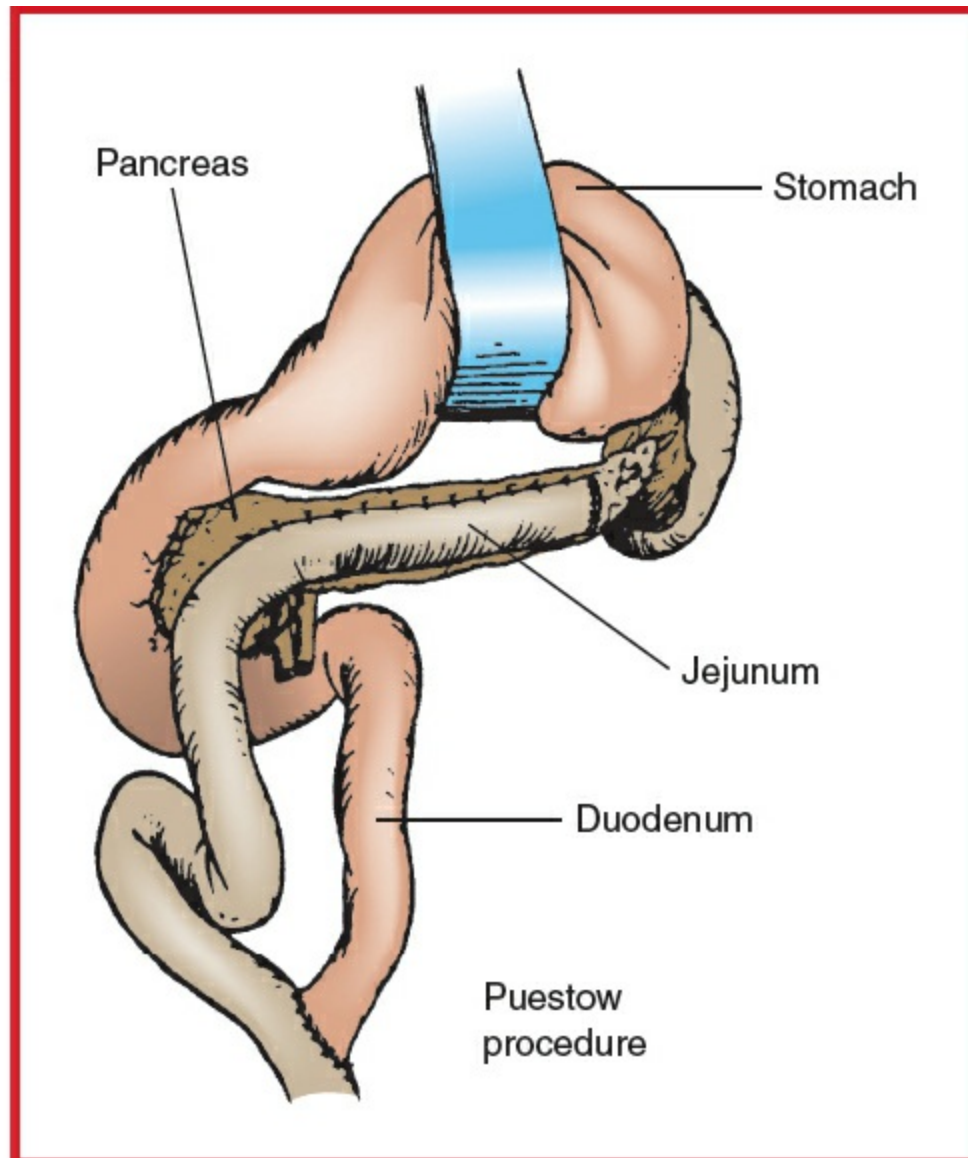


Figure 7.8-3 Operative management of chronic pancreatitis with onlay Roux-en-Y pancreaticojejunostomy (Puestow). (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery. JB Lippincott, Philadelphia: 1988.)

Variant procedure or approaches

A **Whipple resection (pancreaticoduodenectomy)** is an alternative surgical treatment for chronic pancreatitis confined to the head of the gland. Rarely, subtotal or near-total pancreatectomy is indicated. **Fry procedure** is near-total pancreatectomy with Roux-en-Y drainage of the pancreatic head, and **Beger procedure** is duodenum-preserving pancreatic head resection.

Usual preop diagnosis

Abdominal pain with chronic pancreatitis and dilated pancreatic duct (chain of lakes)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting

NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Midline abdominal or transverse
Antibiotics	Cefoxitin 2 g iv
VTE prophylaxis	Heparin 5000 units sq
Surgical time	2–3 h
Closing considerations	Adequate drainage
EBL	300–400 mL
Postop care	Monitor for glucose intolerance
Mortality	1–4%
Morbidity	Failure to relieve pain: 25–50% Pancreatic leak: 5–10% Wound infection: 5%
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	17–72 yr
Male:Female	2.5:1
Etiology	Alcoholism; biliary tract disease; idiopathic; trauma; familial pancreatitis
Associated conditions	Biliary tract disease (25–50%); hyperparathyroidism (<5%); chronic pain; narcotic dependency

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pancreatic Surgery.

Suggested Readings

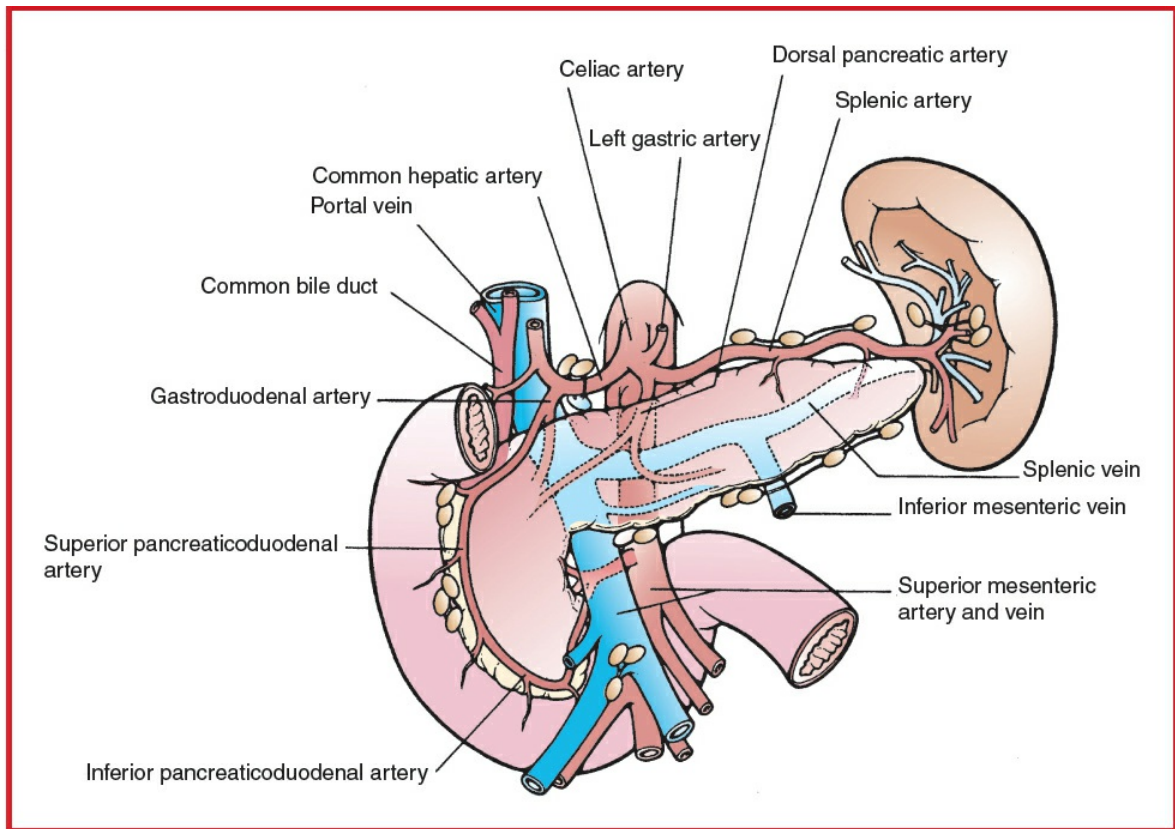
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2. Klaiber U, Alldinger I, Probst P, et al: Duodenum-preserving pancreatic head resection: 10-year follow-up of a randomized controlled trial comparing the Beger procedure with the Berne modification. *Surgery* 2016; 160(1):127–35.
3. Nealon WH, Aatin S: Analysis of success in preventing recurrent acute exacerbations in chronic pancreatitis. *Ann Surg* 2001; 233:793–800.
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PANCREATECTOMY

SURGICAL CONSIDERATIONS

Description

Distal pancreatectomy is performed for tumors in the distal half of the pancreas ([Fig. 7.8-4](#)). It can be done by open or laparoscopic techniques. The lesser sac is opened by dividing the gastrocolic ligament. After entering the lesser sac, the gastrosplenic ligament is divided, ligating the short gastric vessels and the left gastroepiploic vessel. If splenic preserving, the short gastric vessels are not divided. The peritoneum is incised along the inferior surface of the pancreas, with care being taken to avoid injury to the middle colic vessels. Following mobilization of the spleen, the splenic artery is ligated where the pancreas is to be divided. The inferior mesenteric vein is ligated sometimes at the inferior border of the pancreas, and the splenic vein is ligated at the proposed point of transection. The transected pancreas ([Fig. 7.8-5](#)) is usually stapled and drained, although some surgeons suture the cut end and ligate the duct. The spleen may be preserved when operating for benign disease. This surgery is being done laparoscopically. One method to preserve the spleen is not dividing the short gastric vessels but still dividing the splenic artery and vein.



Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

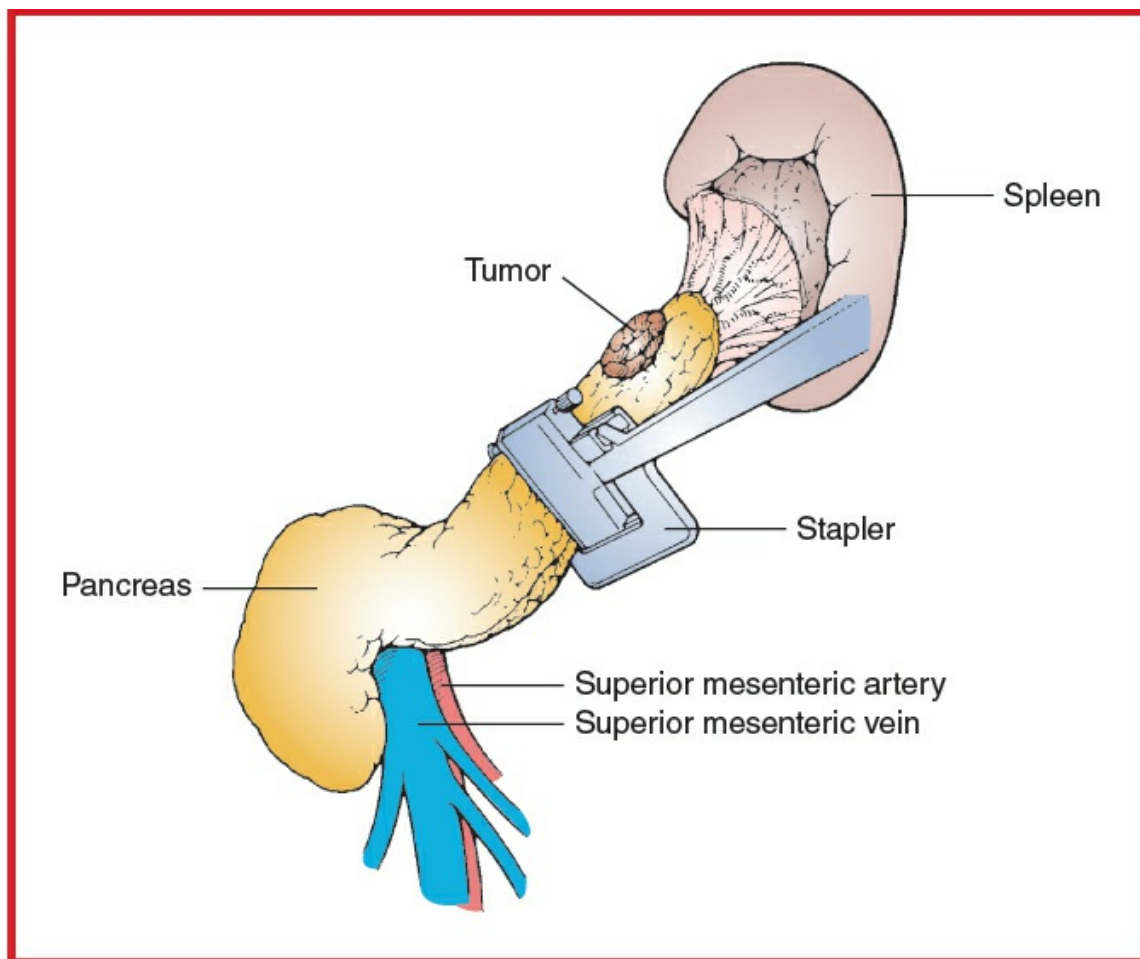


Figure 7.8-5 Resection of distal pancreas. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

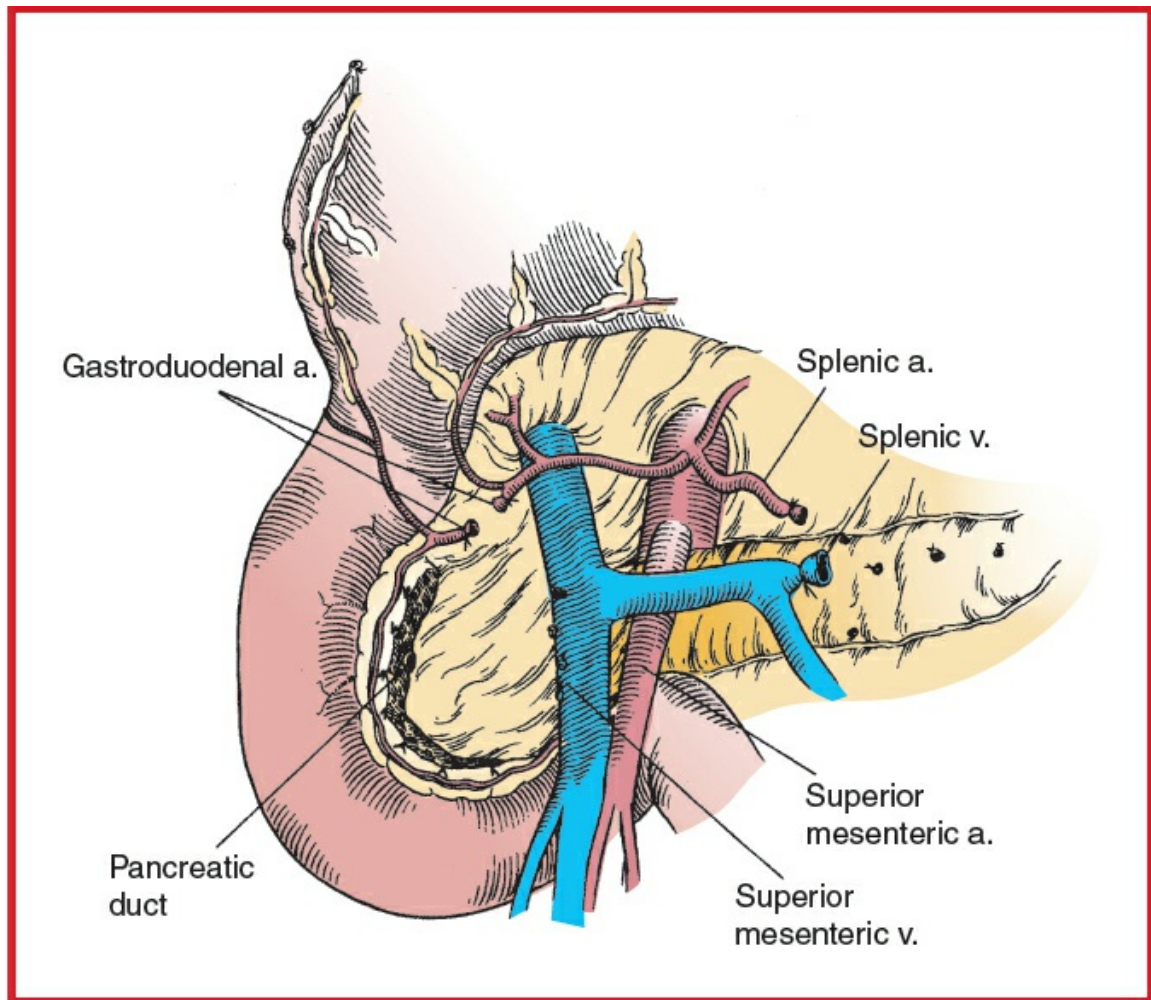


Figure 7.8-6 Near-total pancreatectomy. Shown is the operative field at the conclusion of the procedure. (Reproduced with permission from Baker RJ, Fischer JEMastery of Surgery, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 1997.)

Variant procedure or approaches

Subtotal pancreatectomy usually implies resecting the pancreas from the mesenteric vessels distally, leaving the head and uncinate process intact. This procedure may be performed for tumor or chronic pancreatitis. **Child procedure** (near-total pancreatectomy) consists of removing the entire pancreas except a rim of tissue along the lesser curvature of the duodenum (Fig. 7.8-6); preserving the duodenum makes it unnecessary to reconstruct the bile duct. This procedure is usually reserved for patients with chronic pancreatitis.

Usual preop diagnosis

Carcinoma of pancreas; islet cell tumors; cystic neoplasms; chronic pancreatitis; solid pseudopapillary tumors

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting

NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Distal Pancreatectomy	Subtotal Pancreatectomy	Child Procedure
Position	Supine	⇐	⇐
Incision	Midline abdominal or transverse	⇐	⇐
Unique considerations	NG tube	⇐	Preservation of vasculature of duodenum
Antibiotics	Cefoxitin 2 g iv	⇐	⇐
VTE prophylaxis	Heparin 5000 units sq	⇐	⇐
Surgical time	2–3 h	⇐	3–4 h
Closing considerations	Adequate drainage	⇐	⇐
EBL	300–500 mL	500–750 mL	500–1000 mL
Postop care	NG decompression; PACU	⇐	⇐
Mortality	<5%	⇐	⇐
Morbidity	Diabetes: 5%	⇐	⇐
	Wound infection: 5%	⇐	⇐
	Pancreatic fistula: <5%	⇐	90%
	Common bile duct injury	⇐	⇐
	Hemorrhage	⇐	⇐
		⇐	⇐

	Duodenal necrosis Pancreatic leakage Pancreatic insufficiency		
Pain score	6–8	6–8	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	30–60 yr
Male:Female	1:1
Incidence	~26,000/yr in the United States
Etiology	Adenocarcinoma; chronic pancreatitis; islet cell tumors
Associated conditions	Alcoholism and biliary tract disease with chronic pancreatitis (90%); other endocrine disorders (3–5%)

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pancreatic Surgery.

Suggested Readings

1. Chen BP, Chen M, Bennett S, et al: Systematic review and meta-analysis of restrictive perioperative fluid management in pancreaticoduodenectomy. *World J Surg* 2018; 42(9):2938–50.
2. Ding X, Tan J, Qian J: Laparoscopic spleen preserving distal pancreatectomy. *Hepatogastroenterology* 2013; 60(123):605–10.
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WHIPPLE RESECTION

■ SURGICAL CONSIDERATIONS

Description

A **Whipple resection** is usually done for pancreatic cancer. It consists of a

pancreaticoduodenectomy, followed by a **pancreaticojejunostomy**, an **ampullary or duodenal hepaticojejunostomy**, and a **gastrojejunostomy** (Fig. 7.8-7). On entering the peritoneal cavity, the resectability of the pancreatic tumor is determined. Contraindications to standard resection include liver or peritoneal metastases, involvement of the superior mesenteric vessels, infiltration by tumor into root of the mesentery, and extension into the porta hepatis, with involvement of the hepatic artery. If the tumor is deemed resectable, further mobilization of the head of the pancreas is performed. The common duct is transected above the cystic duct entry, and the gall bladder is removed. After the superior mesenteric vein (SMV) is freed from the pancreas, the neck is transected, with care being taken not to injure the splenic vein, portal vein, or SMV. For a pylorus-preserving Whipple, the duodenum is transected 2 cm distal to the pylorus or, for a standard Whipple, the stomach proximal to the pylorus. The jejunum is transected 20 cm beyond the ligament of Treitz, and the specimen is removed by severing the vascular connections with the mesenteric vessels. Reconstitution is achieved by anastomosing the pancreatic stump, bile duct, and duodenum into the jejunum. Drains are placed adjacent to the pancreatic anastomosis.

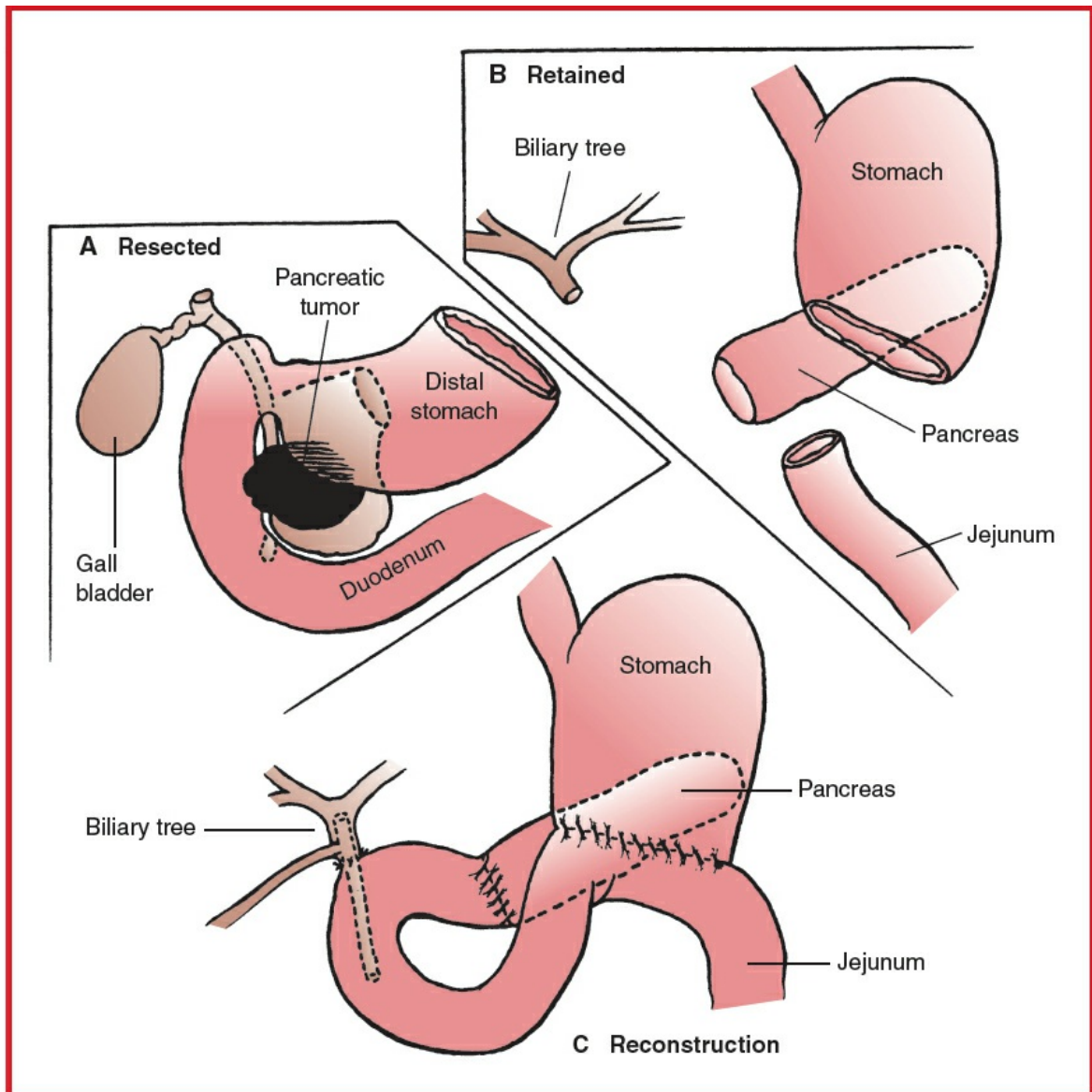


Figure 7.8-7 Standard pancreaticoduodenectomy. **A:** Structures resected, including distal stomach, entire duodenum, head and neck of pancreas with tumor, gallbladder, and distal extrahepatic biliary tree. **B:** Structures retained, including proximal stomach, body and tail of pancreas, proximal biliary tree, and jejunum distal to ligament of Treitz. **C:** Reconstruction: proximal pancreaticojejunostomy, hepaticojejunostomy over T-tube, and distal gastrojejunostomy. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Variant procedure or approaches

There are several variants that consist of extensions of the Whipple procedure: **total pancreatectomy**; **regional pancreatectomy**, involving resection and reconstruction of the retropancreatic SMV and artery; and the pylorus-preserving **pancreaticoduodenectomy** as described above. In addition, the distal pancreatic stump may be anastomosed to the posterior wall of the stomach.

Usual preop diagnosis

Carcinoma of the pancreas; malignant cystadenomas; chronic pancreatitis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Whipple	Total Pancreatectomy	Regional Pancreatectomy
Position	Supine	⇐	⇐
Incision	Midline abdominal or transverse (chevron)	⇐	⇐
Unique considerations	Large-bore iv lines	⇐	⇐
Antibiotics	Cefoxitin 2 g iv	⇐	⇐
VTE prophylaxis	Heparin 5000 units sq		
Surgical time	4–5 h	4–6 h	5–6 h
Closing considerations	Adequate drainage	⇐	⇐
EBL	500–750 mL	750–1000 mL	750–1500 mL
Postop care	NG decompression; PACU/ICU	NG decompression; diabetes management	
Mortality	5–10%	12%	15%
Morbidity	Delayed gastric emptying: 25%	⇐ NA	⇐ 10–20%
	Pancreatic fistula: 10–20%	5% <5%	25% 5%
	Sepsis: 5–15%	⇐	<5%

	Hemorrhage: 5%	<1%	<2%
	MI: 1–3%		
	Biliary fistula:		
	<2%		
Pain score	7–9	7–9	7–9

■ PATIENT POPULATION CHARACTERISTICS (FOR CANCER PANCREAS)	
Age range	40–80 yr
Male:Female	1:1
Incidence	10th most common cancer in the United States (~33,000/yr)
Etiology	Familial and genetic factors (p16, K-ras); tobacco; diabetes; alcohol; diet; pancreatitis
Associated conditions	See Etiology, above.

Suggested Readings

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ANESTHETIC CONSIDERATIONS FOR PANCREATIC SURGERY

(Procedures covered: drainage for pancreatitis; drainage of pancreatic pseudocyst; pancreaticojejunostomy; pancreatectomy; Whipple resection.)

PREOPERATIVE

Patients presenting for pancreatic surgery can be divided into four groups: (a) those with acute pancreatitis, who failed medical treatment and may be extremely ill, presenting for operative excision or drainage of necrotic or infected foci, (b) patients with adenocarcinoma of the pancreas, (c) patients with neuroendocrine active (60–70%) or inactive islet cell tumors (mainly insulinoma and gastrinoma; rarely VIPoma and glucagonoma), and (d) patients suffering from the sequelae of chronic pancreatitis (e.g., pseudocyst or abscess).

Respiratory	Respiratory compromise—such as pleural effusions, atelectasis, and ARDS, progressing to respiratory failure—may occur in patients with acute pancreatitis. Postop mechanical ventilation may be needed for these patients. Tests: Consider CXR; ABG; others as indicated from H&P
Cardiovascular	Patients with acute pancreatitis may suffer from severe intravascular volume depletion 2° plasma exudation and in severe cases 2° hemorrhage (erosion of blood vessels). Aggressive volume resuscitation with crystalloids, colloids, and blood products may be required during surgery. Hypocalcemia is often present (arrhythmias and ↓ myocardial contractility). Serum K⁺ may be elevated (2° acidosis or renal failure associated with acute pancreatitis) or decreased (2° watery diarrhea associated with gastrinoma, VIPoma, prolonged NG suction) and should be corrected before surgery. Hypokalemia resistant to K⁺ replacement may point to ↓ Mg⁺⁺ and warrants replacement. Tests: ECG; electrolytes; others as indicated from H&P
Gastrointestinal	Jaundice and abdominal pain are common presenting Sx in this group of patients. The presence of ileus

(common in acute pancreatitis) or intestinal obstruction should mandate full-stomach precautions and rapid-sequence induction (see Rapid-Sequence Induction, p. B-5). Electrolyte abnormalities are common and may be 2° metabolic acidosis ($\uparrow$ K^+ 2° acute pancreatitis) or alkalosis and intestinal losses ($\downarrow$ K^+ and $\downarrow$ Mg^{++} 2° diarrhea, NG suction). Acute pancreatitis is associated with $\downarrow$ Ca^{++} (omental fat saponification) and $\uparrow$ Na^+ (dehydration). Gastrinoma (Zollinger-Ellison syndrome) is associated with diarrhea, severe peptic ulcer, and GERD. VIPoma often causes massive watery diarrhea (up to 20 L). Electrolyte abnormalities should be treated preop. **Tests:** Electrolytes; glucose; LFTs; others as indicated from H&P

Renal

Patients should be evaluated for renal insufficiency predominantly 2° dehydration, with anesthetic plan adjusted accordingly. Renal insufficiency may also occur in the context of multiorgan failure associated with acute pancreatitis or sepsis.

Tests: BUN; creatinine; others as indicated from H&P

Endocrine

Many patients with acute pancreatitis have diabetes 2° loss of islet cells. Endocrine tumors of the pancreas are linked (10–30%) with multiple endocrine syndrome type I (MEN 1; Wermer syndrome), including adenoma of the parathyroid gland (>90%), the pituitary gland (20–30%), and the pancreas (~50%). Hyperparathyroidism is common ($Ca^{++}\uparrow$), some pituitary adenomas secrete prolactin > growth hormone > ACTH, and some pancreatic islet cell tumors secrete gastrin > insulin $\gg$ VIP, glucagon. Insulinoma can result in severe hypoglycemia necessitating frequent periop blood glucose measurements. Surgical manipulation of the insulinoma may result in massive release of insulin. VIPoma (Verner-Morrison syndrome) is often associated with hypokalemia but can also result in hypercalcemia and hyperglycemia (insulin

	resistance). Glucagonoma can be associated with hyperglycemia. Tests: Electrolytes; glucose; others as indicated from H&P
Hematologic	Hct may be falsely elevated, 2° hemoconcentration, or low, 2° hemorrhage. Coagulopathy may be present (DIC in the context of SIRS or sepsis). Tests: CBC with differential; Plt; consider PT, PTT, fibrinogen, D-dimer, or thromboelastogram (TEG)
Laboratory	Other tests as indicated from H&P
Premedication	Consider midazolam 1–2 mg. Note: Full-stomach precautions in patients with intestinal obstruction, and consider placement and suctioning of a NGT before induction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural for postop analgesia or alternative regional anesthetic techniques including paravertebral and possibly transversus abdominis plane (TAP) blocks (see [p. 1788–1789](#)). However, whether TAP blocks provide meaningful opioid-sparing and analgesic effects in abdominal pancreatic surgery is uncertain. The choice of an appropriate regional anesthetic technique can be significantly restricted by a patient's health status (e.g., coagulopathy or sepsis) (see Adjuvant Regional Anesthetic Techniques, [p. 1738](#)). If postop epidural analgesia is planned, establishing correct catheter placement in the epidural space can be accomplished by injecting 1–2% lidocaine (50–100 mg) via the catheter to elicit a segmental block. The use of epidural anesthetic techniques for postop pain management in patients undergoing major, nonvascular abdominal surgery has been shown to provide superior pain control

compared with IV-PCA (20–30% difference on postop days 1–3). However, adequate pain control can be achieved with the use of IV-PCA.

Epidural anesthetic techniques may be particularly beneficial in patients undergoing upper abdominal procedures and at risk for postop respiratory failure (preop hypoxia and/or hypercarbia, severe chronic lung disease, recent admission for acute respiratory failure). A recent cumulative meta-analysis of all published and unpublished randomized trials examining epidural anesthetic techniques indicates reduced overall mortality in patients undergoing major abdominal and thoracic surgeries (prevention of one death per 60 patients receiving epidural anesthesia). Epidural anesthetic techniques also decreased cardiovascular (arterial fibrillation, supraventricular tachycardia, heart block), pulmonary (emboli, atelectasis, pneumonia), and gastrointestinal (ileus) morbidity, while they increased the incidence of urinary retention and arterial hypotension. Beneficial effects may be particularly pronounced in patients with limited physiologic reserve. This important meta-analysis included trials published between 1971 and 2011, documenting steady mortality risk rates over the entire period. However, whether and to what extent the reduced mortality risk associated with the use of epidural techniques will remain significant when adhering to today's clinical practice guidelines and surgery-specific ERAS protocols (enhanced recovery after surgery) are not certain.

ERAS pathways ([Appendix K](#)) emphasize opioid-sparing and multimodal analgesic techniques for the management of periop pain. These techniques aim at providing superior pain control while reducing the incidence of opioid side effects including dizziness (impaired early mobilization), slow recovery of bowel function (delayed discharge from hospital), nausea and vomiting (decreased patient satisfaction), and potentially life-threatening events (severe respiratory depression). Multimodal approaches include the use of regional anesthesia techniques and nonopioid drug classes, including nonsteroidal anti-inflammatory drugs (NSAIDs), acetaminophen, N-methyl-D-aspartate (NMDA) antagonists, local anesthetics, and gabapentinoids.

Induction

The patient with bowel obstruction or ileus is at risk for pulmonary aspiration, and rapid-sequence induction with cricoid pressure is indicated. If the patient is clinically hypovolemic, restore intravascular volume (colloid, crystalloid, or blood products) before induction and titrate induction dose of sedative/hypnotic agents. Etomidate (0.2–0.4 mg/kg iv) or ketamine (1–3 mg/kg iv) may provide better hemodynamic stability during

	induction of anesthesia. Propofol is associated with a significant reduction of SVR.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. Combined epidural/GA: The epidural catheter ideally is placed at the level of a dermatome corresponding to the surgical site (generally midthoracic spine). This allows the use of both lipophilic and hydrophilic drugs at the lowest possible dose and adds flexibility to the anesthesiologist's drug selection while also minimizing the likelihood of side effects. A continuous infusion (after an initial bolus) is the preferred mode of administering local anesthetics via the epidural catheter because satisfactory analgesia can be achieved without major fluctuations in BP. Lower concentrations (e.g., bupivacaine 0.125–0.25%) are used to provide supplemental analgesia, while higher concentrations (e.g., bupivacaine 0.5%) generally provide optimal surgical conditions (i.e., a complete sensory and motor block). The infusion rate is contingent on the desired segmental spread but often ranges between 5 and 10 mL/h. Lipophilic opioids can effectively be given by continuous infusion (e.g., fentanyl 2 mcg/mL) and offer the advantage of limited segmental spread. Longer-acting hydrophilic opioids (e.g., hydromorphone 0.1–0.4 mg or morphine 1–2 mg for epidural placement at midthoracic spine) can be injected as a bolus along with an initial bolus dose of local anesthetics. However, hydrophilic opioids tend to spread rostrally within the intrathecal space and may cause sedation and respiratory depression if dosed too aggressively. Vulnerable patients include the elderly, patients with obstructive airway disease, and patients with ↑ BMI. The use of epidural local anesthetics is associated with sympatholysis, and ↓ BP has to be anticipated. Critical ↓ BP is treated with fluids and/or vasopressors (e.g., phenylephrine 50–100 mcg iv). In

patients undergoing a surgical procedure with a significant risk for major bleeding, it is prudent to delay administration of epidural local anesthetics until the critical part of surgery has been completed. Systemic sedatives (e.g., opiates and benzodiazepines) should be minimized, particularly in patients receiving opioids by the neuraxial route as they increase the likelihood of postop respiratory depression.

Low-dose ketamine: If the placement of an epidural catheter is not an option, the use of ketamine may be considered as an adjuvant analgesic regimen. Low-dose ketamine provides opioid-sparing effects (30–50%), which can be accompanied by a simultaneous reduction of postop pain (up to ~30%). Administration of ketamine reduces the incidence of opioid-mediated side effects including PONV and decreases postop wound hyperalgesia, which may attenuate the risk for developing persistent pain after surgery. A low-dose ketamine infusion (e.g., 0.5 mg/kg bolus before surgical incision, followed by an infusion of 0.2 mg/kg/h), may be particularly appropriate in patients at risk for developing persistent pain after surgery, including patients on chronic high-dose opioid therapy and patients with widespread pain complaints. For the sole alleviation of acute postop pain, doses >30–50 mg may not provide additional benefits. The risk for occurrence of psychomimetic side effects appears to be low in patients undergoing general anesthesia. Adding ketamine to a postop IV opioid-PCA regimen (typically 1–2 mg per bolus) decreases postop opioid use and pain (~30%), and lowers the incidence of PONV (~40%). At reported doses, the risk of hallucination is not significantly increased.

Magnesium: The intraop or periop use of iv Mg^{++} (typically 2–4 g as a bolus/infusion regimen) has some opioid-sparing (15–30%) and modest analgesic effects (pain 10–20%). However, it is unclear whether administration of Mg^{++} reduces the incidence of opioid-

related side effects.

Maintenance (cont'd)

Gabapentin/pregabalin: Single-dose administration of 100–300 mg pregabalin or 600–1200 mg gabapentin is a consideration in patients not eligible for epidural anesthesia. Pregabalin and gabapentin can provide opioid-sparing effects (15–40%), reduce the incidence of the opioid-related side effects PONV and pruritus (~30%), and exert modest analgesic effects (10–30%). However, administration of pregabalin and gabapentin increases the incidence of sedation (20–45%), and complaints of visual disturbance are much more likely. As such, the periop use of pregabalin and gabapentin may hamper early postop mobilization and increase fall risks in vulnerable patients (e.g., the frail and elderly). Repeated dosing of pregabalin and gabapentin within a 24-h interval does not offer clear benefits but increases the risk of sedation when using higher doses. There is also no compelling evidence to suggest that pregabalin or gabapentin reduce the incidence of persistent pain after surgery.

Nonsteroidal anti-inflammatory drugs and acetaminophen: Perioperative administration of nonselective NSAIDs (e.g., ketorolac 30 iv) COX2-selective NSAIDs (e.g., 200–400 mg celecoxib po) and/or acetaminophen (e.g., 1000 mg po) is associated with moderate opioid-sparing effects. Overall, NSAIDs seem to be about twice as effective (~25%) as paracetamol and only NSAIDs have conclusively been shown to reduce opioid-mediated side effects (e.g., PONV ~30%, sedation ~40%). NSAID but not paracetamol can also significantly reduce postop pain (up to ~30%). The continued use of NSAIDs or paracetamol in the postop period is meaningful to sustain opioid-sparing effects and further reduce the incidence of opioid-related side effects.

Potential benefits of adding NSAIDs need to be balanced against the risk of drug-related side effects including bleeding and renal failure. A large postmarketing study

(>10,000 patients) suggests that nonselective NSAIDs at recommended doses (e.g., ketorolac <100 mg/d) do not increase the risk of surgical site bleeding or GI bleeding in patients <65 yr and exposed for <5 d. However, in the elderly and in patients exposed >5 d, the risk of GI bleeding is increased. A small but elevated risk for surgical site bleeding has also been reported by several randomized clinical trials. NSAIDs should not be withheld from adults with normal preop renal function. However, if patients are at risk for postop renal impairment (e.g., impaired intraop renal perfusion), NSAIDs are best avoided.

Intravenous lidocaine infusion: IV infusion of lidocaine (e.g., 1.5 mg/kg bolus followed by 1–2 mg/kg/min) in abdominal surgery modestly reduces postop pain and opioid consumption (10–30%), decreases the incidence of PONV (~20–30%), accelerates recovery of bowel function, decreases the incidence of postop ileus (2–3×), and moderately reduces the average length of hospital stay. It is uncertain that these benefits translate to other types of surgery. At this time, it is also uncertain whether continuation of the infusion into the postop period provides additional benefits (frequent monitoring of lidocaine plasma concentrations strongly recommended).

Emergence

The decision to extubate a patient at the end of surgery depends on the patient's underlying cardiopulmonary status. Patients undergoing extensive surgery with major fluid shifts may require prolonged intubation until sufficient reduction of soft tissue edema (compromised airway) is achieved. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements	Anticipate large fluid loss IV: 14–16 ga × 2 (particularly Whipple, pancreatectomy, and drainage/debridement for pancreatitis) Warm fluids T&C or T&S	Blood loss can be significant, and blood products should be immediately available. Adequate large-bore iv access is mandatory. Procedures tend to be long and extensive, leading to hypothermia and large 3rd-space fluid loss. If procedure does not involve cancer or infection, cell-saving devices can be utilized. Intraoperative fluid management should be titrated to a patient's particular needs (adequate peripheral perfusion, urine output >0.5 mL/kg/min, no base deficit). Overly generous intraop fluid administration may be associated with increased postop morbidity (e.g., delayed recovery of bowel function) and extended hospital stay. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors UO Arterial line ± CVP or PA catheter	Pancreatic surgery can be associated with major fluid shifts and fluid loss. Availability of an infusion device assisting rapid delivery of intravenous fluids at body temperature is advised. Invasive monitoring is often required. In patients with severe cardiovascular impairment, the use of a PA catheter or TEE to guide

	± TEE	intraop management (the use of pressors and iv fluids) should be considered. Use forced-air warmer to maintain normothermia.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Hypocalcemia Hypovolemia Severe hypoglycemia Sepsis/SIRS	Release of pancreatic lipase → omental fat saponification Extensive 3rd spacing, major hemorrhage during pancreatic dissection Uncontrolled insulin release from insulinoma Manipulation of infected tissue → cardiovascular instability, respiratory deterioration, DIC

▼ POSTOPERATIVE

Complications	Hyperglycemia Electrolyte imbalance Hypovolemia Hypothermia Hypocalcemia PONV VTE	Total pancreatectomy is associated with brittle diabetes that can be very difficult to control. Subtotal resections lead to variable hyperglycemia.
Multimodal pain management	Continuous epidural analgesia PCA	Patient should be recovered in an ICU or hospital ward that is accustomed to treating the side effects of epidural analgesic techniques using opiates and local anesthetics (e.g., arterial hypotension,

respiratory depression, breakthrough pain, nausea, pruritus). Postop pain control with an epidural rather than a PCA regimen is superior in patients undergoing major abdominal surgery. Particularly in high-risk patients suffering from pulmonary impairment, an epidural regimen may reduce the incidence of respiratory failure. The use of epidural (or systemic) local anesthetics can speed up bowel recovery and reduce hospital length of stay. The use of epidural anesthetic techniques in the postop period has also been associated with reduced mortality and morbidity. Widely adapted ERAS protocols emphasize the continued use of multimodal analgesic techniques to alleviate postop pain and minimize the use of opioids (these aspects are discussed in more detail in section Intraoperative Management).

ERAS

PONV protocol (see [Appendix B, p. B-4](#) and [p. B-6](#))

Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.

Tests

CXR (if CVP placed); ABG; Electrolytes; Ca⁺⁺; glucose;

Hct, CBC, metabolic
panel, and coagulation as
indicated

Plts—as indicated for postop
management

Suggested Readings

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7. Mishriky BM, Waldron NH, Habib AS: Impact of pregabalin on acute and persistent postoperative pain: a systematic review and meta-analysis. *Br J Anaesth* 2015; 114:10–31.
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CHAPTER 7.9

Peritoneal Surgery

SURGEON

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ANESTHESIOLOGIST

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EXPLORATORY OR STAGING LAPAROTOMY

SURGICAL CONSIDERATIONS

Description

Exploratory laparotomy is indicated primarily in patients suffering from abdominal trauma or other acute abdominal catastrophes. It is important that a thorough and systematic intra-abdominal examination be carried out to prevent missing significant injuries (e.g., ruptured duodenum or transected pancreas). Any active bleeding should be controlled before a systematic examination. Other indications for laparotomy include certain patients with fever of undetermined origin or those in whom a specific diagnosis cannot be made or for staging of selected patients with Hodgkin disease. A **staging laparotomy** (Fig. 7.9-1) consists of **splenectomy**, **wedge**, and **needle biopsies** of both lobes of the liver and biopsies of the periaortic, celiac, mesenteric, and porta hepatis lymph nodes. In young women, suturing (pexing) the ovaries in the midline protects them from radiation. Indications for staging in Hodgkin disease and lymphomas vary from institution to institution, but PET/CT scans have limited their use.

The procedure begins with a midline abdominal incision; then the abdomen is explored, and both needle and wedge biopsies of the liver may be performed. The spleen may be removed by incising the lateral peritoneal attachment and delivering the spleen into the wound. The short gastric vessels are cut and ligated and the splenic vessels exposed.

These are cut individually and ligated, and the spleen is removed. Paraaortic nodes are exposed through a left paraaortic incision in the retroperitoneum and removed for biopsy. Lymph channels are clipped to prevent lymphatic leakage. The nodes dissected extend to the inferior margin of the duodenum. It may be necessary to cross the aorta and biopsy any enlarged nodes on the right side. More recently, **laparoscopy** is being performed for staging of certain intra-abdominal malignancies (e.g., pancreatic and gastric cancer); however, its use has decreased with improved multiphasic CT scans.

Usual preop diagnosis

Abdominal trauma; Hodgkin disease or other lymphomas

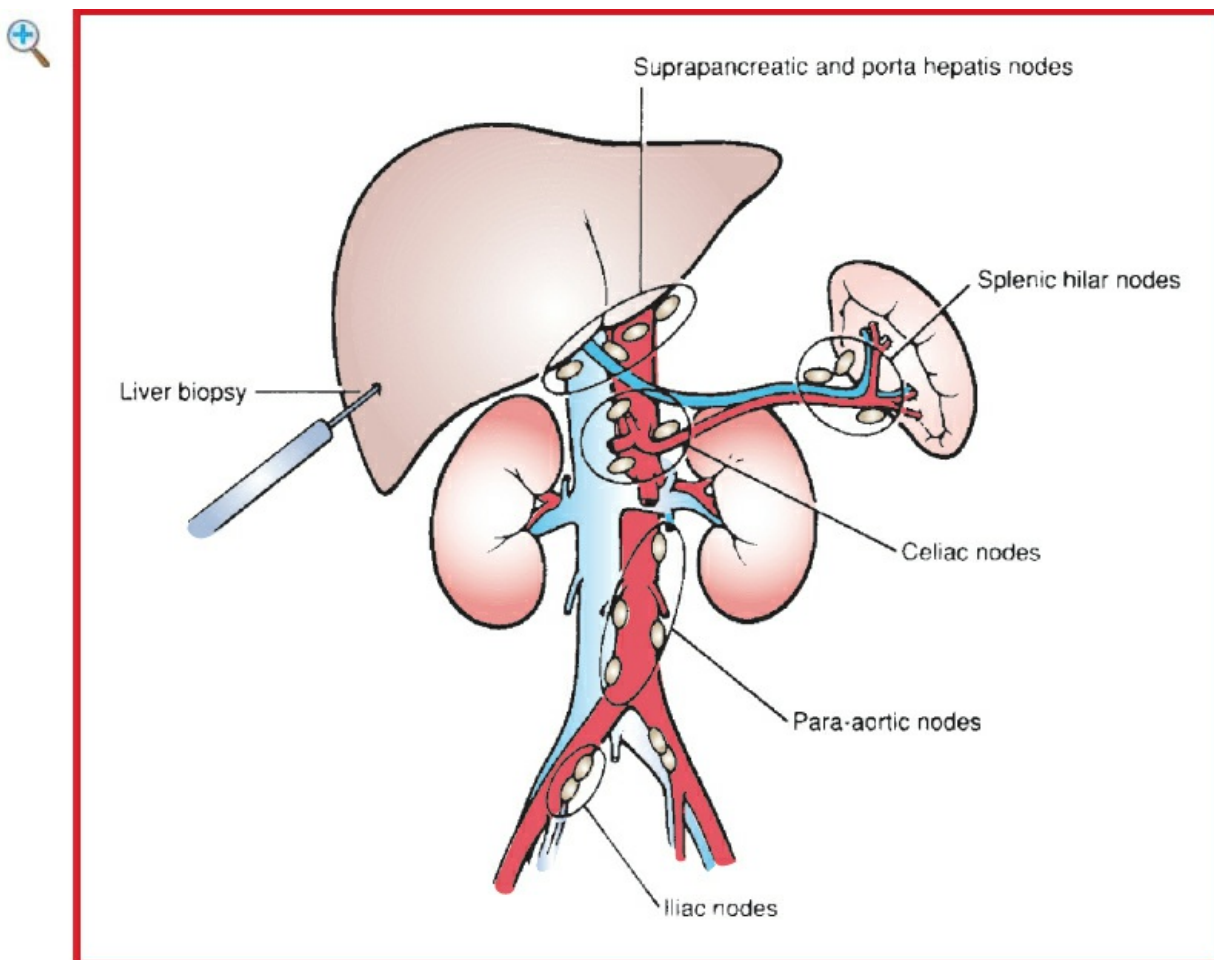


Figure 7.9-1 Staging laparotomy. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Staging Laparotomy	Exploratory Laparotomy
Position	Supine	⇐
Incision	Midline abdominal	⇐ or transverse
Special instrumentation	Abdominal retractor	⇐
Unique considerations	Ovarian pexy	Careful monitoring of VS in trauma patients
Antibiotics	Cefoxitin 2 g iv	Cefazolin 2 g iv
VTE prophylaxis	Heparin 5000 units sq	⇐
Surgical time	1.5–2 h	Variable, 1–2 h+
Closing considerations	Splenic bed hemostasis	Hemostasis
EBL	100–200 mL	Variable, 200–500 mL
Postop care	± NG decompression (especially trauma patients); PACU → ward	ICU for trauma patients
Mortality	<1%	2–5%
Morbidity	Prolonged ileus: 10–15% Pulmonary complications: 5–10% Wound infection: 2–3% Small bowel obstruction: 1% Intraperitoneal bleeding: <1%	⇐ Atelectasis: 5–10% Wound infection: 5–10% Hemorrhage: 1–3% Pneumonia: <1%
Pain score	6–8	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–60 yr	15–75 yr
Male:Female	1:1.5	1:1

Incidence	Common	⇐
Etiology	Unknown	Trauma
Associated conditions	Hodgkin disease (95%); lymphoma (5%)	Other visceral or vascular injuries in trauma

ANESTHETIC CONSIDERATIONS

(Procedures covered: exploratory/staging laparotomy that is not trauma-related; excision of intra-abdominal and retroperitoneal tumors; splenectomy.)

PREOPERATIVE

Patients presenting for staging laparotomy typically have lymphomatous or cancerous disorders. Apart from the primary disease, these patients are often in reasonably good health and will not have had radiation or chemotherapy before the staging laparotomy. Patients undergoing resection of intra-abdominal or retroperitoneal tumors vary widely in their overall health status (e.g., progression of tumor and concomitant diseases). Accordingly, the extent of surgery and the potential for intraoperative complications (e.g., major vascular injury) are case-specific and highly variable. Consequently, a thorough understanding of the patient's pathology and close communication with the surgical team (sometimes multidisciplinary) about planned surgical intervention are essential to devise a compelling anesthetic management plan.”

Patients presenting for splenectomy may be divided into two groups: (a) trauma patients (whose management is described in Trauma Surgery) and (b) a group with myeloproliferative and other disorders resulting in hypersplenism. Patients may have received chemotherapy and/or radiation therapy, which may affect a variety of organ systems. It is incumbent on the anesthesiologist to be aware of the periop implications of these adjunctive treatments.

Respiratory	Patients who have splenomegaly and other intra-abdominal mass lesions may have a degree of atelectasis and compromised ventilation 2° intra-abdominal pathology: ↓ FRC → ↑ A-a gradient + ↓ PaO ₂ . This should be evaluated by physical exam and close evaluation of diagnostic studies. Some may have been treated with chemotherapeutic agents (e.g., bleomycin at a total dose >200 mg/m ²) that
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	cause pulmonary pathology, including fibrosis. Toxic drug effects are potentiated by smoking, XRT, and high FiO₂. Tests: CXR, PFTs, and ABG as clinically indicated
Gastrointestinal	Prolonged diarrhea and/or vomiting can result in electrolyte abnormalities. Tests: electrolytes
Cardiovascular	Patients with systemic disease requiring splenectomy may be chronically ill and have ↓ cardiovascular reserve. Patients who have received certain chemotherapeutic agents (e.g., doxorubicin at a dose >550 mg/m²) may suffer from cardiotoxic side effects that can be worsened by XRT. Manifestations include CHF and dysrhythmias. Tests: ECG, ECHO, and stress test as clinically indicated
Neurologic	Patients may have neurologic deficits from receiving certain chemotherapeutic agents (e.g., peripheral neuropathies caused by vinblastine and cisplatin or CNS pathology caused by 5-fluorouracil and mithramycin). Evidence of neurologic dysfunction should be documented in the preop evaluation.
Hematologic	Patients are likely to present with splenomegaly 2° hematologic disease (e.g., Hodgkin disease, non-Hodgkin lymphoma, chronic leukemia, myeloid metaplasia, thrombotic thrombocytopenic purpura, idiopathic autoimmune hemolytic anemia, and sickle-cell disease). Cytopenia is very common. Preop replacement of platelets is often a consideration in patients with severe thrombocytopenia. However, the effectiveness of preop platelet replacement is limited in patients with hypersplenism because transfused platelets are mostly sequestered by the spleen. Myelosuppression should be anticipated in all patients receiving active chemotherapy. Tests: CBC
Hepatic	Some chemotherapeutic agents (e.g., methotrexate

	and mithramycin) may be hepatotoxic. Evaluation of LFTs should be considered in patients at risk. Tests: LFTs including INR as clinically indicated
Renal	Some chemotherapeutic drugs (e.g., methotrexate and cisplatin) are nephrotoxic, and patients may present with impaired renal function. Tests: BUN, creatinine, and electrolytes
Laboratory	A T&S and T&C are mandatory in all patients with significant relevant intraoperative bleeding risk. Other tests may be indicated based on the H&P.
Premedication	Consider midazolam 1–2 mg iv. In patients with suspected gastrointestinal stasis, the use of full-stomach precautions should be considered to minimize the risk of pulmonary aspiration. Consider placement and suctioning of an NGT before induction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with the surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg ± gabapentin 300 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural for postop analgesia or alternative regional anesthetic techniques including paravertebral and possibly transversus abdominis plane (TAP) blocks. However, whether TAP blocks provide meaningful opioid-sparing and analgesic effects in abdominal surgery is uncertain. The choice of an appropriate regional anesthetic technique can be significantly restricted by a patient's health status (e.g., coagulopathy) see [p. 1788–1789](#). If postop epidural analgesia is planned, placement of catheter before anesthetic induction is helpful to establish correct placement in the epidural space (accomplished by injecting 1–2% lidocaine (50–100 mg) via the epidural catheter to elicit a segmental block). The use of epidural anesthetic techniques for postoperative

pain control in patients undergoing major, nonvascular abdominal surgery has been shown to provide superior pain control compared with IV-PCA (20–30% difference on postoperative days 1–3). However, adequate pain control can be achieved with the use of IV-PCA. ERAS pathways see [Appendix K, p. K-1](#) emphasize opioid-sparing and multimodal analgesic techniques for the management of perioperative pain. These techniques aim at providing superior pain control while reducing the incidence of opioid side effects including dizziness (impaired early mobilization), slow recovery of bowel function (delayed discharge from hospital), nausea and vomiting (decreased patient satisfaction), and potentially life-threatening events (severe respiratory depression). Multimodal approaches include the use of regional anesthesia techniques and nonopioid drug classes, including nonsteroidal anti-inflammatory drugs (NSAIDs) acetaminophen, N-methyl-D-aspartate (NMDA) antagonists, local anesthetics, and gabapentinoids.

Induction	Standard induction except in patients at risk for pulmonary aspiration, who require a rapid-sequence induction (see Appendix B, p. B-5)
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. High inspired O ₂ concentrations (>30%) may aggravate chemotherapy-induced (e.g., bleomycin) lung injuries. Combined epidural/GA: The epidural catheter ideally is placed at a dermatome level corresponding to the surgical site (generally low to midthoracic). This allows the use of both lipophilic and hydrophilic drugs at the lowest possible dose, adds flexibility to the anesthesiologist's choice of agents, and minimizes the likelihood of side effects. A continuous infusion (after an initial bolus dose) is the preferred mode of administering epidural local anesthetics because satisfactory analgesia can be achieved with fewer fluctuations in BP and offer the advantage of limited segmental and rostral spread. Lower concentrations of bupivacaine (0.125–0.25%) can be infused to provide supplemental analgesia, whereas higher concentrations (0.5%) may improve surgical conditions (complete sensory and motor block). The infusion rate is contingent

on the desired segmental spread but often ranges between 5 and 10 mL/h. Lipophilic opioids (e.g., fentanyl) 2 mcg/mL can effectively be given by continuous infusion. Longer-acting hydrophilic opioids (e.g., hydromorphone 0.1–0.4 mg or morphine 1–2 mg for an epidural placement at the lower to midthoracic spine) can be injected as a bolus along with the initial bolus dose of a local anesthetic. However, hydrophilic opioids tend to spread rostrally within the intrathecal space and may cause sedation and respiratory depression if dosed too aggressively. Vulnerable patients include the elderly, patients with obstructive airway disease, and patients suffering from obesity. The use of epidural local anesthetics is associated with sympatholysis, and ↓ BP has to be anticipated. Critical ↓ BP is treated with fluids iv and/or vasopressors (e.g., ephedrine 5–10 mg iv). In patients undergoing a surgical procedure with a significant risk for major bleeding, it is prudent to delay administration of epidural local anesthetics until the critical part of surgery has been completed. Systemic sedatives (e.g., opiates and benzodiazepines) should be minimized, particularly in patients receiving neuraxial opioids as they increase the likelihood of postop respiratory depression.

Low-dose ketamine: If the placement of an epidural catheter is not an option, the use of low-dose ketamine provides opioid-sparing effects (30–50%), which can be accompanied by a simultaneous ↓ of postop pain (up to ~30%). Administration of ketamine ↓ the incidence of opioid-mediated side effects including PONV and ↓ postoperative wound hyperalgesia, which may ↓ the risk for developing persistent pain after surgery. Adding ketamine to a postoperative IV opioid-PCA regimen (typically 1–2 mg per bolus) ↓ postoperative opioid use and pain (~30%) and ↓ the incidence of PONV (~40%). At reported doses, the risk of hallucination is not significantly ↑'d.

Maintenance

Magnesium: The intraop use of iv Mg⁺⁺ (typically 2–4 g

(cont'd)

as a bolus/infusion regimen) has some opioid-sparing (15–30%) and modest analgesic effects (pain 10–20%). However, it is unclear whether administration of Mg^{++} ↓'d the incidence of opioid-related side effects.

Gabapentin/pregabalin: Single-dose administration of 100–300 mg pregabalin or 600–1200 mg gabapentin is a consideration in patients not eligible for epidural anesthesia. Pregabalin and gabapentin can provide opioid-sparing effects (15–40%), ↓ the incidence of the opioid-related side effects PONV and pruritus (~30%), and exert modest analgesic effects (10–30%). However, administration of pregabalin and gabapentin ↑'s the incidence of sedation (20–45%), and complaints of visual disturbance are much more likely. As such, the perioperative use of pregabalin and gabapentin may hamper early postop mobilization and ↑ fall risks in vulnerable patients (e.g., the frail and elderly). Repeated dosing of pregabalin and gabapentin within a 24-h interval does not offer clear benefits but ↑'s the risk of sedation. There is also no compelling evidence to suggest that pregabalin or gabapentin will ↓ the incidence of persistent pain after surgery.

Nonsteroidal anti-inflammatory drugs and acetaminophen: Perioperative administration of nonselective NSAIDs (e.g., ketorolac 30 iv), COX2-selective NSAIDs (e.g., 200–400 mg celecoxib po), and/or acetaminophen (e.g., 1000 mg po) is associated with moderate opioid-sparing effects. Overall, NSAIDs seem to be about twice as effective (~25%) as acetaminophen, and only NSAIDs have conclusively been shown to ↓ opioid-mediated side effects (e.g., PONV ~30%, sedation ~40%). NSAID but not acetaminophen can also significantly ↓ postop pain (up to ~30%). The continued use of NSAIDs and/or acetaminophen postop per is meaningful to sustain opioid-sparing effects and further ↓ the incidence of opioid-related side effects. Potential benefits of adding NSAIDs need to be balanced against the risk of drug-related side effects, including

bleeding and renal failure. A large postmarketing study (>10,000 patients) suggests that nonselective NSAIDs at recommended doses (e.g., ketorolac <100 mg/d) do not ↑ the risk of surgical site bleeding or GI bleeding in patients <65 yr and exposed for <5 d. However, in the elderly and patients exposed >5 d, the risk of GI bleeding is ↑'d. Several clinical trials have also reported a small risk for surgical site bleeding. NSAIDs should not be withheld from adults with normal preop renal function. However, if patients are at risk for postop renal impairment (e.g., impaired intraop renal perfusion),

NSAIDs are best avoided.

Intravenous lidocaine infusion: IV infusion of lidocaine (e.g., 1.5 mg/kg bolus followed by 1–2 mg/kg/min) in abdominal surgery will modestly ↓ postop pain and ↓ opioid consumption (10–30%), ↓ the incidence of PONV (~20–30%), ↑ the recovery of bowel function, ↓ the incidence of postop ileus (2–3×), and moderately ↓ the average length of hospital stay. It is uncertain that these benefits translate to other types of surgery. At this time, it is also uncertain whether continuation of the infusion into the postop period provides additional benefits (frequent monitoring of lidocaine plasma concentrations strongly recommended).

Emergence

Most patients can be extubated at the end of surgery. Patients undergoing extensive surgery with major fluid shifts may require prolonged intubation until cardiovascular stability and sufficient reduction of soft-tissue edema (compromised airway) is achieved. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements

IV: 14–16 ga × 1–2 (particularly major staging laparotomy and excision of tumors)

Potential for major blood loss. Adequate large-bore iv access is mandatory (particularly major staging laparotomy)

	BSS @ 3–10 mL/kg/h (see Goal-Directed Fluid Management, p. K-4)	and excision of tumors). In patients with difficult iv access, postinduction placement of additional access is prudent. In splenectomy patients with hypersplenism, Plt transfusion is best given after ligation of the splenic vessel (avoid sequestration). Availability of a rapid infusion device for delivery of IV fluids at body temperature should be considered in patients at risk for large blood loss.
Blood and fluid requirements (cont'd)	Warm fluids T&C or T&S	Intraop fluid therapy should be titrated to a patient's particular needs (adequate peripheral perfusion, urine output >0.5 mL/kg/h, no base deficit). Overly generous intraop fluid administration may be associated with postop morbidity (e.g., delayed recovery of bowel function) and extended hospital stay.
Monitoring	Standard monitors UO ± Arterial line ± CVP ± TEE	Place an arterial line in patients with hemodynamic instability or those at risk for significant intraop bleeding (major staging laparotomy and excision of tumors). Large volume shifts, need for vasoactive drugs, postop care Assessment of volume status

Positioning	✓ and pad pressure points ✓ eyes	
Complications	Unexpected bleeding	Plt transfusion may be necessary

▼ POSTOPERATIVE

Complications	Bleeding PONV VTE	Patient should be recovered in ICU or hospital ward that is accustomed to treating side effects of epidural local anesthetics and opiates (e.g., arterial hypotension, respiratory depression, breakthrough pain, nausea, pruritus).
Multimodal pain management	Epidural analgesia PCA	Postop pain control with an epidural rather than a PCA regimen is superior in patients undergoing major abdominal surgery. In high-risk patients with pulmonary impairment, an epidural regimen may reduce the incidence of respiratory failure. The use of epidural (or systemic) local anesthetic can speed up bowel recovery and reduce hospital length of stay. The use of epidural anesthetic techniques in the postop period have also been associated with ↓ mortality and morbidity. Emphasize multimodal approach combining nonopioid and

		bolus/PCA opioid analgesics with epidural analgesia, and/or TAP blocks (see p. K-2).
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	CXR CBC, metabolic panel, and coagulation as indicated	If CVP has been placed perioperatively

Suggested Readings

1. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
2. Brogi E, Kazan R, Cyr S, et al: Transversus abdominal plane block for postoperative analgesia: a systematic review and meta-analysis of randomized-controlled trials. *Can J Anaesth* 2016; 63:1184–96.
3. Guay J, Nishimori M, Kopp S: Epidural local anaesthetics versus opioid-based analgesic regimens for postoperative gastrointestinal paralysis, vomiting and pain after abdominal surgery. *Cochrane Database Syst Rev* 2016; (7):CD001893.
4. Ljungqvist O, Scott M, Fearon KC: Enhanced recovery after surgery: a review. *JAMA Surg* 2017; 152:292–8.
5. Nisanevich V, Felsenstein I, Almogy G, et al: Effect of intraoperative fluid management on outcome after intraabdominal surgery. *Anesthesiology* 2005; 103:25–32.
6. Strom BL, Berlin JA, Kinman JL, et al: Parenteral ketorolac and risk of gastrointestinal and operative site bleeding. A postmarketing surveillance study. *JAMA* 1996; 275(5):376–82.
7. Vlassakov KV, Zinner MJ: Putting the patient first. *Ann Surg* 2014; 259:1070–2.
8. For additional references on multimodal pain management, see [Appendix C](#) and [Appendix K](#).

SPLENECTOMY

SURGICAL CONSIDERATIONS

Description

Through a midline abdominal or left subcostal incision or laparoscopically, the spleen is mobilized by dividing the lateral peritoneal attachments while the spleen is retracted medially. (Relevant anatomy is shown in [Fig. 7.9-2](#).) Once the spleen is delivered into the operative wound, the short gastric vessels are clamped, cut, and ligated. This can be done with the LigaSure device. The splenic artery and vein are then exposed with care being taken not to injure the tail of the pancreas. By keeping the splenic hilum between the operator's fingers and thumb, inadvertent bleeding can be controlled easily. Accessory spleens (incidence, 15–30%) also should be looked for if the splenectomy is being done for ITP. They are found along the cephalad and caudad edges of the pancreas behind the stomach and in the area of the gastrohepatic ligament, greater omentum, and the splenic hilum. All patients undergoing splenectomy should receive polyvalent pneumococcal, meningococcal, and *Haemophilus influenza* vaccines.

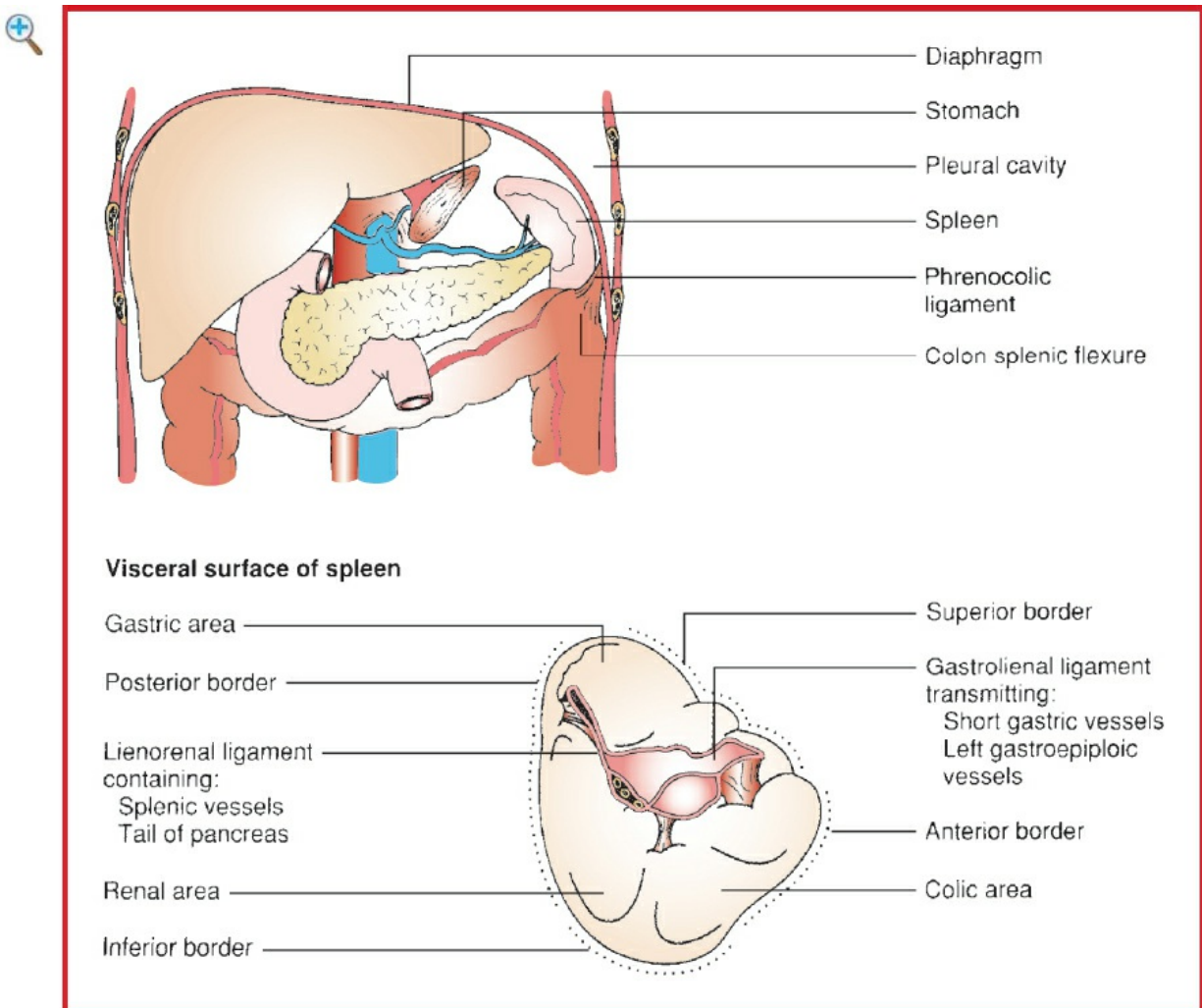


Figure 7.9-2 Anatomic relation of the spleen to the liver, diaphragm, pancreas, colon, and kidney. The

stomach is sectioned to illustrate the anatomic relations in situ. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Variant procedure or approaches

Following trauma, efforts at splenic salvage (**splenorrhaphy**) may be made, if possible, to preserve all or part of the spleen. This may be accomplished by the use of local hemostatic techniques (electrocoagulation, argon beam coagulator, Surgicel or Gelfoam soaked in thrombin, microfibrillar collagen, and the use of fine sutures or mattress sutures with Teflon felt pledgets). Recently, splenectomy has been performed laparoscopically if the spleen is near-normal size (see Laparoscopic Splenectomy).

Usual preop diagnosis

Staging laparotomy results; trauma; immune thrombocytopenic purpura; hereditary spherocytosis; other hereditary hemolytic anemia types; or a variety of myeloproliferative disorders

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURES		
	Splenectomy	Splenorrhaphy
Position	Supine	⇐
Incision	Midline or left subcostal	⇐
Special instrumentation	Suitable abdominal retractors	⇐
Unique considerations	Potential for major blood loss during procedure; avoid splenic laceration and damage to tail of pancreas.	⇐
Antibiotics	Cefoxitin 2 g iv	⇐

VTE prophylaxis	Heparin 5000 units sq	⇐
Surgical time	1–2 h	1–2 h
Closing considerations	Adequate hemostasis	⇐
EBL	50–100 mL	200–500 mL
Postop care	PACU (nontrauma)	⇐
Mortality	0–3%	⇐
Morbidity	Thrombocytosis: >1,000,000 → VTE Pulmonary complications: 3–23% Pancreatitis and/or pancreatic fistula: 1–8% Subphrenic abscess: 0–6% Bleeding: 1–5% Overwhelming sepsis: 0.3–2%	⇐
Pain score	5–7	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	1:1
Incidence	Common
Etiology	See Usual preop diagnosis, above.
Associated conditions	Blood dyscrasia (30–50%); abdominal or thoracic trauma (25%); Hodgkin disease (5–10%); tumors (5%)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Exploratory or Staging Laparotomy.

Suggested Readings

1. Crary SE, Buchanan GR: Vascular complications after splenectomy for hematologic disorders. *Blood* 2009; 114(14):2861–8.
2. Davidson RN, Wall RA: Prevention and management of infection in patients without a spleen. *Clin Microbiol Infect* 2001; 12:657–60.
3. Gamme G, Birch DW, Karmali S: Minimally invasive splenectomy: an update and review. *Can J Surg* 2013; 56(4):280–5.
4. Nurden AT, Nurden P: Increasing the platelet count in chronic ITP. *Lancet* 2008; 371:362–4.

EXCISION OF INTRA-ABDOMINAL AND RETROPERITONEAL TUMORS

SURGICAL CONSIDERATIONS

Description

Intra-abdominal and retroperitoneal tumors, other than those of visceral origin, consist primarily of sarcomas (liposarcoma, fibrous histiocytomas, mesenteric fibromas, and gastrointestinal stromal tumors). They are usually approached through a long midline incision for adequate exposure and to assess their resectability. Resection of such lesions may require excision of adjacent or involved small bowel or large intestine or other involved abdominal viscera. Care must be taken not to injure the ureters or major vessels, particularly at the root of the mesentery to the small bowel. It may be prudent to have ureteral stents placed to avoid injury to the ureters. If residual microscopic tumor remains, intraoperative radiation therapy (IORT) may be indicated. In certain tumors the patient may still benefit from “tumor debulking” (removing as much tumor as possible and treating the remaining tumor with radiation and/or chemotherapy). Operative approaches are dictated by location of tumor. Although most operative approaches are transabdominal, some retroperitoneal tumors may be approached retroperitoneally via oblique incision on either side of the abdomen. Some require thoracoabdominal incisions.

Usual preop diagnosis

Intra-abdominal or peritoneal tumor

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options,

physiotherapy plan, possibility of NG tube, and urinary drain postop (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Midline abdominal or transverse
Unique considerations	Availability of blood
Antibiotics	Cefoxitin 2 g iv preop
VTE prophylaxis	Heparin 5000 units sq
Surgical time	3–4 h
Closing considerations	Hemostasis
EBL	300–1000 mL
Mortality	1–3%
Morbidity	Respiratory problems: 5–10% Wound infection: 2–4% Hemorrhage: 1–3%
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Variable, 20–75 yr
Male:Female	1:1
Incidence	Common
Etiology	Unknown
Associated conditions	Partial bowel obstruction (10–15%); hydronephrosis (10–15%)

 ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Intestinal and Peritoneal Procedures, and for

Exploratory or Staging Laparotomy.

Suggested Readings

1. Gholanai S, Jacobs CD, Kapp DS, et al: The value of surgery for retroperitoneal sarcoma. *Sarcoma* 2009; 2009:605840.
2. Gladdy RA, Gupta A, Catton CN: Retroperitoneal sarcoma: fact, opinion, and controversy. *Surg Oncol Clin N Am* 2016; 25(4):697–711.
3. Shukla PJ, Bareto SG, Shrikhande SV: Retroperitoneal sarcoma. *Br J Surg* 2007; 94:1057–8.

DRAINAGE OF SUBPHRENIC ABSCESS

SURGICAL CONSIDERATIONS

Description

Abscesses may occur in subphrenic spaces, including the right subphrenic, right subhepatic, left subphrenic, lesser sac, or bare area of the liver ([Fig. 7.9-3](#)), following peritonitis, abdominal surgery, or trauma. It is essential to know the anatomy of these spaces for making a correct diagnosis and for treatment. Most commonly abscesses are drained by interventional radiology (85%), but some may require an open surgical approach.

Drainage is accomplished by a **posterior** or **anterior extraperitoneal approach** or by a **transpleural approach**, depending on location of the abscess. Lesser sac abscesses are best approached by an **anterior transperitoneal route**. Abscesses in the bare area of the liver are drained posteriorly. After the abscess is localized, the cavity is entered by finger dissection and drained. Loculations are broken up and the cavity thoroughly irrigated with NS or a suitable antibiotic solution. Appropriate drains are placed, and the wound is closed in a conventional manner. Cultures are routinely obtained.

Variant procedure or approaches

Percutaneous approaches have become more popular as interventional radiologists gain experience. This technique should be reserved for unilocular collections, where sterile cavities are not penetrated, and a safe route is available.

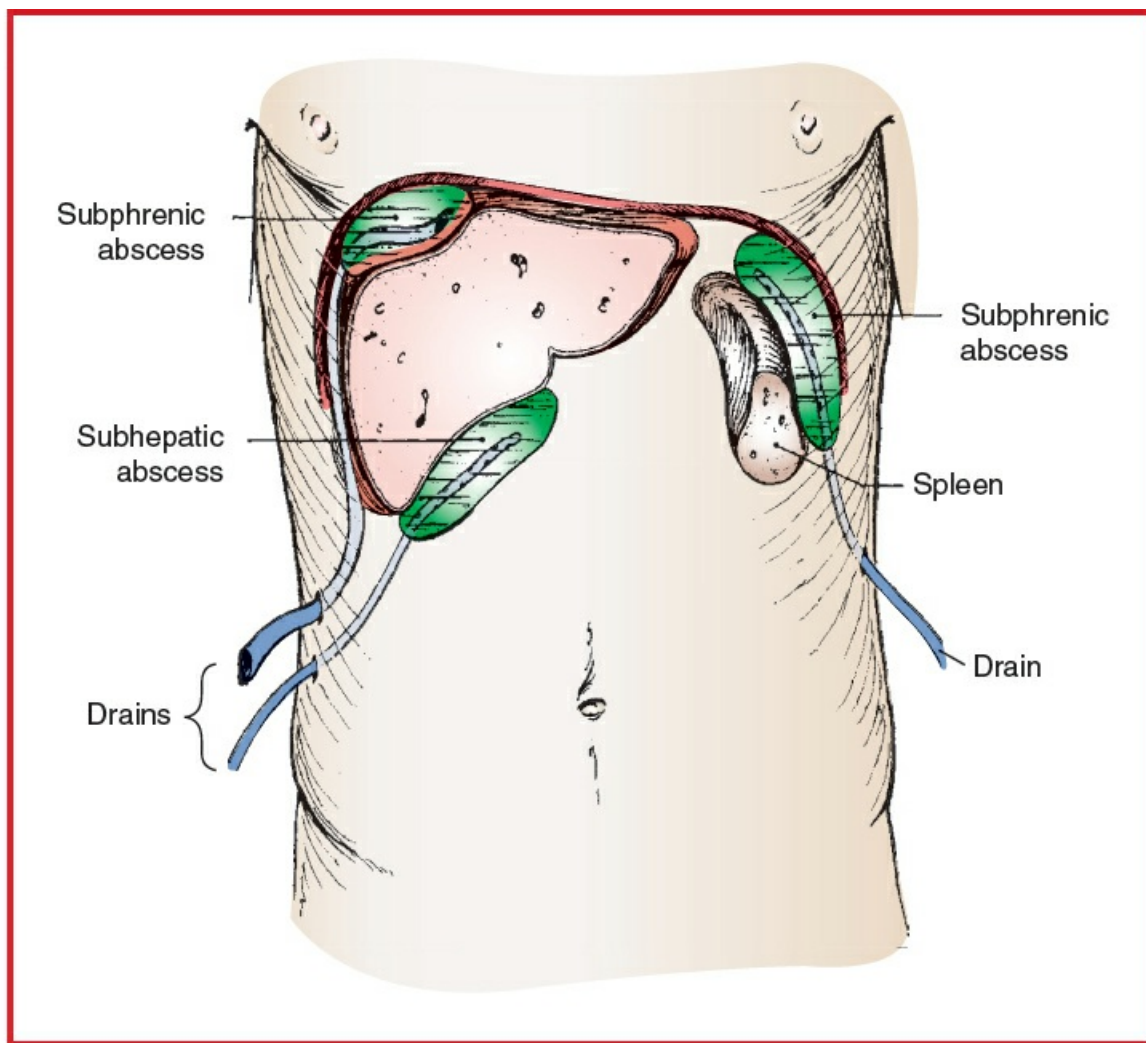


Figure 7.9-3 Anatomy of subphrenic abscess. (Reproduced with permission from Baker RJ, Fischer JE, eds: Mastery of Surgery, Vol. I, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Subphrenic abscess

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURES

**Subphrenic Abscess
Drainage**

Supine or lateral

Percutaneous Approach

←

Position	decubitus, right or left	
Incision	Subcostal or oblique abdominal	None
Special instrumentation	Drainage tubes	Special catheters; CT guidance
Antibiotics	Zosyn® (Piperacillin and Tazobactam) 3.375 g q6h	⇐
VTE prophylaxis	Heparin 5000 units sq	⇐
Surgical time	1–1½ h	⇐
EBL	50–100 mL	10–25 mL
Postop care	Maintain patency of the drainage tubes	⇐
Mortality	<5%	⇐
Morbidity	Inadequate drainage: 5–10% Pulmonary complications: 5–10% Bowel perforation: <2%	⇐
Pain score	7–9	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Variable, 15–80 yr
Male:Female	1:1
Incidence	Common
Etiology	Postop (70–80%); peritonitis (25–30%); trauma (5–10%)
Associated conditions	See Etiology, above.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Intestinal and Peritoneal Procedures.

Suggested Readings

1. Goulet CJ: Endoscopic transgastric drainage of subphrenic abscess. *Gastrointest Endosc* 2007; 65:733–5.
2. Morita S, Kamimura K, Suda T, et al: Endoscopic ultrasound-guided transmural drainage for subphrenic abscess: report of two cases and a literature review. *BMC Gastroenterol* 2018; 18(1):55.

INGUINAL HERNIORRHAPHY

SURGICAL CONSIDERATIONS

Description

Groin hernias are defects in the transverse abdominis layer, where a direct hernia comes through the posterior wall of the inguinal canal, and an indirect hernia comes through the internal inguinal ring (Fig. 7.9-4). Direct hernias are medial to the inferior epigastric artery and vein, whereas indirect hernias are lateral to these vessels. Surgical approach can be either anterior or posterior. In general, an **anterior approach** (Bassini, McVay, Shouldice, or mesh repair) is used for primary repair of an indirect or direct inguinal hernia. The **Bassini repair** consists of ligation of the hernia sac and suturing the conjoint tendon to the shelving edge of Poupart ligament. **McVay repair** sutures the conjoint tendon to Cooper ligament and usually is reserved for femoral inguinal hernias. **Shouldice** emphasizes the closing of the transverse fascia and transversus abdominal muscle layers. Currently, the interposing of Marlex mesh or insertion of a Marlex plug between the conjoint tendon, the internal oblique muscle, and the inguinal ligament is commonly used. Mesh is associated with improved outcomes and less pain. Other modifications are indicated in special situations.

A **posterior approach** is used by some surgeons for the repair of femoral hernias and recurrent inguinal hernias and for treating incarcerated and strangulated hernias. The **posterior preperitoneal approach** is usually performed by suturing the transversus abdominis arch on the superior aspect of the hernia defect to Cooper ligament and the iliopubic tract on the inferior aspect of the defect.

The **laparoscopic approach** is indicated for the repair of recurrent or bilateral inguinal hernias and utilizes a **preperitoneal patch repair** and results in less postop pain and an earlier return to regular physical activity (see Laparoscopic Inguinal Hernia Repair)

Randomized studies suggest that laparoscopic repair has inferior results.

Usual preop diagnosis

Groin pain or lump

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, possibility of NG tube, and urinary drain postop (see [Appendix K, p. K-1](#)).

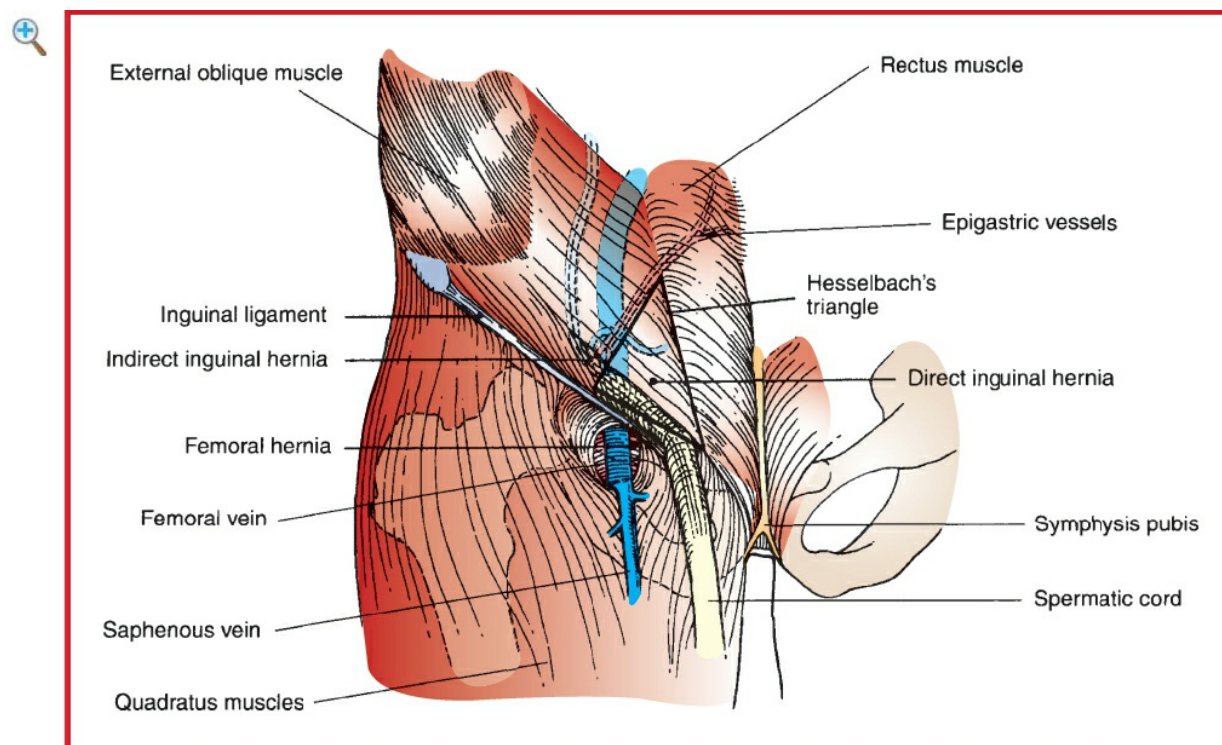


Figure 7.9-4 Inguinal anatomy. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Oblique or transverse
Unique considerations	Avoid damage to nerve structure and spermatic cord. Avoid interfering with blood supply to testes.
Antibiotics	Cefazolin 2 g iv preop
Surgical time	1–1.5 h

Postop care	PACU → ward or d/c to home in some cases
EBL	25–50 mL
Mortality	3/100,000
Morbidity	Wound abscess: <3% Wound hematoma: <2%
Pain score	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–90 yr
Male:Female	85:15
Incidence	15/1000
Etiology	Congenital variants; reduced collagen synthesis in adults
Associated conditions	Chronic cough; urinary retention; chronic constipation

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Repair of Abdominal Dehiscence.

Suggested Readings

1. Aquina CT, Probst CP, Kelly KN, et al: The pitfalls of inguinal herniorrhaphy surgeon volume matters. *Surgery* 2015; 158(3):736–46.
2. Gray SH, Hawn MT, Itani KM: Surgical progress in inguinal and ventral hernia repair. *Surg Clin North Am* 2008; 88:17–26.
3. Hallen M, Bergenfelz A, Westerdahl J: Laparoscopic extraperitoneal inguinal hernia repair versus open mesh repair: long-term follow-up of a randomized control trial. *Surgery* 2008; 143:313–7.
4. Suzuki S, Furui S, Okinaga K, et al: Differentiation of femoral vs. inguinal hernia: CT findings. *AJR* 2007; 189:78–83.

FEMORAL HERNIORRHAPHY

SURGICAL CONSIDERATIONS

Description

The hernia sac is exposed as it exits the preperitoneal space through the femoral canal ([Fig. 7.9-5](#)). If the hernia cannot be reduced, the possibility of strangulation needs to be kept in mind. The peritoneal sac in most cases should be opened proximal to the femoral canal to gain control of the intestine before it reduces itself into the peritoneal cavity. If the bowel is ischemic, it may require resection. The repair consists of suturing the iliopubic tract to Cooper ligament, taking care not to compromise the femoral vein (McVay repair).

Usual preop diagnosis

Bulging of tissues over femoral canal

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Oblique
Antibiotics	Cefazolin 2 g iv
Surgical time	1–1.5 h
EBL	25–50 mL
Postop care	PACU → ward or d/c to home in some cases
Mortality	<1% (6–20%, if strangulated)
Morbidity	Recurrence: 6%
Pain score	5–6

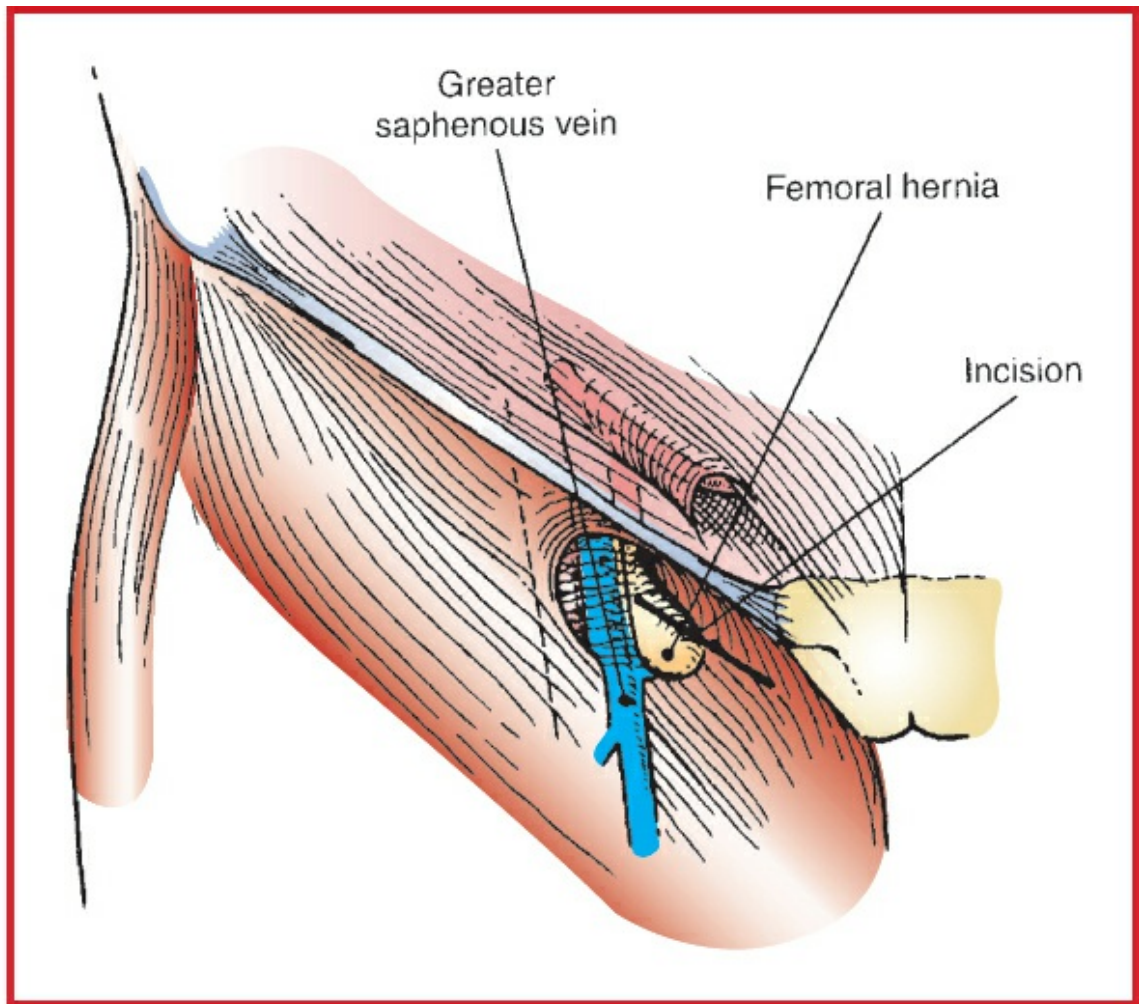


Figure 7.9-5 Femoral hernia repair. (Reproduced with permission from Scott-Conner CEH, Dawson DL: Operative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adults; rare in children
Male:Female	1:4
Incidence	1.5% of all groin hernias
Etiology	Failure of preformed peritoneal sac to obliterate; muscle atrophy in older age group

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Repair of Abdominal Dehiscence.

Suggested Readings

1. Earle DB, McLellan JA: Repair of umbilical and epigastric hernias.Surg Clin North

Am 2013; 93(5):1057–89.

2. Suzuki S, Furui S, Okinaga K, et al: Differentiation of femoral vs. inguinal hernia: CT findings. AJR 2007; 189:78–83.

REPAIR OF INCISIONAL HERNIA

SURGICAL CONSIDERATIONS

Description

Incisional hernias can occur after any abdominal incision but are most common following midline incisions. Factors leading to herniation include wound infection, trauma, inadequate suturing, and ischemia. Following skin incision, the skin edges and subcutaneous fat are retracted, and the dissection is carried down to the hernia defect. The redundant hernia sac is excised, and the fascia is freed up on both sides of the wound. Primary closure is preferred, if possible.

Variant procedure or approaches

In addition to primary repair, the latter may be reinforced by an onlay mesh prosthesis, or the prosthesis may be used to fill the hernial defect or placed behind the muscle layer. In the repair of incisional hernias, laparoscopic tacking of mesh is gaining in popularity.

Usual preop diagnosis

Incisional hernia

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, discussion of carb loading, DVT prophylaxis, and postop pain management options (see [Appendix K, p. K-1](#)).

SUMMARY OF PROCEDURE

Position	Supine
Incision	Vertical or transverse
Special instrumentation	Mesh prosthesis (when indicated)
Antibiotics	Cefoxitin 2 g iv
VTE prophylaxis	Heparin 5000 units sq

Surgical time	1–2 h
Closing considerations	Retention sutures
EBL	100–200 mL
Postop care	± NG decompression; abdominal binder; PACU → ward
Mortality	<1%
Morbidity	Ileus: 5–10% Respiratory complications: 5–10% Wound infection: 1–2%
Pain score	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–70 yr
Male:Female	1:1
Incidence	3–5% of midline abdominal incisions
Etiology	Wound infection; trauma; inadequate suturing; weak tissues
Associated conditions	Obesity; malnutrition

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Repair of Abdominal Dehiscence.

Suggested Readings

1. Gray SH, Hawn MT, Itani KM: Surgical progress in inguinal and ventral hernia repair. *Surg Clin North Am* 2008; 88:17–26.
2. Lockhart K, Dunn D, Teo S, et al: Mesh versus non-mesh for inguinal and femoral hernia repair. *Cochrane Database Syst Rev* 2018; (9):CD011517.
3. Nilsson H, Angerås U, Sandblom G, et al: Serious adverse events within 30 days of groin hernia surgery. *Hernia* 2016; 20(3):377–85.
4. Yeh DD, Alam HB: Hernia emergencies. *Surg Clin North Am* 2014; 94(1):97–130.

REPAIR OF ABDOMINAL DEHISCENCE

SURGICAL CONSIDERATIONS

Description

Dehiscence implies a “splitting apart” or “bursting open” of the abdominal wall fascia. A complete dehiscence is a separation of all layers of the abdominal wall and often is associated with an extrusion of abdominal viscera. If incomplete, the separation of fascial and muscular layers results in an incisional hernia or an obstruction of a herniated loop of intestine. The earliest sign of a wound dehiscence is the presence of serosanguineous drainage from the wound. Minimal disruptions may be treated conservatively with occlusive dressings and an abdominal binder. Major dehiscence requires operative repair using careful fascial closure and retention sutures.

Variant procedure or approaches

Variations in the type of closure depend on surgeon’s preference. Interrupted, nonabsorbable sutures and skin bridges are often used.

Usual preop diagnosis

Wound dehiscence

ERAS: For elective surgery, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, discussion of carb loading, DVT prophylaxis, and postop pain management options (see [Appendix K, p. K-1](#)).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Closure of previous incision
Unique considerations	Adequate muscle relaxation essential
Antibiotics	Cefoxitin 2 g iv
VTE prophylaxis	Heparin 5000 units sq
Surgical time	1–2 h
EBL	50–100 mL

Postop care	Abdominal binder to relieve tension on suture line; PACU → ward
Mortality	5–10%
Morbidity	Recurrent incisional hernia: 5–10% Wound infection: <5% Wound ischemia: 1–2%
Pain score	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	25–90 yr
Male:Female	1:1
Incidence	1.3% in patients <45 yr 5.4% in patients >45 yr
Etiology	Wound infection; excessive coughing or sneezing; excessive abdominal distention; weak tissue; poor nutrition; hematoma formation; poor surgical technique with tissue ischemia
Associated conditions	Malnutrition (25–30%); ascites (20–25%); hypoproteinemia (20%); chronic anemia (5–10%); vitamin C deficiency (<5%)

ANESTHETIC CONSIDERATIONS

(Procedures covered: inguinal herniorrhaphy; femoral herniorrhaphy; incisional hernia repair; repair of abdominal dehiscence.)

PREOPERATIVE

Predisposing factors for hernia often include increased abdominal pressure 2° chronic cough, bladder outlet obstruction, constipation, pregnancy, vomiting, ↑ BMI, and acute or chronic muscular effort. These factors should ideally be managed preop to avoid postop recurrence. The patient population may range from premature infants to the elderly, who have the potential for presenting with multiple medical problems.

Musculoskeletal

Pain is likely to be present in area of hernia;

	evaluate bony landmarks if regional anesthesia is planned.
Gastrointestinal	Hernias may become incarcerated, obstructed, or strangulated, requiring emergency surgery. Fluid and electrolyte imbalance is likely. Tests: Electrolytes, if indicated from H&P
Hematologic	For regional anesthesia, three patient's coagulation status, if indicated from H&P. Tests: As indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	If necessary, standard premedication.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. If appropriate, discuss with surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GA, regional, or local anesthesia $\pm$ iv sedation (MAC) may all be appropriate anesthetic techniques for uncomplicated cases (e.g., without incarceration or obstruction). GA and epidural for postop pain control may be appropriate for patients with unusually large incisional hernias or abdominal dehiscence (see Anesthetic Considerations for Exploratory or Staging Laparotomy). Choice depends on factors such as site of incision, patient physical status, and preference of both patient and surgeon. Profound muscle relaxation may be necessary to facilitate exploration and repair.

Regional anesthesia

Spinal	Single-shot vs. continuous: Patient in sitting or lateral decubitus position (operative site down) for placement of hyperbaric subarachnoid block. Doses of local anesthetics are as follows for T4-T6 level:
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	0.75% bupivacaine in 8.25% dextrose (10–15 mg); 0.5% tetracaine in 5% dextrose (12–16 mg). Consider adding fentanyl (10–20 mcg) and/or morphine (100–200 mcg).
Epidural	Patient in sitting or lateral decubitus position for placement of epidural catheter. After locating the epidural space, administer a test dose (e.g., 3 mL of 1.5% lidocaine with 1:200,000 epinephrine) to determine whether the catheter is subarachnoid or intravascular. Titrate local anesthetic (e.g., 2% lidocaine or 0.5% bupivacaine or ropivacaine) until desired surgical level is obtained (5–7 mL at a time), usually about 20 mL. Consider adding fentanyl (25–100 mcg). Consider adding single-shot morphine (3–5 mg) rather than fentanyl if the epidural catheter is removed shortly after the end of surgery (prolonged duration of action; requires monitoring for 16–24 h after injection).
Local	Requires gentle surgical technique. Surgical field block, plus ilioinguinal and iliohypogastric nerve blocks, using 0.5% bupivacaine with 1:200,000 epinephrine. Often performed by surgeon.

General anesthesia

Induction	LMA vs. ETT: Standard induction. LMA may be suitable for the patient who presents with a uncomplicated chronic hernia. If there is obstruction, incarceration, or strangulation, however, a rapid-sequence induction with ET intubation is indicated. GETA also may be indicated in the patient with wound dehiscence and large incisional hernias.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. Muscle relaxants may be necessary to facilitate surgical repair.

Emergence	Consider extubating the trachea while patient is still anesthetized to prevent coughing and straining. Patients who are at risk for pulmonary aspiration and require awake intubation or rapid-sequence induction are not candidates for deep extubation. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 16–18 ga × 1 BSS @ 2–4 mL/kg/h	See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitoring	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	↓↓ HR + ↓ BP	Vagal reflex evoked by bowel traction

POSTOPERATIVE

Complications	PDPH after neuraxial block Urinary retention, common with regional anesthesia Wound dehiscence with coughing/straining	Patients with urinary retention may require catheterization.
Multimodal pain management	PO analgesics: acetaminophen alone or combined with codeine (Tylenol no. 3, 1–2 tab q4–6h) or oxycodone (Percocet 1 tab q6h)	Surgical field block or regional anesthesia should provide sufficient analgesia postop.
ERAS	PONV protocol (see	Encourage early enteral

Suggested Readings

1. Abbott DE, Dumanian GA, Halverson AL: Management of laparotomy wound dehiscence. *Am Surg* 2007; 73:1224–7.
2. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
3. Cousins MJ, Bridenbaugh PO, eds: *Neural Blockade Pain Management*, 4th edition. Lippincott-Raven Publishers, Philadelphia: 2008.

CHAPTER 7.10

Breast Surgery

SURGEONS

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BREAST BIOPSY

SURGICAL CONSIDERATIONS

Description

Breast biopsy and **lumpectomy** entail the surgical removal of breast tissue for histopathologic examination. The term **excisional biopsy** is usually applied to benign entities and implies the complete removal of the lesion in question (e.g., fibroadenoma). The term **lumpectomy** is used to characterize cancerous lesions that are removed with a rim of normal breast tissue to achieve tumor-free margins.

Breast lesions subjected to definitive open or surgical excision can be classified as palpable or nonpalpable. Accompanying abnormal visual changes in the breast usually denote locally advanced disease. These include edema, redness, ulceration, and brawny discoloration of the skin with or without alterations in the contour or size of the breast. Palpable lesions can be described as nodules or masses having distinct or indistinct borders or subtle parenchymal thickening. Nonpalpable lesions are discovered on routine screening mammography and MRI or diagnostic ultrasound. Microcalcifications masses, densities, architectural distortion, and abnormal enhancement on MRI fall into the category of potentially malignant lesions. Ultrasound can help characterize palpable

or nonpalpable lesions as cystic or solid. MRI is not a routine screening test except in populations at high risk for developing breast cancer. It is particularly useful as an adjunct diagnostic tool in dense breasts, to delineate the extent of cancer and identify other occult lesions in the ipsilateral or contralateral breast.

Most patients undergoing a surgical procedure have had a prior diagnostic biopsy: fine needle aspiration cytology, core needle biopsy, or vacuum-assisted core needle biopsy. Hence, these results guide the extent of surgical resection. Typically, a radiologist will place one or more percutaneous hook wires to delineate the nonpalpable lesions. Instead of a hook wire, radioactive seeds or optical reflectors placed close to the lesion may be used for guiding the surgeon to the desired target. These operative procedures are referred to as **image-guided or wire-localized breast biopsies or lumpectomies**. The aim of these procedures is to remove the breast tissue surrounding the wire including the lesion and biopsy marker. Surgeons may, in some instances, use intraop ultrasound to identify the area of interest instead. A specimen radiograph is performed to confirm that the desired target has been excised.

Although less common, another reason for excisional biopsy is the occurrence of bloody or pathologic **nipple discharge**. The underlying cause of this abnormality typically is a benign intraductal papilloma or, infrequently, carcinoma. The surgeon inserts progressively larger lacrimal probes into the duct with the abnormal discharge and then injects 0.5 mL methylene blue via an ophthalmic needle to internally delineate the duct system. In some institutions, a ductoscope may be used to explore breast ducts.

Most newly diagnosed breast cancers are invasive, but ~20–25% of newly diagnosed breast cancers are noninvasive or ductal carcinoma in situ (DCIS). **Lumpectomy** is typically offered for invasive or in situ cancers that are 5 cm or smaller. **Mastectomy** is also an option (discussed below). Recently, two consensus statements have been published delineating recommendations regarding tumor-free margins. New technologies and contrast agents are under investigation to assess margins intraoperatively.

Commonly, surgeons will demarcate the lumpectomy area with metal clips or insert an interstitial tissue marker to serve as a guide for future radiation therapy planning. **Intraoperative radiation** is combined in some cases with lumpectomy for small breast cancers. Radiation is routinely administered weeks to months after surgery either using partial breast irradiation or whole breast irradiation regimens.

Intravenous sedation/MAC and GA are most commonly used for these procedures. When lumpectomies are coupled with nodal staging operations, GA is the preferred

modality and is facilitated by using a laryngeal mask. The latter is preferred for deeper, larger, or multiple lesions and in cases involving soft-tissue mobilization. Field blocks or regional anesthesia may also be considered.

Usual preop diagnosis

Breast mass; nipple discharge; atypical hyperplasia; known in situ, or invasive cancer; mammographic, sonographic, or MRI abnormalities

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Breast Biopsy/Lumpectomy	Image-Guided Breast Biopsy
Position	Supine with ipsilateral arm abducted (see Fig. 7.10-1)	⇐
Incision	Overlying the lesion, periareolar, radial, or along the inframammary fold	⇐
Special instrumentation	Ultrasound	⇐ Optical reflector to detect biopsy markers Specimen radiograph
Antibiotics	Cefazolin 2 g iv (optional)	⇐
Surgical time	0.5–1.5 h	⇐
Closing considerations	Wound dressings	⇐
EBL	<25 mL	⇐
Postop care	PACU → home	⇐

Mortality	Extremely rare	⇐
	Postop wound	⇐
	seroma: common	⇐
	Ecchymosis or hematoma: <10%	Wire transected; target lesion is missed; metallic marker not removed
Morbidity	Infection: ~1%	
Pain score	2–5	2–5

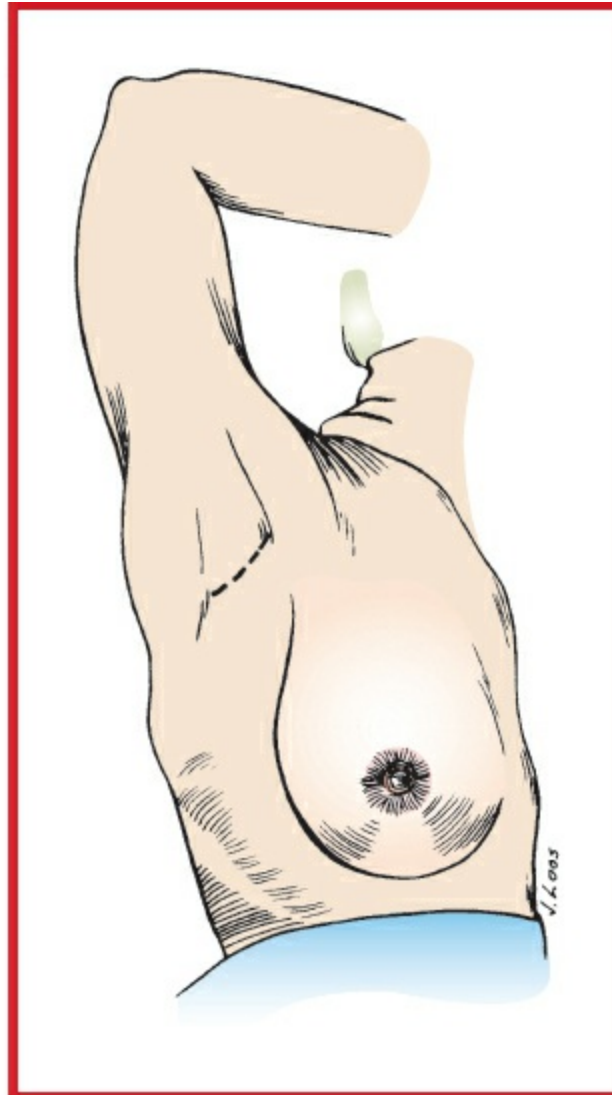


Figure 7.10-1 Arm may be draped into field and positioned on arm board at 90° or suspended overhead as shown. Incision site for lymph node biopsy is indicated by dotted line. (Reproduced with permission from Baker RJ, Fischer JE, eds *Mastery of Surgery*, 6th edition. Lippincott Williams & Wilkins, Philadelphia: 2012.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	>15 yr	>25 yr
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Male:Female	Mainly female	⇐
Incidence	Common	⇐
Etiology	Unknown	⇐

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Breast Biopsy and Sentinel Lymph Node Surgery.

LYMPH NODE SURGERY

SURGICAL CONSIDERATIONS

Description

Sentinel lymph node biopsy (SLNB) is a technique applied to patients with invasive breast cancers who are deemed to have clinically or radiologically negative lymph nodes. SLNB is generally not done for DCIS cancer with the exception of some large tumors or those associated with a mass on imaging. SLNB may be performed in selected patients undergoing **mastectomy** for DCIS where there may be some risk of discovering an occult invasive carcinoma. Of note, SLNB is also used for patients who received neoadjuvant chemotherapy.

Ultrasound interrogation of the axilla and biopsy of abnormal-appearing sentinel nodes has become a routine practice. Subclinical nodal disease is often discovered among patients without palpable adenopathy via this approach. The biopsied lymph nodes may be marked with a clip, radioactive seed, or black ink tattoo. A localizing hook wire may be placed to guide the surgeon to the biopsied node as well as a specimen radiograph to verify the resection of the targeted clip.

Sentinel lymph node(s) (SLNs) are the first nodes to drain afferent lymphatics from the breast and the region where the cancer is located and therefore are most likely to harbor metastatic tumor cells. The confluence of lymphatic channels at the subareolar plexus of Sappey accounts for the fact that periareolar injections are comparable to peritumoral injections. Most frequently lymphatic channels drain to the axilla and less commonly to the internal mammary chain or other extra-axillary sites.

The absence of metastasis on histologic examination of SLNs predicts tumor-free status of non-sentinel lymph nodes. In the context of breast conservation and planned whole

breast radiotherapy, completion axillary node dissection can be safely avoided in clinically negative cases with one to two positive nodes. These techniques have been validated in institutional and multicenter studies and prospective randomized trials.

Two types of agents have been principally tested for lymphatic mapping and sentinel node identification procedures: vital blue dyes (1% isosulfan blue or methylene blue) and ^{99m}Tc -technetium-labeled sulfur colloid (TSC; unfiltered or filtered). Different injection sites of these agents may be used: peritumoral or around biopsy cavity, periareolar/subareolar, or in combination. Isosulfan blue (3–5 mL) is injected in the operating room. The breast can be massaged to promote lymphatic uptake of tracers. The surgeon should inform the anesthesiologist when injecting the dye because a transient drop in the pulse oximeter reading of 2–5% is frequently seen, with maximal drop occurring within 20–25 min after injection. This is artifactual due to effects of blue dye spectral absorption, and blood oxygen remains normal. Patients' skin may retain a bluish hue, most noticeable in the recovery room. The urine, stool, or emesis can be blue for up to 24 h or more (the patient needs to be advised of this prior to discharge). Allergic reactions consisting of "blue hives" to full-blown, life-threatening anaphylaxis have been reported following the injection of isosulfan blue dye. Methylene blue can be used in place of isosulfan blue and has a lower incidence of allergic reactions but can cause skin necrosis if injected undiluted intradermally.

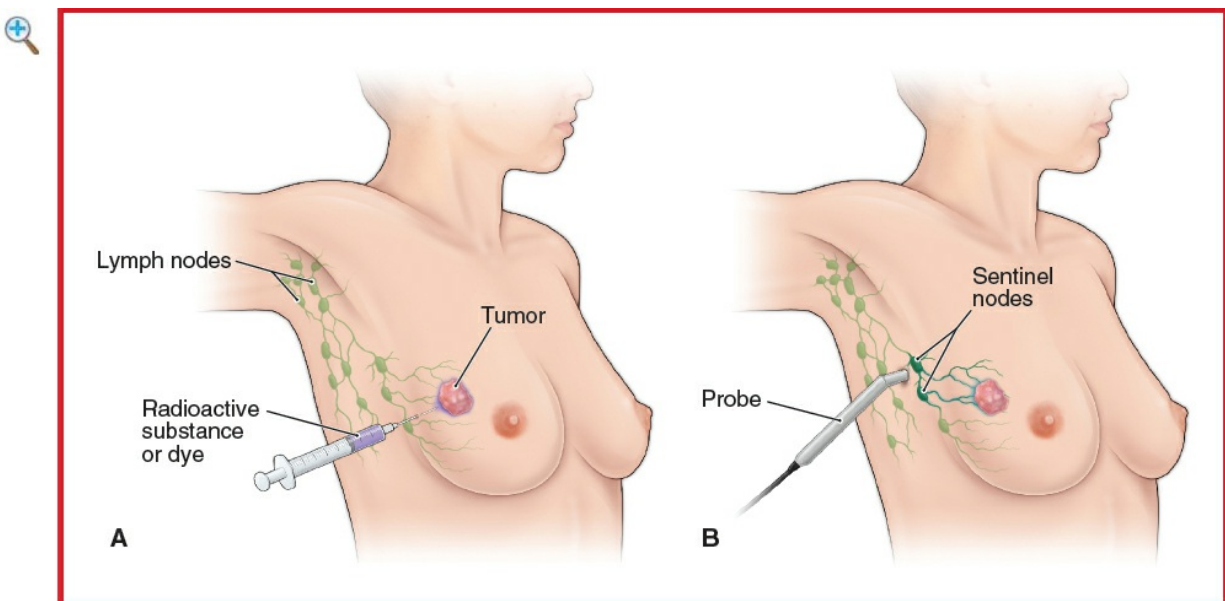


Figure 7.10-2 A radioactive tracer or dye is injected in/around the tumor and an external probe is used to identify the sentinel nodes.

TSC (~1–2 mCi) is usually injected in the nuclear medicine department the day of or the day before the operative procedure. The tracer has very low radioactivity, and no

special protection is required around the patient or in handling tissue specimens. Postinjection lymphoscintigraphy is considered optional. After induction of GA or regional anesthesia, the operative field (chest and lymph node-bearing areas) is surveyed with a handheld gamma probe (a slim rod, similar to a Geiger counter). The “hot” spot(s), denoting accumulated tracer in lymph nodes, are identified (Fig. 7.10-2). A curvilinear incision along the inferior aspect of the hair-bearing area of the axilla is made 3–7 min after injecting the blue dye in the breast. Typically, blue afferent lymphatics and blue nodes are identified below the clavipectoral fascia.

The combination of blue dye and TSC yields the highest sentinel node identification rates. Some surgeons prefer to use one of these two agents alone. Sentinel node mapping should be done with radiotracer alone in pregnant women. Of note, hard, palpable nodes discovered during the axillary exploration are considered sentinel nodes even if not blue or radioactive, as the lymphatic channels may be obstructed.

Sentinel nodes may be examined intraoperatively using touch-prep cytology or frozen section to identify node-positive cases. The number of positive nodes dictates whether the surgeon proceeds to **axillary node dissection** when breast-conserving surgery is performed. In the scenario of mastectomy or when surgery is preceded by neoadjuvant chemo or endocrine therapy, **completion axillary dissection** is performed if sentinel nodes are involved.

Whenever sentinel nodes cannot be identified, a conventional **level I and II axillary dissection** is indicated, although other factors may be taken into consideration influencing this decision. These nodes lie behind and lateral to the edge of the pectoralis minor muscle. The level III, or highest group of axillary lymph nodes, are medial to the pectoralis minor muscle and are not routinely removed unless clinically involved. As part of the axillary dissection, the surgeon preserves the thoracodorsal nerve (innervates the latissimus dorsi muscle) and the long thoracic nerve (innervates the serratus anterior muscle), as well as the blood and nerve supply to the pectoral muscles. The intercostobrachial nerves (sensory to the upper arm) course through the axillary contents, and its preservation usually can be accomplished, avoiding the occurrence of permanent dysesthesias.

Lymphedema can be a consequence of axillary nodal surgery. **Axillary reverse mapping** is a procedure that maps the arm draining nodes and lymphatics in the axilla of the ipsilateral breast. Blue dye (3–5 mL) or indocyanine green (0.5–1 mL) can be used for the procedure. The latter is visualized with the aid of an infrared camera (SPYElite system, Novadaq Inc.). Visualization of these structures allows the surgeon to preserve them, unless suspicious to harbor tumor, and lessen the chances of developing

lymphedema of the arm. Of note, indocyanine green should be avoided in patients with iodine allergies.

Usual preop diagnosis

Invasive breast cancer

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, and physiotherapy plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Small transverse or curvilinear axillary incision
Special instrumentation	Handheld gamma detection probe
Unique considerations	Avoid BP cuff or iv in ipsilateral arm. Possible need to avoid muscle relaxants. Isosulfan blue vital dye → allergic reaction (1–2/100). Methylene blue can cause skin necrosis when injected intradermally but does not cause anaphylaxis.
Antibiotics	Cefazolin 2 g iv (optional)
Surgical time	10–30 min, up to 1.5 h for axillary lymph node dissections
EBL	Minimal
Postop care	PACU → home
Mortality	Rare
Morbidity	Discoloration of urine and stool up to 48 h (usually gone by 24 h) Permanent tattooing of skin with blue dye Transient blue staining of skin Allergic reaction to dye or radioisotope: ~1–2% Anaphylaxis: 1/2000

Pain score	2–5
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■ PATIENT POPULATION CHARACTERISTICS

Age range	>25 yr
Incidence	~1/8 American women will develop breast cancer.
Etiology	Unknown in most cases; familial history may be related to genetic mutation (BRCA 1 and 2).

ANESTHETIC CONSIDERATIONS FOR BREAST BIOPSY AND SENTINEL LYMPH NODE BIOPSY

PREOPERATIVE

Breast masses may vary in size and depth, which will, in part, determine what type of anesthetic is most suitable for the procedure in these often healthy patients. Excisional biopsies can usually be accomplished with iv sedation and local anesthesia; however, patient wishes must be considered in the anesthetic plan. The suitability of local/MAC versus GA is best addressed by preoperative discussion with the surgical team. A paravertebral block (PVB) may be considered for more extensive breast tissue excisions and when a separate incision is made for lymph node removal (see Anesthetic Technique for Mastectomy section).

Psychosocial	Patients are likely to be anxious concerning the possibility of breast cancer and should be supported and premedicated appropriately.
Laboratory	As indicated from H&P. No testing may be appropriate.
Premedication	Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GA or local anesthesia + iv sedation are both appropriate techniques. Choice of anesthetic technique depends on the size and depth of lesion and the wishes of the patient. Surgery is routinely done on an outpatient basis.

MAC	Propofol infusion (25–100 mcg/kg/min), combination of analgesics (e.g., fentanyl/remifentanyl), and anxiolytics (e.g., midazolam), titrated to effect, are most commonly used. Adding Na bicarbonate to 1% lidocaine (1:10) will reduce injection pain. The anesthesiologist may give remifentanyl 0.5–1 mcg/kg 90 sec before initial injection of local anesthetic into the skin.	
Induction	Standard induction. Mask or LMA anesthetic may be appropriate.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Muscle relaxants are not necessary for surgical procedure.	
Emergence	No special considerations. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 20 ga × 1 BSS @ 3–5 mL/kg/h	See Goal-Directed Fluid Management, Appendix K , p. K-4
Monitoring	Standard monitors. If MAC, then monitor adequacy of ventilation and depth of sedation.	Other monitors as clinically indicated. Isosulfan blue vital dye → artifactual ↓ O ₂ sat as low as 92–94%
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Inadequate analgesia	May have to supplement surgical field block with local anesthetic or convert to GA

	Isosulfan dye reaction	Pruritus, localized swelling, blue hives. Rx: diphenhydramine (10–50 mg iv). ↓ BP may require epinephrine.
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▼ POSTOPERATIVE

Complications	No specific complications anticipated	Inform patients that urine, emesis, or stool may be blue for 24–48 h.
Multimodal pain management	PO analgesics	Emphasize multimodal approach (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.
Tests	As clinically indicated	

Suggested Readings

1. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
2. Giuliano AE, Hunt KK, Ballman KV, et al: Axillary dissection vs. no axillary dissection in women with invasive breast cancer and sentinel node metastasis: a randomized clinical trial. *JAMA* 2011; 305(6):569–75.
3. Moran MS, Schnitt SJ, Giuliano AE, et al: Society of Surgical Oncology-American Society for Radiation Oncology consensus guideline on margins for breast-conserving surgery with whole-breast irradiation in stages I and II invasive breast cancer. *Int J Radiat Oncol Biol Phys* 2014; 88(3):553–64.
4. Morrow M, Van Zee KJ, Solin LJ, et al: Society of Surgical Oncology-American Society for Radiation Oncology-American Society of Clinical Oncology Consensus Guideline on margins for breast-conserving surgery with whole-breast irradiation in ductal carcinoma in situ. *Pract Radiat Oncol* 2016; 6(5):287–95.
5. Vokach-Brodsky L, Jeffrey SS, Lemmens HJM, et al: Isosulfan blue affects pulse oximetry. *Anesthesiology* 2000; 93:102–3.

MASTECTOMY ± RECONSTRUCTION

SURGICAL CONSIDERATIONS

Description

The treatment of invasive and in situ breast cancer has significantly evolved. **Modified radical mastectomy** refers to removal of the breast and the axillary lymph nodes. **Total mastectomy** (simple mastectomy) refers to removal of the breast alone. It is done mainly for treatment of extensive duct carcinoma in situ or prophylaxis in high-risk patients such as those that are mutation carriers of breast cancer susceptibility genes.

Modified radical mastectomy is performed through a single elliptical incision, whereas lumpectomy and sentinel lymph node, or axillary lymph node dissection, are normally done through separate incisions. Postop adjuvant radiation therapy is routinely recommended in breast-conserving surgery and is administered following the completion of adjuvant chemotherapy.

Skin-sparing or **nipple-sparing mastectomies** with immediate breast reconstruction (see below) are increasingly performed to preserve the natural appearance of the breast. These operations are offered to women who have had a good response to neoadjuvant therapies. During the mastectomy, the surgeon preserves the entire skin envelope, including the nipple-areolar complex. A variety of incisions can be used, as shown in [Figure 7.10-3](#). Ischemic skin complications are more likely when a large skin envelope is retained. Skin perfusion can be monitored via fluorescent-based skin angiography using indocyanine green. Oncologic safety should not be compromised, and skin flap thickness should be the same as for total mastectomy. Considerations of skin envelope size or potential ischemic complications may prompt two-staged operations. **Breast reduction surgery** with concomitant lumpectomy and nodal staging is one approach to downsize the skin envelope. In other cases, where obesity and breast ptosis are indicators of ischemic complications, devascularization of the central breast skin, primarily the nipple-areolar complex, is used to stimulate collateral circulation and lessen the likelihood of nipple or skin necrosis with second-stage completion mastectomy. The placement of incisions is radial-lateral, inframammary fold, or vertical in lower half of the breast. Sometimes, they may be extended around part of the areolar edge.

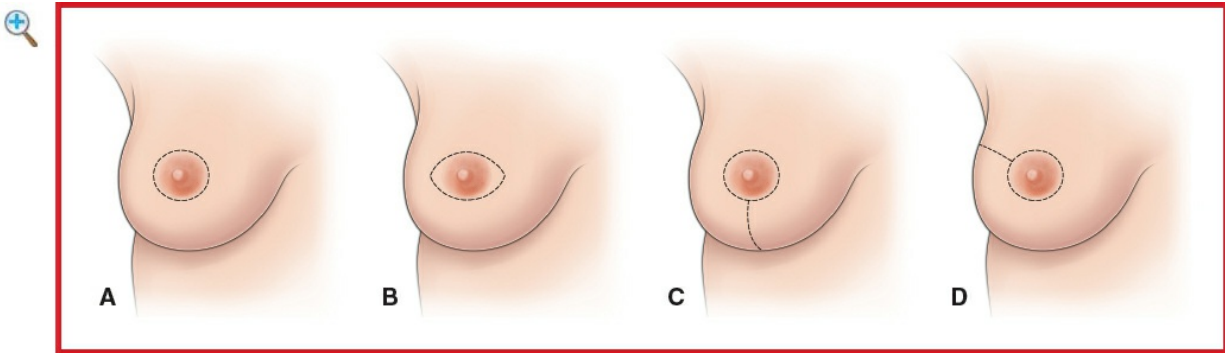


Figure 7.10-3 Examples of skin and nipple-sparing incisions.

Postmastectomy reconstruction is an option for most women undergoing mastectomy. Patients undergoing mastectomy are generally presented with two options for breast reconstruction, that is, implant-based versus autologous reconstruction. Additionally, the issue of timing is discussed, that is, immediate versus delayed reconstruction.

Today, most women opting for postmastectomy reconstruction undergo **immediate reconstruction**. The benefit of immediate reconstruction, particularly in cases of skin- and nipple-sparing mastectomy, is that optimal reconstructive outcomes can be achieved without the patient having to experience the psychological trauma of iatrogenic amastia.

Both reconstructive options have distinct advantages and disadvantages. Implant-based reconstruction is characterized by technical simplicity and absent donor-site morbidity. However, the most commonly chosen approach, that is, 2-stage expander-implant reconstruction, necessitates a period of expansion that separates two surgical procedures. The first stage occurs at the time of mastectomy. Here, a tissue expander is placed either in the submuscular, subpectoral (dual-plane), or prepectoral plane. The advantages and disadvantages of each plane are beyond the scope of this chapter. Of note, the majority of surgeons today use acellular dermal matrices when using the dual-plane or prepectoral approach to optimize outcomes. At the second stage, the tissue expander is removed and the final implant, typically a silicone gel implant, is inserted.

Autologous reconstruction utilizes the patient's tissue for breast reconstruction and as such adheres to the time-honored principle of replacing like with like. Benefits include the ability to recreate a natural breast ptosis as well as absence of implant-related complications (e.g., capsular contracture, or implant malposition.). It is noteworthy that long-term patient satisfaction has been demonstrated to be higher with autologous versus implant-based reconstruction. The most common donor site is the abdomen. However, alternative donor sites include the thighs and buttocks. Advances in surgical technique have resulted in flap survival rates of 98–99% at microsurgical centers. Hence, over the past four decades, focus has shifted away from flap survival rates and toward donor-

site morbidity. This shift is best exemplified when analyzing the various techniques that allow transfer of abdominal tissue. While pedicled transverse rectus abdominis myocutaneous flaps were popular in the 1980s, the development of perforator-based techniques that spare the rectus abdominis muscle entirely (i.e., deep inferior epigastric artery perforator [DIEP] flaps) is considered the gold standard today.

It is important to consider the effects of radiation on postoperative outcome. Radiation is associated with a marked increase of implant-related complications, which is why many surgeons advise against implant-based reconstruction in these cases. However, radiotherapy following autologous reconstruction can likewise be associated with untoward sequelae, including flap fibrosis, and volume loss. Hence, the issue of radiation needs to be taken into account when deciding on the reconstructive plan. An elegant approach in patients who are known to require postmastectomy radiotherapy is a delayed immediate approach. Here, a tissue expander is placed at the time of mastectomy. Subsequent expansion maximizes preservation of native breast skin. The patient then proceeds with radiotherapy. Approximately 6 mo post radiotherapy, the tissue expander is removed and autologous reconstruction performed. The key principle is that preserving native breast skin leads to a better reconstructive outcome.

Usual preop diagnosis

Invasive or in situ breast cancer; high-risk patients (e.g., genetic susceptibility to breast cancer; family history of breast cancer)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Modified Radical Mastectomy	Total Mastectomy	Lumpectomy, Axillary Lymph Node Dissection
Position	Supine with ipsilateral arm abducted and may be prepped on field. Avoid brachial plexus stretch. Repositioning may be	⇐	⇐

	required (latissimus dorsi reconstruction).		
Incision	Elliptical oblique or elliptical transverse to include nipple/areola and previous biopsy; periareolar, or “tennis racquet.”	⇐	Incision over breast mass or previous biopsy site. Separate transverse or oblique incision in axilla.
Unique considerations	Avoid iv and BP cuff on ipsilateral arm; many surgeons prefer no paralytics during axillary portion of surgery.	⇐	⇐
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	1.5–3 h (+1–7 h, if immediate breast reconstruction performed); longer for bilateral surgery	1–2 h (+1–7 h, if immediate breast reconstruction performed)	1–3 h
Closing considerations	Gauze bandage over incision	⇐	⇐
EBL	150–500 mL, depending on whether scalpel or electrocautery is used.	⇐	25–100 mL
Postop care	PACU → 2 d hospitalization	PACU → 1–2 d hospitalization or occasionally → home	PACU → home
Mortality	Rare	⇐	⇐
Morbidity	Lymphedema: 5–30%	⇐	⇐

(depending on extent of axillary dissection)

Seroma: 25%	⇐	⇐
Infection: 2–10%	⇐	⇐
Flap necrosis: <5%	⇐	—
Hematoma: <5%	⇐	<10%
Injury to axillary neurovascular structures: Rare		
Pneumothorax: Rare (may occur with attempts to obtain hemostasis of intercostal perforating vessels)	⇐	⇐

Pain score	4–8	4–6	4–8
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■ PATIENT POPULATION CHARACTERISTICS

Age range	20–90 yr (generally >40 yr)
Incidence	Over their lifetime, ~1/8 of American women develop breast cancer. In 2013, according to American Cancer Society, it is estimated that there will be 232,340 new cases of invasive breast cancers in women and 2240 in men; there are also expected to be 64,640 new cases of carcinomas in situ.
Etiology	Unknown in most cases; about 5–10% of female breast cancers and 4–40% of male breast cancers may be related to a gene such as BRCA1 and BRCA2.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Invasive breast cancer is frequently treated with multiple therapies, including surgery,

radiation therapy, and chemotherapy. Patients often have received neoadjuvant therapy (before surgery) and may include chemotherapy, hormonal therapy, or biologic therapy to induce tumor response before surgery and enable breast conservation. Less than 5% of patients will have distant metastasis at time of diagnosis. In that small group, consider implications of distant spread, typically to the lung, brain, liver, and bone. Consider systemic effects of recent neoadjuvant therapies.

Respiratory	Respiratory compromise can be present if patient has received XRT to the thorax as part of treatment. Tests: Consider CXR if pulmonary Sx or signs of disease present. Others as indicated by H&P
Cardiovascular	Cardiac toxicity may occur from chemotherapy for breast cancer. Anthracyclines (e.g., doxorubicin at doses $>550 \text{ mg/m}^2$) may cause significant and often irreversible cardiomyopathy. The monoclonal antibody trastuzumab may cause $\downarrow$ in LV function that is usually reversible. Tests: ECG if age >65 or if cardiac disease or Sx present. Review any relevant cardiac studies. Consider echo if at risk for cardiomyopathy or HF Sxs present.
Neurologic	Breast cancer may metastasize to the CNS and can present with focal neurologic deficits, $\uparrow$ ICP, or altered mental status. Tests: As indicated by H&P
Hematologic	Look for anemia and thrombocytopenia following chemotherapy. Tests: CBC, with differential and Plt count
Laboratory	Others as indicated by H&P
Premedication	Anxiolysis: Consider midazolam 1–2 mg iv. Analgesia: Consider acetaminophen 1000 mg po, gabapentin 300–600 mg po, and/or celecoxib 400 mg po.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop

pain management. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

GETA or GA with LMA. Regional anesthesia (PVB) in breast surgery is associated with less PONV, less postop pain, earlier discharge from the hospital, and less chronic incisional pain.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. The use of muscle relaxants during axillary dissection should be avoided to permit surgical identification of nerves by nerve stimulator or if electrocautery is used in the axilla.
Emergence	Pressure dressings may be applied with the patient anesthetized and “sitting up” at the end of the procedure. Discuss with surgeons whether they intend to apply this type of dressing, to enable appropriate timing of emergence. Consider PONV prophylaxis. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >46 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Regional anesthesia

Unilateral multiple-level PVB provides satisfactory anesthesia for modified radical mastectomy and lumpectomy with axillary lymph node dissection. A block from T1-T6 is required. Suitable local anesthetics (4–5 mL/level) are 0.5% bupivacaine or 0.5%

ropivacaine with 1:400,000 epinephrine. Sedation is useful during block placement and is continued intraop. (See Anesthetic Considerations for Breast Biopsy) PVB is contraindicated for the following reasons: (a) patient refusal, (b) local anesthetic allergy, (c) pathology or previous surgery → anatomic distortion of paravertebral space, or (d) infection at sites of injection.

Blood and fluid requirements	EBL 25–500 cc depending on extent of surgery. IV: 18 ga × 1–2 (avoid operative side) BSS @ 3–5 mL/kg/h	See Goal-Directed Fluid Management, p. K-4
Monitoring	Standard monitors	Others as indicated by patient status. BP cuff on arm opposite surgical site or leg
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Pneumothorax	Deep surgical exploration may cause inadvertent pneumothorax; monitor patient for Sx (e.g., ↑ PIP, ↓ PaCO ₂ , asymmetric breath sounds, hemodynamic instability). Dx: CXR. Rx: Chest tube and ↑ FiO ₂
Complications 2° PVB	Inadequate block (10%) Pleural puncture Pneumothorax Horner syndrome Inadvertent epidural spread of local anesthetic	

Complications	Pneumothorax PONV Psychological trauma	If index of suspicion for pneumothorax is high, maintain oxygenation (100% FiO ₂) and ventilation; inform surgeons of the likelihood of the Dx. If patient is hemodynamically stable and not hypoxemic, a portable CXR may aid in diagnosis. Consider multimodal intraoperative prophylaxis based on PONV risk: i.e., ondansetron 4–8 mg, dexamethasone 4–8 mg, propofol infusion, scopolamine patch.
Multimodal pain management	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia or TAP blocks (see p. K-5) RA PCA	PO/IV analgesics: Consider appropriate combination of acetaminophen 1000 mg po/iv q6h, gabapentin 100–600 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h. Supplemented if necessary with oxycodone 5–10 mg po q6h or hydromorphone 0.2–.05 mg iv for breakthrough pain.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	Postop portable CXR, if pneumothorax is suspected	

Suggested Readings

1. Andreae MH, Andreae DA: Regional anaesthesia to prevent chronic pain after surgery: a Cochrane systematic review and meta-analysis. *Br J Anaesth* 2013; 111(5):711–20.
2. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
3. Bertoni DM, Nguyen D, Rochlin D, et al: Protecting nipple perfusion by devascularization and surgical delay in patients at risk for ischemic complications during nipple-sparing mastectomies. *Ann Surg Oncol* 2016; 23(8):2665–72.
4. Greengrass RA, Ducloux R: Paravertebral blocks. *Int Anesthesiol Clin* 2012; 50(1):56–73.
5. Wu J, Buggy D, Fleischmann E, et al: Thoracic paravertebral regional anesthesia improves analgesia after breast cancer surgery: a randomized controlled multicentre clinical trial. *Can J Anaesth* 2015; 62(3):241–51.

CHAPTER 7.11

Endocrine Surgery

SURGEONS

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EXCISION OF THYROGLOSSAL DUCT CYST

SURGICAL CONSIDERATIONS

Description

Thyroglossal duct cysts typically are located in the midline at or below the hyoid bone (Fig. 7.11-1). Differential diagnoses after initial physical exam include epidermoid cysts or lymph nodes. Cysts should be removed because of an associated high risk of infection with oral flora and a slight (<1%) risk of either squamous cell or papillary thyroid cancer developing in the cyst itself. A transverse skin incision is made over the cyst, and if the cyst was previously infected and sinus tracts through the skin are present; the skin should be removed along with the cyst. The cyst is identified and followed cephalad to the hyoid bone (Fig. 7.11-2). Then, the midportion of the hyoid bone is resected to minimize recurrence. There may be many small tracts associated with the cyst that tend to attenuate beyond the hyoid bone. The base of the tract or tracts is resected up to the level of the floor of the mouth at the foramen cecum and ligated with absorbable suture (**Sistrunk procedure**). The wound is irrigated copiously and closed in layers. The wings of the hyoid bone are not reapproximated.

Usual preop diagnosis

Thyroglossal duct cyst

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery](#), p. K-1.

■ SUMMARY OF PROCEDURE	
Position	Supine, with neck in hyperextended position
Incision	Transverse skin incision
Unique considerations	Surgeon may request assistance, by placing finger at base of tongue to identify the cephalad extent of the needed dissection.
Antibiotics	Cefazolin 2 g iv
Surgical time	1–1.5 h
Closing considerations	Careful hemostasis. Coughing may be associated with venous congestion and hematoma formation.
EBL	5–10 mL
Mortality	<0.1%
Morbidity	Bleeding: <5% Infection: <5%
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS	
Age range	6 mo to 30 yr (40% present at <10 yr)
Male:Female	1:1
Incidence	~1:3000
Etiology	Persistence of undifferentiated epithelial cells in area of hyoid bone that later become squamous cell epithelium or glandular tissue

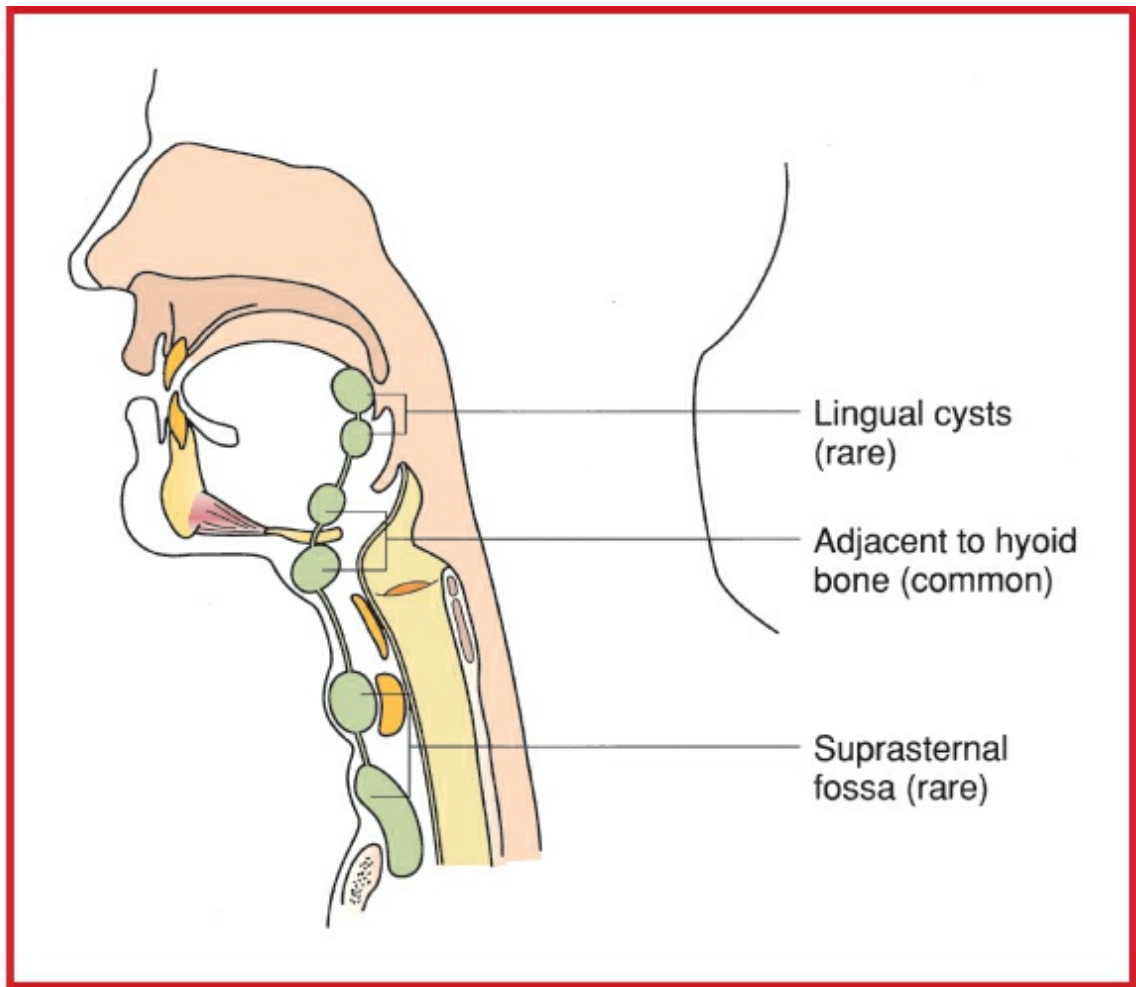


Figure 7.11-1 Location of thyroglossal duct cysts. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

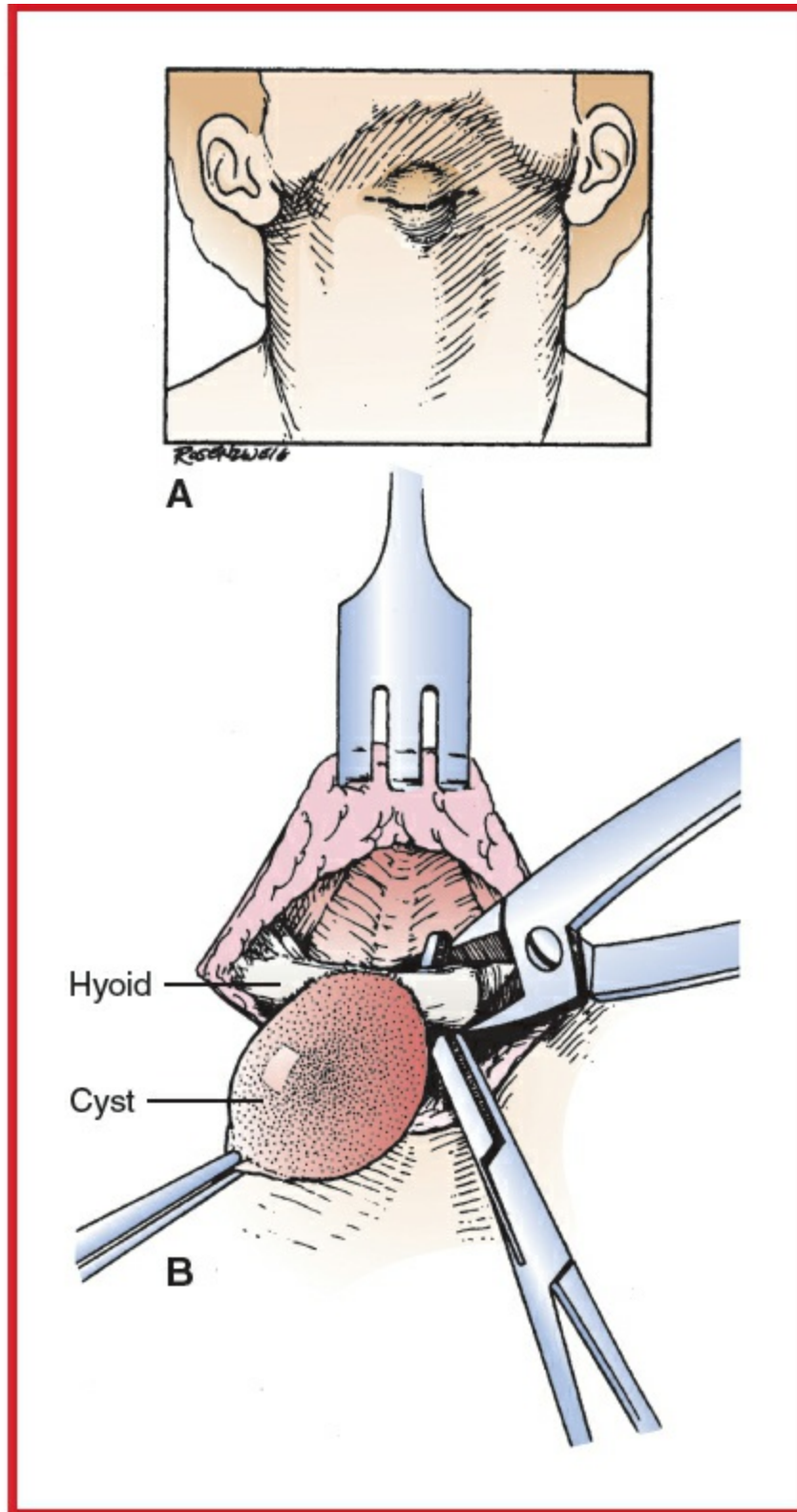


Figure 7.11-2 Thyroglossal duct cyst excision. **A:** Incision placed over presenting cyst; no skin excised. **B:** Cyst has been dissected from surrounding tissues, and hyoid is exposed after division of sternohyoid and thyrohyoid muscles at insertion. The bone is encircled with a short right-angle clamp 1 cm from its midpoint, where it is divided with a bone cutter or cautery. (Reproduced with permission from Baker RJ, Fischer JE: *Mastery of Surgery*, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Thyroidectomy, [p. 748](#).

Suggested Readings

1. Oomen KPQ, Modi VK, Maddalozzo J: Thyroglossal duct cyst and ectopic thyroid surgical management. *Otolaryngol Clin North Am* 2015; 48(1):15–27.
2. Ross J, Manteghi A, Rethy K, et al: Thyroglossal duct cyst surgery: a ten-year single institution experience. *Int J Pediatr Otorhinolaryngol* 2017; 101:132–6.

THYROIDECTOMY

SURGICAL CONSIDERATIONS

Description

Thyroidectomy is performed through a transverse neck incision ([Fig. 7.11-3](#)), usually 6–8 cm long. In the **traditional approach**, the platysma muscle is divided sharply, and subplatysmal flaps are developed superiorly and inferiorly. The two large anterior jugular veins must be avoided and are occasionally a source of blood loss, although rarely of any hemodynamic significance. Once the flaps are adequately developed, a spring or self-retaining retractor may be placed to expose the midline prethyroid fascia (median raphe). This is divided in the midline to expose the strap muscles, which can then be separated from the thyroid gland.

After the thyroid gland is exposed, resection can proceed. Resection may be total, subtotal (lobe + isthmus ± partial remaining lobe), or lobar. Degree of resection depends on diagnosis and may be modified based on operative findings. During this portion of the operation, hemostasis is critical to maintain adequate visualization. Resection of a lobe usually begins with ligation and division of the middle thyroid vein along the midlateral aspect of the gland ([Fig. 7.11-4](#)).

The superior pole is mobilized by controlling and dividing the superior thyroid vessels close to the thyroid capsule to avoid injury to the external branch of the superior laryngeal nerve. As the dissection proceeds, care is taken to identify and preserve the superior and inferior parathyroid glands. The lobe is retracted medially to expose the tracheoesophageal groove. The recurrent laryngeal nerve (RLN) is then visualized and traced along its entire course. Its function may be confirmed using a nerve monitor. The gland is gently dissected away from the nerve and then mobilized off the trachea to

complete the resection. Any enlarged or suspicious lymph nodes are also excised and sent for pathologic examination. Before closing, hemostasis is ensured and can be tested with a Valsalva maneuver. The midline fascia and platysma are closed using absorbable suture and the skin with a running monofilament suture. The use of drains remains controversial and has not been shown to decrease the rate of hematoma formation.

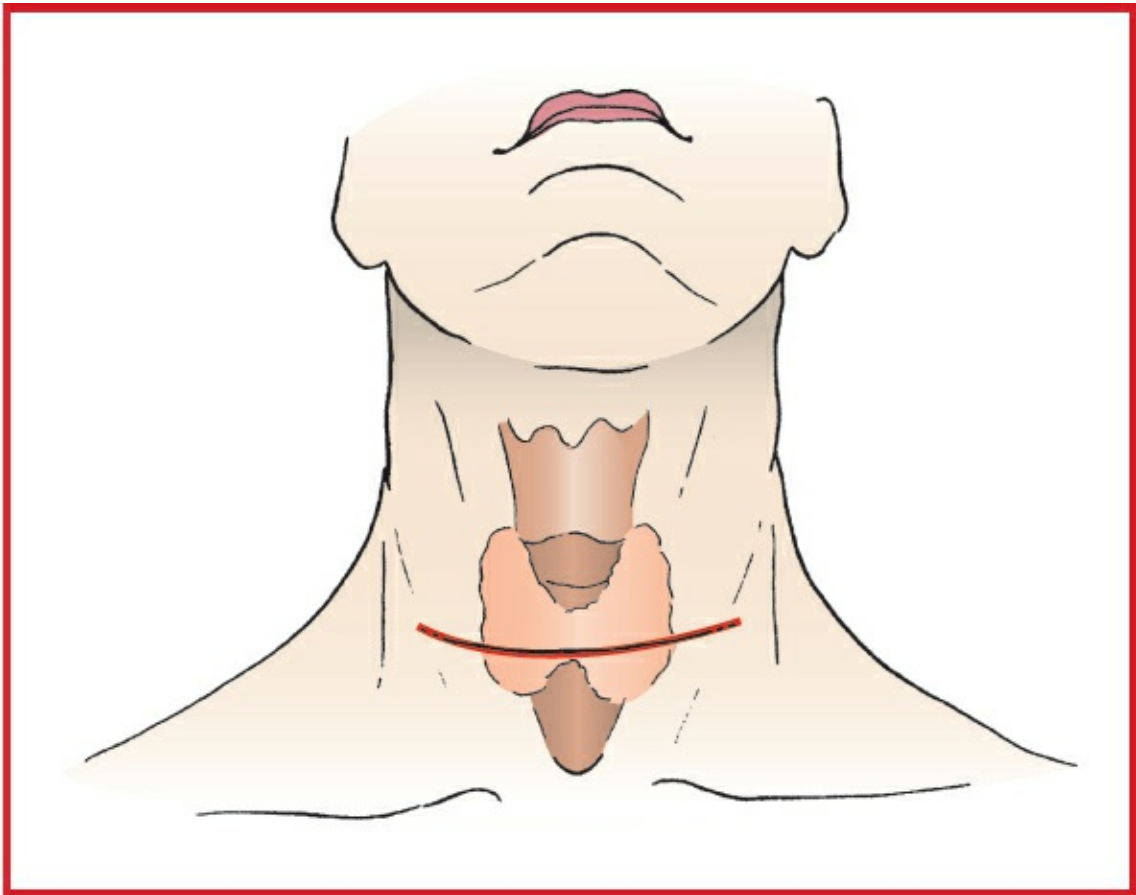


Figure 7.11-3 Transverse incision at base of neck for thyroidectomy. (Reproduced with permission from Baker RJ, Fischer JE *Mastery of Surgery*, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

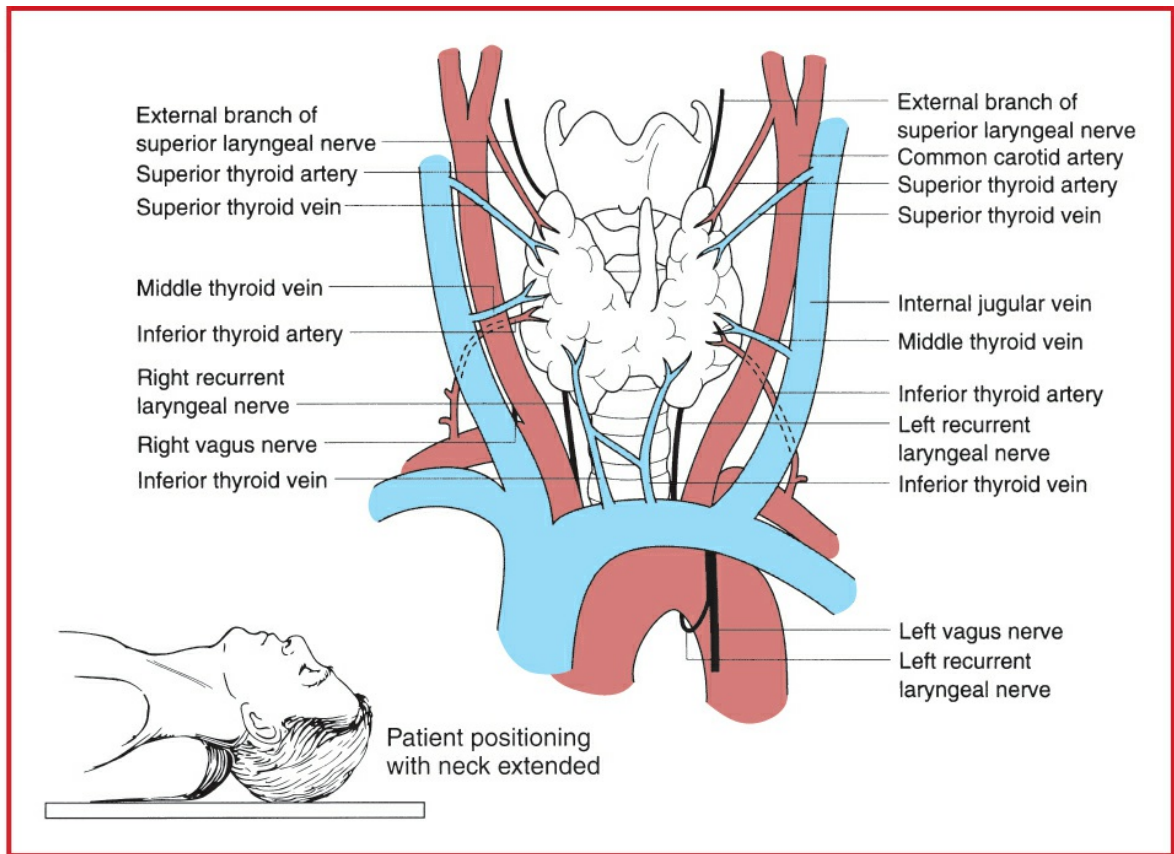


Figure 7.11-4 Vascular relationships to the thyroid gland. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.) **Inset** shows patient positioning with neck extended.

Minimally invasive techniques, such as video-assisted mini-incision or remote-access thyroidectomy via an endoscopic, transoral, or robotic approach, have been described but have not gained widespread use. Video-assisted thyroidectomy has been shown to have similar rates of cure with superior voice preservation, improved cosmesis, shortened hospitalizations, and faster patient recovery when compared with conventional open surgery. Remote-access procedures have utilized various entry sites to reach the thyroid, namely, the mouth, anterior chest wall, breast, axilla, and postauricular area through a facelift incision. Remote-access surgery avoids the presence of a cervical scar altogether while offering enhanced visualization and dexterity as well as comparable postoperative outcomes. Limitations that have prevented widespread adoption of these minimally invasive approaches include narrow patient selection criteria (based on the nature, size, and extent of thyroid pathology as well as body habitus), surgeon inexperience, lengthy operative time, and higher cost.

Intraoperative nerve monitoring (IONM) is increasingly being used as an adjunct to identify, dissect, and confirm the function of the recurrent laryngeal nerve. Specialized endotracheal tubes are inserted with surface electrodes contacting the luminal surface of

the vocal cords on either side. The electrodes are connected to a monitoring device that continually senses EMG activity of the thyroarytenoid muscles. When the surgeon stimulates the RLN using a probe wired to the monitor, the device detects the electrical impulse from the vocal cords and sounds an audible alert, as well as a visual signal in some systems.

Usual preop diagnosis

FNA findings of definite/suspicious/inconclusive for malignancy; goiter; thyroid cancer (papillary, follicular, medullary, anaplastic); thyroid nodule; hyperthyroidism; Graves disease

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine with head elevated to 30° and neck extended (see Fig. 7.11-4, inset)
Incision	Transverse cervical
Unique considerations	Patients with uncontrolled hyperthyroidism are at ↑ risk for developing thyroid storm during surgery. Hyperthyroidism may be controlled preop with either β-adrenergic blockade or propylthiouracil. Preop iatrogenic hypothyroidism may be associated with ↓ BP and circulatory collapse during induction of anesthesia.
Antibiotics	None routinely used
Surgical time	1–2 h
Closing considerations	Adequate hemostasis; minimize coughing
EBL	50–75 mL
Postop care	Observe for postop respiratory problems 2° recurrent laryngeal nerve injury (bilateral), hematoma, or ↓ Ca ⁺⁺ . Admit × 24 h. ✓ Ca ⁺⁺ the night of surgery and the

following a.m. Same-day surgery is gaining ground, especially for thyroid lobectomy. Patients who undergo total thyroidectomy are often kept overnight to monitor for development of delayed hematoma (16 h) and hypocalcemia.

Mortality	<0.5%
Morbidity	Hypoparathyroidism ($\downarrow$ Ca^{++}): 3–5% Hematoma: 1–2% Thyroid storm (usually in association with Graves disease) Wound infection: 0.2–0.5% Recurrent laryngeal nerve damage Unilateral (hoarseness): 0.77% Bilateral (aphonia, respiratory obstruction): 0.39%
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–80 yr
Male:Female	Overall: 1:8 Cancer: papillary (1:2); follicular (1:3); medullary; anaplastic; lymphoma (1:1)
Incidence	Cancer (uncommon) Benign (relatively common)
Etiology	Cancer (50%): papillary (80%); follicular (10%); medullary (5%); anaplastic (1%); lymphoma (1%) Benign lesions (50%): nontoxic goiter (20%); thyrotoxicosis (10%); thyroiditis (5%); benign nodules (5%); other (10%)
Associated conditions	Other endocrine disorders (e.g., pheochromocytoma in association with medullary thyroid carcinoma in patients with MEN 2A and 2B)

ANESTHETIC CONSIDERATIONS

(Procedures covered: excision of thyroglossal duct cyst; thyroidectomy.)

PREOPERATIVE

Hyperthyroidism may be 2° Graves disease (common), toxic multinodular goiter, thyroid adenomas, TSH-secreting tumor (rare), or overdosage of thyroid hormone. Common Sx are fatigue, sweating, intolerance to heat, ↑ appetite, ↑ HR, ↑ BP, ↑ pulse pressure, ↑ T, weight loss or gain, thyroid goiter, and exophthalmos. Some older patients exhibit apathetic thyrotoxicosis, which is often mistaken for hypothyroidism. CHF and AF are common with these patients. Hypothyroidism may be iatrogenic or 2° autoimmune thyroiditis. Common Sx are intolerance to cold, anorexia, fatigue, weight gain or loss, constipation, ↓ HR, ↓ pulse pressure, ↓ DTR, ↓ T, and ↓ mentation. With very few exceptions thyroidectomy is an elective surgery. An important aspect of the preop visit is to ensure that the patient is in a physiologically euthyroid state (normal T, HR, pulse pressure, reflexes). Patients presenting for thyroidectomy usually are made euthyroid before surgery and may be taking one or more of the following medications: propylthiouracil, methimazole, potassium iodide, glucocorticoids, or β-blockers.

According to the 2016 American Thyroid Association Guidelines, patients undergoing thyroid surgery for hyperthyroidism should receive oral Lugol solution (potassium iodide, SSKI) to block iodine uptake and the secretion of thyroid hormone and should receive a synthesis blocker (propylthiouracil or methimazole) at least 1 h before. Lugol solution may also decrease intraoperative bleeding by decreasing vascularity of the gland. Preoperative vitamin D and calcium supplementation are also recommended for the risk of symptomatic hypocalcemia postoperatively.

General

Although cases have been reported describing total thyroidectomy for noneuthyroid patients with worsening cardiac function who have failed aggressive medical therapy.

Respiratory	Beware of tracheal compression with large goiters, → tracheal deviation, stridor, and intolerance of the supine position. Tests: CXR; consider preop ultrasound or CT scan of neck to evaluate possible tracheal involvement, especially in patients with large goiters.			
Endocrine	T4	T3ru	T3	TSH
Hyperthyroid	↑	↑	↑	Normal or ↓
1° hypothyroid	↓	↓	↓ or normal	↑
2° hypothyroid	↓	↓	↓	↓
Tests: Thyroid function; Ca ⁺⁺ ; Mg ⁺⁺ ; phosphate; alkaline phosphatase; glucose				

Hyperthyroidism

Respiratory	↑ BMR → ↑ VO ₂ → rapid desaturation on induction
Cardiovascular	↑ HR, AF (10–40% incidence), palpitations, CHF. A normal resting HR helps determine whether the patient is ready for surgery. If the situation calls for it (e.g., emergency surgery), the patient can be treated with β-blockers to blunt the sympathomimetic effects of the hyperthyroid state. β-blocker therapy can be problematic in patients with CHF (titrate while monitoring filling pressures and CO). Tests: ECG; consider preoperative ECHO for evaluation of LV function.
Neurologic	Warm, moist skin, nervousness, anxiety (may require generous sedation), tremor, ↑ DTRs
Musculoskeletal	Higher incidence of myasthenia gravis and skeletal muscle weakness (↑ sensitivity to muscle relaxants), clubbing of the fingers, weight loss, myopathy Tests: CK, urine myoglobin, BMP
Hematologic	Mild anemia, thrombocytopenia Tests: CBC
Gastrointestinal	Weight loss and diarrhea Tests: BMP + others as indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Midazolam 1–2 mg iv. Continue antithyroid medications preop. Hyperthyroid patients must be made euthyroid before elective surgery and may be on the following drugs: propylthiouracil, methimazole, potassium iodide, β-blockers, and glucocorticoids.
ERAS	Check compliance with preop ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.
Thyroid storm	A life-threatening exacerbation of hyperthyroidism

occurring during periods of stress, which is manifested by hyperthermia ($>40^{\circ}\text{C}$), tachycardia, widened pulse pressure, anxiety, altered mental state $\rightarrow$ psychosis $\rightarrow$ coma, and myopathy (rhabdomyolysis in 50%; severe in 4%). (Thyroid storm has been mistaken intraop for malignant hyperthermia, sepsis, anaphylaxis, and other hypermetabolic reactions, e.g., neuroleptic malignant syndrome.) Thyroid storm is most often associated with Graves disease that has been incompletely treated before surgery. **General Rx:** $\uparrow$ FiO_2 ; fluid resuscitation; electrolyte replacement/correction ($\uparrow$ Ca^{++}); cooling blankets; acetaminophen; maintain diuresis (**maintain euvoemia**) if rhabdomyolysis; treat precipitating event (infection, CHF, DKA, pregnancy). **Specific Rx:** propylthiouracil (block synthesis) (200–250 mg po q4h); sodium iodide (block release; 1–2.5 g iv); steroids (mechanism unclear)—hydrocortisone (100 mg iv q8h), or dexamethasone (4 mg iv q24h); β -blockers (use with caution in patients with reactive airway disease, AV block, or CHF)—propranolol (20–120 mg po qh or 0.25–1.0 mg iv q5min), and/or esmolol (50–300 mcg/kg/min). Note: Block synthesis **before** (1 h is adequate) giving iodides to block release, otherwise “iodine escape” will occur later.

Hypothyroidism

Respiratory

Beware of tracheal compression with large goiters $\rightarrow$ tracheal deviation, stridor. $\downarrow$ ventilatory response to $\uparrow$ CO_2 and $\downarrow$ O_2 (caution with opioids and sedatives). **Tests:** CXR; consider preop CT scan of the neck to evaluate possible tracheal involvement, especially in patients with large goiters.

Cardiovascular

Bradydysrhythmias, diastolic HTN, pericardial effusions, ECG $\rightarrow$ $\downarrow$ voltage, ST-T wave Δ 's, $\uparrow$ QT, occasional VT (Torsades de pointes—pause-

	dependent). This type of VT is treated with MgSO_4, cardioversion; then isoproterenol or pacing to shorten QT. Thyroid replacement must be weighed against the risk of precipitating myocardial ischemia in patients with known CAD. Diastolic dysfunction, $\downarrow$ LV compliance $\rightarrow$ cautious volume expansion. Tests: ECG; consider preop ECHO for evaluation of LV systolic/diastolic function, pericardial effusion/tamponade.
Endocrine	Addison disease occurs in 5–10% of patients with severe hypothyroidism; some patients may receive a “stress dose” of steroids (hydrocortisone 100 mg iv q8h $\times$ 3) in the periop period. Tests: Cortisol stimulation test
Neurologic	$\downarrow$ BMR $\rightarrow$ slow mentation and movement, cold intolerance, $\downarrow$ reflexes with “hang up” (delayed relaxation phase)
Musculoskeletal	Arthralgias and myalgias
Renal	Impaired renal function 2° amyloidosis, urinary retention, oliguria. (50% incidence of $\downarrow$ Na^+) Tests: Consider BUN; Cr; Na^+
Hematologic	Coagulation abnormalities, anemia Tests: CBC
Gastrointestinal	GI bleeding, constipation, ileus Tests: As indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Midazolam 1–2 mg iv. (None in the patient who is clinically hypothyroid and requires emergent surgery.) Hypothyroid patients can safely undergo anesthesia/surgery if they have mild-to-moderate disease. Clinically hypothyroid patients ($\downarrow$ HR, $\downarrow$ T, $\downarrow$ pulse pressure, $\downarrow$ DTRs) should be given thyroid replacement before elective surgery.
ERAS	Check compliance with preop ERAS recommendations (see above) including NPO status. Consider preop

Myxedema comma

acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

Severe hypothyroidism constituting a medical emergency, with mortality of >50%. Manifestations include stupor or coma, hypothermia (which correlates inversely with mortality), hypoventilation with hypoxemia, bradycardia (HR 50–60), hypotension, apathy, hoarseness, and hyponatremia. **Supportive measures:** early intubation and mechanical ventilation; treat ↓ BP with **cautious** volume expansion (risk of pulmonary edema), inotropes (risk of arrhythmias), pacing (carefully, 60–70 b/min), and r/o pericardial effusion, passive rewarming only, especially if ↓ BP (active warming for $T < 30^{\circ}\text{C}$); correct ↓ Na^+ carefully (risk of central pontine myelinolysis); correct ↓ glucose. **Specific Rx:** L-thyroxine (T_4 ; 200–500 mcg iv loading dose, 100–300 mcg iv the next day); or tri-iodothyronine (T_3 ; 20–50 mcg iv q6–12h); hydrocortisone (100–300 mg iv q d). T_4 onset is slow (6 h after iv administration) and has to be converted peripherally (slowed in hypothyroid state) to biologically active T_3 . Also, T_4 may be converted in some critically ill patients to biologically inactive rT_3 (“reverse T_3 ”). ↓ TSH level is the earliest sign of response.

◆ INTRAOPERATIVE

Anesthetic technique

Normally GETA, infrequently under local anesthesia. If using IONM, insert the specialized endotracheal tube, preferably with the aid of a video laryngoscope, and ensure direct contact of the electrodes with the vocal cords on either side. A short-acting, nondepolarizing paralytic may be given to allow for full relaxation during intubation. For inadequately treated hyperthyroid patients, it is key to establish an adequate depth of anesthesia to prevent an exaggerated sympathetic response to surgical stimulation. Avoid agents that stimulate the sympathetic nervous system (e.g., ketamine, pancuronium, meperidine). Hypothyroidism may be associated with ↑

sensitivity to anesthetic agents and muscle relaxants.

Induction	Standard induction for euthyroid patients. If the patient has airway compromise 2° a large thyroid goiter, consider awake fiberoptic intubation. If using IONM, intubation via a video laryngoscope allows the anesthesiologist and surgeon to visualize and confirm the position of the ETT electrodes in relation to the vocal cords.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Maintain muscle relaxation, except if using IONM (avoid paralytic agents). Use nerve stimulation to guide relaxant dosing.	
Emergence	Airway obstruction 2° recurrent laryngeal nerve damage, tracheomalacia (especially in patients with large destructive goiters), or hematoma can occur. Consider visualizing vocal cord function before extubation. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 5–8 mL/kg/h Head-up position	Slight head-up position can help make for a bloodless surgical field without substantially increasing the risk of VAE. See Goal-Directed Fluid Management, Appendix K , p. K-4.
Monitoring	Standard monitors EMG ETT for IONM (e.g., NIM TriVantage®)	+ Others as indicated by patient's status (i.e., pulmonary artery catheter in the setting of emergent surgery in thyrotoxic patients with congestive heart failure). Maintain normothermia , especially in hypothyroid

		patients.
Positioning	✓ pad pressure points ✓ eyes	Supine, with head slightly hyperextended allows for surgical exploration of the neck.
Complications	Cardiorespiratory depression	In hypothyroid patients, marked ↓ BP and ↓ RR may occur with minimal anesthetic doses.

▼ POSTOPERATIVE

Complications	Recurrent laryngeal nerve damage	Bilateral: patient will be unable to speak and will require reintubation. CPAP may temporize situation and make reintubation less emergent. Unilateral: characterized by hoarseness.
	Tracheomalacia or hematoma with airway compromise	Acute airway obstruction may occur immediately postop, and rapid reintubation may be lifesaving. If airway compromise is 2° hematoma, reopen incision and drain remaining blood; if patient still requires artificial airway, consider CPAP or awake reintubation.
	Acute hypoparathyroid state (hypocalcemia)	Acute hypocalcemia can present as laryngeal stridor (24–48 h postop), although it most often presents with tingling in the fingertips and lips. If untreated and severe, this can progress to tetany and seizures. Administering 1

		amp Ca ⁺⁺ gluconate given iv over 20 min usually alleviates symptoms. Rx: measure Ca ⁺⁺ ; replace if necessary. CPAP is effective for airway compromise. Can mimic MH (see Rx above).
Multimodal pain management	PCA hydromorphone or fentanyl Oral narcotic analgesics	Emphasize nonopioid analgesics, including acetaminophen and NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.
Tests	Vocal cord function	Ability to phonate “e” implies continued vocal cord function.

Suggested Readings

1. Abbas G, Dubner S, Heller KS: Re-operation for bleeding after thyroidectomy and parathyroidectomy. *Head Neck* 2001; 23(7):544–6.
2. Khadra H, Mohamed S, Hauch A, et al: Safety of same-day thyroidectomy: meta-analysis and systematic review. *Gland Surg* 2017; 6(4):292–301.
3. Kitahara CM, Sosa JA: The changing incidence of thyroid cancer. *Nat Rev Endocrinol* 2016; 12:646–53.
4. Lee C, Stack BC: Intraoperative neuromonitoring during thyroidectomy. *Expert Rev Anticancer Ther* 2011; 11(9):1417–27.
5. Ross DS, Burch HB, Cooper DS, et al: 2016 American Thyroid Association Guidelines for diagnosis and management of hyperthyroidism and other causes of thyrotoxicosis. *Thyroid* 2016; 26(10):1343–421.
5. Scotto L, Citron L, Mihm FG, et al: Airway ultrasonography and its clinical application to airway management. In: Orloff LA, ed. *Head and Neck Ultrasonography*, 2nd Edition. Plural Publishing, Inc., San Diego: 2017, 283–304.
7. Sinclair CF, Bumpous JM, Haugen BR, et al: Laryngeal examination in thyroid and parathyroid surgery: an American Head and Neck Society consensus statement: AHSN Consensus Statement. *Head Neck* 2016; 38(6):811–9.

PARATHYROIDECTOMY

SURGICAL CONSIDERATIONS

Description

The traditional approach for primary hyperparathyroidism requires **four-gland visualization** with removal of any abnormally large glands. Newer methods of localization and confirmation of adenomas have ushered in an era of **minimally invasive parathyroid surgery**. Minimally invasive techniques involve a focused operative dissection via **mini-incision, endoscopic, or robotic approach**. **Focused parathyroidectomy** is aided by the preop localization studies, the mainstays of which are ultrasound and sestamibi scanning. If localization fails to identify the correct gland, then the minimally invasive approach is converted to the standard incision, and all four glands are explored until the diseased gland is identified.

For the standard open operation, an incision in the lower neck ([Fig. 7.11-5](#)) is made, and the platysma is divided. Subplatysmal flaps are created superiorly and inferiorly to allow for increased working space and to expose the prethyroid fascia. The midline between the strap muscles is opened, exposing the thyroid gland. The superior parathyroids are located behind the upper pole, which may be mobilized to reveal the parathyroid. The inferior glands are located near the junction of the inferior thyroid artery and the recurrent laryngeal nerve ([Fig. 7.11-6](#)). Hemostasis is crucial to maintaining adequate visualization in the surgical field. In the traditional approach, all four glands are identified and biopsied for confirmation of parathyroid tissue. Abnormal glands are removed. If four-gland hyperplasia is found, which may occur with multiple endocrine neoplasia (MEN) 1 or 2A, secondary or tertiary hyperparathyroidism, then either a 3.5-gland parathyroidectomy or total parathyroidectomy with forearm reimplantation is performed.

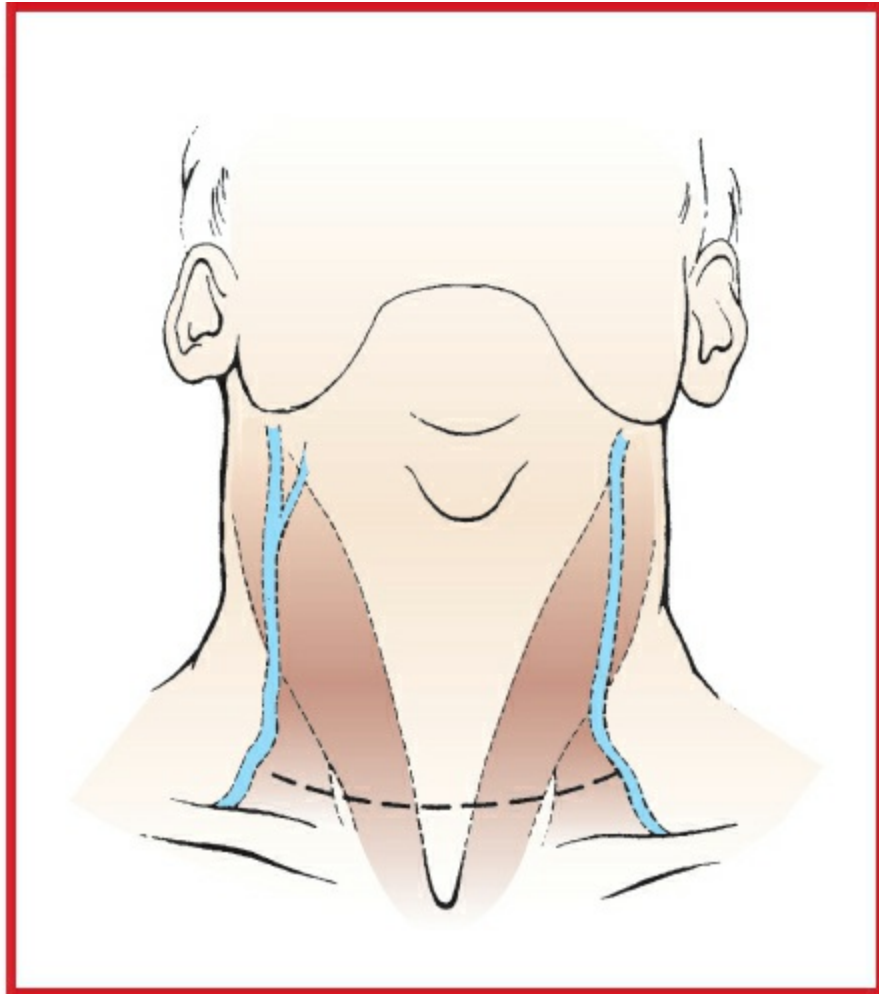


Figure 7.11-5 Skin incision for parathyroid exploration, made 1–2 fingerbreadths superior to sternal notch, as far lateral as the external jugular veins on both sides. (Reproduced with permission from Baker RJ, Fischer JE: *Mastery of Surgery*, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

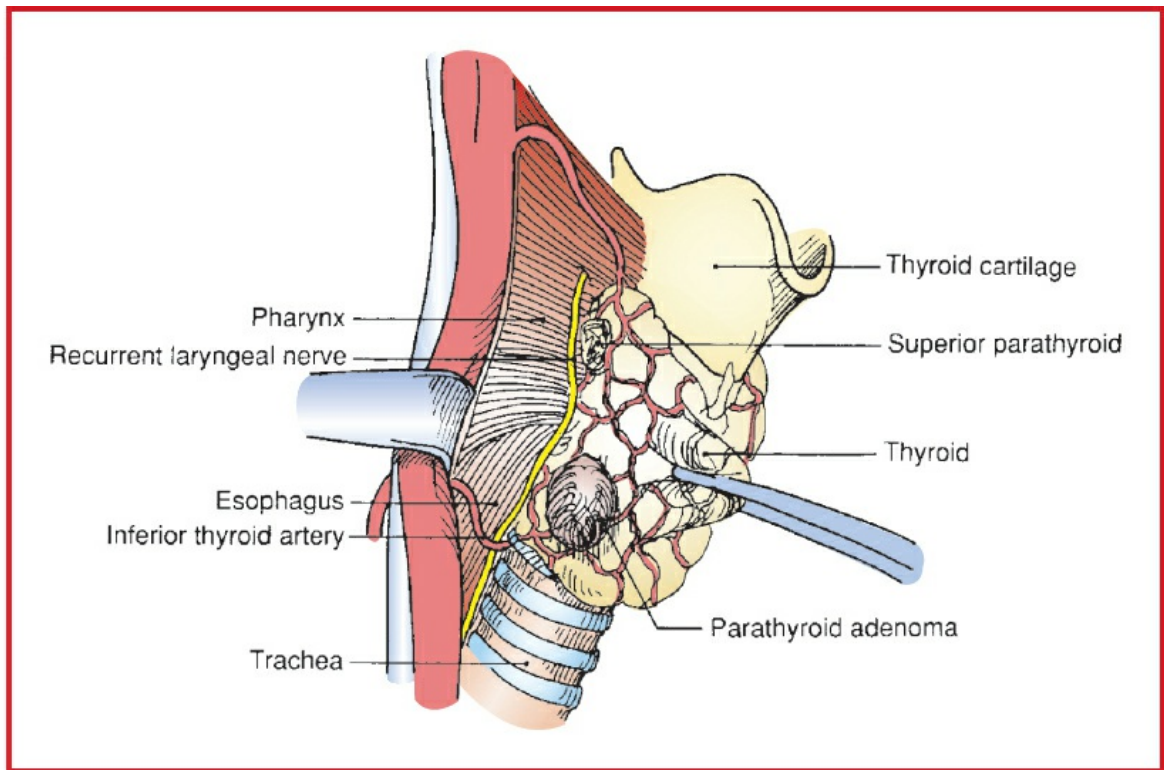


Figure 7.11-6 Identification of upper parathyroid gland on right side. (Reproduced with permission from Scott-Conner CEH, Dawson DLOperative Anatomy, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2003.)

With preop imaging studies, dissection may be limited to the area of suspicion, and the abnormal gland or glands are removed. Successful resection is confirmed by observing a 50% decrease in or normalization of the parathormone level 5–10 min after removal, as compared to preop value. This very sensitive assay is rapidly becoming the standard of care in the treatment of primary hyperparathyroidism. The administration of preop methylene blue or radioactive tracers may aid in localizing parathyroid tissue. If the parathyroid adenoma cannot be found, the surgeon should explore other areas of the neck, including the paraesophageal and retroesophageal space, carotid sheath, and thymus. Unilateral thyroid lobectomy may be performed when three normal glands have been found, and the fourth gland is missing, as the adenoma may be within the thyroid substance. Mediastinal exploration should not be done on primary exploration.

IONM can be used during parathyroid surgery to assist in identifying and preserving the recurrent laryngeal nerves. IONM is particularly helpful in the reoperative setting for persistent or recurrent hyperparathyroidism.

Variant procedure or approaches

Variations of minimally invasive parathyroid surgery utilizing endoscopic and robotic techniques have been described. Those involving a mini-incision in the neck include

video-assisted and endoscopic parathyroidectomy. “Scarless” remote-access endoscopic and robotic procedures via extracervical sites are also being performed in select centers. Several studies have demonstrated similar morbidity and cure rates between the video-assisted and traditional approach, with improved visualizations of neck structures, less postop pain, and better cosmetic results. However, cost, lengthy operative time, and surgeon inexperience have limited wide adoption of many of these techniques.

Usual preop diagnosis

Primary hyperparathyroidism due to a single adenoma (most common, 85%), double adenoma, hyperplasia, or carcinoma (very rare, <1%); secondary or tertiary hyperparathyroidism in patients with a history of renal failure.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine; shoulder roll; reverse Trendelenburg or 30° tilt; headrest (gel donut)
Incision	Transverse cervical (4–8 cm; see Fig. 7.11-5)
Unique considerations	Methylene blue (7.5 mg/kg in 500 mL of NS) may be administered in the preop holding area 30 min before surgery → spurious ↓ SpO ₂ . Radioactive technetium sestamibi (20 mCi) may be administered 60–90 min before operation. Methylene blue and radioactive tracers may be used independently or simultaneously to aid in identification of parathyroid tissue.
Antibiotics	None routinely used
Surgical time	1–2 h
Closing considerations	Coughing may be associated with venous congestion and hematoma formation.
EBL	25–50 mL
Postop care	Monitor serum Ca ⁺⁺ (nl = 8.5–10.5 mg% total Ca; 1–

	1.3 mM ionized Ca ⁺⁺). Patients typically admitted for 24 h observation to check serum Ca ⁺⁺ levels and monitor for hematoma/airway compromise.
Mortality	<0.5%
Morbidity	Hypocalcemia: <15% Hypoparathyroidism: <5% Hematoma: 1% Infection: 1% Recurrent laryngeal paralysis: <1%
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Increases with age
Male:Female	4:1
Incidence	50–100/100,000 (1.5/100 in elderly)
Etiology	Single adenoma (85%); double adenoma (2–3%); hyperplasia (10–15%); cancer (<1%)
Associated conditions	Bone disease (5–15%); duodenal ulcer (5–10%); renal calculi (60–70%); MEN-1 (parathyroid + hyperplasia + pituitary adenoma + pancreatic neuroendocrine tumor); MEN-2A (parathyroid hyperplasia + medullary thyroid cancer + pheochromocytoma); HTN (20–50%)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

These patients typically present with hypercalcemia (hyperparathyroidism), which must be controlled before surgery. Although 25–50% of cases are asymptomatic, many will present with a variety of Sx, including fatigue, muscle weakness, depression, anorexia, nausea, constipation, abdominal and bone pain, HTN, renal stones, polydipsia, muscle cramps, bronchospasm, and anxiety. Differential diagnosis for the hypercalcemic patient includes metastatic disease, multiple myeloma, milk-alkali syndrome, vitamin D intoxication, sarcoidosis, hyperthyroidism, thiazide diuretics, adrenal insufficiency, Paget disease, immobilization, or an exogenous parathyroid hormone-producing tumor.

Respiratory	Hyperparathyroidism is associated with ↓ clearance of secretions from the tracheobronchial tree → postop atelectasis. Avoid respiratory or metabolic acidosis, which will ↑ the free fraction of Ca^{++} → hypercalcemia (↑ BP, ↑ muscle weakness, ↑ HR). Tests: As indicated from H&P
Airway	Hyperparathyroidism and thyroid nodules often coexist. Being aware of their presence, size, extension, and effect on the airway is crucial for airway management (Eur J Anaesthesiol 2008; 25(9):772–4).
Cardiovascular	HTN (usually resolves with treatment; if severe or episodic, r/o pheochromocytoma); ECG may show tachycardia with ↓ PR and ↓ QT intervals. Patients may be hypovolemic (2° anorexia, N/V, and polyuria) and have ↑ sensitivity to digitalis + resistance to catecholamines. ↓ BP may result from polyuria → hypovolemia. Preop management includes correction of intravascular volume and electrolyte abnormalities. Preop treatment of hypercalcemia includes aggressive expansion of intravascular volume, followed by diuresis (maintain euvoemia), to ↑ renal Ca^{++} excretion, usually accomplished with iv NS, and then furosemide. Calcium channel blockers (verapamil, nifedipine, diltiazem) can be used. Hypophosphatemia can impair myocardial contractility and should be corrected; hemodialysis or peritoneal dialysis can be performed to lower dangerously elevated serum Ca^{++} levels. Mithramycin, calcitonin, bisphosphonates, cisplatin, or steroids are not useful for acute Rx of ↑ Ca^{++} (too slow). Tests: ECG; electrolytes; others as indicated from H&P
Neurologic	Patient may present with seizures, hyporeflexia, mental status changes (somnolence, depression,

	memory loss, psychosis, coma), or peripheral neuropathy. Significant improvement often follows correction of hypercalcemia.
Musculoskeletal	These patients may have muscle atrophy and weakness, osteopenia, arthralgia, pathologic fractures (take care during laryngoscopy and positioning), and osteitis fibrosa cystica. Response to NMBs may be enhanced 2° $\uparrow$ Ca^{++} → muscle weakness.
Hematologic	Patients also tend to be hypophosphatemic and may show signs of hemolysis, plt dysfunction, impaired ventricular contractility, and leukocyte dysfunction. Tests: CBC with Plt count, PO_4^{-2}
Endocrine	Primary hyperparathyroidism is most commonly due to benign parathyroid adenoma (90%) or hyperplasia (9%) and rarely carcinoma. It may be associated with MEN syndrome. MEN-1 consists of tumors of the pituitary, parathyroid, and pancreatic islets; MEN-2 is characterized by parathyroid tumors, medullary thyroid carcinoma, pheochromocytoma, and mucosal neuromas. *NB: Anesthetizing a patient with an unrecognized pheochromocytoma could result in a fatality. Tests: As indicated from H&P
Renal	Patients may have renal dysfunction 2° nephrolithiasis, nephrocalcinosis, renal tubular disorders, and glomerular disorders. Polyuria 2° $\uparrow$ Ca^{++} → electrolyte disturbances. Tests: BMP, Mg^{++}
Gastrointestinal	These patients may have constipation, anorexia, N/V, and epigastric pain.
Laboratory	Symptoms not related as well to absolute Ca^{++} level but to rate of change. Serum Ca^{++} <12 mg/dL, likely asymptomatic; 12–14 mg/dL, mild symptoms; >16 mg/dL, life-threatening. Albumin ($\uparrow$ albumin by 1 g/dL will $\uparrow$ total serum Ca^{++} by 0.8 mg/dL); electrolytes; Mg^{++} ; phosphate (usually low).

Premedication	All medications to lower hypercalcemia should be continued unless Ca^{++} levels have normalized. Often, this strategy includes a combination of hydration and diuresis to increase urinary secretion of calcium. If patient has been treated with steroids in the preop period, administer a stress dose (hydrocortisone 100 mg iv q8h $\times$ 24 h) before induction of anesthesia and continue into the early postop period. Standard premedications are usually appropriate in this patient group, unless patient has mental status changes.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA, with head slightly hyperextended, allows for adequate exposure of the neck. Cervical plexus blocks may be appropriate in selected patients; however, phrenic nerve block $\rightarrow$ respiratory compromise. For minimally invasive surgery, a superficial cervical block may be as effective as the combination of a superficial/deep block.

Induction	Standard induction. If patient is clinically hypovolemic, restore intravascular volume before induction, and titrate induction dose of sedative/hypnotic agents. If using IONM, insert specialized ETT (CMAC helpful), and ensure direct contact of the electrodes with the vocal cords on either side. A short-acting, nondepolarizing paralytic may be used to facilitate intubation; however, long-acting neuromuscular blocking agents should be avoided.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of

	nerve blocks.	
Maintenance	Standard maintenance, with muscle relaxant titrated to effect. Avoid hyperventilation or hypoventilation (acidosis will ↑ Ca^{++} levels, while alkalosis will ↓ Ca^{++} levels). Maintain adequate hydration and UO throughout the procedure. There is no evidence to support the avoidance of propofol-based anesthetic (TIVA) for parathyroid surgery.	
Emergence	No special considerations (see Postop Complications, below). Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 5–8 mL/kg/h	Avoid Ca^{++} containing iv solution (e.g., LR), use Normosol. Surgeon may request blood samples to be drawn for pre- and postexcision PTH levels. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors ± arterial line	Others as indicated by patient status. Frequent blood samples for PTH are required during a parathyroidectomy. Arterial line is helpful for frequent blood draws.
Positioning	✓ and pad pressure points ✓ eyes	Patients should be positioned carefully as they tend to be osteopenic and are prone to pathologic bone fractures. Slight head-up positions may ↓ blood loss and improve surgical visibility without significantly ↑ risk of VAE.

POSTOPERATIVE

Complications	Hypocalcemia Hypocalcemic tetany	Hypocalcemia may occur in the immediate postop period. Sx include paresthesias, muscle spasm, tetany, laryngospasm, bronchospasm, and apnea. Rx: oral Ca^{++} supplementation; if severe, 10–20 mL Ca^{++} gluconate 10% over 10 min. Follow levels and repeat therapy until the clinical signs of hypocalcemia are controlled. CPAP is effective for airway obstruction.
	Recurrent laryngeal nerve injury Hematoma with airway compromise	Having the patient vocalize the letter “e” can monitor recurrent laryngeal nerve dysfunction. Unilateral vocal cord dysfunction results in hoarseness, whereas bilateral vocal cord dysfunction results in aphonia. CPAP is effective for airway obstruction.
Multimodal pain management	Oral narcotic analgesics	Emphasize nonopioid analgesics, including acetaminophen and NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.
Tests	Serial measurements	The lowest Ca^{++} level usually

of: Ca⁺⁺
phosphate
Mg⁺⁺
others as clinically
indicated

is seen after 4–5 d postop.
Follow clinical Sx of
hypocalcemia: Trousseau sign
(carpopedal spasm in
response to application of a
BP cuff at a pressure > SBP
for 3 min); Chvostek sign
(contracture of the facial
muscles produced by tapping
on the facial nerve).

Suggested Readings

1. Betea D, Potorac I, Beckers A: Parathyroid carcinoma: challenges in diagnosis and treatment. *Ann Endocrinol (Paris)* 2015; 76(2):169–77.
2. Callender GG, Udelsman R: Surgery for primary hyperparathyroidism. *Cancer* 2014; 120(23):3602–16.
3. Dumitras M, Strambu V, Radu PA, et al: Technical factors involved in parathyroid surgery. *Chirurgia (Bucur)* 2015; 110(5):425–9.

ADRENALECTOMY

SURGICAL CONSIDERATIONS

Description

Adrenalectomy can be performed by different approaches, each with its own merits. Traditionally, the adrenal glands have been removed with an open incision through either the **transperitoneal** or **extraperitoneal** (flank) approach ([Fig. 7.11-8](#)). With some exceptions (see below), **laparoscopic** approaches are becoming the favored methods. Indication in malignant disease or metastasis remains controversial and is currently being evaluated. Relative contraindications include large adrenal adenocarcinomas (>5 cm), malignant pheochromocytoma, invasive adrenal mass, large adrenal mass (>8–10 cm), or other contraindications to abdominal laparoscopic surgery (e.g., multiple previous abdominal surgeries). Each approach is discussed individually, but their underlying principles remain the same. The anatomy of the retroperitoneum is shown in [Figure 7.11-7](#).

Open transperitoneal approach

This approach allows for easy access to both adrenals and is preferred for large tumors.

A midline or bilateral subcostal incision ([Fig. 7.11-8](#)) is used, with the patient in the supine position. The left adrenal is accessed by incising the lateral peritoneal attachments of the spleen. The spleen and tail of the pancreas are reflected medially, Gerota fascia is incised, and the upper pole of the kidney is separated from the adrenal gland. The left adrenal vein is identified arising from the renal vein and is ligated and divided between ties or using an endovascular stapler. A combination of blunt and sharp dissection is used to fully mobilize and resect the gland. The right gland is exposed by retracting the liver cephalad and taking down the hepatic flexure inferiorly. The duodenum and pancreas are rotated medially to expose the inferior vena cava (IVC). Downward traction is placed on the right kidney, and the right adrenal vein is identified entering the IVC. The adrenal vein is circumferentially controlled and divided, and the gland is excised.

Open extraperitoneal approach (flank)

Before the advent of laparoscopic surgery, this approach was favored to minimize pain and improve postop recovery of adrenalectomy through the use of a smaller incision and by remaining extraperitoneal. In general, this approach is best used for unilateral, smaller tumors. The patient is placed in the prone jackknife position, and a dorsal curved flank incision is made, exposing the 12th rib. The rib is resected, Gerota fascia is identified and incised, and the adrenal gland is removed.

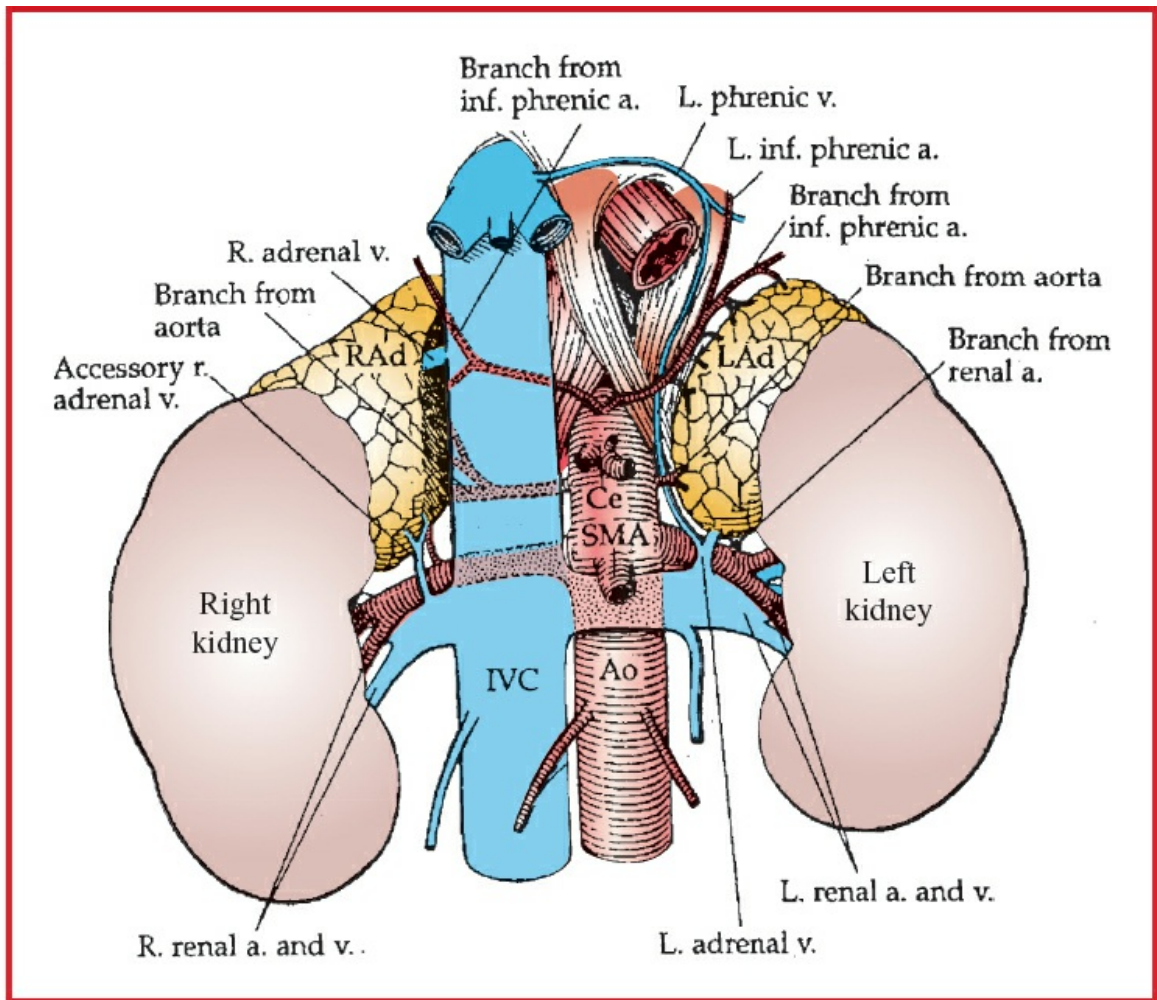


Figure 7.11-7 Anatomic relationships of the adrenal glands. Note the origins of the three main arteries: inferior phrenic, aortic, and renal branches. Note also the single draining veins (except for a small accessory right adrenal vein): the right-located superior and medial and the left-found inferior and medial. (Ao, aorta; Ce, celiac; LAd, left adrenal gland; RAd, right adrenal gland; SMA, superior mesenteric artery.) (Reproduced with permission from Baker RJ, Fischer JEMastery of Surgery, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Laparoscopic anterior transperitoneal approach

This approach may be used for **bilateral adrenalectomy** with the patient in the supine position. The surgical plan is similar to that of the transabdominal open approach but is associated with longer operating times and additional trocar sites for placement of more retractors to mobilize the intra-abdominal organs.

Laparoscopic lateral transperitoneal approach

This is the preferred approach for adrenalectomy in most centers. The patient is positioned in the lateral decubitus position with the operative site up and is immobilized with a carefully placed beanbag to minimize nerve compression injuries. The patient is prepped for either an open or laparoscopic approach (so that conversion

to an open procedure may be done swiftly, if necessary). When adrenal glands are removed, the right is done because there is a higher risk of converting to an open procedure due to the proximity of the adrenal vein to the IVC.

For **right adrenalectomy**, pneumoperitoneum is established, and four ports are inserted several fingerbreadths below the costal margin between the mid abdomen and midaxillary line (Fig. 7.11-9). Attachments to the right lobe of the liver are divided, and the liver is reflected medially to bring the adrenal and IVC into view. The retroperitoneum is incised too close to the IVC to expose the right adrenal vein, which is carefully isolated, clipped, and divided. Dissection of the gland then proceeds out laterally and up superiorly using a laparoscopic energy device. Once fully mobilized, the gland is removed through a laparoscopic retrieval bag.

The **left adrenal gland** is approached similarly. After placement of the trocars, the lateral splenic attachments are incised, and the spleen and tail of the pancreas are rotated medially to expose the anterior surface of the kidney (Fig. 7.11-10); the inferior and medial edges of the gland are mobilized first, allowing for the left adrenal vein to be identified. After the vein is controlled, the remaining gland is dissected free, and the gland is withdrawn.

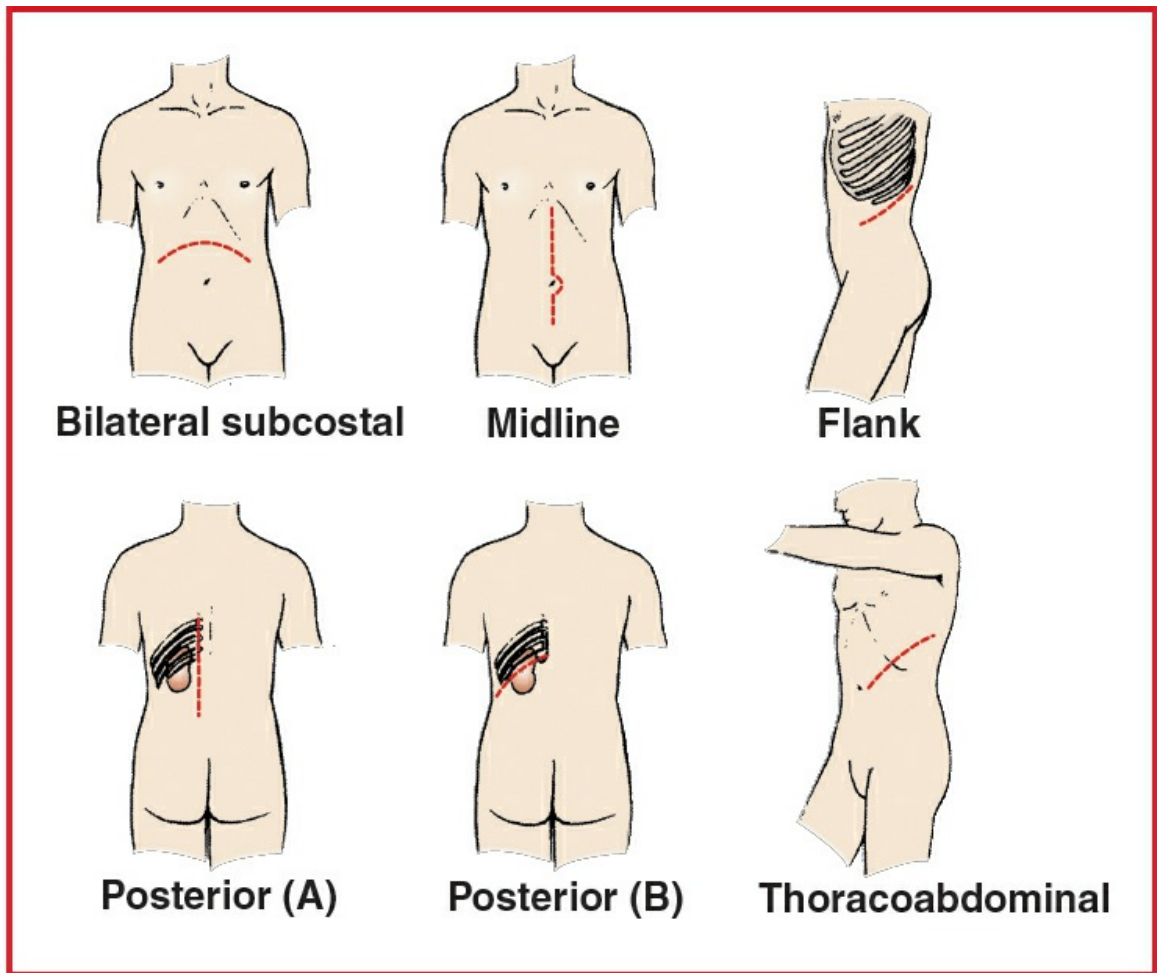


Figure 7.11-8 Potential incisions for adrenalectomy. (Reproduced with permission from Baker RJ, Fischer JE: Mastery of Surgery, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Laparoscopic retroperitoneal approach

This technique is particularly good for patients who have undergone previous abdominal surgery as it avoids the peritoneal cavity and adhesions altogether. The patient is intubated on the stretcher and positioned prone on the operating table with slight flexion at the hips and knees. Bolsters are placed under the hips and chest to allow space for the abdomen and pannus. Three ports are placed horizontally below the costal margin into the retroperitoneal space. The posterior retroperitoneal plane is developed, separating perinephric and periadrenal fat away from the peritoneum and paraspinous muscles. The inferior border of the adrenal is divided from the superior pole of the kidney, which is rotated inferiorly and medially to help expose the adrenal vein. Medial dissection of the adrenal gland from the paraspinous muscles will reveal the inferior phrenic vein on the left and IVC on the right. These structures further facilitate identification of the adrenal vein, which is ligated and divided with clips or an energy sealant device. Once controlled, the adrenal gland is separated from the filmy attachments to the peritoneum and removed via an extraction bag.

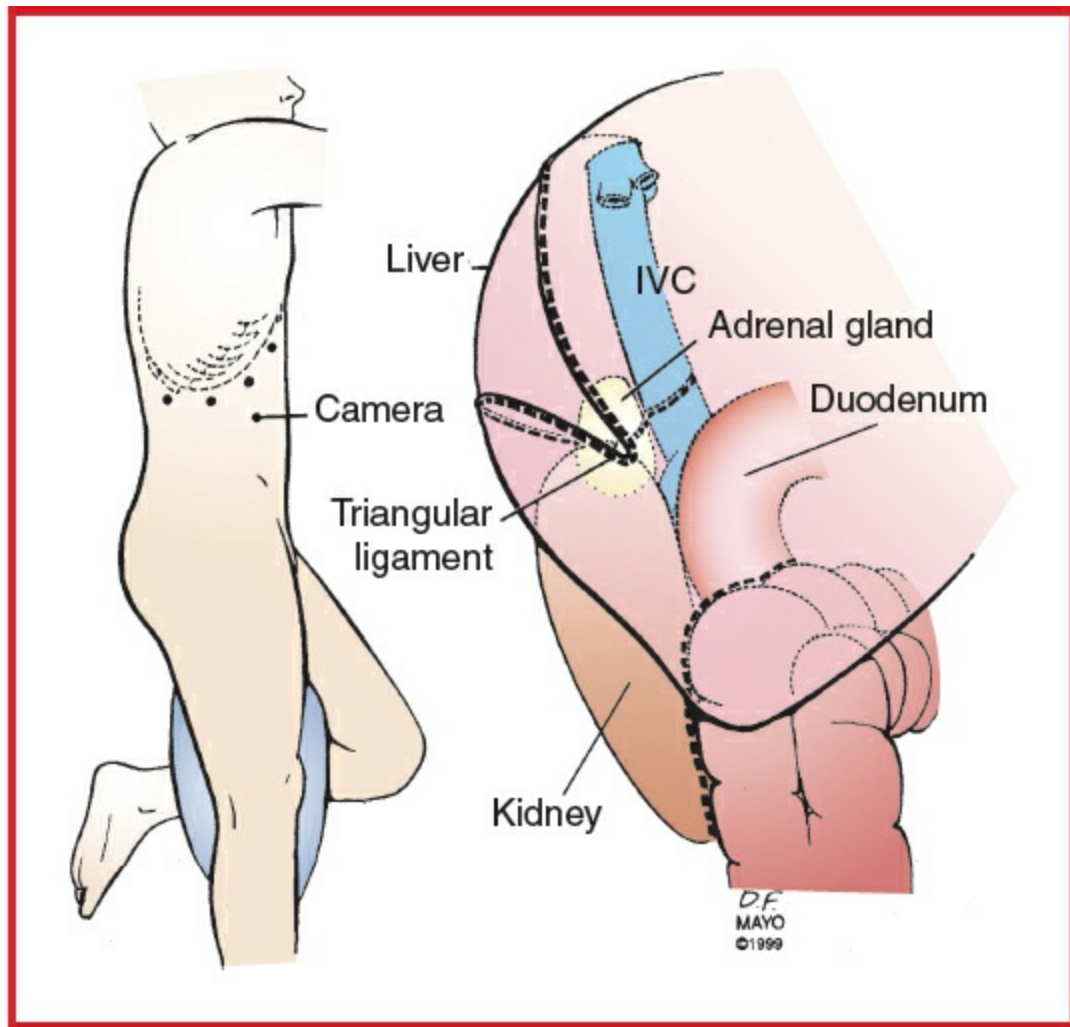


Figure 7.11-9 Port placement for laparoscopic right adrenalectomy. Relevant anatomy is displayed through the right lobe of the liver. (Reproduced with permission from Baker RJ, Fischer JEMastery of Surgery, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

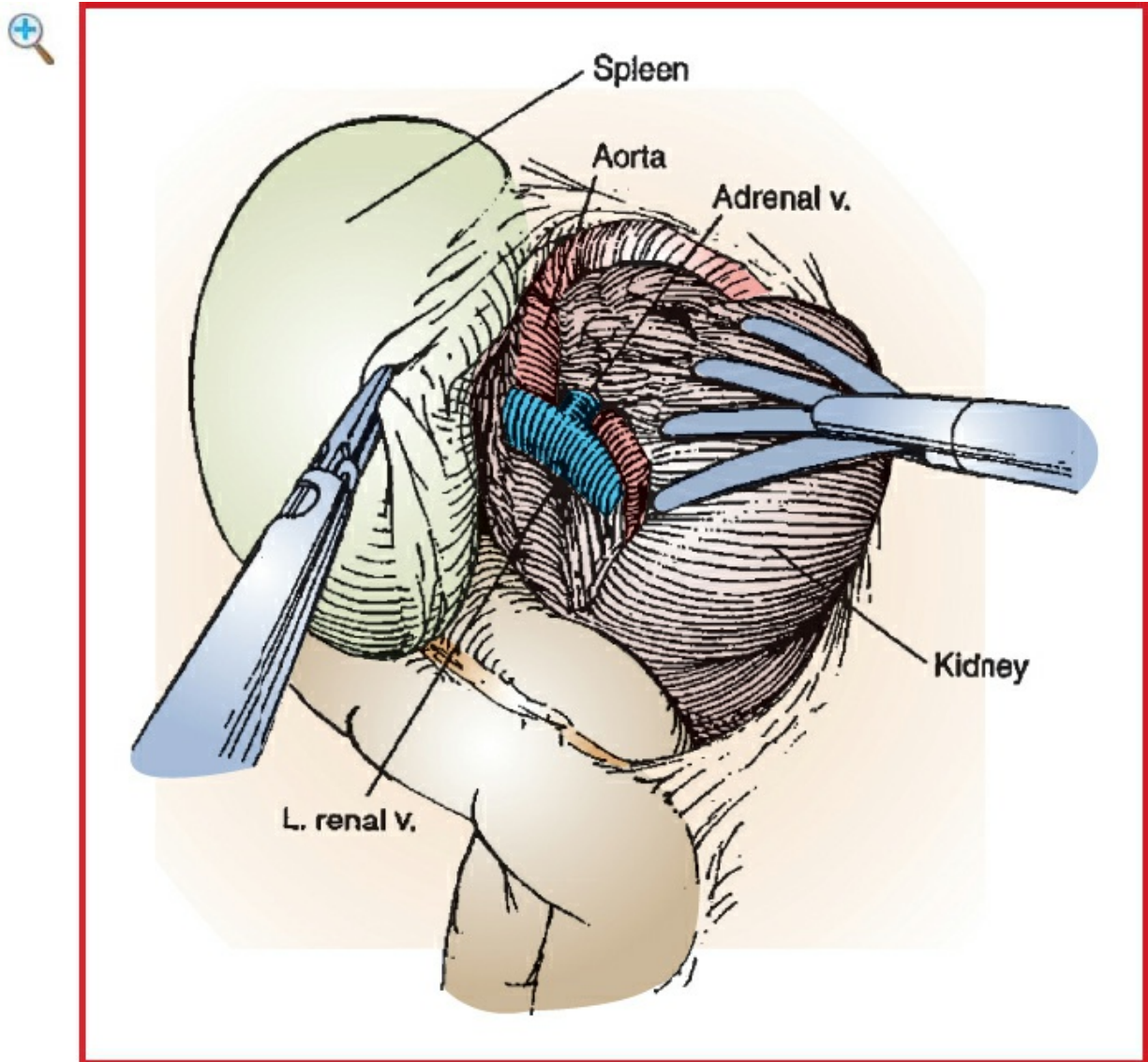


Figure 7.11-10 Exposure of left adrenal after mobilization of the spleen. (Reproduced with permission from Baker RJ, Fischer JE *Mastery of Surgery*, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Robot-assisted approach

Both **robotic transperitoneal and retroperitoneal adrenalectomy** have been described. Advantages of the robot include a three-dimensional view of the operative field, superior ergonomics, and a stable camera platform. Although patient outcomes have been comparable to that of laparoscopic adrenalectomy, training and cost are significant drawbacks to the approach.

Usual preop diagnosis

Functioning adrenal tumors (i.e., aldosteronoma, pheochromocytoma, cortisol-producing adrenal tumor); nonfunctioning adrenal lesions (i.e., incidentaloma, angiomyolipoma); and solitary adrenal metastasis with otherwise negative metastatic

survey, adrenocortical carcinoma.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPC duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Open Adrenalectomy	Laparoscopic Adrenalectomy
Position	Transperitoneal, supine; extraperitoneal, prone jackknife	Transperitoneal anterior, supine; transperitoneal lateral, decubitus; retroperitoneal, prone
Incision	Transperitoneal, midline or bilateral subcostal (Fig. 7.11-8); extraperitoneal, dorsal flank oblique or curved posterior	Transperitoneal anterior, 4–5 trocar incisions (Fig. 7.11-9); transperitoneal lateral, 3–4 trocar incisions; retroperitoneal, 3 trocar incisions
Special instrumentation	Transperitoneal, Denier or Bookwalter retractor; extraperitoneal, none	Transabdominal anterior/lateral; retroperitoneal, video setup
Unique considerations	BP monitoring essential for pheochromocytoma patients	⇐
Antibiotics	Cefazolin 2 g IV	⇐
Surgical time	1–2 h	1–3 h
Closing considerations	Hemostasis; hemodynamic stability	⇐
EBL	100–250 mL	⇐
Postop care	Continue monitoring BP; PACU → ward; ± ICU,	⇐

	based on hemodynamic stability	
Mortality	<0.5%	⇐
Morbidity	Overall: 1–12%	⇐
	Bleeding: <10%	⇐
	Pancreatic fistula: <1%	⇐
	DVT: 0.8%	⇐
	PE: 0.5%	⇐
	Renovascular HTN: <1%	⇐
	Peroneal nerve palsy: <1%	⇐
	Venous thrombosis with embolism (Cushing's): 2%	
Pain score	6–8	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	13–75 yr
Male:Female	1:2.8
Incidence	Pheochromocytoma: 0.4–2% of all hypertensive patients
	Primary hyperaldosteronism: ~4% of all hypertensive patients
	Cushing syndrome: 6/1 million
Etiology	Functional cortical adenoma (aldosteronoma, cortisol-secreting tumor), adrenocortical carcinoma, pheochromocytoma (benign or malignant), large nonfunctioning incidentaloma, solitary adrenal metastasis; palliation for persistent Cushing disease, ectopic ACTH syndrome. Idiopathic bilateral hyperplasia is usually treated medically.
Associated conditions	HTN (75–80%); diabetes mellitus (10–15%)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Cushing syndrome

Hyperadrenocorticism can be due to adrenal hyperplasia, adrenal carcinoma, pituitary hypersecretion (Cushing disease), hypersecretion from exogenous tumor, or exogenous steroid administration (most common). Adrenalectomy is the traditional treatment for hyperadrenocorticism 2° adrenal carcinoma. These typically moon-faced patients present with one or more of the following: HTN 2° glucocorticoids, renin (usually not severe); renal calculi; osteoporosis; glucose intolerance; personality changes; and myopathy. In addition, a fragile vasculature predisposes these patients to easy bruising and difficult vascular access.

Respiratory	Patient may be obese with all attendant problems of morbid obesity (see Preoperative Considerations in Operations for Morbid Obesity). Tests: As indicated from H&P
Cardiovascular	HTN and hypervolemia due to Na ⁺ retention, dysrhythmias 2° hypokalemia Tests: ECG; orthostatic vital signs; consider ECHO to evaluate LV function, if indicated from H&P
Renal	Excess steroid production → Na ⁺ retention, K ⁺ excretion, and insulin intolerance (hyperglycemia) Tests: UA; creatinine; glucose; urine concentrations of catecholamine metabolites; others as appropriate
Neurologic	Psychiatric changes, headache
Musculoskeletal	Striae, muscular wasting, buffalo hump, truncal obesity, thin skin, easy bruisability, osteopenia (compression fractures), weakness
Hematologic	Polycythemia
Laboratory	Tests as indicated from H&P
Premedication	Midazolam (1–2 mg iv), consider withholding if morbidly obese
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip

of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

Conn syndrome

Hyperaldosteronism can be primary (Conn syndrome—adrenal adenoma or hyperplasia) or secondary (caused by excess renin secretion related to renal dysfunction). These patients are typically hypokalemic and alkalotic → muscle weakness, paresthesias, tetany, and polyuria. They may also be hypervolemic → CHF, hypernatremic, and hypertensive (diastolic).

Respiratory	Respiratory muscle weakness. Tests: As indicated from H&P
Cardiovascular	HTN, dysrhythmias, T wave, + U wave Tests: ECG; orthostatic vital signs; consider ECHO to evaluate LV function, if indicated from H&P
Renal	Renal HTN, polyuria, polydipsia, ↓ K, ↑ Na. Correction of hypokalemia requires >24 h supplemental K infusion (e.g., 5–20 mEq/h). Tests: UA; creatinine; glucose; urine concentrations of catecholamine metabolites; others as appropriate
Musculoskeletal	Conn syndrome: muscle weakness, tetany, sensitivity to muscle relaxants, osteoporosis
Laboratory	Tests as indicated from H&P
Premedication	Spironolactone often given to inhibit excess aldosterone effects. Midazolam (1–2 mg iv); hydrocortisone 100 mg q8h.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ($\pm$ **epidural for postop analgesia**). If postop epidural analgesia is planned, placement of catheter before anesthetic induction is helpful to establish correct placement in the epidural space and ensuring a bilateral block (accomplished by placing 5–7 mL of 1% lidocaine via the epidural and eliciting a segmental block). Epidurals cannot be used for a posterior approach because they are in the operative field.

Induction	Gentle iv induction and muscle relaxation. The patient should be adequately anesthetized before to any stimulation. Unopposed parasympathetic response to laryngoscopy may $\rightarrow$ bradycardia/asystole.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Volatile anesthetic (sevoflurane), opiate, muscle relaxant. N ₂ O may cause bowel distention. Local anesthetic (2% lidocaine 3–5 mL q60min) can be injected into the epidural catheter to provide both anesthesia and optimal surgical exposure (contracted bowel and profound muscle relaxation). A continuous infusion of local anesthetic (e.g., 2% lidocaine or 0.25% bupivacaine) at 3–5 mL/h may enhance hemodynamic stability. Some anesthesiologists will not use the epidural catheter intraop because chemical “sympathectomy” is more difficult to reverse. If epidural opiates are used for postop analgesia, a loading dose (e.g., hydromorphone 0.4–1.0 mg) should be administered at least 1 h before the conclusion of surgery. Systemic sedatives (opiates and benzodiazepines) should be minimized during this type of anesthetic as they increase the likelihood of postop respiratory depression.
Emergence	Depends on ease of the surgical procedure and the hemodynamic stability of the patient intraop. If patient is hemodynamically unstable, hypothermic, or has a large third-space fluid requirement consider

	postop ventilation. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Anticipate large fluid loss. IV: 14–16 ga × 2 BSS @ 10–15 mL/kg/h Warm all fluids. ± Cell saver central line	As blood loss can be significant, blood should be immediately available. If procedure does not involve cancer, cell-saving devices can be utilized. Guide fluid management by EBL, known volume deficits, UO, and filling pressures/CO if available. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors UO	Others as clinically indicated. Forced air warmer useful for maintaining body temperature.
Complications	Labile HTN	Surgical manipulation of the adrenal may cause ↑↑ BP and dysrhythmias. Rx: Alert surgeon and control BP with esmolol/SNP/clevidipine
Positioning	✓ and pad pressure points ✓ eyes	Strict attention to patient positioning, padding, and taping is important in patients with glucocorticoid excess because of osteopenia and thin, easily traumatized skin.

POSTOPERATIVE

Complications	Pneumothorax Hypoglycemia Hypoadrenocorticism	Dx: pleuritic chest pain, dyspnea, ↑ RR, ↓ breath sounds, hypoxemia,
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	after tumor resection Cushing syndrome: hypoventilation 2° to obesity (hypoxemia, hypercarbia) HTN	lethargy, or altered mental status. Frequent glucose checks and CXR. Treat with glucose, O ₂ , chest tube, and reintubation as necessary. Consider glucocorticoid and mineralocorticoid replacement— hydrocortisone 100 mg q8h.
Multimodal pain management	Epidural analgesia PCA	Patient should be recovered in an ICU or ward accustomed to treating the side effects of epidural opiates (e.g., respiratory depression, breakthrough pain, nausea, pruritus). Emphasize nonopioid analgesics, including acetaminophen and NSAIDs (see Appendix K , p. K-5).
ERAS	PONV protocol (see Appendix B , p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.
Tests	CXR; ECG; electrolytes; glucose	

ANESTHETIC CONSIDERATIONS FOR PHEOCHROMOCYTOMA

PREOPERATIVE

Pheochromocytoma

Tumors of chromaffin tissue origin release massive amounts of catecholamines (norepinephrine > epinephrine) and are responsible for the patient's clinical presentation. The tumor is usually found unilaterally in one of the adrenal glands but also can be found anywhere in the body that chromaffin tissue arises (e.g., urinary bladder, sympathetic chain). There is an increased incidence of pheochromocytoma in certain inherited diseases (multiple endocrine neoplasia type II, neurofibromatosis type I, tuberous sclerosis, Sturge-Weber syndrome, von Hippel-Lindau disease) and with specific gene mutations involving mitochondrial succinate dehydrogenase SDH (B, C, I subtypes). These patients require extensive preop evaluation for cardiac injury (myocarditis, cardiomyopathy, arrhythmias) and preparation, consisting of α -blockade (phenoxybenzamine 10–400 mg/d). Ideally, this care should be monitored closely by the anesthesiologist/intensivist who can provide the preoperative, intraoperative management, and intensive care as needed.

In contrast to its use in essential HTN, β -blockade is not indicated to control hypertension in patients with pheochromocytoma and is dangerous when used alone. Alone, β -blockade may cause marked hypertension, $\downarrow$ CO, intense vasoconstriction $\rightarrow$ CHF, organ ischemia, and shock (unopposed α -agonism). Patients with tachydysrhythmias may require β -blockade to control persistent reflex tachycardia but only after institution of α -blockers. Titration of adrenergic blockade may take 2–8 wk and should be titrated to relief of all episodic symptoms (HAs, sweating, palpitations) and control of blood pressure. Patients should receive **effective** α -blocker dose for 1–2 wk before surgery. Inadequate preop preparation will increase the perioperative morbidity of patients with pheochromocytomas. The adequacy of medical therapy is assessed by the absence of symptoms of catecholamine excess and BP in the normal range measured multiple times/day. The point of delaying surgery for several weeks is to allow recovery, recovery from catecholamine-induced ECG changes, myocarditis, and cardiomyopathy and recovery from other organ injuries (brain, kidney, and bowel) caused by ischemia. These patients may also need recovery time to reset abnormal shifts in autoregulation. All of these “recovery” events are considered essential to enable the patient to tolerate surgery and the post-tumor removal shock period.

Anticipate that some degree of postural hypotension will be observed during titration of phenoxybenzamine, and patients should be educated to avoid sudden postural movements. There is no rush to surgery.

Respiratory	Cardiogenic pulmonary edema
Cardiovascular	Paroxysmal/continuous HTN, tachydysrhythmias, orthostatic hypotension, hypovolemia, myocardial

	dysfunction, cardiomyopathy, ventricular ectopy, ↓ intravascular volume, ↓ sensitivity of α- and β (cardiac)-receptors to normal levels of catecholamines, CHF and acute myocarditis. Tests: ECG; ± CK/TP enzymes, orthostatic vital signs; ECHO to evaluate LV function and mitral valve, and r/o cardiomyopathy (hypertrophic, takotsubo, or dilated).
Renal	HTN from excess catecholamine state may damage kidneys; hyperglycemia. Tests: UA; creatinine; glucose; HbA_{1c} urine concentrations of catecholamine metabolites; others as appropriate
Neurologic	Sx include: Tremulousness, headache, anxiety, nervousness, personality changes, psychosis (rare), paresthesia in arms, hypertensive retinopathy, dilated pupils.
Musculoskeletal	Weight loss, weakness, fatigue
Hematologic	Polycythemia 2° hemoconcentration (common) or tumor erythropoietin production (rare)
Laboratory	Tests as indicated from H&P
Premedication	± Midazolam (0.025–0.05 mg/kg iv). Preop steroid replacement if bilateral adrenalectomy is contemplated.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA: Establish invasive hemodynamic monitoring (arterial and PA lines) BEFORE to induction using iv fentanyl (50–200 mcg) and generous local anesthesia. These procedures are not difficult and can be done safely with ultrasound. This not only

establishes the patient's true hemodynamic baseline but also, equally important, allows the anesthesiologist to pay full attention to the patient at the start of anesthesia/surgery, rather than being distracted with placing invasive lines after induction. Femoral or axillary arterial pressure monitoring is preferred over radial artery because of the need to monitor “**central**” arterial pressures in a patient who may experience high catecholamine levels during surgery with intense vasoconstriction. The use of a PA catheter with continuous CO and SvO₂ delivers valuable “hands-free” information during the case. If postop epidural analgesia is planned (open surgical procedures), placement of catheter before anesthetic induction helps establish correct placement in the epidural space and ensuring a bilateral block (accomplished by placing 5–7 mL of 2% lidocaine **without** epinephrine) via the epidural and eliciting a segmental block. Administering epidural narcotic (hydromorphone 0.4–1.0 mg) before skin incision will enhance preemptive analgesia effect if lidocaine test dose is not used. Epidural catheters cannot be used for a posterior approach because they are in the operative field. Epidurals are not indicated for laparoscopic procedures.

Epidurals block neurally mediated sympathetic responses which are amplified by sympathetic nerve endings that are supersaturated with catecholamines and may → ↓ tumor release of catecholamines but do not block catecholamine release resulting from direct surgical manipulation of tumor. Local anesthetic-induced sympathetic block after tumor resection may compound ↓ BP at this important stage in the operation. If the patient's BP is stable at the time of skin closure and is not on vasopressors, the epidural can be bolused with long-acting local anesthetic to allow for a comfortable emergence.

Induction	Gentle iv induction (titration to effect with propofol) and muscle relaxation (vecuronium 0.1 mg/kg or rocuronium 0.8 mg/kg). Deep mask GA before laryngoscopy. Double DL with LTA spray (4% lidocaine, 4 mL) will minimize the response to intubation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Volatile anesthetic (isoflurane, sevoflurane), opiate (hydromorphone), and muscle relaxant. Most patients who undergo laparoscopic surgery can be extubated at the end of the case, so use of sedatives

(opiates, and benzodiazepines. etc.) should be minimized during this type of anesthetic to avoid postop respiratory depression.

Intraop HTN Rx:

1. Nitroprusside 0.5–3.0 mcg/kg/min
Note: This drug has the best profile but is now prohibitively expensive.
2. Clevidipine 0.1–4 mcg/kg/min or 1–10 mg/h
3. Fenoldopam 0.1–1.5 mcg/kg/min
4. Nitroglycerin 0.5–10 mcg/kg/min

MgSO₄, (1–2 g iv) may provide some additional BP control. The surgeon may need to interrupt the tumor manipulation to control BP.

Note: Titrating volatile anesthetic depth for BP control is too slow.

Intraop Arrhythmia Rx:

Esmolol is **not** used to control htn but can be used (50–300 mcg/kg/min) for controlling reflex tachycardia from the above agents or excessive β stimulation from tumor-released catecholamines. Amiodarone (150 mg bolus, then infusion 1 mg/min) may be needed for supraventricular/ventricular ectopy (especially with epinephrine-secreting tumors).

Excellent communication with surgical team is essential especially when the adrenal gland is being mobilized and when the tumor veins are ligated. Relax BP control (i.e., accept MAP 90–110), and lighten volatile depth nearing tumor removal.

Hypotension (after tumor removal) Rx:

1. Anticipation → fluid loading before tumor removal
2. Fluid resuscitation based on PA pressures, TEE, stroke volume response, SvO₂, pulse pressure variation (PPV)
3. Inotropes (epinephrine) if ↓ CO continuous CO and S_vO₂ will guide effective dosing of inotrope.
4. Vasopressor (vasopressin) if ↓ MAP with normal

	or ↑ CO monitoring “central” MAP will guide effective dosing of vasopressor.	
Emergence	Depends on ease of the surgical procedure and the hemodynamic stability of the patient after tumor removal. If patient is hemodynamically unstable, hypothermic, or has a large third-space fluid requirement, consider postop ventilation, as with any other surgical case. Most cases done laparoscopically can be extubated. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Prepare for large fluid loss (unusual). IV: 14–16 ga × 2 BSS @ 10–15 mL/kg/h Warm all fluids. ± Cell Saver	As blood loss may be significant (especially with R-sided tumors near IVC), blood should be available. Guide fluid management by UO, filling pressures, CO, TEE, SvO₂, PPV, SVV. See Goal-Directed Fluid Management, Appendix K, p. K-4. Normosol (with K, Mg) is ideal fluid for these patients + can be given with blood. ? safety of cell saver blood if tumor margins breached.
Monitoring	Standard monitors UO Arterial line (femoral/axillary) CVP/± PA catheter with continuous ScvO₂ or SvO₂ ± TEE	Forced air warmer useful for maintaining body temperature. The use of TEE may help establish fluid

		balance depending on other monitors used.
Complications	↑↑↑ BP and dysrhythmias	Surgical manipulation of the tumor will cause ↑↑ BP ± dysrhythmias. Rx: Control BP with SNP/fenoldopam/TNG and HR with esmolol, and alert surgeon if you need him to interrupt dissection.
	↓↓ BP (postexcision—most dangerous part of anesthetic)	↓↓ BP common after removal of tumor. May be severe and life-threatening. Rx: anticipate event; fluids, vasopressin, or epinephrine infusions depending on need for inotropic support. Standard vasopressors (phenylephrine) may not be effective in face of residual α-blockade.
Positioning	✓ and pad pressure points ✓ eyes	

POSTOPERATIVE

Complications	Pneumothorax	Dx: pleuritic chest pain, dyspnea, ↑ RR, ↓ breath sounds, hyperresonance to percussion on side of ptx, hypoxemia. ✓ CXR. Rx: O ₂ ; chest tube and reintubation as necessary.
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	Hypoglycemia	Return of normal insulin sensitivity can cause profound hypoglycemia. Rx: anticipate + use D5LR as maintenance fluids.
	Hypoadrenocorticism after tumor resection (usually only with bilateral tumors)	Consider glucocorticoid and mineralocorticoid replacement → hydrocortisone 100 mg q8h.
	HTN	Patient may have other cause for HTN (e.g., essential HTN).
	Hypotension	Continue vasopressor/inotropes, fluids as indicated.
	Myocardial dysfunction	Patient should be recovered in ICU overnight.
	Pulmonary edema	
Multimodal pain management	Epidural analgesia PCA	After leaving an ICU, patients with epidurals should be recovered on a ward accustomed to treating the side effects of epidural opiates (e.g., respiratory depression, breakthrough pain, nausea, pruritus). Emphasize nonopioid analgesics, including acetaminophen and NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and p.	Encourage early enteral nutrition and

	B-5)	mobilization. Consider DVT prophylaxis.
Tests	CXR; ECG; electrolytes; close glucose monitoring q1–2h × 12 h	

Suggested Readings

1. Brunt LM: Laparoscopic adrenalectomy. In: Cameron JL, ed. *Current Surgical Therapy*, 7th edition. Mosby Inc., St. Louis: 2001, 1460–7.
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CHAPTER 7.12

Liver/Kidney/Pancreas Transplantation

SURGEON

Marc L. Melcher, MD, PhD (Kidney, pancreas, liver transplantation)

ANESTHESIOLOGISTS

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We gratefully acknowledge the past contributions of surgeon, Amy E. Gallo, MD.

KIDNEY TRANSPLANTATION: CADAVERIC AND LIVE DONOR

SURGICAL CONSIDERATIONS

Description

Kidney transplantation offers patients with end-stage renal disease (ESRD) freedom from dialysis. The source of the renal graft may be a deceased donor, a relative (e.g., parent, sibling), or a genetically unrelated donor such as a spouse, friend, or person through paired donor exchange.

After induction of anesthesia, a 3-way Foley catheter is placed into the bladder. The

kidney allograft is placed in the extraperitoneal iliac fossa. A curvilinear incision is made in the right or left lower quadrant. The retroperitoneal space is developed by retracting the peritoneum medially and cephalad, exposing the iliac vessels. A self-retaining retractor is placed to maintain exposure. The lymphatics surrounding the external iliac artery and vein are ligated and divided. Several centimeters of the vessels are mobilized.

The external iliac vein is clamped first, and the renal vein-to-iliac vein anastomosis is performed. When a right deceased donor kidney is used, the renal vein will often have been extended using the donor cava, since the right renal vein is often quite short. Then, the external iliac artery is clamped, and a renal artery-to-iliac artery anastomosis is performed. Usually, deceased donor kidneys will come with a “Carrel” patch of the aorta to allow for a larger anastomosis. The living donors do not have the patch and therefore have a smaller arterial anastomosis. The patient should be euvolemic at this point; mannitol and/or furosemide can be given. The clamps are then released starting with the renal vein ([Fig. 7.12-1](#)).

The bladder is filled with an antibiotic irrigation solution to facilitate its identification and exposure for the implantation of the ureter. The spatulated ureter is anastomosed to the mucosa of the bladder with fine monofilament absorbable sutures, often over a stent. The detrusor muscle is then reapproximated over 3–4 cm of the ureter to create an antireflux valve. The wound is closed, normally leaving native kidneys intact. Many surgeons will leave a closed-suction drain around the kidney.

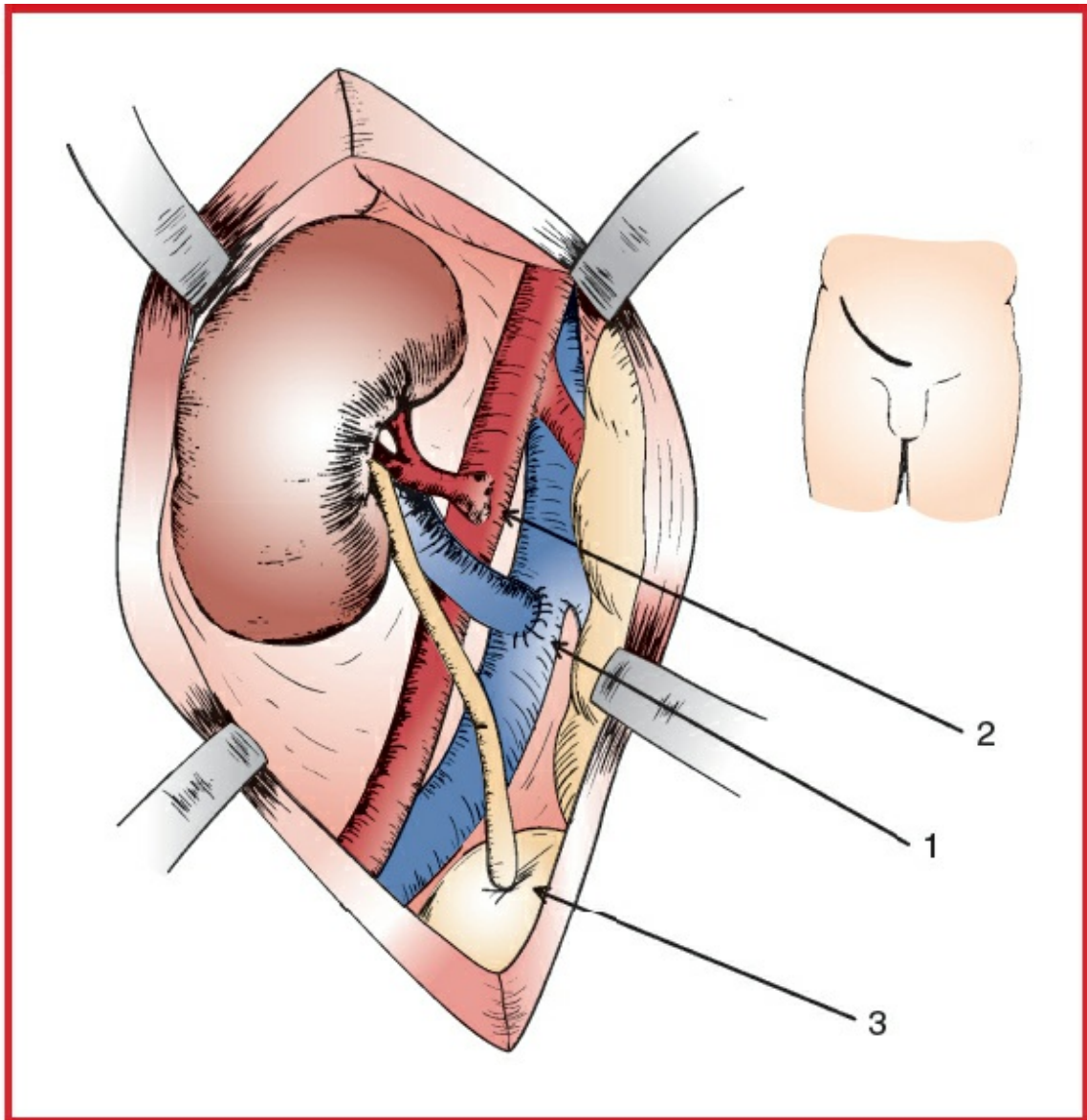


Figure 7.12-1 Sketch of a kidney transplant labeled 1, 2, 3 in the order of the surgical anastomoses. 1, renal vein to external iliac vein; 2, renal artery to external iliac artery, and 3, ureter to bladder. The **inset** shows the lower quadrant curvilinear incision (Paul Gallo, DDS).

Variant procedure or approaches

Cadaveric or live-donor transplantation

Usual preop diagnosis

ESRD

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, urinary drain, pain management options, and postop recovery plan. See [Appendix K: Enhanced Recovery After](#)

■ SUMMARY OF PROCEDURES

	Cadaveric Kidney	Live-Donor Kidney
Position	Supine	⇐
Incision	Lower quadrant curvilinear (Fig. 7.12-1 inset)	⇐
Special instrumentation	Self-retaining retractor; vascular instruments; CVP; Foley catheter (3-way)	⇐
Unique considerations	CVP 10–12 mm Hg; mannitol 12.5–25 g and furosemide 40–80 mg; intraop immunosuppression before reperfusion (steroids, antilymphocyte preparation); potassium-free iv fluid; protection of shunt or fistula is important.	⇐
Antibiotics	Piperacillin/tazobactam 3.375 g iv, within 1 h preincision	Cefazolin 2–3 g iv, 1 h preop
Surgical time	1.5–3 h	⇐
EBL	100–200 mL	⇐
Postop care	Replace urine output mL for mL with iv fluids; may have delayed graft function 2° prolonged cold storage; monitored bed	⇐ Delayed graft function unlikely
Mortality	1–2%	⇐
Morbidity	Lymph or serous leak/stenosis: 3–5%	⇐ 1–2%
	Postop bleeding: 3–5%	⇐
	MI: 2–3%	⇐
	Ureteral leak/stenosis: 2–3%	⇐

	Wound infection: 2–3%	⇐
	Arterial thrombosis: 1–2%	⇐
	Venous thrombosis: 1–2%	⇐
	Wound hematoma: 1–3%	⇐
	Other infectious complications: 15–40%	
Pain score	5	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–80 yr
Male:Female	1:1
Incidence	60/1,000,000
Etiology	Diabetes mellitus (45%); HTN (20%); polycystic disease and others (20%); glomerulonephritis (15%)
Associated conditions	CAD (40%); HTN (25%); uremic and/or diabetic neuropathy (25%); hyperparathyroidism (15–20%); SVC syndrome

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Cadaveric Kidney/Pancreas Transplantation.

Suggested Readings

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CADAVERIC KIDNEY/PANCREAS TRANSPLANTATION

SURGICAL CONSIDERATIONS

Description

Pancreas transplantation: Combined kidney and pancreas transplantation not only provides renal replacement for type I diabetes patient with ESRD but also controls diabetes. Over 80% or more of pancreas transplants are performed in combination with kidney transplantation from the same donor (**simultaneous pancreas-kidney [SPK] transplant**). Pancreas transplantation also can be performed for patients who have already received a kidney transplant (**pancreas after kidney [PAK]**). Less commonly, pancreas transplantation is done for patients with brittle diabetes or with impending complications while they still enjoy normal or near-normal kidney function. Immunosuppression regimen for pancreas transplantation is generally more aggressive than that used for kidney transplantation, and induction therapy with antilymphocyte preparation (ATG, IL-2 blockers) is commonly used.

The pancreas transplant is placed in the right iliac fossa, and the kidney transplant is placed in the left iliac fossa. This can be done through a transperitoneal lower midline incision or through two separate extraperitoneal lower quadrant incisions in the same manner as kidney transplantation. The graft is first prepared on the back table. For arterial inflow, a Y-graft is fashioned using the donor iliac artery bifurcation to provide circulation to both the superior mesenteric and the pancreatic artery. The portal vein coming off the pancreatic graft is anastomosed to the external iliac vein. The Y extension arterial graft is then anastomosed to the recipient external or common iliac artery. The donor duodenum is anastomosed to a loop of small bowel or to the urinary bladder to drain the exocrine secretions (Fig. 7.12-2A). With pancreas transplantation, there may be significant blood loss if the graft mesenteric vessels are not occluded properly. After the pancreas is implanted, the kidney transplant is placed into the opposite iliac fossa (as described in Kidney Transplantation).

Variant procedure or approaches

The pancreas may be placed in the upper abdomen with the portal vein anastomosed to the superior mesenteric vein. The exocrine secretions are bowel-drained. This more physiologic approach, however, is associated with a higher technical failure rate and requires a long upper midline incision (Fig. 7.12-10). Pancreatic islet cells may be infused via a radiologic **portal vein approach**, a procedure that is usually performed in the radiology/angiography suite.

Usual preoperative diagnosis

ESRD 2° diabetes mellitus (DM)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and preop supplementation, discussion of carb loading, DVT prophylaxis, urinary drain, pain management options, and postop recovery plan. See [Appendix K, p. K-1](#).

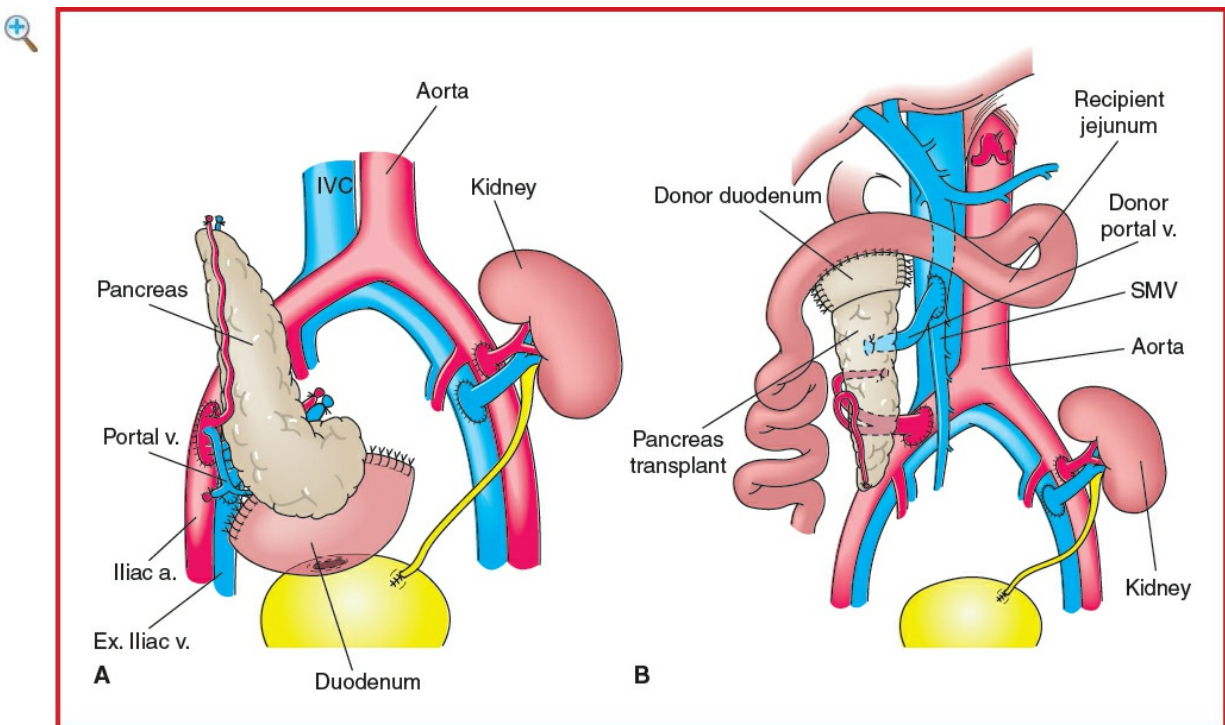


Figure 7.12-2 A: SPK transplantation, with drainage of pancreatic exocrine secretions into the bladder. Note that portal vein drains into iliac vein (systemic venous [SV] drainage). In normal individuals, 50% of secreted insulin is extracted from the circulation in the first pass through the liver. Transplant recipients with SV have peripheral insulin levels 2–2½ × higher than normal. **B:** SPK transplantation with drainage of pancreatic exocrine secretions into the proximal jejunum (enteric drainage [ED]). This technique has been adopted for SPK by most transplant centers in the United States for solitary pancreas transplantation; most centers still utilize ED to allow monitoring of the urinary amylase. Note that donor portal vein drains into the recipient superior mesenteric vein (portal venous [PV] drainage) preventing peripheral hyperinsulinemia. Most centers continue to place the pancreas in the pelvis, combining ED and SV, which requires enteric anastomosis to a more distal segment of the jejunum or ileum. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ SUMMARY OF PROCEDURE

Position	Supine; cushion heels
Incision	Midline or bilateral lower quadrant
Special instrumentation	Thompson retractor; vascular instruments; Foley catheter; NG tube; CVP; arterial line
Unique	Do not correct hyperglycemia <300 mg/dL. Maintain

considerations	adequate hydration, CVP 10–12 mm Hg; mannitol 0.25–0.5 g/kg on unclamping; intraop immunosuppression before reperfusion (125–250 mg methylprednisolone, monoclonal or polyclonal preparation [e.g., ATG, Zenapax, OKT3, Simulect]).
Antibiotics	Piperacillin/tazobactam 3.375 g iv q6h × 5 d; fluconazole 200–400 mg (nl creatinine clearance)
Surgical time	4–6 h
EBL	250–500 mL
Postop care	ICU × 1–2 d, hourly monitoring of glucose; serum glucose should decline by 50 mg/dL each hour and remain <200 mg/dL.
Mortality	2%
Morbidity	Thrombosis of graft: 5–10% Postop bleeding: 5–10% Wound infection: 2–6% Pancreatitis: 2–4% Anastomotic leak: 1%
Pain score	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–55 yr
Male:Female	1:1
Incidence	15–20% of all patients with ESRD
Etiology	Type I diabetes
Associated conditions	Retinopathy (100%); uremic and/or diabetic neuropathy (50%); CAD (25–50%); gastropathy (25%); hyperparathyroidism (15–20%)

Suggested Reading

1. Sharples EJ, Mittal SM, Friend PJ: Challenges in pancreas transplantation *Acta Diabetol* 2016; 53(6):871–8.

ANESTHETIC CONSIDERATIONS FOR KIDNEY AND KIDNEY/PANCREAS TRANSPLANTATION

PREOPERATIVE

Typically, patients presenting for renal transplantation fall into two patient populations: (a) the young and relatively healthy (following dialysis) or (b) an older, more chronically ill group. Rarely, patients will present for transplant surgery without adequate preparation (e.g., $\uparrow K^+$, $\downarrow pH$, hypervolemia). Patients presenting for pancreas transplantation are usually severe diabetics with many of the associated problems, such as CAD, autonomic neuropathy, gastroparesis, and stiff joint syndrome (difficult intubation).

Respiratory	Pleuritis and pleural effusions may occur in this patient population. Increased susceptibility to infection is common in the patient with chronic uremia.
Cardiovascular	Pericarditis (acute or constrictive), HTN, CHF, dysrhythmias, and pericardial effusion are common, especially in the undialyzed patient. Diabetes, a common cause of ESRD, is often associated with PVD, CAD, and autonomic neuropathy. Tests: ECG (rhythm, CAD, electrolyte abnormalities, pericarditis, LVH). Other tests (ECHO, stress, etc.) as indicated by the H&P
Gastrointestinal	Gastroparesis may occur, especially in diabetic patients with autonomic neuropathy. Consider full stomach precautions. Ranitidine (50 mg iv) and metoclopramide (10 mg iv) should be given 60 min preop to aid gastric emptying and $\downarrow$ acidity.
Renal	Patients are usually on dialysis. Postdialysis goals include $K^+ = 4\text{--}5$ mEq/L, BUN < 60 mg%, creatinine < 10 mg%. Metabolic acidosis, hypocalcemia, and hypermagnesemia often present and may require preop correction. Patients may be hypovolemic following dialysis; ✓ pre- and postdialysis weight (> 2 kg loss is significant). Rapid correction of severe hyperkalemia can be achieved by giving 50 mL of

	50% glucose iv, together with 10 U regular insulin and 50 mEq NaHCO₃. Further correction can be obtained by coadministration of an inhaled β-agonist (e.g., albuterol, 5–10 puffs). Tests: Cr; BUN; creatinine clearance; electrolytes
Hematologic	These patients are frequently anemic. Preop correction usually not needed. A coagulation disorder may be present with abnormal platelet function (improved by dialysis) as well as thrombocytopenia, resulting in a prolonged bleeding time. There is a high incidence of posttransfusion hepatitis in this patient population. Tests: Hct; PT; PTT; platelet count; hepatic screen, consider platelet function assay.
Endocrine	Assess glycemic control preop. Individualize diabetic Rx for the day of surgery. In general, hold oral agents 12–24 h, and $\downarrow$ or hold usual insulins, and consider sliding scale insulin on the day of surgery. Favor hyperglycemia (≥ 120–200 mg/dL) over tight glucose control. Preop/intraop corticosteroid immunosuppressive Rx will likely $\rightarrow$ hyperglycemia. Postop glycemic management complex and dynamic Tests: Serial glucose levels
Neurologic	Peripheral neuropathy may occur, and specific deficits should be documented. Autonomic neuropathy can $\rightarrow$ cardiac problems (e.g., orthostatic hypotension, $\uparrow$ HR, or $\downarrow$ HR), silent MI, and GI problems.
Premedication	Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

Anesthetic technique

GETA favored. Spinal, epidural, or combined spinal-epidural anesthesia may be considered for renal transplantation, if coagulation and platelet function are acceptable. Ilioinguinal-iliohypogastric and intercostal nerve blockade can be utilized as an alternative method for postop pain control, although their efficacy in this patient population has yet to be demonstrated. Avoidance of hypotension after organ(s) transplantation is an important consideration that may limit effectiveness of RA techniques.

Induction	Rapid sequence induction. ET intubation is aided by succinylcholine (1 mg/kg), if $K^+ < 5.5$ mEq/L; otherwise, use cisatracurium (0.2–0.5 mg/kg) or rocuronium (1.2 mg/kg). Fentanyl (2–5 mcg/kg) may be used to suppress the cardiovascular response to intubation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance w/ potent inhalational agent, though their effects on ischemia-reperfusion injury are being assessed in multiple clinical studies. Maintain muscle relaxation with cisatracurium or rocuronium, titrated to effect using a nerve stimulator. Avoid meperidine (accumulation of normeperidine → CNS toxicity). Anticipate prolonged drug effects, and avoid agents that are primarily excreted by the kidney.
Emergence	Usually extubated in the OR after protective laryngeal reflexes have returned. Ensure adequate reversal of NMB drugs. Sugammadex can be considered for reversal; however, the rocuronium/sugammadex complex is renally excreted and may be a problem in an anuric patient. However, it can be removed by hemodialysis. Pancreatic transplant patients (e.g., brittle diabetics, hemodynamically unstable) are sent to the ICU for

	close glycemic management. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14–16 ga × 1 NS/BSS/colloid to keep CVP = 10–15 mm Hg Warm fluids	Preop fluid status is highly variable (hypovolemia→hypervolemia). Give fluids to maintain CVP 10–15 mm Hg. Important to maintain adequate vascular volume and BP. Mannitol (0.25–1 g/kg), furosemide (5–20 mg), and norepinephrine can be used. Empiric dopamine is no longer recommended. Recent evidence suggests that balanced crystalloid solutions containing K ⁺ can be used without concern about elevating K ⁺ levels, instead of normal saline, to better preserve acid-base profiles. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors. Arterial line CVP/PA line	Arterial pressure is often monitored. Avoid the side of AV fistulae. The axillary artery is a useful alternative. CVP is essential, and a PA line is needed occasionally (severe cardiac disease). CVP is kept at 10–15 mm Hg, especially after the new kidney is reperfused, to ensure adequate renal blood flow. Dynamic parameters such as pulse pressure or systolic volume variation can

		be easily assessed in real time and utilized instead.
	Hct, K ⁺ , and glucose	In pancreatic transplant patients, glucose should be checked q30min and then q10min for the first hour following reperfusion. Keep glucose <300 mg/dL prior to reperfusion but do not fully correct to <150 mg/dL.
	Neuromuscular	Monitor neuromuscular block to avoid excessive use of neuromuscular relaxants; anticipate prolonged effects.
Positioning	✓ and pad pressure points ✓ eyes - Protect/pad AV fistulas	Pressure on AV fistula may lead to thrombosis. Document bruit q15min.
Complications	- Disruption of renal anastomoses - Hemorrhage - Low UO - Reperfusion injury (pancreas transplant)	Bucking or coughing during emergence or at any time after kidney implanted may lead to "popping" and vascular disruption and need for urgent repair.

POSTOPERATIVE

Complications	- Fluid overload and CHF - Femoral neuropathy - Hemorrhage - Electrolyte abnormalities	Monitor UO. Dialysis may be needed until renal function returns, in cases with delayed graft function. Sudden cardiac arrest can complicate pancreatic transplantation
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	Hypo-/hyperglycemia PONV VTE	(due to autonomic neuropathy). Especially dynamic in pancreas tx
Multimodal pain management	PCA Epidural	Anticipate prolonged effect of some opiates. Strive to avoid hypotension. Emphasize nonopioid analgesics including acetaminophen (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization. DVT prophylaxis
Tests	Hct Electrolytes Cr, BUN Amylase Glucose	A rise in amylase and blood glucose may indicate failure of the pancreatic transplant.

Suggested Readings

1. Barton JG: Enhanced recovery pathways in pancreatic surgery. *Surg Clin North Am* 2016; 96(6):1301–12.
2. Calixto Fernandes MH, Schricker T, Magder S, et al: Perioperative fluid management in kidney transplantation: a black box. *Crit Care* 2018; 22(1):14.
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LIVE-DONOR NEPHRECTOMY: LAPAROSCOPIC AND OPEN

SURGICAL CONSIDERATIONS

Description

Use of a kidney donated by a healthy genetically or emotionally related donor greatly increases the number and quality of kidneys available for transplantation. Kidney transplantation from living donors is associated with a better patient and graft survival rate. A **laparoscopic (LSC) approach** for kidney donation was introduced in 1995 as an alternative that would reduce postop pain, wound morbidity, and recovery time associated with open nephrectomy. Now the LSC approach is the procedure of choice for more than 75% of the live kidney donations in the United States. Initial concerns regarding ureteral complications and longer warm ischemic time have mostly subsided with the improvement of the surgical technique and greater experience. The left kidney is preferred for the LSC approach, as the renal vein is longer. Some centers use the LSC approach for the right kidney with comparable results.

The patient is positioned in lateral decubitus over a cushioned beanbag, the kidney rest is slightly elevated, and pillows and an axillary roll are used to prevent compression injuries. Three or four ports are used for the pure laparoscopic approach. The hand-assisted approach, however, has gained popularity over the years and is currently the preferred technique of the majority of transplant centers in the United States. In this approach, two or three ports are used with a 6–8-cm incision made at the level of the umbilicus or infraumbilical. The pneumoperitoneum is kept <15 mm Hg to avoid decreased perfusion to the kidney. Aggressive hydration and intermittent use of iv mannitol help improve kidney perfusion.

On the left side, the descending colon and spleen are mobilized medially; the renal vessels are exposed; the adrenal, lumbar, and gonadal veins are clipped and divided; the ureter is mobilized en bloc, along with the gonadal vein, down to the pelvic inlet. The artery is freed from surrounding lymphatic and neural tissue as it comes off the aorta ([Fig. 7.12-3](#)). Gerota fascia is mobilized to completely free the kidney. The ureter is transected distally.

If the procedure is done purely laparoscopically, a 6-cm suprapubic incision is then made, the peritoneum is exposed in the midline, and an 18-mm port is used to insert a 15-mm Endocatch retrieval bag. The kidney is placed in the bag as it continues to be perfused, avoiding warm ischemia. With the hand-assisted approach, the incision is already made for retrieval with the surgeon's hand. The patient is fully heparinized. An endo TA vascular stapler is used to staple and transect the artery close to the aorta and the vein close to the vena cava. An endo GIA vascular stapler can also be used. The

kidney is brought to the suprapubic incision and gently extracted. The kidney is immediately immersed in cold slush. The kidney is perfused with preservation solution in the usual manner. The heparin is reversed with protamine, the suprapubic incision is closed, and homeostasis is verified before extracting the ports. For a right nephrectomy, the right colon and duodenum are mobilized medially and the liver is retracted upward. The remainder of the operation is as described for the left kidney.

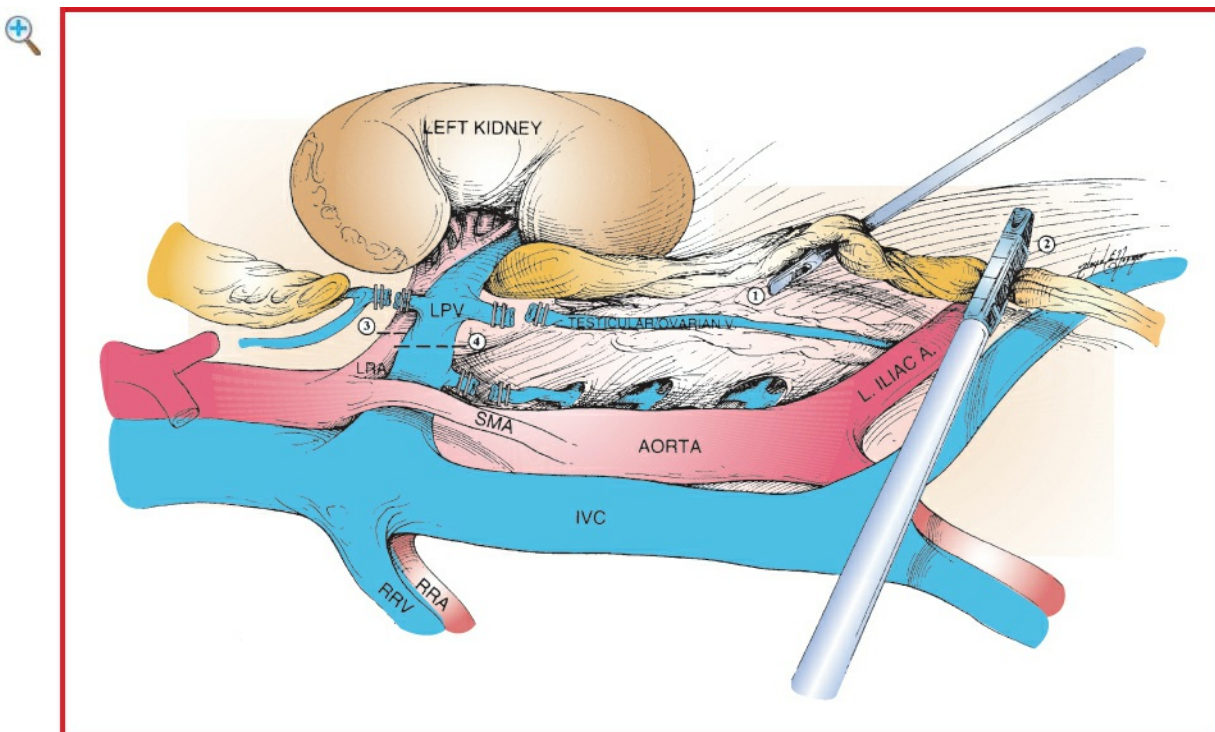


Figure 7.12-3 Anatomy for laparoscopic live-donor nephrectomy. (Reproduced with permission from Cho ES, Flowers JL: Laparoscopic live-donor nephrectomy. In: Zuker KA, ed *Surgical Laparoscopy*, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

In **open nephrectomy**, the **donor/patient** is placed in a lateral decubitus position on a flexible OR table with a kidney rest. A beanbag or sandbags are also helpful for positioning. An incision is made from the rectus muscle, angling slightly cephalic to cross into the flank just below the tip of the 12th rib. The retroperitoneum is exposed using a Thompson retractor. The kidney is then mobilized, and the ureter is transected. A clamp is placed across the renal artery at the aorta and the renal vein at the IVC. Just before clamping the renal artery, furosemide and/or mannitol may be given to stimulate diuresis. It is important to keep the vascular volume expanded in these patients before kidney removal. The kidney is removed and taken to the back table, where it is flushed with a cold preservation solution. It is then transported to the recipient room for reimplantation. Some surgeons use a full dose of heparin (75 U/kg) before clamping and use protamine for reversal. Smaller incisions and muscle-sparing incisions are now

used to improve postop recovery.

Usual preop diagnosis

Donor nephrectomy

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, pain management options, and postop recovery plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Laparoscopic Nephrectomy	Open Nephrectomy
Position	Lateral decubitus	⇐
Incision	Three to four ports; retrieval incision, Pfannenstiel's or vertical suprapubic	Flank; may require 12th rib resection
Special instrumentation	Foley catheter; flexible OR table with kidney rest; beanbag; SCDs for DVT prophylaxis; harmonic scalpel	Foley catheter; Thompson retractor; flexible OR table with kidney rest; beanbag or sandbags; SCDs for DVT prophylaxis
Unique considerations	Avoid ETT dislodgement when turning patient from supine to flank position; vigorous hydration	Possible pneumothorax; avoid ETT dislodgement when turning patient from supine to flank position. Vigorous hydration to encourage urine production
Antibiotics	Cefazolin 2–3 g iv	Cefazolin 2–3 g iv
Surgical time	2.5–4.5 h	2–2.5 h
Closing considerations	—	Deflex table to facilitate closure
EBL	Minimal	100 mL

Postop care	PACU → ward; PCA for pain management	PACU → ward; PCA for pain management
Mortality	<0.1%	⇐
Morbidity	Ileus: 2–5% Urinary retention: 2–5% Wound infection: 1–2% Bleeding: 0.1–0.5%	5–10% 5–10% 1–3% Pneumothorax: 1%
Pain score	3–5	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–70 yr	⇐
Male:Female	1:1	⇐
Incidence	Up to 50% of all kidney transplants at some centers	⇐
Etiology	N/A	⇐
Associated conditions	Good health is mandatory for kidney donation.	⇐

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Kidney Transplant Nephrectomy.

KIDNEY TRANSPLANT NEPHRECTOMY

■ SURGICAL CONSIDERATIONS

Description

With improvements in graft survival and immunosuppressive therapy, the necessity of removing a kidney transplant graft for uncontrolled rejection has decreased significantly. This operation is divided into categories: **early nephrectomy**, performed during the first month post transplant, and **late nephrectomy**, thereafter. **Early transplant nephrectomy** may be required for primary nonfunction, vascular thrombosis, and, rarely, refractory rejection. In these cases, an **extracapsular**

approach, through the original transplant incision, is used. The kidney is freed from the surrounding adhesion to obtain vascular control of the renal artery and renal vein. These structures are clamped and oversewn individually. The ureter is ligated as close as possible to the bladder and excised completely, with primary repair of the bladder. A suction drain is used if minimal oozing or lymph drainage is present. **Late transplant nephrectomy** is performed most commonly for acute, irreversible rejection with failure of the renal allograft. Most of these patients have returned to dialysis, and the immunosuppressive medications are stopped. Chronic infection and hypertension (HTN) associated with nonfunctional grafts are also an indication for surgical removal of the kidney allograft. It may be a difficult operation, as intense inflammatory adhesions are present between the renal capsule and the surrounding tissue. In the setting of acute late rejection, the graft is usually swollen and enlarged. Hematuria may be present, and the graft is friable. Spontaneous rupture and hemorrhage have been reported. The surgical approach is through the same incision as the implantation. In contrast to early transplant nephrectomy, extracapsular dissection may not be possible with late nephrectomy. To avoid injury to extrarenal structures, such as the iliac vessels, an **intracapsular approach** may be preferred. The kidney is mobilized gently from within the capsule toward the hilum. The capsule is reopened on the medial side to have access to the renal vessel high in the hilum. When the hilum is sufficiently mobilized, a strong vascular clamp is applied high in the hilum of the kidney away from the iliac vessels. After clamping the hilum en bloc, confirmation of distal pulses is obtained; then the kidney is excised over the vascular clamp. A running suture is used over the clamp, which is then released, and hemostasis is obtained. The ureter is identified and excised as close as possible to the bladder. The intracapsular dissection of the kidney may be associated with significant bleeding, as the kidney may fracture. This step should be done expeditiously to avoid excessive bleeding. The patient must have good vascular access for fluid resuscitation. Blood must be available for transfusion. After hemostasis is obtained, a low-pressure suction drain may be placed before closing.

Usual preop diagnosis

Transplant rejection

ERAS: Time permitting, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Previous incision used for kidney transplant
Instrumentation	Self-retaining retractor; vascular instruments
Unique considerations	Large-bore vascular access; PRBCs available; stress dose of steroids, if chronic usage; protection of shunt or fistula
Antibiotics	Cefazolin 2–3 g iv
Surgical time	1–2.5 h
EBL	200–1000 mL
Postop care	PCA for pain management → PACU → ward
Mortality	1–3%
Morbidity	Overall: 3–5% Wound infection: 3–5% Abscess formation: 1–2% Exsanguinating hemorrhage: <1%
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–75 yr
Male:Female	1:1
Incidence	60/1,000,000
Etiology	Failed kidney transplant
Associated conditions	CAD (40%); HTN (25%); uremic and/or diabetic neuropathy (25%); hyperparathyroidism (15–20%)

■ ANESTHETIC CONSIDERATIONS FOR LIVE-DONOR AND POSTTRANSPLANTATION NEPHRECTOMY

■ PREOPERATIVE

In order to be a live donor, one must be in good health with bilaterally functional kidneys. Diabetes, HIV infection, liver disease, and malignancy are all

contraindications to kidney donations.

Cardiovascular	Assess for HTN and CAD.
Renal	Normal bilateral renal function is required. Tests: IVP; Cr, creatinine clearance
Fluid status	Adequate hydration is important, and UO should be >1.5 mL/kg/h. Various regimens are used to ensure adequate hydration, usually with iv fluid starting the night before.
Premedication	Consider midazolam 1–2 mg iv. Organ donors are making a great sacrifice and should be treated with special care. Standard premedication.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

GETA ± epidural for postop pain management. Epidurals are not necessary for most laparoscopic donor nephrectomies.

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. Avoid long-acting, renally excreted drugs. Ventilate to maintain normocapnia to avoid possible renal artery vasoconstriction. Use of an epidural with local anesthetic and/or narcotic may aid both intraop and postop pain relief, but ↓ BP should be avoided.
Emergence	Routine extubation in OR. Discuss with surgeon local

	wound infiltration. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 2 NS/BSS @ 6–8 mL/h Warm fluids UO ≥ 1.5 mL/kg/h	Aim for a minimum of 1.5 mL/kg/h UO. Mannitol (0.25–1 g/kg) given iv once the kidney is being manipulated and if UO decreases. Consider dopamine infusion to ↑ BP as needed. Limit use of direct vasoconstrictors. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	CVP or invasive arterial monitoring is rarely required. Lower extremity NIBP measurements on the same side as implant will be inaccurate during period of iliac vessel clamping.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Hemorrhage Pneumothorax	Because the vessels are tied close to the aorta and IVC, the possibility of severe hemorrhage exists. Pneumothorax is always possible, especially when the 12th rib is resected.

POSTOPERATIVE

Complications	Pneumothorax/pulmonary problems Hemorrhage Infection PONV
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	Electrolyte abnormalities VTE Ileus	
Multimodal pain management	Epidural narcotics. Avoid hypotension. PCA	Emphasize nonopioid analgesics including acetaminophen and NSAIDs (see Appendix K, p. K-5).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.
Tests	CXR Hct, electrolytes, BUN/Cr	

Suggested Readings

1. Ghyselen L, Naesens M: Indications, risks and impact of failed allograft nephrectomy. *Transplant Rev (Orlando)* 2019; 33(1):48–54.
2. Giacomoni A, Di Sandro S, Lauterio A, et al: Robotic nephrectomy for living donation: surgical technique and literature systematic review. *Am J Surg* 2016; 211(6):1135–42.
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LIVER TRANSPLANTATION

SURGICAL CONSIDERATIONS

Description

Liver transplantation is the treatment of choice for patients with acute and chronic end-stage liver disease (ESLD). Metabolic orders can also be cured or improved with live transplant but are more often performed in the pediatric population and will not be discussed below. Patients with ESLD, besides intrinsic liver dysfunction, also may have other organ system diseases including hepatorenal and hepatopulmonary syndrome resulting in oliguria and hypoxia, respectively, and hepatic hydrothorax. Patients with alcohol-mediated cirrhosis and Wilson disease are at risk for significant cardiomyopathy, whereas those with fulminant hepatic failure may have significantly elevated intracranial pressures. These additional comorbidities present an added level of complexity to the anesthetic and surgical management of the liver transplant recipient. The liver transplant operation can be divided into three stages: (a) **hepatectomy**; (b) **anhepatic phase**, which involves the implantation of the liver; and (c) **post revascularization**, which includes hemostasis and reconstruction of the hepatic artery and common bile duct. Many variations in the technical aspects of the liver transplant operation contribute to physiologic changes during anesthesia. The anesthesiologist must be aware of these technical variations to optimize the intraoperative management of the liver transplant recipient. Examples of these variations include cross-clamping of the vena cava during the implantation of the liver, which results in impairment of the systemic venous return, with possibility of profound hypotension; utilization of the venovenous bypass, which may be associated with thrombus or air embolism, and/or fibrinolysis; and the use of a “cutdown liver,” which may result in significant bleeding from the cut surface following revascularization.

Table 7.12-1 Contributing Factors Associated with Increased Blood Loss in Liver Transplantation
1. Severe coagulopathy 2. Severe portal HTN 3. Portal or splenic vein thrombosis 4. Previous surgery in the RUQ 5. Renal failure

6. Uncontrolled sepsis
7. Retransplantation
8. Transfusion reaction
9. Venous bypass–induced fibrinolysis
10. Primary graft nonfunction
11. Intraop vascular complications

The **hepatectomy** may be a formidable task in patients with severe portal hypertension (HTN), coagulopathy, and previous surgery in the upper abdomen. In such circumstances, blood loss is significant and may be minimized by placing the patient on venovenous bypass or by creating a temporary portacaval shunt to relieve the portal HTN. [Table 7.12-1](#) lists factors that may be associated with significant blood loss during the transplant operation. The hepatectomy is usually much easier in patients with acute fulminant hepatitis, primary biliary cirrhosis, or inborn errors of metabolism than in patients with shrunken cirrhotic livers, such as in postnecrotic cirrhosis from hepatitis B or C, alpha-1 antitrypsin deficiency, or Wilson disease, among others. The liver transplant incision is traditionally subcostal, extending from the left midclavicular line across the midline to just medial of the right 12th rib, with a vertical midline extension from the xiphoid process to the transverse incision. This provides wide exposure to the upper abdomen.

The hepatectomy begins with a manual exploration of the abdomen to ensure there are no occult malignancies, abscesses, or other abdominal processes that may contraindicate proceeding with the transplant. The liver is then mobilized by freeing the falciform and left cardinal ligaments, followed by entering the lesser sac through the division of the gastrohepatic ligament. The mobilization of the liver and the subsequent dissection of the porta hepatis may be significantly complicated and a tedious process due to large, thin-walled varices that require careful dissection and ligation. The dissection of the porta hepatis begins with identification and ligation of the hepatic artery, followed by the common bile duct. The portal vein is carefully dissected from its bifurcation into left and right branches, proximally to its emergence from behind the pancreas. If the degree of portal HTN is severe—such that mobilization of the liver may result in significant blood loss—or the patient is hemodynamically unstable, then portal vein mobilization may be performed early so that a temporary portacaval shunt or venous bypass may be instituted to allow decompression of the varices and enhance venous return to the heart.

After the portal dissection is complete, the right lobe of the liver is mobilized. The infrahepatic vena cava is carefully dissected to prevent injury to the right renal and adrenal veins, followed by mobilization of the suprahepatic vena cava. The liver can be removed easily by cross-clamping and dividing the supra- or infrahepatic vena cava, with or without the use of venous bypass. Alternatively, the recipient vena cava may be left in situ (piggyback technique) by further mobilization of the liver by dividing the short hepatic veins that run from the anterior surface of the vena cava directly into the posterior aspect of the liver. To gain access to the short hepatic veins, the liver must be lifted and rotated to the left. This maneuver may result in partial occlusion of the inferior vena cava, which may impair venous return causing a temporary drop in blood pressure. The piggyback technique, where the recipient vena cava is left in situ, has the advantage that venous return is not compromised during the anhepatic phase and thus precludes the need for venous bypass; however, the hepatectomy phase is longer and can be associated with greater bleeding.

The **anhepatic phase** may be associated with significant hemodynamic changes, depending on the technique used for vascular control. This stage of the operation consists of implantation of the liver allograft, with or without venous bypass. The use of venous bypass is particularly helpful in coagulopathic patients with severe portal HTN. In these high-risk patients, the goal of the venous bypass system is to relieve the portal HTN by “bypassing” the liver. Cannulae, placed in the femoral and portal veins, draw the blood out of the systemic and splanchnic venous systems into a pump that delivers the blood into the axillary or jugular vein, maintaining the venous return ([Fig. 7.12-4](#)). This system allows the interruption of the vena cava with mild-to-moderate hemodynamic changes, depending on the blood flow rate through the system. The benefits and potential complications of the venous bypass system are listed in [Table 7.12-2](#).

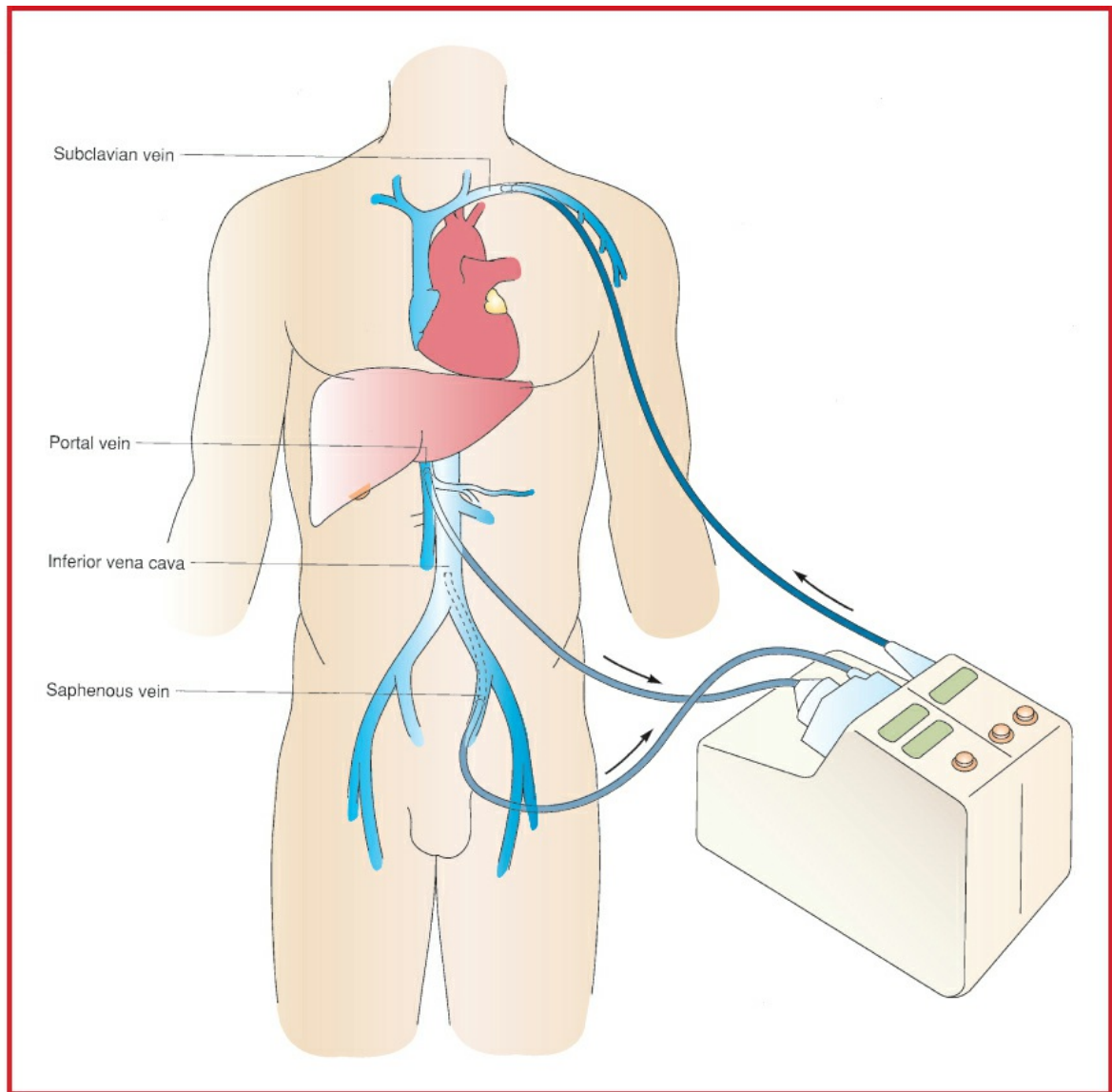


Figure 7.12-4 Setup for venovenous bypass during hepatic transplantation. Cannulae are placed into the portal vein to decompress the splanchnic bed and inferior vena cava (through the greater saphenous vein) to decompress the lower extremities and kidneys during the anhepatic phase of the transplant. A peristaltic pump is used to deliver bypassed blood to the central circulation by means of a cannula introduced into the axillary veins. Cannulae also may be placed percutaneously directly into the femoral and subclavian veins. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Wound complications and nerve injuries may be prevented by introducing the bypass cannulae percutaneously, rather than approaching the vessels through a surgical incision. An internal jugular bypass cannula can be placed preoperatively using the **Seldinger technique**. A femoral line can also be placed preoperatively and replaced by a bypass cannula intraoperatively. When venous bypass is utilized, vascular control is obtained by placing vascular clamps across the supra- and infrahepatic vena cava or the confluence of the hepatic veins (piggyback technique) and the portal vein. The

splanchnic venous return is interrupted during the anhepatic phase. The systemic venous return is interrupted in the case of formal cross-clamping of the supra- and infrahepatic vena cava and mildly diminished when the piggyback or venous bypass techniques are used. These can lead to significant hypotension unless preventive measures, as reviewed in Anesthetic Considerations, are taken (Fig. 7.12-5).

In a **standard orthotopic liver transplant**, with or without venous bypass, the recipient's vena cava is removed, leaving two cuffs—one just below the diaphragm and the other distal to the confluence of the renal veins. The deceased donor liver, with the corresponding segment of the vena cava, is used for restoring the continuity of the recipient's vena cava. The first vascular anastomosis consists of an end-to-end anastomosis of the allograft suprahepatic vena cava and the cuff of the recipient's infradiaphragmatic vena cava. This is followed by the reconstruction of the infrahepatic vena cava with an end-to-end anastomosis. Immediately prior to completion of the infrahepatic vena caval anastomosis, the liver is purged with chilled or room temperature albumin and/or crystalloid solution via the allograft portal vein to remove the preservative solution, which may contain high concentrations of potassium (~145 mEq/L K⁺). In addition, flushing the liver also removes a significant amount of the air that gets introduced during the procurement and preparation of the allograft for transplantation. Finally, the portal vein reconstruction is completed with an end-to-end anastomosis. At this point, the clamps are removed, ending the anhepatic phase of the operation.

Table 7.12-2 Benefits and Potential Complications of the Venovenous Bypass System	
Benefits	Complications
Improved hemodynamics during anhepatic phase	PE
↓ blood loss	Air embolism
May improve perioperative renal function*	Brachial plexus injury Wound seroma/infection Vascular injury

*In a prospective randomized trial comparing venovenous bypass with no bypass, no difference was found in the periop renal function between the two groups.

Venous bypass is not necessary when the **piggyback technique of liver transplantation** is utilized because the diseased liver is separated from the vena cava (systemic venous

return remains unimpaired), and vascular control is obtained by placing a clamp across the confluence of the hepatic veins as they join the vena cava (Fig. 7.12-6). A temporary portacaval shunt may be created to minimize bleeding in cases of severe portal HTN. The first anastomosis is between the suprahepatic vena cava of the liver allograft and the cuff created from the hepatic veins. The infrahepatic vena cava of the liver allograft is ligated, and the portal vein reconstruction is then completed. The clamps are then removed and the liver is revascularized.

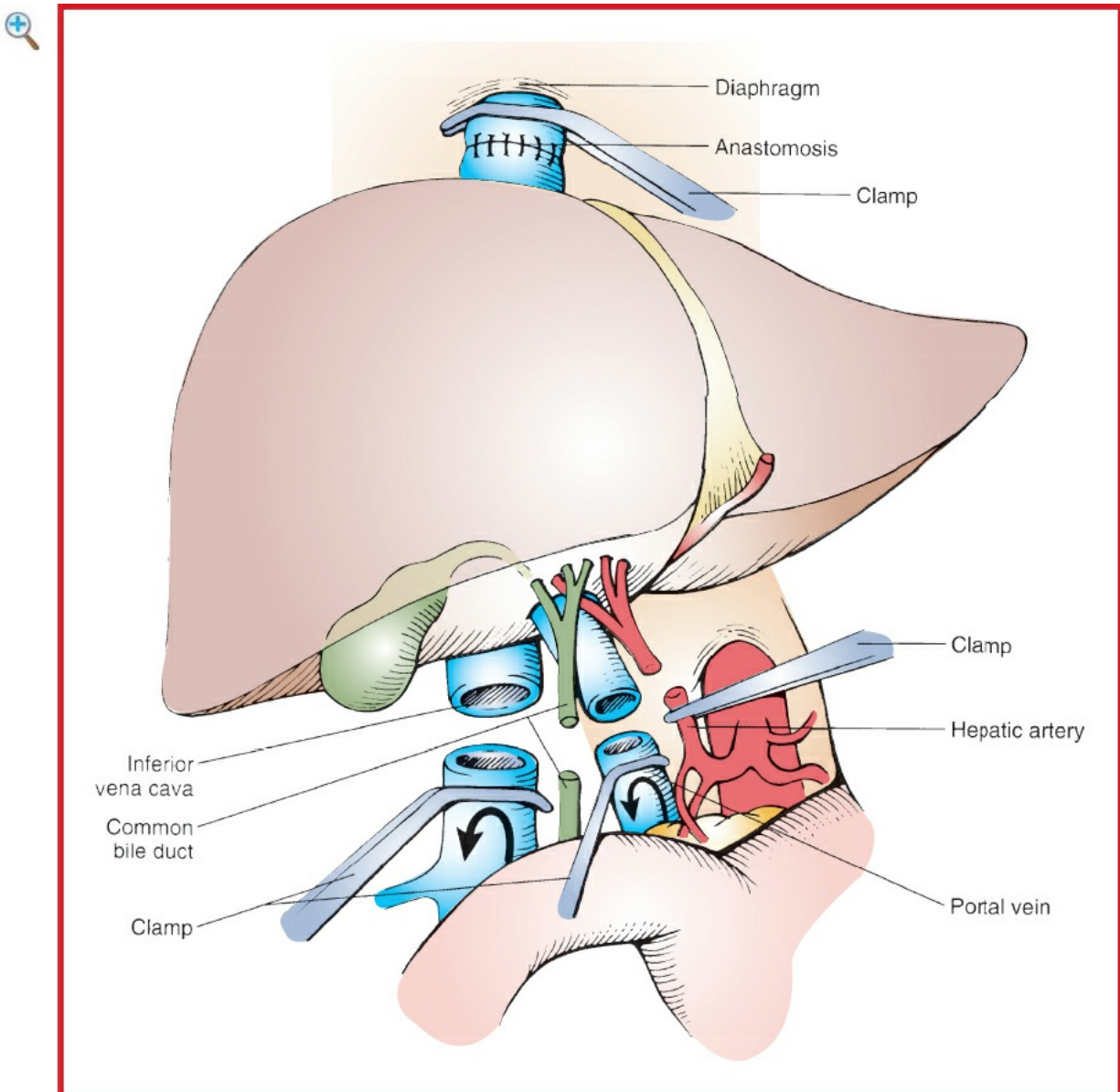


Figure 7.12-5 Standard liver transplantation without venovenous bypass. Venous return is significantly impaired.

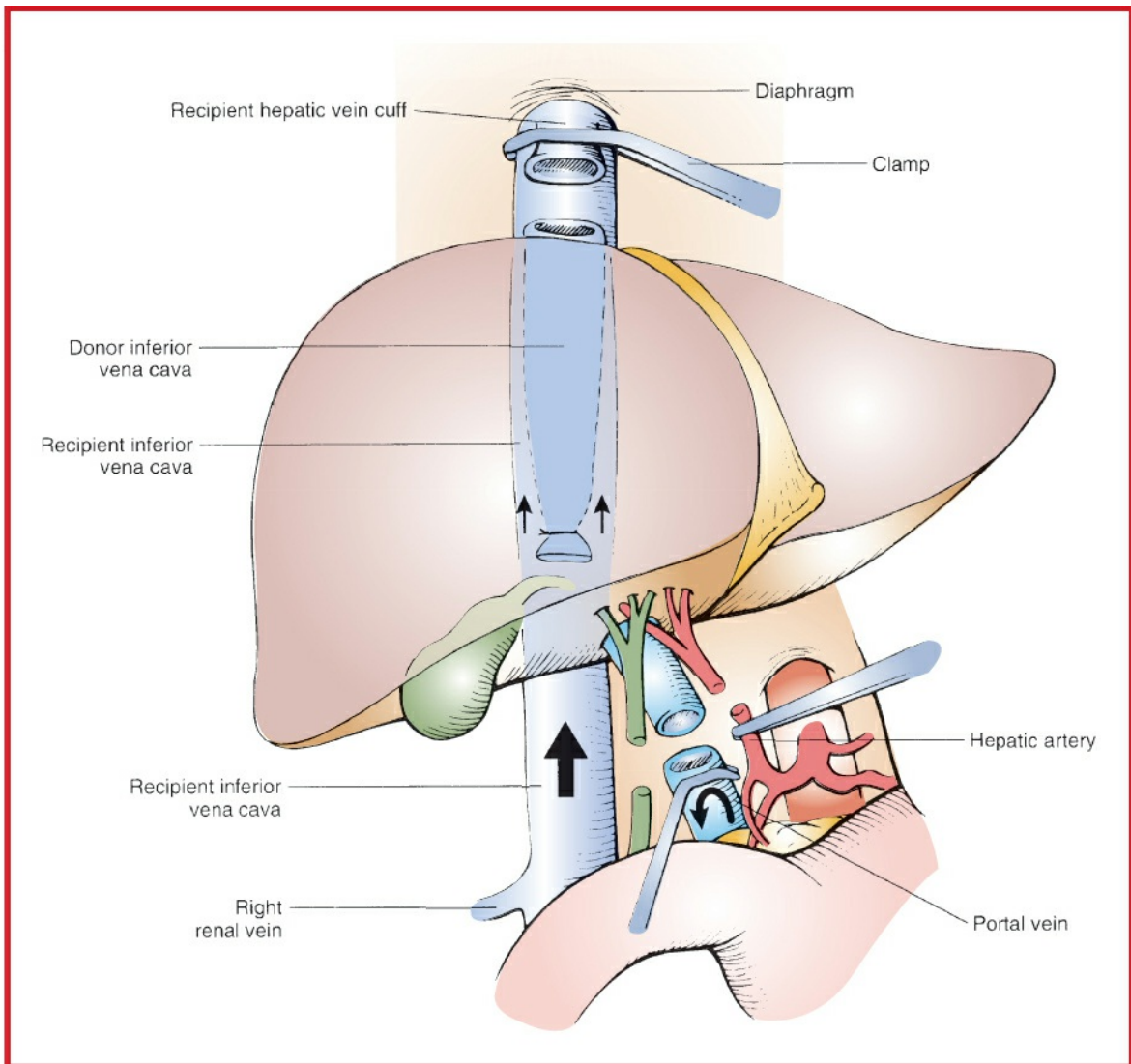


Figure 7.12-6 Piggyback liver transplantation. Note that the recipient's vena cava is left intact and systemic venous return is unimpaired.

The **postrevascularization stage** of the transplant begins with the removal of the vascular clamps. The reperfusion of the liver may be the most critical part of the operation. Despite flushing the liver to remove the high K^+ -containing organ preservation solution, hyperkalemia may cause cardiac arrhythmias following liver reperfusion, particularly with livers that sustained significant injury during preservation and reperfusion. In addition, massive air embolism is an immediate concern following revascularization, as it may quickly lead to cardiac arrest. It is also during this stage that the patient may experience increased right heart pressures, which can lead to right heart failure and severe systemic hypotension. Pulmonary HTN and right heart failure must be treated aggressively with inotropic agents; otherwise, the liver is subjected to high outflow resistance resulting in congestion and worsening of the allograft preservation injury. The cause of this phenomenon is not well understood; fortunately, it is seen in

very few patients. Another reperfusion phenomenon is that of systemic hypotension secondary to peripheral vasodilation. This may be due to the release of systemic inflammatory mediators, which include kinins, cytokines, and free radicals from the liver allograft. Reperfusion of the liver also can have dramatic effects on coagulation, such as fibrinolysis resulting in severe hemorrhage or hypercoagulation that can result in venous thrombosis and a massive pulmonary embolism with cardiovascular collapse.

Immediately prior to revascularization, the patient is usually given methylprednisolone (250–1000 mg) as part of the immunosuppressive regimen, as well as an adjunct to counteract the systemic effects of ischemia-reperfusion injury of the liver. At this point, all of the vascular anastomoses, the peritoneum, and the liver (especially the cut surface in segmental or reduced-size grafts) are inspected for surgical bleeding.

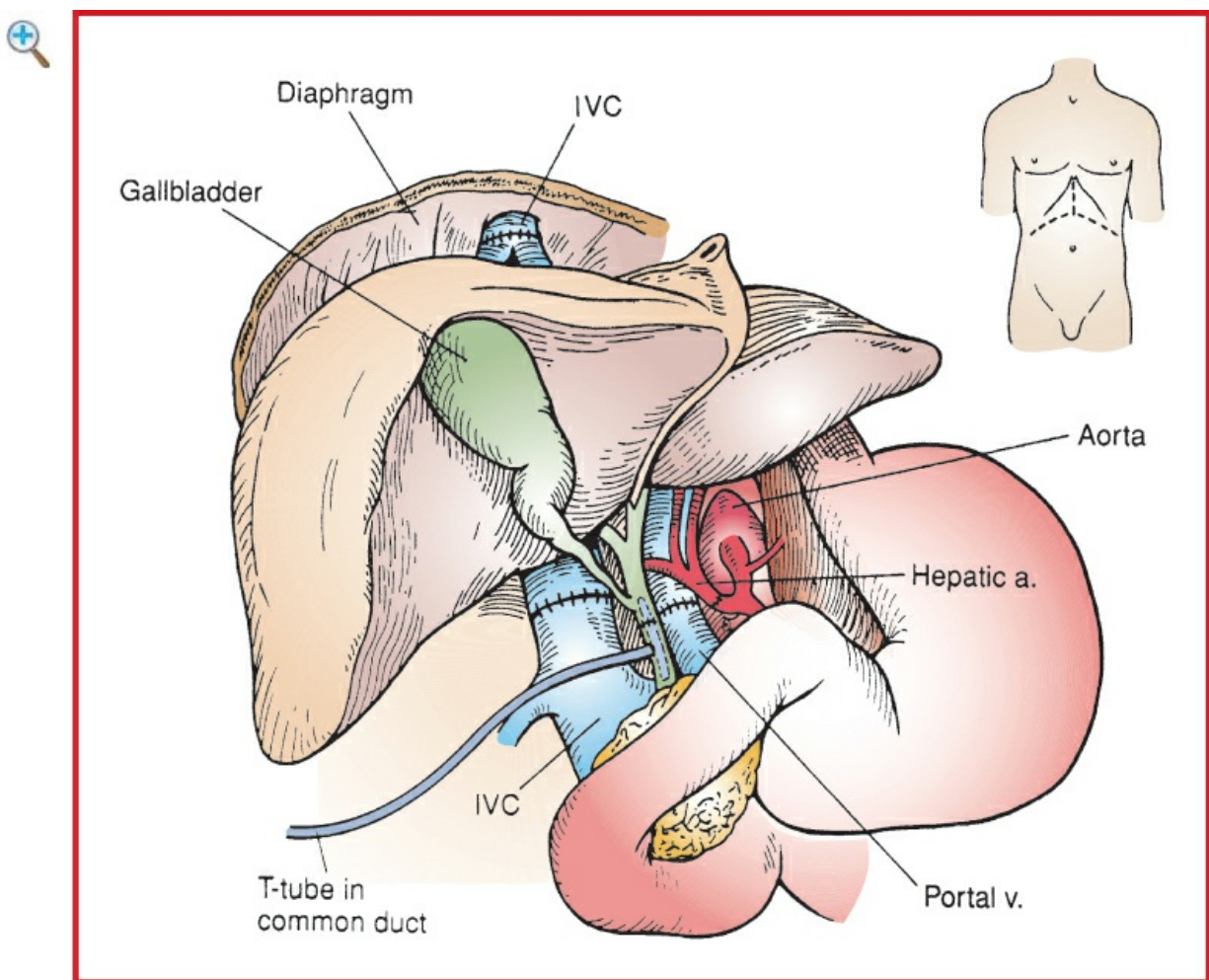


Figure 7.12-7 Liver transplantation. Anastomoses—including suprahepatic and infrahepatic IVC, portal vein, hepatic artery, and common bile duct—are complete as shown here. Roux-en-Y loop of the small intestine is an alternative biliary drainage conduit. **Inset** shows a chevron incision with midline extension. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

The hepatic artery reconstruction is performed after stabilization of the patient following revascularization of the liver. After the hepatic arterial anastomosis is completed, it is important to maintain adequate arterial BP (MAP > 65 mm Hg) to prevent hepatic artery thrombosis. This is especially critical in pediatric transplant recipients, where the hepatic artery diameter ranges from 1 to 3 mm. The last part of the procedure involves hemostasis, removal of the gallbladder, and reconstruction of the bile duct ([Fig. 7.12-7](#)). There are three basic methods for the **bile duct reconstruction**: (a) an end-to-end anastomosis, with or without a T-tube (in patients with normal common bile ducts), and (b) a choledochojejunostomy to a Roux-en-Y limb of the jejunum ([Fig. 7.12-8](#); in patients with biliary atresia, primary sclerosing cholangitis, and diseased common bile ducts or when there is a size discrepancy between the donor and recipient common bile duct), (c) a choledochoduodenostomy (in patients with congested bowel or extensive bowel adhesions and an indication for a Roux-en-Y). In cadaveric or live-donor segmental transplantation, the technique for the recipient's hepatectomy and the implantation of the allograft is not different from that of full-size liver transplantation; however, the technique of piggyback liver transplantation must be used with live donors because the allograft segment does not include the vena cava. The anesthesiologist must be alert during the reperfusion of a segmental graft because significant bleeding may ensue from the raw surface of the liver.

After the biliary reconstruction is completed and bleeding controlled, a feeding jejunostomy tube and two to three closed-suction drains may be placed. The position of an OG or NG tube (placed at the beginning of the case) is confirmed and the abdomen is closed.

Usual preoperative diagnosis

ESLD

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status and possible optimization, discussion of carb loading, DVT prophylaxis, urinary drain, pain management options, and postop recovery plan. See [Appendix K, p. K-1](#).

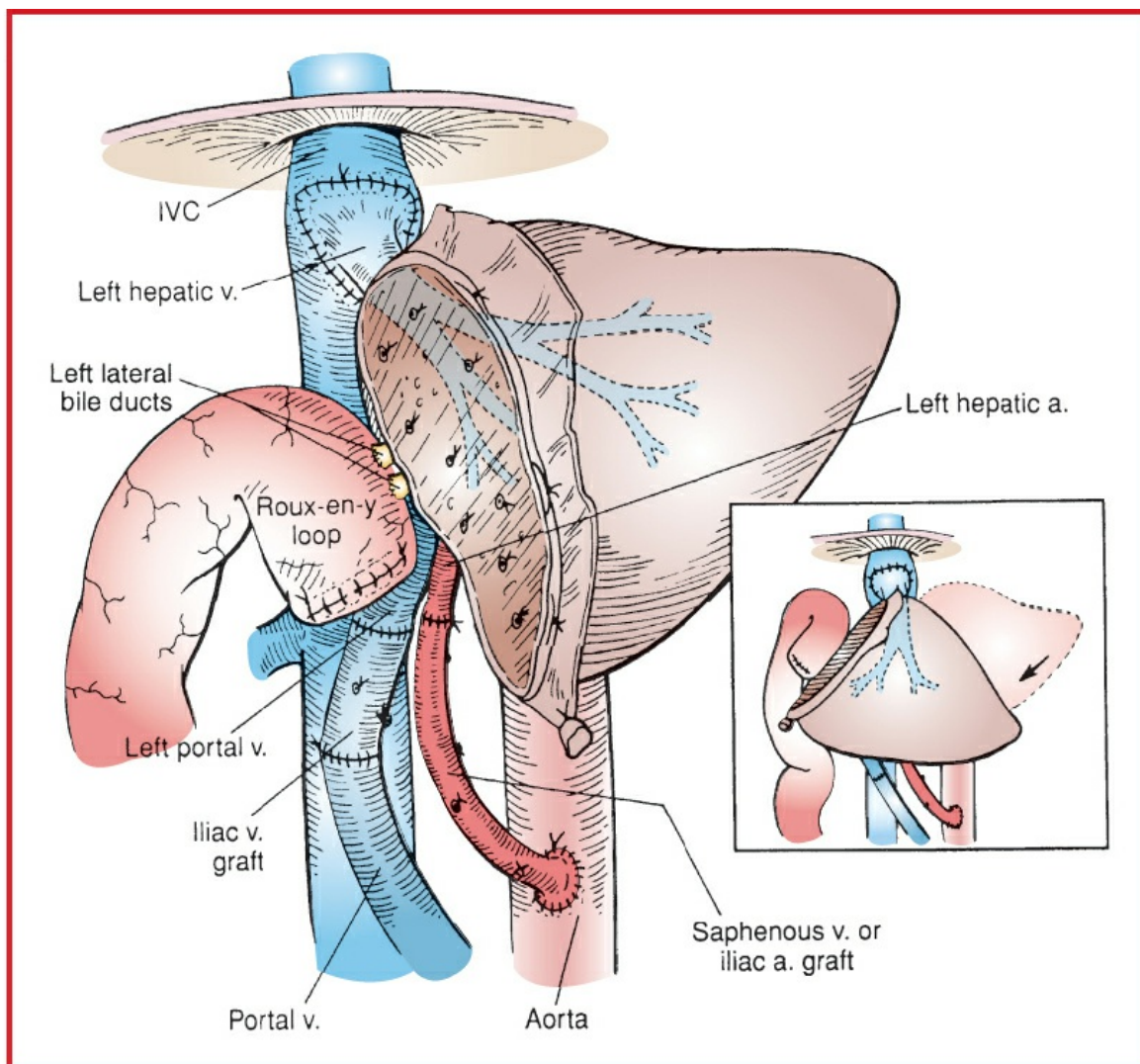


Figure 7.12-8 Liver transplantation (child) using left lateral segment from an adult liver. The hepatic artery and portal vein are extended with donor iliac artery and vein, respectively. The final position of the graft is shown (**inset**). A Roux-en-Y loop of the small intestine is used to drain the bile duct(s). The IVC is left intact. The cut surface of the liver can bleed excessively if the central venous pressure is too high. (Reproduced with permission from Broelsch CE, Whittington PF, Emond JC, et al: Liver transplantation in children from living related donors: surgical techniques and results. *Ann Surg* 1991; 214(4):432.)

■ SUMMARY OF PROCEDURE

Position	Supine; arms tucked. Left arm and left groin area out for access to the axillary and femoral veins if venous bypass is anticipated
Incision	Bilateral subcostal, in children; in adults, incision must extend cephalad to the xiphoid process.
Special instrumentation	Upper hand or Thompson retractor; venous bypass pump; rapid-infusion system; cell salvage system;

	argon beam coagulator, ultrasound
Unique considerations	Thrombus or air embolism may occur during removal of clamps from vena cava or with the use of venovenous bypass. Right heart failure, ↓ BP, and ↓ SVR may be observed after revascularization. Continuous AV hemofiltration may be required if renal failure is present. Head and extremities should be covered with plastic to maintain core body temperature, particularly in children. Gastric tube is required.
Antibiotics/drugs	Third-generation cephalosporin (e.g., cefotaxime 1 g q3h intraop) or broad-spectrum penicillin/beta-lactamase prior to incision. Methylprednisolone and antilymphocyte antibody preparations for immunosuppression
Surgical time	4–12 h
EBL	6 U average blood loss (range 0–100 U)
Postop care	ICU: 1–2 d. HTN commonly seen
Mortality	3% at 1 yr
Rejection	10–30% during the first yr
Morbidity	Infectious complications: 20–50% Biliary stenosis or leaks: 5–15% Retransplantation: 6–14% Primary graft nonfunction: 2–5% Hepatic artery thrombosis: 0–6% Portal vein thrombosis: 1–4%
Pain score	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Neonate–70 yr
Male:Female	1:1
Incidence	10/million/yr (15% pediatrics)
Etiology	Adult: hepatitis C cirrhosis; alcoholic cirrhosis; primary

Associated conditions

biliary cirrhosis; primary sclerosing cholangitis; hepatitis B cirrhosis; hepatocellular carcinoma. Pediatric: biliary atresia; inborn errors of metabolism, hepatoblastoma

Coagulopathy; hypoalbuminemia; ascites; cardiomyopathy (in alcoholic patients, hemochromatosis, and Wilson disease); hepatorenal syndrome; hepatopulmonary syndrome, portopulmonary hypertension/GI bleed, hepatic encephalopathy, and hypoglycemia in acute fulminant hepatitis

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients needing liver transplantation represent a formidable challenge to the anesthesiologist. Frequently, these patients present for surgery with multiorgan system failure.

Respiratory

Patients are often hypoxic because of ascites, pleural effusions, atelectasis, V/Q mismatch, pulmonary AV shunting, or hepatopulmonary syndrome. As a result, they are usually tachypneic and have a respiratory alkalosis. Evidence of pulmonary infection is usually a contraindication to surgery, but ARDS that may occur with hepatic failure is not.

Tests: ABG; PFTs, as indicated. CXR: infection, effusions, atelectasis

Cardiovascular

These patients demonstrate a hyperdynamic state with $\uparrow$ plasma volume, $\uparrow$ CO, and $\downarrow$ SVR 2° arteriolar vasodilation in the splanchnic circulation, with intense vasoconstriction in other vascular territories (e.g., renal, brain, muscle, spleen). The SVR usually is not responsive to α -agents. AV fistulae may occur across the pulmonary circulation, so that precautions to prevent air embolism are important. Ejection fraction (EF) is usually high ($>60\%$), but some

patients have cirrhotic cardiomyopathy → ↓ contractility 2° ↓ β-receptors, alterations in myocardial cell membrane properties, and ↑ myocardial depressant substances. This cardiac dysfunction, however, usually is masked by a reduction in afterload. Pericardial effusions may be present and should be drained at surgery. Many of these patients will have dysrhythmias, HTN, pulmonary HTN (very high risk), valvular disease, cardiomyopathy (alcoholic disease, hemochromatosis, Wilson disease), and CAD. These patients will require appropriate preop consultation and workup.

Tests: ECG; stress ECHO: EF, contractility, pulmonary HTN, wall motion abnormalities, valve problems. If abnormal, right and left heart catheterization with coronary artery angiography should be performed.

Neurologic

Patients are often encephalopathic and may be in hepatic coma; however, other organic causes of coma should be ruled out. In fulminant hepatic failure, ↑ ICP is common, accounting for 40% of mortality (herniation) and may require prompt treatment (mannitol, hyperventilation, etc.).

Tests: Continuous ICP monitoring in fulminant hepatic failure

Hepatic

Hepatitis serology and the cause of hepatic failure should be determined. Vascular abnormalities, previous RUQ surgery, or portal vein decompressive surgery places the patient in a high-risk group. Albumin is usually low, with consequent low plasma oncotic pressure → edema, ascites. The magnitude and duration of drug effects may be unpredictable, but generally, these patients have ↑ sensitivity to all drugs and their actions are prolonged.

Tests: Bilirubin; PT; ammonia level; SGOT; SGPT; albumin

Gastrointestinal	Portal HTN, esophageal varices, and coagulopathies ↑ risk of GI hemorrhage. Gastric emptying is often slow and, together with the emergent nature of this surgery, warrants rapid-sequence induction. H ₂ antagonists are indicated preop.
Renal	Renal function ↓, especially in fulminant hepatic failure (hepatorenal syndrome). The kidneys often recover after transplantation, but simultaneous kidney transplantation may be justified. These patients are often hypervolemic, hyponatremic, and possibly hypokalemic. Ca ⁺⁺ is usually normal. Metabolic alkalosis may be present. Consider preop dialysis and intraop continuous venovenous hemofiltration (CVVH). CVVH corrects fluid overload, acidosis, and electrolyte abnormalities and results in the washout of released cytokines after reperfusion of the graft liver. Mannitol (0.5–1 g/kg) may be used intraop to maintain renal function. Tests: BUN; Cr, creatinine clearance; electrolytes; ABG
Endocrine	Patients are often glucose intolerant or frankly diabetic, although acute hypoglycemia may be seen in acute hepatic failure. Hyperaldosteronism may be present. Tests: Glucose; electrolytes
Hematologic	These patients are often anemic 2° either blood loss or malabsorption. Coagulation is impaired because of ↓ hepatic synthetic function (all factors except VIII and fibrinogen are ↓), abnormal fibrinogen production, ↓/impaired Plt, fibrinolysis, and low-grade DIC. Tests: PT; PTT; Plt count; bleeding time; fibrinogen; fibrin split products (FSP); TEG
Premedication	Low doses of benzodiazepines may be used judiciously, but often nothing is given prior to surgery. Usually good preop evaluation and discussion suffice. Intramuscular injection should be

ERAS

avoided. Full-stomach precautions are justified. Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane or thoracic epidural block for postop pain management. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

GETA: These patients are extremely complex to manage because of the hemodynamic instability, massive blood loss, coagulopathy, and metabolic problems. It is convenient to divide the operation into three stages: preanhepatic, anhepatic, and neohepatic (discussed later).

Induction

Often, a narcotic (e.g., fentanyl 2–5 mcg/kg) is given just before induction; and rapid-sequence induction is preferred. Etomidate (0.3 mg/kg) with succinylcholine (1–2 mg/kg), together with cricoid pressure

ERAS

Verify preincision antibiotics. Assess volume status (see [Appendix K](#)). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.

Maintenance

Standard maintenance with fentanyl 10–50 mcg/kg. A benzodiazepine (e.g., midazolam 0.1–0.3 mcg/kg) often is given to ensure amnesia during periods of hemodynamic instability when the volatile agent may need to be off. N₂O is avoided because of bowel distension and possible air embolism. Ventilation with FiO₂ > 0.5 and PaCO₂ = ~35 mm Hg. Occasionally, PEEP (5 cm H₂O) is added. Antibiotics and immunosuppressants should be given per surgeon's direction. Muscle relaxation usually is maintained.

Preanhepatic phase

The **preanhepatic phase** starts at skin incision and ends with removal of the recipient liver. Pleural and pericardial effusions are drained, which may improve

oxygenation. Hyperglycemia is common during this period. A drop in filling pressures may be 2° hemorrhage or vascular compression. Hemorrhage can be severe 2° portal HTN. Coagulation problems usually increase during this stage, although fibrinolysis is not usually a problem. Blood loss replacement is accomplished with blood (PRBC) and FFP.

Cryoprecipitate and Plts are given as needed, but a hypercoagulable state should be avoided, particularly if venovenous bypass is contemplated. Hemodynamic instability is not uncommon during the hepatic vascular dissection 2° manipulation of the liver and ↓ venous return.

Venovenous bypass relieves most of the complications of portal and IVC cross-clamping (↓ venous return, low CO, tachycardia, acidosis, ↓ renal function, intestinal swelling). Blood is pumped from the femoral vein and the portal system (either portal vein or inferior mesenteric) via a centrifugal pump to the right internal jugular vein. Generally, no heparin is used. Bypass flow needs to be at least 1 L/min to avoid possible thromboembolism. Bypass flow depends on venous inflow and is drawn into the pump by negative pressure. Low flows may be caused by hypovolemia or obstructed cannulae. Complications include unexpected decannulation, thromboembolism, and air embolism, all of which may need rapid termination of bypass and treatment of ↓ BP; fibrinolysis is seen with prolonged venovenous bypass.

Massive blood transfusion is associated with ↓ Ca⁺⁺, and replacement is usually needed (±500 mg/1000 mL of blood/FFP/Plasma-Lyte mixture). If hyperkalemia occurs, it should be treated aggressively. Metabolic acidosis >5 mEq/L should be treated with bicarbonate or THAM acetate (tris-hydroxymethyl aminomethane) to avoid a rapid increase in sodium. Typical loading dose: mL of 0.3M THAM = lean body weight [kg] × base deficit [mmol/L]. Occasionally, inotropic support

is needed, but α -adrenergic agents should be avoided because of $\downarrow$ renal and peripheral perfusion. UO needs to be maintained by ensuring adequate intravascular volume; occasionally mannitol may be needed.

Anhepatic phase

The **anhepatic phase** begins with clamping of the hepatic vessels and vena cava and removal of the liver; it ends with the reperfusion of the donor liver. Problems during this period include hemorrhage, increasing coagulopathy and fibrinolysis, acidosis, hypothermia, and $\downarrow$ renal function. The hemodynamic instability associated with clamping of the hepatic vessels and the congestion of the bowel that occurs can be decreased by venovenous bypass (see the previous text). Care should be taken to maintain intravascular volume, while avoiding volume overload, because this would worsen fluid overloading on reperfusion. At the completion of vena caval anastomoses, the liver is flushed via the portal vein to remove air, preservation fluid, and metabolites. Reperfusion may take place after completion of the portal vein anastomosis or after both portal vein and hepatic artery anastomoses are completed. As in the preanhepatic phase, acidosis, $\downarrow$ Ca^{++} , glucose, coagulation, and other electrolyte abnormalities should be treated. Fibrinolysis usually starts in this period but is not usually treated unless severe because of the potential for embolism during venovenous bypass.

Neohepatic phase

The **neohepatic phase** begins with the unclamping of the portal vein, hepatic artery, and vena cava and reperfusion of the donor liver. Preparation for this phase is important because this may be a period of great hemodynamic instability. Before removal of the clamps, acidosis should be corrected, ionized Ca^{++} should be normal, and K^+ should be <4.5 mEq/L. CaCl_2 , NaHCO_3 , and epinephrine should be readily available. Fluid overload prior to declamping should be avoided. High venous filling pressures decrease hepatic perfusion, especially prior to hepatic artery

	anastomosis. Declamping can be attended by ↓ BP, ↓ HR, dysrhythmias, hypothermia, lactic acidosis, coagulopathy, hyperglycemia, and thromboembolism.
Reperfusion syndrome	The reperfusion syndrome (which can occur in this phase) is characterized by ↓ HR, ↓ BP (30% of patients develop MAP < 70% of baseline), conduction defects, and ↓ SVR in the face of acutely ↑ RV filling pressures. Cause unknown. A rapid ↑ K⁺ can → cardiac arrest. (Rx: ensure normal pH and electrolytes prior to unclamping, rapid therapy when it occurs.) ↓ BP and ↓ HR are treated with epinephrine (10 mcg increments), whereas CaCl₂ and NaHCO₃ are used to correct hyperkalemia and acidosis. Pulmonary edema may occur as a result of fluid overload and can be treated with diuretics, inotropes, and phlebotomy. A high venous pressure will cause graft congestion and should be avoided. Reperfusion is associated with severe coagulopathy due to fibrinolysis (usually primary), release of heparin, and hypothermia. As liver function returns, there should be an improvement in coagulation, acid-base status (metabolic alkalosis may occur), ↓ lactic acidosis, return of glucose to normal, and bile production. Hypokalemia may occur 2° uptake by the liver. Graft failure is associated with coagulopathy, ↑ lactic acid, citrate intoxication, hyperglycemia, and ↓ bile formation.
Emergence	Extubation is deferred. These patients are generally ventilated postop in ICU until stable and able to be weaned from ventilatory support. Apart from the usual tests, monitor hepatic function. Also ensure immunosuppression provided, infection controlled, analgesia adequate (usually fentanyl), and peptic ulcer prophylaxis given (ranitidine preferred). Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. PONV prophylaxis may include dexamethasone and ondansetron.

Blood and fluid requirements	Massive blood loss IV: 10 Fr × 2 BSS UO > 1 mL/kg/h Warm all fluids Humidify gases Rapid-infusion system Cell salvage system 20 U PRBC 20 U FFP 2 U PLT	Generally, ivs are placed in the right antecubital fossa, left IJ or EJ. The right IJ is reserved for venovenous bypass. Plasma-Lyte A™ and Normosol™ are preferred (absence of glucose, Ca⁺⁺, and lower Na⁺ content) over NS or LR. Hypernatremia may occur due to administration of NaHCO₃. The ability to infuse up to 1.5 L/min of blood should be available. Usually a mixture of Normosol (250 mL), PRBC (1 U), and FFP (1 U) is used, yielding Hct = 26–30%. Actual blood loss estimation is extremely difficult, and usually replacement is judged by hemodynamic status, UO, and S_vO₂. Cell salvage systems are used to conserve blood. Anticoagulation is achieved with citrate solution to avoid heparin contamination, and cells are washed with Normosol/Plasma-Lyte A. Discontinue use before biliary reconstruction (infection) or in neoplasms, hepatitis B, or spontaneous bacterial peritonitis. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors ECG (5-lead) Temp bladder	Include 5-lead ECG and bladder T. (These patients sustain significant heat loss.) A full-time anesthesia technologist and lab/blood bank runner are useful. Lab and blood

Arterial line(s)

bank should be notified of the expected transplant. One or two arterial lines are placed at the outset—one in the right radial, for ongoing lab and blood gas sampling; another line, in the right femoral artery, is utilized for continuous pressure measurement.

PA catheter/ S_vO_2 /CO

A PA catheter is essential for management of hemodynamics in these patients because of the rapid changes in VS and the fact that a significant number of patients have recognized or unrecognized pulmonary hypertension. A catheter capable of measuring mixed venous O_2 sat is very useful because it gives early clues to impending decompensation. Coagulopathy complicates the placement of central lines, and the use of ultrasound guidance is recommended.

TEE

TEE is useful to monitor cardiac filling and function and to diagnose problems such as PE or air embolism. Care needs to be taken in placing the TEE because many of these patients have esophageal varices.

ICP

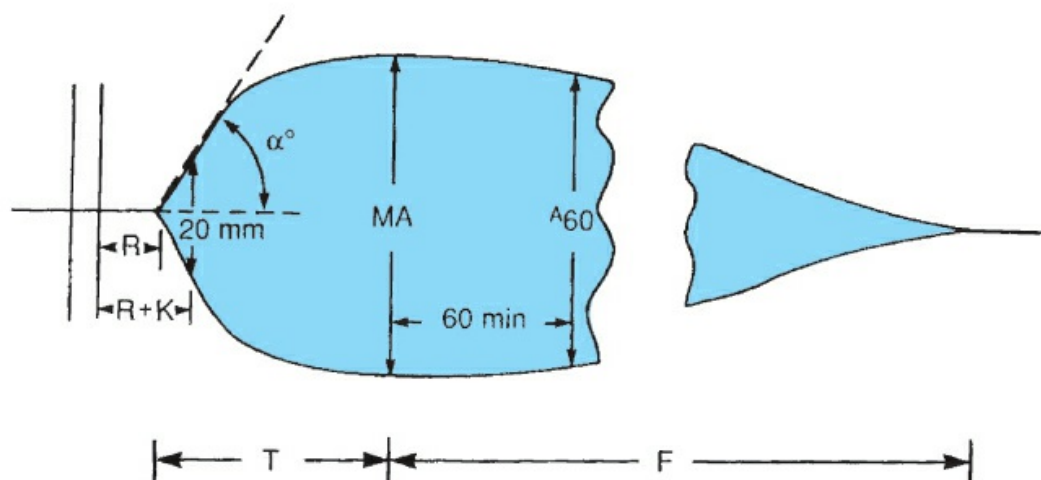
ICP should be measured in patients with fulminant hepatic failure if $\uparrow$ ICP is a concern.

Laboratory

ABG, acid-base status, electrolyte, lactate, osmolality, Ca^{++} , PT, PTT, Plt, Hct—all

	Thromboelastograph (TEG)	should be monitored on a regular basis (hourly or half-hourly; occasionally, more frequently). TEG is useful for monitoring coagulation (see the subsequent discussion of coagulation management).
Coagulation management	PT PTT Plt counts Fibrinogen FSP TEG	Patients are prone to a variety of coagulopathies (↓ Plt, ↓ coagulation factors, DIC, fibrinolysis, etc.) because of preop factors, massive hemorrhage, anhepatic period, and reperfusion of the new liver; therefore, monitoring and treatment are necessary. Also, states of hypercoagulopathy need to be avoided because of unheparinized venovenous bypass. Although PT, PTT, Plt counts, fibrinogen, and FSP may provide relevant information, they may not reflect the true coagulability of patient's blood and tend to take considerable time to perform. Thus, in some centers, TEG has gained in popularity. It measures whole blood coagulability, not specific factors. TEG works by measuring viscoelastic properties of blood as it forms clot (fibrin connections) between a rotating cuvette and a spindle. Characteristic patterns are formed by the various coagulopathies with the common

		types shown in Figure 7.12-9. Evaluation of the TEG leads to more rational transfusion therapy, reducing the number of units of blood products used. Comparing specimens of native whole blood vs. blood mixed with EACA or protamine can guide pharmacologic therapy of coagulopathies. Table 7.12-3 gives specific recommendations.
Positioning	✓ and pad pressure points ✓ eyes	Table and arm boards should be very well padded. The head should be placed on a foam rest. Particular care should be taken to pad the retractor supports where they may impinge on the arms and on the radial nerve as it curls around the humerus.
Temperature control	Warming blanket Humidifier	Patient's arms, head, and legs should be wrapped in plastic to protect against heat loss. Plastic drapes and the use of a cesarean section-type drape to protect the ECG electrodes and direct fluid flow off the table are useful to prevent the patient from lying in a pool of fluid. A warming blanket under the patient and over the lower legs is very useful.
Complications	Coagulopathy Hemorrhage Air embolism RV failure Metabolic acidosis	



Qualitative Interpretation

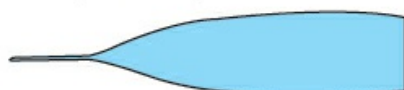
Normal

$R/K/MA/\text{Angle} = \text{Normal}$



Heparin

$R/K = \text{Prolonged}$; $MA/\text{Angle} = \text{Decreased}$



Thrombocytopenia

$R = \text{Normal}$; $K = \text{Prolonged}$; $MA = \text{Decreased}$



Fibrinolysis (e.g., streptokinase)

$R = \text{Normal}$; $MA = \text{Continuous decrease}$



Hypercoagulation

$R/K = \text{Decreased}$; $MA/\text{Angle} = \text{Increased}$



No Platelet Function (DIC)

$R = \text{Prolonged}$; $MA/\text{Angle} = \text{Decreased}$



Figure 7.12-9 Variables and normal values measured by TEG: R = reaction time, 6–8 min; $R + k$ = coagulation time, 10–12 min; α = clot formation rate, $>50^\circ$; MA = maximum amplitude, 50–70 mm; A_{60} = amplitude 60 min after MA ; $A_{60}/MA-100$ = whole blood clot lysis index, $>85\%$; F = whole blood clot lysis time, >300 min. (Reproduced with permission from Kang YG, Martin DJ, Marquez J, et al: Intraoperative changes in blood coagulation and thromboelastographic monitoring in liver transplantation. *Anesth Analg* 1985; 64:891.)

Table 7.12-3 Coagulation Therapy Guided by TEG Monitoring

1. Maintenance fluid

RBC:FFP:Plasma-Lyte A = 300:200:250 mL

2. Replacement therapy

- FFP (2 U) for prolonged reaction time ($R > 15$ min)
- Plt (10 U) for small MA ($MA < 40$ mm)
- Cryoprecipitate (6–12 U) for persistent slow clot formation rate ($\alpha < 40^\circ$) with normal MA

3. Pharmacologic therapy

- Compare coagulability of whole blood, blood treated with protamine sulfate, and blood treated with epsilon aminocaproic acid.
- Epsilon aminocaproic acid (1 g) for severe fibrinolysis ($F < 60$ min)
- Protamine sulfate (50 mg) for severe heparin effect

- Heparin (1000–2000 U) for hypercoagulable state

▼ POSTOPERATIVE

Monitoring of hepatic function	Serial LFTs PT, PTT Ammonia level Lactate TEG Bile output	Initial LFTs often show very high liver enzymes, which subside over a period of days. PT generally improves to normal levels, while lactic acidosis usually corrects quickly. Often, a metabolic alkalosis follows and may need HCl treatment.
Complications	Bleeding Partial vein thrombosis Hepatic artery thrombosis Biliary tract leaks Primary nonfunction Rejection Infection Pulmonary complication	This is not a complete list. Feared complications that may → graft loss include portal vein thrombosis, hepatic artery thrombosis, bile leaks, and rejection. These are attended by ↑ LFTs, lactic acidosis, coagulopathy, hypoglycemia, ↓ renal function, and poor bile formation.

HTN
Electrolyte
abnormalities
(hypokalemia, ↓ Ca⁺⁺,
↑ Na)
Alkalosis
Renal failure
Peptic ulceration
Neurologic

Suggested Readings

1. Adelman D, Kronish K, Ramsay MA: Anesthesia for liver transplantation. *Anesthesiol Clin* 2017; 35(3):491–508.
2. Black CK, Termanini KM, Aguirre O, et al: Solid organ transplantation in the 21(st century. *Ann Transl Med* 2018; 6(20):409.
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4. Lemon K, Al-Khafaji A, Humar A: Critical care management of living donor liver transplants. *Crit Care Clin* 2019; 35(1):107–16.
5. Melloul E, Hübner M, Scott M, et al: Guidelines for perioperative care for live surgery: Enhanced Recovery After Surgery (ERAS) Society recommendations. *World J Surg* 2016; 40(10):2425–40.

LIVING-DONOR LIVER TRANSPLANTATION

SURGICAL CONSIDERATIONS

Description

The success of deceased donor liver transplantation has resulted in an ever-increasing number of patients with ESLD waiting for transplantation; however, the number of deceased donors has remained relatively constant. Consequently, the waiting time to receive an organ has increased significantly, and ~15% of patients will die while waiting. The success of **living-donor renal transplantation**, coupled with the experience in adult-to-pediatric living-donor liver transplantation, as well as advances in surgical and postsurgical care of patients undergoing major liver resections, has led to the implementation of adult-to-adult living-donor liver transplantation.

Potential living liver donors undergo extensive medical and psychosocial evaluation to ensure psychological as well as physical fitness to undergo a major surgical procedure that provides no medical benefits to the donor. Donors must have full blood typing to ensure compatibility with the recipient and then fill out an extensive medical questionnaire, followed by a complete physical exam and screening lab tests. Any evidence of diabetes, HTN, or renal, pulmonary, living-donor liver transplantation, cardiovascular, or hepatic abnormalities usually is a contraindication to donation. After the potential donor is medically and psychosocially cleared, they undergo a detailed magnetic resonance imaging study of the liver to assess liver segment size, anastomosis, and possible anatomical contraindications.

The donor and recipient operations are usually conducted simultaneously to minimize the ischemic injury to the donor liver segment. The donor operation, however, is initiated first, with the recipient operation commencing only after the donor liver has been directly examined and no unforeseen anatomic barriers to donation are found. The living donor operation is similar to a right or left hepatic lobectomy undertaken for hepatic disease, although there are some differences that can have a significant impact on anesthetic management.

The donor may elect to have an epidural catheter for postop analgesia, and this usually is placed before surgery. Living donor liver resection has been successfully performed laparoscopically; however, due to limited experience and utilization of this technique, it will not be detailed here. A vertical midline incision is made from the xiphoid to just above the umbilicus and extended transversely to the right anterior axillary line. Occasionally, bisubcostal incisions are required. Following exploration of the abdomen, intraoperative ultrasound may be performed to map the hepatic venous anatomy so the plane of resection can be delineated. In addition, an intraoperative cholangiogram is performed via the cystic duct (a cholecystectomy is performed in either right or left hepatic lobectomy) or the common bile duct, to define the biliary anatomy. After this is performed, the corresponding portal vein and hepatic artery are isolated. Unlike in a hepatic lobectomy for tumor, the venous and arterial inflow to the liver segment is not ligated; thus, the transaction of the liver parenchyma may result in significant hemorrhage. The respective lobe of the liver is mobilized from its attachments, and the liver is dissected from the retrohepatic vena cava, with ligation of the short hepatic veins. This maneuver can cause transient hypotension secondary to torque and compression of the IVC and hepatic veins, as well as potential bleeding from the vena cava itself. Next, the hepatic vein is isolated, and the liver parenchyma is then divided, which can be a slow and tedious process. Following division of the hepatic parenchyma, the liver segment is ready to be removed. Heparin (80–100 U/kg) is given

to prevent intrahepatic clot formation. Following heparinization, the hepatic artery and portal vein are ligated and divided, followed by the hepatic vein. The donated hepatic lobe is immediately placed in ice and flushed with organ preservation solution. Once the donated liver segment has been removed, the abdomen and cut surface of the remaining liver are inspected for hemostasis and bile leak. Closed drains are placed, after which the abdomen is closed. Essentially, the same procedure is followed for a left lateral segmentectomy (adult to child), except that the amount of donor liver removed is about 25%, compared with 40% and 60% for a left or right hepatic lobectomy, respectively.

Usual preop diagnosis

Healthy, living liver donor

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine; arms tucked
Incision	Vertical midline with a right and/or left subcostal extension, depending on liver segment being utilized. Laparoscopic or laparoscopically assisted technique described
Special instrumentation	Retractor; laparoscopic or laparoscopically assisted technique described ultrasonic or hydrojet dissection/aspirator; irrigating bipolar cautery; argon beam coagulator; cell salvage system; rapid-infusion system
Unique considerations	Intraop cholangiogram and/or ultrasound. Hemodilution immediately before surgery, with removal of 1 U whole blood if Hct $\geq$ 40. Patients also may have donated 1–2 U of autologous blood. Heparin 80–100 U/kg
Antibiotics	Ampicillin 1 g and ceftriaxone 1 g iv before skin incision

Surgical time	3–5 h
EBL	250–500 mL
Postop care	ICU overnight or surgical floor. Hospital stay 5–7 d
Mortality	34 deaths reported worldwide (as of August 2010) following ~11,553 donor operations
Morbidity	Infectious complications: 15% Biliary leak: 8% Reoperation: 3–5% Acute hepatic failure: <0.1%
Pain score	7–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–55 yr
Male:Female	1:1
Incidence	Uncommon

ANESTHETIC CONSIDERATIONS

See Anesthesia for Hepatic Resection.

Suggested Readings

1. Abecassis MM, Fischer RA, Olthoff KM, et al: Complications of living donor hepatic lobectomy—a comprehensive report. *Am J Transplant* 2012; 12(5):1208–17.
2. Au KP, Chok KSH: Minimally invasive donor hepatectomy, are we ready for prime time? *World J Gastroenterol* 2018; 24(25):2698–709.
3. Baker TB, Jay CL, Ladner DP, et al: Laparoscopy-assisted and open living donor right hepatectomy: a comparative study of outcomes. *Surgery* 2009; 146(4):817–23.
4. Brige P, Hery G, Chopinet S, et al: Morbidity and mortality of hepatic right lobe living donors: systematic review and perspectives. *J Gastrointest Liver Dis* 2018; 27(2):169–78.
5. Chen KH, Siow TF, Chio UC: Laparoscopic donor hepatectomy. *Asian J Endosc Surg* 2018; 11(2):112–7.

MULTIORGAN PROCUREMENT

SURGICAL CONSIDERATIONS

Description

The families of brain-dead patients may allow donation of the patient's functioning organs—which may include, but is not limited to, the heart, lungs, liver, kidneys, pancreas, and small intestine. The process of organ procurement can be chaotic, with multiple operative teams and technicians working simultaneously. Moreover, brain-dead patients may be hemodynamically unstable, sometimes requiring multiple pressors, and the possibility of acute decompensation is ever present.

The donor patient's chest and abdomen are opened in the midline from the sternal notch to pubis (Fig. 7.12-10). The chest is opened with a sternal saw, and generally an extra-large Balfour retractor is used to widely retract the abdomen. The aorta and IVC are dissected first to allow rapid placement of a flush line in the event that the patient experiences a cardiovascular collapse. Following this, the liver vasculature is identified in the hepatoduodenal ligament and is dissected out. This part of the procedure generally takes about 1.5 h. In cases where the pancreas is procured, an additional 45–60 min is required for mobilization of the pancreas. During pancreas procurement, a Betadine/amphotericin B solution is administered through an NG tube into the stomach and duodenum. A total of about 300–500 mL of the Betadine solution is passed in two divided aliquots. After the heart, liver, pancreas, and kidneys have been mobilized, the supraceliac aorta just below the diaphragm is dissected for placement of the aortic cross-clamp. Immediately before cross-clamping, 30,000 U of heparin (300 U/kg) is given systemically, with the α -antagonist phentolamine in some cases. The supraceliac aorta is then clamped, and the organs are perfused with an organ preservation solution. At this point, the ventilator can be turned off, except in cases where the lungs are procured. The lungs must be inflated with 100% O₂ just before removal. The heart is the first organ to be removed, followed by the lungs.

If the lungs are procured, an extra 20–30 min of perfusion time is required. After removal of the heart and/or lungs, the liver can be removed, followed by the pancreas, small intestines, and kidneys. After the organs are removed, spleen and lymph nodes for tissue typing are obtained from the abdominal and thoracic cavities; and because of the possible need for vascular reconstruction in the recipients, bilateral iliac veins and arteries are removed. The total time for multiorgan procurement is ~4 h, although the anesthesia time typically ends with aortic cross-clamping.

Variant procedures

During the **rapid-flush technique** used by some procurement teams, the dissection of

individual abdominal organs is minimized, and an en bloc resection is done after clamping the aorta below the diaphragm and flushing preservation solution through the distal aorta. The en bloc organs are then dissected ex vivo, often at the transplant center, in preparation for transplantation. This technique is more rapid, requiring only about 1.5 h.

A second variant procedure, developed in response to the chronic shortage of transplantable organs, is the use of non-heart-beating donors. Patients, who are not brain dead, but who have no hope of recovery (due to irreversible brain injury or pulmonary or cardiac failure), may be suitable donors. In this case, the patient is taken to the OR and, under the supervision of a physician who is not part of the transplant team, the life-sustaining treatment (e.g., pressors, ventilator) is D/C'd. When it has been determined that the donor has suffered cardiac death, the body is rapidly cooled with ViaSpan through a cannula that is located in the aorta at the level of the renal arteries. Depending on the regional regulations, this cannula may be placed before cardiac death or immediately following the declaration of cardiac death. After the preservative flush is initiated, the abdomen is entered as rapidly as possible, the peritoneal cavity is packed with ice, and the organs are removed expeditiously. The anesthesiologist's role is ended when cardiac death has been declared. The disadvantage of this approach is that there can be significant warm ischemia from the time the life-sustaining treatment is stopped to the time that cardiac death is reached.

Usual preop diagnosis

Brain death

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Midline only, neck to pubis, ± bilateral transverse extensions
Special instrumentation	Chest and abdominal retractors
Unique considerations	Maintain oxygenation and BP as if a live patient. May require pressors and/or blood transfusion. Temporarily deflate lungs for sternal sawing.
Antibiotics	Ampicillin (1 g iv), ceftriaxone (1 g iv); Betadine/amphotericin B via NG tube for pancreas (with duodenal segment) procurement

Surgical time

4 h; rapid-flush technique: 1.5 h

EBL

200 mL

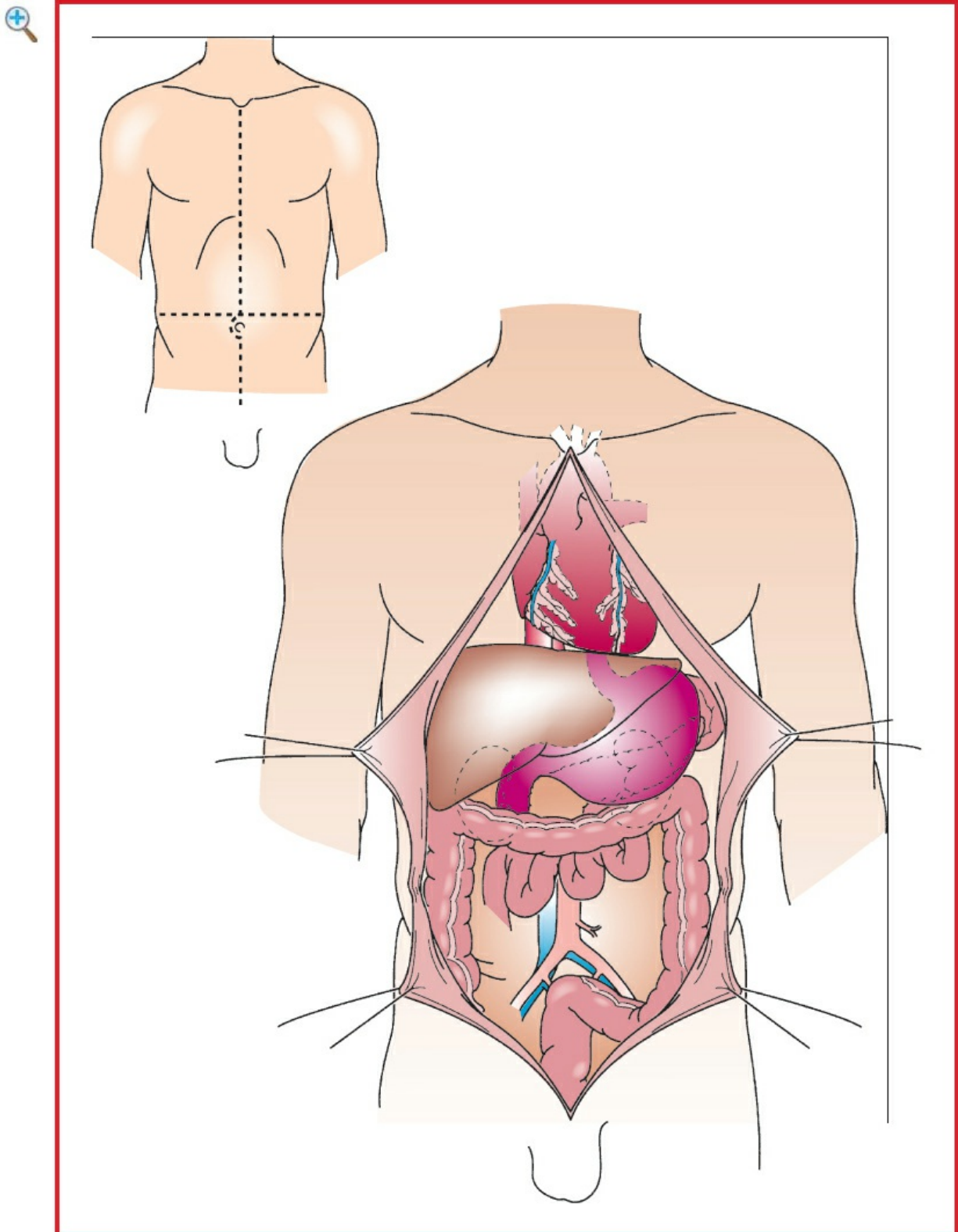


Figure 7.12-10 A complete midline incision, from the suprasternal notch to pubis, is made for multiple organ procurement, and the sternum is split. If necessary, cruciate abdominal incisions are added to

facilitate exposure of the intra-abdominal organs. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds: *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Neonate–70 yr
Male:Female	N/A
Incidence	~5000/yr in the United States
Etiology	Usually head trauma (e.g., motor vehicle accidents, gunshot wounds to the head) or intracranial bleeding
Associated conditions	Vasomotor instability; diabetes insipidus; intracranial HTN

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

In general, organ donors are previously healthy individuals who have suffered catastrophic, irreversible brain injury of known etiology, most commonly due to blunt head trauma, penetrating head injury, or intracranial hemorrhage. A declaration of brain death by physicians not participating in the organ procurement must be documented. This documentation, together with certification of death and familial consent, should be verified by the anesthesiologist before organ procurement. The United Network for Organ Sharing (UNOS) has produced the Critical Pathway for the Organ Donor to facilitate the administrative aspects of organ procurement (http://www.unos.org/docs/Critical_Pathway.pdf). There should be no evidence of disease or trauma involving the organs targeted for donation, and in general, the patient should be hemodynamically stable with minimal inotropic requirements. After brain death has been declared, it is important to shift the emphasis away from cerebral resuscitation efforts and to focus instead on the maintenance of adequate tissue perfusion and oxygenation. Brain death is frequently followed by a series of pathophysiologic events that may complicate the management of these patients.

Recently, Donation after Circulatory Death (DCD) has been instituted at many medical centers, where a non-brain-dead patient is brought to the operating room for withdrawal of life support and subsequent removal of organs for transplant. Although the ethics of such organ donation had been debated, the ASA and other organizations have developed

protocols to separate the teams caring for the patient during the withdrawal of life support and the operating room team responsible for organ retrieval. However, in most institutions, an anesthesiologist is present in the operating room as a member of the latter team. After withdrawal of life support has been initiated and the patient meets the criteria for cessation of cardiopulmonary function, organ removal is initiated as quickly as possible to limit warm ischemia time and possible damage to the organs to be removed. DCD patients often do not have as serious pathophysiologic alterations as patients who meet brain-death criteria.

Respiratory

Pulmonary dysfunction following brain death has many possible etiologies: aspiration, atelectasis, pneumonia, and pulmonary edema. In addition, trauma may cause pulmonary dysfunction related to contusion, pneumothorax, or hemothorax. Meticulous pulmonary toilet is essential to prevent atelectasis and pneumonia. Maintenance of adequate oxygenation is requisite to ensure preservation of other organs for transplantation. Use mechanical ventilation with TVs of 8–10 mL/kg and a minute ventilation that maintains PaCO₂ 30–35 mm Hg and pH 7.35–7.45. The FiO₂ should ensure a PaO₂ 75–150 mm Hg and arterial saturation >95%. PEEP usually is applied at 3–5 cm H₂O and should not exceed 7.5 cm H₂O because of the deleterious effects on CO and regional blood flow and possible barotrauma. The FiO₂ generally should be increased to 100% before transport to the OR. An important exception is in the case of heart-lung or lung retrieval, where it is important to maintain FiO₂ < 40% to minimize possible effects of O₂ toxicity. Ideally, PIP should be <30 cm H₂O to minimize possible barotrauma to the lungs. The use of lung protective ventilation (LPV = tidal volume of 6–8 mL/kg with PEEP 8–10 cm H₂O) strategies may be considered to improve donor lung function.

Tests: Frequent ABGs, including immediate preop period. Proper position of the ETT should be confirmed preop.

Cardiovascular

Hypotension should be anticipated in all organ donors. However, shortly after brain death, a prolonged period of $\uparrow$ sympathetic tone and cardiac output ("catecholamine storm") may lead to hypertension. This results most commonly from neurogenic shock (derangement of descending vasomotor control $\rightarrow$ progressive $\downarrow$ SVR and venous pooling) and hypovolemia. Hypovolemia is usually the result of dehydration therapy for cerebral edema, hemorrhage, DI, or osmotic diuresis 2° hyperglycemia. Hypothermia, LV dysfunction, and endocrine abnormalities also can contribute to $\downarrow$ BP. Fluid resuscitation with crystalloid, colloid, and PRBCs to maintain Hct $> 30\%$ should be initiated preop. Hemodynamic goals are (a) CVP 4–8 cm H₂O; (b) MAP > 70 mm Hg; (c) SBP > 100 mm Hg; (d) HR 60–120 bpm; (e) PCWP ≤ 12 mm Hg; (f) SVR 800–1200 dynes/sec $\times$ cm⁻⁵ (or 800–1200 woods units); and (g) UO > 1 mL/kg/h. Donors are often placed on inotropic therapy to maintain these parameters; however, following adequate volume resuscitation, preop inotropic therapy often may be gradually decreased or D/C'd. If inotropic therapy remains necessary, typically it would consist of dopamine (2–10 mcg/kg/min), followed by dobutamine (3–15 mcg/kg/min) or epinephrine (0.1–1.0 mcg/kg/min), then norepinephrine. The latter three agents may be combined with dopamine (2–3 mcg/kg/min) in an attempt to augment or preserve renal, mesenteric, and coronary arterial blood flow. It should be noted that brain death may be accompanied initially by a transient hypertensive crisis that may require short-term treatment with SNP and/or esmolol. ECG abnormalities are common in patients with intracranial injury and are of no pathologic consequence. Atrial and ventricular dysrhythmias and various degrees of conduction block occur frequently in organ donors; the etiology may be electrolyte

imbalance, ABG disturbance, $\uparrow$ ICP, loss of the vagal motor nucleus, inotropic therapy, hypothermia, or myocardial contusions or ischemia. Antidysrhythmic therapy should follow the usual guidelines except for $\downarrow$ HR, which is resistant to atropine in this setting. Bradycardia, if accompanied by $\downarrow$ BP, should be treated with isoproterenol, dopamine, epinephrine, or temporary cardiac pacing.

Tests: ECG; ECHO (to assess wall motion abnormalities) and possibly coronary angiography (if CAD is suspected)

Neurologic

Diabetes insipidus (DI) frequently occurs in brain-dead donors; it is likely the result of destruction of the hypothalamic-pituitary axis. Untreated, it may cause marked hypovolemia and electrolyte disturbances ($\uparrow$ Na^+ , $\uparrow$ Mg^{++} , $\downarrow$ K^+ , $\downarrow$ PO_4^- , $\downarrow$ Ca^{++}). Therapy with iv vasopressin (titrated from 2 mcg/kg/min) or DDAVP (titrated from 0.3 mcg/kg/min) often is initiated to maintain $\text{UO} < 1.5\text{--}3$ mL/kg/h. Many believe that the benefits of minimizing electrolyte imbalance, fluid shifts, and reduction of core T outweigh the risks of vasopressin or desmopressin therapy, including coronary and renal vasoconstriction and possible organ ischemia or uneven distribution of the preservation solutions during flushing. It may be prudent, however, to D/C vasopressin or DDAVP infusions for at least 1 h prior to aortic cross-clamping and infusion of preservation solutions. Thermoregulation is abnormal in brain-dead donors due to hypothalamic dysfunction, and core T should be monitored (bladder or esophageal). Aggressive warming techniques may have to be employed early to maintain a core T $> 34\text{--}35^\circ\text{C}$, as there are numerous undesirable consequences of significant hypothermia ($< 32^\circ\text{C}$) in the organ donor (e.g., cardiac dysrhythmia, cardiac instability, $\downarrow$ GFR and cold diuresis, a left shift in the oxyhemoglobin dissociation curve, and pancreatitis). Although other

	endocrine or metabolic disturbances may exist as a result of destruction of the hypothalamic-pituitary axis, currently there is no consistent recommendation for any other hormonal replacement therapy. Tests: Serum electrolytes and osmolality q4h
Hematologic	Donors may be anemic from hemodilution and/or hemorrhage. To ensure adequate tissue O₂ delivery, PRBCs are transfused to maintain Hgb > 7–8 g/dL. Some donors may exhibit a coagulopathy; clinically significant bleeding should be treated with clotting factors and Plts. Persistent or severe primary fibrinolysis or DIC may require rapid transfer of the donor to the OR for organ retrieval. Administration of epsilon-aminocaproic acid to treat fibrinolysis is avoided for fear of microvascular thrombosis in the donor organs. Tests: Hgb/Hct; PT; PTT; Plt count; DIC screen as clinically indicated
Others	The role of oxygen-free radicals, with regard to reperfusion injury, has prompted the suggested use of mannitol and steroids (and other compounds) as scavengers.

INTRAOPERATIVE

Anesthetic technique

Although anesthesia is unnecessary in brain-dead organ donors, both visceral and somatic reflexes can lead to physiologic responses during the procedure. The goals of intraoperative management with regard to respiratory, cardiovascular, hematologic, and neurologic status are identical to those discussed under preop considerations earlier.

Induction	Settings for mechanical ventilation parallel those of the ICU, although it may be advisable to begin with an FiO₂ of 100% until the first ABG result is obtained. The exception is when procurement of the lungs or heart-lungs is anticipated; then FiO₂ should not
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	exceed 40%. To eliminate reflex neuromuscular activity and to facilitate surgical retraction, a longer-acting neuromuscular blocking agent should be given at the beginning of the procedure and supplemented as necessary.	
Maintenance	Reflex hypertensive responses to surgical stimulation occur frequently and may → excessive intraop blood loss and damage to donor kidneys; management should include the weaning of vasopressors and the initiation of vasodilator therapy with isoflurane, SNP, or NTG. Anesthetic care continues until the proximal aortic cross-clamp is applied. D/C all monitoring and supportive therapy at this point. The notable exception is the case of heart-lung or lung procurement; in this situation, all monitoring except FiO₂ should cease with proximal aortic cross-clamping. All supportive care is terminated, with the exception of mechanical ventilation of the lungs at 4 breaths/min or as directed by the transplant team and suctioning of the ETT after cessation of mechanical ventilation just prior to removal of the tube. Extubation marks the termination of anesthetic care of the heart-lung or lung donor.	
Blood and fluid requirements	IV: 14–16 ga × 1–2 BSS @ 2–4 mL/kg/h	Significant third-space losses may require large volumes of crystalloid, colloid (up to 1 L), and PRBCs (not uncommon to transfuse 2 or more U to maintain Hct > 30). Central venous access is necessary for monitoring and for vasoactive drug delivery. Volume repletion can be guided by arterial waveform analysis (pulse pressure or systolic pressure variation).
Monitoring	Standard monitors Art line	If a PA catheter is in place, it may be used or removed based on concerns of catheter-related, right-side

	CVP line UO	endocardial lesions. Rarely is insertion of a PA catheter warranted in these operations. ABG, Hgb/Hct, serum electrolytes, and glucose should be monitored hourly; for operations involving procurement of the lungs or heart-lungs, ABGs should be obtained at least every 30–60 min.
Complications	Hypotension	Most commonly 2° hypovolemia and neurogenic shock (loss of descending vasomotor control). Ensure adequate volume repletion as described previously, then institute or increase inotropic/vasopressor therapy as previously outlined.
	Dysrhythmias	Multiple possible etiologies as described previously. Standard treatment and diagnosis should be employed, with the exception of bradycardia, which is atropine-resistant and should be treated with isoproterenol, dopamine, epinephrine, or transvenous pacing.
	Cardiac arrest	CPR should be instituted in an effort to maintain the viability of the liver, kidneys, and other abdominal viscera intended for transplantation. Procurement of the liver and kidneys should proceed rapidly to aortic cross-clamping at the diaphragm and administration of cold preservation fluid into the aorta and portal vein. This series of events will undoubtedly preclude use of the heart and lungs for transplantation.
	Oliguria	Ensure adequate volume replacement and BP as outlined, then

add dopamine (2–3 mcg/kg/min), if not previously instituted to promote renal vasodilation and to increase renal blood flow, glomerular filtration rate, and UO. If these measures are ineffective at restoring adequate UO (>1 mL/kg/h), then furosemide or mannitol may be used, after consultation with the transplant team.

Diabetes insipidus (DI)

Fluid and electrolyte therapy as determined by filling pressures and hourly serum electrolyte values. Adjustment of vasopressin or DDAVP infusion to maintain UO < 1.5 – 3.0 mL/kg/h; initiation of this infusion should be done in consultation with the transplant team. As previously discussed, it may be advisable to D/C vasopressin or DDAVP at least 1 h before aortic cross-clamping.

Coagulopathy

Transfuse Plt, FFP, and cryoprecipitate as necessary for clinical bleeding in the setting of abnormal coagulation studies. Avoid EACA due to risk of microvascular thrombosis in the donor organs.

Hyperglycemia

Avoid dextrose-containing solutions, which may aggravate existing hyperglycemia and contribute to osmotic diuresis and electrolyte abnormalities.

Hypothermia

Early aggressive attempts to minimize intraop heat loss are essential and include warming the OR, use of a forced-air warming blanket, insulating exposed areas,

		warming all fluids, and using heated, humidified inspired gases.
Special considerations	Heart-lung procurement	Division of the mediastinal pleura and tracheal dissection with manipulation of each lung outside the mediastinum may result in ↓↓ BP and may cause problems with oxygenation and ventilation. Adequate intravascular volume is essential, and inotropic therapy may be required during this period. Problems with ventilation and oxygenation must be communicated immediately to the transplant team. Following aortic cross-clamping and infusion of cardioplegia solution, the lung preservation fluid will be infused via the right and left PAs. During this period, the lungs should be ventilated manually with 4 bpm or as otherwise directed by the transplant team. It is prudent early on in the procurement procedure to verify the position of the ETT with the transplant surgeon to ensure that the tube does not contribute to mucosal injury at the site of the anticipated suture line.
	Organ preservation	Therapy aimed at improving organ preservation may require several pharmacologic manipulations, as directed by the transplant team. Agents commonly used during organ procurement include dopamine (2–3 mcg/kg/min), furosemide, mannitol, allopurinol (free-radical scavenger), chlorpromazine and phentolamine (vasodilators), heparin (prevents microvascular thrombosis and

promotes reperfusion), and PGE₁ (vasodilator, membrane stabilizer, antiplatelet effect). Systemic infusion of PGE₁ prior to aortic cross-clamping (commonly used in heart-lung or lung procurement) will lead to predictable and profound ↓ BP; efforts at volume resuscitation toward optimal CVP should continue until the aortic cross-clamp is applied. If heparin is to be administered iv, a catheter should be used after verifying the ability to freely aspirate blood.

Methylprednisolone (30 mg/kg) is commonly administered at least 2 h before organ retrieval in an effort to protect the heart and kidneys from ischemic injury. Recent trials suggest that administration of thyroid hormone, cortisol, and insulin may improve cardiovascular status and improved donor viability.

Suggested Readings

1. Anderson TA, Bekker P, Vagefi PA: Anesthetic considerations in organ procurement surgery: a narrative review. *Can J Anesth* 2015; 62:529–39.
2. Van Norman GA: Another matter of life and death: what every anesthesiologist should know about the ethical, legal and policy implications of the non-heart-beating cadaver organ donor. *Anesthesiology* 2003; 98:763–73.
3. Xia VW, Braunfeld M: Anesthesia management of organ donors. *Anesthesiol Clin* 2017; 35:395–406.

CHAPTER 7.13

Trauma Surgery

SURGEONS

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INITIAL ASSESSMENT AND AIRWAY MANAGEMENT FOR TRAUMA SURGERY

INITIAL ASSESSMENT

The Advanced Trauma Life-Support (ATLS) Program, developed by the American College of Surgeons Committee on Trauma, provides a safe and reliable method for the immediate treatment of injured patients. The crucial components of ATLS are the rapid and accurate assessment of a patient's condition, the resuscitation, and stabilization of the patient according to a standard sequence of priorities, determining the need for and arranging interhospital transfer, and ensuring optimal care throughout the entire process. The initial assessment and management consist of the primary survey and resuscitation.

During the **primary survey**, life-threatening conditions are identified and treated following the ABCDE sequence: **a**irway, **b**reathing, **c**irculation, **d**isability, and **e**xposure.

In critically injured patients, cervical spine clearance is performed after securing airway. Thus, **airway** patency should be assessed while maintaining strict cervical spine protection. In situations where a patient's airway integrity is unclear or at risk, a

definitive airway should be established. High-flow supplemental oxygen should be delivered and, oxygenation should be monitored with pulse oximetry. **Breathing** and ventilation are assessed, paying attention to any chest injuries that can impair adequate gas exchange. Conditions such as tension pneumothorax, massive hemothorax, flail chest, and open pneumothorax should be identified and treated. **Circulation** is assessed and vascular access established. Shock should be recognized early if present. In trauma, the presence of shock is usually due to hemorrhage, and definitive control of bleeding and replacement of intravascular volume are crucial. **Disability** from neurologic injury should be assessed. Patients with severe neurologic injury may require definitive airway management or urgent neurosurgical evaluation. **Exposure** involves undressing the patient to identify any other life-threatening injuries while keeping the patient warm.

Extended Focused Assessment with Sonography for Trauma (eFAST)

In addition to AP chest x-ray and pelvic x-ray, eFAST (see [Fig. 7.13-1](#)) is performed as an adjunct to the primary survey. eFAST is an ultrasound screening examination of the trauma patient. eFAST allows for the rapid detection of pericardial fluid (pericardial tamponade), pneumothorax, hemothorax, and peritoneal free fluid and can speed triage and treatment of trauma patients. It is important to note that many other potentially life-threatening injuries cannot be detected by ultrasound (e.g., retroperitoneal bleeding, bowel perforations, torn diaphragms). eFAST examinations have a low but significant false-negative rate. Ultrasound is not a substitute for more definitive diagnostic procedures.

AIRWAY MANAGEMENT

In trauma, indications for definitive airway can be divided into conditions that require airway protection and conditions that require control of ventilation or oxygenation. Airway patency can be compromised by obtundation, severe facial injuries, bleeding or vomiting, or obstruction from neck or airway injuries. The need for ventilation or oxygenation is indicated by apnea, respiratory distress, severe closed head injury, or hemodynamic instability.

Although airway management in injured patients does not differ fundamentally from airway management in other situations, special attention must be paid to cervical spine protection, high risk of vomiting and aspiration, and recognition of maxillofacial, neck, laryngeal, or head injuries that can cause airway compromise.

Airway maneuvers such as the jaw-thrust maneuver are useful techniques to improve airway patency in unconscious or obtunded patients, although they must be performed without extending the neck and potentially exacerbating a cervical spine injury. Other

maneuvers such as chin lift should ideally be avoided in trauma with suspected cervical spine injuries.

Airway adjuncts such as oropharyngeal airway, nasopharyngeal airway, or laryngeal mask airway are useful for providing temporary ventilation and oxygenation until a definitive airway (oral or nasal endotracheal tube [ETT] or surgical airway) can be established.

Orotracheal intubation, using rapid-sequence intubation and cricoid pressure, is preferred. The approach is rapid, but at least three people are required to perform it safely in the patient with suspected C-spine injury.

In-line stabilization of the neck is performed to minimize neck and spine movements. Because a failed intubation may force operative airway intubation, equipment for cricothyrotomy should be immediately accessible. It is often beneficial to have several types of intubation devices available in addition to traditional laryngoscopes, such as video laryngoscopes, in the case of traumatic airways. Fiberoptic assistance may be used in the stable patient with a difficult airway. Patients in respiratory distress with severe facial or neck trauma or unstable cervical spine injury require a surgical airway. An airway placed in transport should be immediately assessed for position and changed to a definitive airway when appropriate.

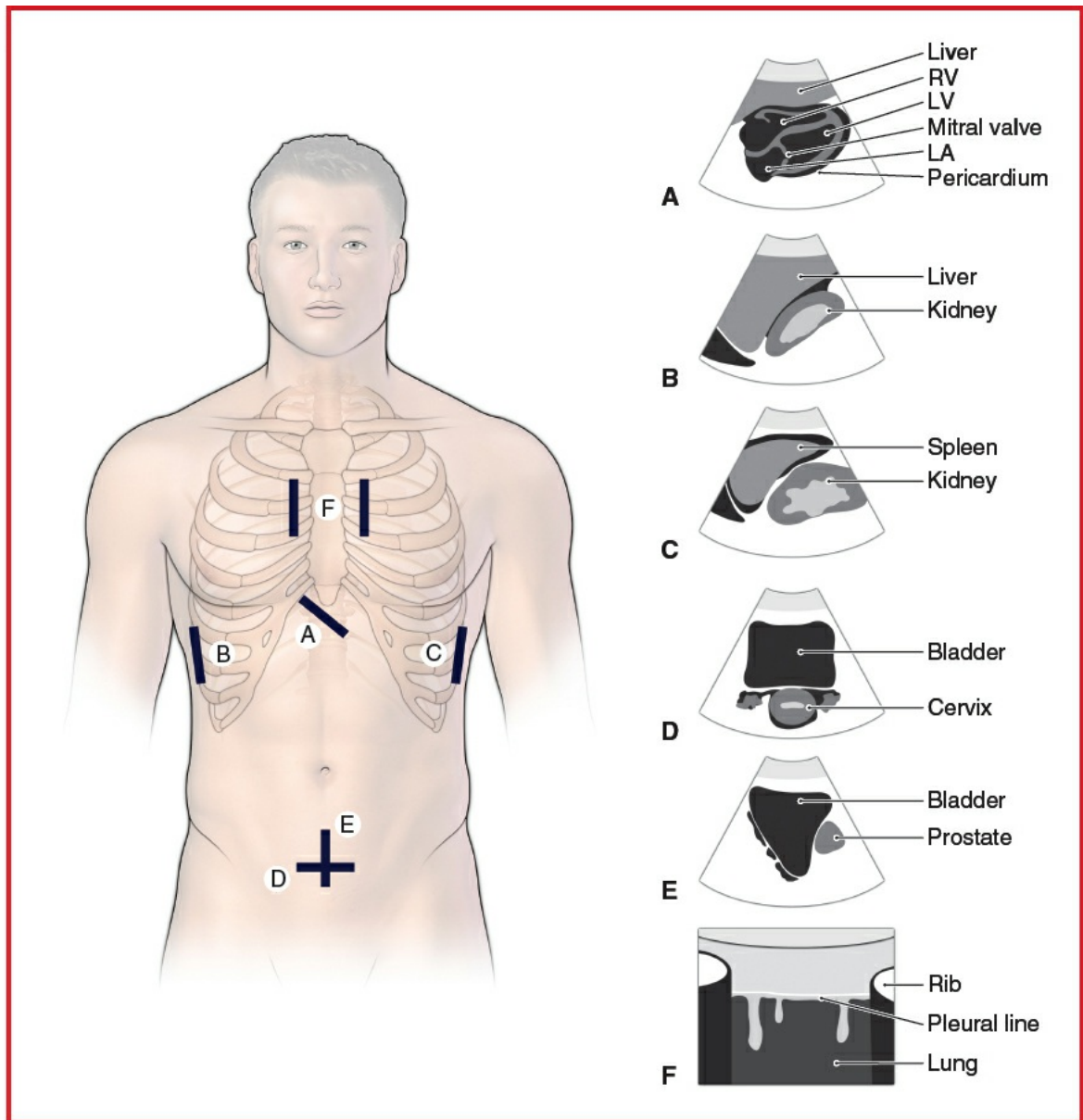


Figure 7.13-1 eFAST transducer placements. A: subxiphoid; B: perihepatic; C: perisplenic; D and E suprapubic; F: thoracic.

Nasotracheal intubation, used only in spontaneously breathing trauma patients, can be performed without the use of pharmacologic agents or special equipment. It is, however, associated with higher incidence of vomiting and aspiration. In the intoxicated patient with a depressed level of consciousness, the success rate may be as low as 65%. Blind nasal intubation is contraindicated in patients with unstable midface fractures, penetrating neck trauma, or significant neck hematomas.

Surgical airway, cricothyrotomy (Fig.7.13-2), is the preferred method in adults who require a surgical airway. The important anatomic landmarks of the superior and inferior borders of the thyroid and cricoid cartilages are palpated. With the nondominant hand, the thyroid cartilage is stabilized, using the index finger to palpate

the cricothyroid membrane. A vertical skin incision is made over that area. With a hemostat or a blade, a transverse incision is made through the cricothyroid membrane, and a size 5–6 ETT or tracheostomy tube is rapidly advanced through the passage. The cricothyroid membrane lies very superficially, covered only by the skin and platysma muscle. In emergency situations, a standard small-caliber ETT is generally easier to insert than a tracheostomy tube (Fig. 7.13-2). Cricothyroidotomies should be converted to tracheotomies within 72 h after the initial injury, provided the patient's condition permits.

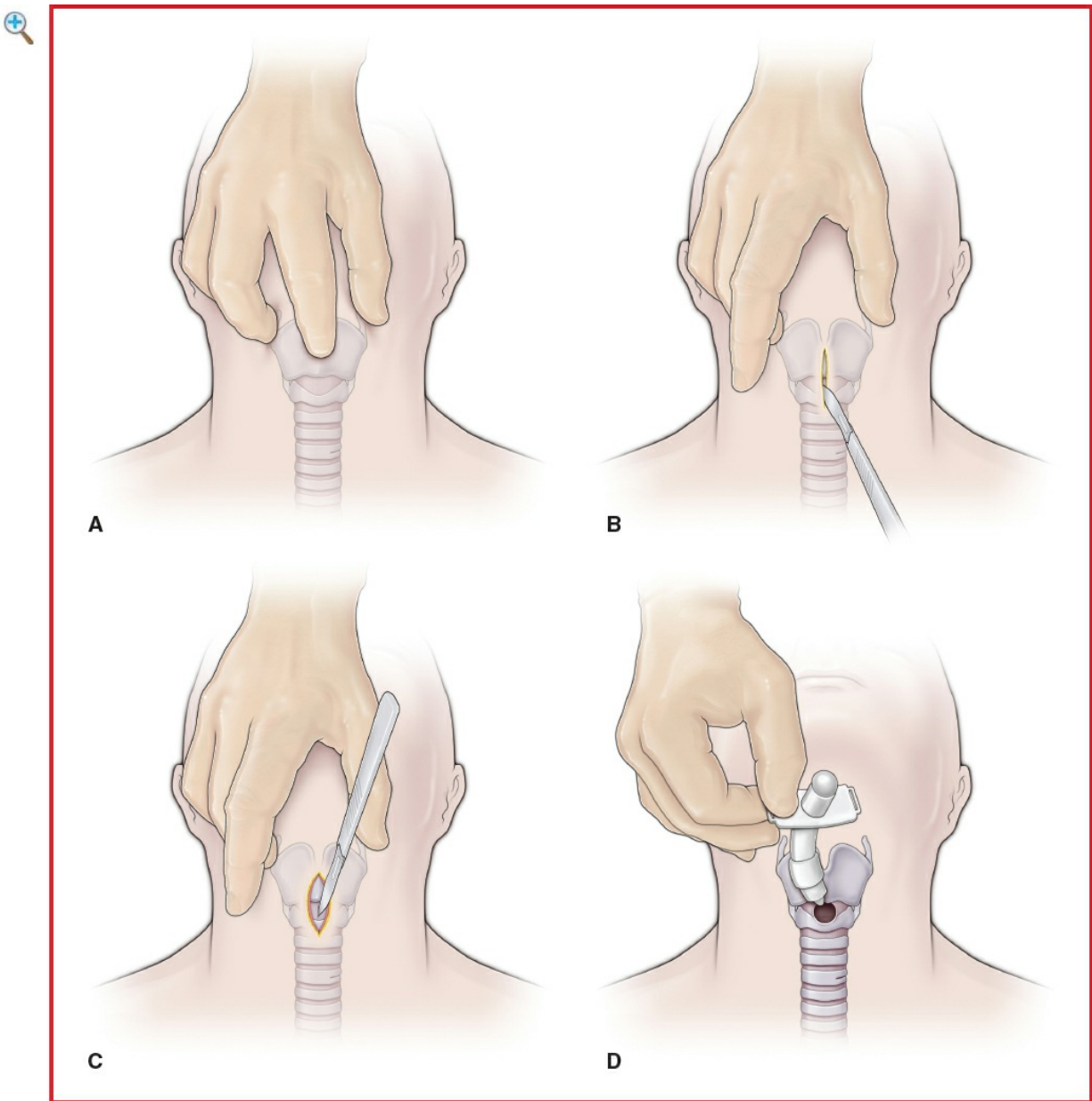


Figure 7.13-2 **A:** Immobilize the larynx and palpate the cricothyroid membrane with the index finger of the nondominant hand. **B:** Make a midline vertical skin incision, 3–5 cm in length. **C:** Incise the cricothyroid membrane transversely. **D:** Insert the tracheostomy tube into the trachea.

Percutaneous cricothyroidotomy is also another alternative to the above technique. There are commercially available kits for that use, with both cuffed and uncuffed cricothyroidotomy tubes. A percutaneous Seldinger technique is employed. If all fails, jet ventilation may be performed through the cricothyroid membrane by using a wide-bore 14- to 16-ga Angiocath and attaching it to high-flow oxygen tubing. A key principle is to incise the side of the oxygen supply tube and periodically cover the slit with a finger using a 1:4 second ratio to help with CO₂ elimination until additional help is available.

A **tracheostomy** is indicated for patients requiring surgical airway in less emergent situations or if a cricothyrotomy cannot be performed due to direct laryngeal injury. A tracheostomy can be accomplished through the same incision, extended caudally if laryngeal injury is found. Rarely, the injury is in the distal cervical or proximal intrathoracic trachea. In such cases, it may be necessary to intubate the distal end of the airway through the wound. A subsequent median sternotomy may be required to expose the injury. Right thoracotomy provides access to the distal intrathoracic trachea (see Chest Trauma).

Usual preop diagnosis

Airway compromise

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Midline 3- to 4-cm longitudinal incision in the neck at the area of the cricothyroid membrane
Unique considerations	The large number of legal claims involving failed intubation suggests that surgical cricothyrotomy remains an underutilized technique.
Antibiotics	Usually not given until clear indications related to the primary injury are apparent.
Surgical time	2–5 min
EBL	Minimal
Postop care	Mechanical ventilation
Mortality	Related to the primary injury

Morbidity	Cricothyrotomy is more likely to result in airway stricture or damage to more proximal structures in the larynx. On this basis, cricothyrotomy is converted to tracheostomy within 48–72 h of admission if patient's general condition permits.
Pain score	3

EMERGENCY TUBE THORACOSTOMY

SURGICAL CONSIDERATIONS

Description

In the United States, trauma is the most common cause of death in young people; 25% of these deaths (~16,000 per yr) are the result of thoracic trauma. Most of these are due to lethal injuries at the scene (e.g., cardiac rupture, free aortic transection). For patients who reach the hospital, proper management is crucial because many deaths can be prevented. Early deaths are due to airway obstruction, tension pneumothorax, massive hemothorax, flail chest, cardiac tamponade, and open pneumothorax. Later deaths are due to respiratory failure, sepsis, and unrecognized injuries.

Eighty percent of blunt thoracic injuries are caused by motor vehicle collisions (MVCs). Penetrating injuries to the chest are almost as common as blunt trauma. The death rate in hospitalized patients with isolated chest injury is 4–8%; this increases to 10–15% when one other organ system is involved and to 35% if multiple additional organs are injured. Eighty-five percent of chest injuries do not require thoracotomy, and the patient can be managed with relatively simple measures, such as airway control, tube thoracostomy, and pain management. Blunt trauma can induce injury by three mechanisms: direct blow, deceleration injury, and compression injury. Rib fracture is the most common sign of blunt thoracic trauma. Fracture of the upper ribs (1st–3rd), clavicle, or scapula implies high-energy impact and is associated with a higher likelihood of major vascular injury.

Life-threatening injuries caused by penetrating trauma are distinctly different from those caused by blunt trauma. In penetrating chest injuries, pneumothorax is almost always present, and hemothorax is present in 80% of cases. Hemodynamic instability in penetrating chest injury generally heralds a major intrathoracic vascular injury.

Tension pneumothorax may be caused by blunt or penetrating trauma. Venous return to

the heart is impaired by the increased intrathoracic pressure and compression of the vena cava → ↓ BP and distended neck veins. Loss of lung volume on the ipsilateral side and subsequent compression of the contralateral side lead to impaired ventilation and hypoxia. The diagnosis of tension pneumothorax is made clinically. The presence of respiratory distress and absent or diminished breath sounds warrant immediate needle decompression (14- to 16-ga catheter through the 2nd intercostal space, midclavicular line), followed by subsequent tube thoracostomy. Treatment should not be delayed for radiographic confirmation. In the hemodynamically stable patient, however, suspicion of a pneumothorax should be confirmed by x-ray. On the CXR, a 20% loss of lung dimension corresponds to ~50% loss of lung volume. A small, simple pneumothorax (<10%) with no respiratory compromise may be observed. These patients require close observation in the hospital and a repeat CXR in 4–6 h. Tube thoracostomy should be performed for a large pneumothorax (>10%), for patients with respiratory compromise or multiple injuries, or when it is not possible to adequately monitor the patient (e.g., during extended transport). Recent developments in emergency ultrasonography have led to the adoption of the “eFAST” examination, which includes ultrasound examination of the anterior chest to detect pneumothoraces. An abnormal eFAST exam may alert the team to the presence of a significant pneumothorax even before a CXR can be obtained. In the presence of an anterior pneumothorax, the eFAST may even be superior in sensitivity to a supine CXR.

As much as 40% of the circulating blood volume can accumulate in a **hemothorax**. The most frequent sources of bleeding are the intercostal and internal mammary vessels. Some degree of hemothorax is present in almost every patient with chest injury. A supine CXR may miss up to 1 L of blood. Although an upright CXR is more sensitive this is generally impractical in a multiple-trauma patient (whose spine often has not been fully evaluated). Following chest tube placement, blood loss >1200–1500 mL or an ongoing loss of 250 mL/h for 4 h suggests the need for surgical intervention.

Simple pneumothorax without associated hemothorax can be treated with a 20- to 22-Fr chest tube placed in the 4th intercostal space in the midaxillary line or in the 5th intercostal space in the anterior axillary line. Hemothorax and tension pneumothorax require a large-bore, 28- to 36-Fr chest tube placed in the midaxillary line through the 5th intercostal space (Fig. 7.13-3). A 20-mL syringe with 1% lidocaine can be used not only to provide local anesthesia but also to locate the upper edge of the rib in the obese patient. A generous 3-cm incision should be made one interspace below the targeted level. The subcutaneous tissues are dissected bluntly, creating a tunnel that is directed upward. The pleural space should be entered just above the upper edge of the rib to avoid injury to the intercostal neurovascular bundle, located just below the lower edge

of the rib. After the pleural space has been entered bluntly, it should be explored with the operator's finger swept around to ensure proper location and to free potential adhesions. The chest tube should be inserted and advanced in the posterior and superior direction. The tube then should be connected to a suction/collection system under 20 cm of negative pressure water, preferably through an autotransfusion device. Repeat chest exam and CXR should immediately follow chest tube placement to evaluate decompression and assess for other injuries. If the pleural space still contains blood, another chest tube could be inserted. Hemothoraces require drainage within the first 24–48 h, to minimize the risk of retained hematoma leading to clotted hemothorax and subsequent empyema.

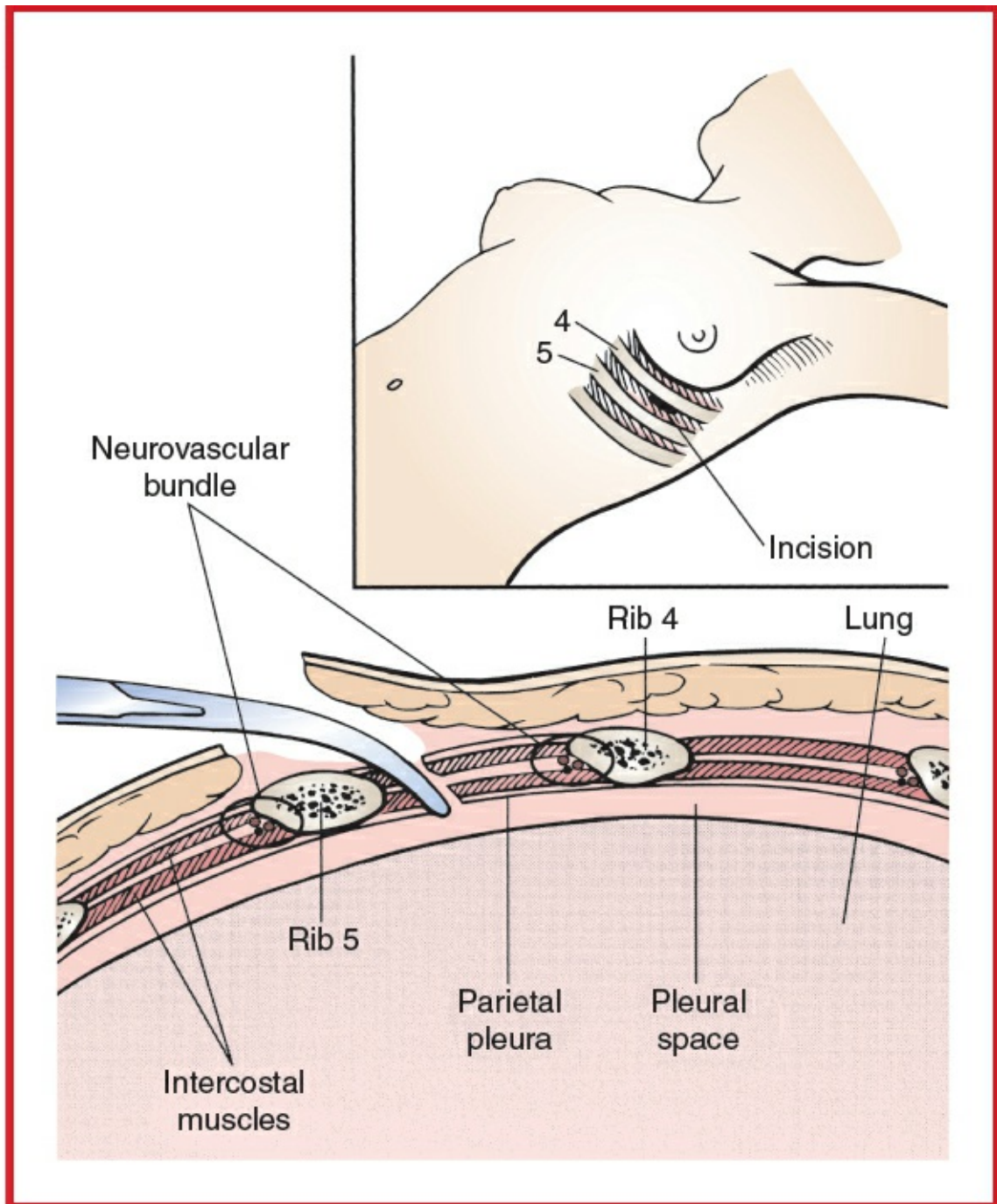


Figure 7.13-3 Tube thoracostomy. Incision through the 4th or 5th interspace at anterior axillary line. Forceps are used to tunnel over the superior edge of the rib and to enter the pleural space bluntly.

Usual preop diagnosis

Tension pneumothorax; pneumothorax; hemothorax

■ SUMMARY OF PROCEDURE

Position Supine with arm abducted 90°

Incision	3 cm in interspace below the 5th intercostal space in midaxillary line
Special instrumentation	28- to 36-Fr chest tube should be used in most patients. Small, 20- to 22-Fr tube reserved for simple pneumothorax in stable patients. Autotransfusion device should be available in ED.
Unique considerations	In tension pneumothorax, a large-bore needle inserted in the 2nd intercostal space of the midclavicular line can relieve tension and save a patient's life.
Antibiotics	Cefazolin 2 g iv
Surgical time	5 min
EBL	Minimal from tube placement. Variable amounts drained out, depending on the extent of hemothorax and associated injuries
Postop care	Chest tube may be removed after at least 48 h when there is no air leak from the lung and <100–150 mL of fluid drainage per 24 h.
Mortality	Isolated chest injury: 4–8% When one other organ system is involved: 10–15% When two or more organ systems involved: >35%
Morbidity	Clotted hemothorax Empyema Lung abscess Lung contusion
Pain score	5

EMERGENCY DEPARTMENT THORACOTOMY

SURGICAL CONSIDERATIONS

Description

Rarely, emergency department **thoracotomy** may offer the only chance of survival in some trauma patients. Its use should be restricted largely to patients showing signs of

life in the emergency department (ED) but lose such signs shortly after that. The usual indications are (a) massive exsanguination in the left chest, usually due to cardiac, vascular, or pulmonary injuries, and (b) pericardial tamponade. For patients with penetrating cardiac injuries who show signs of life in the ED, survival may be as high as 15%. For blunt trauma in this setting, survival is <1%, regardless of presentation. With either mechanism, functional survival is almost unprecedented if the patient arrives without vital signs and reactive pupils.

A **left anterolateral thoracotomy** is the preferred approach because pericardiotomy, open cardiac massage, and aortic occlusion are best achieved by this means. This incision can be extended easily across the sternum and into the right chest to improve exposure and to control massive blood loss and from the right lung. The entire chest is prepped liberally, and left anterolateral thoracotomy is performed rapidly in the 5th intercostal space using a large-blade scalpel. Heavy scissors can be used to divide the intercostal muscles quickly and to cut across the sternum. If pericardial tamponade is encountered, the pericardium is opened longitudinally, anterior to the phrenic nerve. Blood and clot are evacuated and bleeding sites controlled with gentle digital pressure. Large, full-thickness lacerations that extend into the chambers may be controlled by inserting a Foley catheter, inflating the balloon, and pulling it snug against the myocardium. The open end of the Foley can be clamped or used as an infusion line for resuscitation. During ED thoracotomy, placement of clamps on the atria or ventricles should be avoided as they may extend the laceration. Attempts to repair cardiac lacerations should be delayed until resuscitative measures have been completed. In the nonbeating heart, suturing is performed before defibrillation. If coronary or systemic air embolism is present, the appropriate hilum is cross-clamped, and air is aspirated from the left ventricle through the elevated apex. Cardiac arrest is an indication for immediate internal massage. The two-hand method is preferred, and internal defibrillation should be instituted. If internal defibrillation does not restore proper cardiac activity, cross-clamping of the aorta will improve coronary perfusion. To cross-clamp the aorta, the left lung is retracted anteriorly and superiorly, and the posterior pleura is dissected under direct vision. An orogastric or nasogastric tube placed in the esophagus aids in the identification of the esophagus by feel, allowing for cross-clamping of the aorta without risking injury to the adjacent esophagus.

Usual preop diagnosis

Penetrating chest trauma with cardiac arrest and recently recorded signs of life

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Left anterolateral thoracotomy. Extension of incision across the sternum could be considered for improved exposure.
Special instrumentation	ED thoracotomy tray; internal defibrillator paddles; suction device; Foley catheter; rapid-infusion device; O(-) blood
Unique considerations	Airway should be secured first. OG/NGT tube should be placed if possible. Typically, patients are not anesthetized for this procedure.
Antibiotics	Cefazolin 2 g iv
Surgical time	10 min
EBL	1–2 L
Postop care	ICU
Mortality	Blunt trauma: 98% All penetrating chest injuries requiring ED thoracotomy: 85% Penetrating cardiac injury: 50%
Morbidity	Arrhythmias Acute MI
Pain score	10

EXPLORATORY SURGERY FOR NECK TRAUMA

SURGICAL CONSIDERATIONS

Description

The neck contains vital structures in a relatively unprotected region of the body (Fig. 7.13-4). The cardiovascular, respiratory, digestive, endocrine, and CNS systems are all represented in the neck; injury to any of these can be fatal.

Injuries to the neck may result from blunt or penetrating trauma. In blunt trauma, <5% of cases will have C-spine injuries. Because the consequences are so grave, however, all

patients should be evaluated for C-spine injuries (H&P, x-rays, CT), and full spinal precautions should be maintained until the C-spine is cleared. Blunt airway injuries can be devastating and present significant management difficulties; however, the majority of blunt neck trauma consists of minor soft tissue injuries that can be managed nonoperatively. Blunt cerebrovascular injuries can occur from a direct blow to the carotid or vertebral arteries or from a severe hyperextension or rotation mechanism.

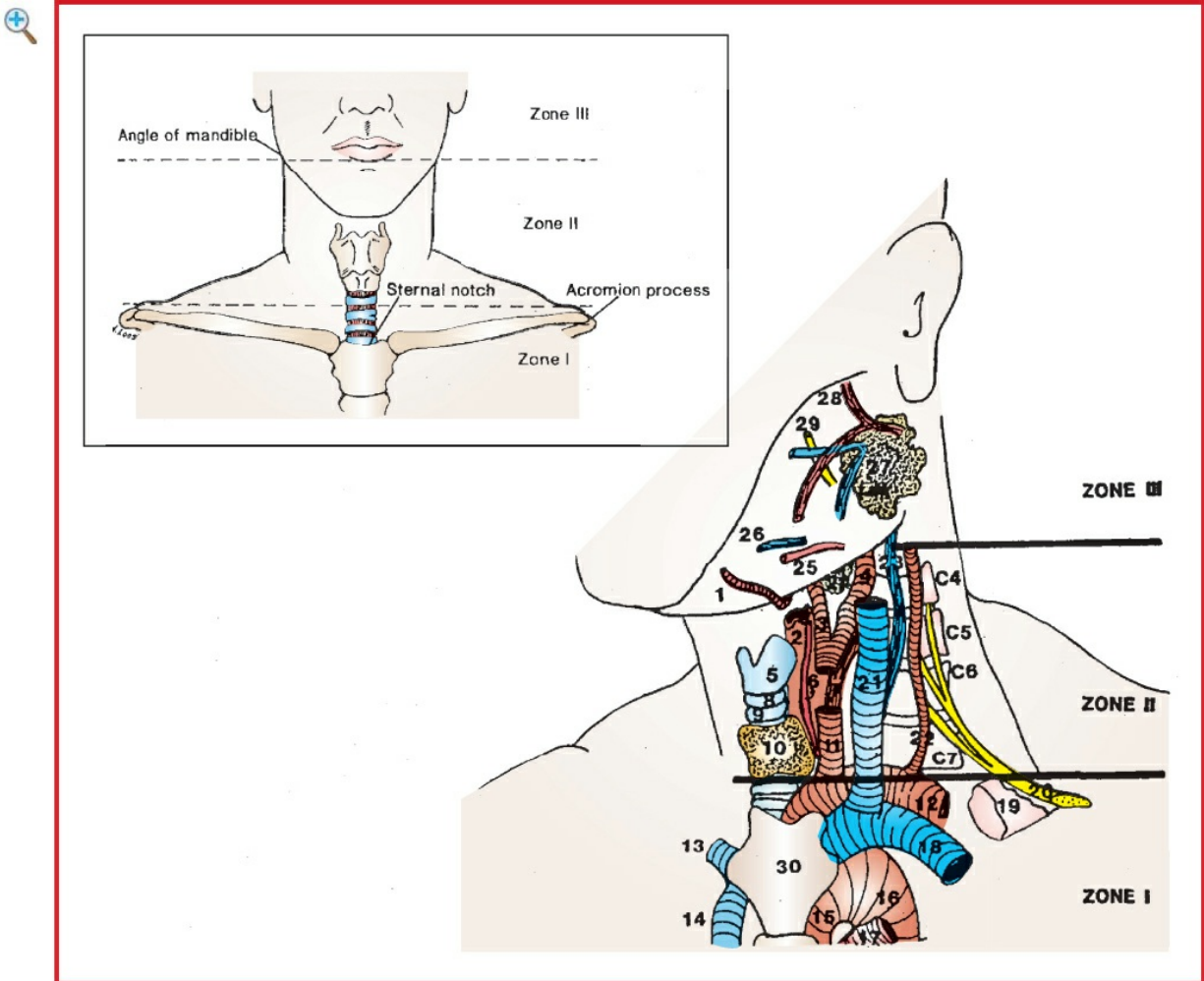


Figure 7.13-4 Cervical structures contained in Zones I, II, and III: (1) facial artery; (2) esophagus; (3) internal carotid artery; (4) external carotid artery; (5) thyroid cartilage; (6) sympathetic trunk; (7) vagus nerve; (8) cricothyroid membrane; (9) cricoid cartilage; (10) thyroid cartilage; (11) common carotid artery; (12) subclavian artery; (13) right innominate vein; (14) SVC; (15) ascending aorta; (16) descending aorta; (17) PA; (18) subclavian vein; (19) clavicle; (20) brachial plexus; (21) IJ vein; (22) vertebral artery; (23) phrenic nerve; (24) submandibular gland; (25) lingual artery; (26) hypoglossal nerve; (27) parotid gland and duct; (28) facial nerve and its branches; (29) maxillary artery; (30) sternal manubrium. The thoracic duct is not shown in this figure. (Adapted with permission from Ordog GJ, Albin D, Wasserberger J, et al: 110 bullet wounds to the neck. *J Trauma* 1985; 25:238–46). (Inset reproduced with permission from Baker RJ, Fischer JEMastery of Surgery, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Penetrating neck trauma is defined as penetration of the platysma muscle. For these

injuries, the neck is usually divided into three horizontal zones (Fig. 7.13-4, inset). Zone I extends from the sternal notch to the cricoid cartilage. Penetrating injuries in this area are associated with a high mortality. Zone II extends from the cricoid cartilage to the angle of the mandible. Because this is the most exposed region of the neck, injuries can be evaluated and explored using standard operative techniques. Zone III extends from the angle of the mandible to the base of the skull. Because of anatomic constraints, injuries to Zones I and III can be difficult to identify and repair. In patients with a Zone II injury, iv access should be established in the contralateral upper extremity because of the possible injury to the ipsilateral internal jugular vein or subclavian vein.

Until recently, evaluation and management of stable patients with neck injuries that penetrated the platysma depended on location of injury. Injuries to Zones I and III were evaluated radiographically, whereas injuries to Zone II were indications for mandatory exploration. With the increased availability of CT and arteriography, however, a selective exploration strategy is now favored. The presence of so-called hard signs, including airway compromise, shock, active bleeding, an expanding hematoma, an evolving stroke, or extensive subcutaneous air, generally mandates operative exploration. “Soft signs,” which include hoarseness, hemoptysis, hematemesis, dysphagia, or prominent ecchymosis, can be evaluated with CT and endoscopy.

Management of hemodynamically unstable or symptomatic patients with penetrating neck injuries should be limited to applying direct pressure, protecting the airway, establishing iv access, and obtaining a CXR. Exploration and definitive management of the injury in the OR should follow as soon as possible. In stable patients with penetrating injuries and no indications for exploration, CT is generally the first diagnostic test.

A **median sternotomy incision** may be used for patients with right neck Zone I injuries or if injuries to the mediastinum involving the innominate artery or right subclavian artery are suspected. Exposure of injuries to the proximal left subclavian artery is extremely difficult via median sternotomy. In this case, a left anterior thoracotomy is preferred. The chest and left arm should be prepped and draped to allow arm manipulation.

Usual preop diagnosis

Penetrating neck injury

■ SUMMARY OF PROCEDURE

Position	Supine with head turned away from side of exploration
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	and neck extended. If both sides of neck require exploration, head should be in midposition. It is helpful to clear the C-spine before exploration.
Incision	Sternocleidomastoid anterior border
Unique considerations	Monitor BP and HR carefully during carotid sinus manipulation.
Antibiotics	Cefazolin 2 g iv. If pharyngoesophageal injury, give ampicillin 1 g and gentamicin 80 mg iv.
Surgical time	1 h
Closing considerations	Wound requires drainage, especially in Zone I exploration and if esophagus or airway was violated.
EBL	Variable
Postop care	ICU 12–24 h for neurologic and airway monitoring. After carotid injury repair, angiography should be considered.
Mortality	1%
Morbidity	Hemorrhage: acute from injuries; delayed from pseudoaneurysms Airway injury: acute loss of airway; delayed tracheal stenosis Damage to neural or vascular structures Esophageal fistula
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Typically young adult
Etiology	Typically male
Incidence	5–10% of all trauma involves neck structures.
Associated conditions	Intrathoracic injuries; spinal cord injuries; recurrent laryngeal nerve injury; phrenic nerve injuries; thoracic duct injury

ANESTHETIC CONSIDERATIONS FOR NECK TRAUMA SURGERY

PREOPERATIVE

Patients with neck injuries often present unique challenges for endotracheal (ET) intubation. Extensive internal damage may be present despite minimal external signs. High-velocity deceleration events, or “seatbelt” sign injuries, are often associated with airway compromise. Vascular injuries, particularly involving the carotid artery, can cause a significant neck hematoma that compresses vital structures and markedly distorts airway anatomy, preventing visualization of laryngeal structures or passage of an ETT. Known or suspected C-spine injuries will restrict optimal positioning of the head and neck for laryngoscopy. Facial or dental injuries may impede access for laryngoscopy because of limited mouth opening or the presence of blood in the oropharynx. The use of alternative methods to secure the airway (e.g., fiberoptic bronchoscopy or video laryngoscopy) or a surgical approach may be the safest option. Equipment for a surgical airway should be available before attempting ET intubation.

Airway and Respiratory

Preop assessment of the airway and neck is of utmost importance. The extent of mouth opening and presence of jaw or dental injuries should be noted. Stridor and hoarseness may be present with laryngeal injury or compression of the trachea; therefore an assessment of the patient’s ability to phonate as an important part of the airway evaluation. Damage to the recurrent laryngeal nerve results in changes in voice and ↑ aspiration risk. Presence of C-spine injury (though not common with penetrating neck injuries) should be evaluated by x-ray or CT scan. Additionally, any distortion of internal and external structures due to tissue swelling or hematoma should be determined before intubation. The CXR or CT should be examined carefully for evidence of tracheal deviation, compression, or pneumothorax. Airway injury can be associated with subcutaneous emphysema or pneumothorax.

Tests: x-ray; CT scan; ABG

Cardiovascular	The patient should be evaluated for the extent of blood loss based on their method and location of injury, hemodynamics (HR, BP), physical exam (capillary refill, peripheral pulse), and imaging results (eFAST results). Notably, facial and dental injuries may result in significant occult blood loss that may accumulate in the stomach. Venous lacerations can allow air embolism.
Neurologic	Physical exam should evaluate for deficits associated with acute compromise of cerebral arterial blood flow. C-spine injuries should be assessed by physical exam (neck pain, neurologic examination) and radiologic studies (lateral C-spine x-rays [including C-7], CT scan, MRI).
Premedication	Premedication may compromise a tenuous airway or lead to respiratory decompensation; use extreme caution when administering anxiolytics or sedatives. Patients are considered to have a full stomach. Sedative premedication should be avoided in patients who are hemodynamically unstable and those with probable head injuries.

INTRAOPERATIVE

Anesthetic technique

GETA.

Induction	The induction technique will depend on the associated injuries and physical exam. If C-spine injury is suspected, in-line stabilization of the patient's head and neck should be provided by an assistant. Patients should be presumed to have a full stomach, and consideration should be given for a rapid-sequence iv induction with the use of cricoid pressure. However, in the presence of an unstable C-spine, cricoid pressure may be associated with further injury. If a difficult intubation is anticipated, the anesthesiologist should consider the risks and
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benefits of an “asleep” vs. an “awake” intubation. A variety of intubation tools (laryngoscope blades, video laryngoscopes, laryngeal mask airways) and a range of ETT sizes (5.9–8.0 mm) should be available. LMA may be useful for providing temporary airway control without C-spine manipulation if direct laryngoscopy is difficult. Fiberoptic intubation or a surgical airway (cricothyrotomy or tracheotomy) should also be considered. In patients with penetrating neck injuries, “blind” intubation techniques should be avoided.

The choice of induction medications should take into account the preop hemodynamics and airway concerns. In the hypovolemic patient, induction doses of propofol may be reduced by 50–75% to minimize ↓↓ BP. Consider etomidate (0.2–0.3 mg/kg) or ketamine (0.5–1 mg/kg iv) as alternative induction agents. In the presence of a vascular injury, it is important to avoid rapid ↑BP to prevent expansion of the hematoma. If rocuronium is used to facilitate intubation, sugammadex (e.g., 4–16 mg/kg iv) should be available for rapid reversal of paralytics in case of unanticipated difficult ventilation and intubation.

Maintenance

Standard maintenance is usually appropriate for the normotensive neck trauma patient. In cases of vascular injury, careful control of BP in the low normal range is advantageous. When nerve testing is anticipated, paralytics may need to be avoided.

Emergence

If only minimal distortion of the airway or residual swelling remains, awake extubation at the end of the procedure can be considered. Full reversal of muscle paralytics is of the utmost importance to avoid postop respiratory distress. Postop intubation and mechanical ventilation may be prudent for patients with residual neck or oropharyngeal swelling.

Blood and fluid

IV access: 14–16 ga Resuscitation with balanced

requirements	× 2	crystalloid solutions is recommended (i.e., Normosol®). Blood transfusion may be necessary for vascular injuries. Type and screen should be sent as soon as possible.
Monitoring	ASA standard monitors ± Arterial line	Invasive monitoring may be indicated for patients with major vascular injuries; however, the placement should not necessarily delay the start of emergency surgery.
Positioning	✓ and pad pressure points ✓ eyes	With suspected or known C-spine injuries, stabilization of the head and neck in the neutral position is required. Bed will often be turned 180° for surgical access to the injured site.

POSTOPERATIVE

Complications	Hemorrhage Hematoma Airway compromise	BP control at low normal levels (using SNP/NTG infusion, labetalol) and elevation of the head of the bed (10–20°) will help minimize tissue edema and hematoma formation, which could lead to airway compromise.
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Suggested Reading

1. American College of Surgeons Committee on Trauma: Advanced trauma life support

for doctors. In: Student Course Manual, 9th edition. American College of Surgeons, Chicago: 2012.

CHEST TRAUMA: PERICARDIAL WINDOW, RELEASE OF TAMPONADE, AND REPAIR OF CARDIAC LACERATION

SURGICAL CONSIDERATIONS

Description

Cardiac contusions may occur in patients with blunt chest trauma. The degree of injury varies from localized contusion to cardiac rupture, but most are clinically insignificant if the patient survives in the hospital. Autopsy studies of victims of immediately fatal accidents show that as many as 65% have rupture of one or more cardiac chambers and 45% have pericardial lacerations. Most of the patients with cardiac rupture die at the scene; however, survivors are reported if vital signs are present during transport. Early clinical findings in cardiac contusion are most commonly dysrhythmias, but occasionally, patients can develop cardiac failure. The initial ECG in the ED is the most sensitive diagnostic test. The most common abnormalities are atrial arrhythmias and RBBB. Serial cardiac monitoring is indicated if the initial ECG is abnormal. Serum enzymes are not helpful. Management is symptomatic. Arrhythmias should be treated in the standard fashion and are not a contraindication to surgery. In the presence of blunt cardiac injury, hemodynamic abnormalities mandate echocardiographic examination to evaluate for structural lesions such as valve or myocardial rupture. Pericardial lacerations from stab wounds tend to seal and cause tamponade, present in 80–90% of patients with stab wounds to the heart. Accumulation of 150 mL of blood in the pericardium may impair preload and cause shock. **Beck triad** of distended neck veins, muffled heart sounds, and ↓ BP is present in only 30% of patients with tamponade. Pulsus paradoxus is even less reliable. The diagnostic test of choice is ultrasound, performed in the ED as part of the focused assessment by sonography for trauma (eFAST) exam.

Treatment of penetrating cardiac injuries has gradually changed from initial management by **pericardiocentesis** to prompt **thoracotomy** and **pericardial decompression**. Pericardiocentesis, with ultrasound guidance, may be used to stabilize a patient until **sternotomy** or thoracotomy can be performed. A **subxiphoid pericardial window** is an option but is best performed in the OR.

Gunshot wounds (GSWs) can produce extensive myocardial damage, multiple perforations, and massive bleeding into the pleural space. Hemothorax, shock, and exsanguination occur in nearly all cases of cardiac GSWs. Pericardial tamponade is often absent. Hemodynamically stable patients with penetrating injuries close to the heart should have ECHO evaluation in the ED only if equipment is immediately available. The presence of pericardial fluid should be evaluated with an eFAST scan. If hemopericardium is confirmed, blood should be evacuated and injury treated in the OR.

Patients who are stable enough to be transported to the OR have an excellent prognosis with reported survival of 97% for stab wounds and 71% for GSWs. In contrast, patients who require ED thoracotomy have only a 15% survival rate for penetrating injuries.

Subxiphoid pericardial window under local or GA: The entire chest is prepped for potential sternotomy. A vertical midline incision is made over the xiphoid process and upper epigastrium. The xiphoid is elevated or excised, allowing access to the pericardiophrenic membrane and anterior mediastinum. The pericardium is opened between two stay sutures and inspected for the presence of blood. With the use of eFAST scanning, this diagnostic test is most useful when the patient is already undergoing an **exploratory laparotomy**, and there is a suspicion for a cardiac injury. If blood is present, a median sternotomy should be performed.

Median sternotomy provides excellent exposure to the heart, great vessels, and pulmonary hila. This approach is ideal for anterior injuries and for the unstable patient but is less well suited for posterior injuries and left subclavian injuries.

Left anterior/anterolateral thoracotomy in the 5th intercostal space can be used in the stable patient or in the newly unstable ED patient with penetrating injury. The pericardium is opened anterior to the phrenic nerve, and tamponade is relieved. The bleeding heart is controlled with digital occlusion, and the laceration is closed with mattress sutures, with care being taken not to occlude coronary flow. Small coronary branches can be ligated, whereas others should be repaired and may even require **CPB**.

Usual preop diagnosis

Cardiac laceration

■ SUMMARY OF PROCEDURE

Position	Supine, L arm abducted 90°
Incision	Median sternotomy or left anterior or anterolateral thoracotomy in the 5th intercostal space (see Fig.

7.13-5). Further exposure may be obtained by transsternal extension into the R chest.

Special instrumentation	Rapid-infusion device; active rewarming system; internal defibrillator
Unique considerations	Massive blood loss is expected on opening pericardium and during repair of laceration. Large-bore peripheral lower extremity iv and central venous access are recommended for monitoring of volume replacement. A Foley catheter can be used to occlude laceration and as access for infusion. Arrhythmias are common.
Antibiotics	Cefazolin 2 g iv
Surgical time	45 min
Closing considerations	Hypothermia and coagulopathy contribute to mortality.
EBL	1–2 L
Postop care	ICU
Mortality	Without tamponade: ~90% With tamponade: ~30%
Morbidity	Overall: 85% Pulmonary complications Acute MI Arrhythmias Intracardiac shunts Ventricular aneurysms Valvular lesions Retained foreign bodies
Pain score	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Usually young adult
Etiology	GSWs, stab wounds, blunt trauma Hemothorax; pneumothorax; great vessel injury; lung

**Associated
conditions**

contusion

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Chest Trauma.

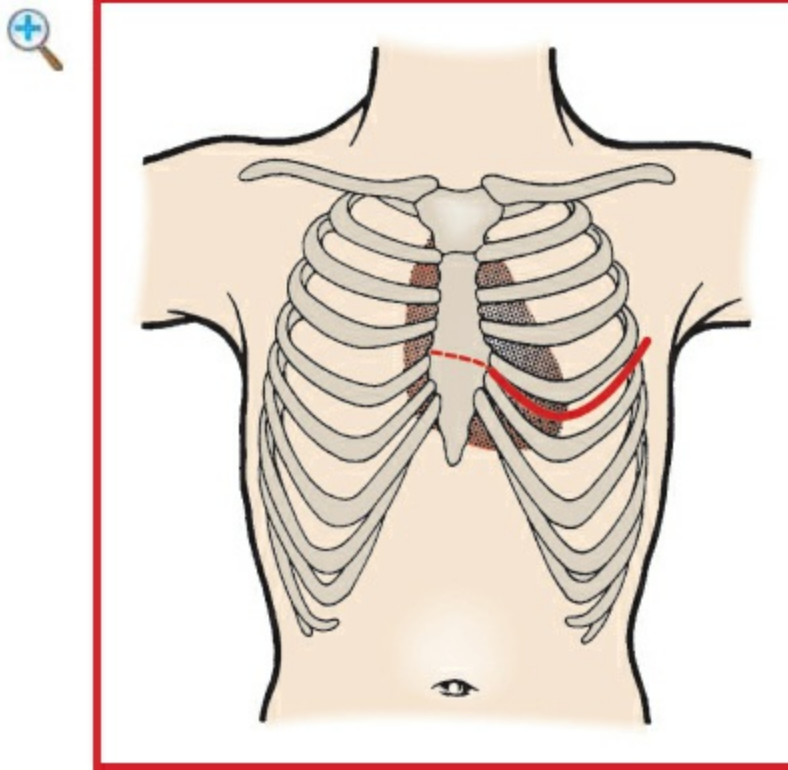


Figure 7.13-5 Incision (dashed/solid line) for ED thoracotomy: made immediately below the nipple, extending from the sternum as far laterally as possible. (Reproduced with permission from Baker RJ, Fischer JE, eds: Mastery of Surgery, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

CHEST TRAUMA: REPAIR OF GREAT VESSELS

SURGICAL CONSIDERATIONS

Description

Thoracic great vessel injury accounts for 8–9% of all vascular injuries seen in the trauma center. The subclavian artery and descending thoracic aorta are the vessels injured most often (21% of cases), followed by pulmonary artery (16%), subclavian vein (13%), vena cava (11%), and innominate artery and pulmonary veins (9%).

Rupture of the thoracic aorta is the most lethal injury following blunt chest trauma and

causes up to 50% of fatalities in MVCs. Proposed mechanism of injury is a deceleration force causing flexion or torsion of the aortic arch and subsequent disruption of the aortic wall at the ligamentum arteriosum immediately distal to the left subclavian artery. Survival of the patient depends on retention of the hematoma by the adventitial layer of the aorta. The mechanism of injury and the mediastinal silhouette on CXR are the two most sensitive markers for an injured thoracic aorta. Depressed left mainstem bronchus, apical capping, and a deviated trachea or esophagus (OG/NT tube) seen on CXR may be suggestive of aortic injury. CT angiogram of the chest has high sensitivity and is the preferred modality of evaluation in most trauma centers as a screening test. Arteriography is the gold standard for evaluating aortic injuries and may also allow therapeutic intervention. TEE can also aid in the evaluation and diagnosis of aortic injury in intubated patients, with a sensitivity and specificity similar to CT angiography in the hands of a skilled provider.

Smaller dissections in blunt trauma may be managed in the ICU with tight control of heart rate and systolic blood pressure. Injuries requiring operative intervention are traditionally managed with **thoracotomy** and **graft interposition**, but not without significant morbidity. Newer endovascular techniques are increasingly used as an adjunct to surgery or for definitive repair.

Investigation of these techniques is ongoing.

Ascending aorta

Rupture requires median sternotomy, CPB, and repair with a Dacron® graft. Penetrating injury to the anterior aspect of the aorta can be repaired primarily; if there is additional posterior injury, CPB is required for successful repair.

Aortic arch

Complete exposure of the great vessels is required; median sternotomy with extension to the neck and division of the innominate vein may be utilized. Complex injuries may require CPB.

Innominate artery, right carotid artery

Approach is via a median sternotomy with right cervical extension and division of the innominate vein, if necessary. Blunt trauma typically involves the proximal innominate artery, in contrast to penetrating trauma, which usually involves the distal portion of the artery near the carotid or subclavian bifurcation. Injuries are repaired using an interposition graft of Dacron® or Gore-Tex®. Because cerebral perfusion is maintained through the L carotid and subclavian arteries, shunting is not necessary.

Descending thoracic aorta

Ninety-seven percent of patients with great vessel injury who arrive alive at the hospital will have an injury at the isthmus. Clamping and direct reconstruction are required under temporary bypass shunt or pump-assisted shunt. Left posterolateral thoracotomy via the 4th intercostal space is the preferred access. Distal control of the descending aorta is obtained first; then the transverse aortic arch is exposed, and umbilical tape is applied between the L carotid and L subclavian arteries. Vascular clamps are applied at the proximal aorta, distal aorta, and subclavian artery. Graft interposition is utilized in 85% of the cases. An aortic cross-clamp time of <30 min minimizes the incidence of paraplegia.

Subclavian artery or vein

Approach is via a cervical extension of the median sternotomy for right-sided injuries. Exposure of injuries to the proximal, left, subclavian artery via median sternotomy is usually inadequate. In patients with such injuries, a high left thoracotomy, often with either a supraclavicular or trap-door incision, is needed. Graft interposition is often necessary.

Pulmonary artery and veins

In major hilar injury, **pneumectomy** may be necessary, despite high mortality. Cross-clamping of the hilum prevents air embolus and hemorrhage. Simultaneous cross-clamping of the vessels and bronchus may reduce mortality in those patients requiring pneumectomy. Fluid administration should be kept to a minimum to prevent right heart failure.

Thoracic vena cava

Intrathoracic IVC injury may cause hemopericardium and tamponade. Repair often is performed through the R atrium. SVC repair often is performed via a lateral venorrhaphy.

Usual preop diagnosis

Thoracic great vessel injury

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Anterolateral thoracotomy is preferred as it provides access to the descending aorta for cross-clamping and good access to heart and may be extended

	across the sternum into the right chest.
Special instrumentation	Intraop blood recovery device; rapid-infusion device; graft materials for any vessels larger than 5 mm. CPB may be necessary.
Unique considerations	Use of vasodilators (e.g., SNP) prevents cardiac strain during cross-clamping of the aorta. Large amount of fluid is required to prevent hypotension after clamp removal.
Antibiotics	Cefazolin 2 g iv
Surgical time	1–3 h
EBL	≥1–2 L
Postop care	ICU. Careful hemodynamic monitoring is critical because underlying pulmonary contusion may be worsened by inappropriate fluid administration. PA catheter may be necessary to optimize hemodynamic parameters, particularly in older patients. Coagulation studies must be carefully monitored and corrected with appropriate blood products.
Mortality	In general, mortality is associated with multisystem trauma and usually 2° concomitant head injury, infection, respiratory insufficiency, and renal failure. Pulmonary artery or vein, suprahepatic IVC, or SVC injury: >70% Ascending thoracic aorta, aortic arch (in patients with stable vital signs on arrival): 50% Blunt injury to the descending thoracic aorta: 5–25% Subclavian artery injury: 5%
Morbidity	Paraplegia: ~8% (with descending aortic injuries)
Pain score	10

■ PATIENT POPULATION CHARACTERISTICS

Age range	Typically young adult
Incidence	Thoracic great vessel injury accounts for 8–9% of all

Etiology	vascular injuries seen in the trauma center. Blunt trauma from MVC; penetrating trauma from stab wounds or GSWs
Associated conditions	Pneumothorax; hemothorax; head injury

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Chest Trauma.

CHEST TRAUMA: PNEUMONECTOMY, LOBECTOMY, AND REPAIR OF TRACHEOBRONCHIAL INJURY

SURGICAL CONSIDERATIONS

Description

Penetrating injuries to the chest can be divided into high- and low-velocity injuries. All stab wounds are classified as low-velocity injuries. GSWs may be considered low- or high-velocity injuries, depending on the energy of the bullet. The majority of GSWs seen in the ED are low velocity; however, higher-velocity injuries are being seen with increased frequency. The extent of tissue destruction in high-velocity injuries is related to blast effect, tumbling, and fragmentation of the missile, as well as secondary missiles, such as bone fragments. Patients with such injuries are more likely to require **thoracotomy** and **pulmonary resection**. In general, pulmonary resection is required in 1% of stab wounds and 2% of GSWs. The indications for early operation are continued shock, prolonged bleeding, a larger air leak with inability to oxygenate or ventilate the patient, and suspected concomitant injuries to the vital intrathoracic structures. In most patients with stab wounds or low-velocity GSWs, bleeding from the lung tissue stops spontaneously after evacuation of the hemothorax and reexpansion of the lung. Only 5–10% of such patients will require thoracotomy to control bleeding as compared to a thoracotomy rate of 70% for high-velocity injuries. Pulmonary contusion is a frequent sequela of GSWs or blunt trauma to the chest. Pulmonary contusion occurs in 75% of patients with flail chest but also can occur following blunt trauma without rib fracture. Alveolar rupture with fluid transudation and extravasation of blood are early findings.

The incidence of tracheobronchial injuries reported in the autopsy series from MVC is 1%. Mechanisms of injury include rapid deceleration, direct blow, or sudden increase of intratracheal pressure against a closed glottis. The most common tracheobronchial injury is transverse rupture, occurring in 74%, followed by longitudinal rupture in 18%, and complex in the remaining 8% of cases. Approximately 80% of cases with tracheal rupture occur within 2.5 cm of the carina. Patients with injury to the airway may present with severe dyspnea and with massive subcutaneous emphysema. About 90% of these patients will have an abnormal CXR, showing pneumothorax, pneumomediastinum, subcutaneous emphysema, or pleural effusion.

Tracheobronchoscopy should be performed in all patients with suspected tracheobronchial injuries to establish the diagnosis and plan operative treatment.

Parenchymal lacerations are repaired by the simplest method available to stop bleeding or air leak. If pulmonary resection is required, formal segmental resection is not necessary. A stapling device should be used to preserve as much lung tissue as possible.

Anatomic pulmonary resections are indicated when bronchial injury repair is not feasible or may lead to complete lobar collapse. **Pneumonectomy** may be required for major hilar injuries but is associated with a mortality rate of 75%. **Primary repair of tracheobronchial injuries** should be performed as soon as possible. Transverse rupture may require the placement of a sterile tracheal tube into the distal trachea through the operative field. After posterior sutures are placed, an orotracheal tube is advanced beyond the area of injury. **Mainstem bronchial repair** is performed under one-lung ventilation. High-frequency jet ventilation may be required to maintain oxygenation. Occasionally, total **CPB** is necessary for repair of complex airway injuries. Before closing, the suture line is pressure-tested and evaluated by **fiberoptic bronchoscopy**.

Usual preop diagnosis

Tracheobronchial injury; penetrating/blunt chest trauma

■ SUMMARY OF PROCEDURE

Position	Supine, with neck extended for proximal tracheal injury, or lateral decubitus
Incision	Transverse cervical for almost all proximal tracheal injuries Right posterolateral thoracotomy in the 5th intercostal space for thoracic trachea and right bronchial wounds

	Left posterolateral thoracotomy for L bronchial injury
Special instrumentation	Fiberoptic bronchoscope, intrabronchial tube, jet ventilator/oscillator
Unique considerations	In patients with lobar resection or after pneumonectomy, PEEP may result in bronchopleural fistula.
Antibiotics	Cefazolin 2 g iv
Surgical time	2–3 h
EBL	Variable
Postop care	ICU (typically)
Mortality	Pneumonectomy for trauma: 75% Penetrating tracheobronchial injuries: ~15%
Morbidity	Bronchopleural fistula: 10% Empyema: 5% Hemothorax: 2–5% Pneumothorax ARDS (depends on extent of injuries)
Pain score	Cervical approach: 5–7 Thoracic approach: 8–10

ANESTHETIC CONSIDERATIONS FOR CHEST TRAUMA SURGERY

Blunt and penetrating chest trauma may present as any of the following conditions: hemo-pericardium, hemo-/pneumothorax, blunt cardiac injury, airway injury, pulmonary contusion with or without associated flail chest syndrome, and diaphragmatic injury. Hemo-pericardium may rapidly progress to pericardial tamponade, requiring immediate pericardiocentesis or pericardial window, followed by surgical exploration and repair of the cardiac or vascular laceration. These patients typically present with shock (↓↓ BP, ↑↑ HR) with distended neck veins (↑↑ venous pressure) and distant heart sounds (**Beck triad**) without evidence of tension pneumothorax. Blunt cardiac trauma may present with findings ranging from isolated ECG changes to cardiac rupture. The right ventricle and interventricular septum are most commonly affected. Arrhythmias and ventricular dysfunction are frequently observed. Hemothorax may present as respiratory failure, shock, and absent breath sounds over the affected hemithorax. Pulmonary

contusion often presents with hypoxia and tachypnea; hemoptysis may also be present. Large flail chest defects may require mechanical ventilation. Intrathoracic tracheobronchial injuries are less frequent than upper airway injuries, although they are associated with a high mortality and usually require operative intervention. Video-assisted thoracic surgery (VATS) has emerged as the leading diagnostic tool for detection of diaphragmatic injury. Preoperatively, and particularly during the induction of GA, the patient's intravascular volume must be expanded, and myocardial contractility, HR, and SVR maintained. Inotropes and antiarrhythmic drugs may be required if hemodynamic instability occurs and patient does not respond to iv fluid administration. CPB is usually not required.

PREOPERATIVE

Respiratory

Injuries such as hemothorax and pneumothorax may be present, requiring thoracostomy tube placement. Fractures of the 1st or 2nd rib should prompt a detailed search for associated intrathoracic injuries, especially injuries to the great vessels. Injuries to the great vessels may present on imaging with a widened mediastinum or apical pleural capping. Multiple rib fractures are often associated with pulmonary contusions, which may not be apparent on the initial CXR but can progressively impair oxygenation. In cardiac tamponade, spontaneous respiration is preferred over PPV ($\rightarrow$ $\downarrow$ venous return + $\downarrow$ CO). Significant intrathoracic tracheobronchial disruption may require rigid bronchoscopy or a surgical airway.

Tests: CXR (PA + lateral views)—Upright inspiratory films best delineate chest structures; expiratory films enhance visualization of pneumothorax but are difficult to obtain in multiple-trauma patients; ultrasound can identify the presence of a pneumothorax or pleural effusion. CT of the chest if the patient's condition permits. ABG

Cardiovascular

Intravascular volume should be restored. BP and HR should be followed, and responses to fluid resuscitation noted. Given the risk of myocardial

contusion, RV failure (due to the substernal location of the RV), and hemopericardium, assessment of cardiac function is also important; cardiac output can be significantly reduced despite normal BP measurements. TTE or TEE is useful for evaluating cardiac function, the presence of pericardial effusion or tamponade, or aortic injuries. In the setting of suspected cardiac dysfunction or tamponade, it may be necessary to support myocardial contractility with inotropes (e.g., epinephrine or dopamine).

Myocardial contusion can also be associated with atrial or ventricular arrhythmias, which may require supportive therapy with antiarrhythmic medications (e.g., lidocaine 1–1.5 mg/kg bolus or 1–2 mg/min infusion or amiodarone 150–300 mg bolus or 1 mg/min infusion) or electrical cardioversion/defibrillation. Electrolyte abnormalities should be corrected. Large-bore intravascular access should be obtained, with strong consideration given to the need for invasive monitoring (arterial line, CVP, PAC etc.).

Tests: ABG, continuous telemetry monitoring or serial ECGs. Ultrasound—eFAST, TTE, TEE

Neurologic

↓ BP and ↓ CO may compromise cerebral perfusion. Associated head injury also can contribute to alterations in mental status. Seek physical evidence of open head or closed head injuries, such as palpable depressions of the skull or scalp lacerations, abrasions, or contusions. Pupil size and reactivity should be noted. Intubation in the ER is necessary for patients who are unable to protect their airway, require hyperventilation, or are combative and unable to cooperate with medical staff for exam and treatment. In general, any patient with a GCS ≤ 8 (no spontaneous eye opening, inappropriate or incomprehensible speech, and only reflexive motor responses) requires intubation. GCS score ≤ 8 indicates significant brain injury. Motor or sensory

	deficits may reflect spinal cord injury and may be associated with neurogenic (“spinal”) shock, particularly with upper thoracic or cervical cord injuries. Tests: C-spine x-rays (lateral view, including C7) are good screening exams— ✓ for altered vertical alignment and unequal disc interspaces. CT scan or MRI of the head: ✓ for gross asymmetry, hemorrhage, or midline shift.
Hematologic	Preop T&C appropriate. Depending on estimations of prior, ongoing, and anticipated surgical blood losses, blood products should be available (preferably cross-matched units, but O-neg or type-specific blood if cross-matching not complete). Tests: Serial Hct; T&C; ✓ Ca⁺⁺ and K⁺ following massive transfusions
Musculoskeletal	Known or suspected C-spine fractures require intubation precautions, with laryngoscopy performed while an assistant provides in-line stabilization of the patient’s head in the neutral position. Tests: Lateral C-spine x-rays, including C7; CT; MRI
Premedication	Premedication is rarely useful due to the urgency of the procedures and the need to have an alert, responsive patient for serial evaluations of mental status. Sedative premedication should be avoided in patients who are hemodynamically unstable and those with probable head injuries. Virtually all patients are considered to have full stomachs, and any compromise of the ability to protect the airway is inappropriate.

INTRAOPERATIVE

Anesthetic technique

GETA with full-stomach precautions.

Induction and intubation

Rapid-sequence induction with full-stomach precautions is often preferred. **NB:** In the patient with an unstable C-spine, cricoid pressure (~10 lb) may cause spinal cord injury, and consideration should be given to the establishment of a surgical airway under local anesthesia.

Induction medications:

- Ketamine (0.5–2.0 mg/kg): associated with ↑ HR, ↑ BP, and ↑ CO. Careful consideration should be given to the catecholamine-depleted trauma patient, in whom the direct myocardial depressant effects of ketamine may be unmasked.
- Etomidate (0.2–0.6 mg/kg): useful alternative for patients with head injuries
- Agents that tend to decrease SVR, such as inhalational anesthetics and narcotics, should be introduced with caution.
- Succinylcholine is attractive due to the quick return of spontaneous ventilation; alternatively, RSI-dose rocuronium (1 mg/kg) can now be immediately reversed with sugammadex (16 mg/kg).
- Inotropic agents (ephedrine, dopamine, epinephrine) should be immediately available to treat ↓ BP.

GETA is required for exploration through a median sternotomy or left thoracotomy. In the latter case, one-lung ventilation will likely be necessary, with a DLT or bronchial blocker (BB). If a difficult airway is anticipated, a single-lumen ETT may be placed initially with a DLT placed with an airway exchange catheter. Fiberoptic scopes aid in the placement of DLTs and BBs and assist in the confirmation of appropriate positioning.

The potential for sudden and substantial blood loss exists, and PRBCs (cross-matched, type-specific or O-neg) should be available in the OR before induction.

Maintenance

Initially, low-dose inhalational agents and low-dose

	narcotics may be used, as tolerated. ↓ BP may require resuscitation with crystalloid, vasoconstrictor, or inotropic agents. Muscle relaxation and PPV are used. N₂O should be avoided if there will be a need for one-lung ventilation or there is associated pulmonary contusion. If the patient is hypotensive and acidotic, scopolamine (0.2–0.4 mg iv) will ↓ risk of recall. As cardiac and vascular injuries are repaired, BP and CO often will improve, permitting increased depth of anesthesia and decreased inotropic support. Indicators of improved tissue perfusion should be followed, such as resolution of lactic acidosis, central or mixed venous oxygen saturation, and improved urine output.	
Emergence	Trauma patients often require prolonged intubation; however, hemodynamically stable patients with limited injuries may be extubated awake. Patients who received significant blood replacement (>50% of blood volume), those requiring inotropic support, intoxicated patients, and those with head injuries should remain intubated and mechanically ventilated postop. Consideration for continued mechanical ventilation should be given to those patients with large pulmonary contusions or flail chest segments.	
Blood and fluid requirements	Be prepared for large blood loss. IV: 14 ga × 2 Fluid warmer Rapid-infusion device	Large blood loss may occur, depending on the severity of the injury. Blood products (pRBC, FFP, platelets) should be available for balanced resuscitation in the setting of hemorrhage. Crystalloid solutions should also be available for resuscitation (e.g., BSS such as Normosol®).
Monitoring	ASA standard monitors	ECG should be observed for changes associated with

	Arterial line CVP ± PA catheter ± TEE	myocardial contusion (atrial/ventricular dysrhythmias, RBBB, ST-T wave changes). Arterial line useful in unstable patients or when frequent lab sampling is anticipated. CVP line or PA catheter may be useful if vasoactive infusions are needed or if ventricular dysfunction is apparent. Placement of additional monitoring should be accomplished without delaying the surgical control of hemorrhage or without interrupting aggressive volume resuscitation. (With release of tamponade, CVP rapidly drops toward normal, and CO and BP will markedly improve.) TEE offers continuous cardiac monitoring of contractility, volume status, and assessment of regional wall motion.
Positioning	✓ and pad pressure points ✓ eyes ± C-spine: neutral position	Axillary roll, if lateral decubitus position; chest roll if median sternotomy
Complications	Hypothermia Awareness Renal failure	Hypothermia → dysrhythmias + ↓ CO → acidosis. Rx: warm OR, warm iv fluids, patient warming devices A concern in unstable patients. Rx: Consider

Coagulopathy

scopolamine (0.2–0.4 mg iv). Usually 2° ↓ renal perfusion (prerenal). Rx: Restore CO with volume and inotropes. Most commonly 2° dilutional thrombocytopenia or hypothermia. DIC may require replacement therapy with Plts, FFP, and cryoprecipitate. Massive transfusion may benefit from a 1:1 ratio of PRBC to FFP. Maintain nl Ca⁺⁺ (1.05–1.3 mM/L).

POSTOPERATIVE

Complications

Hypotension
Cardiogenic shock
Arrhythmias

Hypothermia

Atelectasis, V/Q mismatch, ARDS

Coagulopathy, VTE

Cardiogenic shock may be 2° prolonged ↓ BP or cardiac contusion. Neurogenic shock may be seen with associated spinal cord injuries. Hemodynamic instability with atrial/ventricular dysrhythmias may occur. Treatment with inotropic infusions (epinephrine 10–150 ng/kg/min or norepinephrine 1–20 mcg/min or dopamine 5–10 mcg/kg/min) and antiarrhythmic medications. Active warming of iv fluids and warming blanket should be continued in ICU or PACU if hypothermia persists. If neurologic injury is present, targeted temperature

management is recommended (33–36°C, with avoidance of hyperthermia). Pulmonary compliance may decrease with large volumes of fluid replacement. Interstitial and alveolar edema → progressively worsening pulmonary function requiring prolonged mechanical ventilation. Pulmonary contusions may aggravate this problem and severely compromise oxygenation and ventilation, requiring high-inspired O₂ concentration, high PIP, and PEEP. Institution of lung-protective ventilation strategies should be considered. Based on clinical assessment and laboratory data (PT, Plt count, fibrinogen), transfusion of Plts, FFP, or cryoprecipitate may be indicated.

Pain management

PCA or parenteral narcotics
Epidural

Epidural infusions of narcotic and/or low-dose local anesthetics can be effective in patients without coagulopathy. Continuous epidural anesthesia may be beneficial in the setting of multiple rib fractures.

ABDOMINAL TRAUMA: DAMAGE CONTROL

SURGICAL CONSIDERATIONS

Description

The incidence of abdominal injuries requiring laparotomy approaches 25% for penetrating and 5–8% for blunt abdominal trauma. In blunt trauma, liver and spleen injuries occur with an incidence of ~50%. In penetrating trauma, the most common injuries are in the small bowel (29%), liver (28%), colon (23%), and stomach (13%). Many preventable deaths in trauma patients are related to shock from unrecognized intra-abdominal hemorrhage caused by solid organ injury.

Victims of severe multisystem trauma are particularly susceptible to the development of a fatal coagulopathic state 2° hypothermia, acidosis, dilution, and consumption. Replacement of two or more blood volumes with BSS and PRBCs will decrease the level of coagulation factors to 15%. Because of delays in obtaining coagulation profile results, coagulation factors should be replaced empirically in the setting of a large transfusion requirement (e.g., 1 U FFP/1.5 U PRBC). Metabolic acidosis affects both the circulatory system and coagulation, $\rightarrow \downarrow$ CO and triggering DIC. Chances of salvaging a patient with pH < 7.0 approach zero. To stop this self-perpetuating downward cycle, the concept of “damage control” has evolved. This involves **rapid laparotomy** to control hemorrhage and GI spillage, followed by temporary closure of the abdomen and subsequent exploration after the patient has been warmed and stabilized. With the use of this technique, ~40% of critically injured patients can be saved from otherwise fatal injuries.

With the patient on a heated operating table, the patient is prepped from the thighs to the neck and draped, and the abdomen is entered through a midline incision. This critical moment can be associated with significant blood loss and may require rapid blood transfusion. **Four-quadrant packing** with laparotomy pads is performed in the abdominal cavity, and manual compression of the subdiaphragmatic aorta may be instituted if packing alone does not control the hemorrhage. If necessary, the operation is stopped, and blood/fluid resuscitation is performed. After consultation with the anesthesiologist, the surgeon proceeds with the sequential unpacking of each of the four quadrants and identifying injuries. Vascular injuries are controlled with clamping and ligation or shunting, bowel injuries are stapled across, but no attempt is made to restore bowel continuity. Liver and retroperitoneal injuries are controlled with packing alone. When damage control is performed, the abdomen is closed with a running skin suture, if appropriate; otherwise, a temporary vacuum dressing is used. PIP should be monitored during closure, as patients may develop **abdominal compartment syndrome** ($\uparrow\uparrow$ intra-abdominal pressure $\rightarrow \uparrow$ PIP, $\downarrow$ UO, normal filling pressures). The patient is then

transported to the ICU and actively warmed and resuscitated. Reoperation should be performed at 24–48 h when the patient is rewarmed, and acidosis and coagulopathy have resolved. Definitive repair of the injured organs should be completed.

Usual preop diagnosis

Intra-abdominal trauma and hemorrhage

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Midline abdominal
Special instrumentation	Fluid-warming, rapid-infusion device
Unique considerations	Autotransfusion device (e.g., Cell Saver) may be of use if no contamination of abdominal cavity.
Antibiotics	Cefotetan 1–2 g iv
Surgical time	45–90 min
Closing considerations	Even temporary closure may not be possible because of severe bowel edema. A silo made from a plastic iv bag may be used to cover the bowel. Monitor PIP and UO during abdominal wall closure because abdominal compartment syndrome may develop.
EBL	Average transfusion requirement: 12 L crystalloids, 20 U PRBCs, 10–15 U FFP, 6 U Plts
Postop care	ICU; patient remains intubated. Active rewarming is of primary importance.
Mortality	70%
Morbidity	Intra-abdominal abscess: 30% Fistulas: 10%
Pain score	10

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Typically young adult

Male:Female	9:1
Incidence	25% for penetrating and 5% for blunt abdominal trauma
Etiology	MVC; penetrating injury (e.g., GSW or stab wound)
Associated conditions	Chest trauma; closed head trauma; pelvic fracture

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Abdominal Trauma Surgery.

ABDOMINAL TRAUMA: HEPATIC AND SPLENIC INJURIES

SURGICAL CONSIDERATIONS

Description

The liver is the most commonly injured organ in patients with blunt trauma and the second most commonly injured organ in penetrating trauma. The management strategy and decision to operate depends upon the hemodynamic status of the patient, grading of the liver injury, and presence of associated injuries. The majority of injuries can be managed nonoperatively if the patient is hemodynamically stable unless other injuries mandate laparotomy. The adjunctive use of angioembolization has significantly increased nonoperative success rates. Thus, most liver injuries that require operation are complex (grades IV–V), with large blood loss and high mortality (up to 30%).

In a patient with massive **intra-abdominal hemorrhage**, sudden cardiovascular collapse is predictable when the abdomen is opened. Laparotomy with manual compression of the aorta at the aortic hiatus is recommended. Access for subsequent aortic clamping is rapidly obtained by blunt finger dissection of the lesser sac. After removing all clots and free blood, four-quadrant packing is used to control bleeding. Significant liver bleeding should be controlled with manual compression, the Pringle maneuver, and perihepatic packing.

Several maneuvers can be used to facilitate **repair of liver injuries: manual compression and packing** temporarily controls bleeding and allows time for volume

resuscitation. The use of topical hemostatic agents and direct suturing of the liver parenchyma provide direct control.

Portal triad occlusion (Pringle maneuver; Fig. 7.13-6) decreases blood loss and identifies the patient who might benefit from selective hepatic artery ligation. It also allows for differentiation from hepatic venous or retrohepatic caval bleeding if clamping is unsuccessful in arresting hemorrhage.

In persistently hemodynamically unstable patients, damage control techniques provide time for resuscitation. **Perihepatic packing and planned reexploration** is a lifesaving maneuver and should be used early for patients with severe injuries, before they become hypothermic, coagulopathic, and acidotic. Extensive liver mobilization, parenchymal disruption (i.e., finger fracturing), and atriocaval shunts are rarely indicated. Hepatic angiogram and embolization in the immediate or early postop phase may also be very useful for patients with severe injuries. At reexploration, intrahepatic omental packing may be used for obliterating dead space. Closed suction drains should be used in all patients.

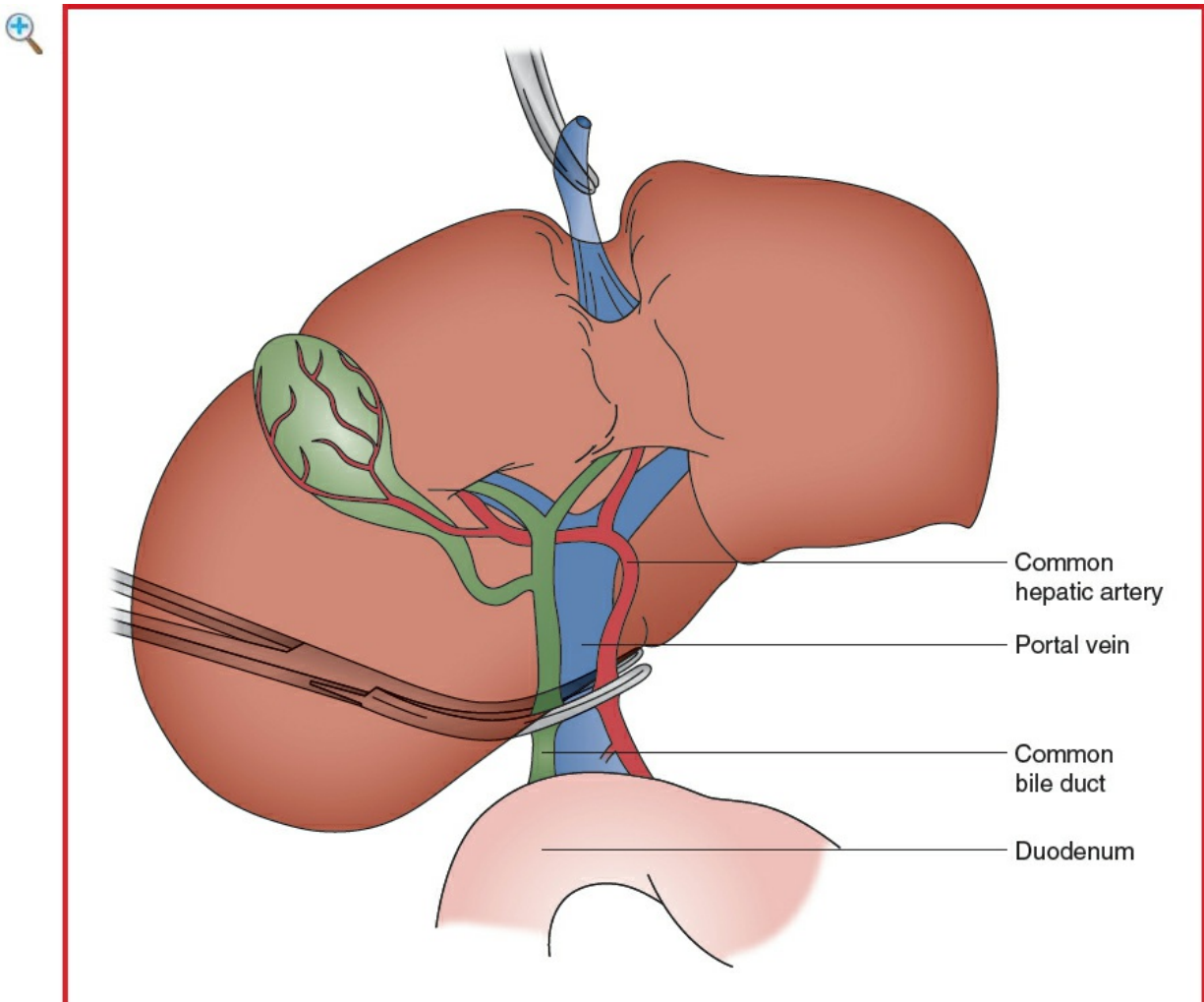


Figure 7.13-6 Pringle maneuver compression of the portal triad structures with a non-crushing vascular clamp for hepatic inflow control. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al, eds: Surgery: Scientific Principles and Practice 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Portal triad (portal vein, hepatic artery, common bile duct) injuries, although rare, are associated with extremely high mortality. Simple lacerations to the common bile duct may be repaired using interrupted absorbable sutures. More destructive injuries are best repaired over a T-tube to minimize risk of stricture and facilitate postop contrast studies. Complete transections and tissue defects should be evaluated for reconstruction by a hepatobiliary surgeon. Isolated portal vein injuries are associated with a 70% mortality. The portal vein should be repaired if possible; however, ligation can be tolerated. Careful volume resuscitation should follow portal vein ligation to avoid ↓ BP 2° fluid sequestration in the splanchnic bed. Simple ligation of the hepatic artery, preferably at one of its branches, is recommended for major hepatic artery injuries and patients in extremis. Ligation of the hepatic artery proper carries a significant risk of hepatic necrosis that may necessitate liver resection or transplantation. Shock and transfusion-related coagulopathy occurring in the immediate postop period are responsible for 80% of the deaths in liver injury patients. Control of hemorrhage remains the critical component in the successful management of liver injuries.

More than 90% of splenic injuries are caused by blunt trauma. Approximately 70% of these patients can be managed nonoperatively. Active contrast extravasation or contrast blush on CT scan warrants angioembolization. Hemodynamic instability or failure of nonoperative management (up to 20% failure rate) necessitates **splenectomy**. Grades IV (active intraperitoneal bleeding) and V (shattered/avulsed injuries) are associated with the highest rates of splenectomy. Massive bleeding from the LUQ is probably caused by splenic injury. In trauma, the spleen is delivered into the wound by blunt dissection. The spleen is removed by cross-clamping the hilum and dividing the short gastric vessels. The LUQ is packed and reinspected for hemostasis once better resuscitation is provided. The splenic salvage rate in the pediatric population approaches 90%.

Usual preop diagnosis

Hepatic injury; splenic injury

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Midline abdominal

Special instrumentation	Active rewarming device; cell salvage
Unique considerations	Most emergency laparotomies require close cooperation between anesthesiologist and surgeon. The procedure may need to be interrupted for fluid or blood resuscitation if patient becomes hypotensive. The entire chest and abdomen, including both groins, need to be accessible to the surgeon; ECG electrodes are placed preferably on the patient's back.
Antibiotics	Cefazolin 2 g iv
Surgical time	1–2 h
Closing considerations	After massive fluid resuscitation, concern over compartment syndrome may preclude primary fascial closure. The abdomen may be left open for later closure.
EBL	2–10 L (transfusion requirement 6–20 U)
Postop care	ICU: shock and coagulopathy management; monitoring for possible rebleeding and abdominal compartment syndrome
Mortality	Liver: 10% Spleen: 10% (usually due to associated injuries)
Morbidity	Liver—perihepatic abscess: 10% Spleen—septic complications: 7%
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	Any age
Male:Female	3:1
Incidence	Liver: 30% of patients requiring laparotomy for trauma Spleen: 10% of patients requiring laparotomy for trauma
Etiology	Liver: penetrating wound (80%) with gunshot wounds responsible for 60% and stab wounds for 40%

Associated conditions

Spleen: blunt trauma (90%), mostly due to MVC
Liver: isolated hepatic injury (30%); injury to one or two other organs (50%), with the diaphragm, major vascular structures, stomach, lung, and colon being most common

ANESTHETIC CONSIDERATIONS

See Anesthetic Conditions for Abdominal Trauma Surgery.

ABDOMINAL TRAUMA: VASCULAR INJURIES

SURGICAL CONSIDERATIONS

Description

The incidence of abdominal vascular injuries in blunt abdominal trauma patients is ~5–10%. Penetrating abdominal trauma results in major abdominal vascular injury for 10% of patients, compared with 20–25% for gunshot injuries. Blunt injuries typically lead to avulsion of vessels, intimal tears with secondary thrombosis, or traumatic false aneurysm. Penetrating abdominal vascular injuries can additionally lead to blast injuries and free hemorrhage or arteriovenous fistula formation.

Initial resuscitation of the patient with abdominal vascular injuries depends on the patient's condition. In blunt trauma, hypotensive patients with a positive eFAST or hypotensive patients with penetrating injury, time in the emergency room should be minimized and transfer to the operating room should be near immediate. Multiple large-bore catheters should be inserted in the upper extremities, or if necessary, central venous access should be obtained. Because of probable intra-abdominal venous injury, lower extremity venous access is not indicated. Blood replacement during resuscitation is done preferably with type-specific blood. It is a good practice to have 2 U of O-neg blood available immediately in the ED in case there is no time for a limited cross-match. Efforts to limit hypothermia should start as soon as the patient arrives (use of warmed fluids and high-flow blood warmers and covering the patient with warm blankets or a forced-air warming blanket). Emergency department resuscitative thoracotomy and descending aorta compression or cross-clamping should be considered in select situations, such as witnessed loss of vital signs and penetrating injury. Even then, survival rates are dismal.

Injuries to the abdominal vessels can be grouped into four regions, which require different surgical approaches:

Midline supramesocolic hemorrhage or hematoma (superior to the transverse mesocolon) is usually 2° injury of the suprarenal aorta, celiac axis, proximal superior mesenteric artery, or proximal renal artery. Proximal aortic control should be obtained at the hiatus by either aortic compression or manually by entering the lesser sac and digitally splitting the muscle fibers of the crura. Once this is done, direct access to the vessels is achieved through medial visceral rotation of all left-sided viscera (**Mattox maneuver**). Injuries to the aorta are then repaired directly or with the appropriate graft. An injured celiac axis probably can be ligated safely if the remaining visceral vessels are intact. Superior mesenteric artery injuries must be repaired, usually with a jump graft. Repair of the superior mesenteric vein is preferred, but the vein may be ligated if complex injuries are present. These patients require substantial fluid resuscitation postop and are at high risk for abdominal compartment syndrome. Injuries to the IVC or right side of the aorta can be exposed by **duodenal mobilization (Kocher maneuver)** and **right-sided medial visceral rotation (Cattell-Braasch)**.

Midline inframesocolic hemorrhage or hematoma results from infrarenal aorta or IVC injury. Exposure is obtained by incising the posterior peritoneum in the midline after displacement of the small bowel and cephalic retraction of the transverse mesocolon. A proximal aortic clamp is then placed just below the left renal vein, with a distal clamp near the aortic bifurcation. The defect is repaired primarily, using patch aortoplasty, end-to-end anastomosis, or a graft. If the aorta is intact and an inframesocolic hematoma seems to be more extensive on the right side, or if there is active bleeding coming through the base of the mesentery, then injury to the IVC should be suspected. Access to the infrahepatic IVC is preferably obtained by mobilization of the right colon and duodenum. Proximal and distal controls are best obtained by either digital compression or two sponge sticks. The injury is then repaired directly. Blind clamping should be avoided, but occasionally, with good exposure, a **Satinsky clamp** can be placed. In young patients with exsanguinating hemorrhage, the infrarenal IVC can be ligated, providing time for appropriate fluid management. These patients require significant fluid postop, and **leg fasciotomies** should be performed.

Lateral perirenal hematoma or hemorrhage suggests injury to the renal vessels or kidney. In patients with blunt abdominal trauma who have a negative abdominal CT, IVP, or arteriogram, surgery is not required. A penetrating injury usually requires surgical exploration. Vascular control of the ipsilateral renal artery is obtained before the hematoma is entered. If there is active bleeding from the kidney or overlying

retroperitoneum, then the kidney is exposed via a lateral incision, and a vascular clamp is applied to the renal vessel. This usually is followed by **nephrectomy** (after palpation and a one-shot IVP to verify function of a normal contralateral kidney). If the contralateral kidney is missing or nonfunctional, then **back-table salvage surgery and autotransplantation** of the injured kidney should be attempted, despite low salvage rates.

Lateral pelvic hematoma or hemorrhage indicates injury to the iliac vessels. Pelvic hematoma 2° blunt trauma and pelvic fracture should not be explored. The majority of severe bleeding due to pelvic fractures is venous in origin and therefore amenable to a combination of preperitoneal pelvic packing techniques and external fixation of the pelvis. Preperitoneal pelvic packing employs laparotomy sponges in the preperitoneal space to tamponade bleeding and reduce the volume of retroperitoneal space. Primary control of arterial bleeding is by **angiography/embolization**. For penetrating injuries, vascular control is obtained at the aortic bifurcation proximally and close to the inguinal ligament distally. The internal iliac artery is best visualized by elevating common and external iliac arteries on vascular tapes. Unilateral internal iliac artery injuries can be ligated. Common or external iliac artery injuries can be repaired, or a graft can be inserted. Grafts may be used even in the presence of GI contamination, provided the abdomen is thoroughly irrigated, and the retroperitoneum is closed over the graft. A temporary intravascular shunt should be used in patients requiring damage control surgery. Injuries to the iliac veins are treated with lateral venorrhaphy or ligation.

Additional Considerations and Techniques

Resuscitative Endovascular Balloon Occlusion of the Aorta (REBOA)

While hemorrhage from extremity trauma is compressible, torso hemorrhage is difficult to control and a leading cause of trauma-related mortality. Significant bleeding from the major vessels can lead to rapid exsanguination. In addition to damage control techniques such as open cross-clamping of the aorta (discussed above), REBOA is a useful and less invasive alternative for temporary hemorrhage control. It acts to reduce the amount of bleeding distal to the site of occlusion to allow time for resuscitation and definitive management.

In the context of available resources, REBOA should be considered for those patients with a pulse whose systolic blood pressure is persistently below 80 mm Hg. The level at which the balloon is placed depends upon the suspected location of hemorrhage (see Fig 7.13-7). Zone 1 level aortic occlusion is useful in circumstances of abdominal exsanguination. Severe pelvic fracture bleeding might warrant infrarenal, or Zone III occlusion. Another advantage of this modality is that femoral access can be obtained

preemptively during trauma resuscitation and subsequently used for balloon catheter access if needed. Placement requires proper endovascular technique and fluoroscopic guidance. Once common femoral arterial access is obtained and the sheath secured, the balloon catheter is inserted and judiciously inflated until operative control of bleeding is obtained. The balloon is then deflated and the sheath removed. Every attempt should be made to minimize occlusion time, as thromboembolic complications and end-organ damage can result.

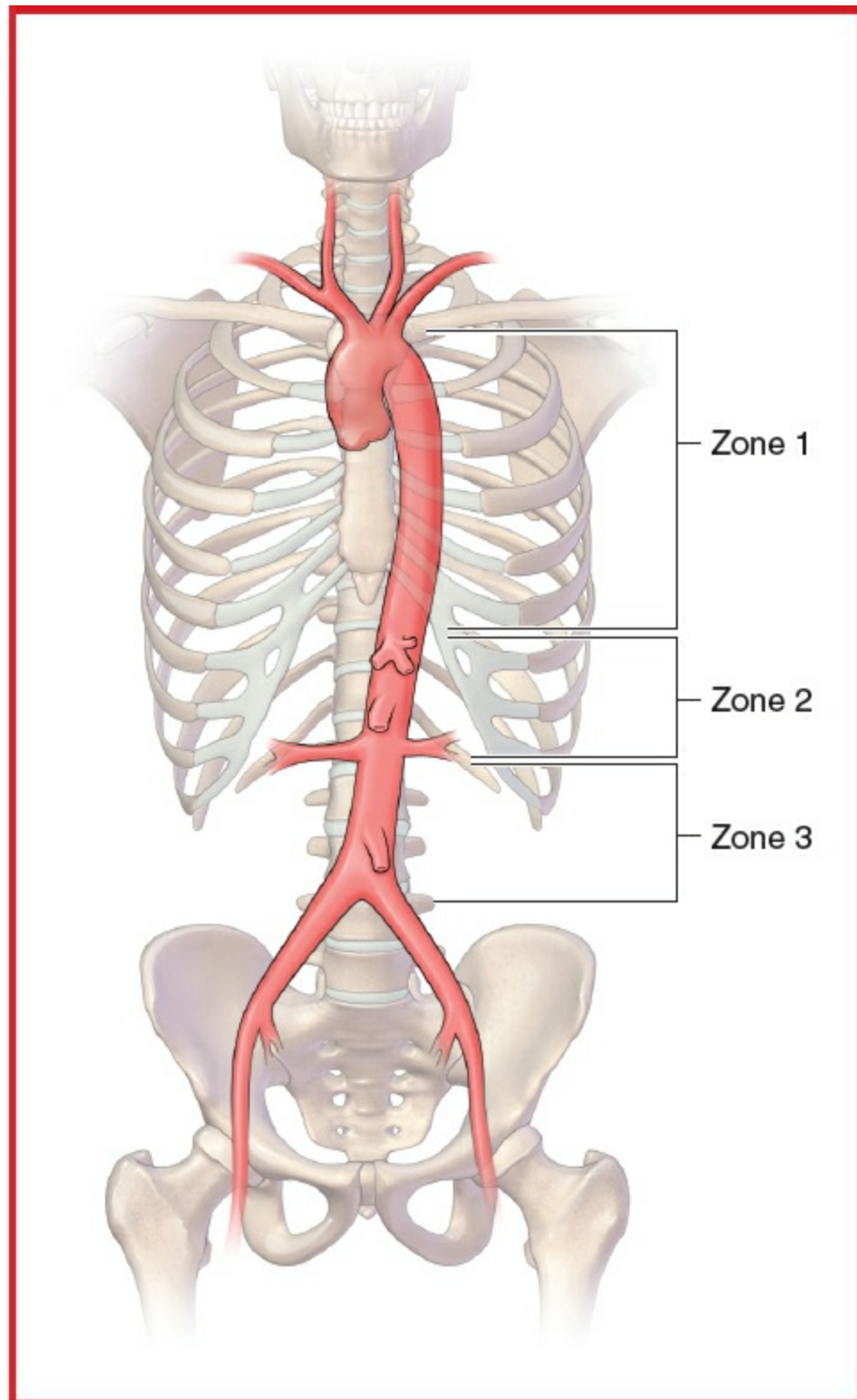


Figure 7.13-7 REBOA zones.

Suggested Reading

1. See [p. 839](#) and end of this chapter.

Usual preop diagnosis

■ SUMMARY OF PROCEDURE

Position	Supine, with left arm abducted 90°
Incision	Midline abdominal
Special instrumentation	Autotransfusion device (e.g., Cell Saver®); thoracotomy tray; aortic compressor
Unique considerations	Prevent heat loss and warm all infusions and irrigation solutions. Ventilator warming should be used for cold patients. For the patient requiring massive transfusion, ratio of 1 U FFP:1.5 U PRBCs is useful. After prolonged aortic cross-clamp, prophylactic administration of NaHCO ₂ (1–2 mEq/kg) may be indicated to prevent “washout” acidosis.
Antibiotics	Cefazolin 2 g iv
Surgical time	Variable
Closing considerations	Once vascular injuries are repaired, hepatic injuries are controlled with packing, bowel injuries are closed with staplers, and the abdominal wall is closed temporarily.
EBL	5–10 L
Postop care	ICU; patients after infrarenal vena cava ligation require volume expansion and prevention of lower extremity venous pooling. Elastic wraps and lower extremity elevation should be maintained for at least 1 wk. Similarly, after superior mesenteric vein ligation, splanchnic hypervolemia requires vigorous fluid resuscitation and lasts ~3 d.
Mortality	Combined injury to the suprarenal aorta and IVC: 100% Aorta: 60% Infrarenal abdominal aorta: 50% Superior mesenteric artery: 40–80% Iliac artery: 40%

	Iliac vein: 30% Infrarenal vena cava: 30% Superior mesenteric vein: 20% Renal artery: 15%
Morbidity	Abdominal compartment syndrome (↑↑ abdominal pressure → ↑ PIP + ↓ UO) Acute renal failure Intra-abdominal infection Fistula
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	Typically young adult
Male:Female	Male > female
Incidence	15% of patients with abdominal trauma
Etiology	10% of penetrating stab wounds and 25% of gunshot wounds to the abdomen will cause a major vascular injury.
Associated conditions	Multiple vascular injuries; hollow viscus perforation; fecal contamination

ANESTHETIC CONSIDERATIONS FOR ABDOMINAL TRAUMA SURGERY

Abdominal injuries range from relatively simple penetrating injuries (e.g., a stab wound) to severe blunt trauma (e.g., liver lacerations and pelvic fractures) and are often associated with massive hemorrhage. Head injuries and spinal fractures may complicate management plans. The ability to provide rapid, aggressive volume replacement is often the key to survival. Coagulopathy (DIC) and hypothermia present additional challenges.

PREOPERATIVE

Respiratory	Injuries such as hemothorax and pneumothorax may be present, requiring thoracostomy tube placement. Fractures of the 1st or 2nd rib should prompt a
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detailed search for associated intrathoracic injuries, especially injuries to the great vessels. Injuries to the great vessels may present on imaging with a widened mediastinum or apical pleural capping. Multiple rib fractures are often associated with pulmonary contusions, which may not be apparent on the initial CXR but can progressively impair oxygenation.

Tests: CXR (PA + lateral views). Ultrasound can identify the presence of a pneumothorax or pleural effusion. CT of the chest if the patient's condition permits. ABG

Cardiovascular

BP and HR should be followed, and responses to fluid resuscitation noted. Tachycardia can maintain an adequate BP with reduced pulse pressure, despite 25–30% loss of blood volume. Attempt to quantify overt blood loss (e.g., scalp lacerations, open fracture sites). Blunt chest trauma (e.g., steering wheel contact) may result in myocardial contusion with various dysrhythmias, most often premature ventricular or atrial complexes.

Tests: Serial Hct; ABG, continuous telemetry monitoring or serial ECGs. Ultrasound—eFAST, TTE, TEE

Neurologic

↓ BP and ↓ CO may compromise cerebral perfusion. Associated head injury also can contribute to alterations in mental status. Seek physical evidence of open head or closed head injuries, such as palpable depressions of the skull or scalp lacerations, abrasions, or contusions. Pupil size and reactivity should be noted. Intubation in the ER is necessary for patients who are unable to protect their airway, require hyperventilation, or are combative and unable to cooperate with medical staff for exam and treatment. In general, any patient with a GCS ≤ 8 (no spontaneous eye opening, inappropriate or incomprehensible speech, and only reflexive motor responses) requires intubation. GCS ≤ 8 indicates

	significant brain injury. Motor and sensory deficits may reflect spinal cord injury and may be associated with neurogenic (“spinal”) shock, particularly with upper thoracic or cervical cord injuries. Tests: C-spine x-rays (lateral view, including C7) is a good screening exam—✓ for altered vertical alignment and unequal disk interspaces. CT scan or MRI of the head: ✓ for gross asymmetry, hemorrhage, or midline shift.
Musculoskeletal	Known or suspected C-spine injuries require intubation precautions. In urgent cases, intubation without neck extension is achieved with an assistant providing in-line stabilization of the head in the neutral position. Basilar skull fractures contraindicate nasal ET or NG tubes. Pelvic and femur fractures may represent sources of significant (>1000 mL) occult blood loss. Tests: Radiographs of the C-spine (see above), skull, pelvis/extremities
Hematologic	Preop T&C appropriate. Depending on estimations of prior, ongoing, and anticipated surgical blood losses, blood products should be available (preferably cross-matched units, but O-neg or type-specific blood if cross-matching not complete). Tests: Serial Hct; T&C; ✓ Ca⁺⁺ and K⁺ following massive transfusions
Laboratory	Other tests as indicated from H&P or suspected injuries, including electrolytes, liver panel, toxicology screen, and blood alcohol level.
Premedication	Premedication is rarely useful due to the urgency of the procedures and the need to have an alert, responsive patient for serial evaluations of mental status or abdominal pain. Sedative premedication should be avoided in hemodynamically unstable patients and those with probable head injuries. Virtually all patients are considered to have full stomachs.

◆ INTRAOPERATIVE

Anesthetic technique

GETA with full-stomach precautions

Induction

Before induction, a variety of laryngoscope blades, ETT tube sizes, and ideally intubating adjuncts including video laryngoscope and FOB in preparation for potential difficult airway should be ready. LMA may be useful for providing temporary airway control without C-spine manipulation, if direct laryngoscopy is difficult. LMA may be used as a conduit for ET intubation (fiberoptic or fast-track LMA).

Most often, preoxygenation is followed by a rapid-sequence iv induction with cricoid pressure (Sellick maneuver) using propofol (1–2.5 mg/kg) and succinylcholine (1.0–1.5 mg/kg). If hypotension is present or a concern, alternate induction agents (e.g., ketamine 0.5–2.0 mg/kg iv or etomidate 0.1–0.3 mg/kg iv) may be used. Axial head and neck stabilization are necessary if C-spine injury is present or suspected.

Some trauma patients are intubated in the ED. Verify ETT placement by auscultation and ETCO₂ monitoring. Very low ETCO₂ values may be obtained in patients with ↓↓CO. Ventilation with 100% O₂ and paralysis with muscle relaxant are appropriate. Ongoing fluid resuscitation should be continued during this time. Consider sending lab studies/ABG at this point and ensuring availability of blood products.

Maintenance

Titrate O₂/air, muscle relaxants, narcotics, and volatile agents. Avoid N₂O in the presence of pneumothorax, pneumocephalus, bowel distension, or prolonged procedures. Shorter-acting agents (e.g., volatile agents, remifentanyl, rocuronium), carefully titrated, may be preferred in patients with head injuries to facilitate early postop assessment of neurologic status. If ↓ BP precludes use of volatile

	anesthetics and opiates, consider low-dose scopolamine (0.1–0.3 mg iv) or ketamine (0.25 mg/kg/15–30 min) for amnesia. Use forced-air warming blanket(s) and warmed iv fluids, and consider elevated OR temperatures to maintain normothermia.	
Emergence	Before extubation, patient should be awake and able to protect his/her airway and should be hemodynamically stable and spontaneously ventilating with adequate tidal volumes and respiratory rates. Consider postop mechanical ventilation and ICU care for trauma patients with pulmonary compromise/trauma, ongoing acidosis, hemodynamic instability, ongoing or anticipated significant fluid and blood product resuscitation, coagulopathies, intoxication, or neurologic impairment.	
Blood and fluid requirements	Anticipate large blood loss IV: 14–16 ga × 2 or 7–9 Fr × 2 BSS Fluid warmer Rapid-infusion device T&C PRBCs, ± other blood products (FFP, platelets, cryo)	Large blood losses may have already occurred and continue to occur, depending on the mechanism of injury (e.g., liver lacerations, major vascular injury, pelvic fractures). Early determination of availability of blood products advised. BSS, colloid, and blood products should be given to reestablish euvolemia (following BP, HR, CVP, urine output, Hct, ABGs). With massive transfusion, Plts and FFP will be needed. Consider a transfusion ratio of 1 U PRBC:1 U FFP:1 U platelet during massive and ongoing hemorrhage. Vigilant assessment of bleeding, EBL,

		and hemodynamic status is essential during ongoing massive bleeding. Postop hypothermia is best minimized by warming all iv and irrigating fluids, maintaining OR temperature @ 78–80°F, humidifying inspired gases, and using forced-air and circulating water warming blankets.
Monitoring	ASA standard monitors Urinary catheter ± Arterial line ± CVP line ± PA catheter ± TEE	Monitor patient during transport to OR and apply standard monitoring as soon as patient enters the OR. Arterial line useful in unstable patients or when frequent lab sampling is anticipated. CVP line or PA catheter may be useful if vasoactive infusions needed or if ventricular dysfunction apparent. Placement of additional monitoring should be accomplished without delaying the surgical control of hemorrhage or without interrupting aggressive volume resuscitation.
Positioning	✓ and pad pressure points ✓ eyes	If C-spine has not been cleared by both radiographically and clinically, the neck should remain immobilized intraop and postop.
Complications	Hypothermia	Preexisting hypothermia

	Awareness	common in trauma patient. Measures to manage hypothermia early are key and include warming the OR, use of forced-air warming, and administering warm iv fluids. May be unavoidable in unstable or hypotensive patients. Rx: Consider scopolamine 0.2–0.4 mg iv or midazolam 1–2 mg iv in such situations.
	Coagulopathy	Causes include dilutional, hypothermia, and brain injury. Assess status and treat early.
	Aspiration	Assume full stomach; aspiration may have occurred before OR.

POSTOPERATIVE

Complications	Hypothermia	Active warming of fluids and blood products and forced-air warming blankets and warm room temperatures should be continued in PACU/ICU if hypothermia persists.
	Atelectasis, V/Q mismatch	Pulmonary compliance may decrease with large volumes of fluid replacement. Pulmonary contusions may aggravate this problem and severely compromise oxygenation and ventilation, requiring high-inspired O ₂

	Coagulopathy, VTE	concentration, high PIP, and PEEP. Increased abdominal pressures may also affect ventilation and increase risk of V/Q mismatch. Causes of coagulopathy include dilutional, hypothermic, brain injury, and development of SIRS/sepsis. Trauma is a risk factor for VTE.
Pain management	PCA or parenteral narcotics Epidural narcotic	Patients with rib fractures benefit from epidural narcotic and/or low-dose local anesthetic infusions.
Tests	Hgb/Hct Coags/TEG CXR	PT/PTT, Plt counts, TEG if unexplained bleeding postop Fibrinogen, fibrin split products, if DIC suspected

PEDIATRIC TRAUMA

AIRWAY AND VASCULAR ACCESS

Description

Trauma remains the leading cause of death in children and young adults. Children younger than 15 yr are victims in about 25% of all trauma occurring in the United States. The leading causes of injury and death vary by age group. According to the National Pediatric Trauma Registry, the leading cause of pediatric injury/death among children 5–19 yr old is a motor vehicle collision. Falls remain the most common cause of severe nonfatal injury in children under the age of 15. The majority of pediatric injuries that occur are 2° blunt trauma, and infants <2 yr of age are known to have higher mortality rates for the same level of injury compared to older children.

Children have unique differences in physiology and anatomy, which must be acknowledged at the time of resuscitation. Due to their smaller FRC, children tolerate much shorter periods of hypoxia. Confirmation of a patent airway is the essential first

step. The best method for restoring airway patency is the **jaw-thrust maneuver** and removal of any debris from the mouth. In infants, the head is relatively large compared to the body, larger tongue size and a floppy epiglottis impede visualization of the airway, and the subglottic airway is narrower with less developed cartilage. Padding the torso and allowing the occiput to rest on the supporting surface allows for more favorable airway alignment. The Broselow tape measure (relates body length to weight and precalculated medication doses) can be used to estimate endotracheal tube size and facilitate weight-based medication dosing. The most common reason for intubation in the pediatric trauma patient is loss of consciousness or as part of resuscitation from shock. Only 2% of children sustaining trauma will present with complete mechanical obstruction to the airway. In the rare child who presents with acute airway obstruction, **needle cricothyrotomy** is the preferred method of securing the airway until definitive airway control can be achieved. This technique of ventilation uses the principle of jet insufflation. **Surgical cricothyrotomy** in children results in a high incidence of subglottic stenosis, but it is still a viable option in children >10 if needle cricothyrotomy fails to be effective.

Because infants are obligatory nasal and diaphragmatic breathers, fractures and soft tissue injuries that occlude the nostrils may actually obstruct the airway. Because air swallowed by the infant or insufflated into the stomach may cause acute gastric distension and restrict diaphragmatic excursion, the stomach should be decompressed with an OG tube.

Once the airway is secured and breathing is ensured, attention should be given to the circulation. Differences in cardiovascular physiology also exist. Children may compensate for as much as 25–30% of circulating blood volume loss without a change in BP. In the noncrying child, the SBP should be $\sim 80 + \text{their age in years} \times 2$. The most sensitive indicator of shock in children is heart rate. Poor peripheral perfusion, decreased level of consciousness, and $\downarrow$ UO are also suggestive of hypovolemia. IV access must be obtained rapidly to begin crystalloid resuscitation in any child with impending signs of shock. If the peripheral iv access is difficult to obtain, saphenous vein cut down at the saphenofemoral junction should be performed. In infants, if iv access cannot be obtained within 2 min, intraosseous access should be attempted (see below and [Fig. 7.13-8](#)). After iv access has been obtained, as many as three boluses of crystalloid, using a volume of 20 mL/kg, can be given. If the hypovolemic shock state has not been reversed after the second bolus, and other causes of shock—such as spinal injury, cardiac tamponade, or pneumothorax—are excluded, blood (10 mL/kg) should be administered without delay.

Another important problem in the management of pediatric trauma is related to a high ratio of body surface area to body mass and a lack of substantial subcutaneous tissue. A small infant who is hypothermic may be refractory to therapy; therefore, every attempt should be made to prevent heat loss, and all iv fluids should be warmed.

Needle cricothyrotomy: With the head in neutral position (which may require placement of towels under the shoulders), the neck should be prepped from the jaw to the chest. The neck is protected by in-line immobilization. The cricothyroid membrane should be identified and the thyroid cartilage immobilized with the surgeon's left hand. The cricothyroid membrane is punctured perpendicularly with a 14- to 16-ga iv catheter over a needle. The needle is then redirected caudally, the catheter slid off into the trachea, and jet insufflation initiated. Placement of a permanent airway should follow.

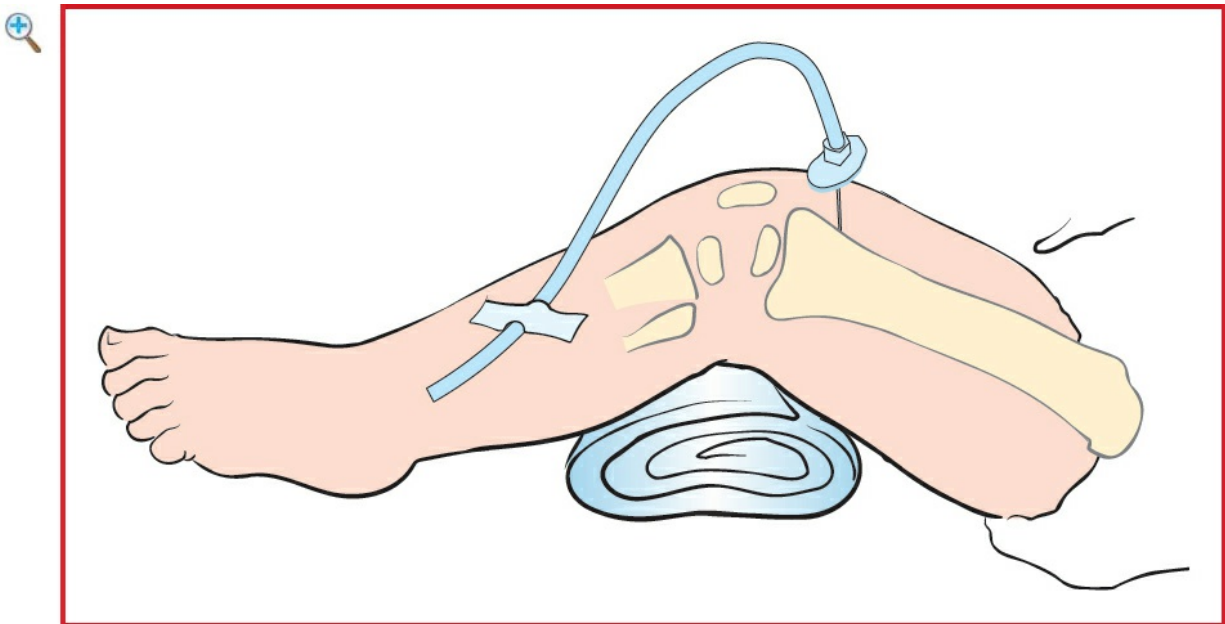


Figure 7.13-8 Intraosseous infusion. (Reproduced with permission from Hodge Dee III: Intraosseous infusion. In: King C, Henretig FM, eds: Textbook of Pediatric Emergency Procedures, 2nd edition. Lippincott Williams & Wilkins, Philadelphia: 2008.)

Intraosseous infusion: After skin preparation, an incision is made 2 cm distal to the tibial tuberosity on the flattened medial aspect of the tibia. An 18- to 20-ga spinal needle (with obturator) can be used in children <18 mo of age. Older patients may require the use of a 13- to 16-ga bone marrow biopsy needle. Pressure and rotational motion are applied in a direction perpendicular to the bone until a decrease in resistance is felt. The position of the needle can be confirmed by bone marrow or blood aspiration. This route can be used for rapid fluid infusion, and most resuscitation medications can be given this way. Intraosseous infusion, however, is only an emergency maneuver and should be used to restore circulating volume to the level that

enables more permanent iv access.

Saphenous cutdown: The groin should be prepped and draped and a curvilinear incision made 1–2 cm below and parallel to the inguinal ligament. The saphenous vein is identified at the saphenofemoral junction medial to the femoral artery. Two ligatures are passed underneath; the distal ligature is tied and used to apply tension to the vein. The vein is punctured with a scalpel blade (No. 11) and cut, creating a small flap. Tension applied to the proximal ligature reduces back bleeding during cannulation. A catheter is then introduced, the proximal ligature is tied, and the distal ligature is used to secure the catheter in place.

Usual preop diagnosis

Airway obstruction; hypovolemic shock; hypovolemic shock with difficult iv access

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